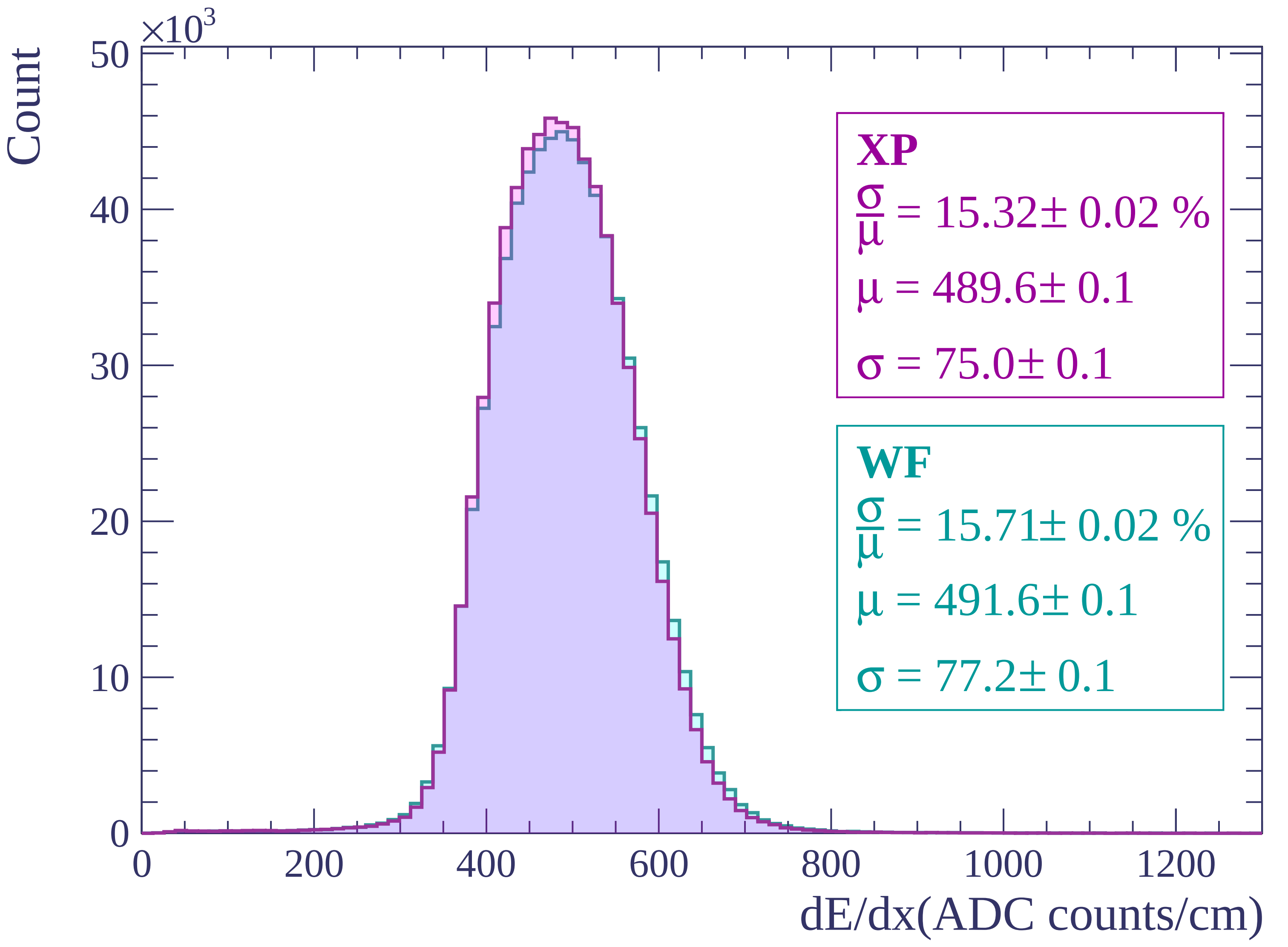
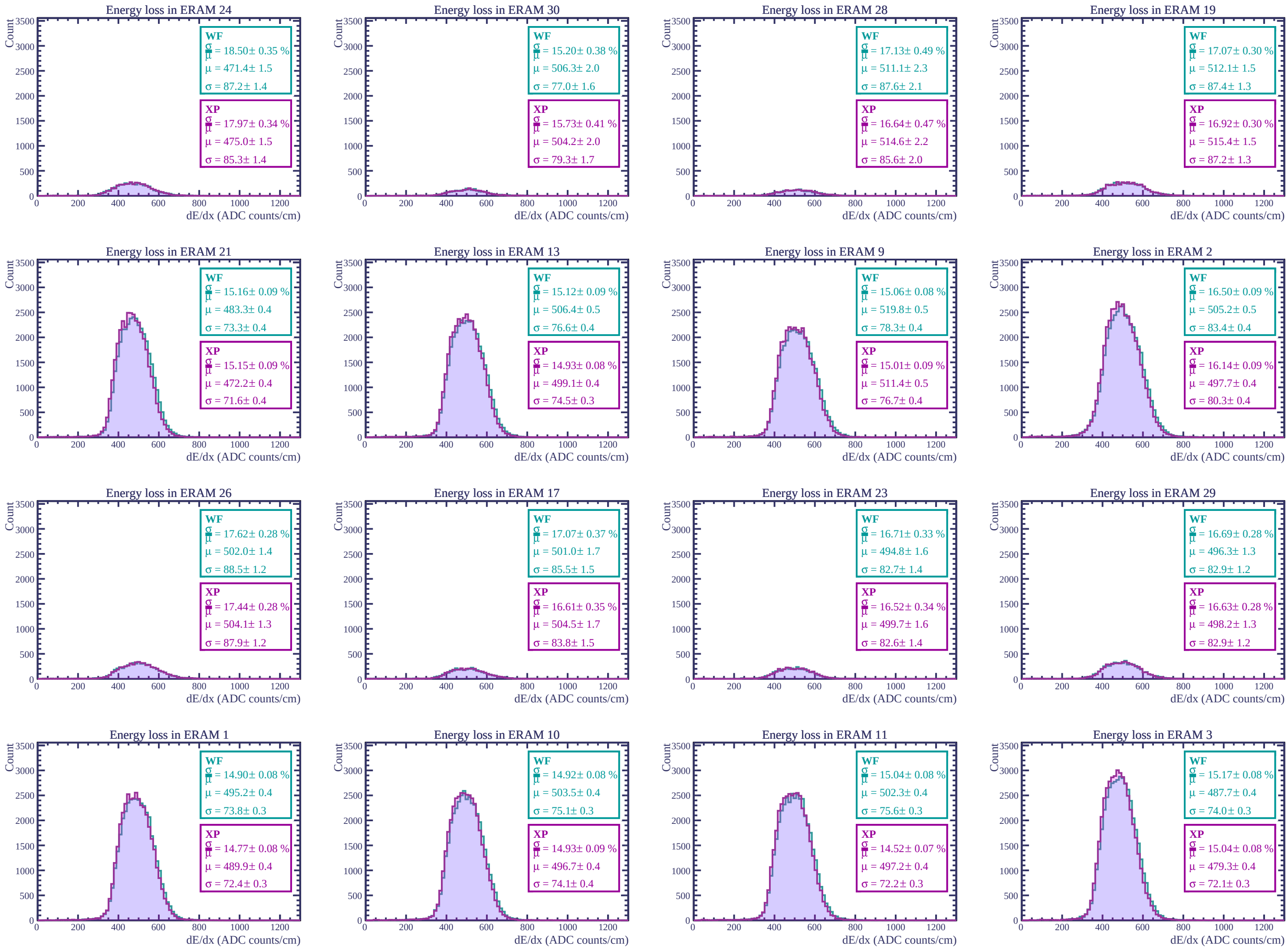
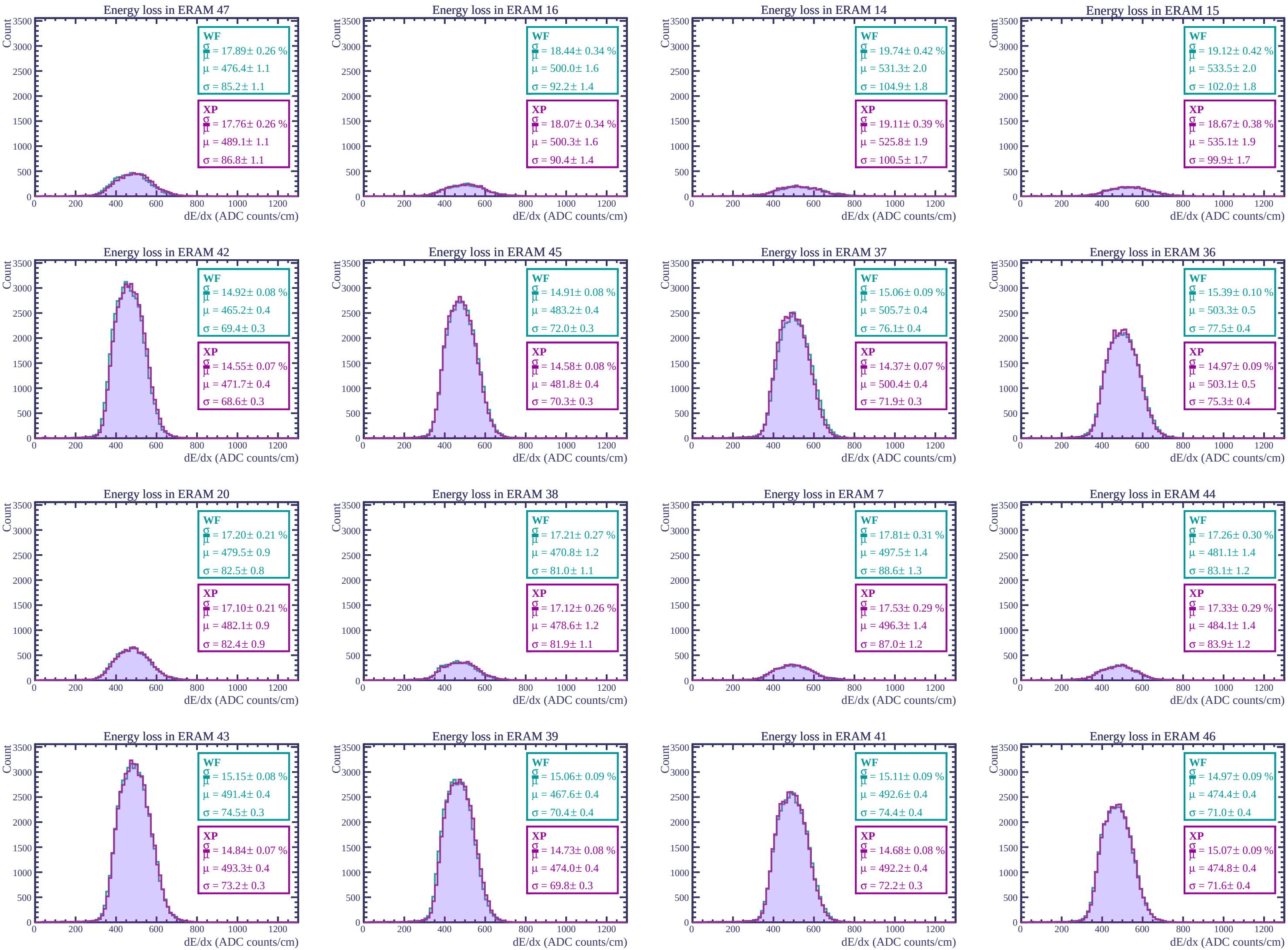
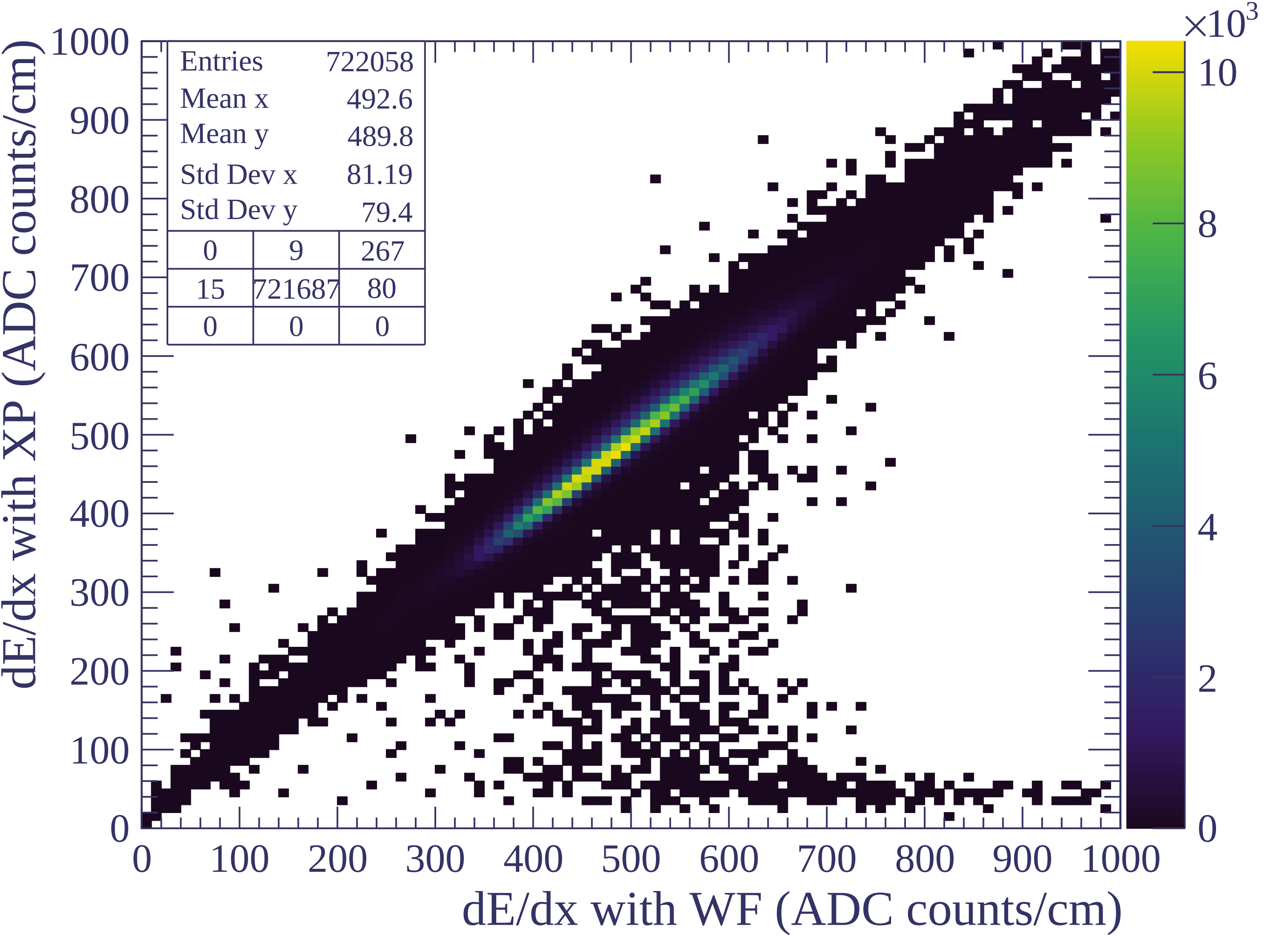


Count
×10^3
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx(ADC counts/cm)
XP
σ/μ = 15.32± 0.02 %
μ = 489.6± 0.1
σ = 75.0± 0.1
WF
σ/μ = 15.71± 0.02 %
μ = 491.6± 0.1
σ = 77.2± 0.1

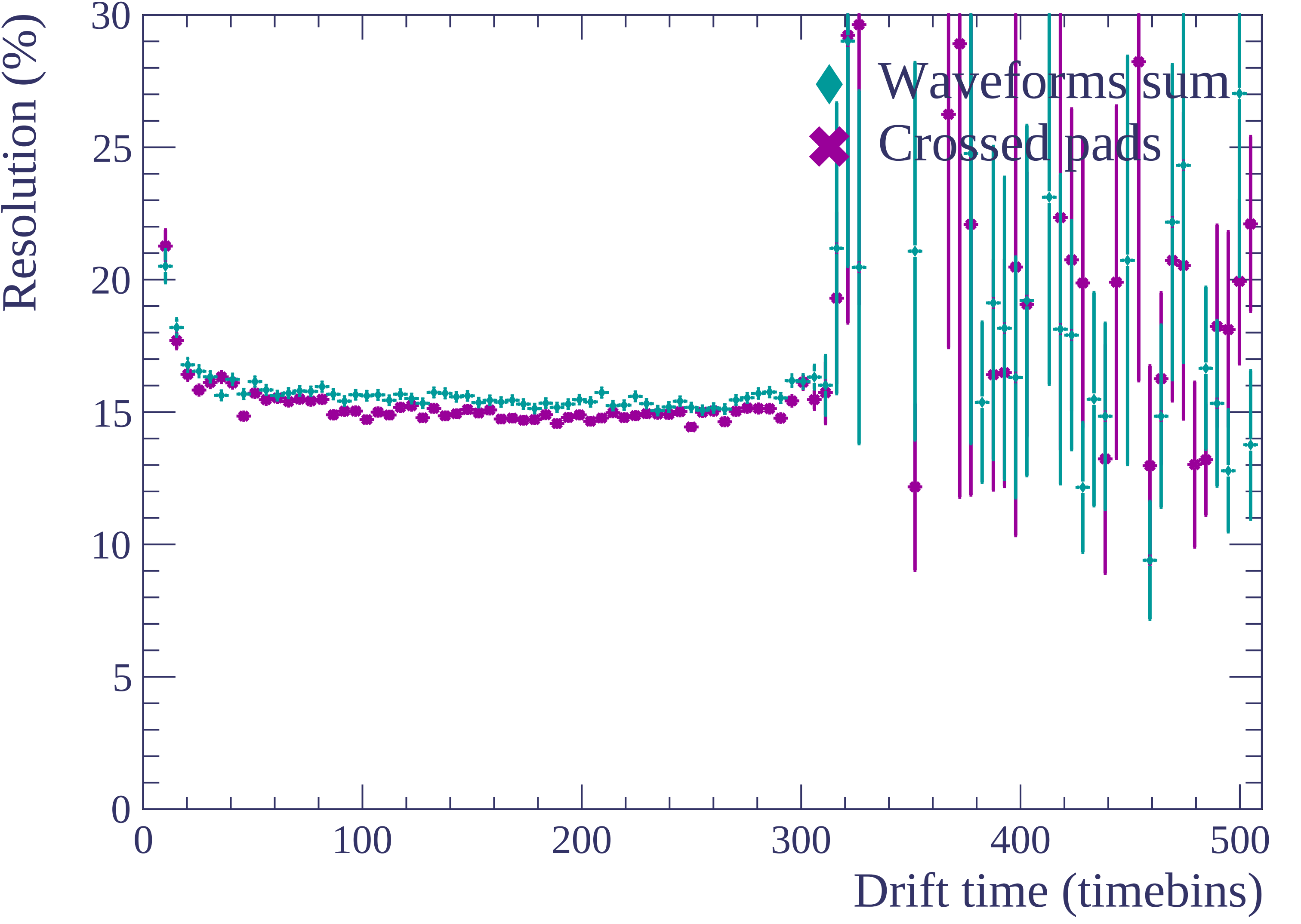


Energy loss in ERAM 24
WF σ/μ = 18.50± 0.35 % μ = 471.4± 1.5 σ = 87.2± 1.4
XP σ/μ = 17.97± 0.34 % μ = 475.0± 1.5 σ = 85.3± 1.4
Energy loss in ERAM 30
WF σ/μ = 15.20± 0.38 % μ = 506.3± 2.0 σ = 77.0± 1.6
XP σ/μ = 15.73± 0.41 % μ = 504.2± 2.0 σ = 79.3± 1.7
Energy loss in ERAM 28
WF σ/μ = 17.13± 0.49 % μ = 511.1± 2.3 σ = 87.6± 2.1
XP σ/μ = 16.64± 0.47 % μ = 514.6± 2.2 σ = 85.6± 2.0
Energy loss in ERAM 19
WF σ/μ = 17.07± 0.30 % μ = 512.1± 1.5 σ = 87.4± 1.3
XP σ/μ = 16.92± 0.30 % μ = 515.4± 1.5 σ = 87.2± 1.3
Energy loss in ERAM 21
WF σ/μ = 15.16± 0.09 % μ = 483.3± 0.4 σ = 73.3± 0.4
XP σ/μ = 15.15± 0.09 % μ = 472.2± 0.4 σ = 71.6± 0.4
Energy loss in ERAM 13
WF σ/μ = 15.12± 0.09 % μ = 506.4± 0.5 σ = 76.6± 0.4
XP σ/μ = 14.93± 0.08 % μ = 499.1± 0.4 σ = 74.5± 0.3
Energy loss in ERAM 9
WF σ/μ = 15.06± 0.08 % μ = 519.8± 0.5 σ = 78.3± 0.4
XP σ/μ = 15.01± 0.09 % μ = 511.4± 0.5 σ = 76.7± 0.4
Energy loss in ERAM 2
WF σ/μ = 16.50± 0.09 % μ = 505.2± 0.5 σ = 83.4± 0.4
XP σ/μ = 16.14± 0.09 % μ = 497.7± 0.4 σ = 80.3± 0.4
Energy loss in ERAM 26
WF σ/μ = 17.62± 0.28 % μ = 502.0± 1.4 σ = 88.5± 1.2
XP σ/μ = 17.44± 0.28 % μ = 504.1± 1.3 σ = 87.9± 1.2
Energy loss in ERAM 17
WF σ/μ = 17.07± 0.37 % μ = 501.0± 1.7 σ = 85.5± 1.5
XP σ/μ = 16.61± 0.35 % μ = 504.5± 1.7 σ = 83.8± 1.5
Energy loss in ERAM 23
WF σ/μ = 16.71± 0.33 % μ = 494.8± 1.6 σ = 82.7± 1.4
XP σ/μ = 16.52± 0.34 % μ = 499.7± 1.6 σ = 82.6± 1.4
Energy loss in ERAM 29
WF σ/μ = 16.69± 0.28 % μ = 496.3± 1.3 σ = 82.9± 1.2
XP σ/μ = 16.63± 0.28 % μ = 498.2± 1.3 σ = 82.9± 1.2
Energy loss in ERAM 1
WF σ/μ = 14.90± 0.08 % μ = 495.2± 0.4 σ = 73.8± 0.3
XP σ/μ = 14.77± 0.08 % μ = 489.9± 0.4 σ = 72.4± 0.3
Energy loss in ERAM 10
WF σ/μ = 14.92± 0.08 % μ = 503.5± 0.4 σ = 75.1± 0.3
XP σ/μ = 14.93± 0.09 % μ = 496.7± 0.4 σ = 74.1± 0.4
Energy loss in ERAM 11
WF σ/μ = 15.04± 0.08 % μ = 502.3± 0.4 σ = 75.6± 0.3
XP σ/μ = 14.52± 0.07 % μ = 497.2± 0.4 σ = 72.2± 0.3
Energy loss in ERAM 3
WF σ/μ = 15.17± 0.08 % μ = 487.7± 0.4 σ = 74.0± 0.3
XP σ/μ = 15.04± 0.08 % μ = 479.3± 0.4 σ = 72.1± 0.3
Count
dE/dx (ADC counts/cm)

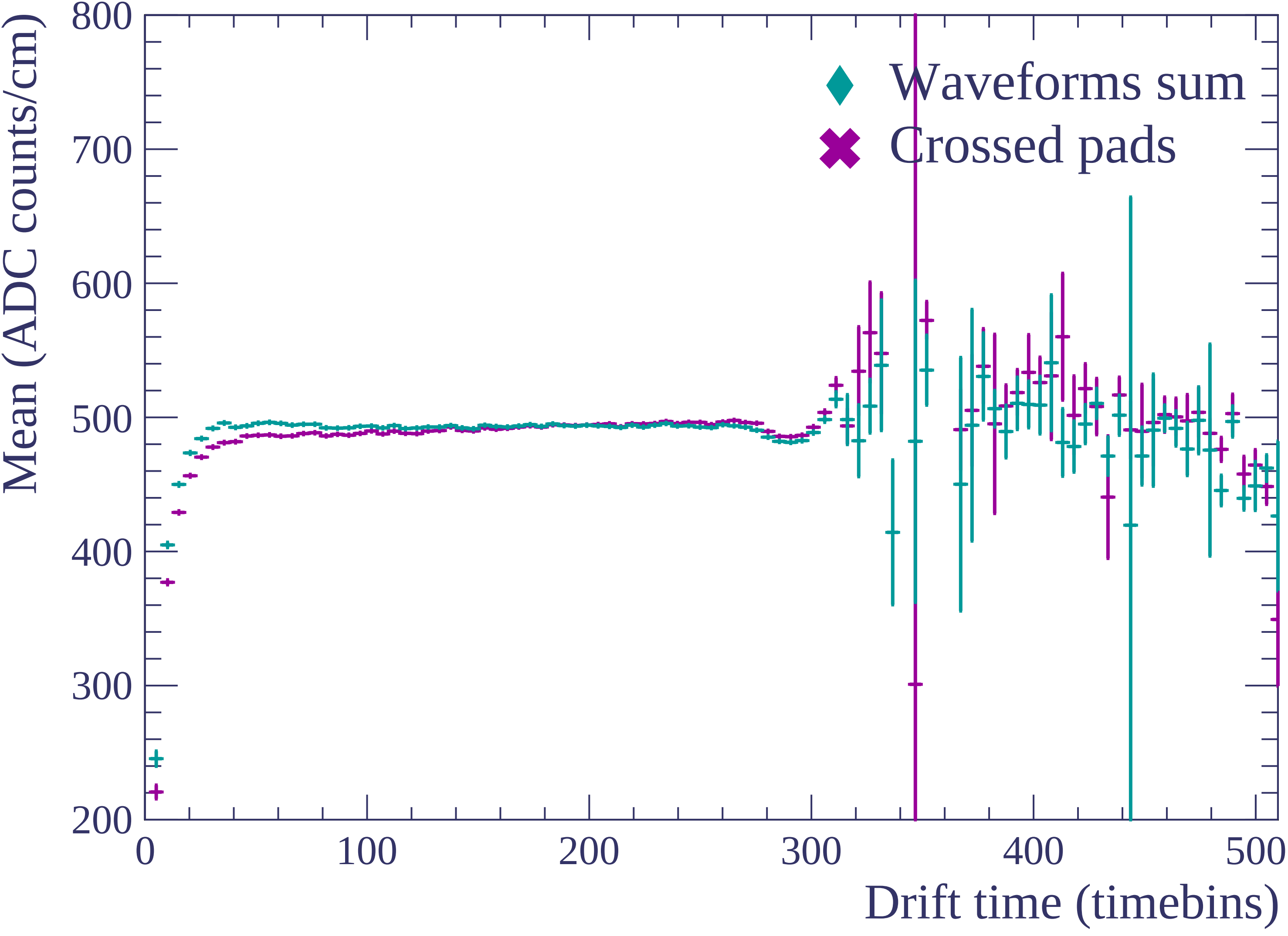

Energy loss in ERAM 47
WF
σ/μ = 17.89± 0.26 %
μ = 476.4± 1.1
σ = 85.2± 1.1
XP
σ/μ = 17.76± 0.26 %
μ = 489.1± 1.1
σ = 86.8± 1.1
Energy loss in ERAM 16
WF
σ/μ = 18.44± 0.34 %
μ = 500.0± 1.6
σ = 92.2± 1.4
XP
σ/μ = 18.07± 0.34 %
μ = 500.3± 1.6
σ = 90.4± 1.4
Energy loss in ERAM 14
WF
σ/μ = 19.74± 0.42 %
μ = 531.3± 2.0
σ = 104.9± 1.8
XP
σ/μ = 19.11± 0.39 %
μ = 525.8± 1.9
σ = 100.5± 1.7
Energy loss in ERAM 15
WF
σ/μ = 19.12± 0.42 %
μ = 533.5± 2.0
σ = 102.0± 1.8
XP
σ/μ = 18.67± 0.38 %
μ = 535.1± 1.9
σ = 99.9± 1.7
Energy loss in ERAM 42
WF
σ/μ = 14.92± 0.08 %
μ = 465.2± 0.4
σ = 69.4± 0.3
XP
σ/μ = 14.55± 0.07 %
μ = 471.7± 0.4
σ = 68.6± 0.3
Energy loss in ERAM 45
WF
σ/μ = 14.91± 0.08 %
μ = 483.2± 0.4
σ = 72.0± 0.3
XP
σ/μ = 14.58± 0.08 %
μ = 481.8± 0.4
σ = 70.3± 0.3
Energy loss in ERAM 37
WF
σ/μ = 15.06± 0.09 %
μ = 505.7± 0.4
σ = 76.1± 0.4
XP
σ/μ = 14.37± 0.07 %
μ = 500.4± 0.4
σ = 71.9± 0.3
Energy loss in ERAM 36
WF
σ/μ = 15.39± 0.10 %
μ = 503.3± 0.5
σ = 77.5± 0.4
XP
σ/μ = 14.97± 0.09 %
μ = 503.1± 0.5
σ = 75.3± 0.4
Energy loss in ERAM 20
WF
σ/μ = 17.20± 0.21 %
μ = 479.5± 0.9
σ = 82.5± 0.8
XP
σ/μ = 17.10± 0.21 %
μ = 482.1± 0.9
σ = 82.4± 0.9
Energy loss in ERAM 38
WF
σ/μ = 17.21± 0.27 %
μ = 470.8± 1.2
σ = 81.0± 1.1
XP
σ/μ = 17.12± 0.26 %
μ = 478.6± 1.2
σ = 81.9± 1.1
Energy loss in ERAM 7
WF
σ/μ = 17.81± 0.31 %
μ = 497.5± 1.4
σ = 88.6± 1.3
XP
σ/μ = 17.53± 0.29 %
μ = 496.3± 1.4
σ = 87.0± 1.2
Energy loss in ERAM 44
WF
σ/μ = 17.26± 0.30 %
μ = 481.1± 1.4
σ = 83.1± 1.2
XP
σ/μ = 17.33± 0.29 %
μ = 484.1± 1.4
σ = 83.9± 1.2
Energy loss in ERAM 43
WF
σ/μ = 15.15± 0.08 %
μ = 491.4± 0.4
σ = 74.5± 0.3
XP
σ/μ = 14.84± 0.07 %
μ = 493.3± 0.4
σ = 73.2± 0.3
Energy loss in ERAM 39
WF
σ/μ = 15.06± 0.09 %
μ = 467.6± 0.4
σ = 70.4± 0.4
XP
σ/μ = 14.73± 0.08 %
μ = 474.0± 0.4
σ = 69.8± 0.3
Energy loss in ERAM 41
WF
σ/μ = 15.11± 0.09 %
μ = 492.6± 0.4
σ = 74.4± 0.4
XP
σ/μ = 14.68± 0.08 %
μ = 492.2± 0.4
σ = 72.2± 0.3
Energy loss in ERAM 46
WF
σ/μ = 14.97± 0.09 %
μ = 474.4± 0.4
σ = 71.0± 0.4
XP
σ/μ = 15.07± 0.09 %
μ = 474.8± 0.4
σ = 71.6± 0.4
Count
dE/dx (ADC counts/cm)



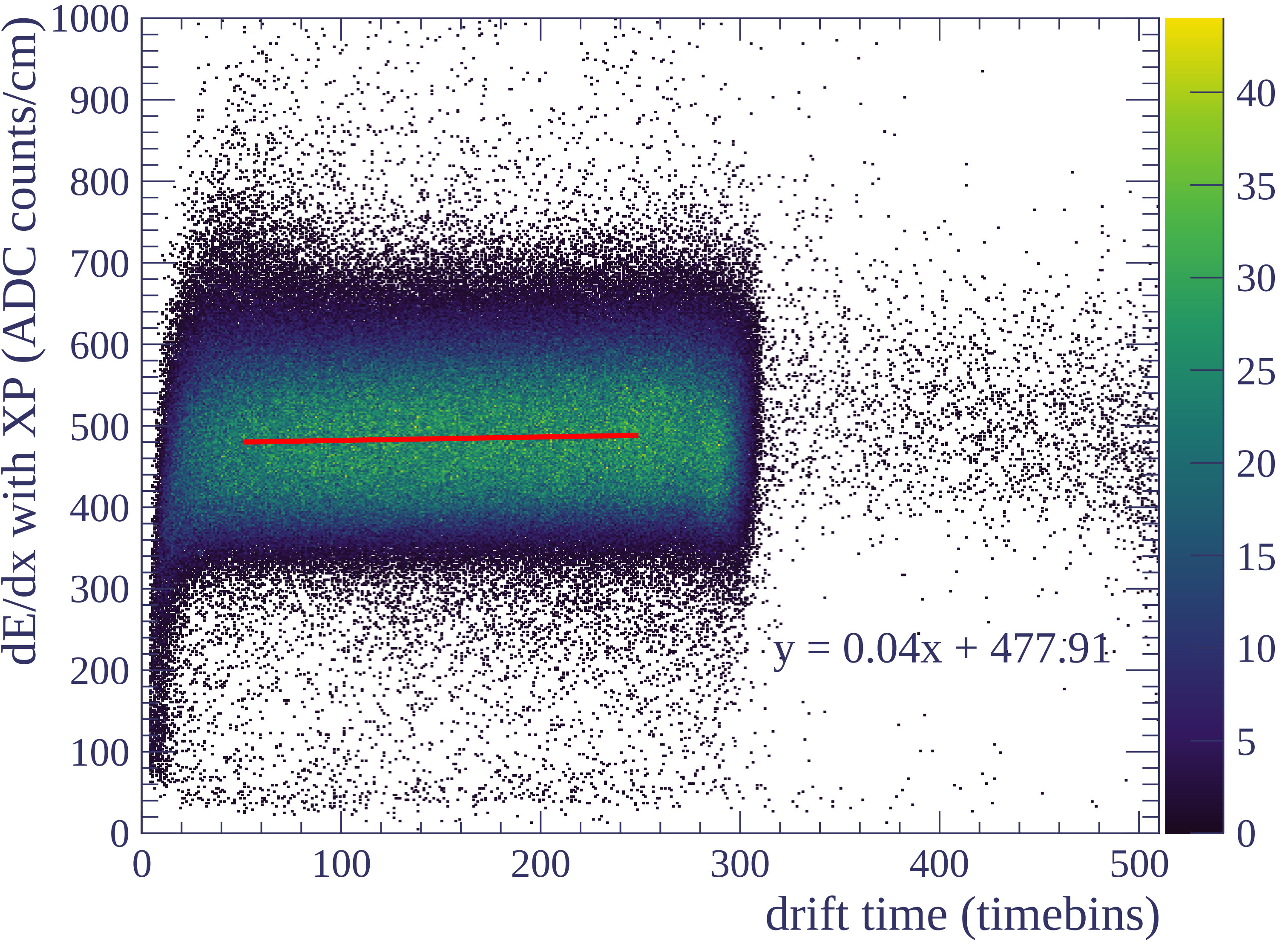
Entries 722058
Mean x 492.6
Mean y 489.8
Std Dev x 81.19
Std Dev y 79.4
0 9 267
15 721687 80
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3



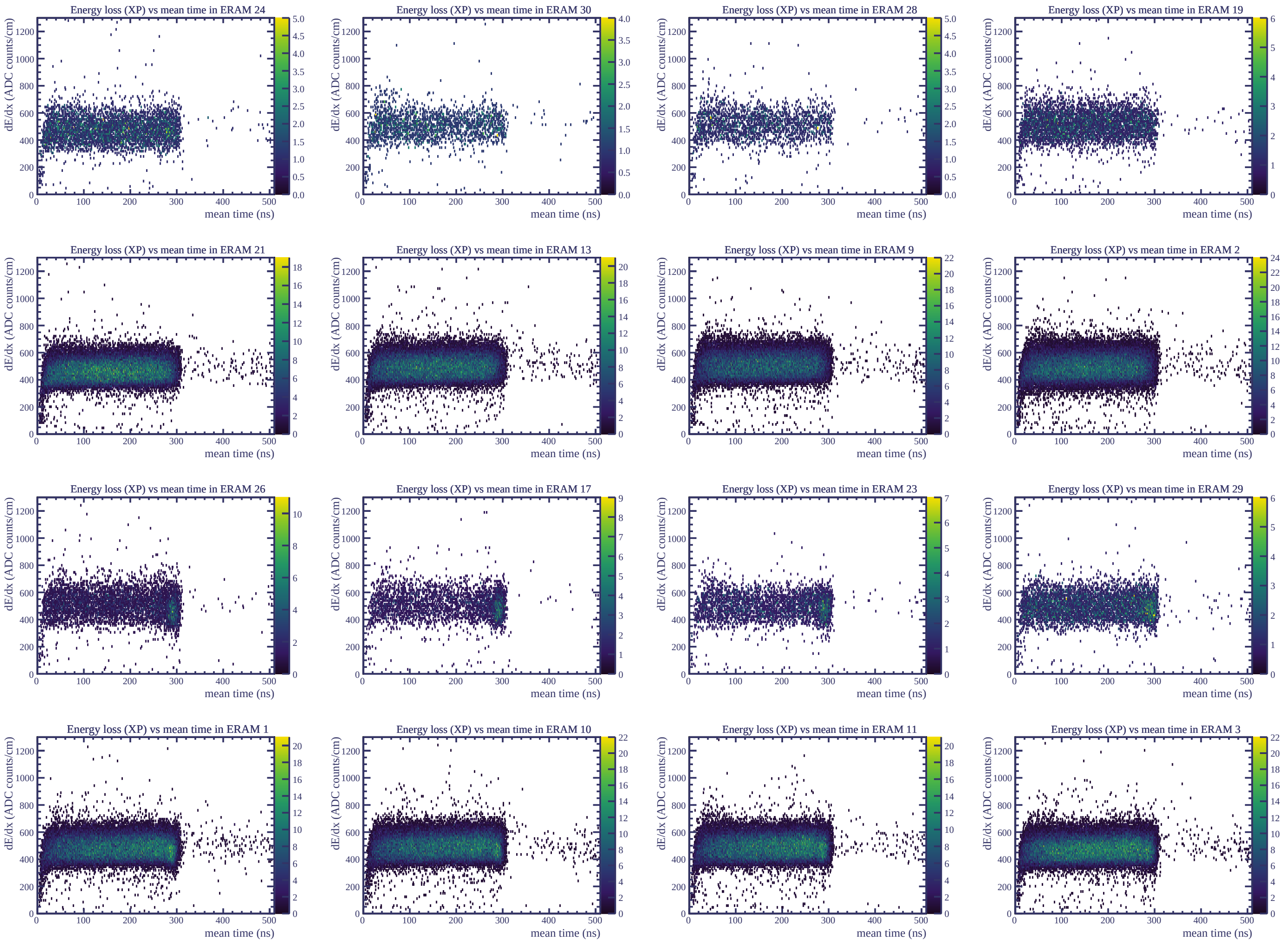
Waveforms sum
Crossed pads
Resolution (%)
Drift time (timebins)
0
5
10
15
20
25
30
0
100
200
300
400
500

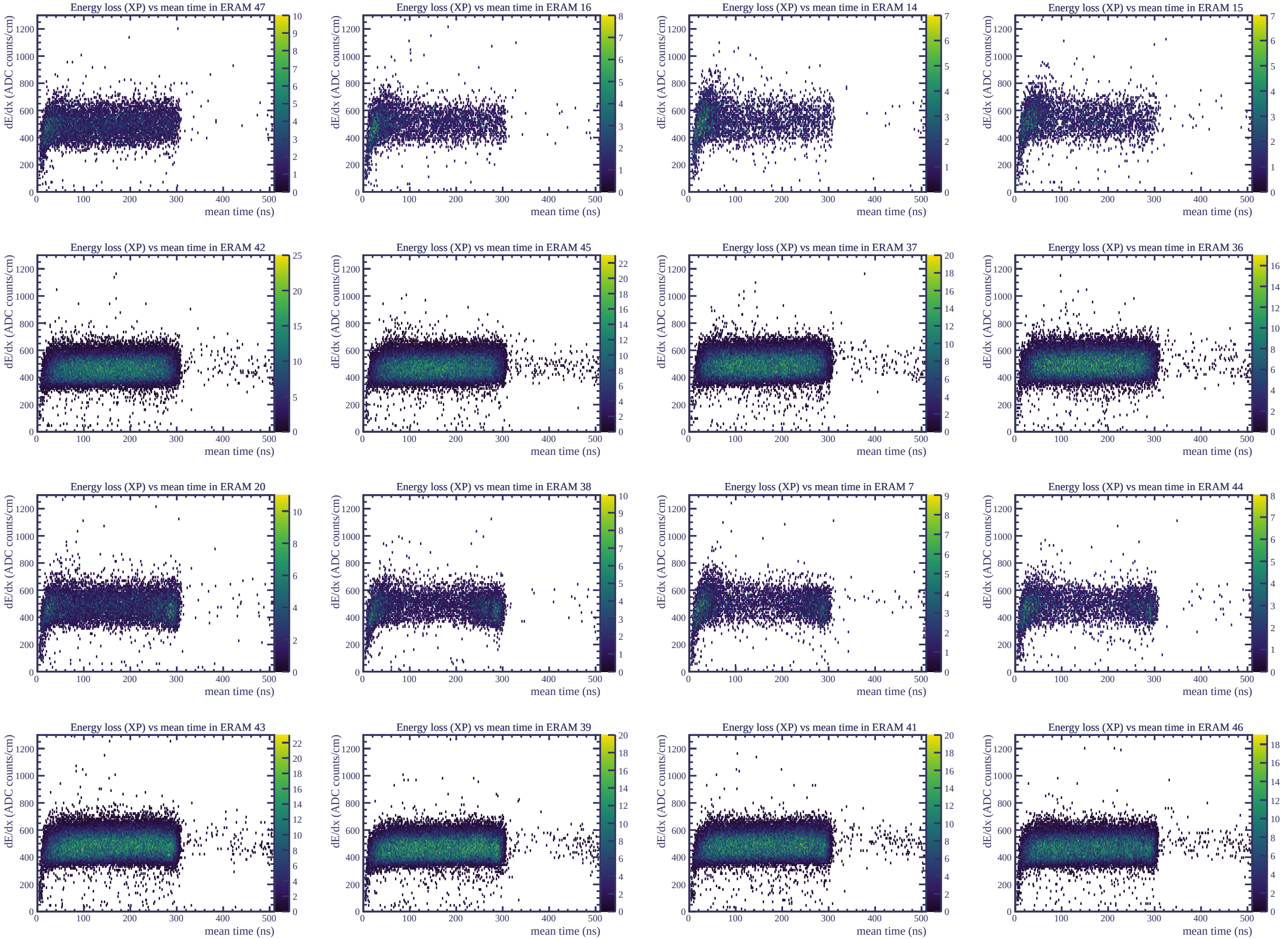


Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Drift time (timebins)
800
700
600
500
400
300
200
0
100
200
300
400
500

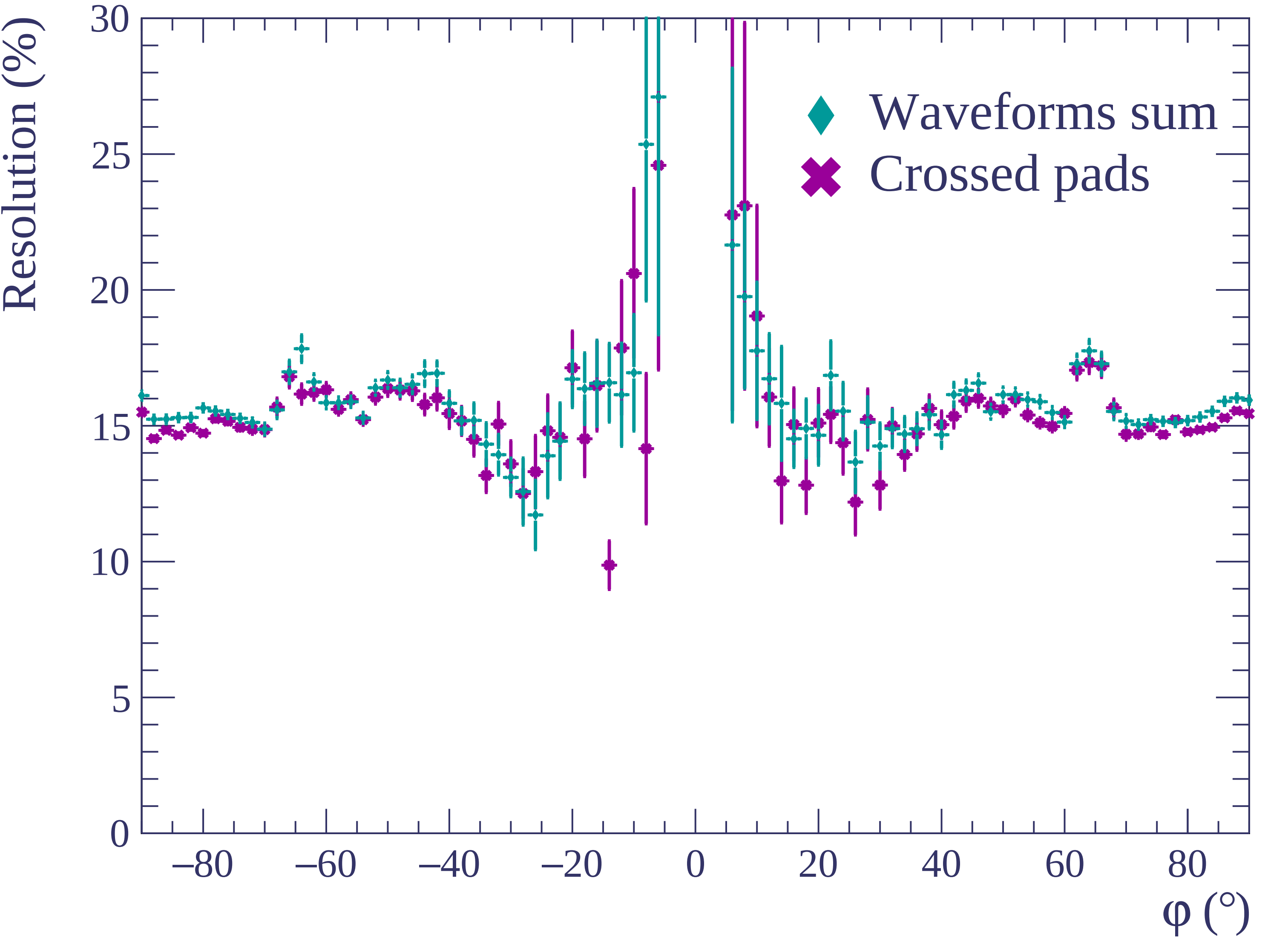


dE/dx with XP (ADC counts/cm)
drift time (timebins)
y = 0.04x + 477.91
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
5
10
15
20
25
30
35
40

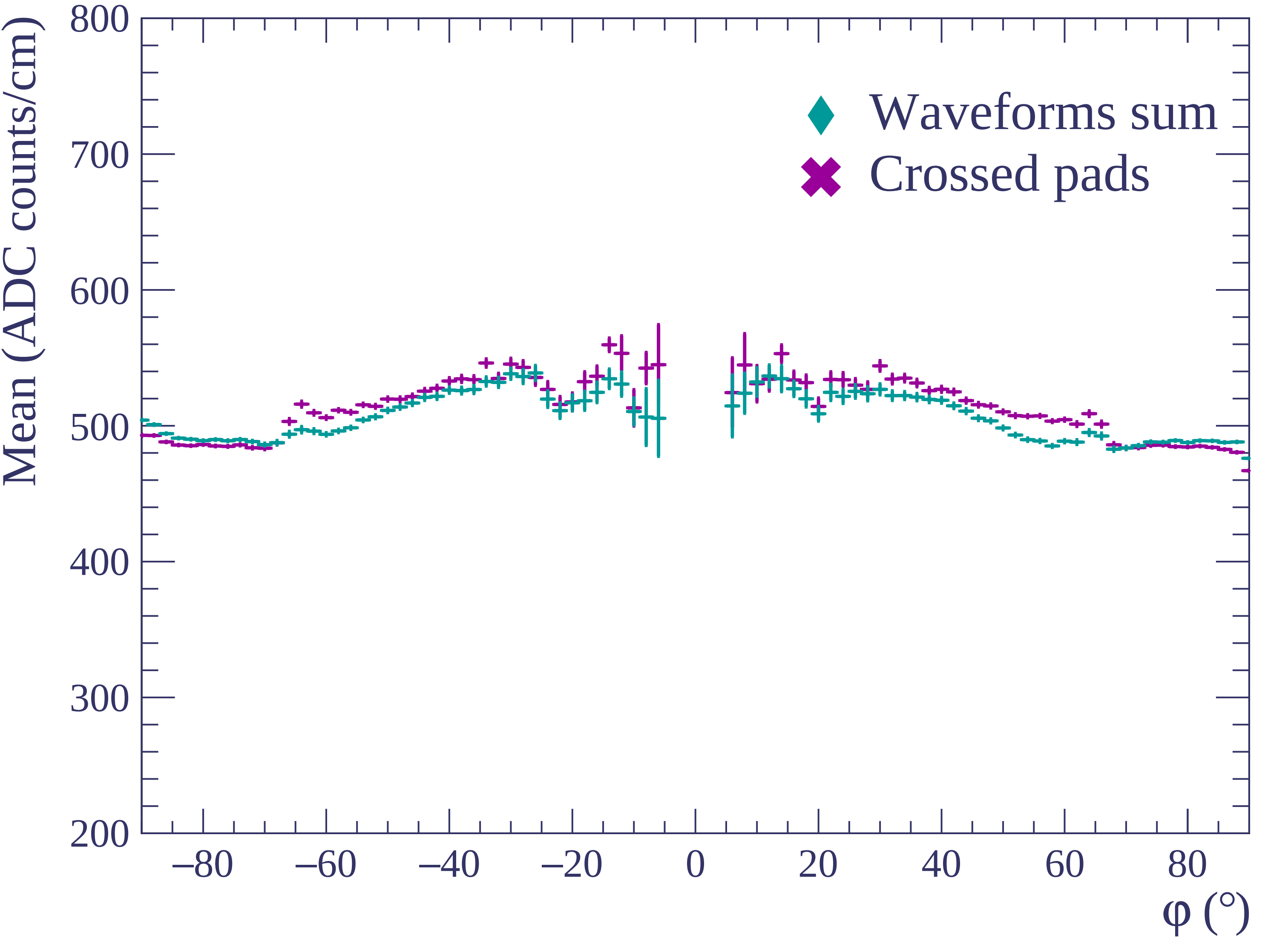

Energy loss (XP) vs mean time in ERAM 24
Energy loss (XP) vs mean time in ERAM 30
Energy loss (XP) vs mean time in ERAM 28
Energy loss (XP) vs mean time in ERAM 19
Energy loss (XP) vs mean time in ERAM 21
Energy loss (XP) vs mean time in ERAM 13
Energy loss (XP) vs mean time in ERAM 9
Energy loss (XP) vs mean time in ERAM 2
Energy loss (XP) vs mean time in ERAM 26
Energy loss (XP) vs mean time in ERAM 17
Energy loss (XP) vs mean time in ERAM 23
Energy loss (XP) vs mean time in ERAM 29
Energy loss (XP) vs mean time in ERAM 1
Energy loss (XP) vs mean time in ERAM 10
Energy loss (XP) vs mean time in ERAM 11
Energy loss (XP) vs mean time in ERAM 3
dE/dx (ADC counts/cm)
mean time (ns)



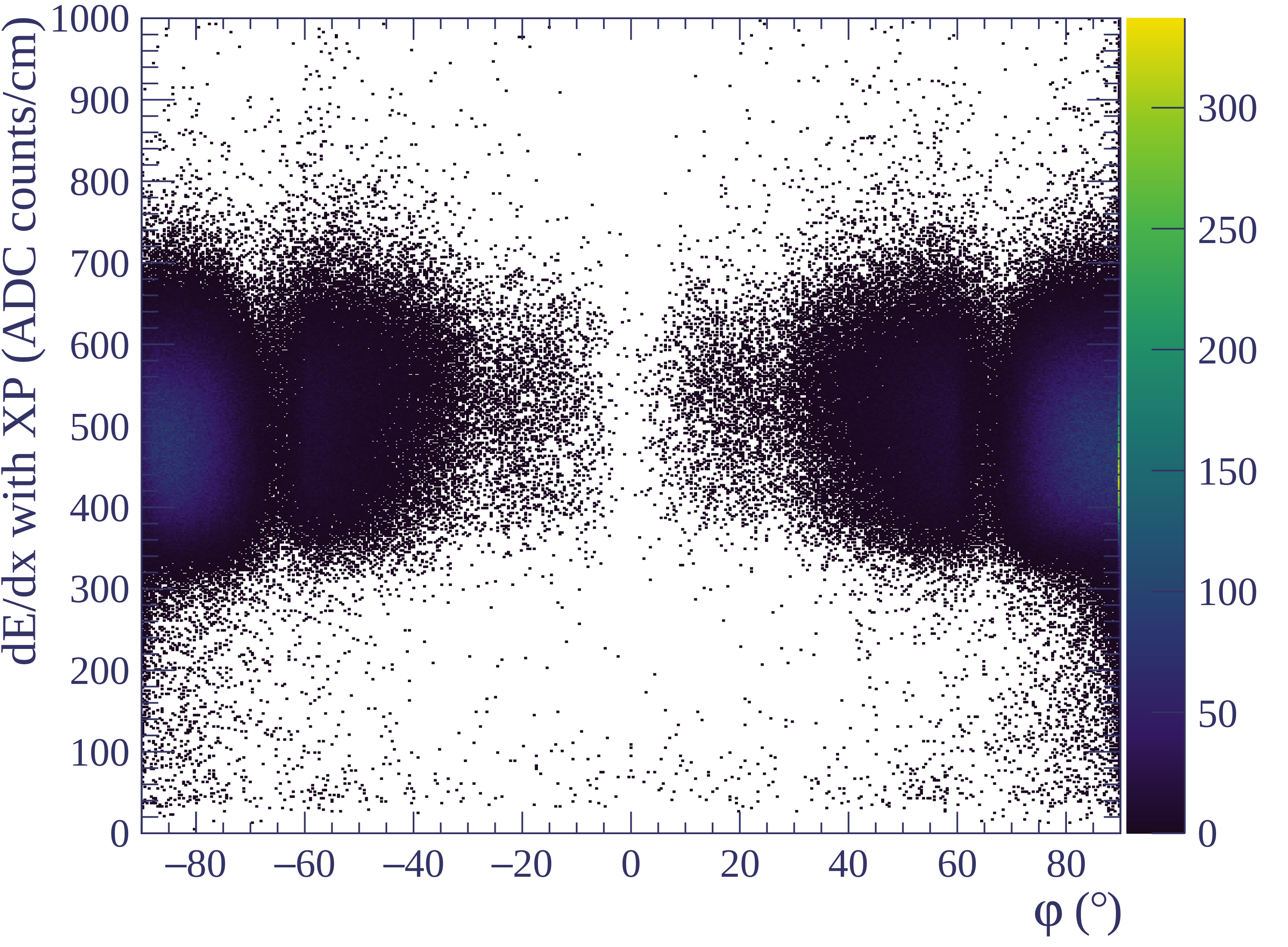
Energy loss (XP) vs mean time in ERAM 47
Energy loss (XP) vs mean time in ERAM 16
Energy loss (XP) vs mean time in ERAM 14
Energy loss (XP) vs mean time in ERAM 15
Energy loss (XP) vs mean time in ERAM 42
Energy loss (XP) vs mean time in ERAM 45
Energy loss (XP) vs mean time in ERAM 37
Energy loss (XP) vs mean time in ERAM 36
Energy loss (XP) vs mean time in ERAM 20
Energy loss (XP) vs mean time in ERAM 38
Energy loss (XP) vs mean time in ERAM 7
Energy loss (XP) vs mean time in ERAM 44
Energy loss (XP) vs mean time in ERAM 43
Energy loss (XP) vs mean time in ERAM 39
Energy loss (XP) vs mean time in ERAM 41
Energy loss (XP) vs mean time in ERAM 46
dE/dx (ADC counts/cm)
mean time (ns)



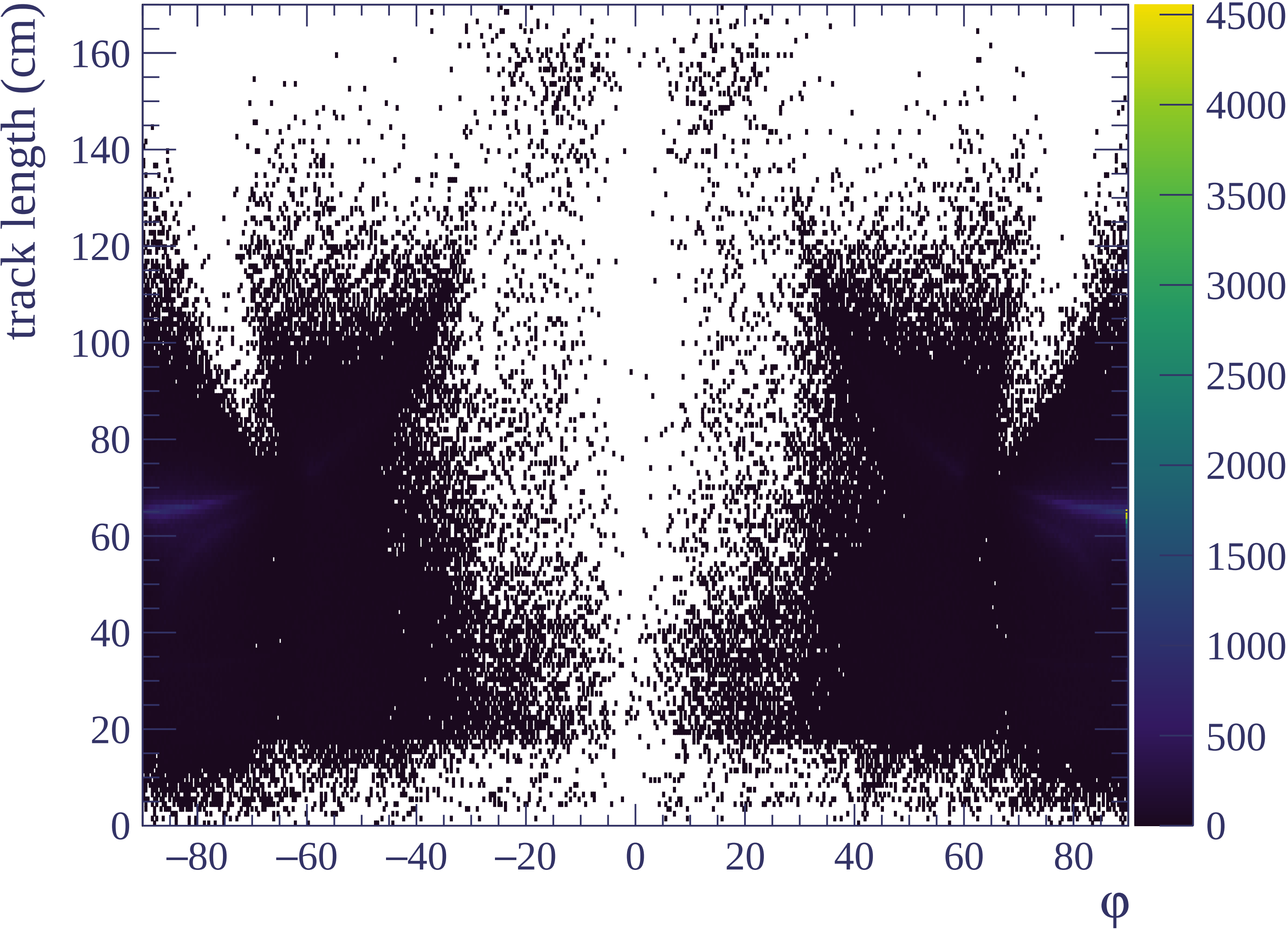
Resolution (%)
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads



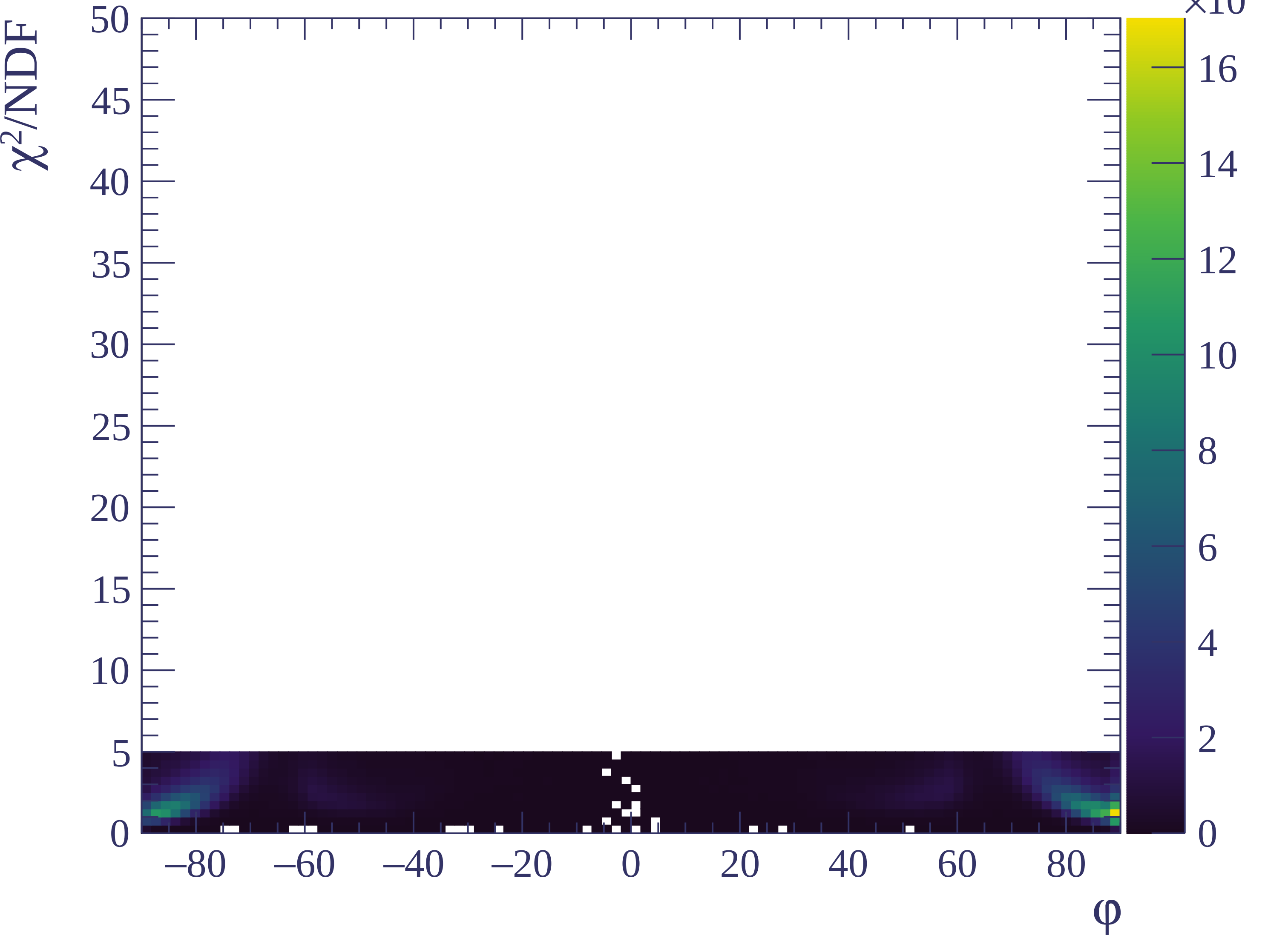
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
φ (°)
Waveforms sum
Crossed pads



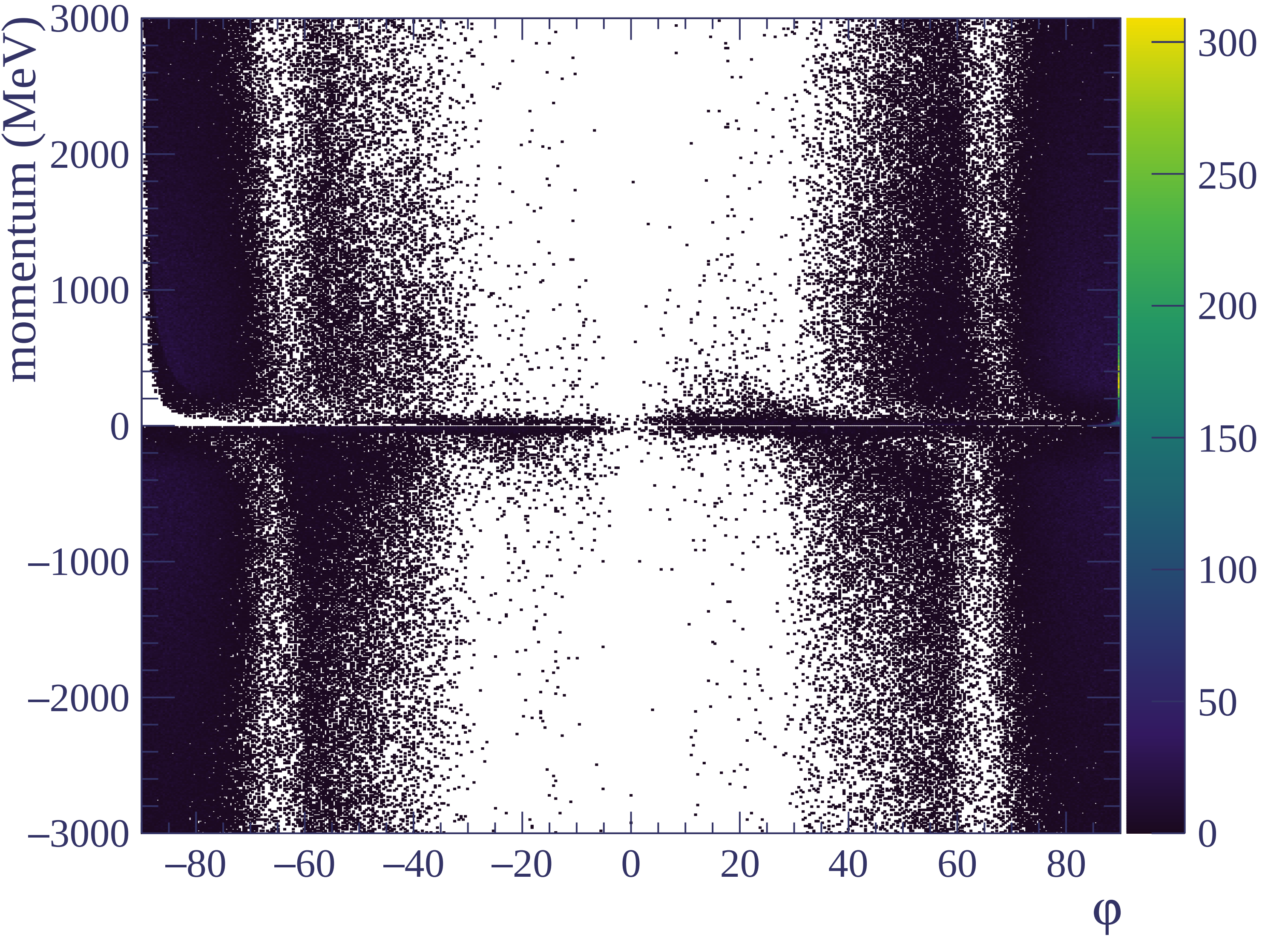
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
300
250
200
150
100
50
0



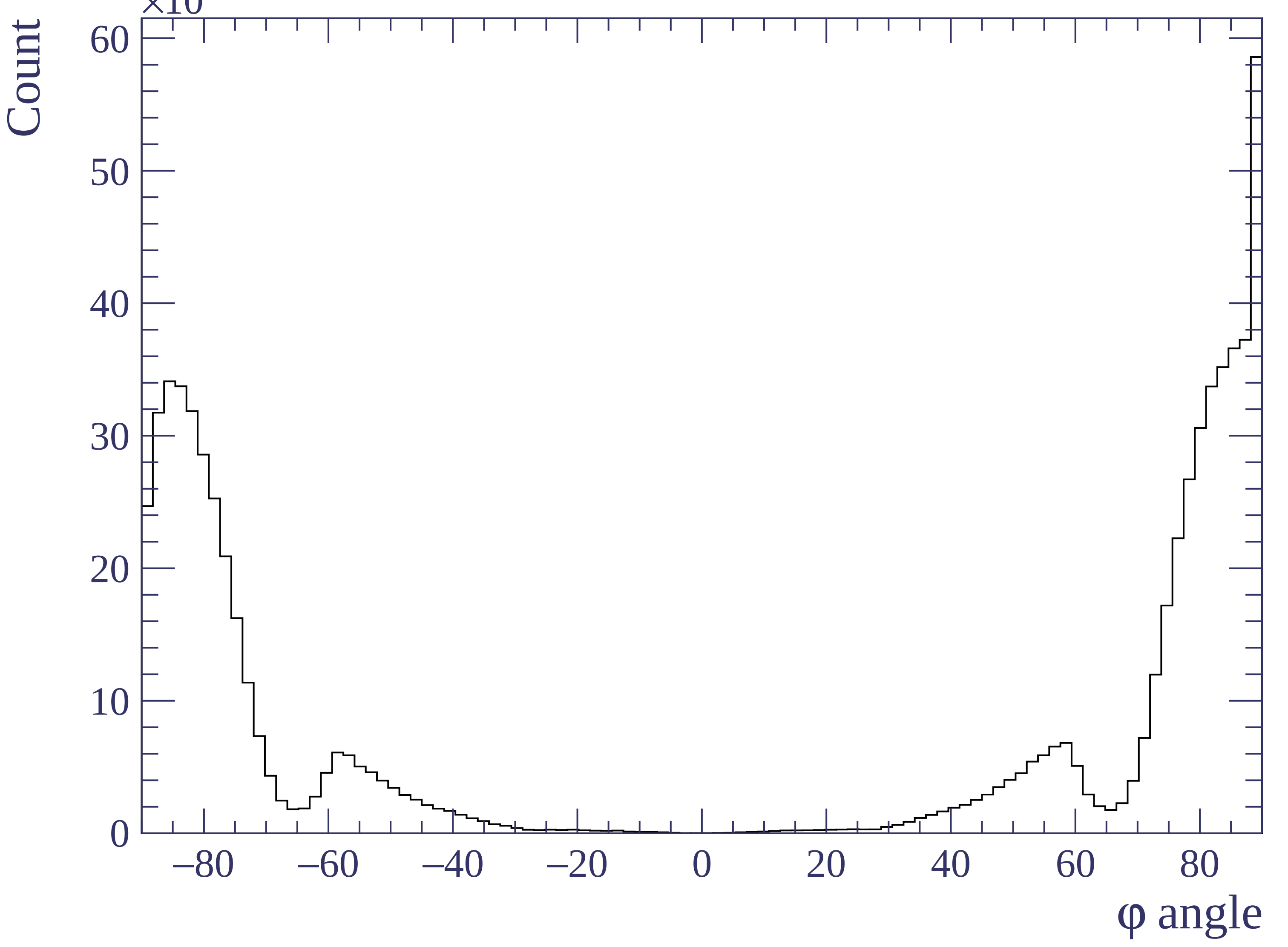
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
4500
4000
3500
3000
2500
2000
1500
1000
500
0



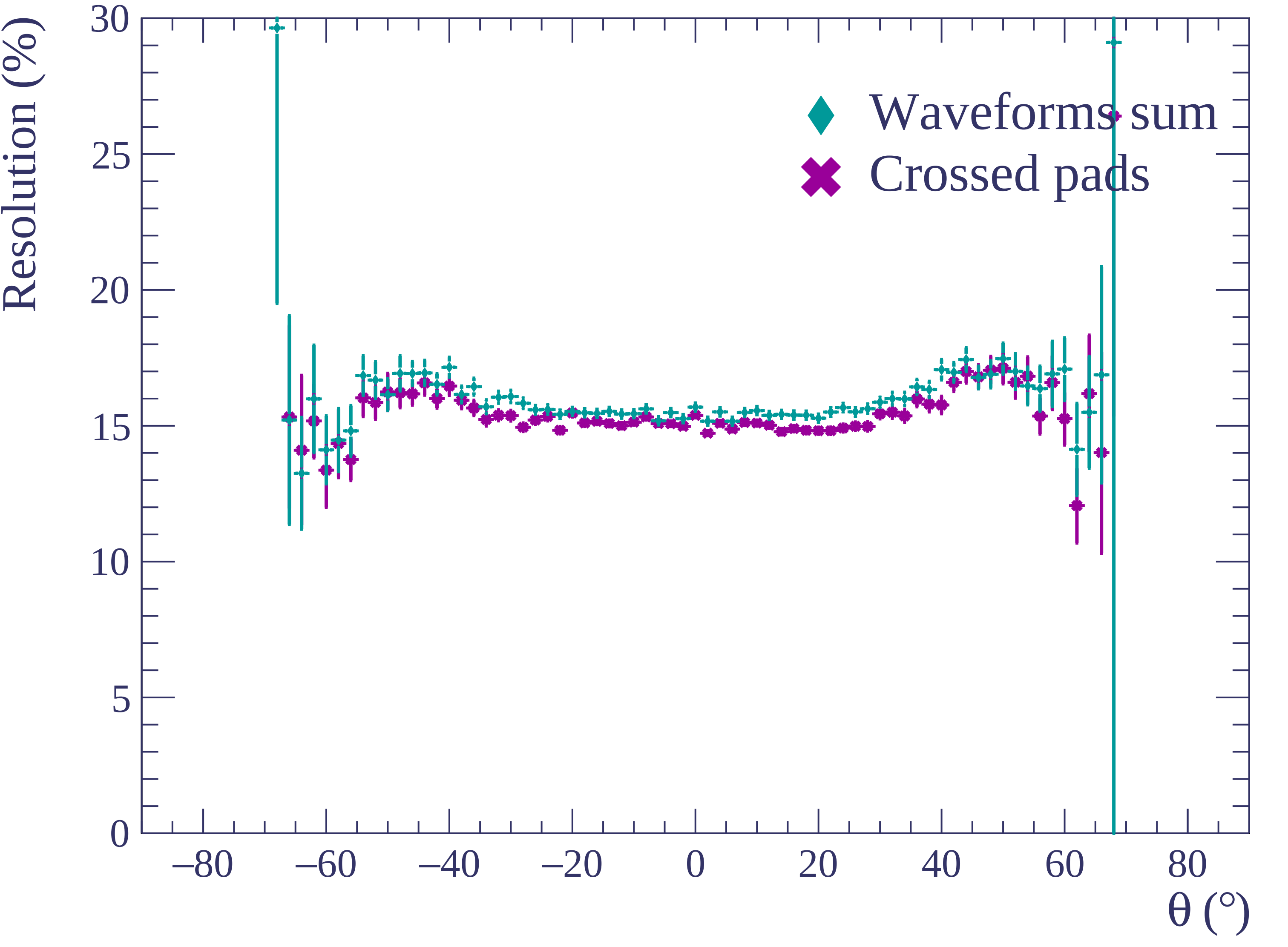
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
φ
×10
16
14
12
10
8
6
4
2
0

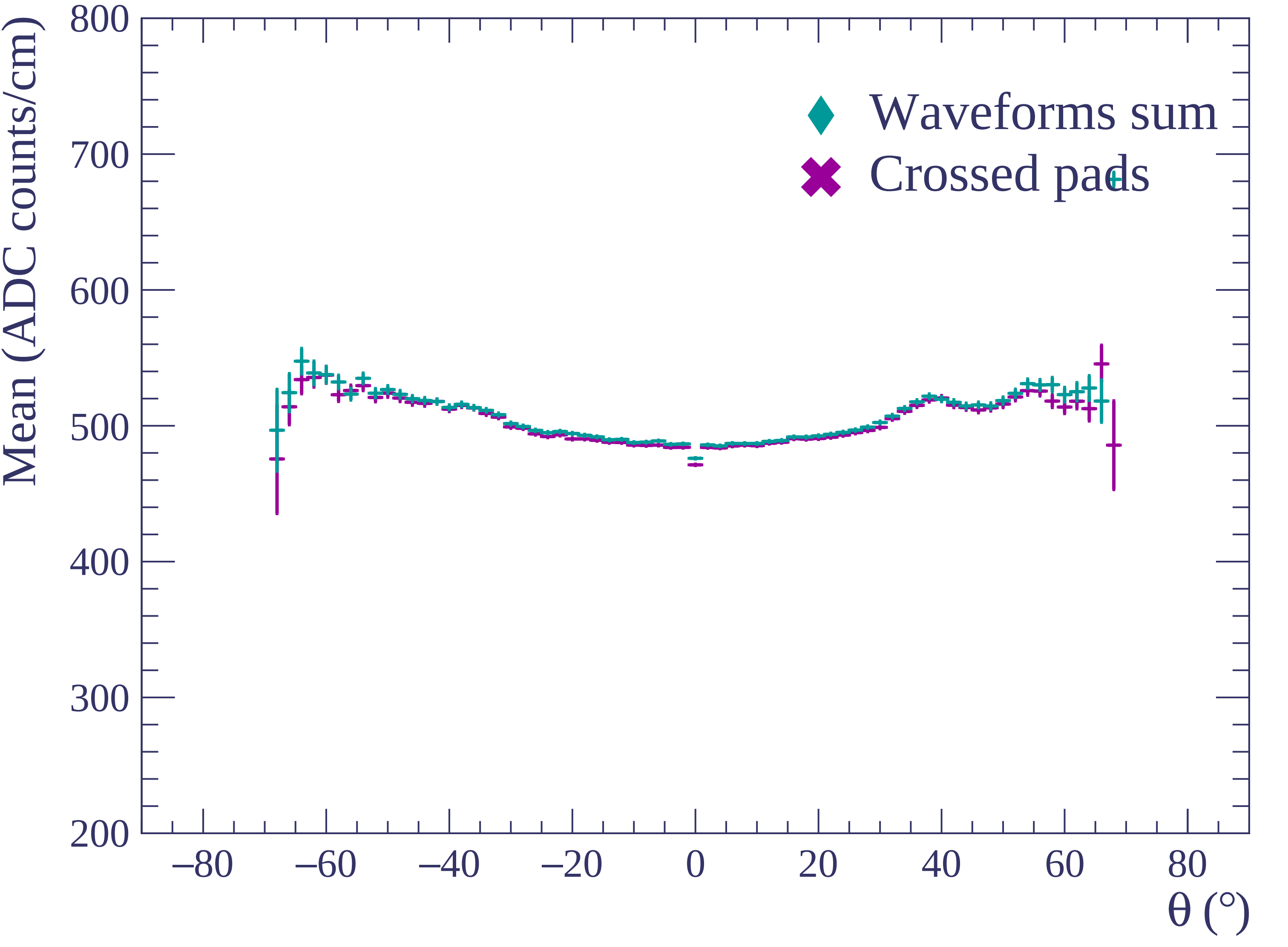


momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−80
−60
−40
−20
0
20
40
60
80
φ
300
250
200
150
100
50
0

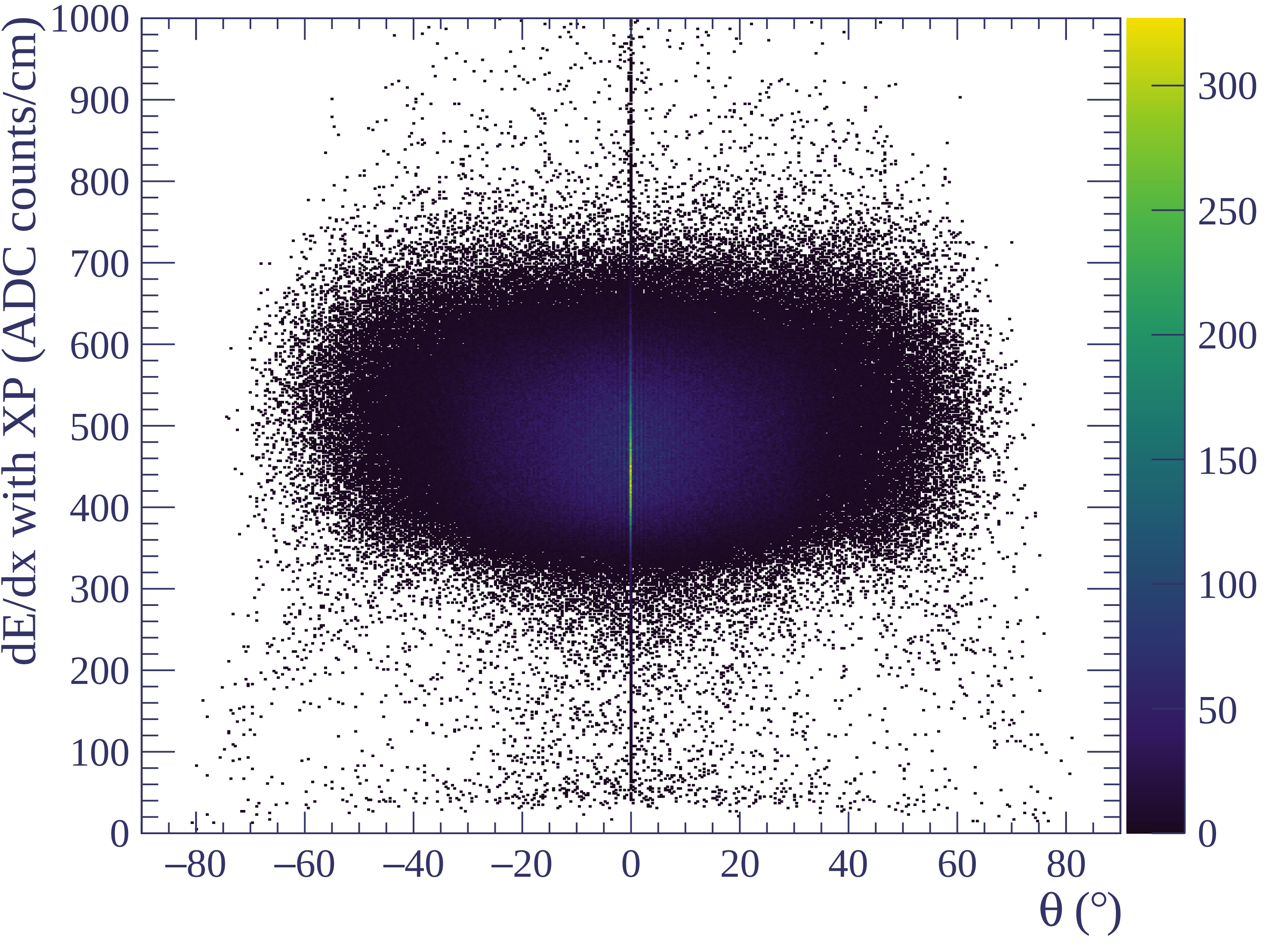


×10
Count
60
50
40
30
20
10
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

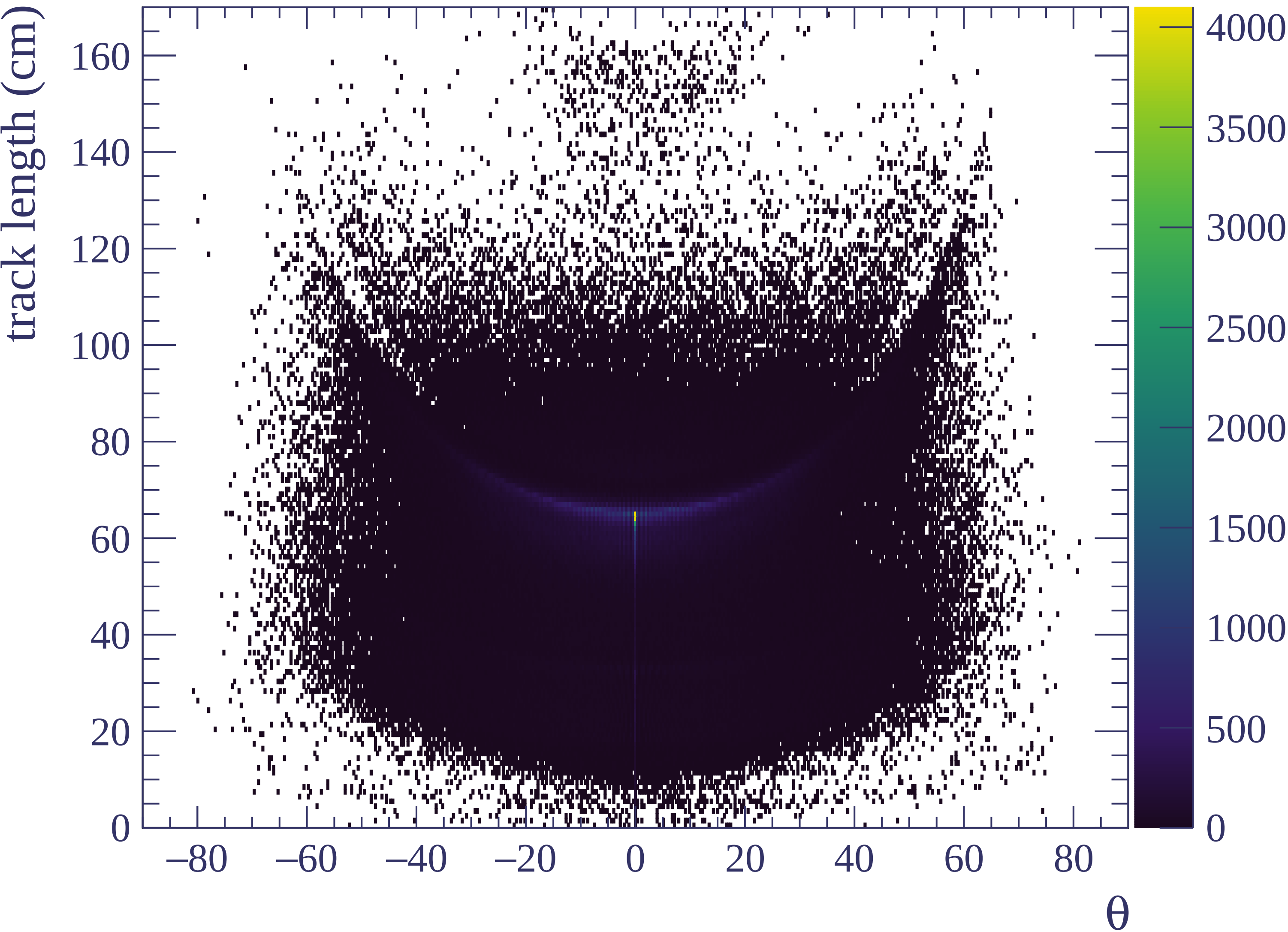

Resolution (%)
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
Waveforms sum
Crossed pads



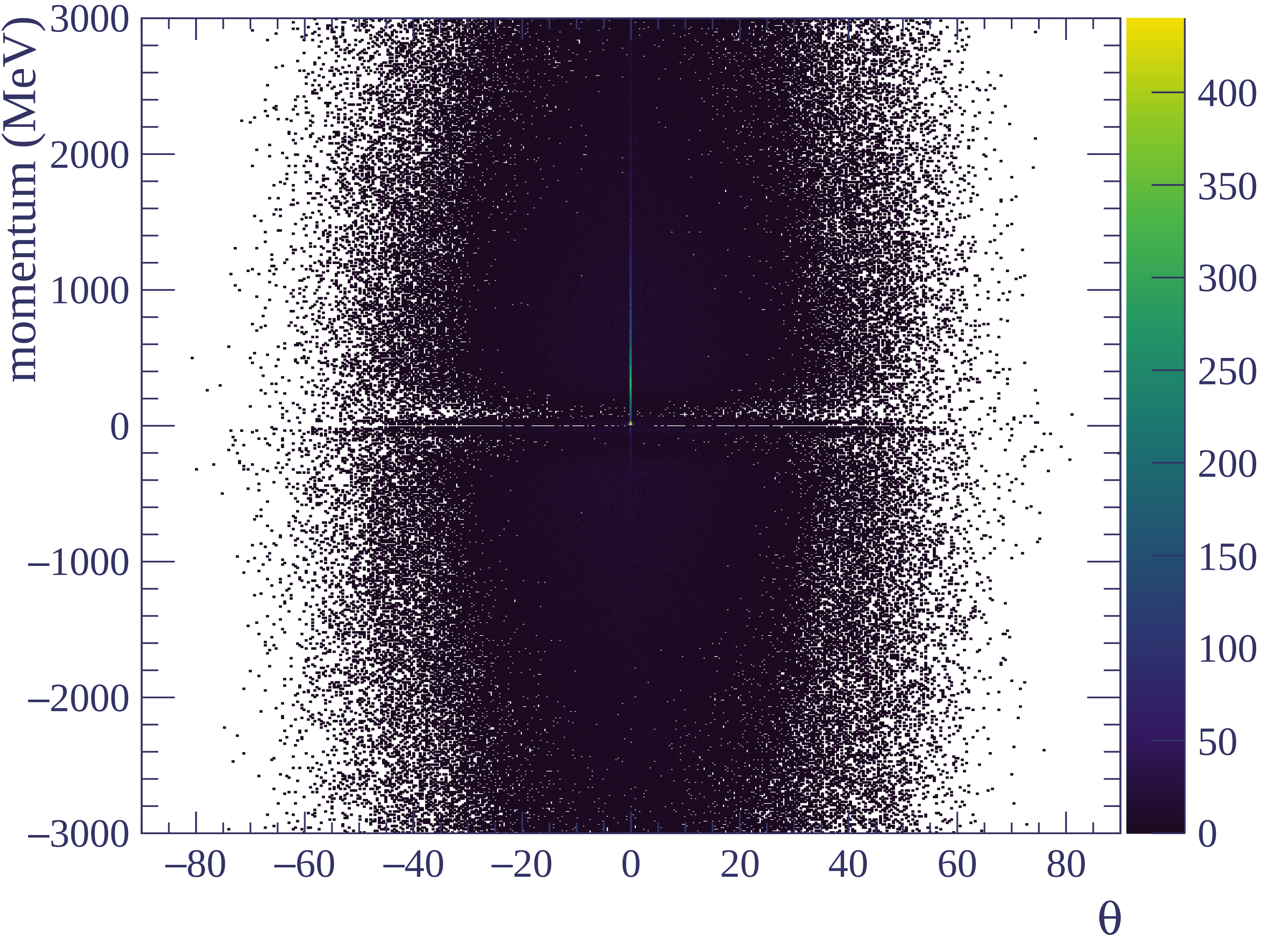
Mean (ADC counts/cm)
800
700
600
500
400
300
200
−80
−60
−40
−20
0
20
40
60
80
θ (°)
Waveforms sum
Crossed pads



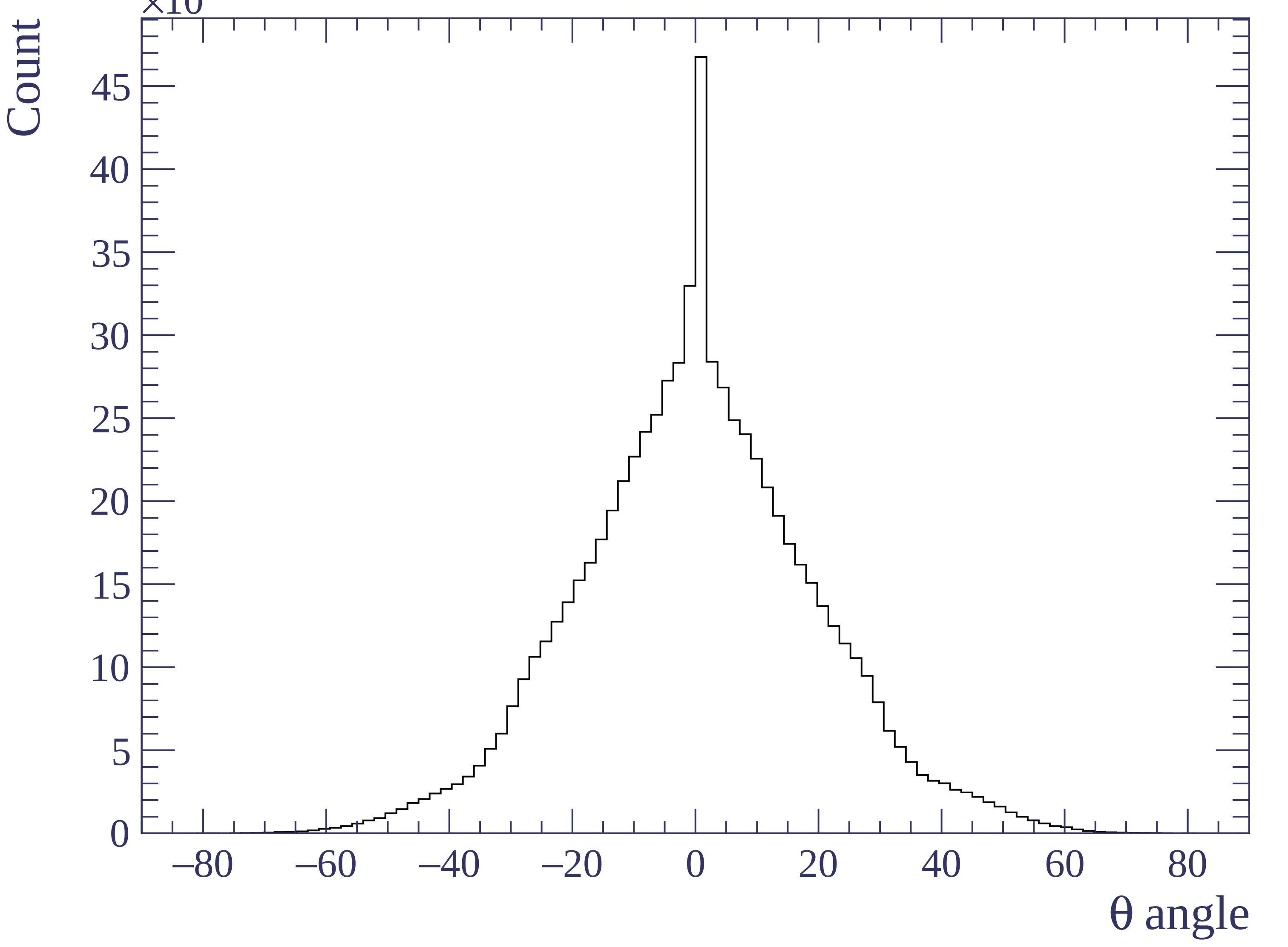
dE/dx with XP (ADC counts/cm)
θ (°)
0
100
200
300
400
500
600
700
800
900
1000
−80
−60
−40
−20
0
20
40
60
80
0
50
100
150
200
250
300



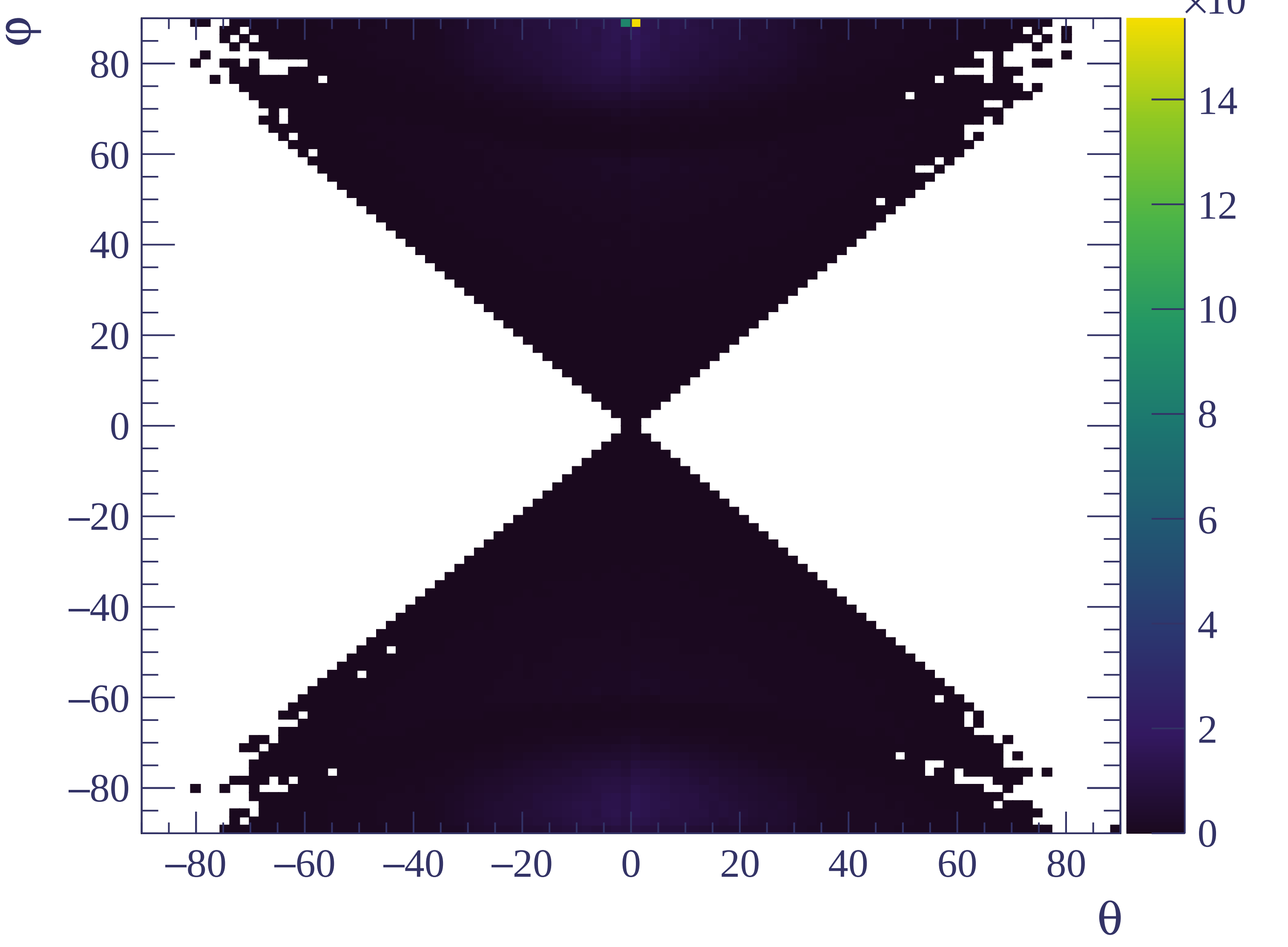
track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
θ
0
500
1000
1500
2000
2500
3000
3500
4000



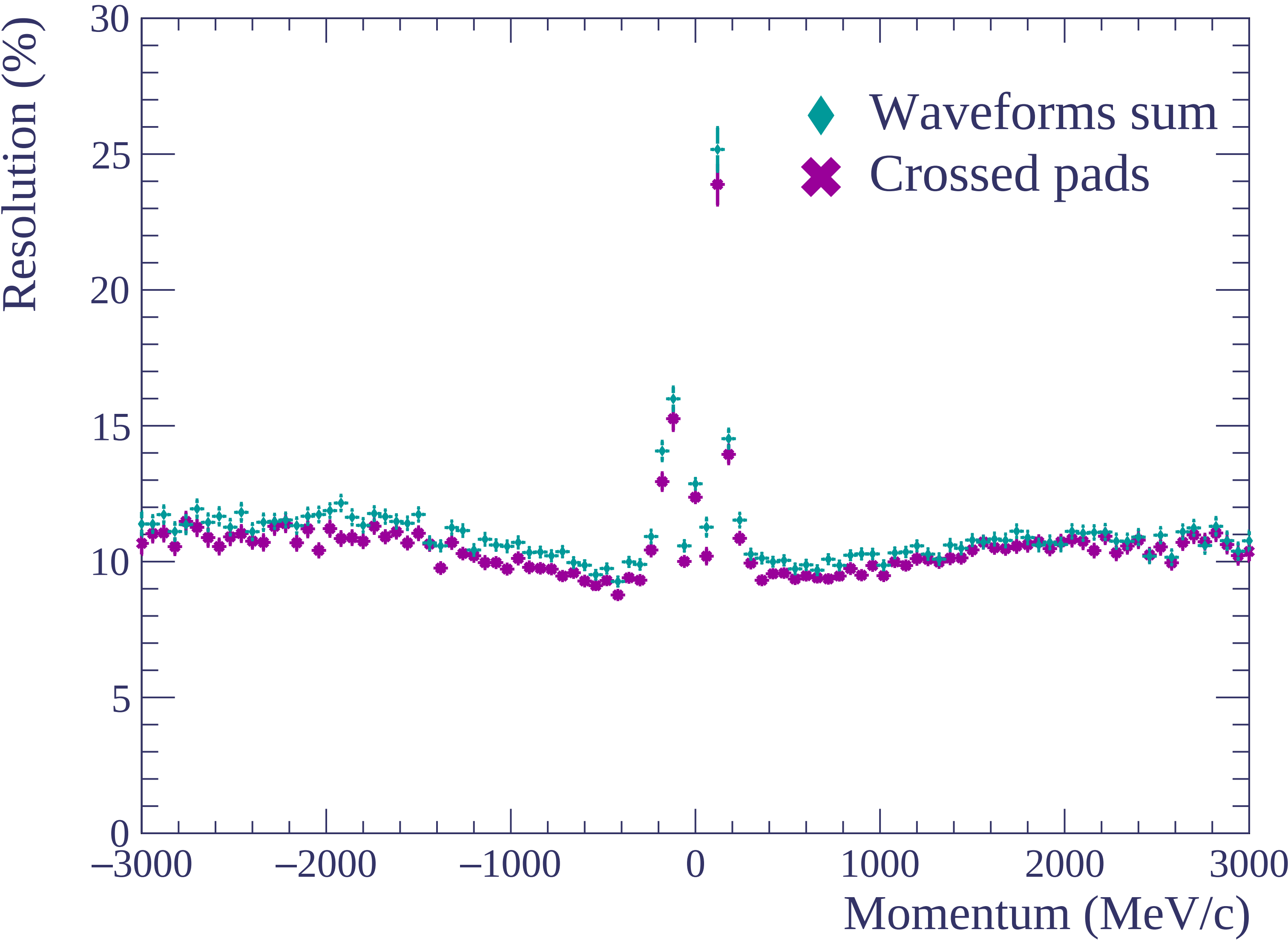
momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−80
−60
−40
−20
0
20
40
60
80
θ
400
350
300
250
200
150
100
50
0



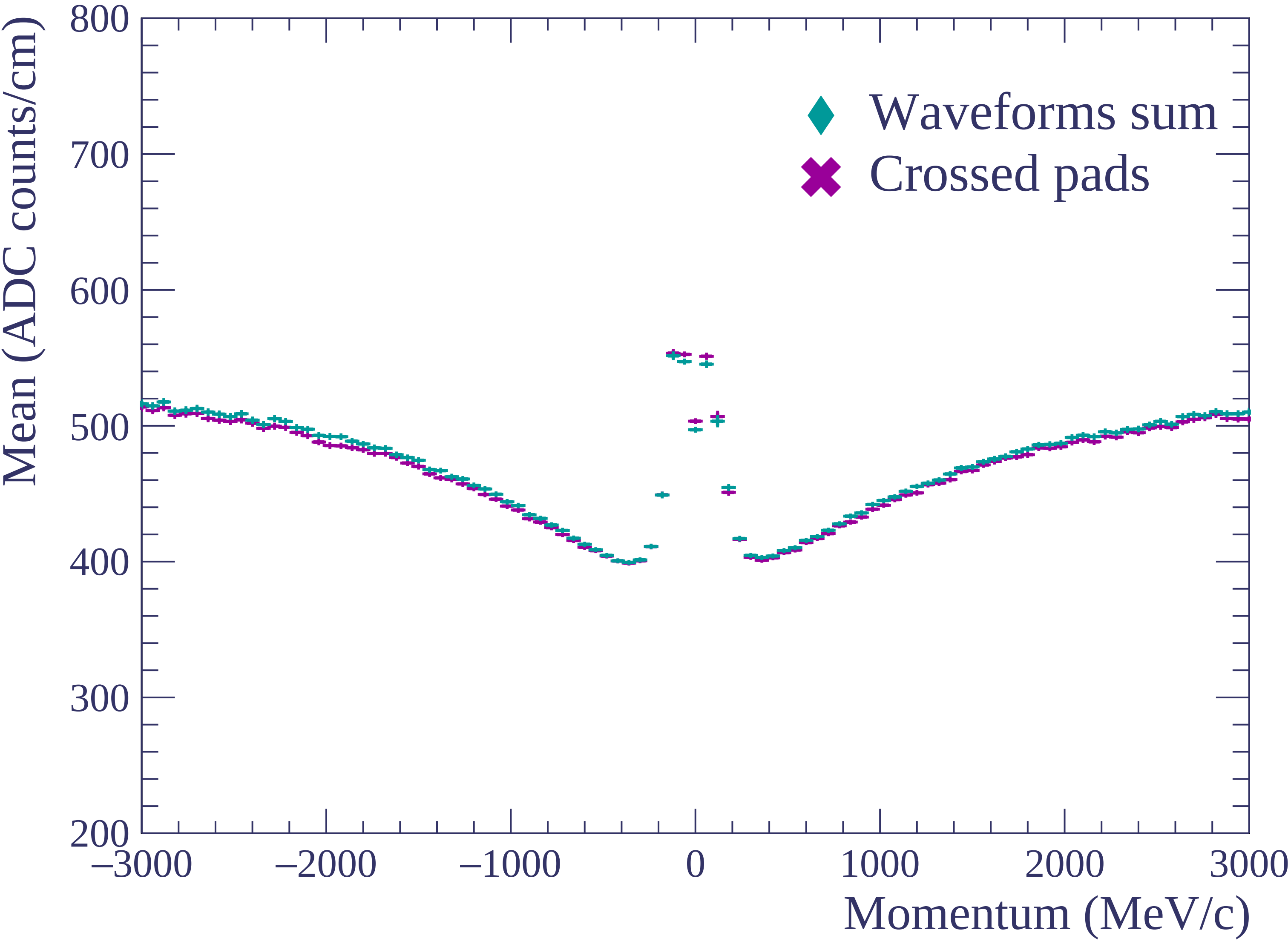
×10
Count
45
40
35
30
25
20
15
10
5
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

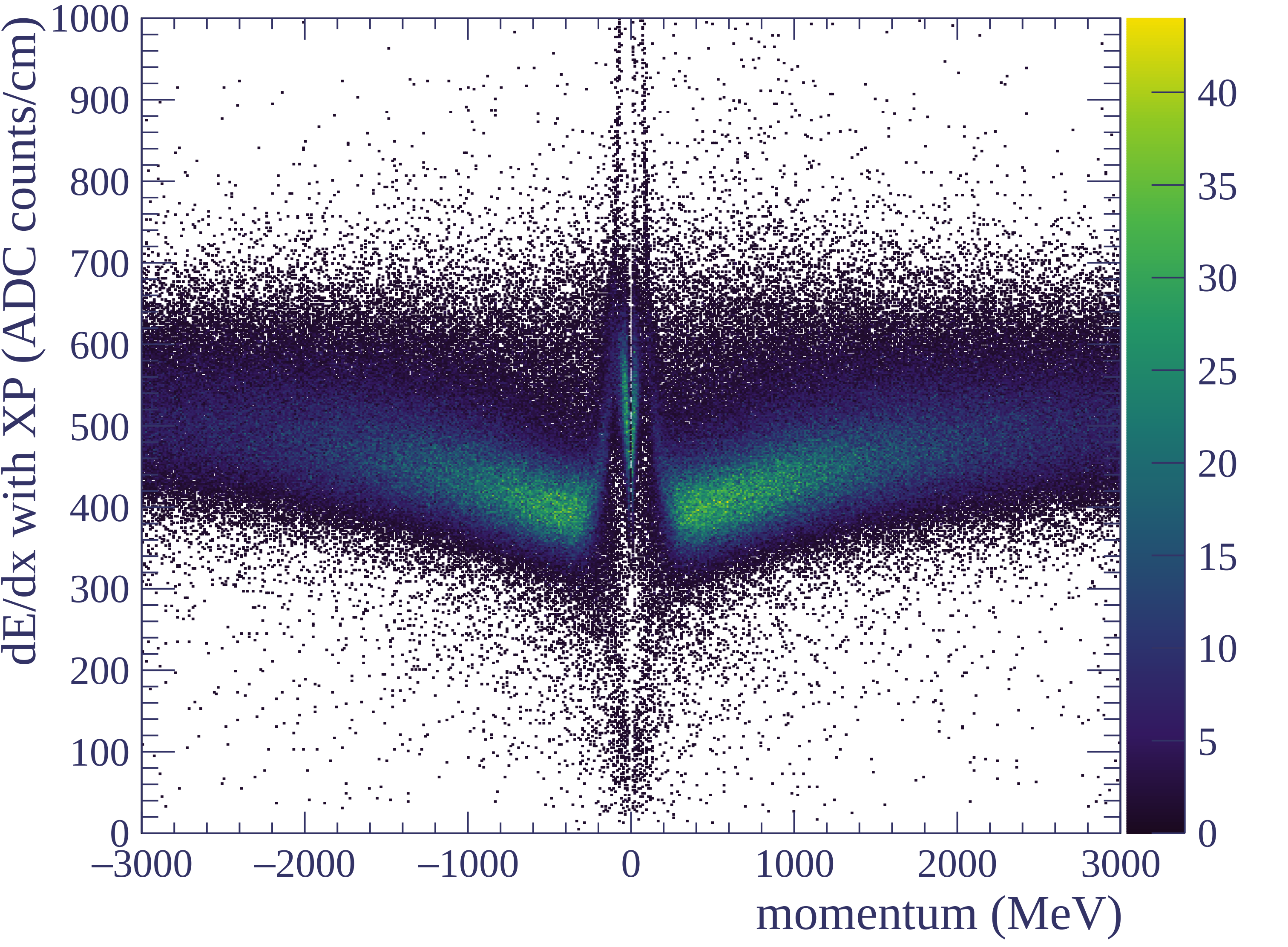


φ
80
60
40
20
0
−20
−40
−60
−80
θ
−80
−60
−40
−20
0
20
40
60
80
14
12
10
8
6
4
2
0

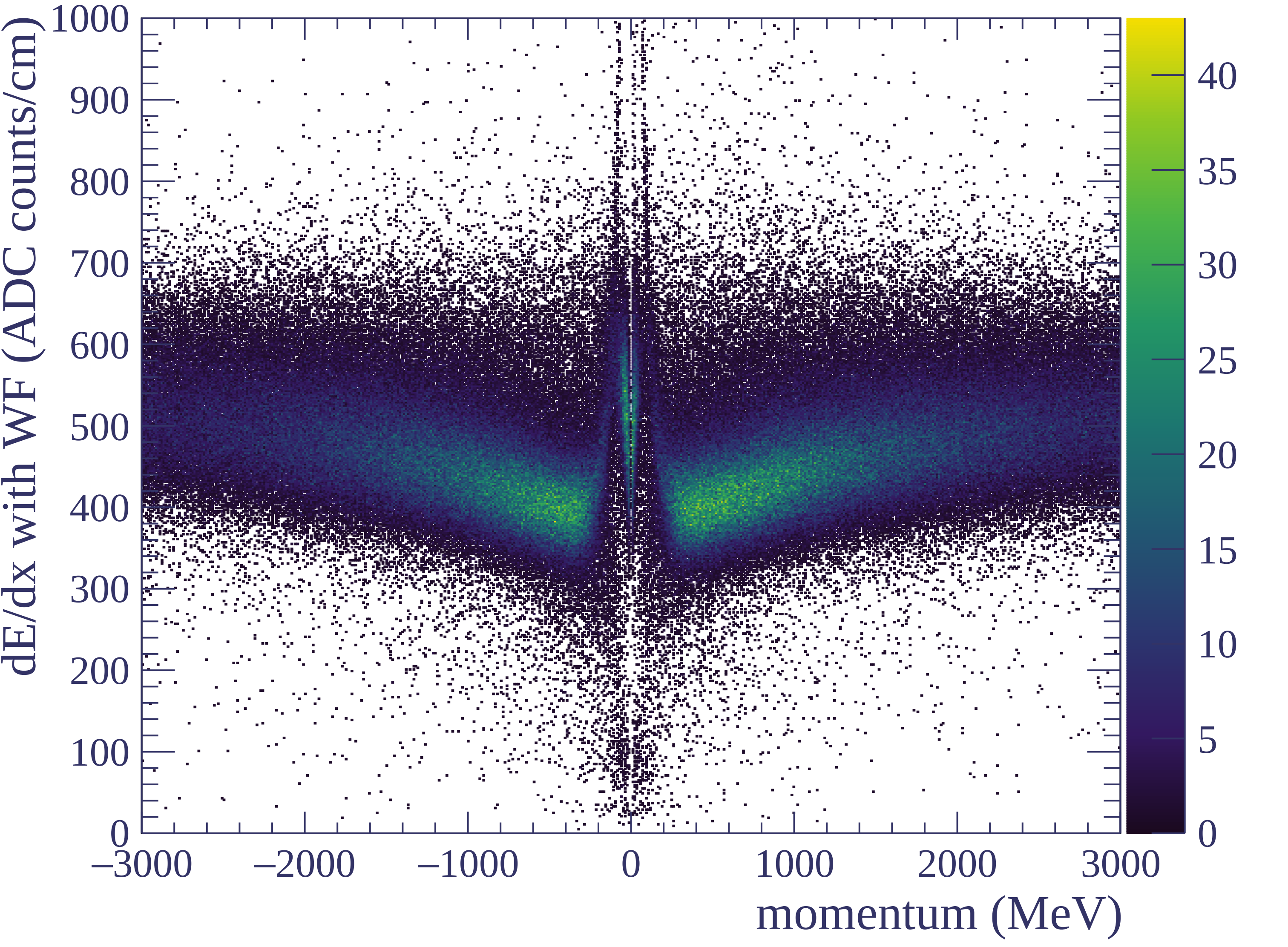


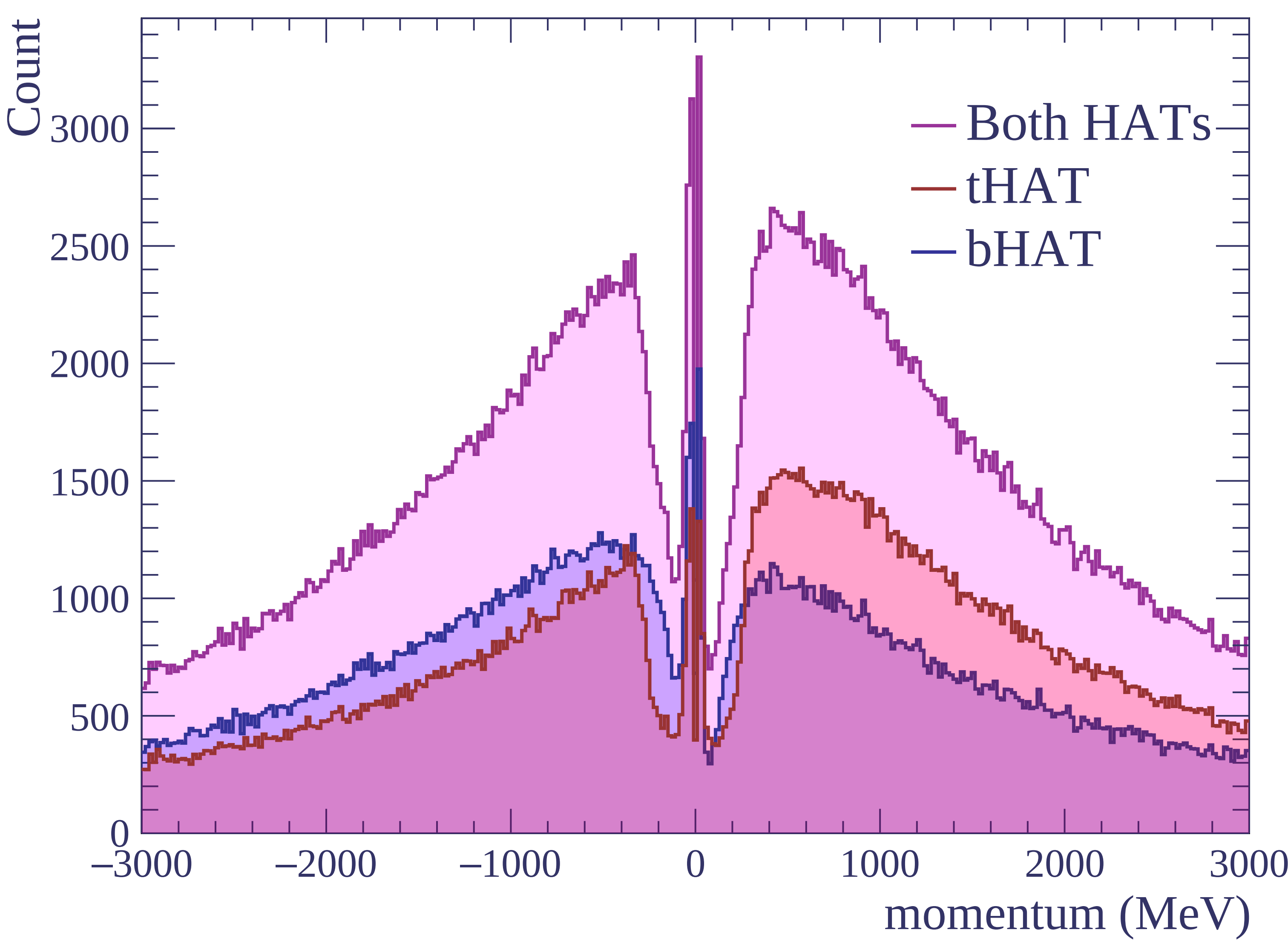
Resolution (%)
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
Waveforms sum
Crossed pads


Waveforms sum
Crossed pads
Mean (ADC counts/cm)
Momentum (MeV/c)
800
700
600
500
400
300
200
−3000
−2000
−1000
0
1000
2000
3000

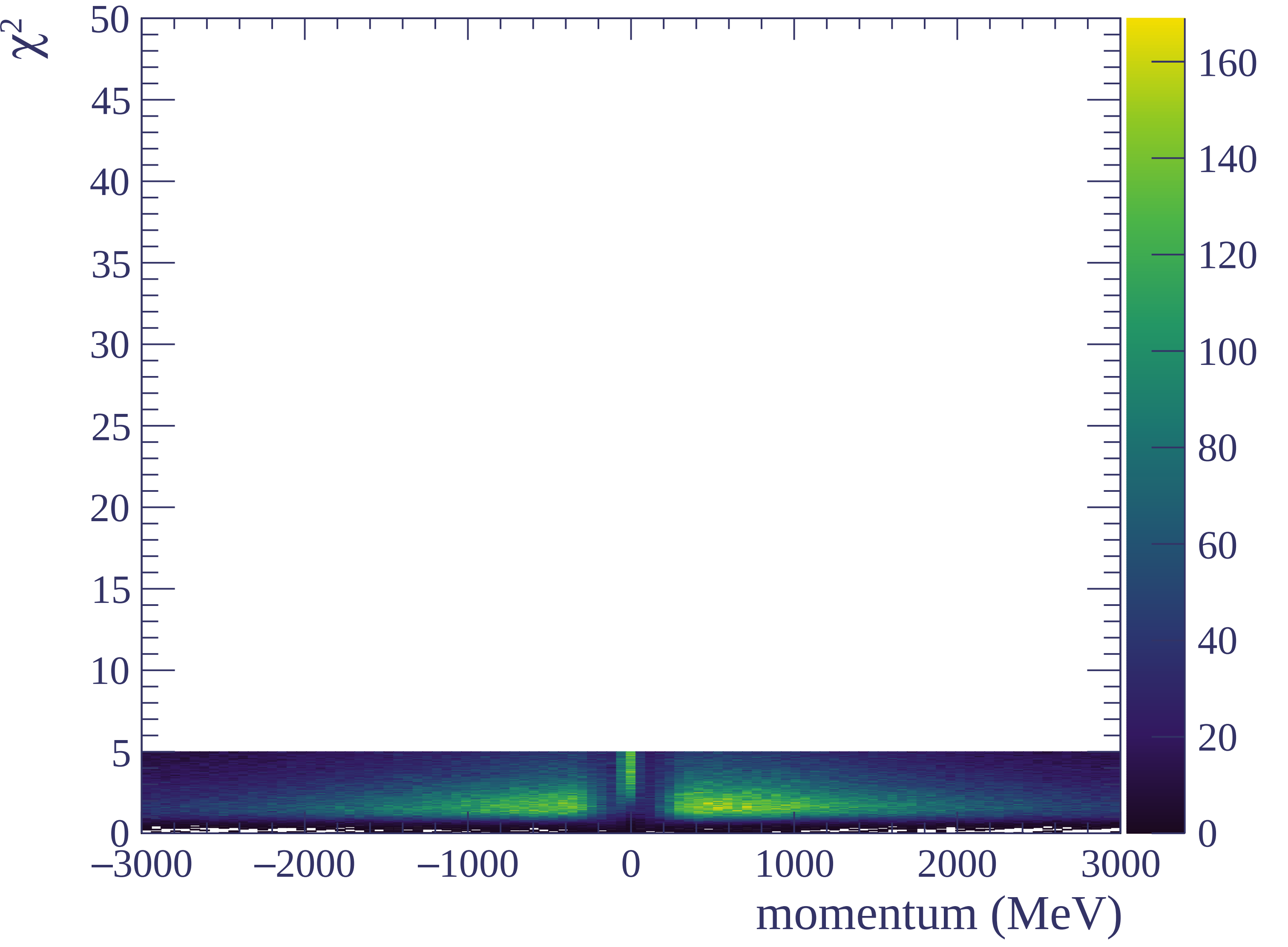


dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−3000
−2000
−1000
0
1000
2000
3000
0
5
10
15
20
25
30
35
40

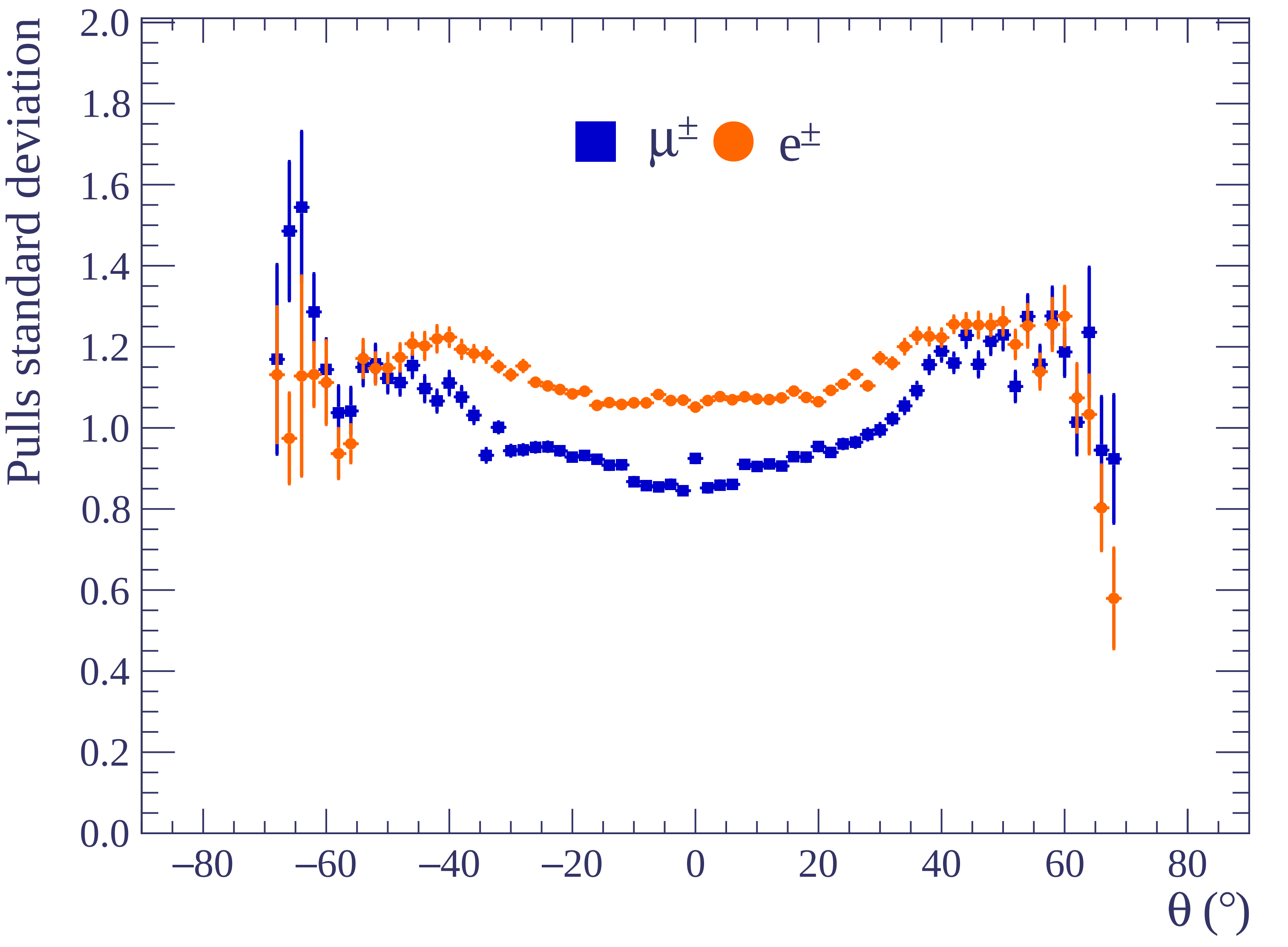

dE/dx with WF (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
40
35
30
25
20
15
10
5
0



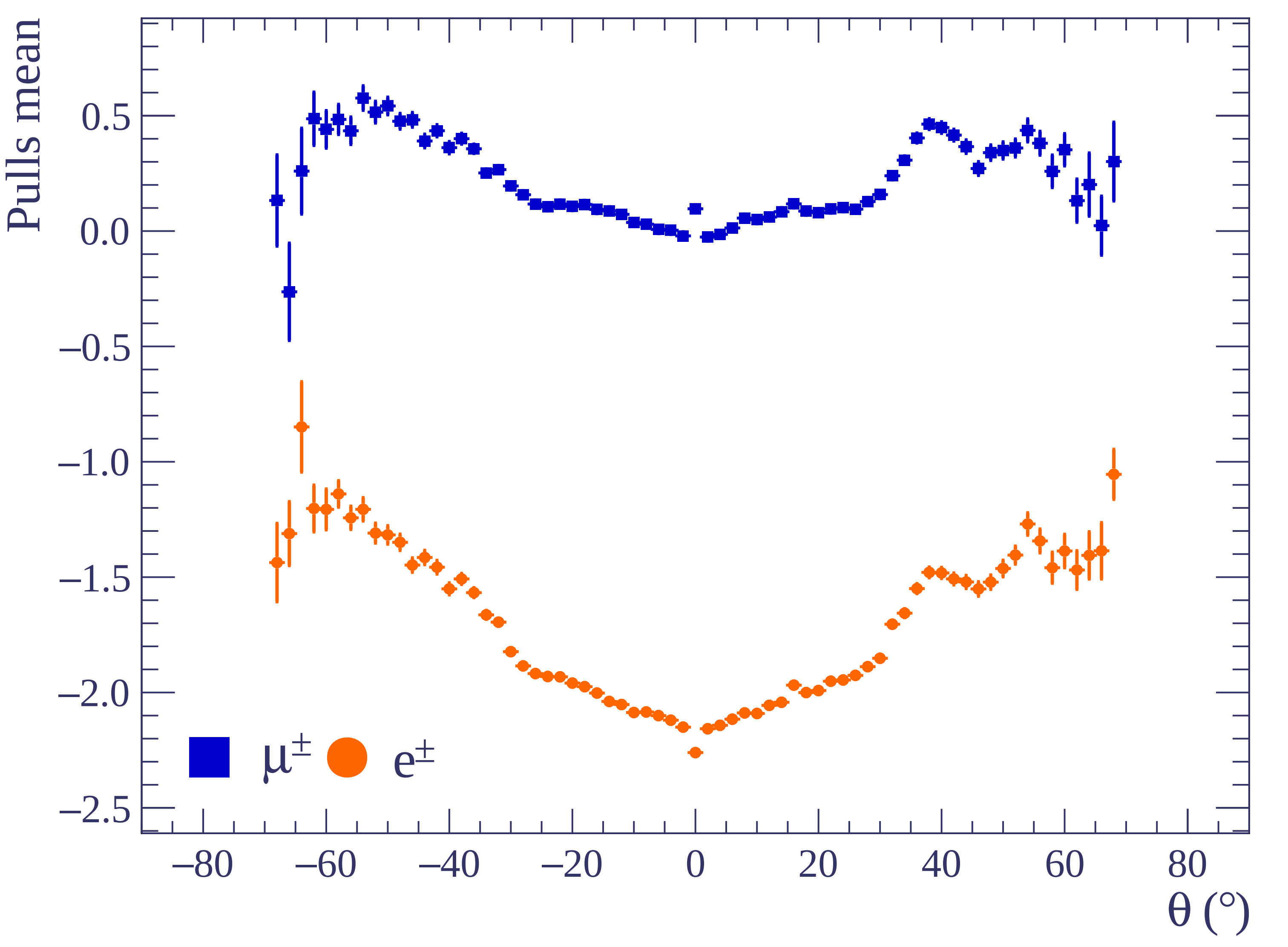
Count
0
500
1000
1500
2000
2500
3000
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
Both HATs
tHAT
bHAT



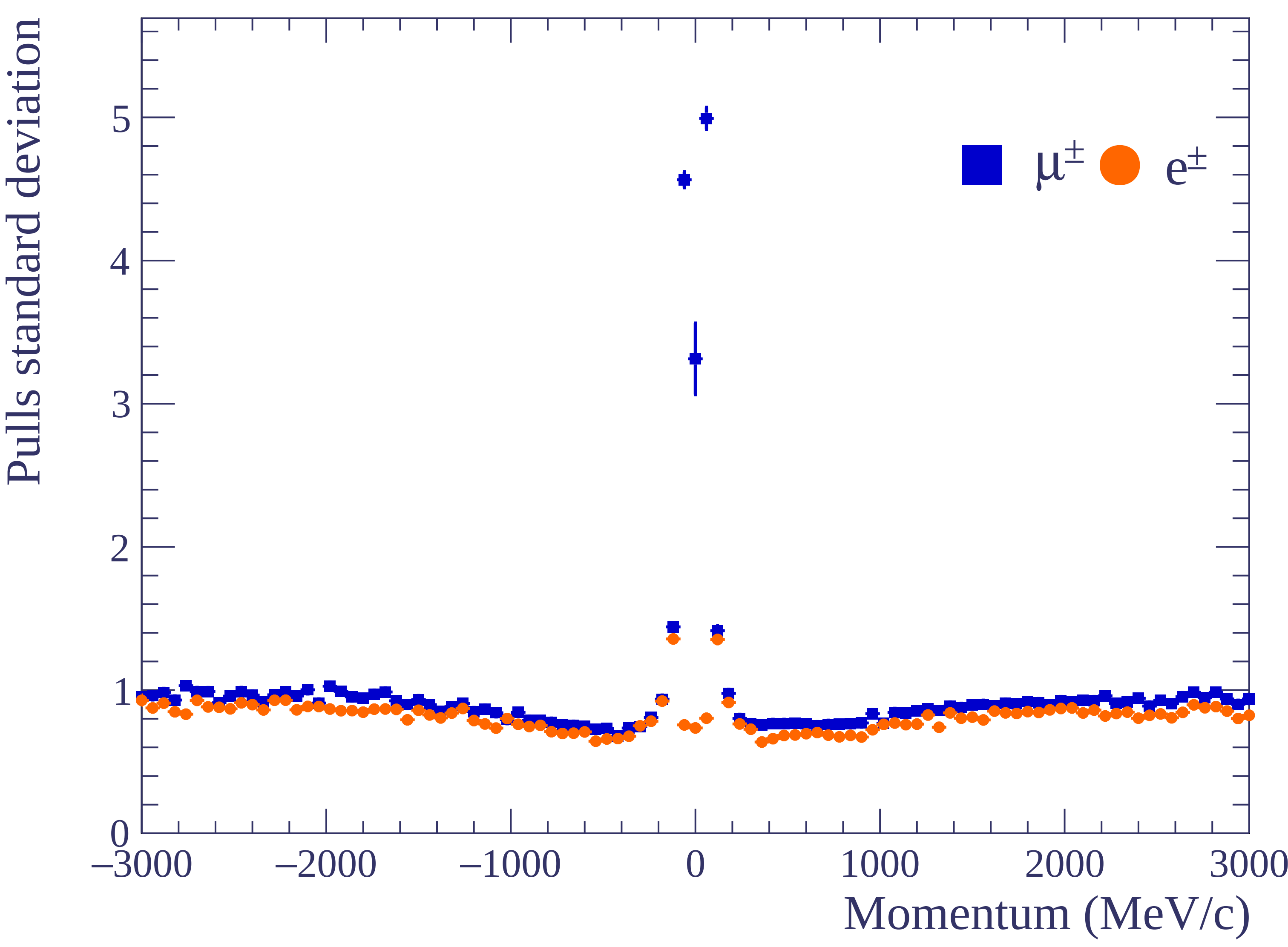
χ^2
50
45
40
35
30
25
20
15
10
5
0
−3000
−2000
−1000
0
1000
2000
3000
momentum (MeV)
160
140
120
100
80
60
40
20
0



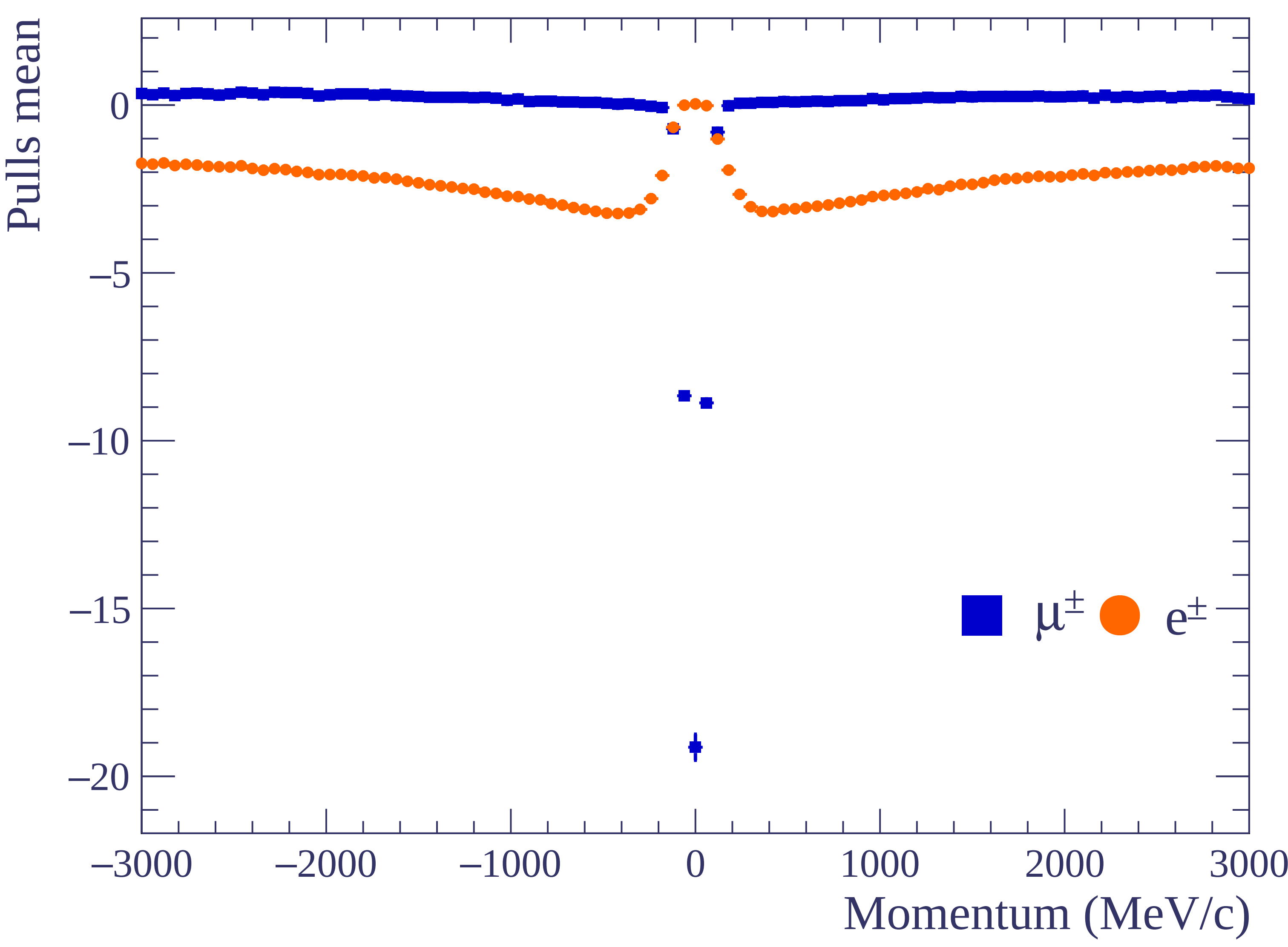
Pulls standard deviation
2.0
1.8
1.6
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±



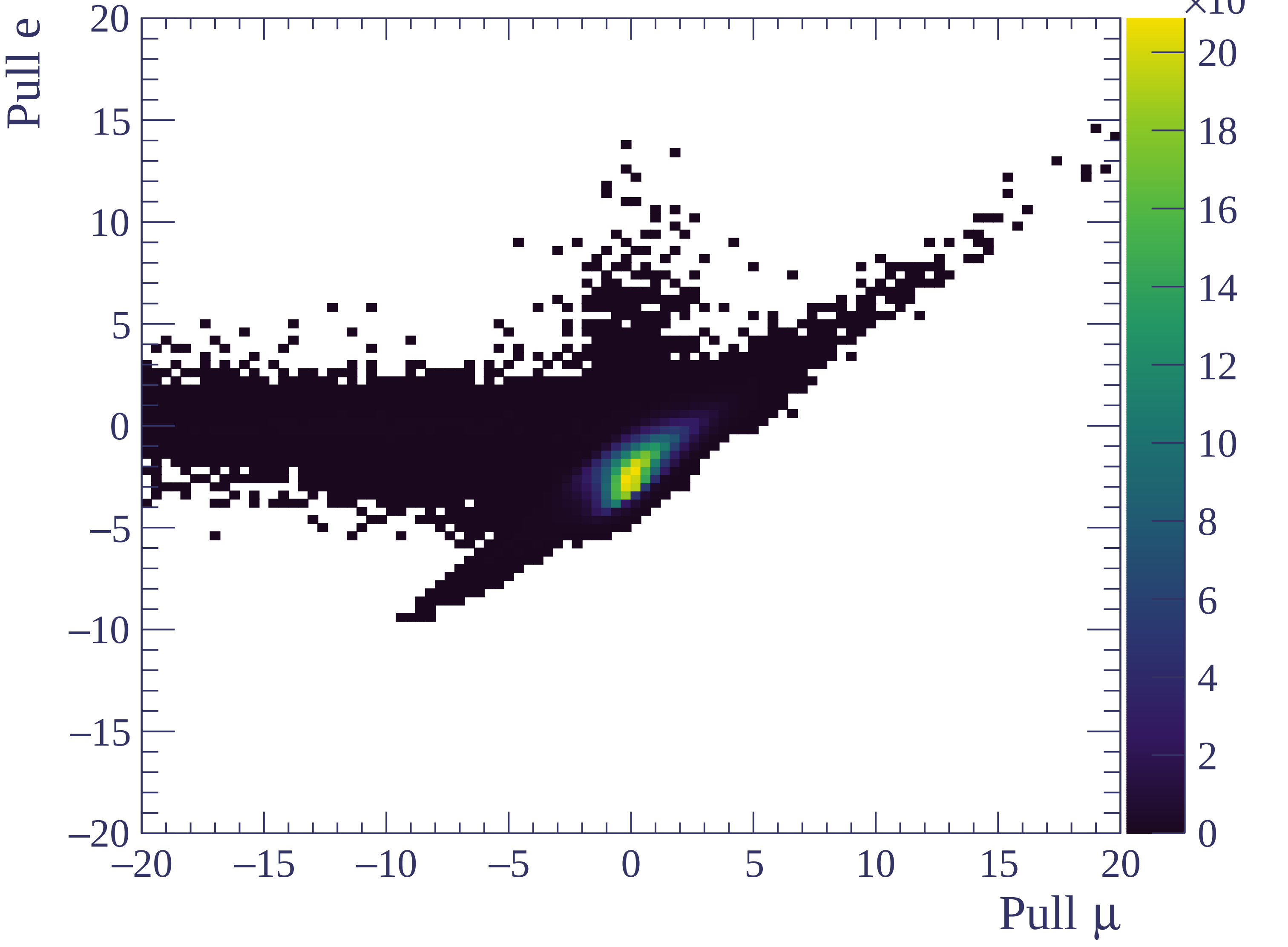
Pulls mean
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±



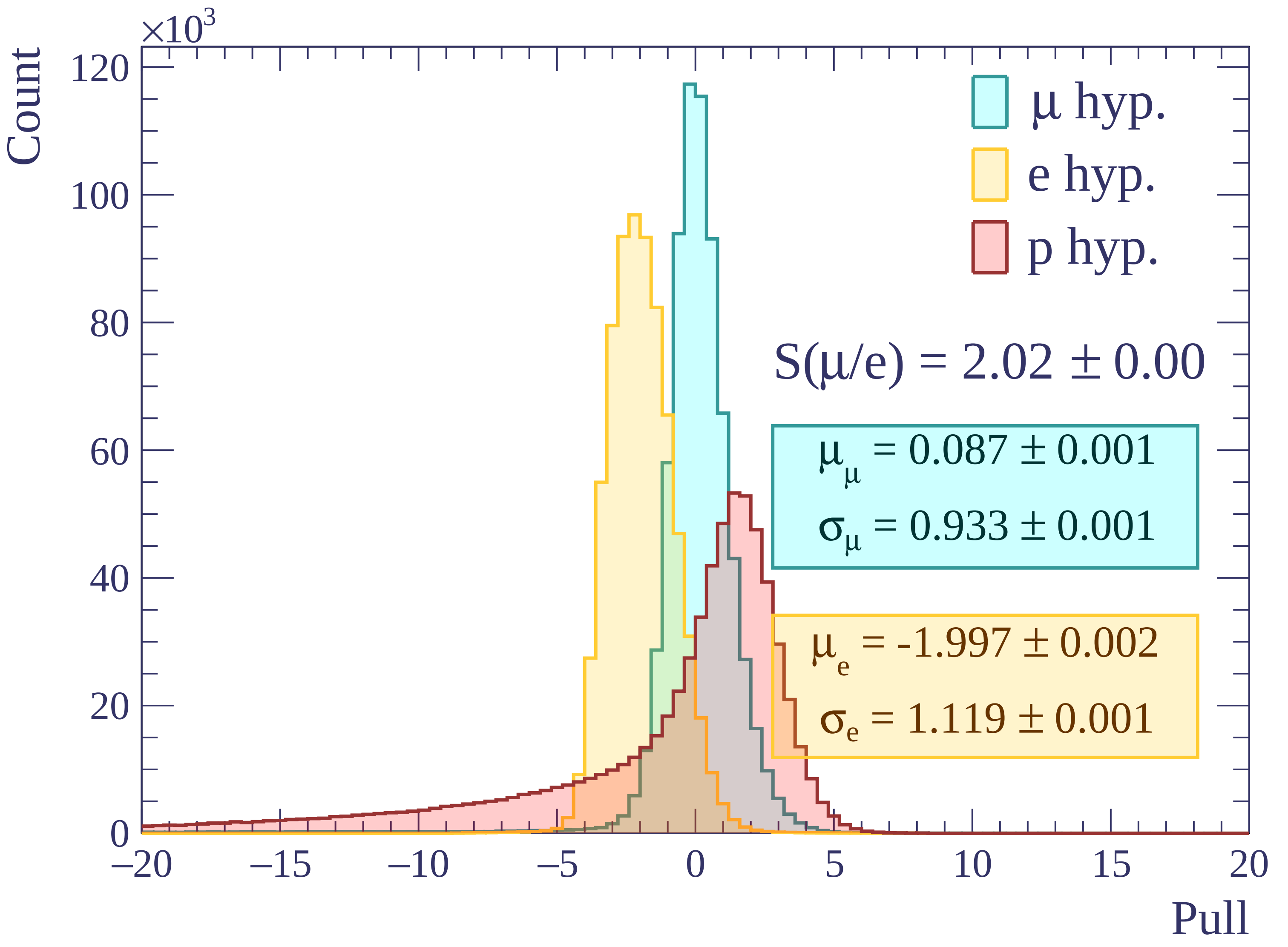
Pulls standard deviation
0
1
2
3
4
5
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
μ±
e±



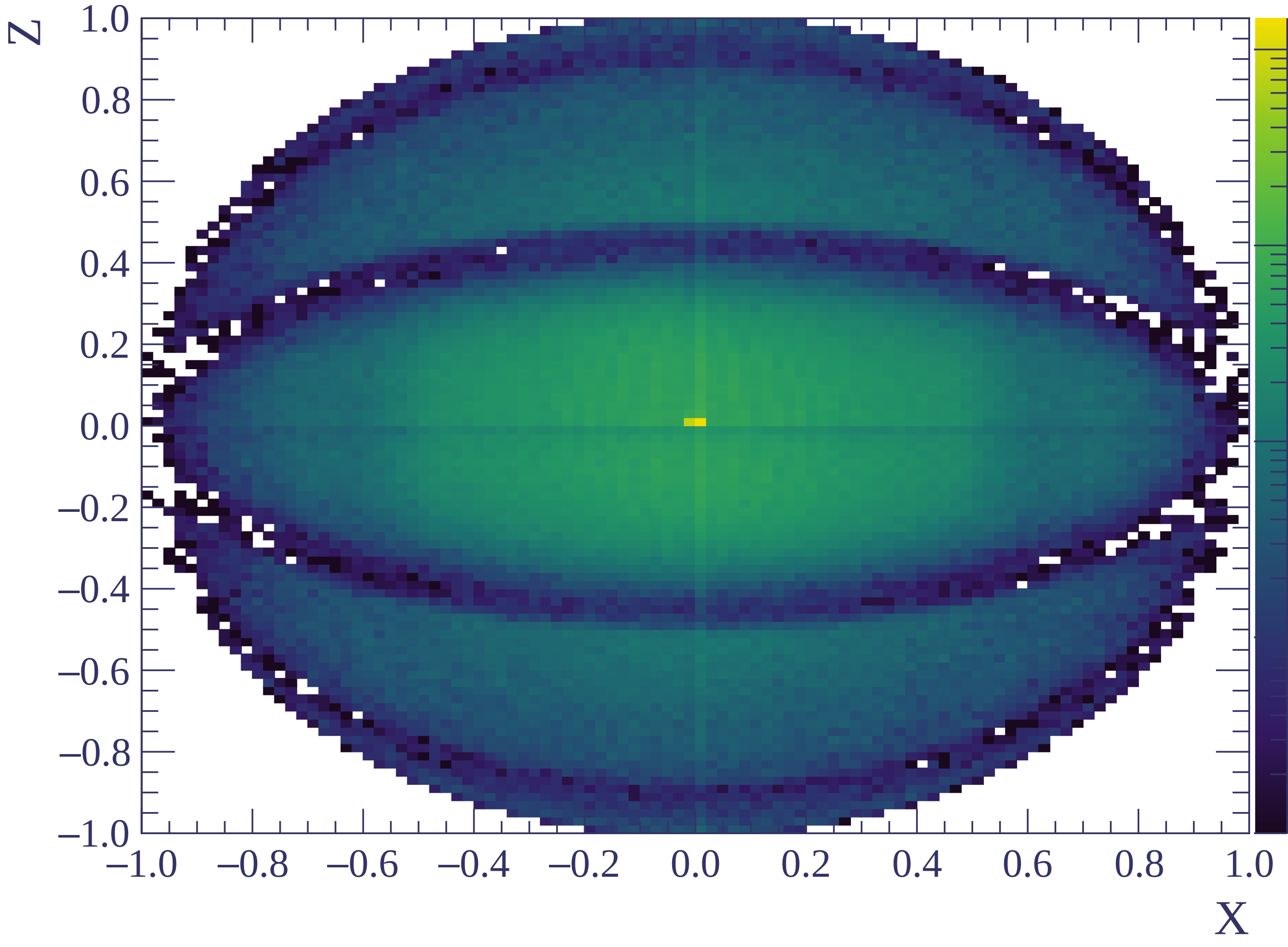
Pulls mean
0
−5
−10
−15
−20
−3000
−2000
−1000
0
1000
2000
3000
Momentum (MeV/c)
$\mu^\pm$
$e^\pm$

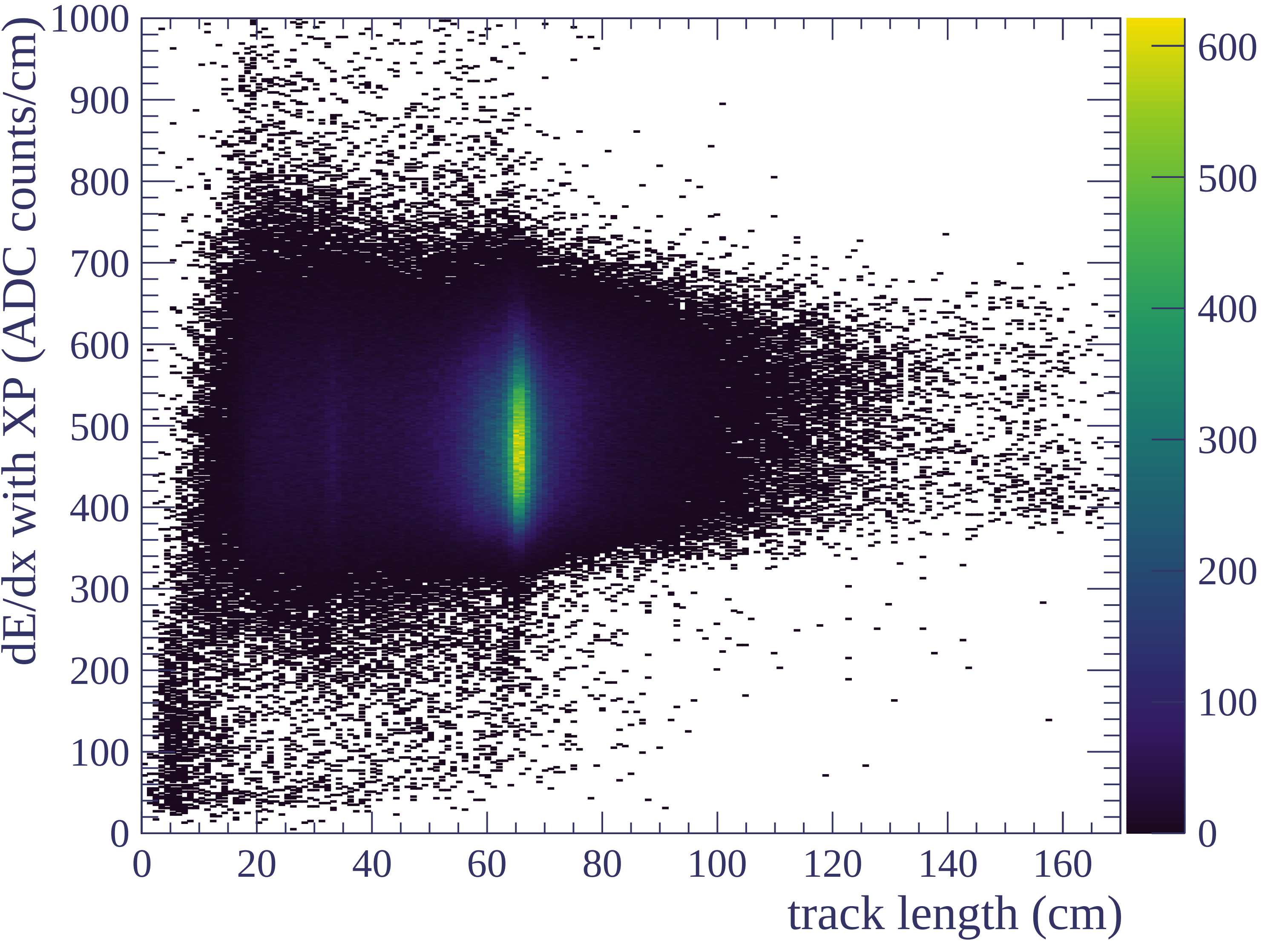


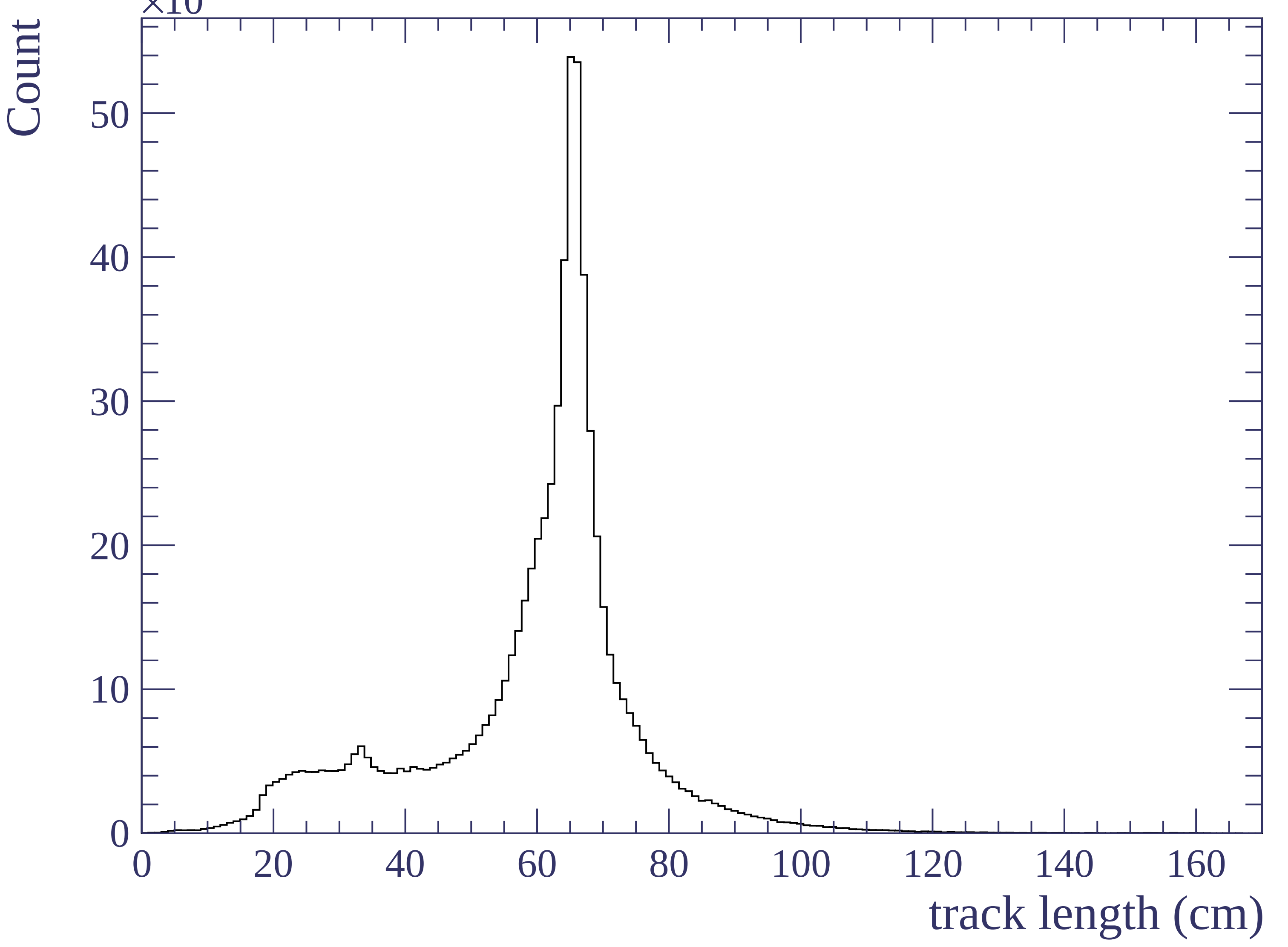
Pull e
Pull μ
×10
20
15
10
5
0
−5
−10
−15
−20
20
18
16
14
12
10
8
6
4
2
0



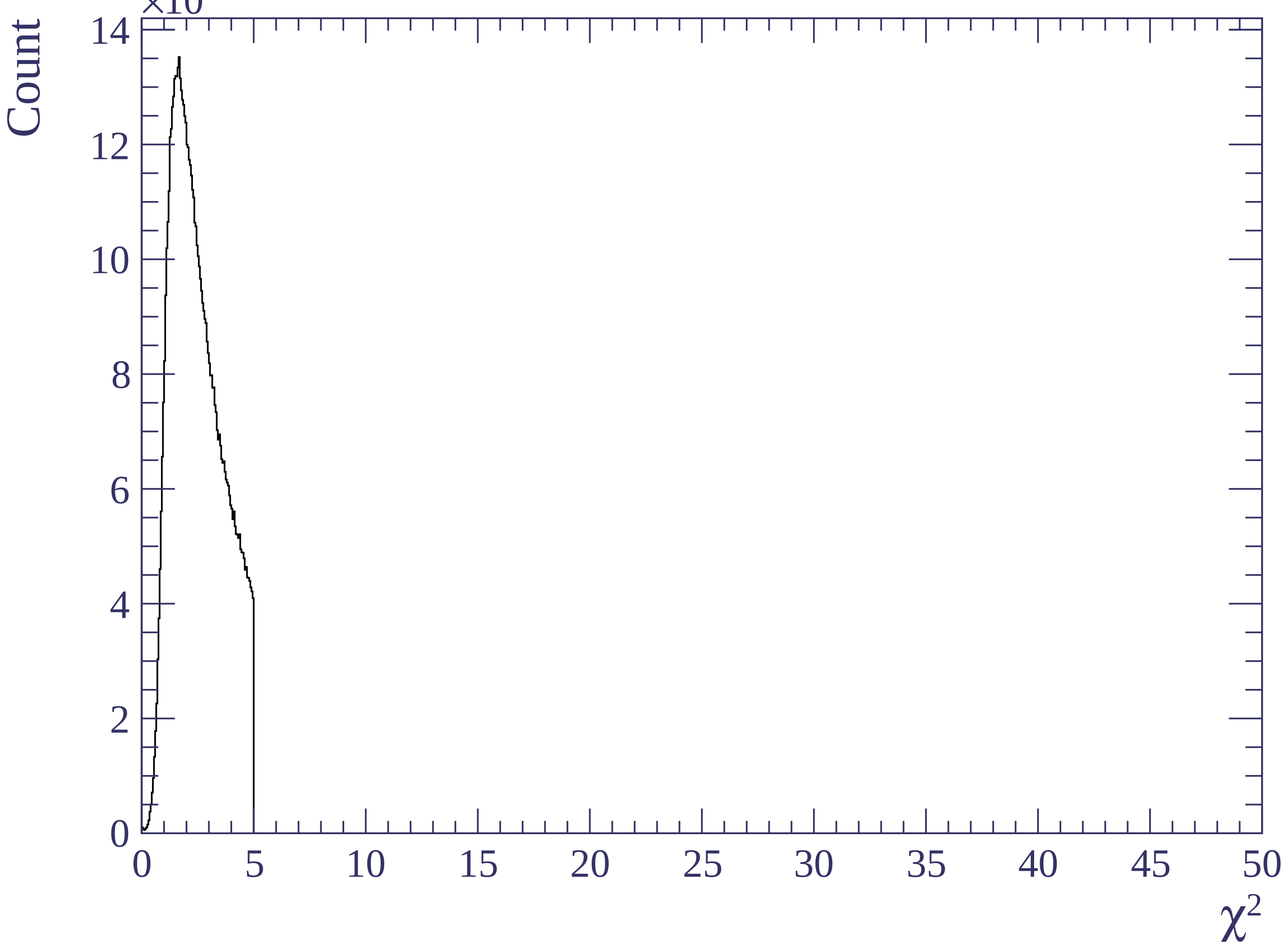
×10³
Count
120
100
80
60
40
20
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 2.02 ± 0.00
μ_μ = 0.087 ± 0.001
σ_μ = 0.933 ± 0.001
μ_e = -1.997 ± 0.002
σ_e = 1.119 ± 0.001


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

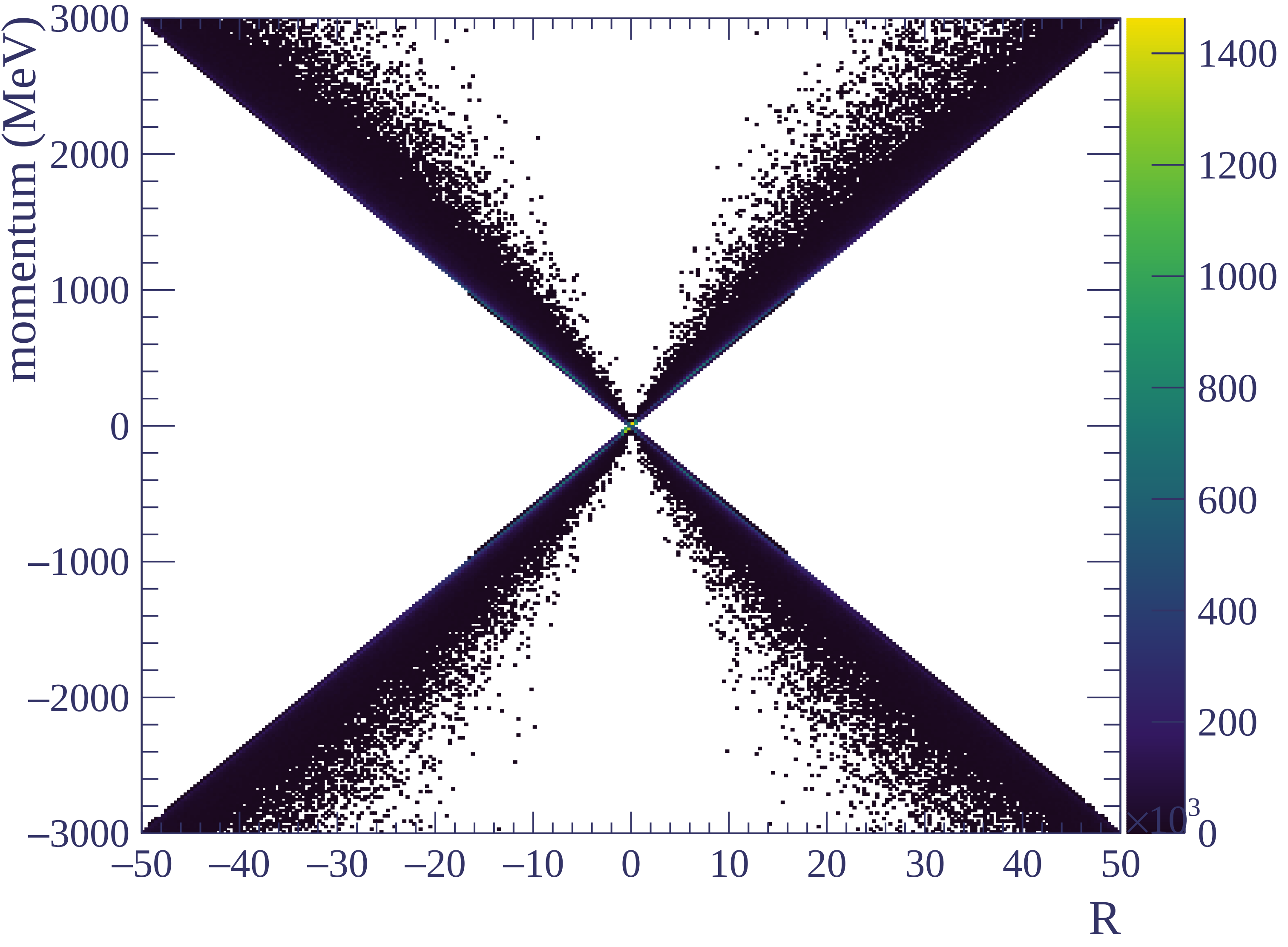

dE/dx with XP (ADC counts/cm)
track length (cm)
0
100
200
300
400
500
600
700
800
900
1000
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600



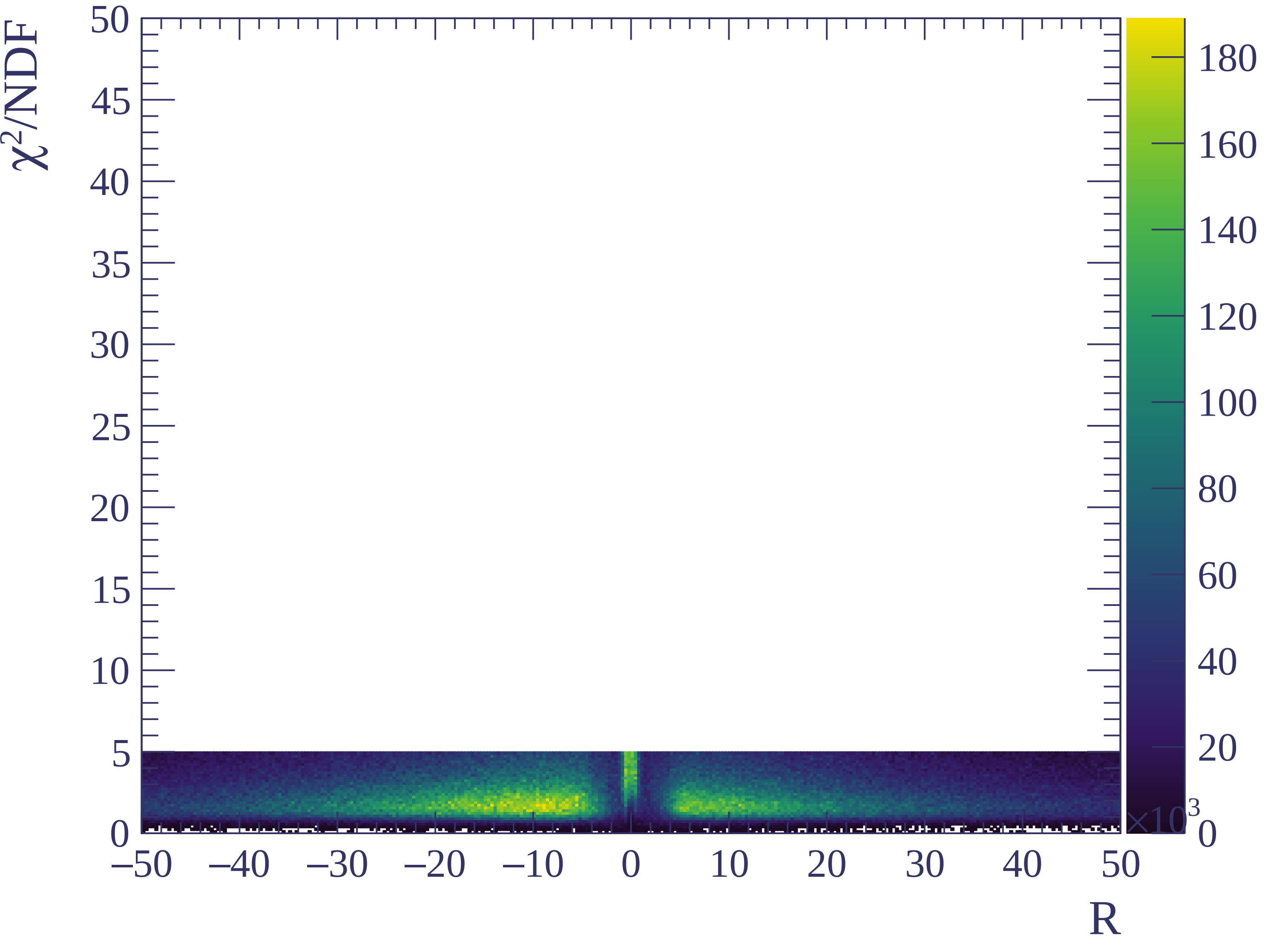
×10
Count
0
10
20
30
40
50
0
20
40
60
80
100
120
140
160
track length (cm)



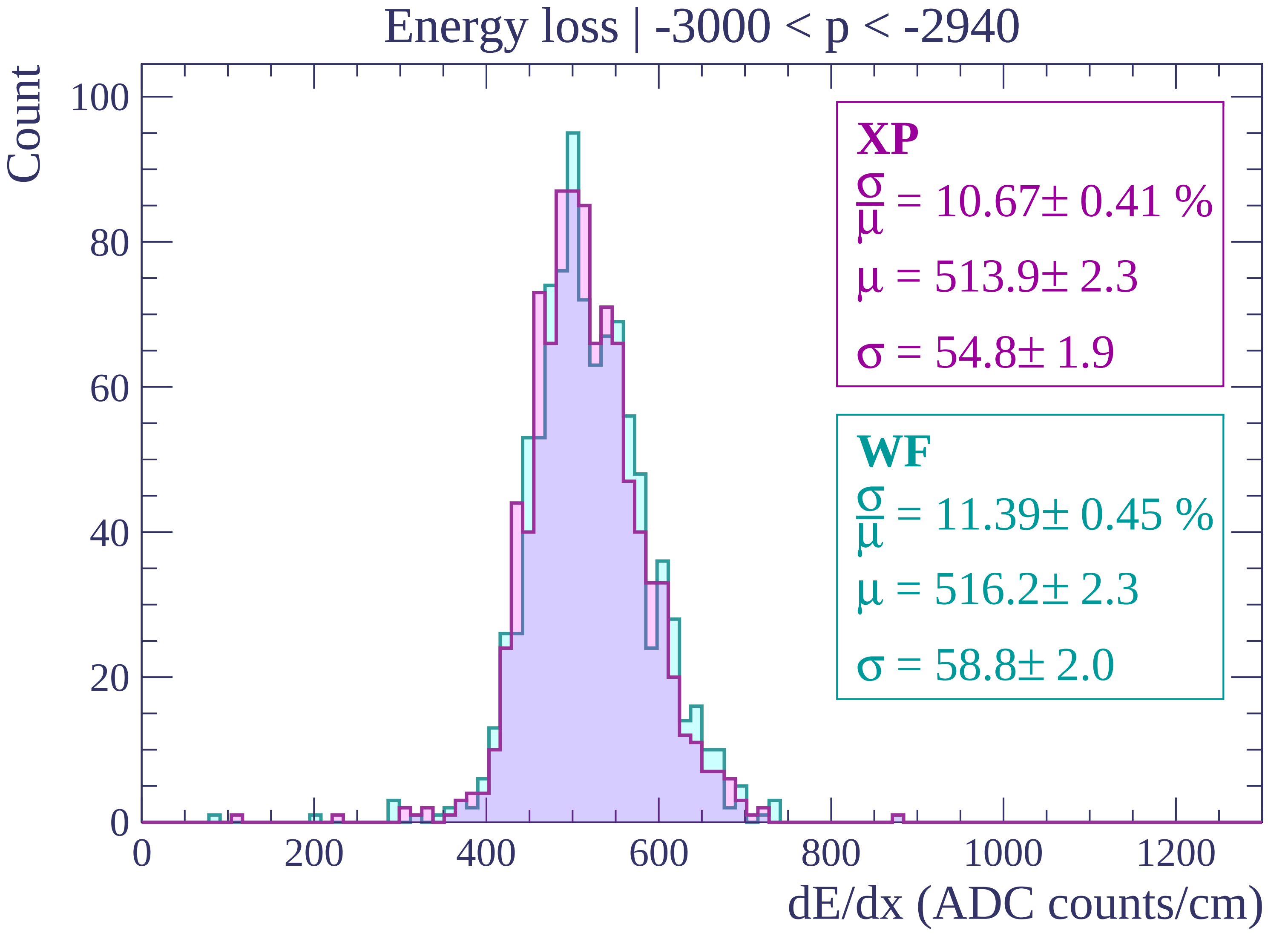
×10
Count
0
2
4
6
8
10
12
14
0
5
10
15
20
25
30
35
40
45
50
χ^2



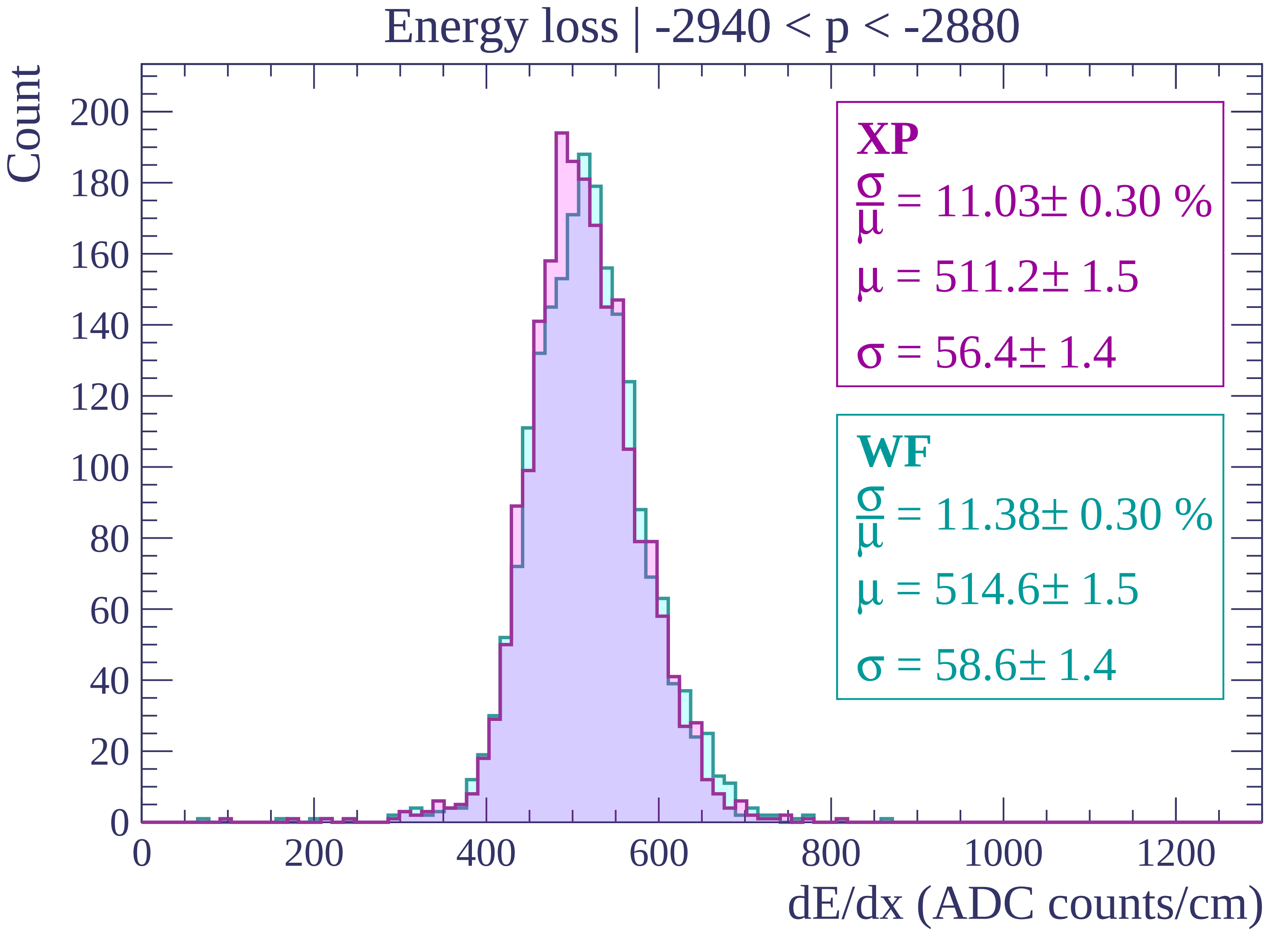
momentum (MeV)
3000
2000
1000
0
−1000
−2000
−3000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
1400
1200
1000
800
600
400
200
0
×10^3

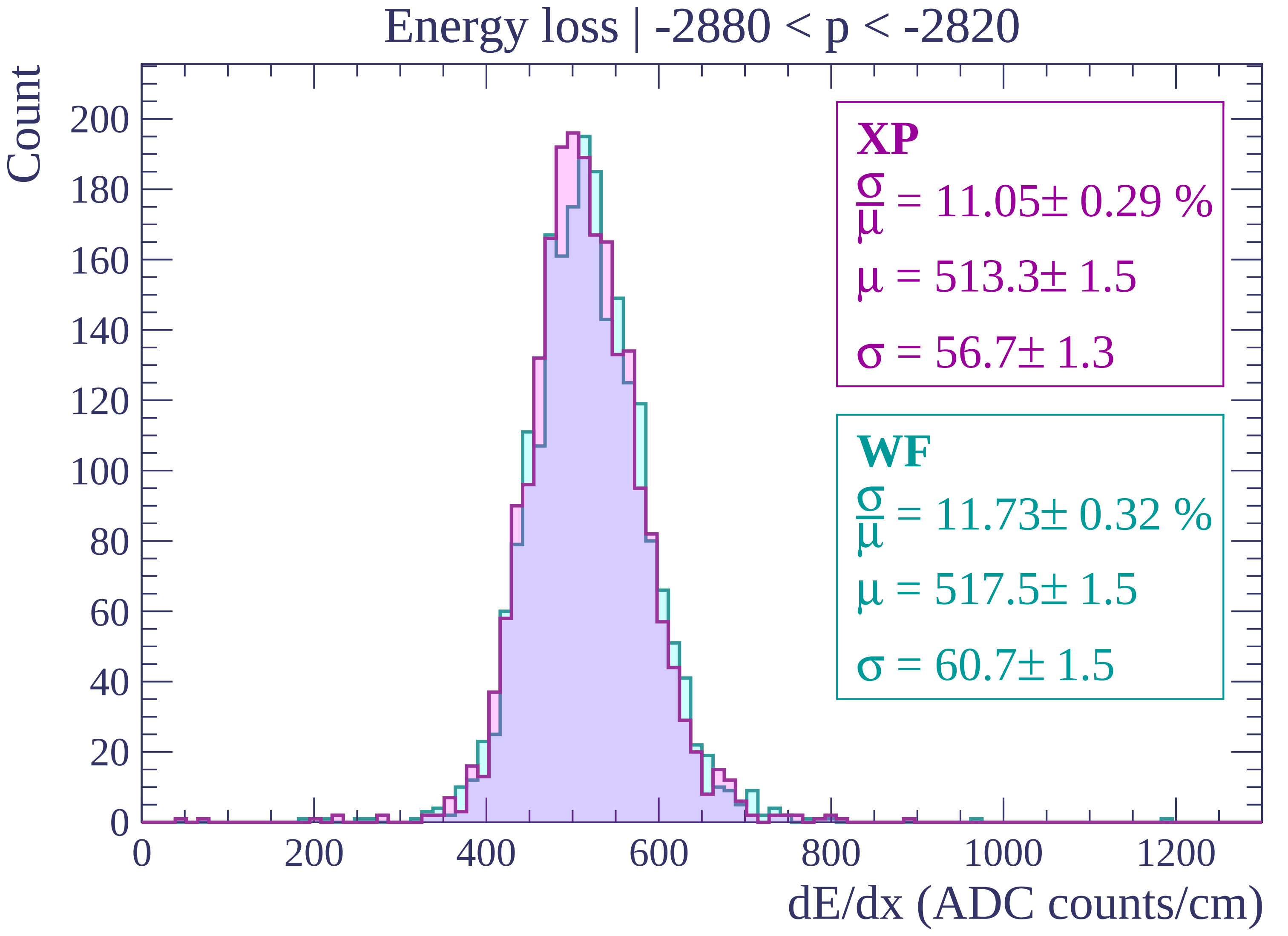


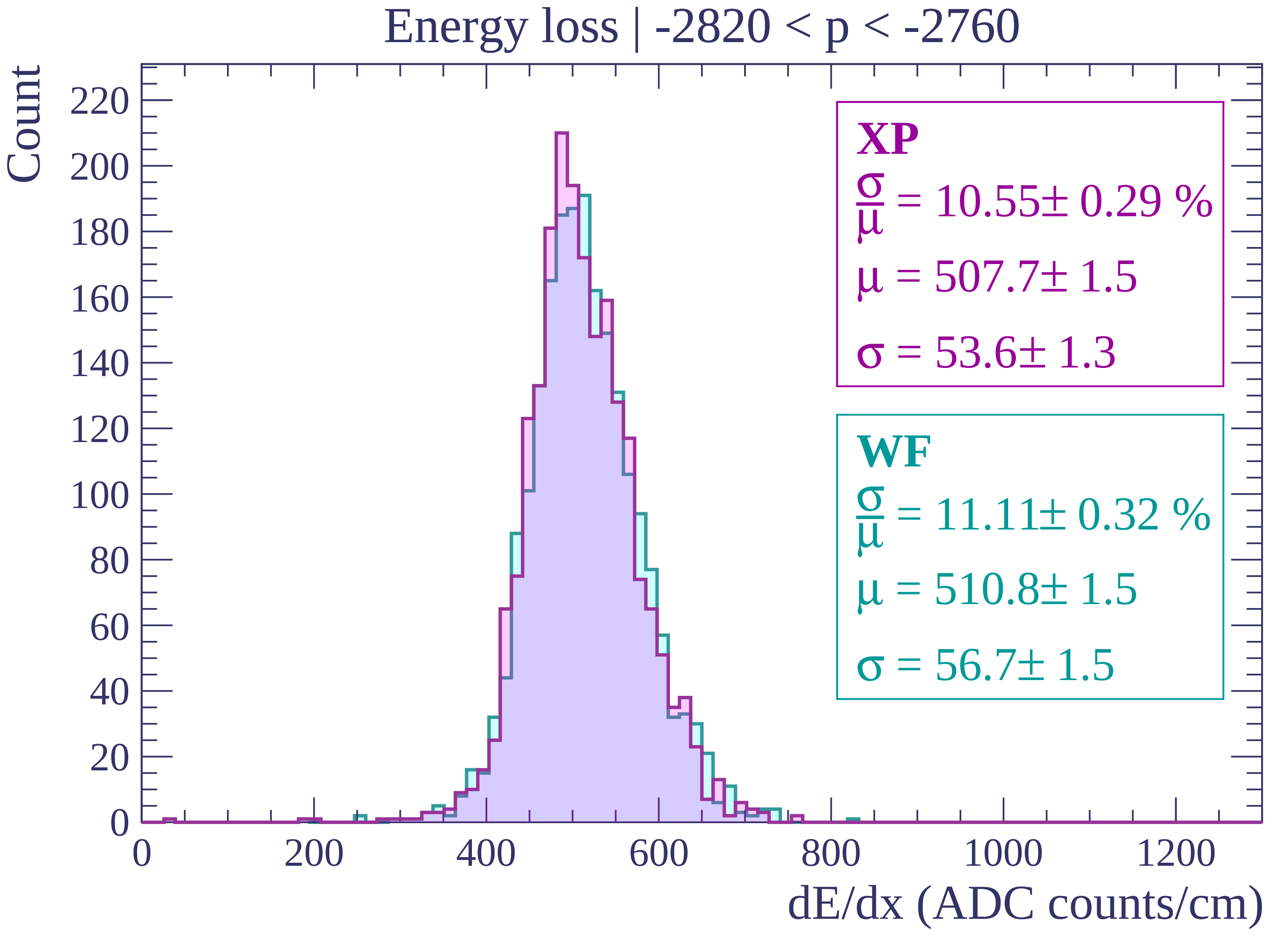
χ^2/NDF
R
$\times 10^3$



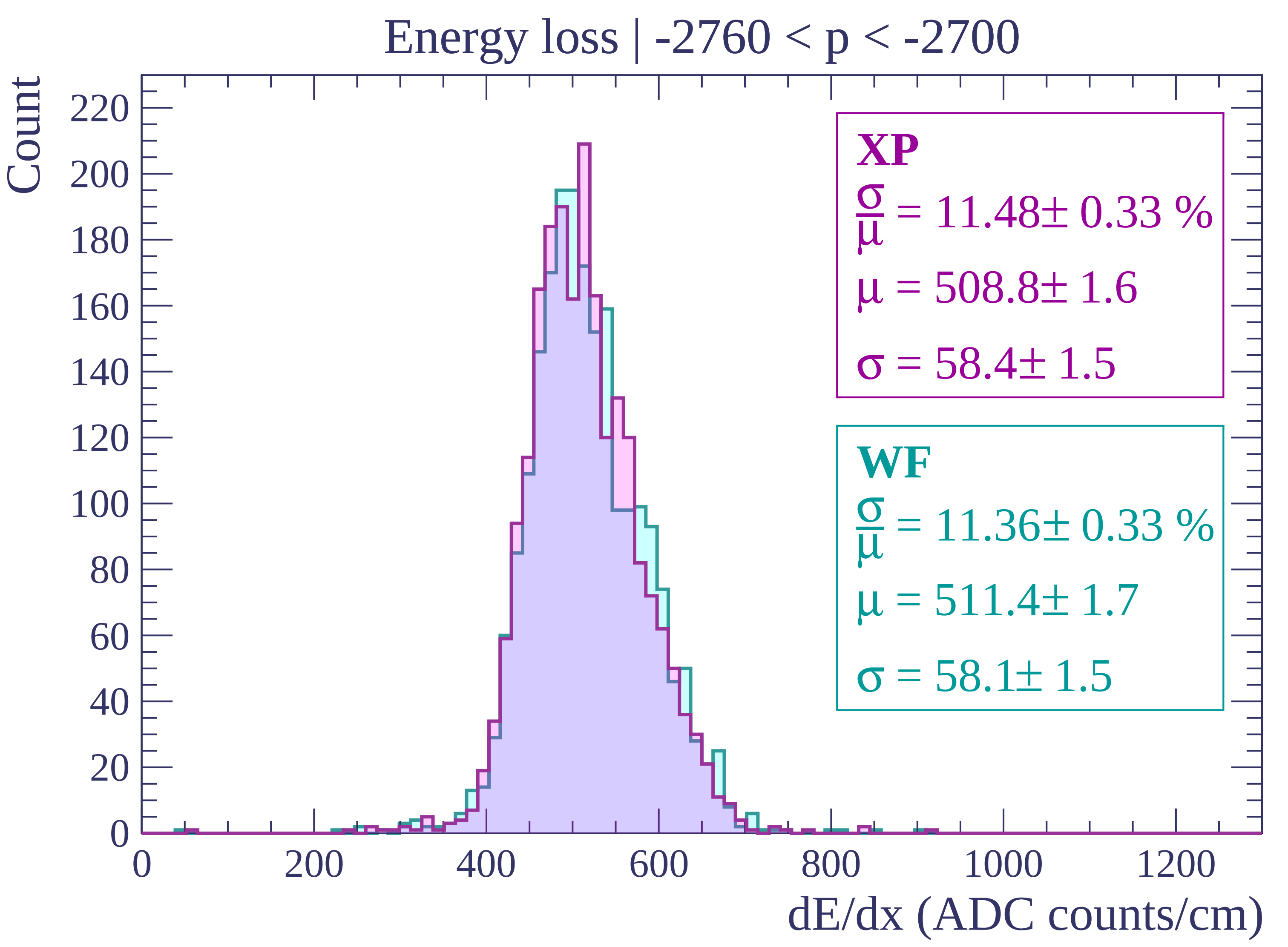
Energy loss | -3000 < p < -2940
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
XP
σ/μ = 10.67± 0.41 %
μ = 513.9± 2.3
σ = 54.8± 1.9
WF
σ/μ = 11.39± 0.45 %
μ = 516.2± 2.3
σ = 58.8± 2.0


Energy loss | -2940 < p < -2880
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.03± 0.30 %
μ = 511.2± 1.5
σ = 56.4± 1.4
WF
$\frac{\sigma}{\mu}$ = 11.38± 0.30 %
μ = 514.6± 1.5
σ = 58.6± 1.4

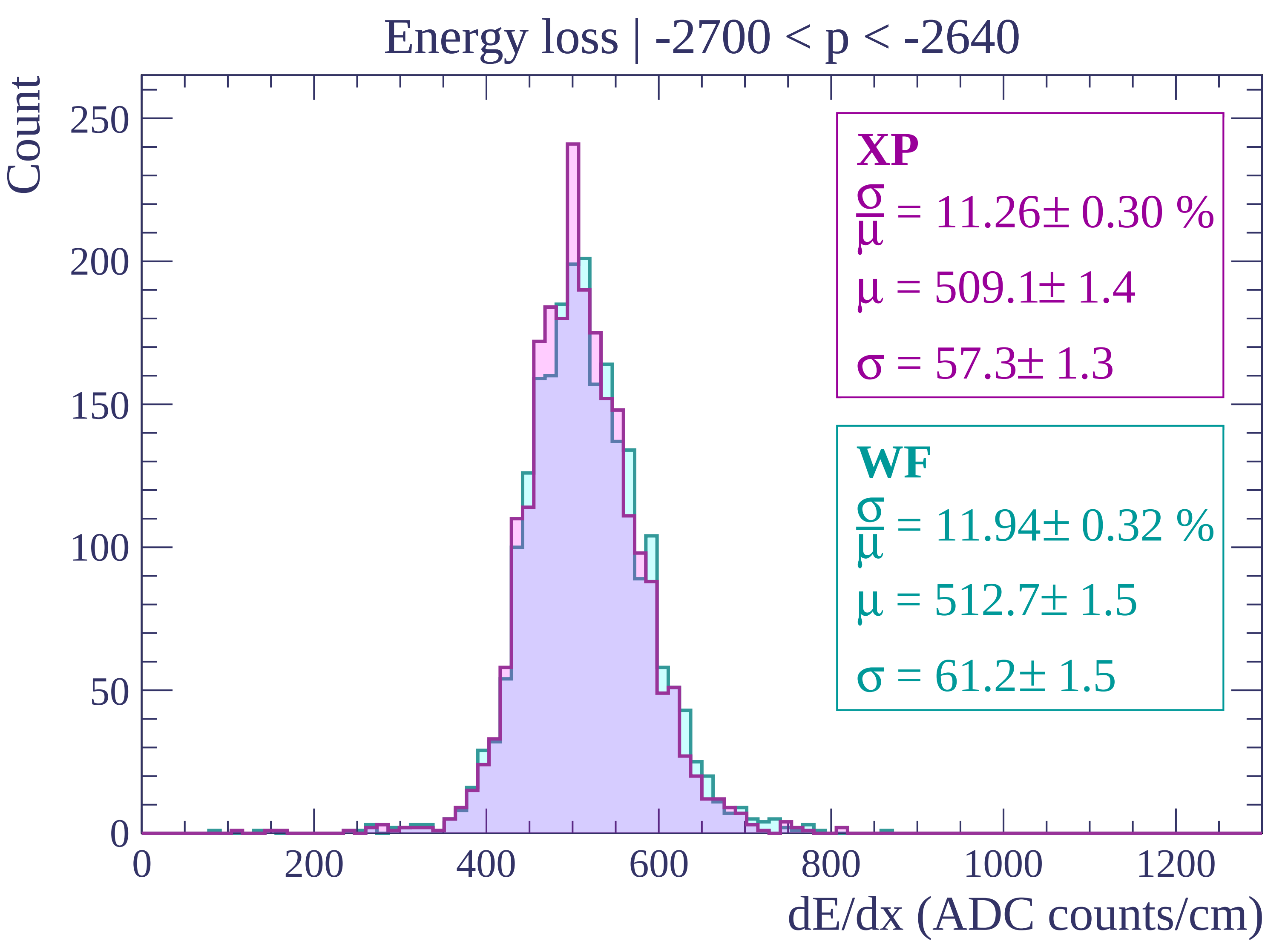

Energy loss | -2880 < p < -2820
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
$\frac{\sigma}{\mu} = 11.05 \pm 0.29$ %
$\mu = 513.3 \pm 1.5$
$\sigma = 56.7 \pm 1.3$
WF
$\frac{\sigma}{\mu} = 11.73 \pm 0.32$ %
$\mu = 517.5 \pm 1.5$
$\sigma = 60.7 \pm 1.5$



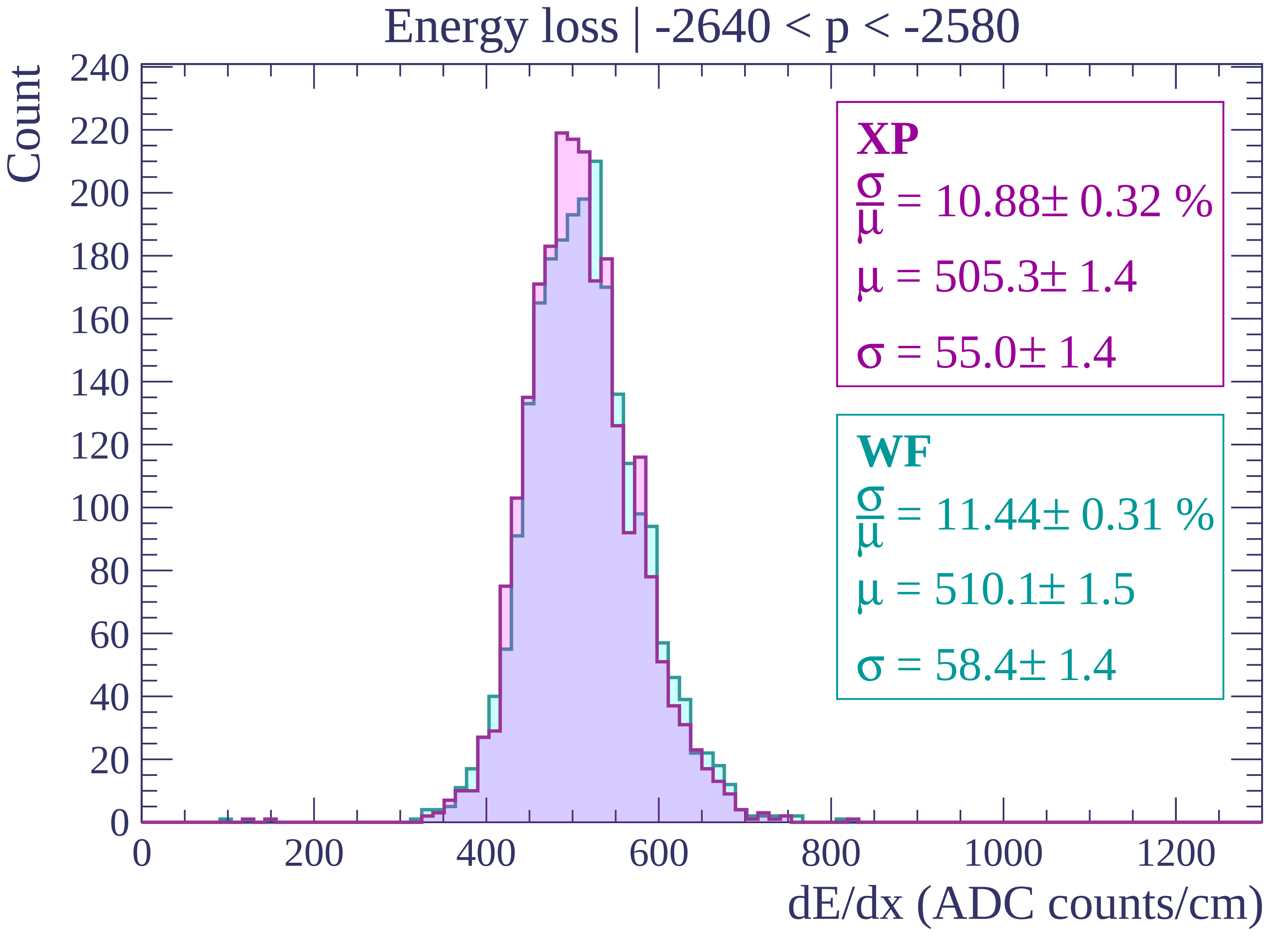
Energy loss | -2820 < p < -2760
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.55± 0.29 %
μ = 507.7± 1.5
σ = 53.6± 1.3
WF
$\frac{\sigma}{\mu}$ = 11.11± 0.32 %
μ = 510.8± 1.5
σ = 56.7± 1.5

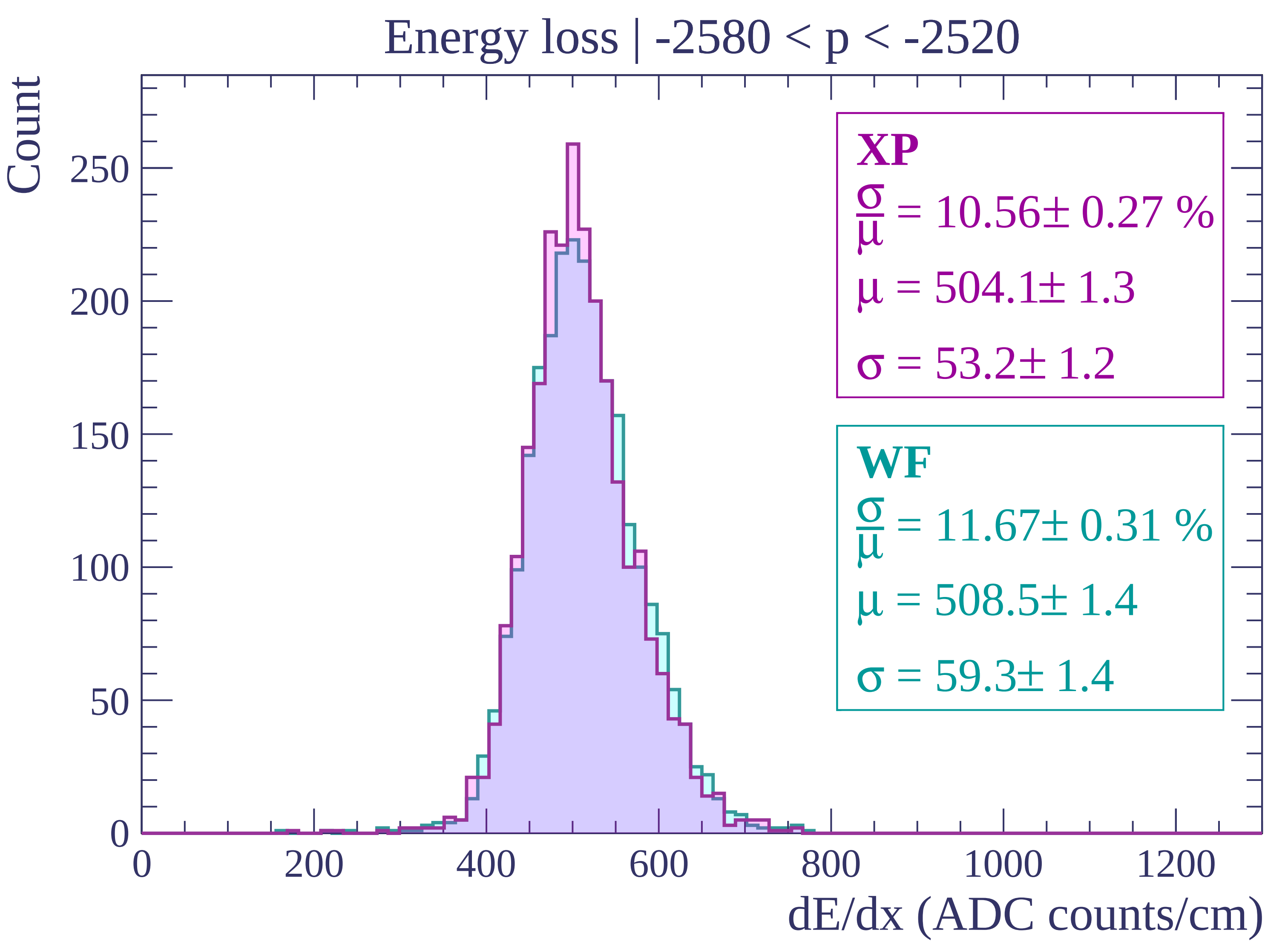


Energy loss | -2760 < p < -2700
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 11.48± 0.33 %
μ = 508.8± 1.6
σ = 58.4± 1.5
WF
σ/μ = 11.36± 0.33 %
μ = 511.4± 1.7
σ = 58.1± 1.5
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200

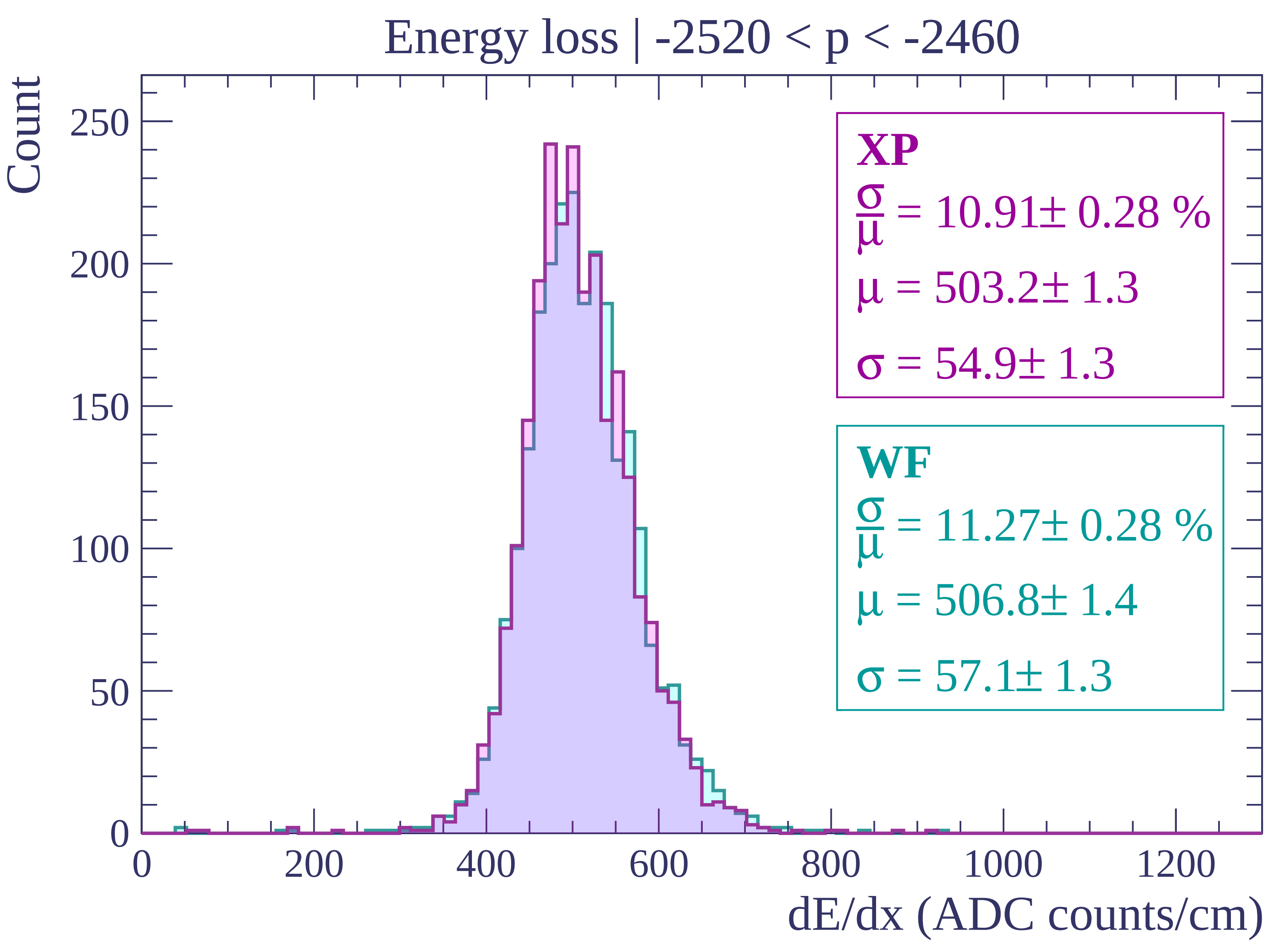


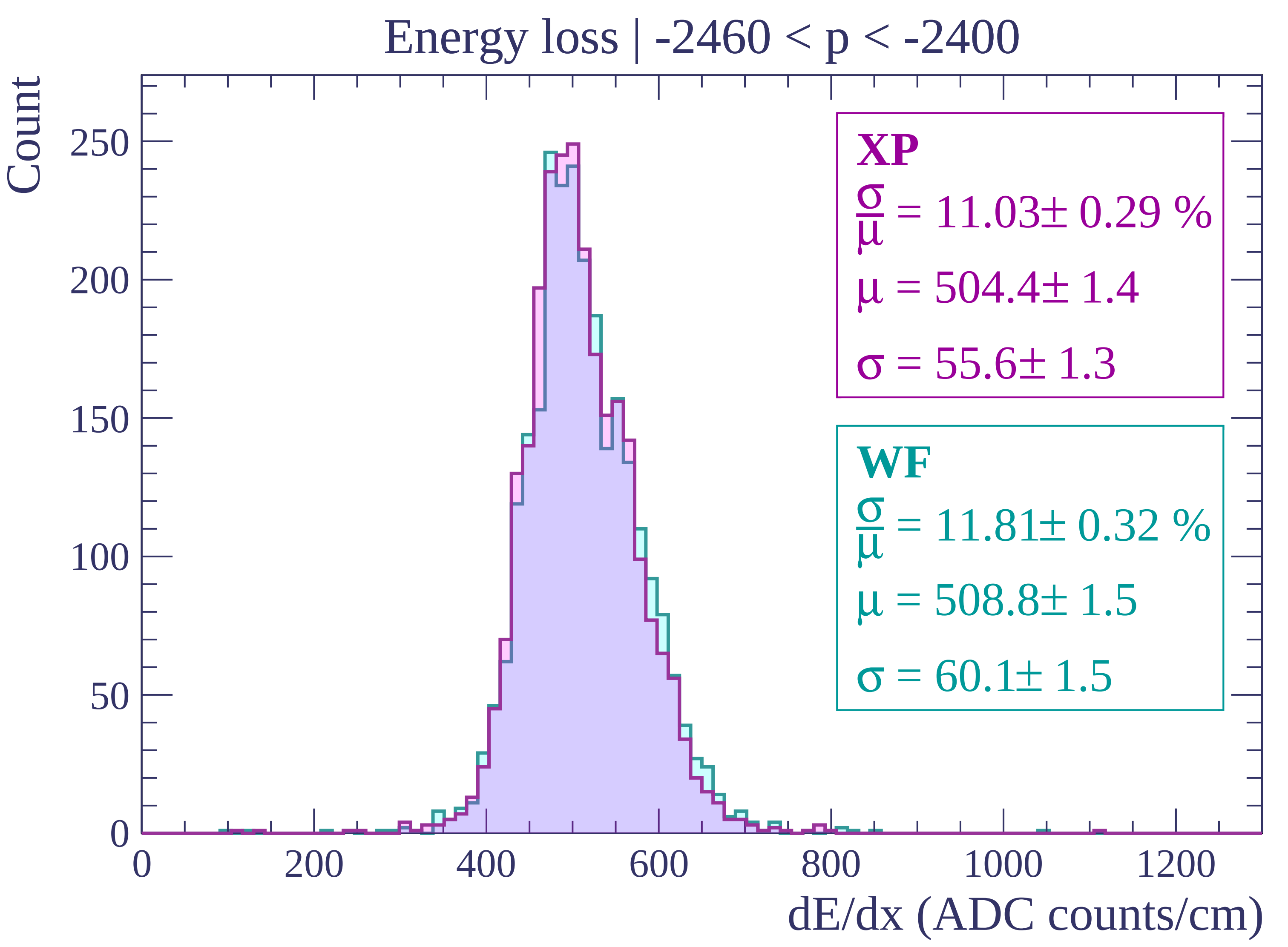
Energy loss | -2700 < p < -2640
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.26± 0.30 %
μ = 509.1± 1.4
σ = 57.3± 1.3
WF
$\frac{\sigma}{\mu}$ = 11.94± 0.32 %
μ = 512.7± 1.5
σ = 61.2± 1.5

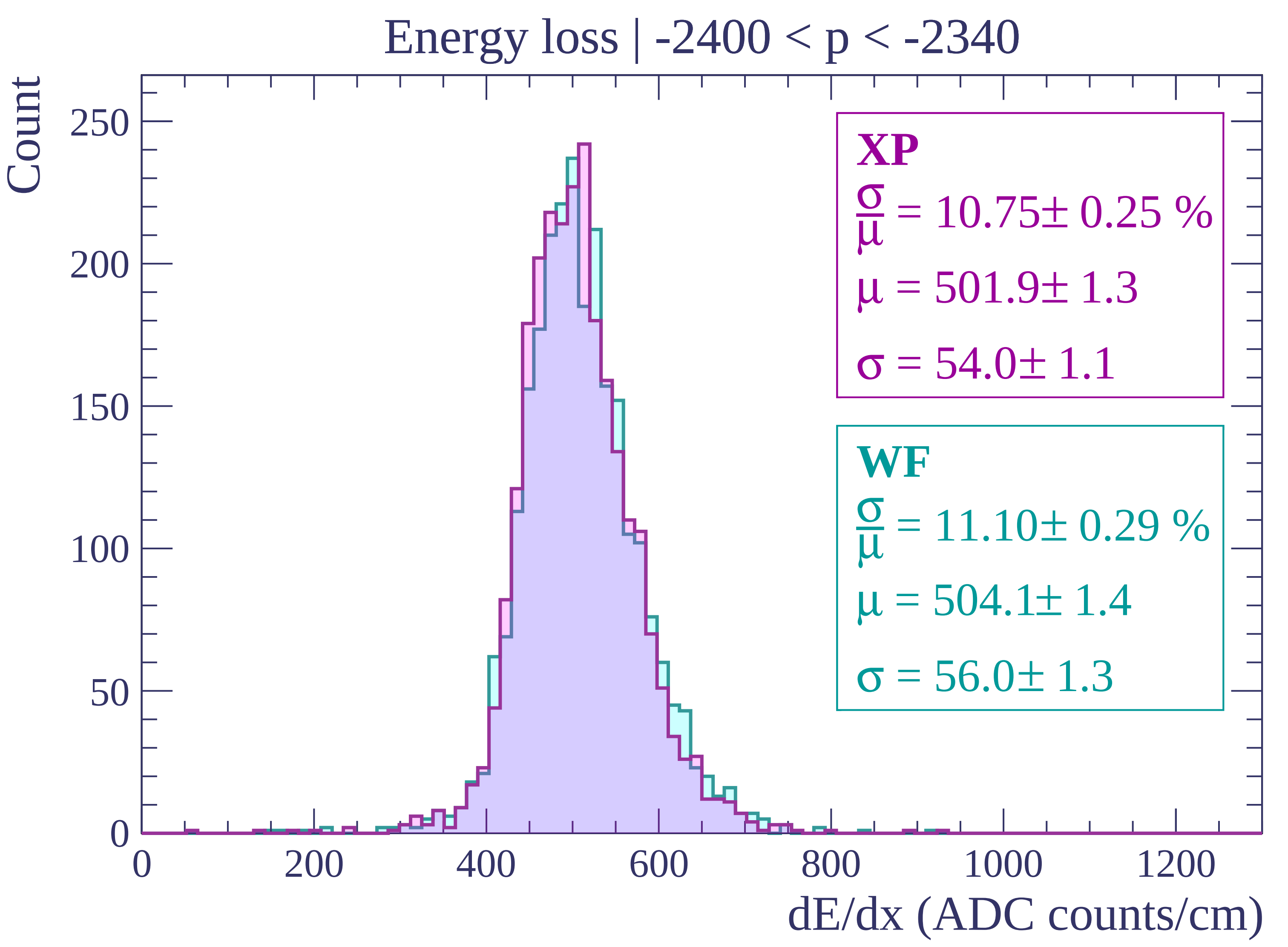

Energy loss | -2640 < p < -2580
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
240
0
200
400
600
800
1000
1200
XP
σ/μ = 10.88± 0.32 %
μ = 505.3± 1.4
σ = 55.0± 1.4
WF
σ/μ = 11.44± 0.31 %
μ = 510.1± 1.5
σ = 58.4± 1.4



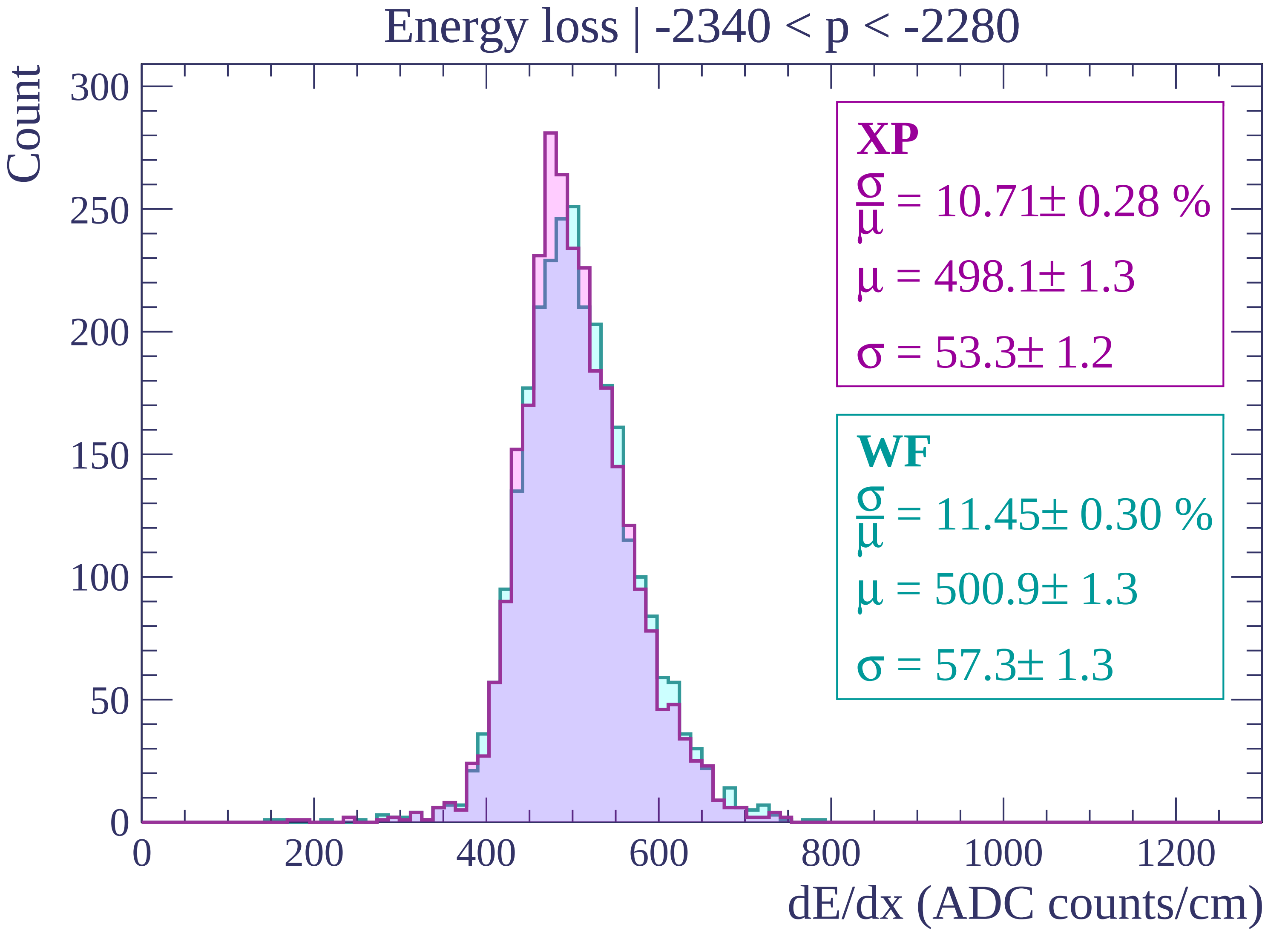
Energy loss | -2580 < p < -2520
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
σ/μ = 10.56± 0.27 %
μ = 504.1± 1.3
σ = 53.2± 1.2
WF
σ/μ = 11.67± 0.31 %
μ = 508.5± 1.4
σ = 59.3± 1.4


Energy loss | -2520 < p < -2460
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.91± 0.28 %
μ = 503.2± 1.3
σ = 54.9± 1.3
WF
$\frac{\sigma}{\mu}$ = 11.27± 0.28 %
μ = 506.8± 1.4
σ = 57.1± 1.3

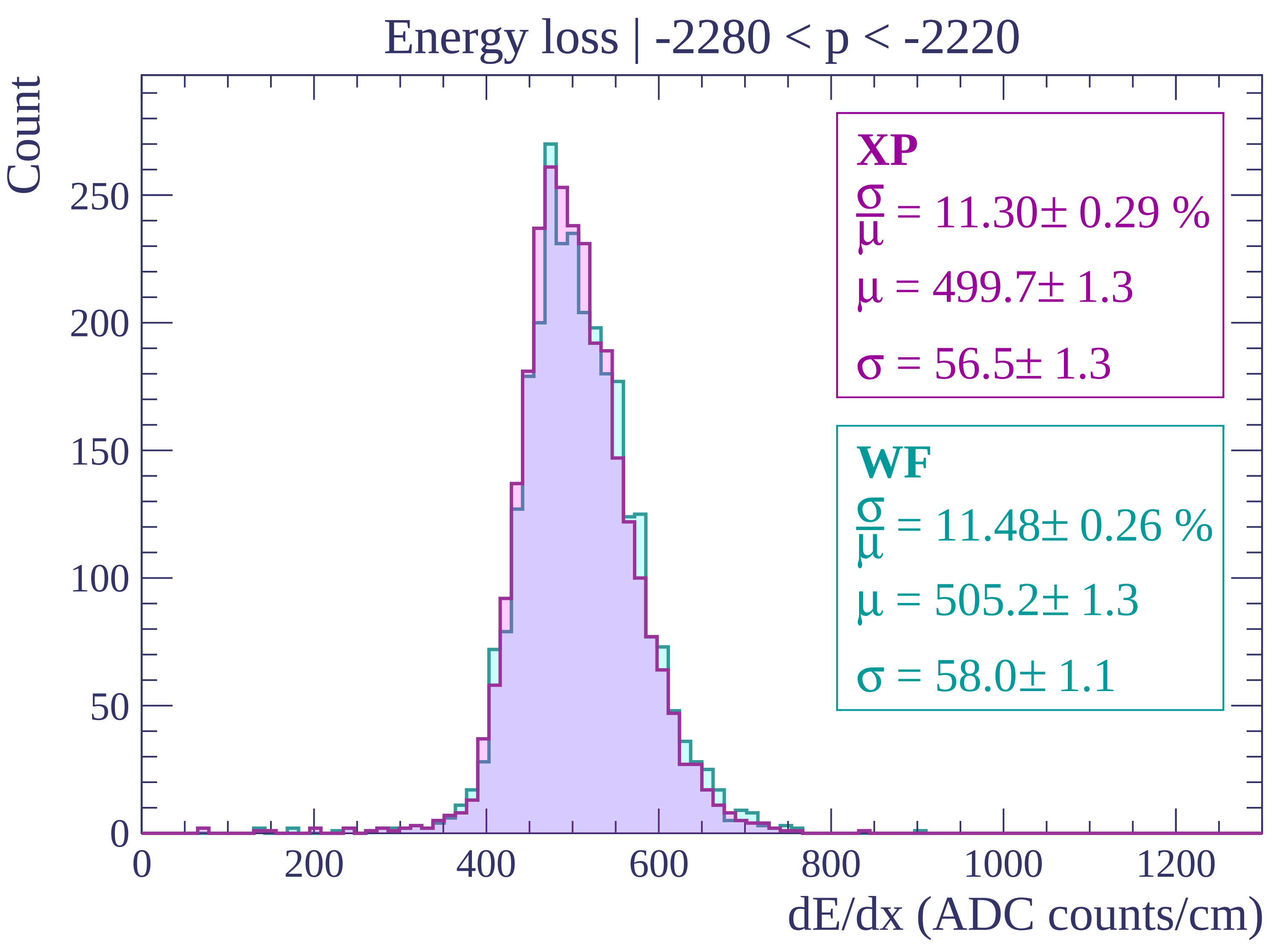

Energy loss | -2460 < p < -2400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 11.03± 0.29 %
μ = 504.4± 1.4
σ = 55.6± 1.3
WF
σ/μ = 11.81± 0.32 %
μ = 508.8± 1.5
σ = 60.1± 1.5

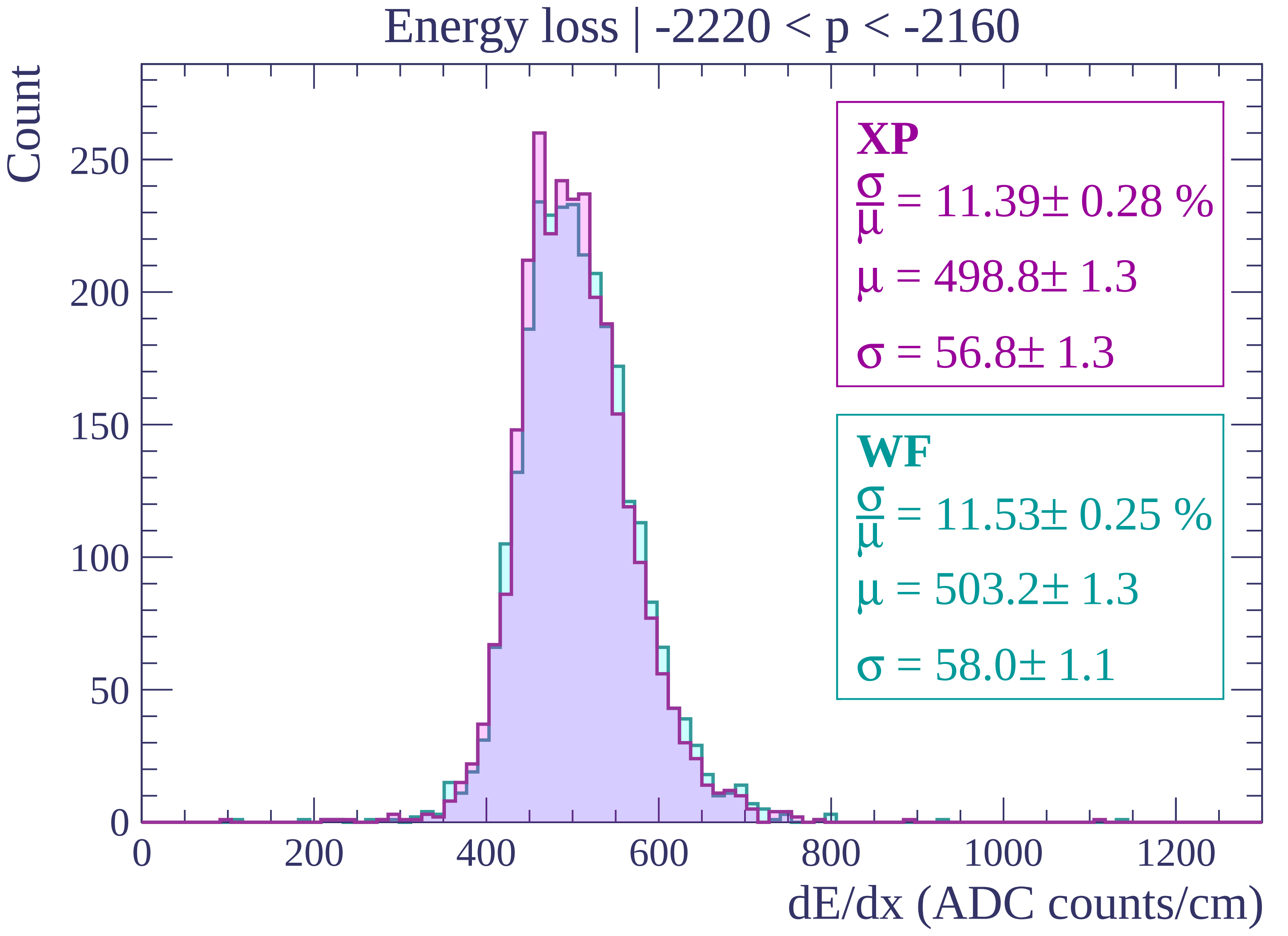


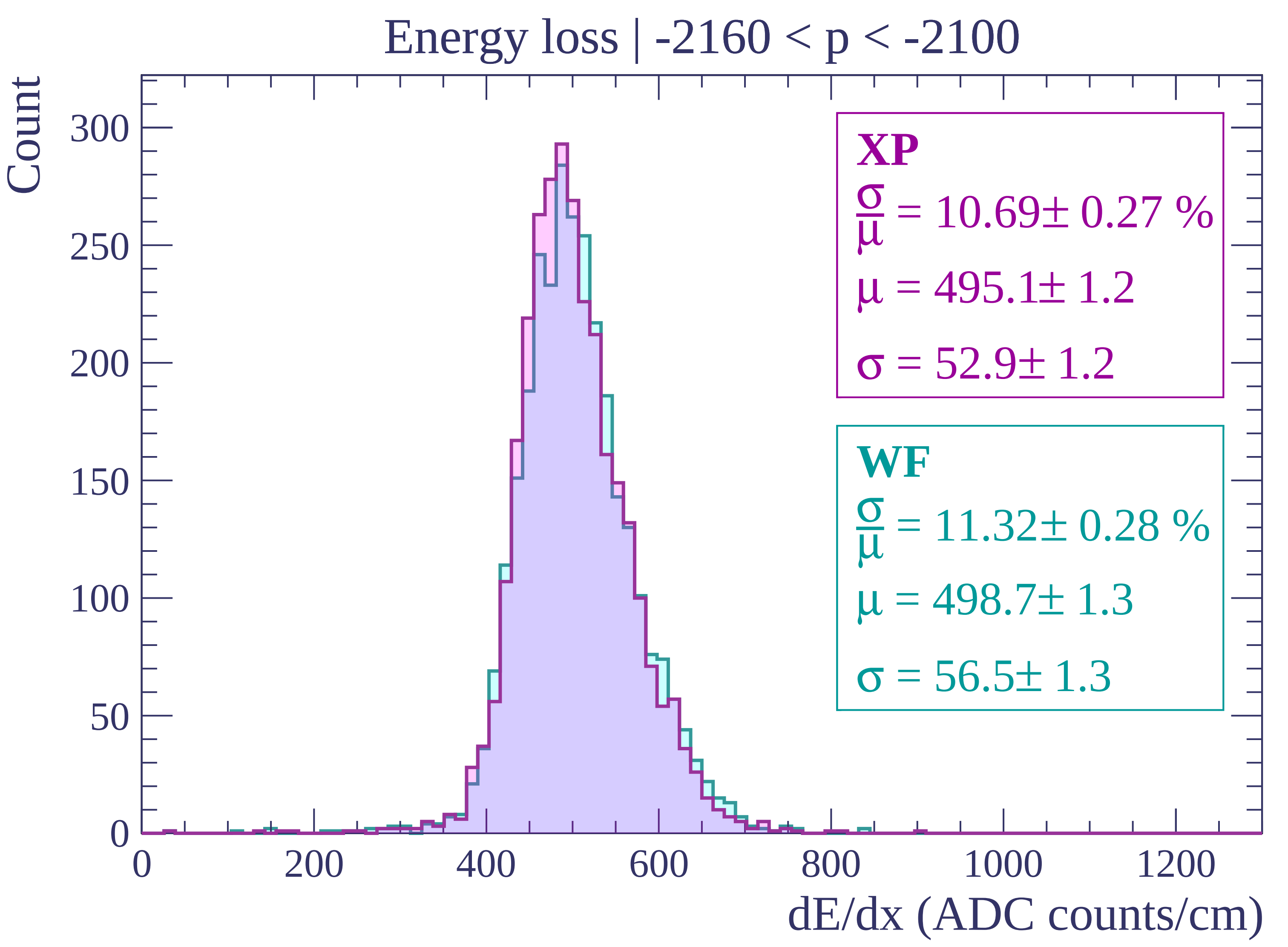
Energy loss | -2400 < p < -2340
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.75± 0.25 %
μ = 501.9± 1.3
σ = 54.0± 1.1
WF
$\frac{\sigma}{\mu}$ = 11.10± 0.29 %
μ = 504.1± 1.4
σ = 56.0± 1.3

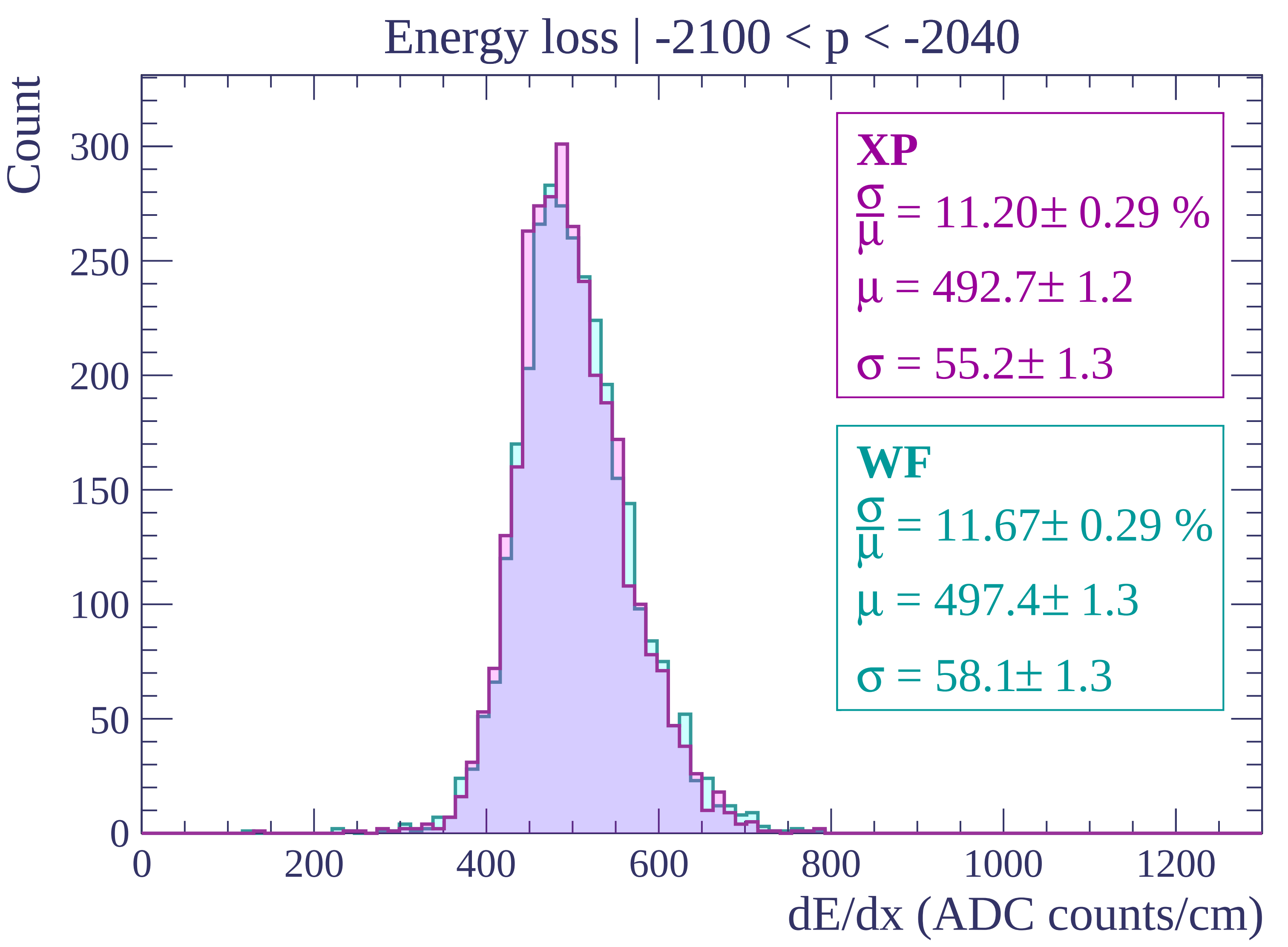


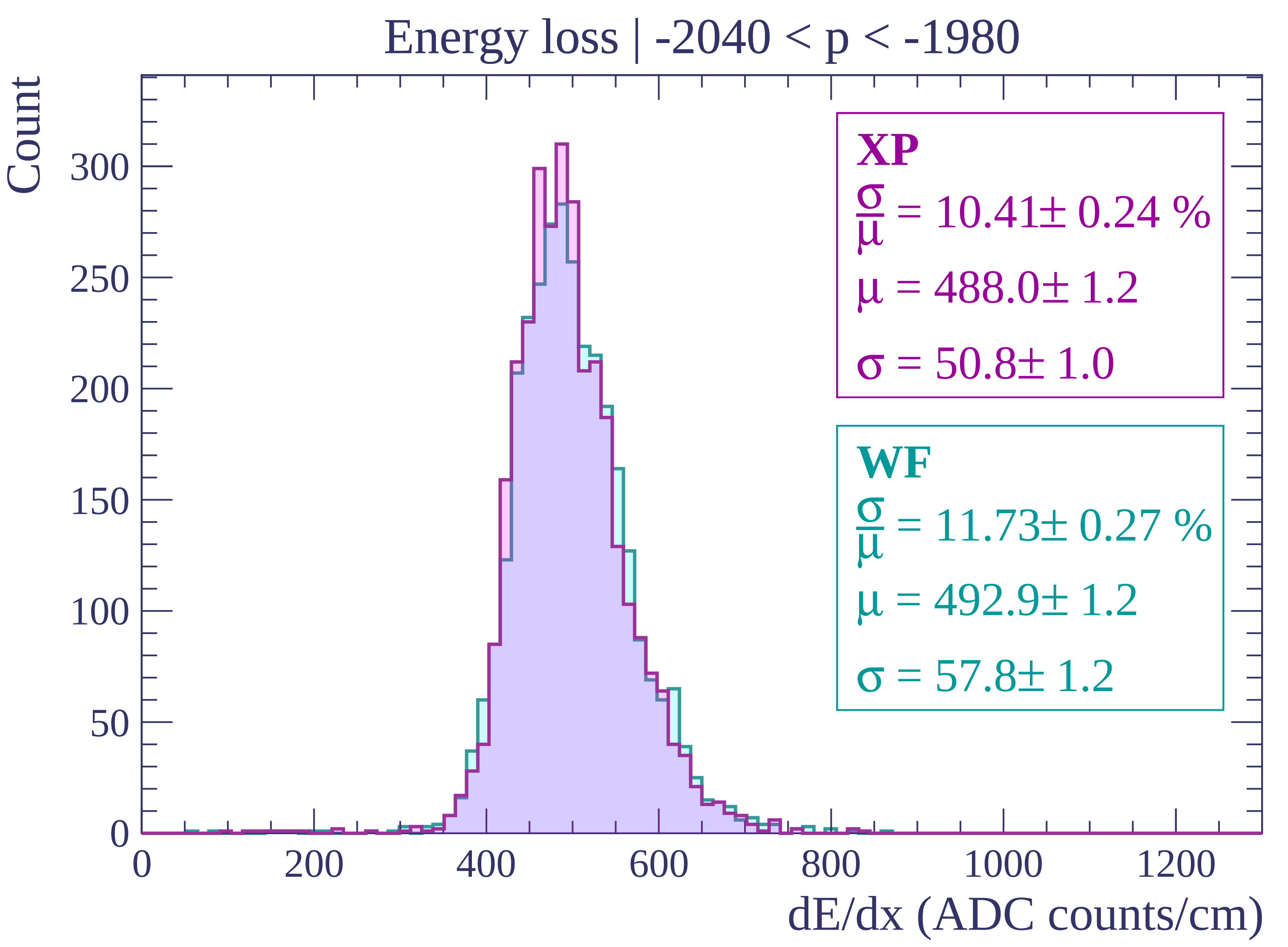
Energy loss | -2340 < p < -2280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.71± 0.28 %
μ = 498.1± 1.3
σ = 53.3± 1.2
WF
$\frac{\sigma}{\mu}$ = 11.45± 0.30 %
μ = 500.9± 1.3
σ = 57.3± 1.3


Energy loss | -2280 < p < -2220
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 11.30± 0.29 %
μ = 499.7± 1.3
σ = 56.5± 1.3
WF
σ/μ = 11.48± 0.26 %
μ = 505.2± 1.3
σ = 58.0± 1.1

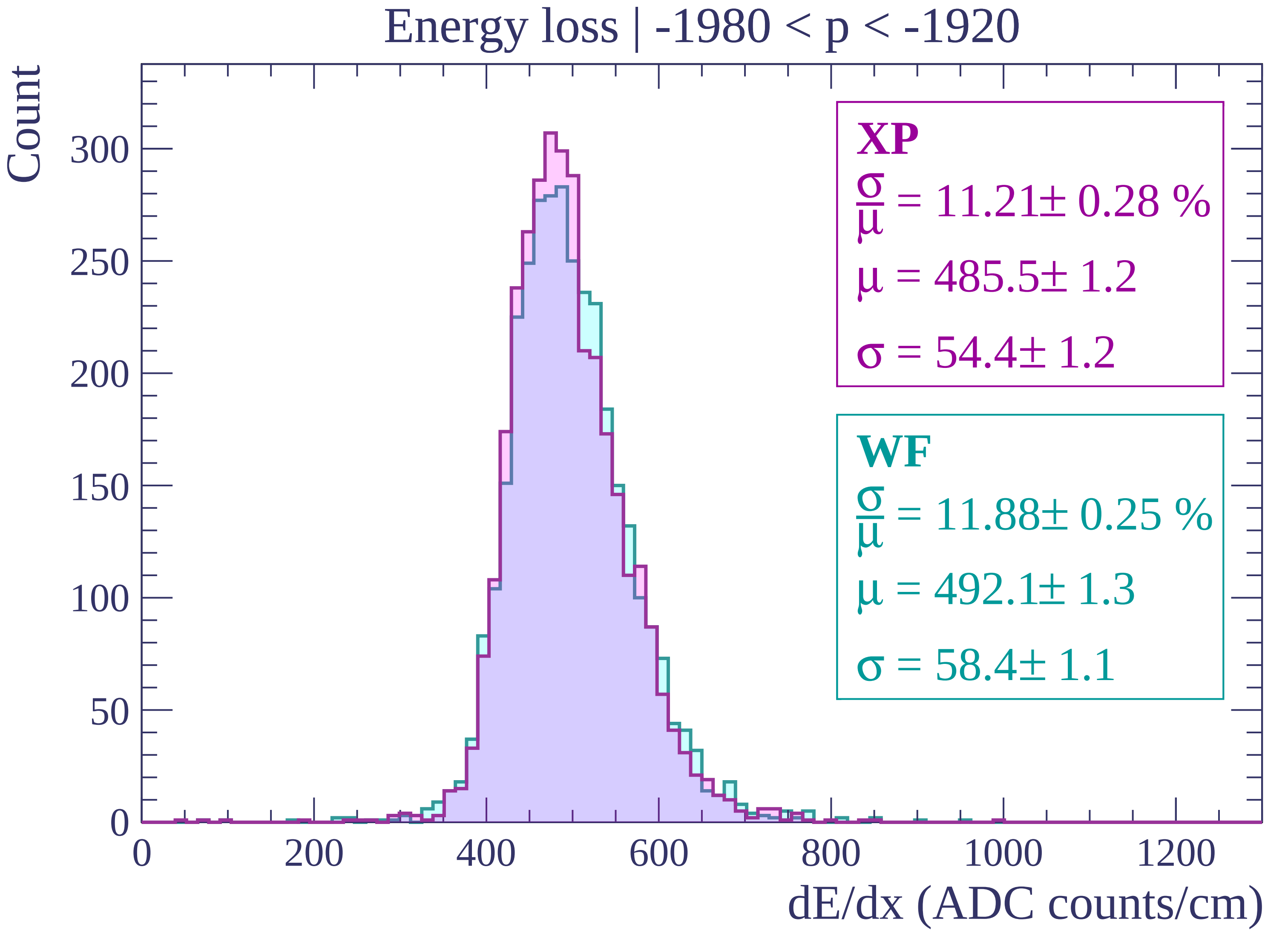

Energy loss | -2220 < p < -2160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 11.39± 0.28 %
μ = 498.8± 1.3
σ = 56.8± 1.3
WF
σ/μ = 11.53± 0.25 %
μ = 503.2± 1.3
σ = 58.0± 1.1

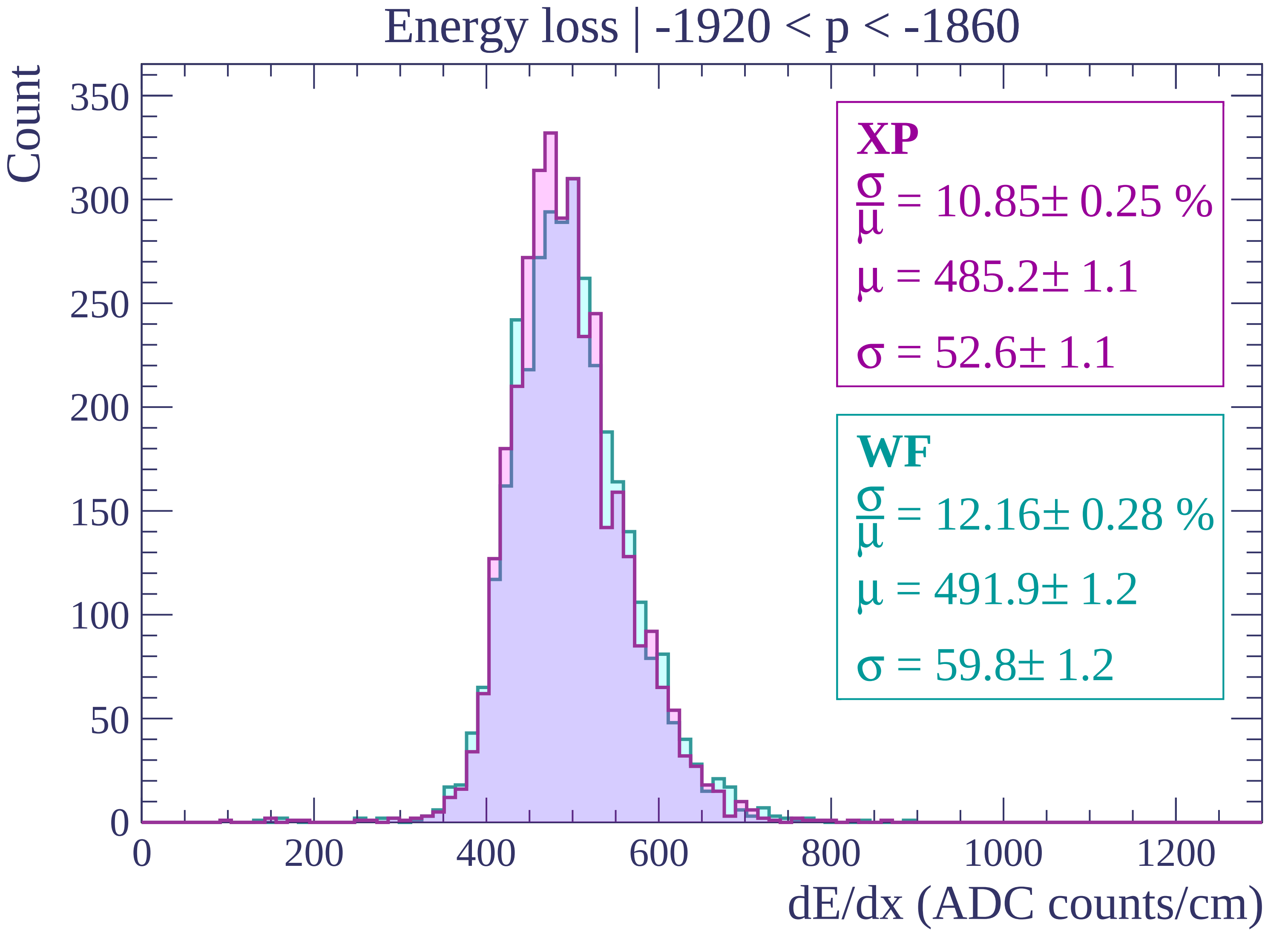

Energy loss | -2160 < p < -2100
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.69± 0.27 %
μ = 495.1± 1.2
σ = 52.9± 1.2
WF
σ/μ = 11.32± 0.28 %
μ = 498.7± 1.3
σ = 56.5± 1.3
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200


Energy loss | -2100 < p < -2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
σ/μ = 11.20± 0.29 %
μ = 492.7± 1.2
σ = 55.2± 1.3
WF
σ/μ = 11.67± 0.29 %
μ = 497.4± 1.3
σ = 58.1± 1.3

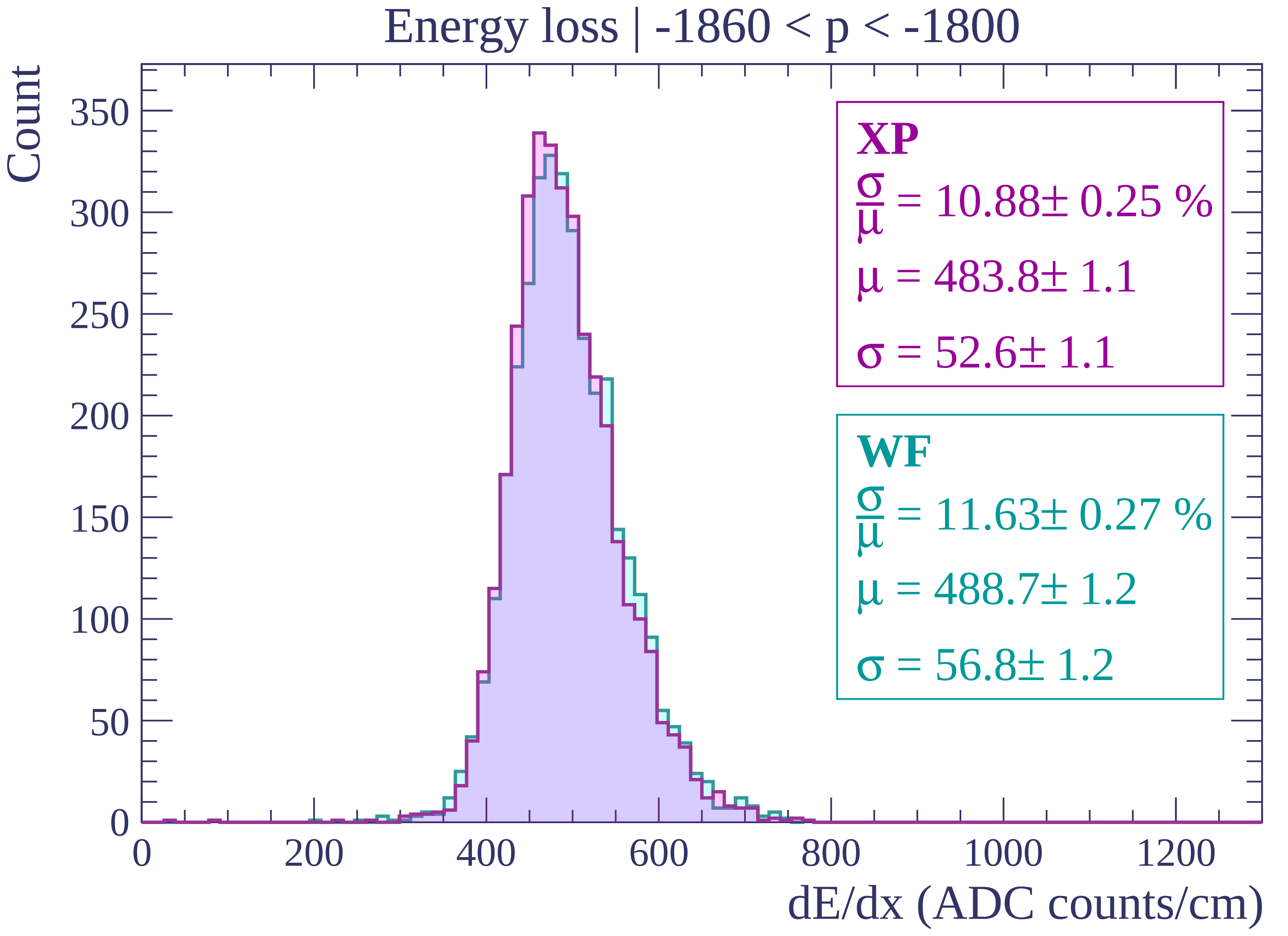


Energy loss | -2040 < p < -1980
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.41± 0.24 %
μ = 488.0± 1.2
σ = 50.8± 1.0
WF
σ/μ = 11.73± 0.27 %
μ = 492.9± 1.2
σ = 57.8± 1.2

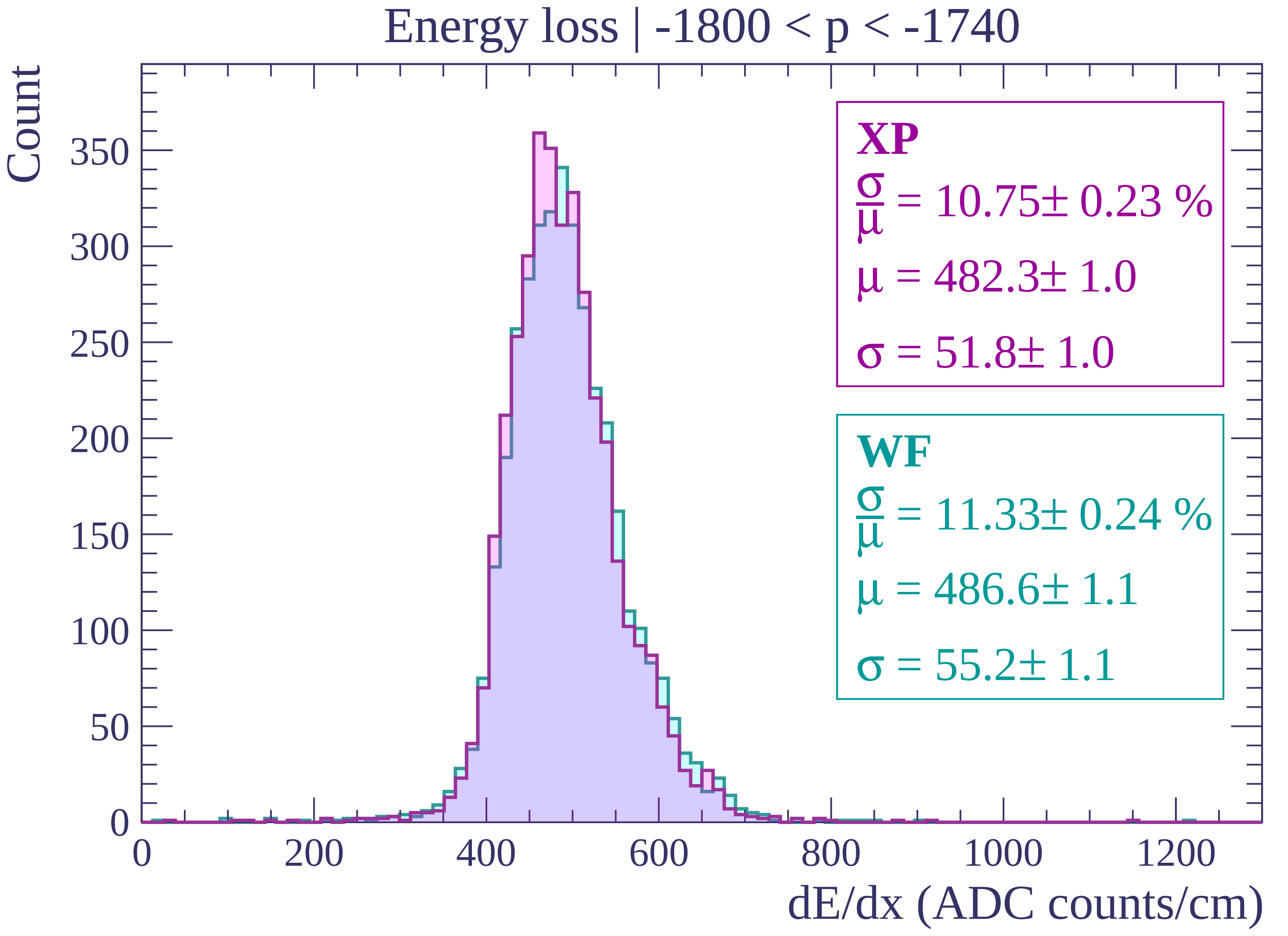

Energy loss | -1980 < p < -1920
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 11.21± 0.28 %
μ = 485.5± 1.2
σ = 54.4± 1.2
WF
$\frac{\sigma}{\mu}$ = 11.88± 0.25 %
μ = 492.1± 1.3
σ = 58.4± 1.1



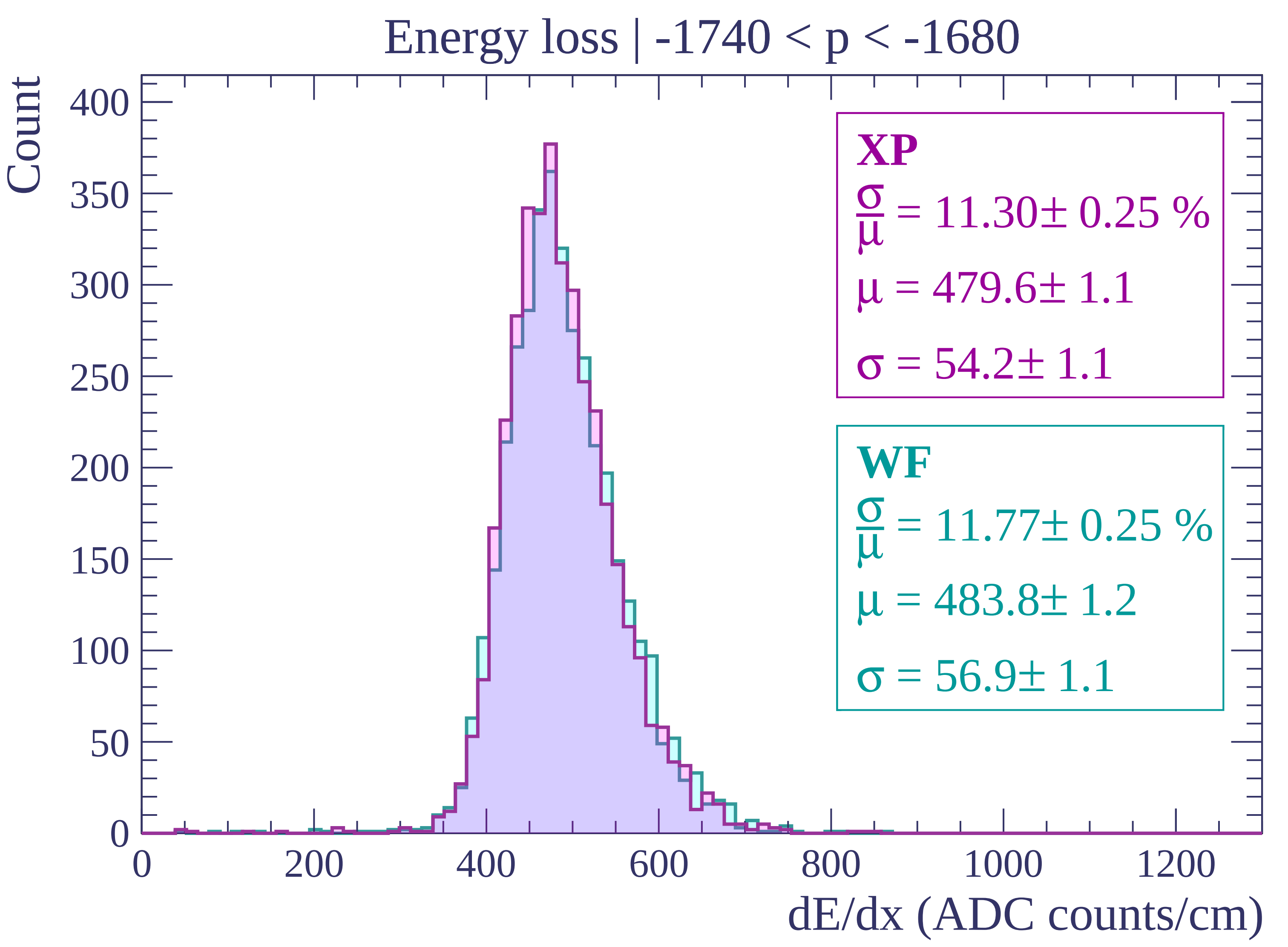
Energy loss | -1920 < p < -1860
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.85± 0.25 %
μ = 485.2± 1.1
σ = 52.6± 1.1
WF
$\frac{\sigma}{\mu}$ = 12.16± 0.28 %
μ = 491.9± 1.2
σ = 59.8± 1.2



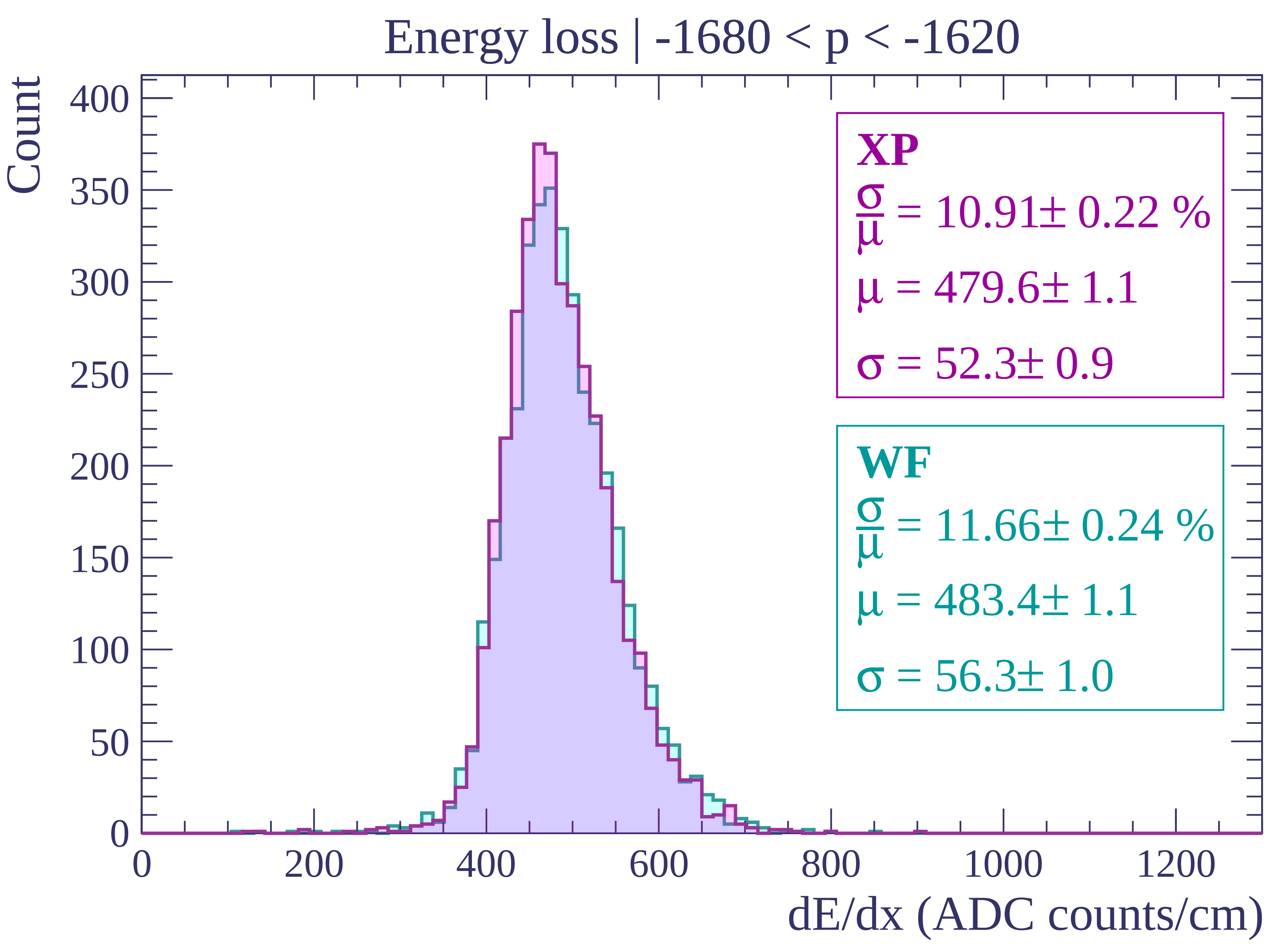
Energy loss | -1860 < p < -1800
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.88± 0.25 %
μ = 483.8± 1.1
σ = 52.6± 1.1
WF
$\frac{\sigma}{\mu}$ = 11.63± 0.27 %
μ = 488.7± 1.2
σ = 56.8± 1.2

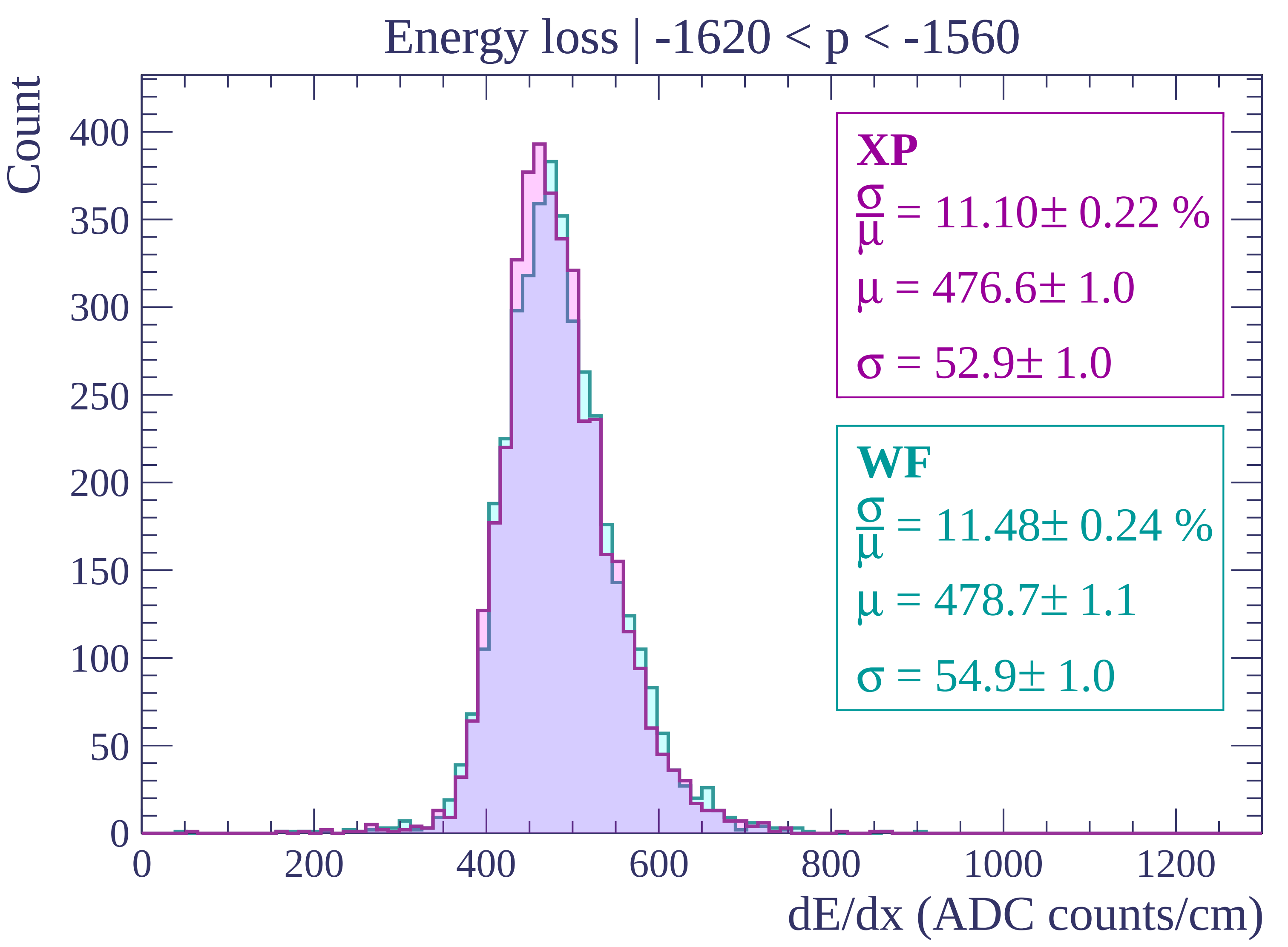


Energy loss | -1800 < p < -1740
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 10.75± 0.23 %
μ = 482.3± 1.0
σ = 51.8± 1.0
WF
σ/μ = 11.33± 0.24 %
μ = 486.6± 1.1
σ = 55.2± 1.1

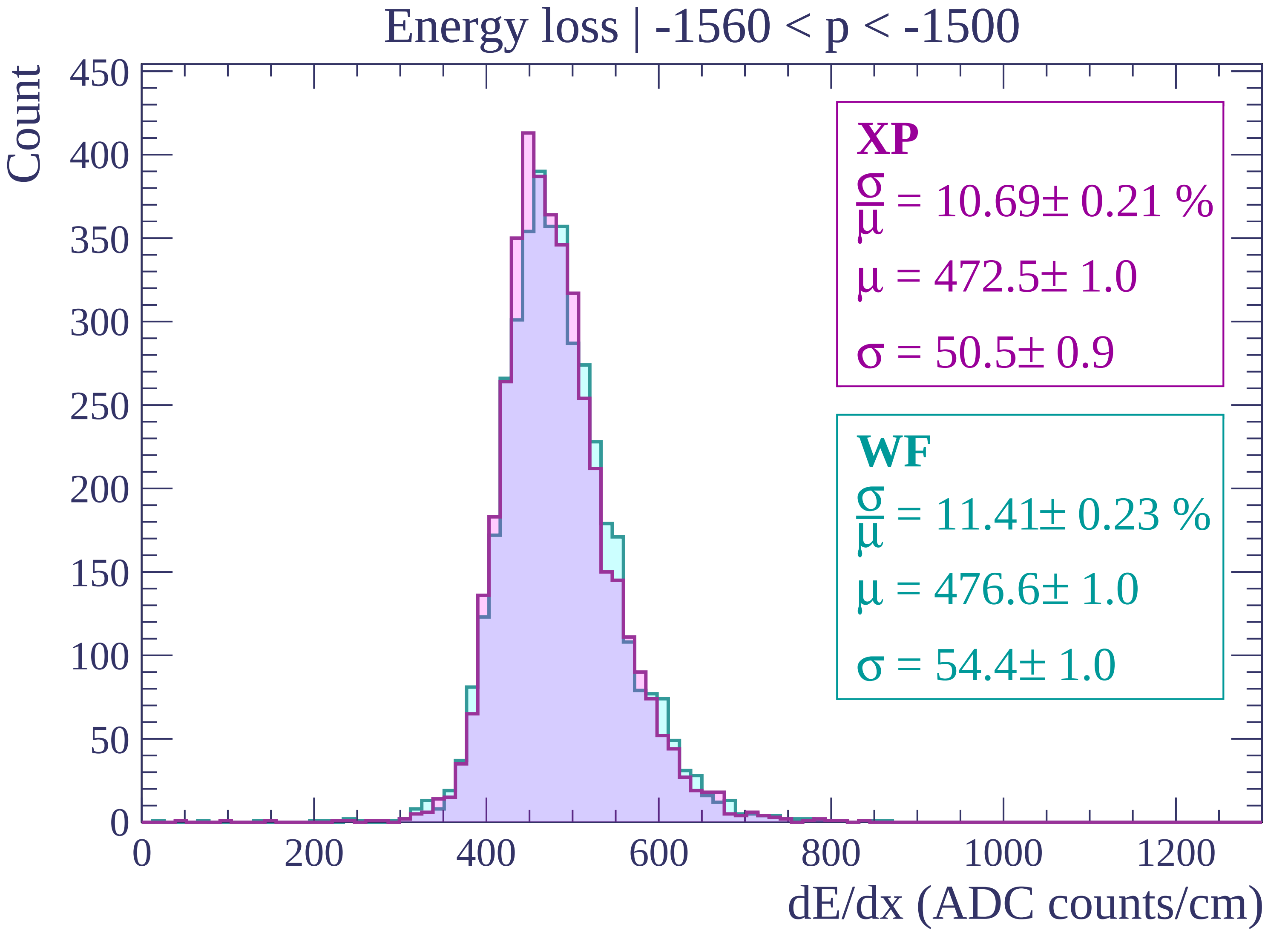


Energy loss | -1740 < p < -1680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 11.30± 0.25 %
μ = 479.6± 1.1
σ = 54.2± 1.1
WF
$\frac{\sigma}{\mu}$ = 11.77± 0.25 %
μ = 483.8± 1.2
σ = 56.9± 1.1

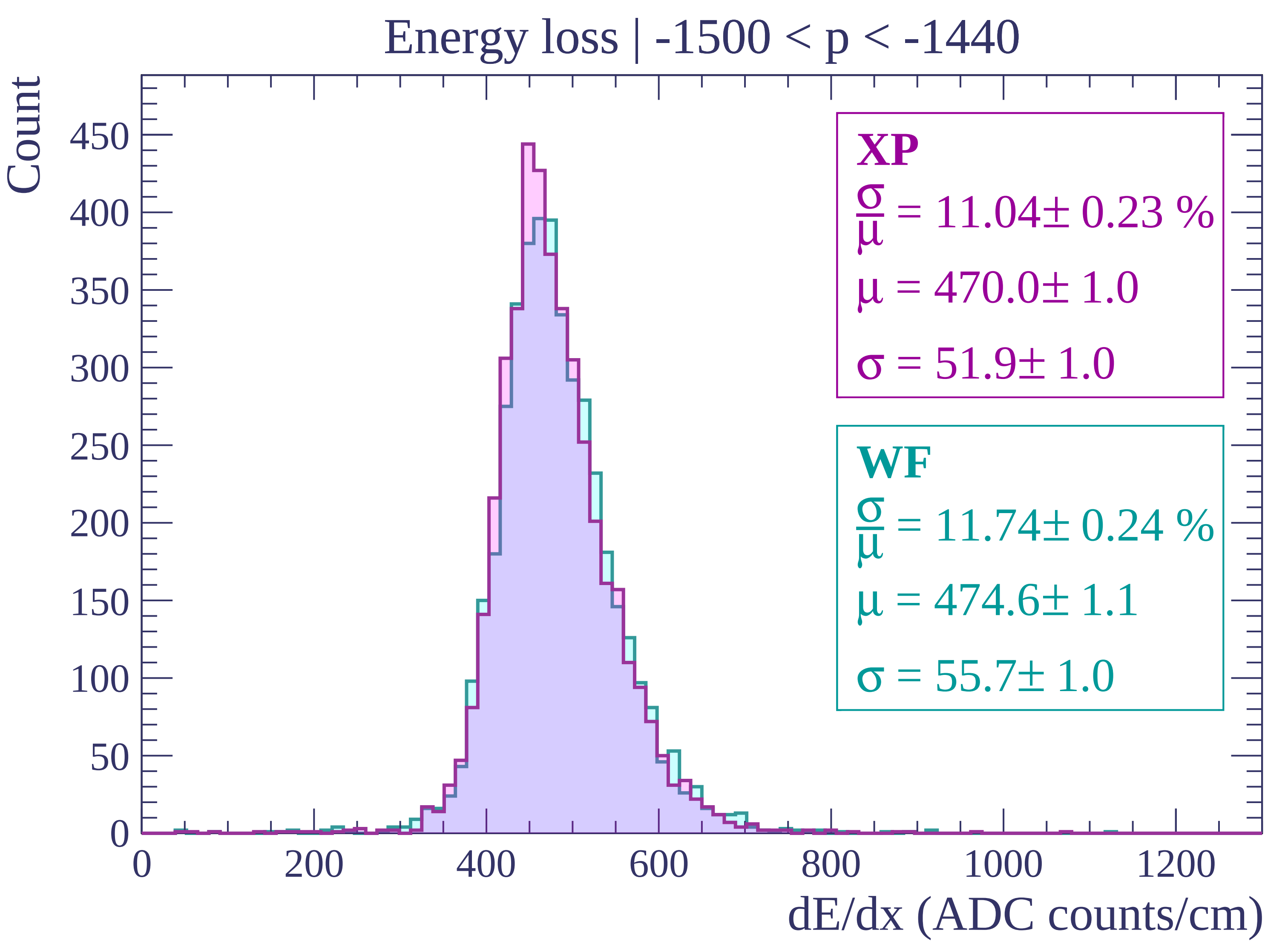

Energy loss | -1680 < p < -1620
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 10.91± 0.22 %
μ = 479.6± 1.1
σ = 52.3± 0.9
WF
σ/μ = 11.66± 0.24 %
μ = 483.4± 1.1
σ = 56.3± 1.0

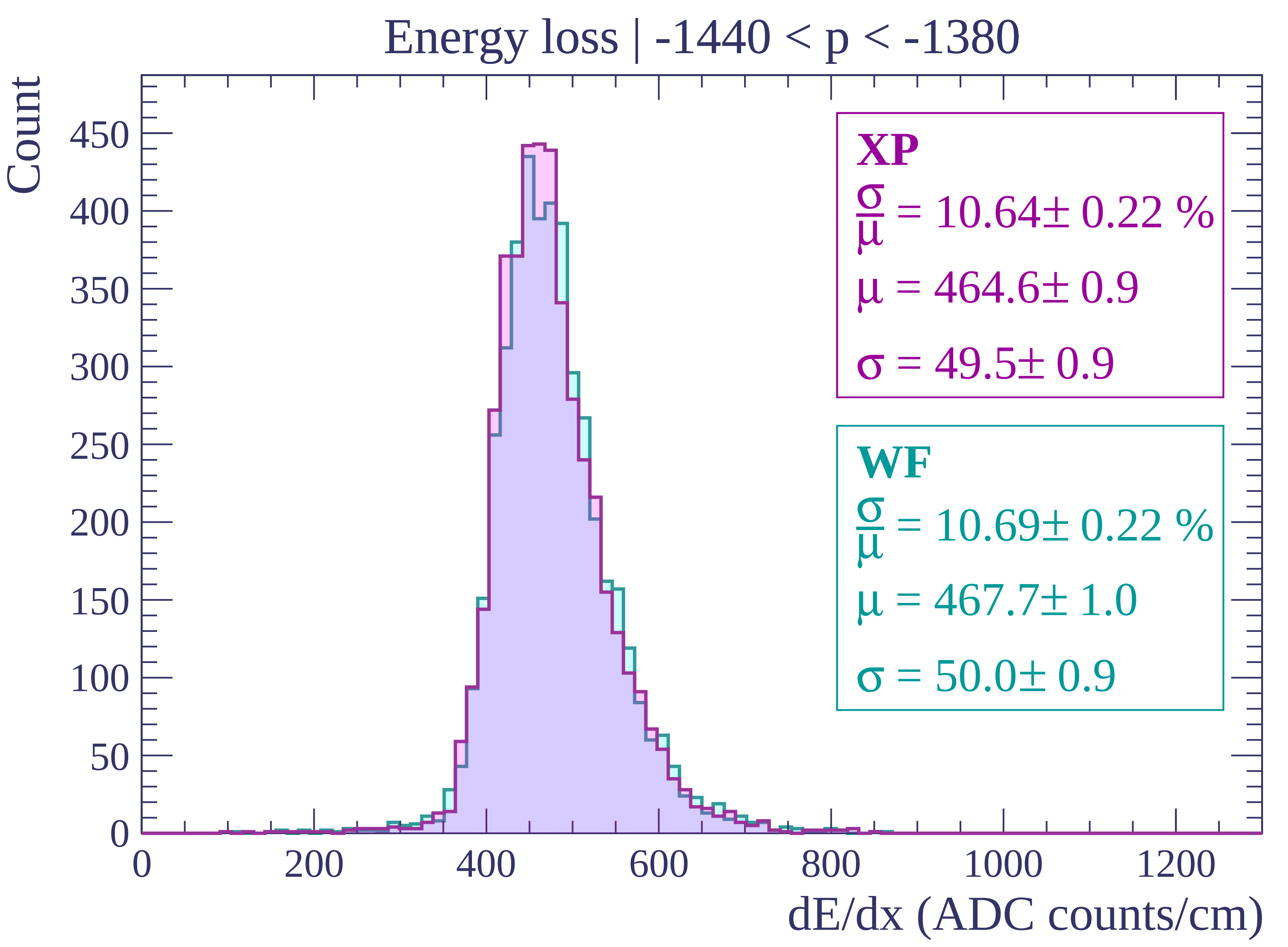


Energy loss | -1620 < p < -1560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.10± 0.22 %
μ = 476.6± 1.0
σ = 52.9± 1.0
WF
$\frac{\sigma}{\mu}$ = 11.48± 0.24 %
μ = 478.7± 1.1
σ = 54.9± 1.0

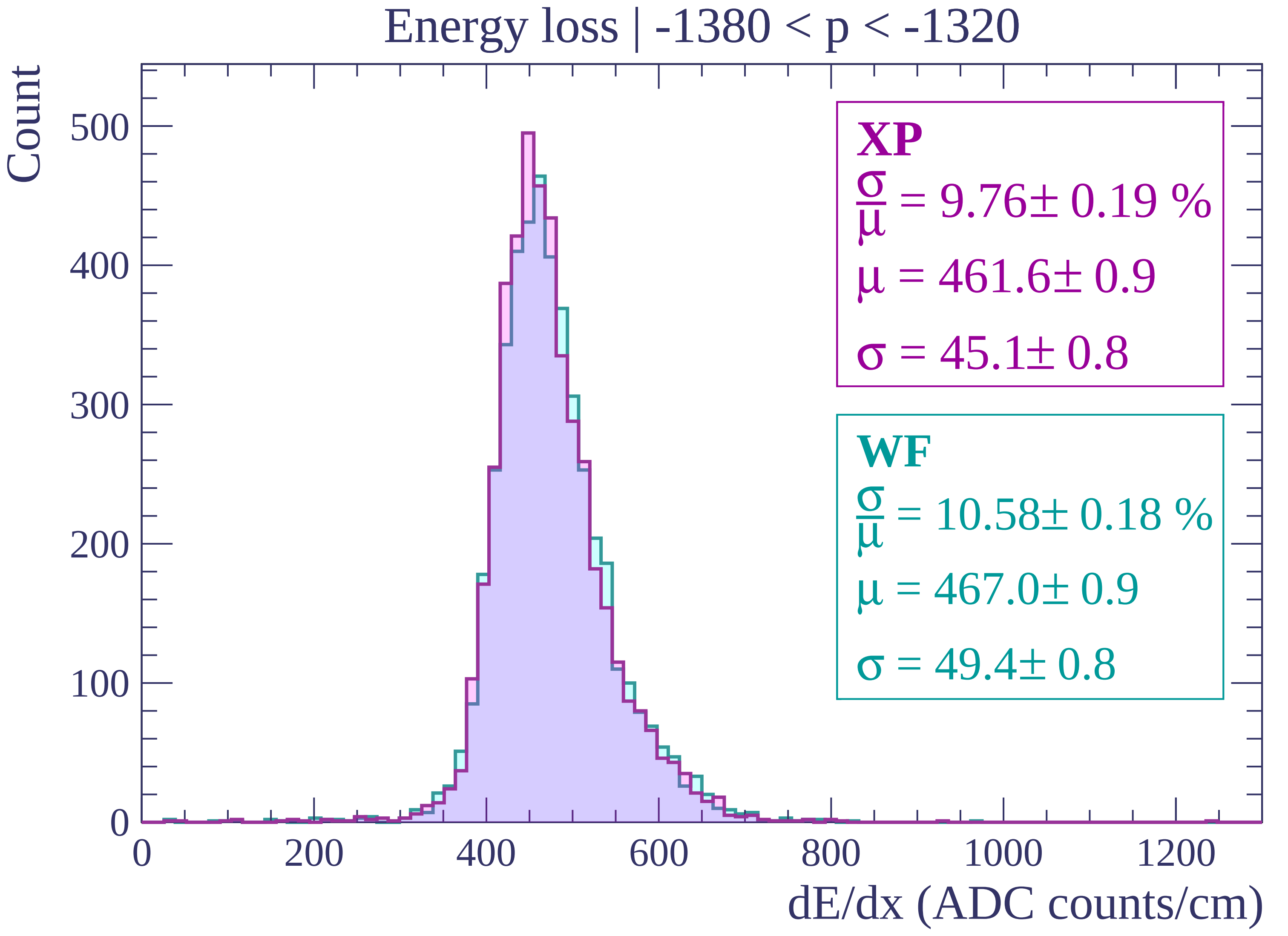


Energy loss | -1560 < p < -1500
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 10.69± 0.21 %
μ = 472.5± 1.0
σ = 50.5± 0.9
WF
σ/μ = 11.41± 0.23 %
μ = 476.6± 1.0
σ = 54.4± 1.0

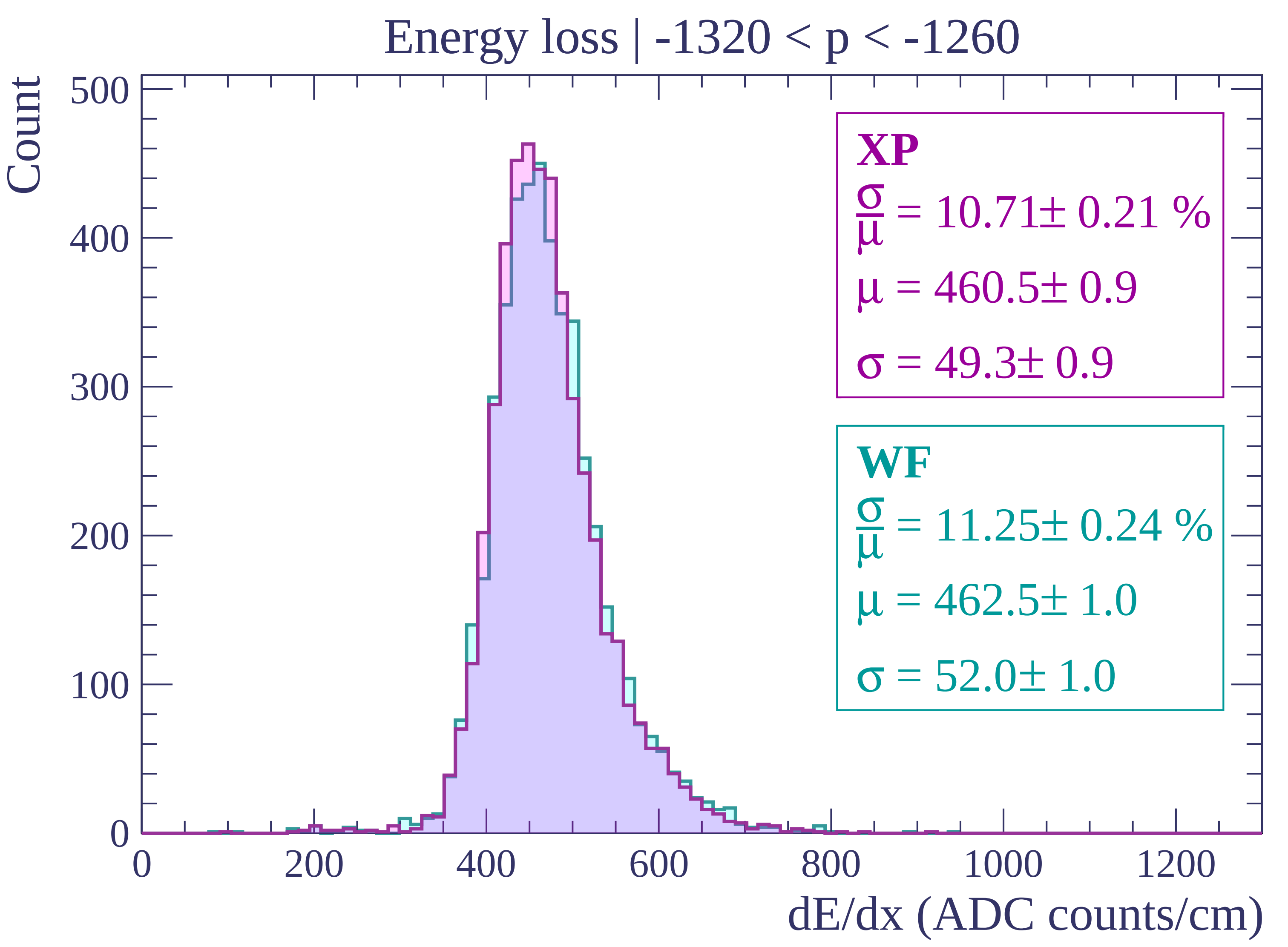

Energy loss | -1500 < p < -1440
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 11.04± 0.23 %
μ = 470.0± 1.0
σ = 51.9± 1.0
WF
σ/μ = 11.74± 0.24 %
μ = 474.6± 1.1
σ = 55.7± 1.0



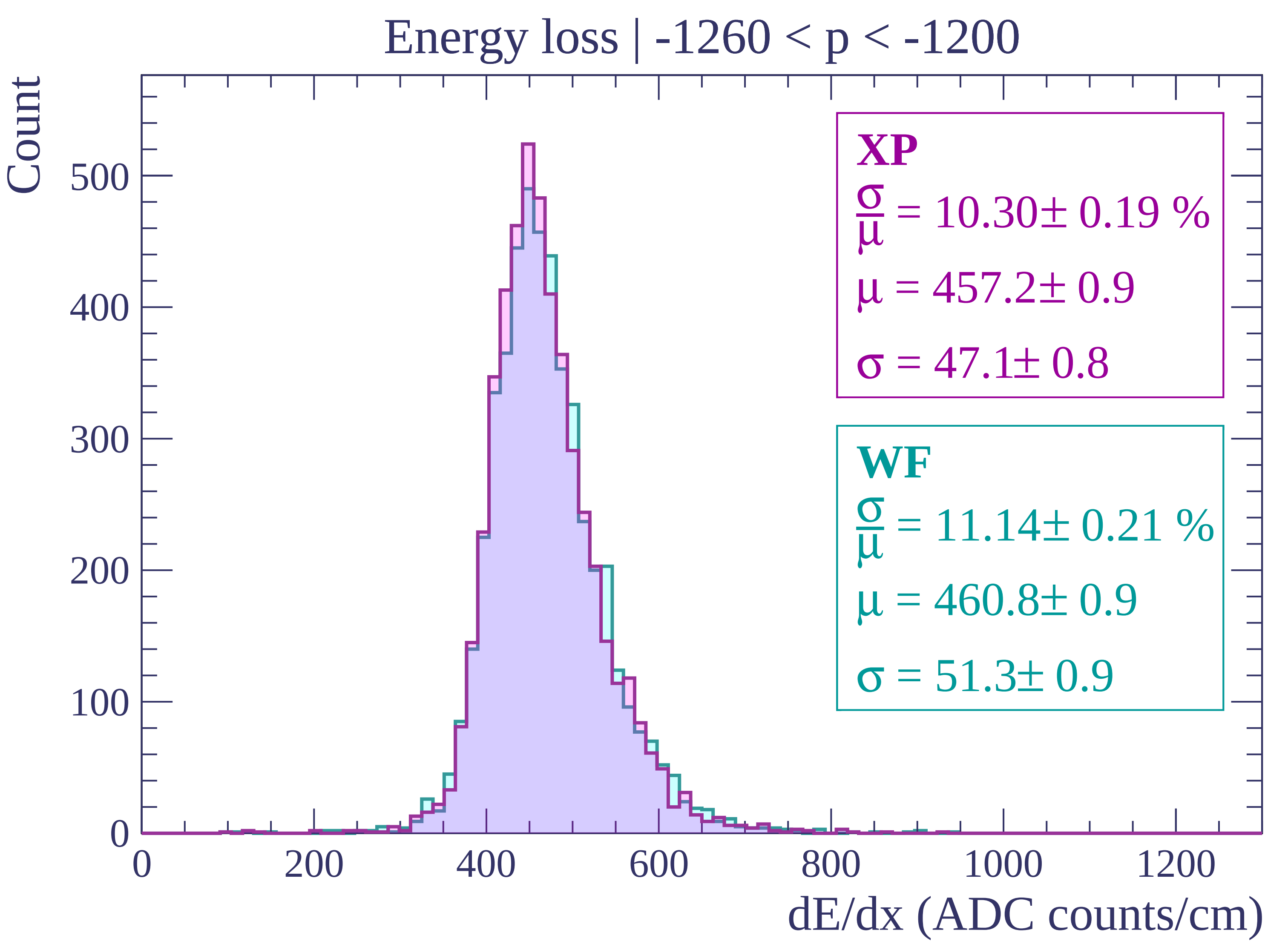
Energy loss | -1440 < p < -1380
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.64± 0.22 %
μ = 464.6± 0.9
σ = 49.5± 0.9
WF
σ/μ = 10.69± 0.22 %
μ = 467.7± 1.0
σ = 50.0± 0.9

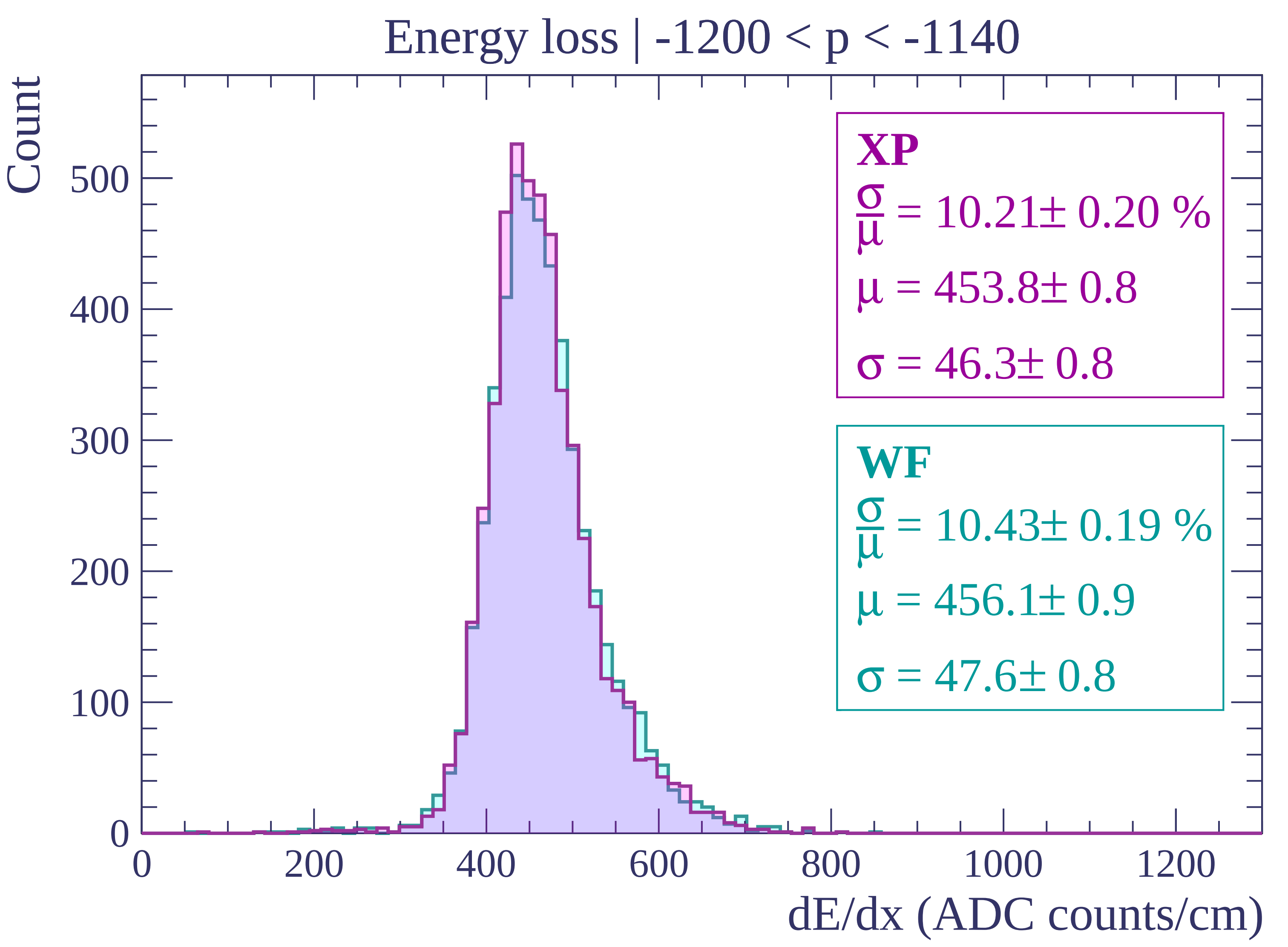


Energy loss | -1380 < p < -1320
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.76± 0.19 %
μ = 461.6± 0.9
σ = 45.1± 0.8
WF
σ/μ = 10.58± 0.18 %
μ = 467.0± 0.9
σ = 49.4± 0.8

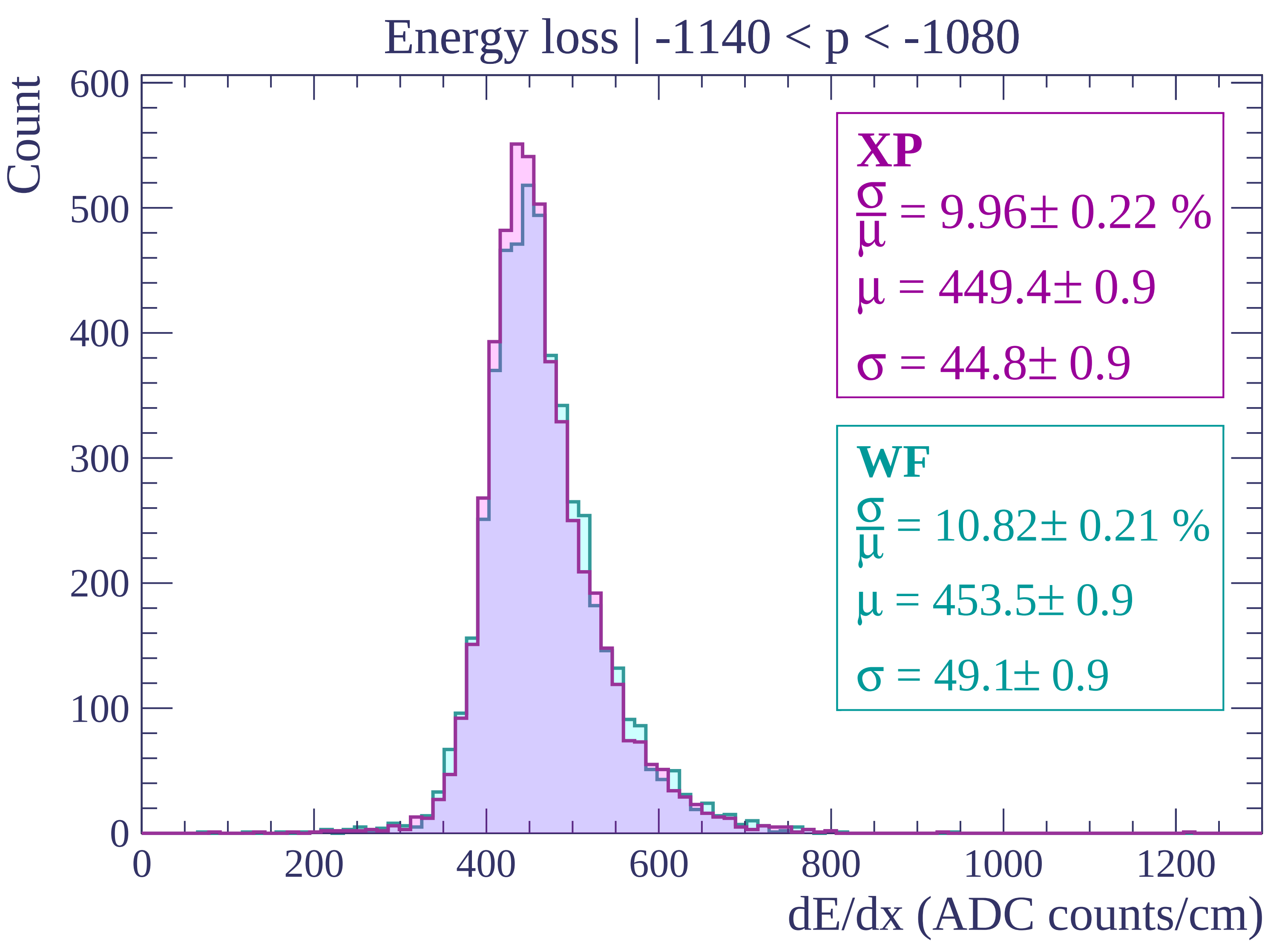


Energy loss | -1320 < p < -1260
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 10.71± 0.21 %
μ = 460.5± 0.9
σ = 49.3± 0.9
WF
σ/μ = 11.25± 0.24 %
μ = 462.5± 1.0
σ = 52.0± 1.0

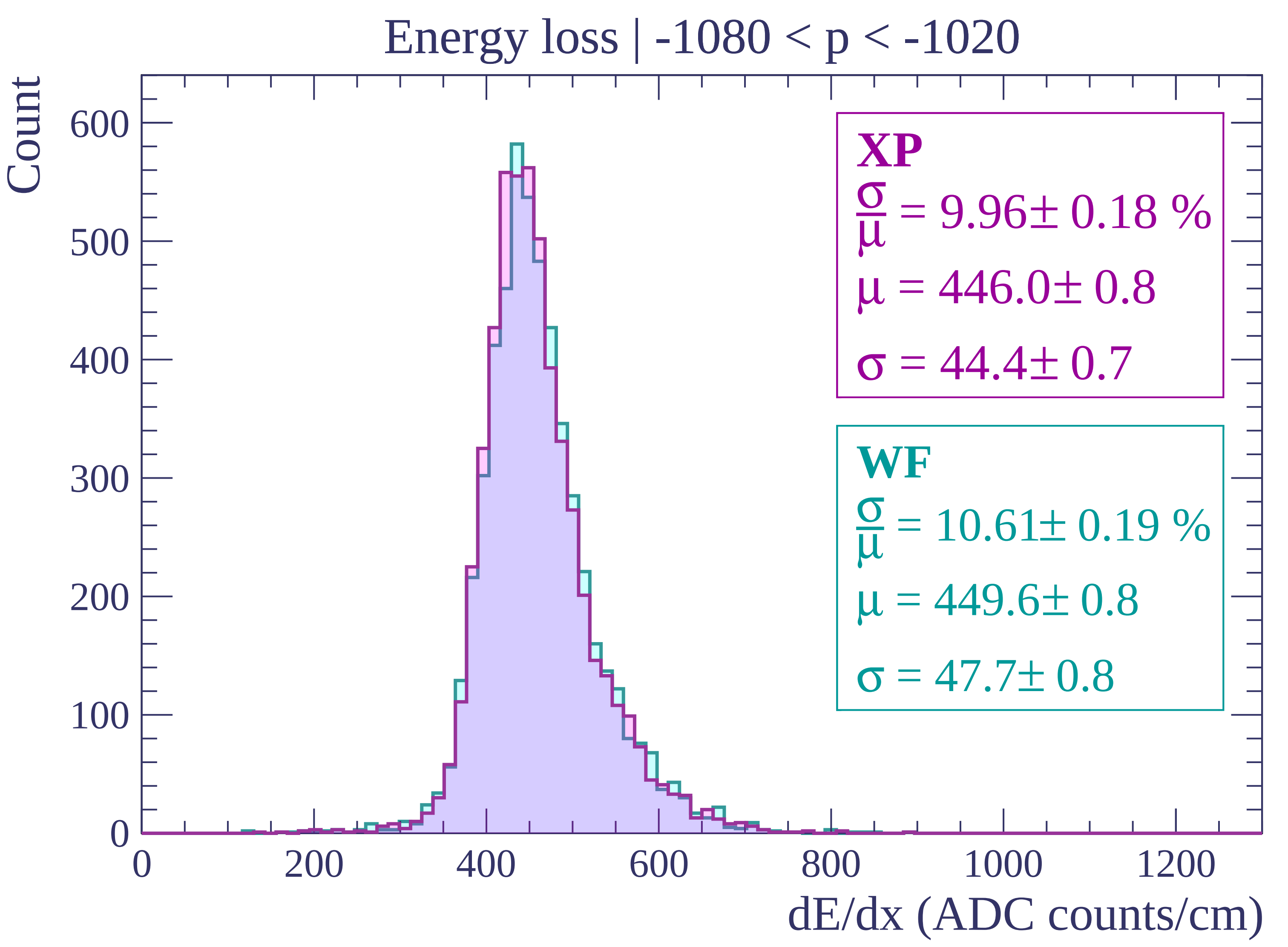

Energy loss | -1260 < p < -1200
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 10.30± 0.19 %
μ = 457.2± 0.9
σ = 47.1± 0.8
WF
σ/μ = 11.14± 0.21 %
μ = 460.8± 0.9
σ = 51.3± 0.9

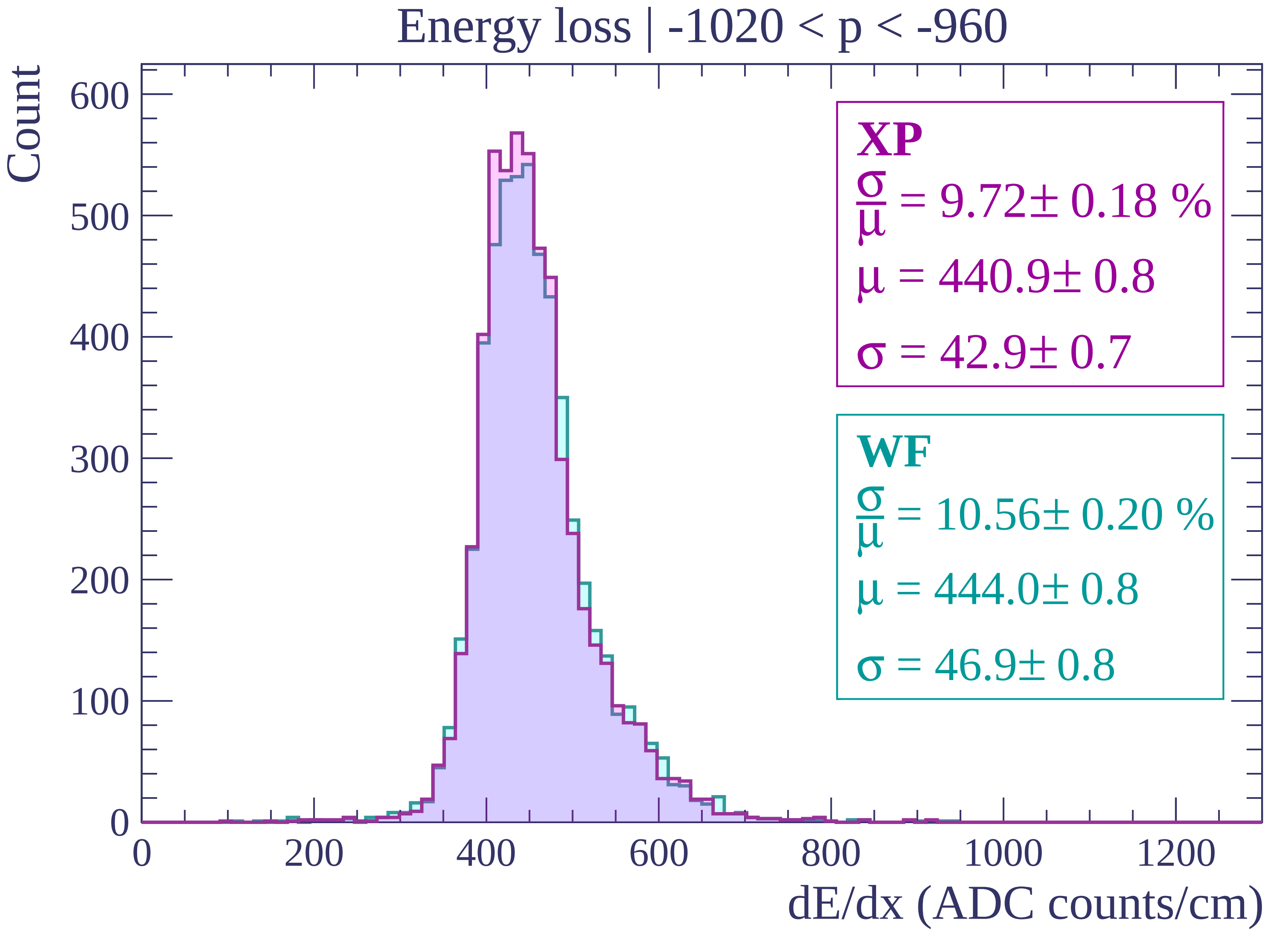


Energy loss | -1200 < p < -1140
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 10.21± 0.20 %
μ = 453.8± 0.8
σ = 46.3± 0.8
WF
σ/μ = 10.43± 0.19 %
μ = 456.1± 0.9
σ = 47.6± 0.8

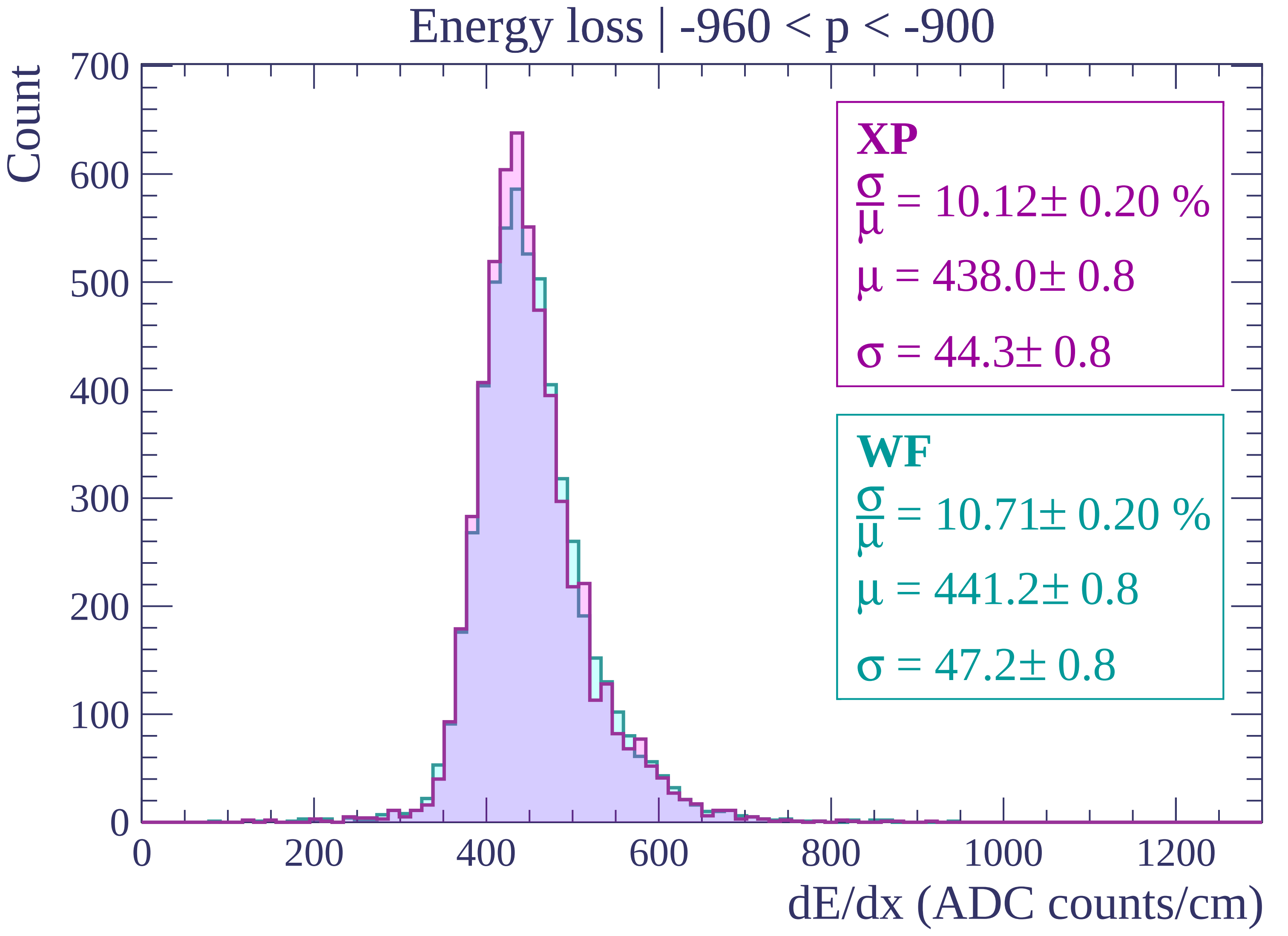


Energy loss | -1140 < p < -1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP
$\frac{\sigma}{\mu}$ = 9.96± 0.22 %
μ = 449.4± 0.9
σ = 44.8± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.82± 0.21 %
μ = 453.5± 0.9
σ = 49.1± 0.9

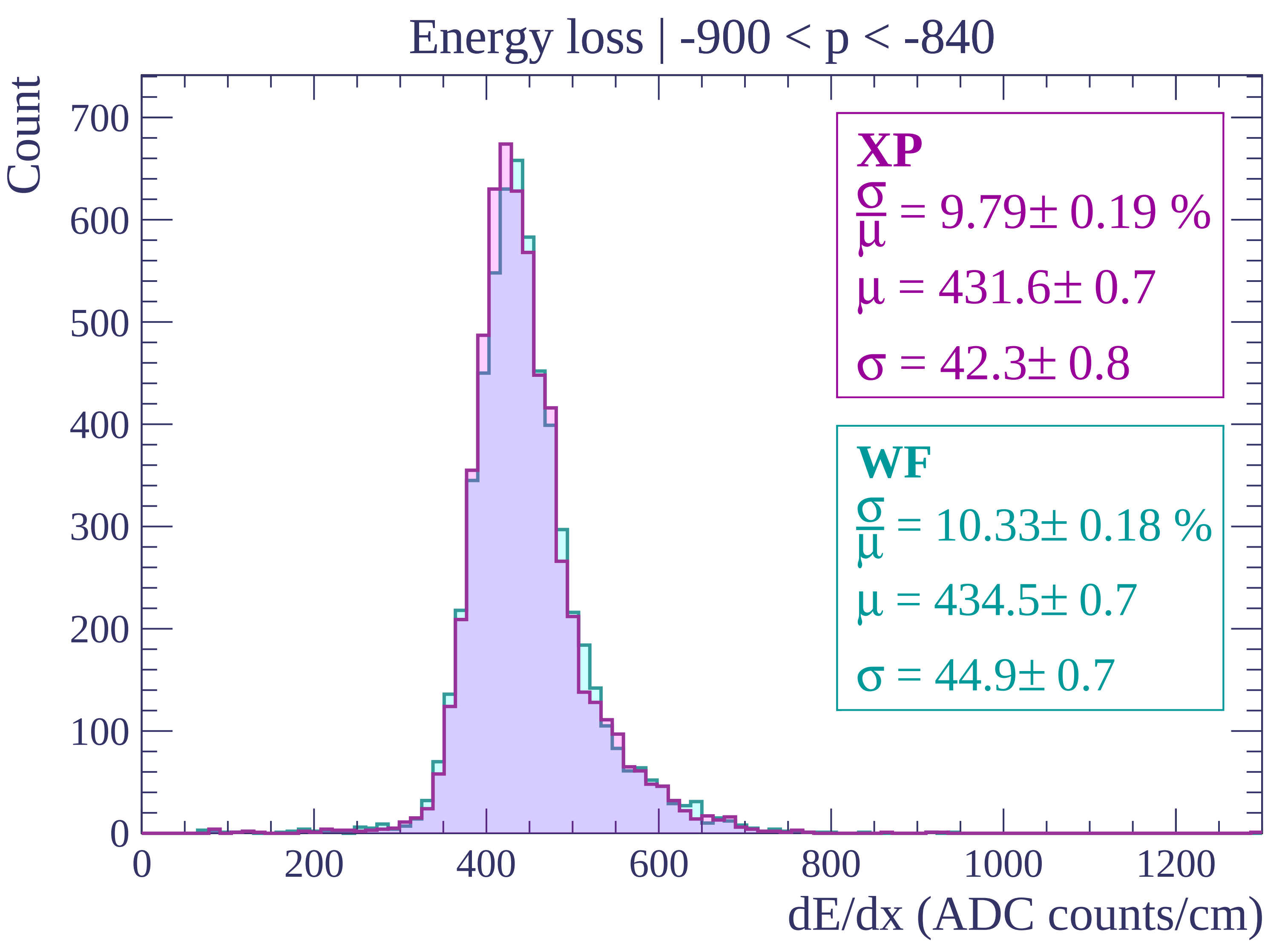

Energy loss | -1080 < p < -1020
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.96± 0.18 %
μ = 446.0± 0.8
σ = 44.4± 0.7
WF
$\frac{\sigma}{\mu}$ = 10.61± 0.19 %
μ = 449.6± 0.8
σ = 47.7± 0.8



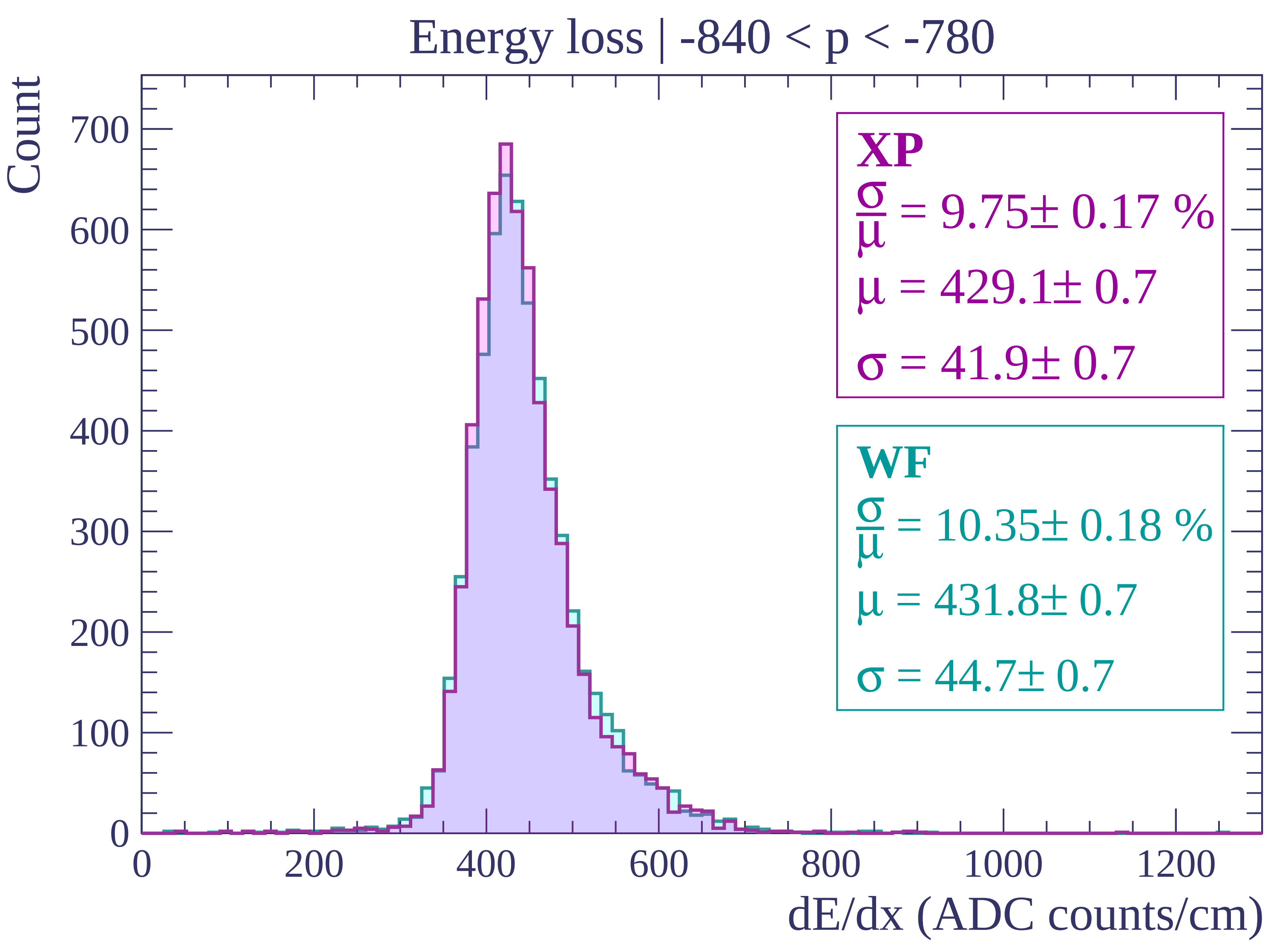
Energy loss | -1020 < p < -960
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.72± 0.18 %
μ = 440.9± 0.8
σ = 42.9± 0.7
WF
σ/μ = 10.56± 0.20 %
μ = 444.0± 0.8
σ = 46.9± 0.8



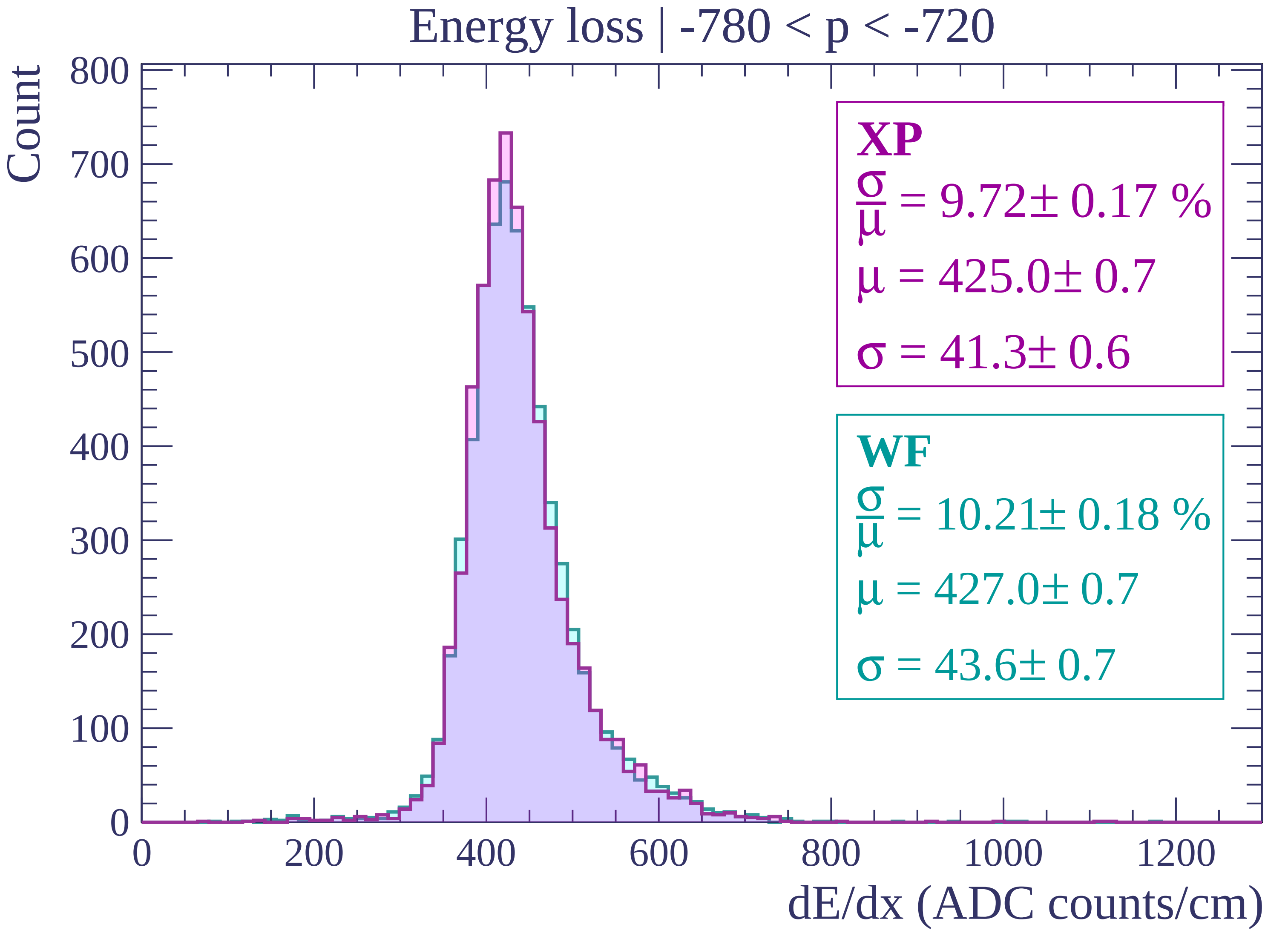
Energy loss | -960 < p < -900
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.12± 0.20 %
μ = 438.0± 0.8
σ = 44.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.71± 0.20 %
μ = 441.2± 0.8
σ = 47.2± 0.8

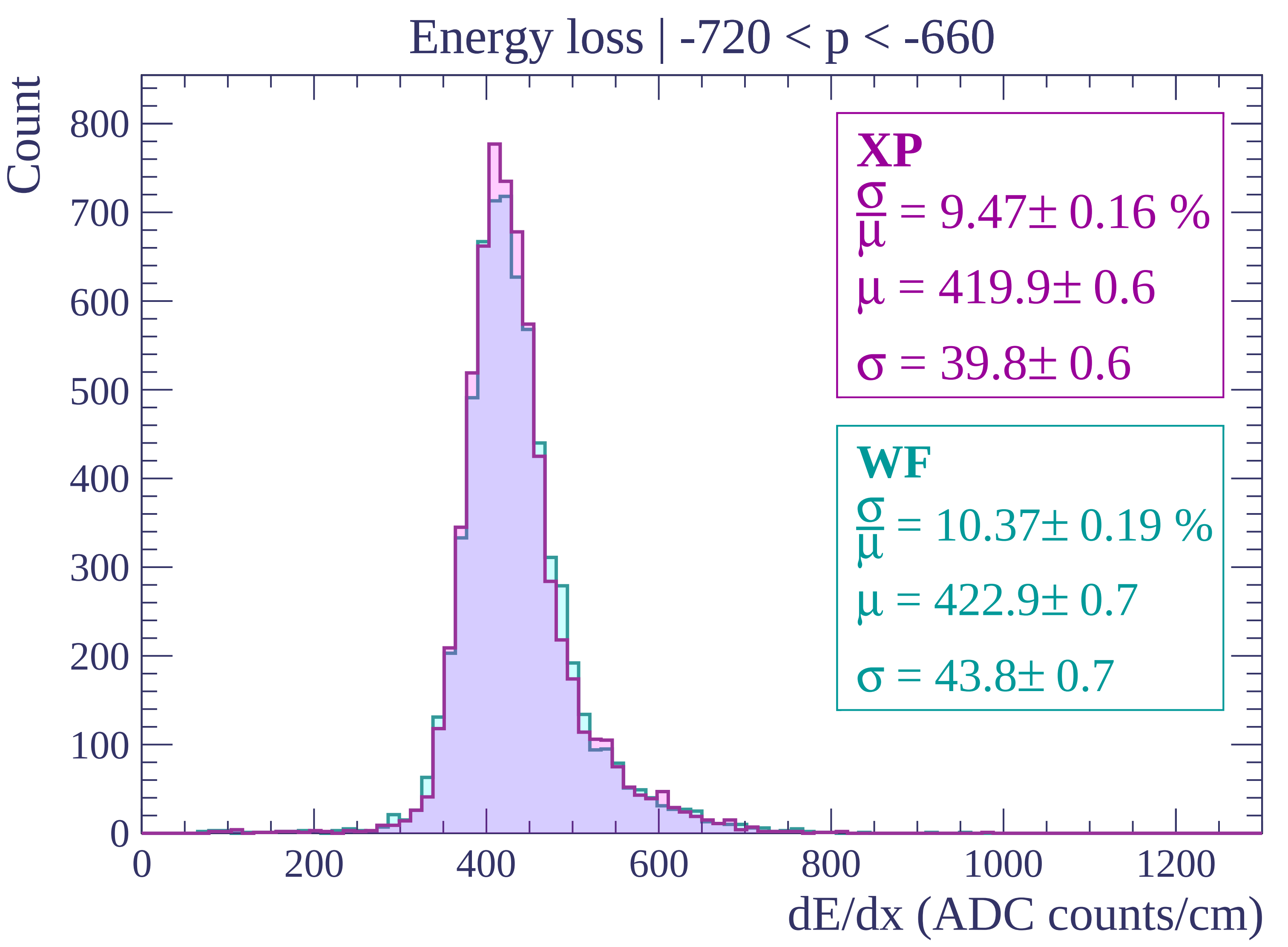


Energy loss | -900 < p < -840
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.79± 0.19 %
μ = 431.6± 0.7
σ = 42.3± 0.8
WF
σ/μ = 10.33± 0.18 %
μ = 434.5± 0.7
σ = 44.9± 0.7

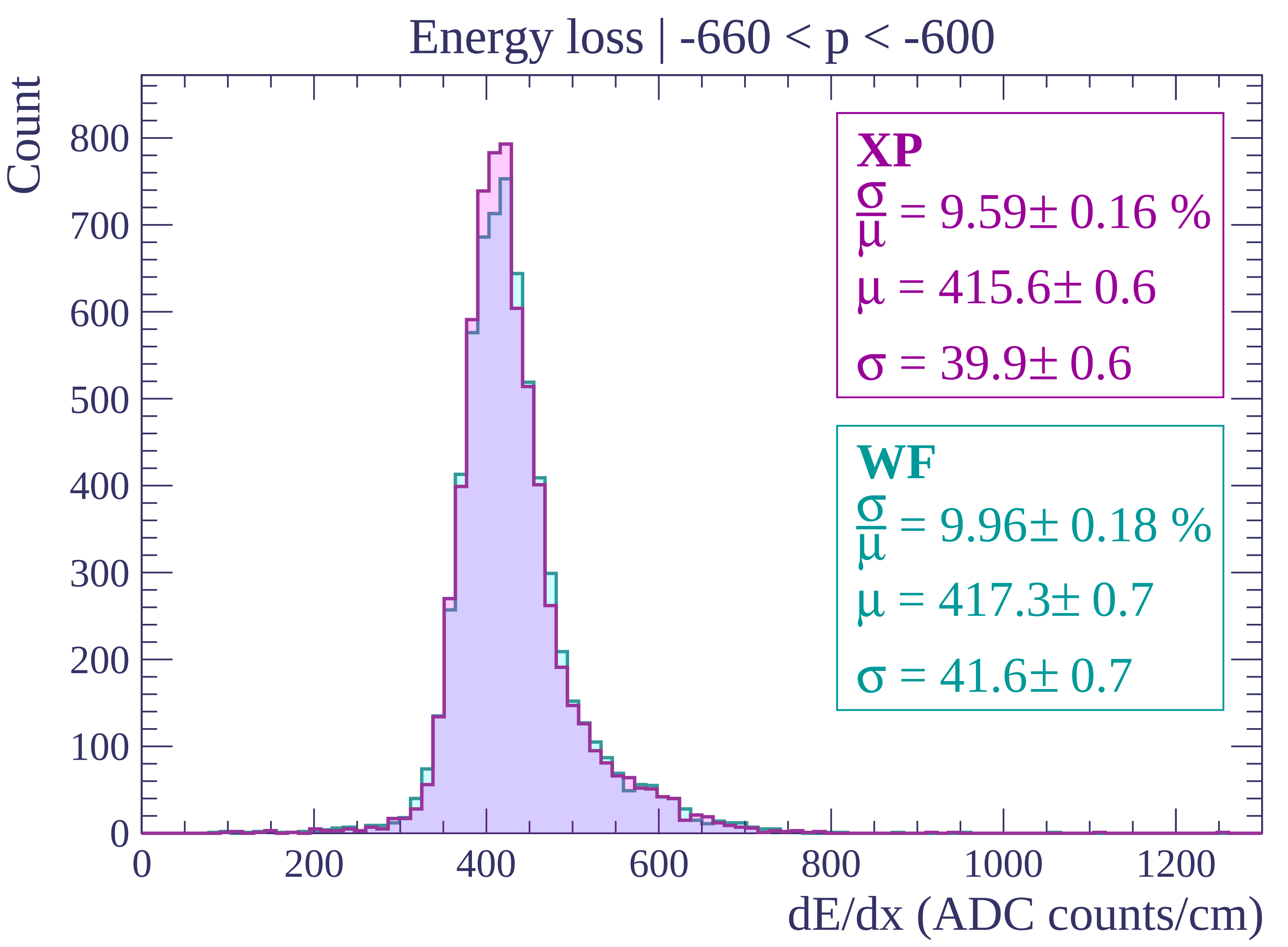


Energy loss | -840 < p < -780
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.75± 0.17 %
μ = 429.1± 0.7
σ = 41.9± 0.7
WF
σ/μ = 10.35± 0.18 %
μ = 431.8± 0.7
σ = 44.7± 0.7

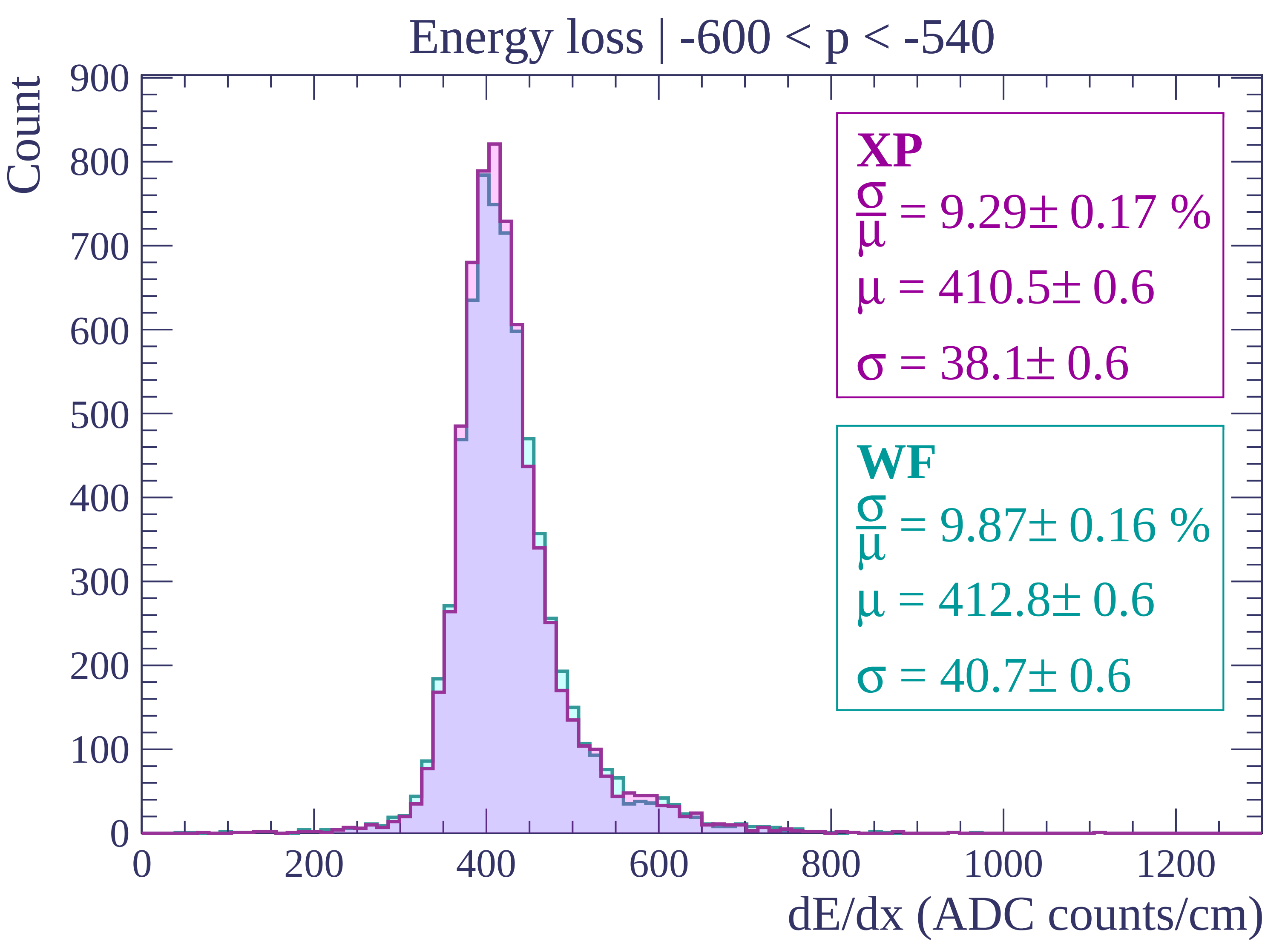

Energy loss | -780 < p < -720
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.72± 0.17 %
μ = 425.0± 0.7
σ = 41.3± 0.6
WF
σ/μ = 10.21± 0.18 %
μ = 427.0± 0.7
σ = 43.6± 0.7
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200



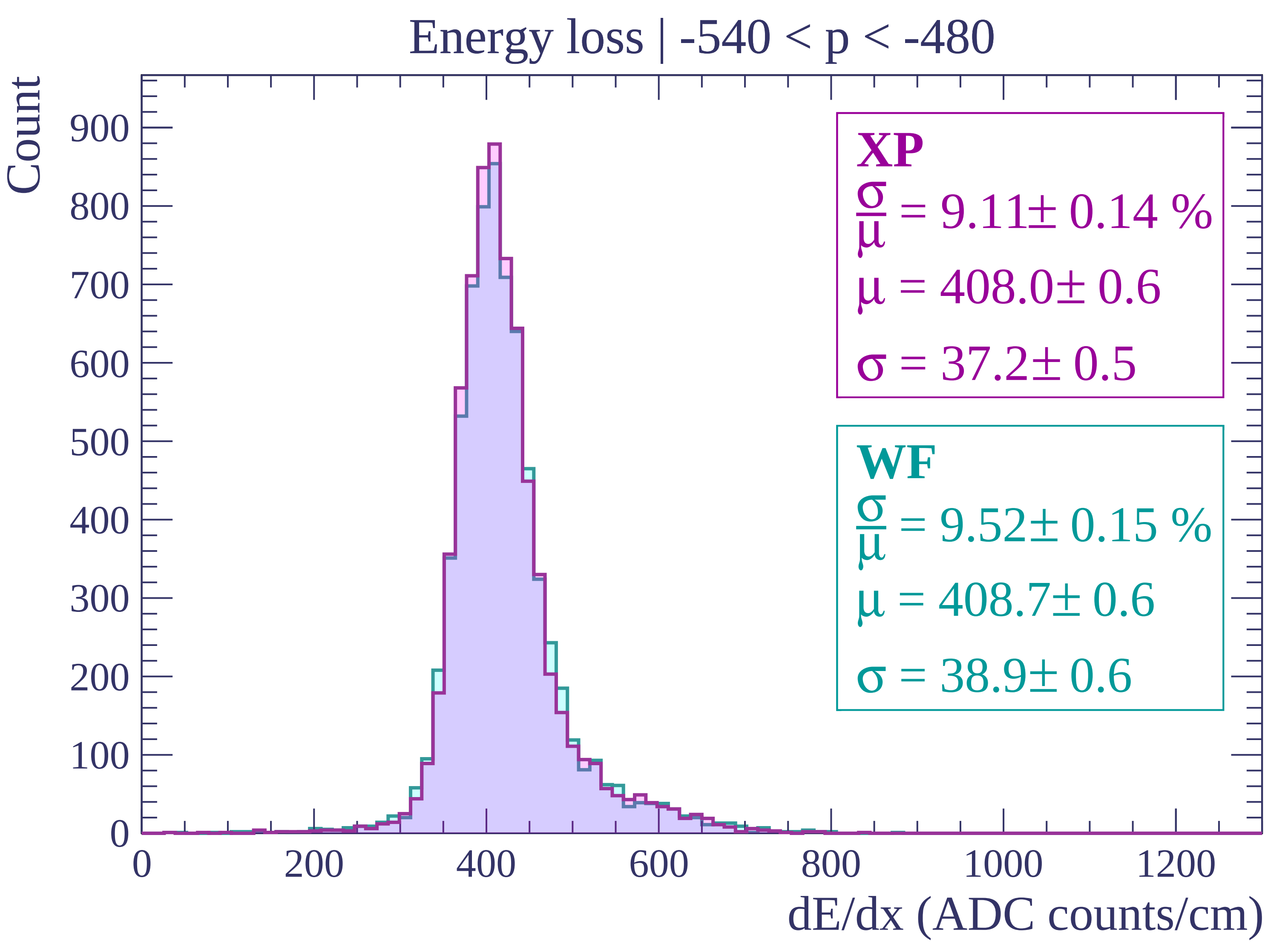
Energy loss | -720 < p < -660
Count
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.47± 0.16 %
μ = 419.9± 0.6
σ = 39.8± 0.6
WF
σ/μ = 10.37± 0.19 %
μ = 422.9± 0.7
σ = 43.8± 0.7



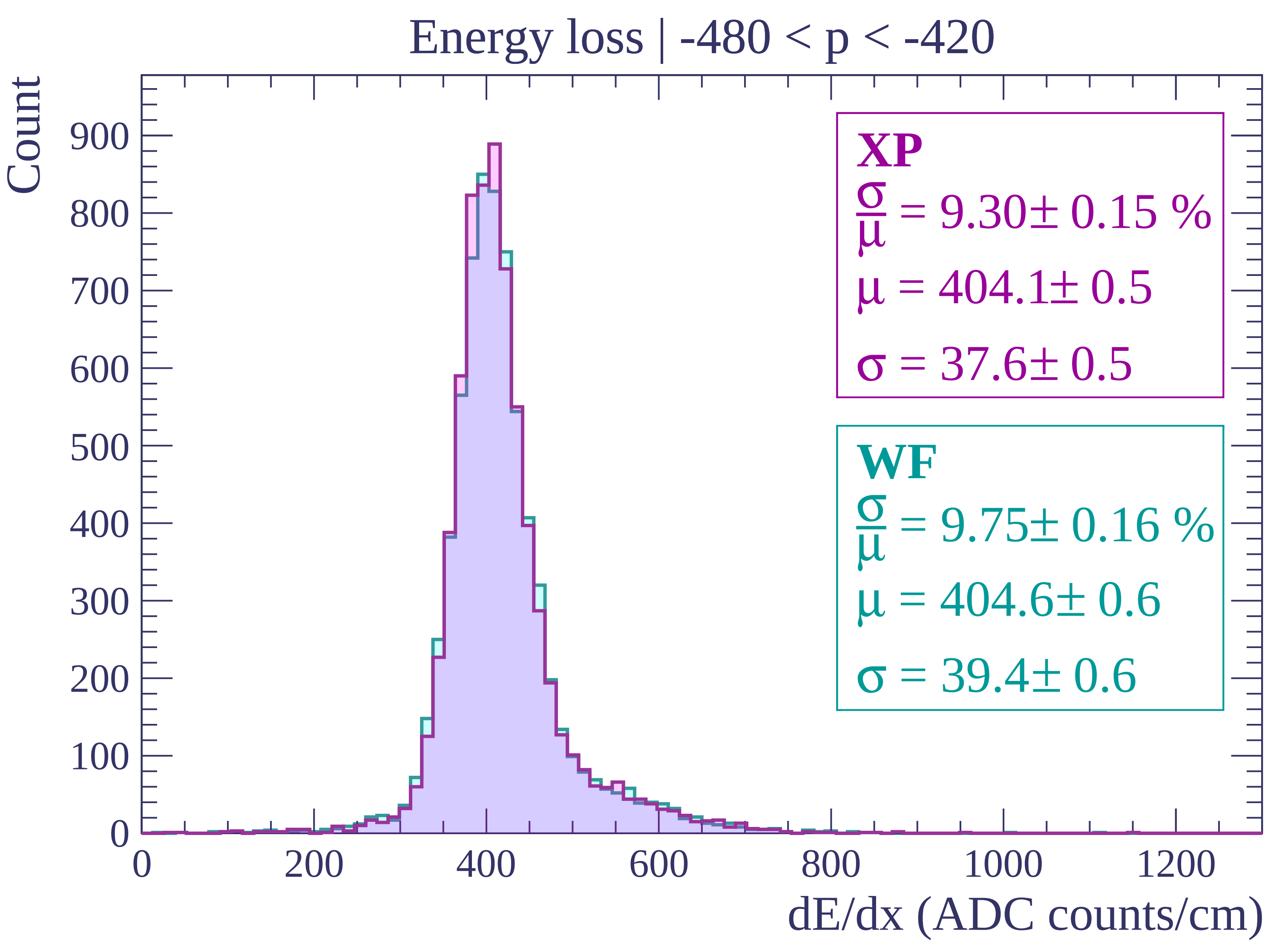
Energy loss | -660 < p < -600
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 9.59± 0.16 %
μ = 415.6± 0.6
σ = 39.9± 0.6
WF
σ/μ = 9.96± 0.18 %
μ = 417.3± 0.7
σ = 41.6± 0.7



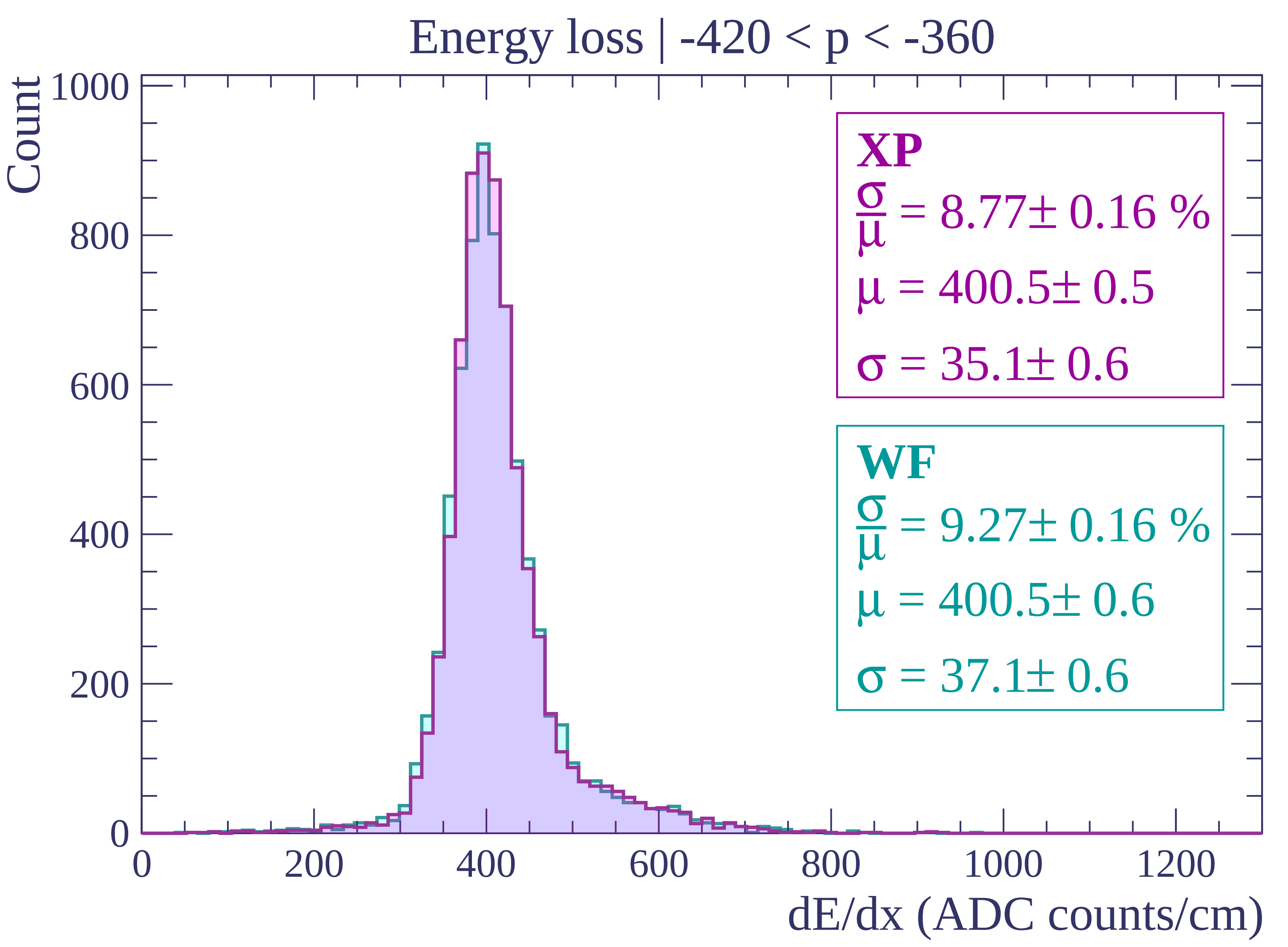
Energy loss | -600 < p < -540
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.29 \pm 0.17$ %
$\mu = 410.5 \pm 0.6$
$\sigma = 38.1 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 9.87 \pm 0.16$ %
$\mu = 412.8 \pm 0.6$
$\sigma = 40.7 \pm 0.6$



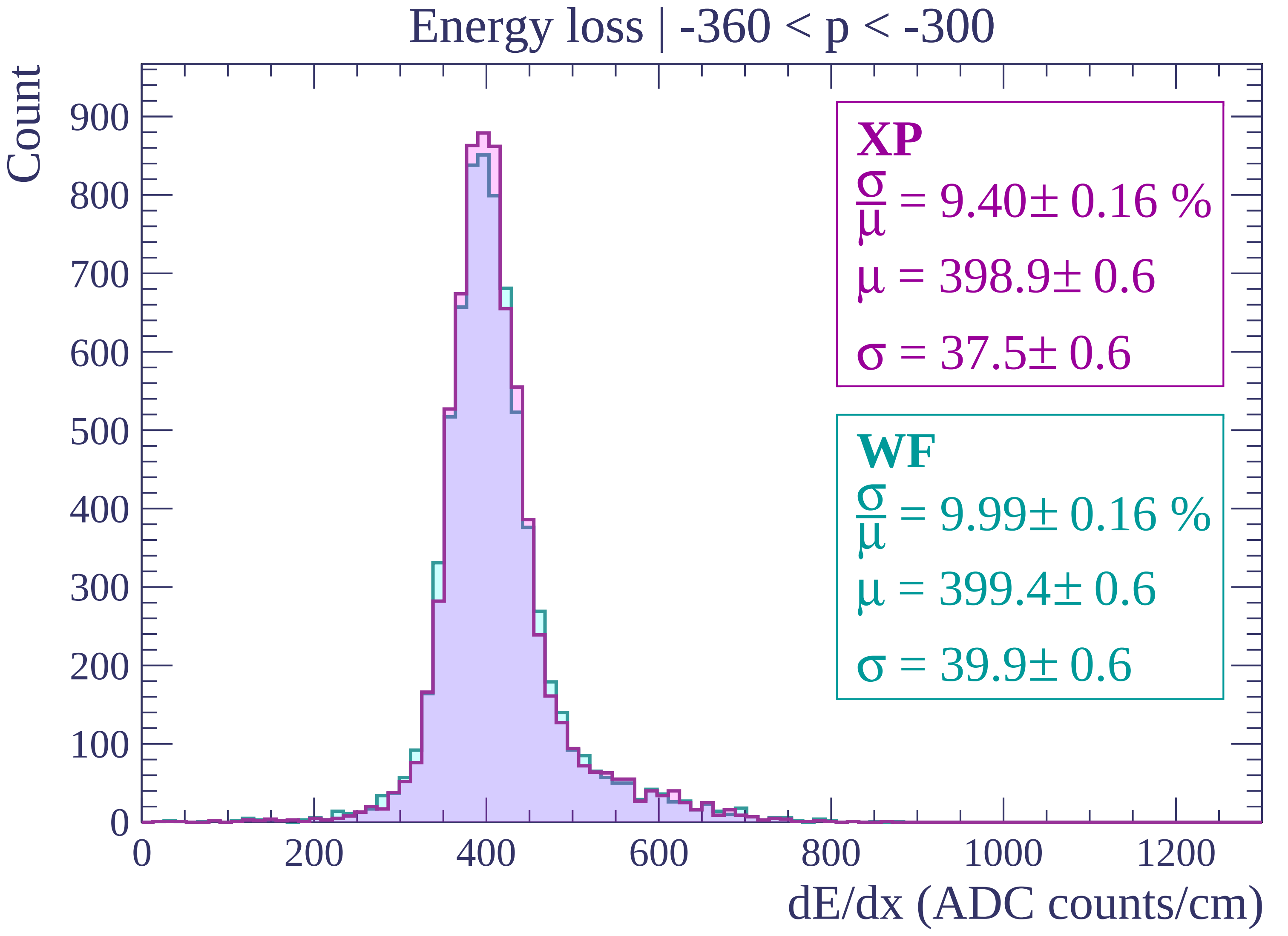
Energy loss | -540 < p < -480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.11± 0.14 %
μ = 408.0± 0.6
σ = 37.2± 0.5
WF
$\frac{\sigma}{\mu}$ = 9.52± 0.15 %
μ = 408.7± 0.6
σ = 38.9± 0.6

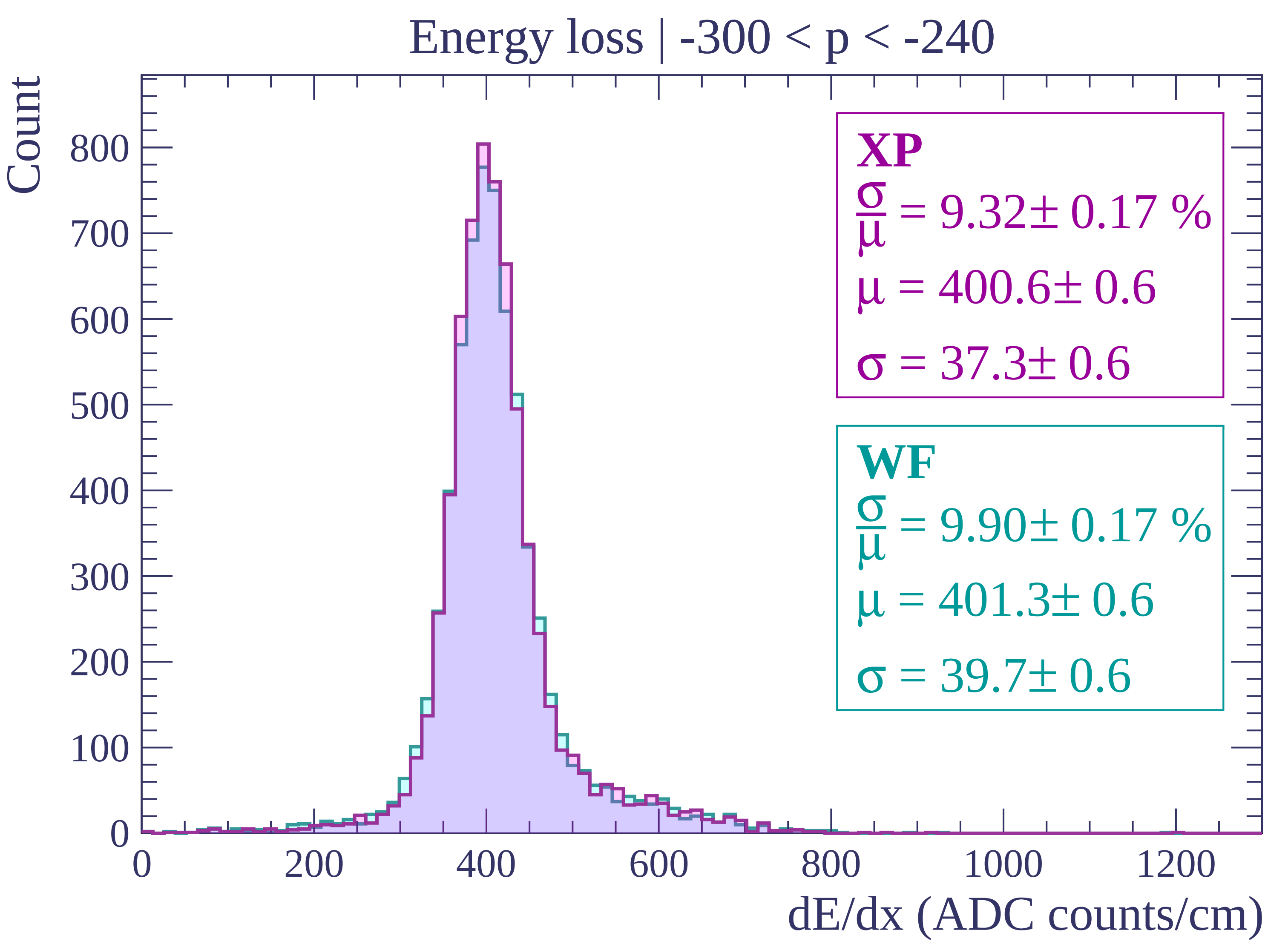


Energy loss | -480 < p < -420
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 9.30± 0.15 %
μ = 404.1± 0.5
σ = 37.6± 0.5
WF
σ/μ = 9.75± 0.16 %
μ = 404.6± 0.6
σ = 39.4± 0.6

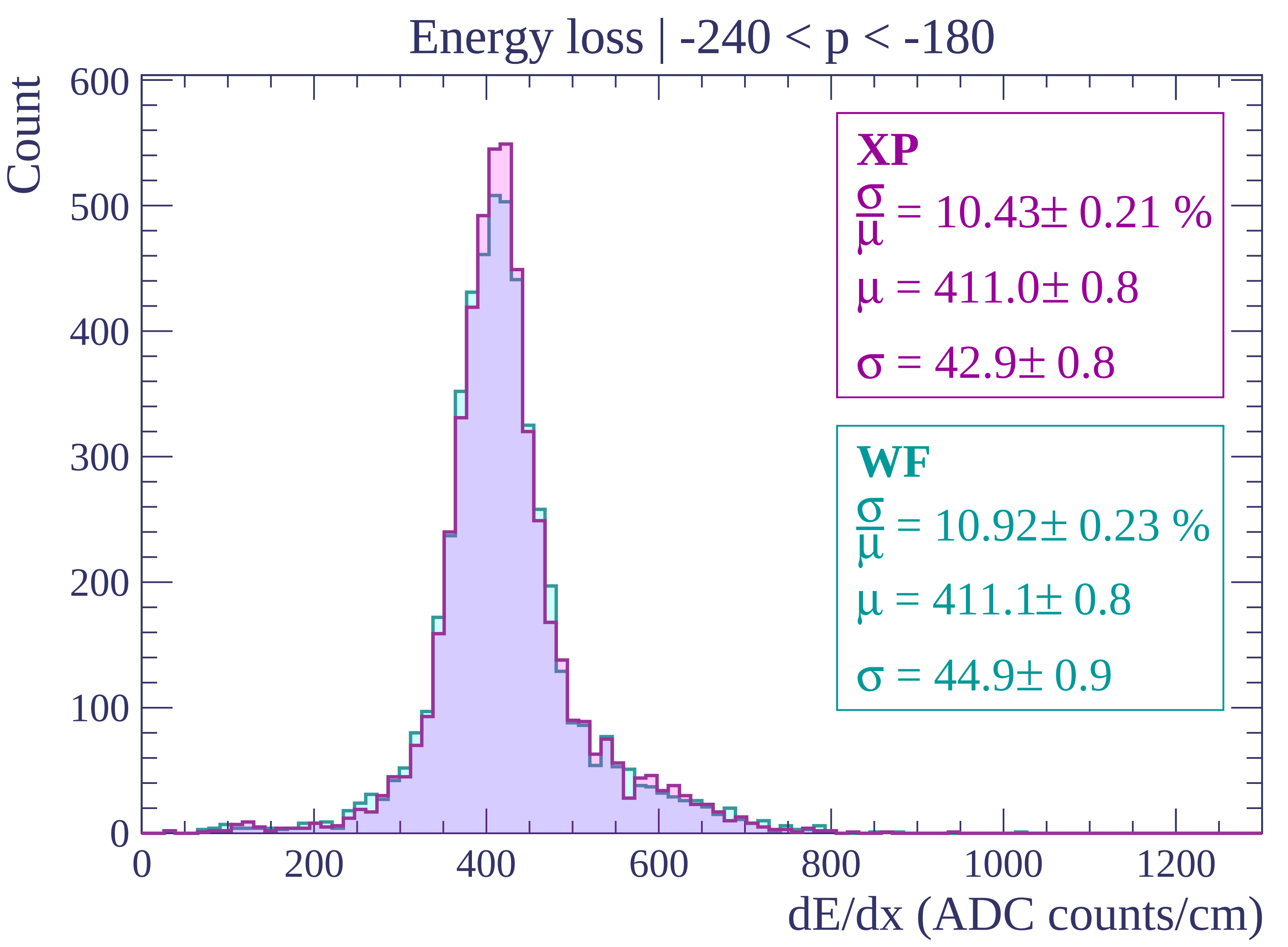


Energy loss | -420 < p < -360
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.77± 0.16 %
μ = 400.5± 0.5
σ = 35.1± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.27± 0.16 %
μ = 400.5± 0.6
σ = 37.1± 0.6
0
200
400
600
800
1000
1200

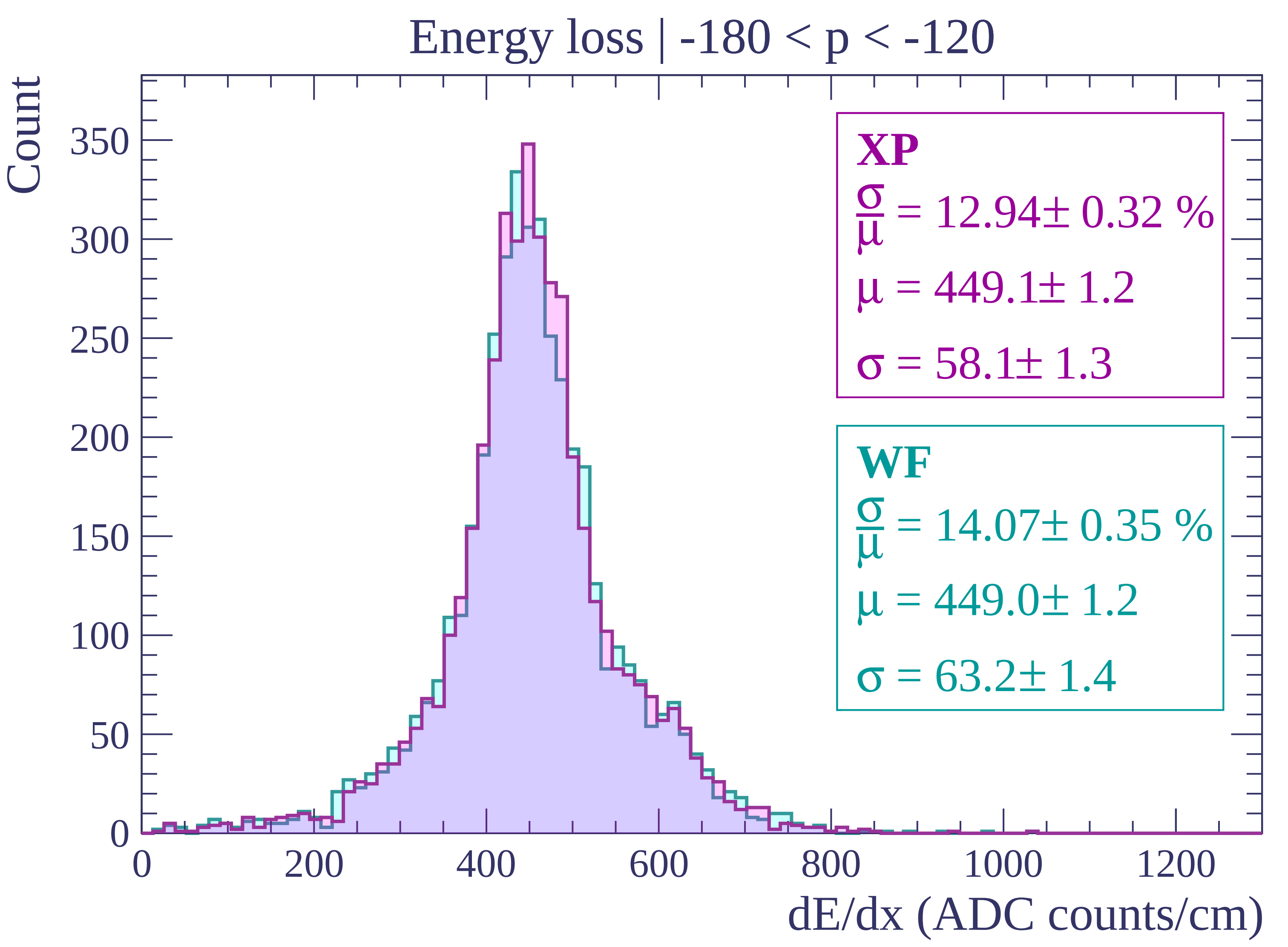

Energy loss | -360 < p < -300
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.40± 0.16 %
μ = 398.9± 0.6
σ = 37.5± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.99± 0.16 %
μ = 399.4± 0.6
σ = 39.9± 0.6
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200



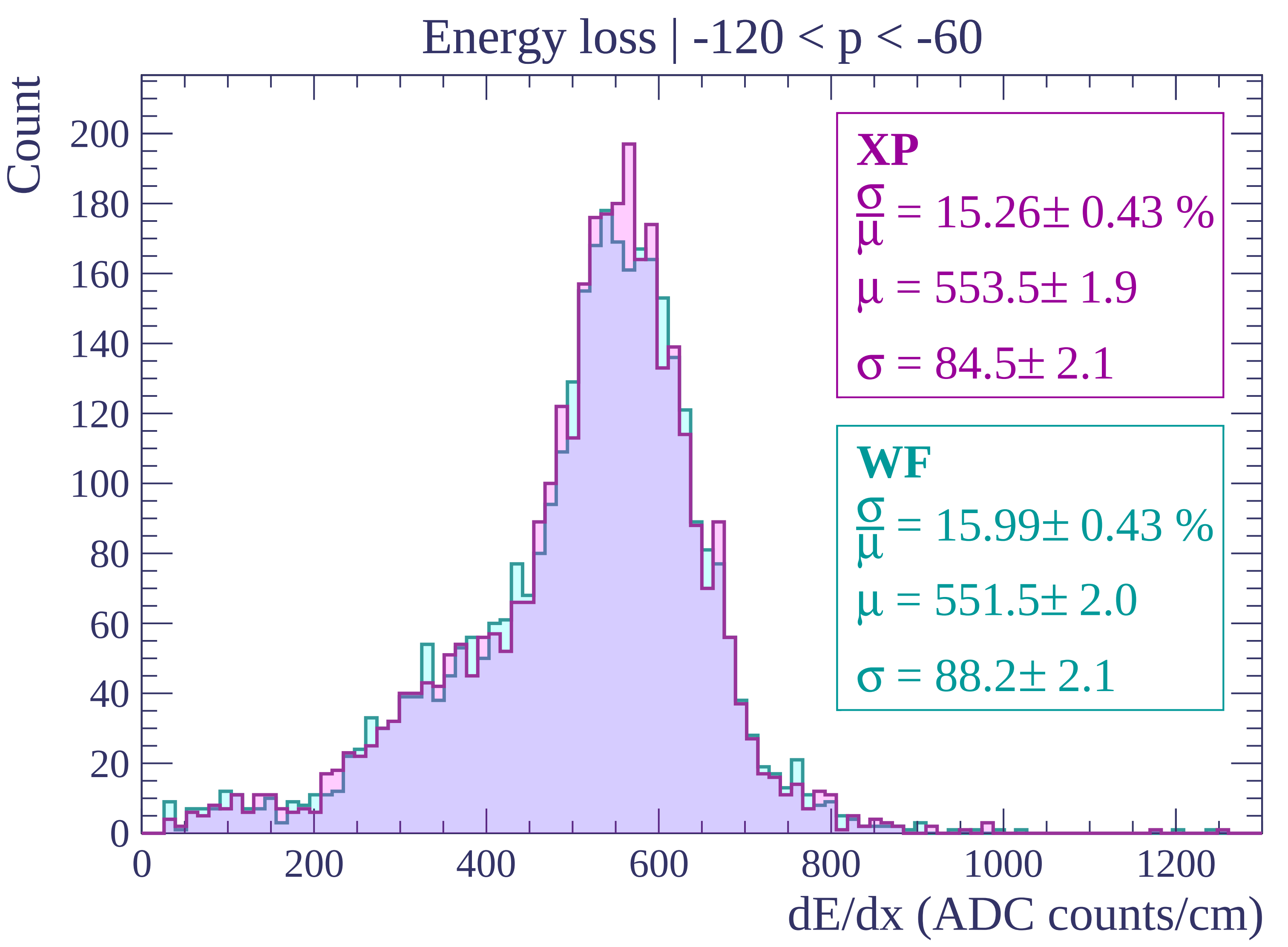
Energy loss | -300 < p < -240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP
$\frac{\sigma}{\mu} = 9.32 \pm 0.17$ %
$\mu = 400.6 \pm 0.6$
$\sigma = 37.3 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 9.90 \pm 0.17$ %
$\mu = 401.3 \pm 0.6$
$\sigma = 39.7 \pm 0.6$



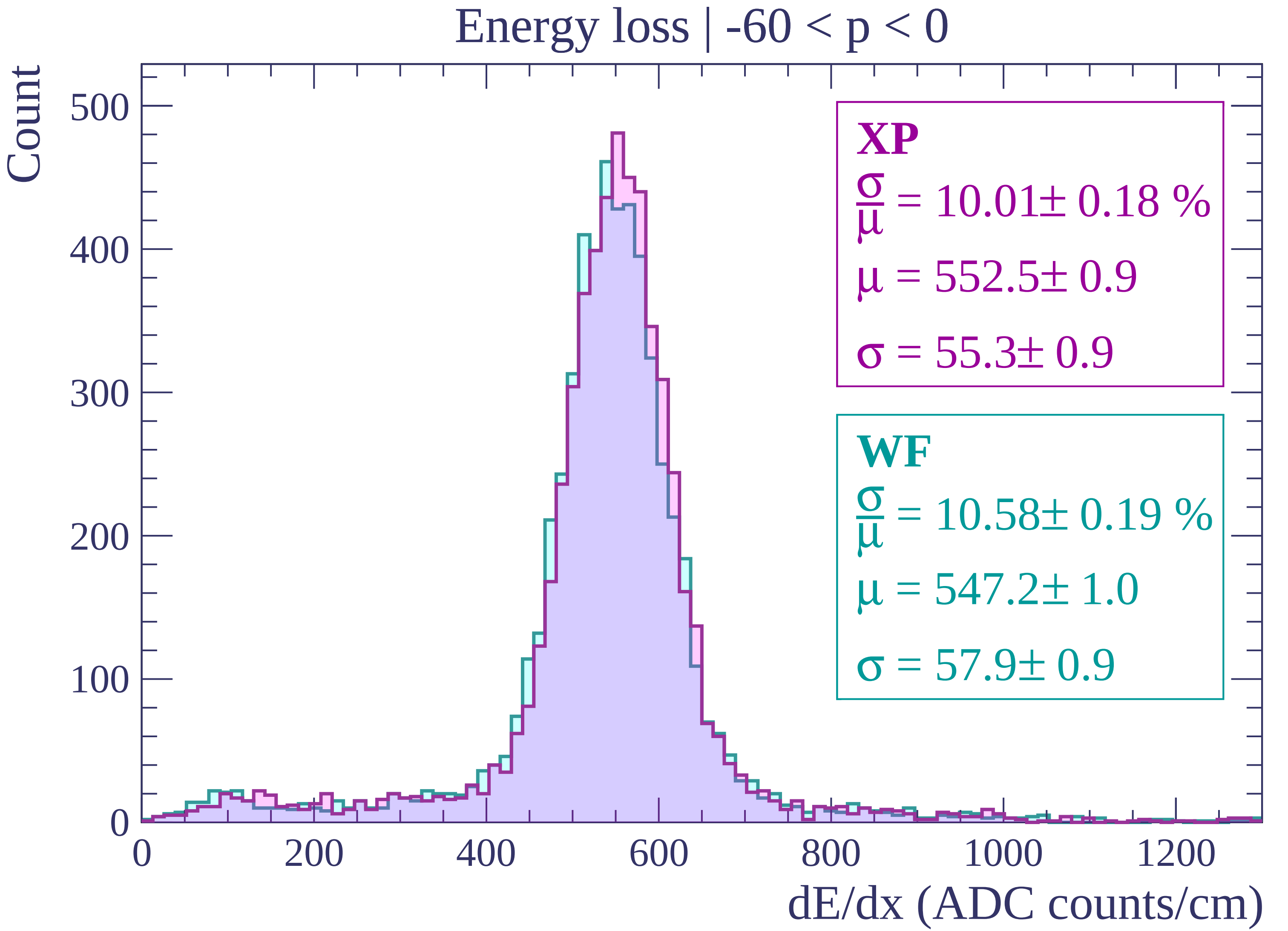
Energy loss | -240 < p < -180
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.43± 0.21 %
μ = 411.0± 0.8
σ = 42.9± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.92± 0.23 %
μ = 411.1± 0.8
σ = 44.9± 0.9



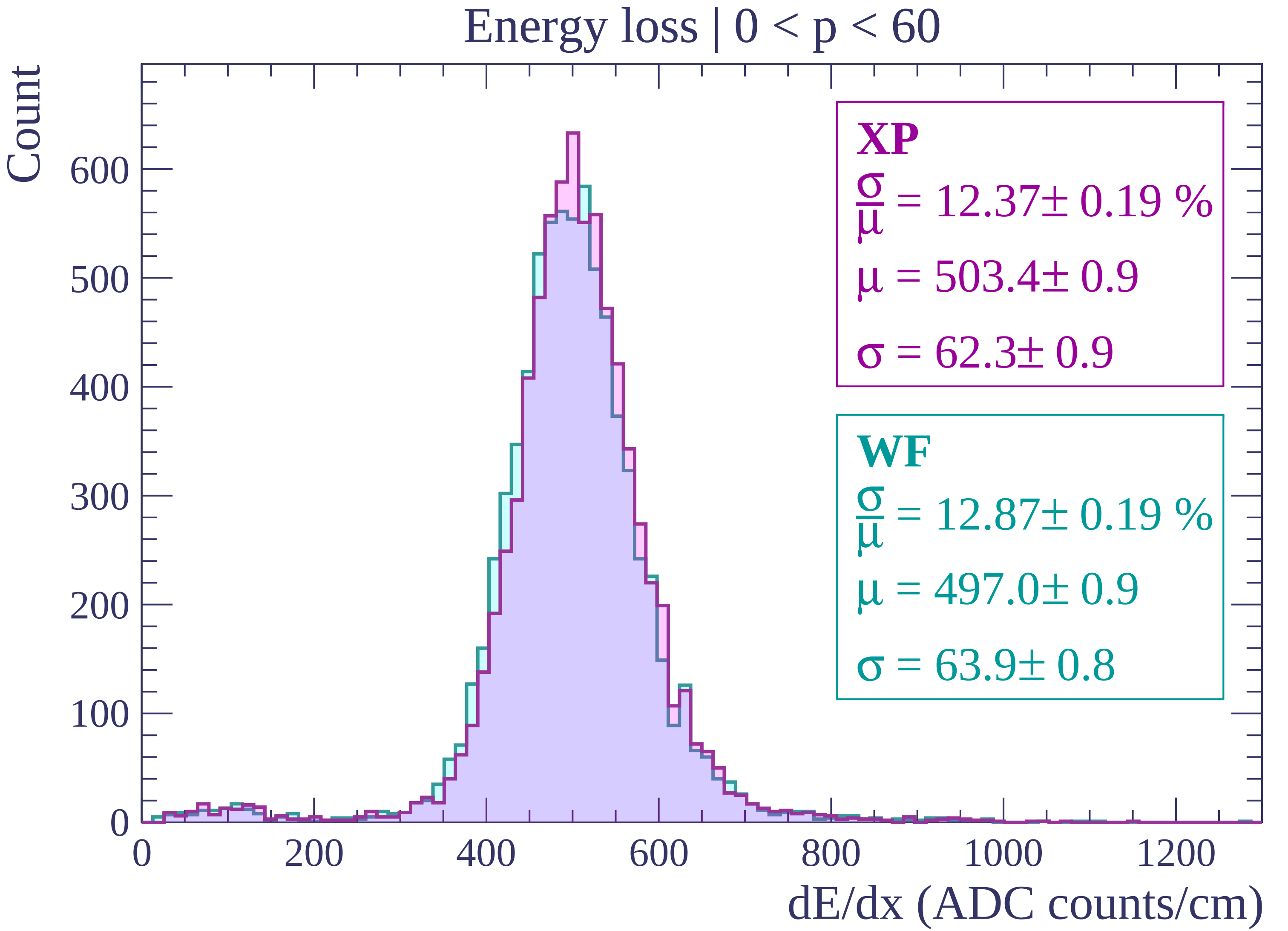
Energy loss | -180 < p < -120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 12.94± 0.32 %
μ = 449.1± 1.2
σ = 58.1± 1.3
WF
σ/μ = 14.07± 0.35 %
μ = 449.0± 1.2
σ = 63.2± 1.4



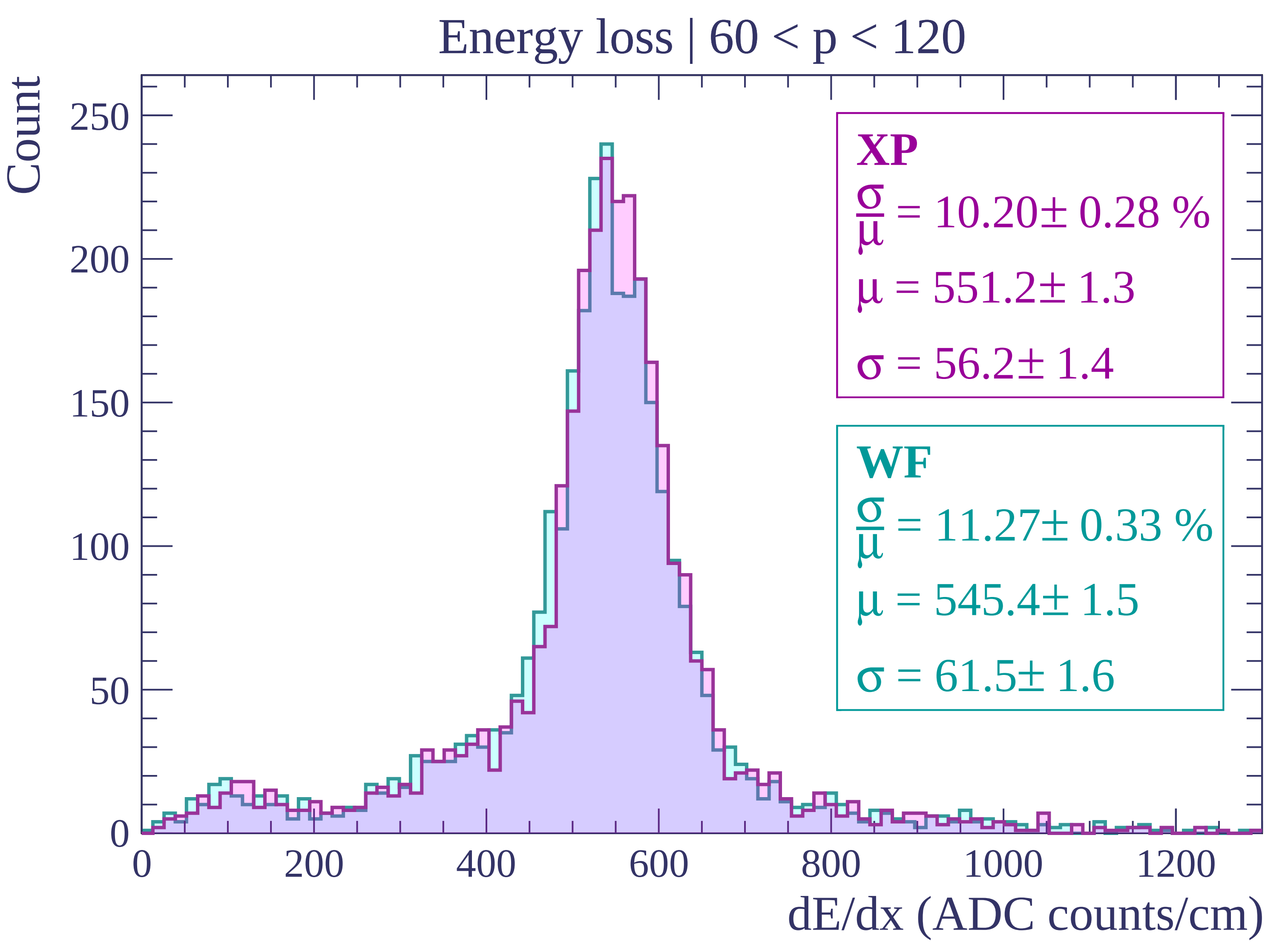
Energy loss | -120 < p < -60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 15.26± 0.43 %
μ = 553.5± 1.9
σ = 84.5± 2.1
WF
σ/μ = 15.99± 0.43 %
μ = 551.5± 2.0
σ = 88.2± 2.1

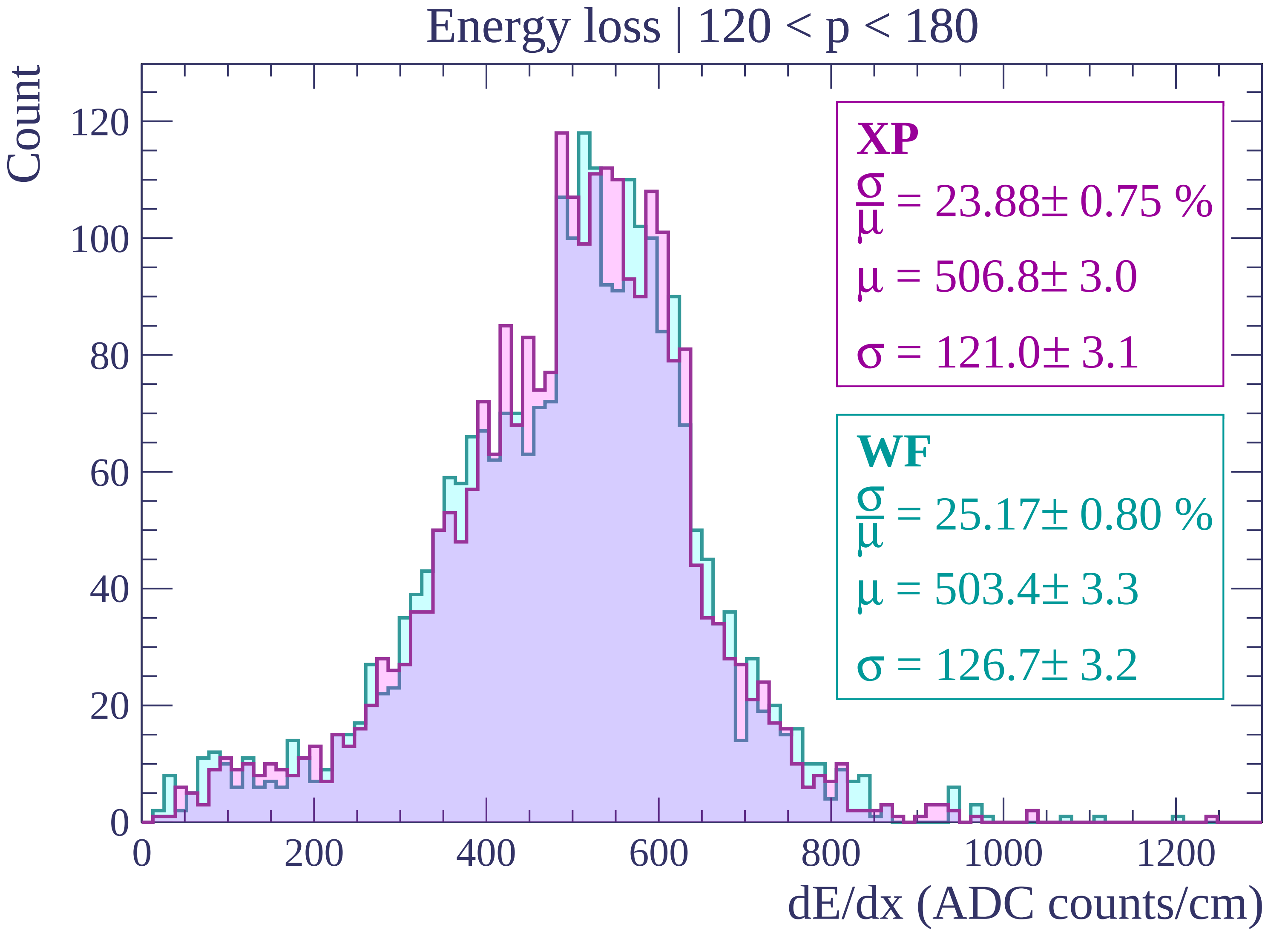


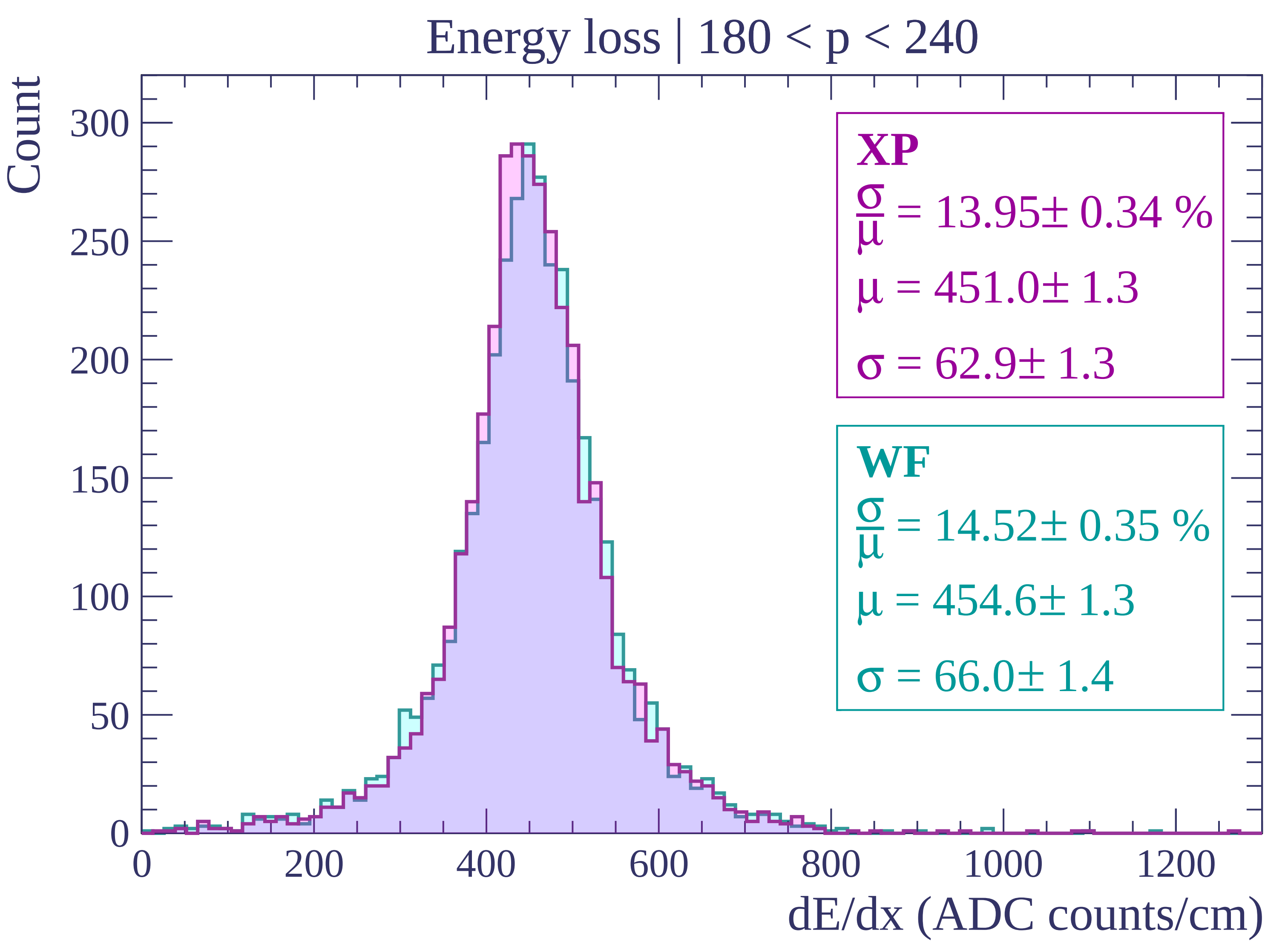
Energy loss | -60 < p < 0
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.01± 0.18 %
μ = 552.5± 0.9
σ = 55.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.58± 0.19 %
μ = 547.2± 1.0
σ = 57.9± 0.9



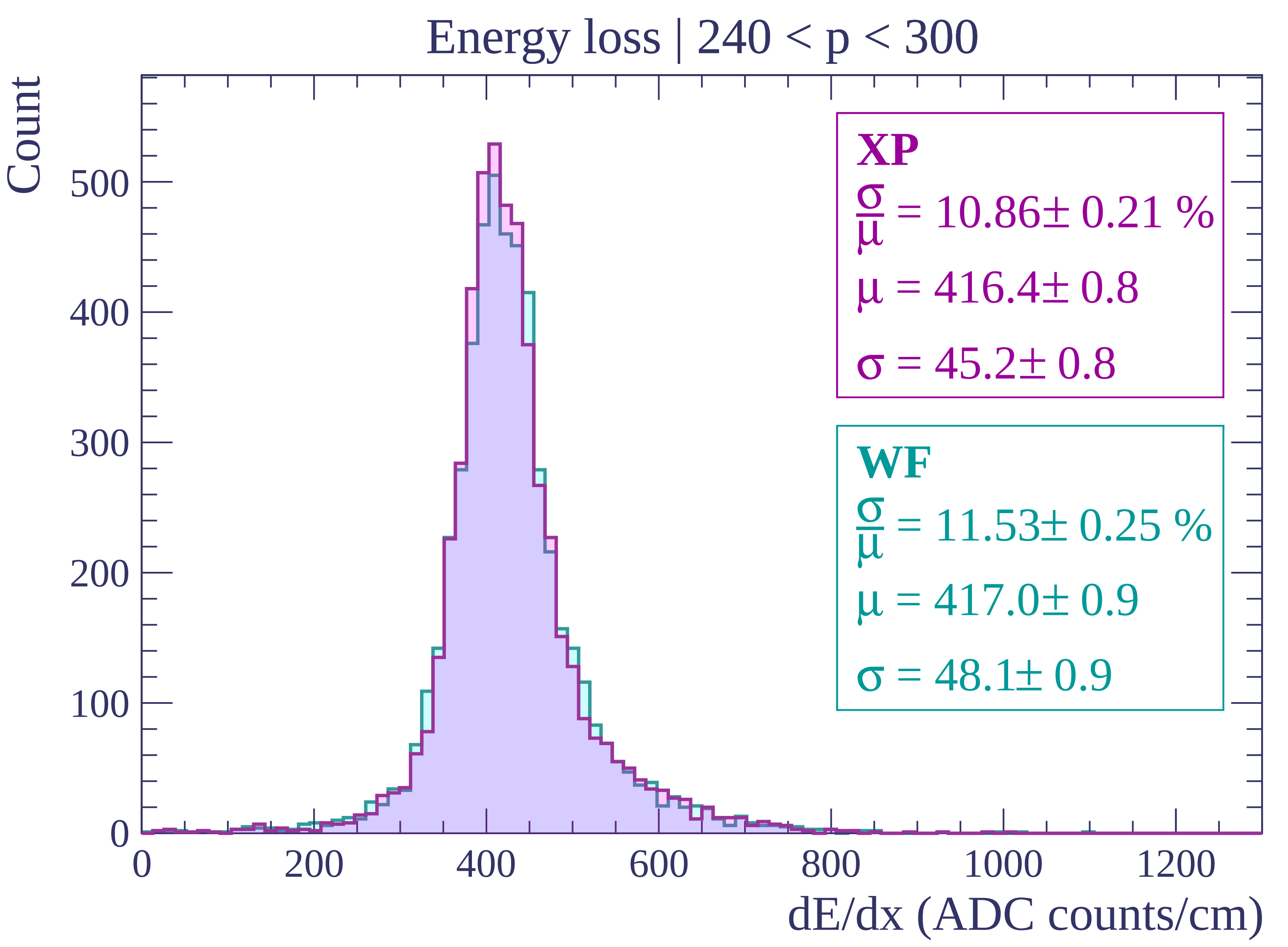
Energy loss | 0 < p < 60
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 12.37± 0.19 %
μ = 503.4± 0.9
σ = 62.3± 0.9
WF
$\frac{\sigma}{\mu}$ = 12.87± 0.19 %
μ = 497.0± 0.9
σ = 63.9± 0.8
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200

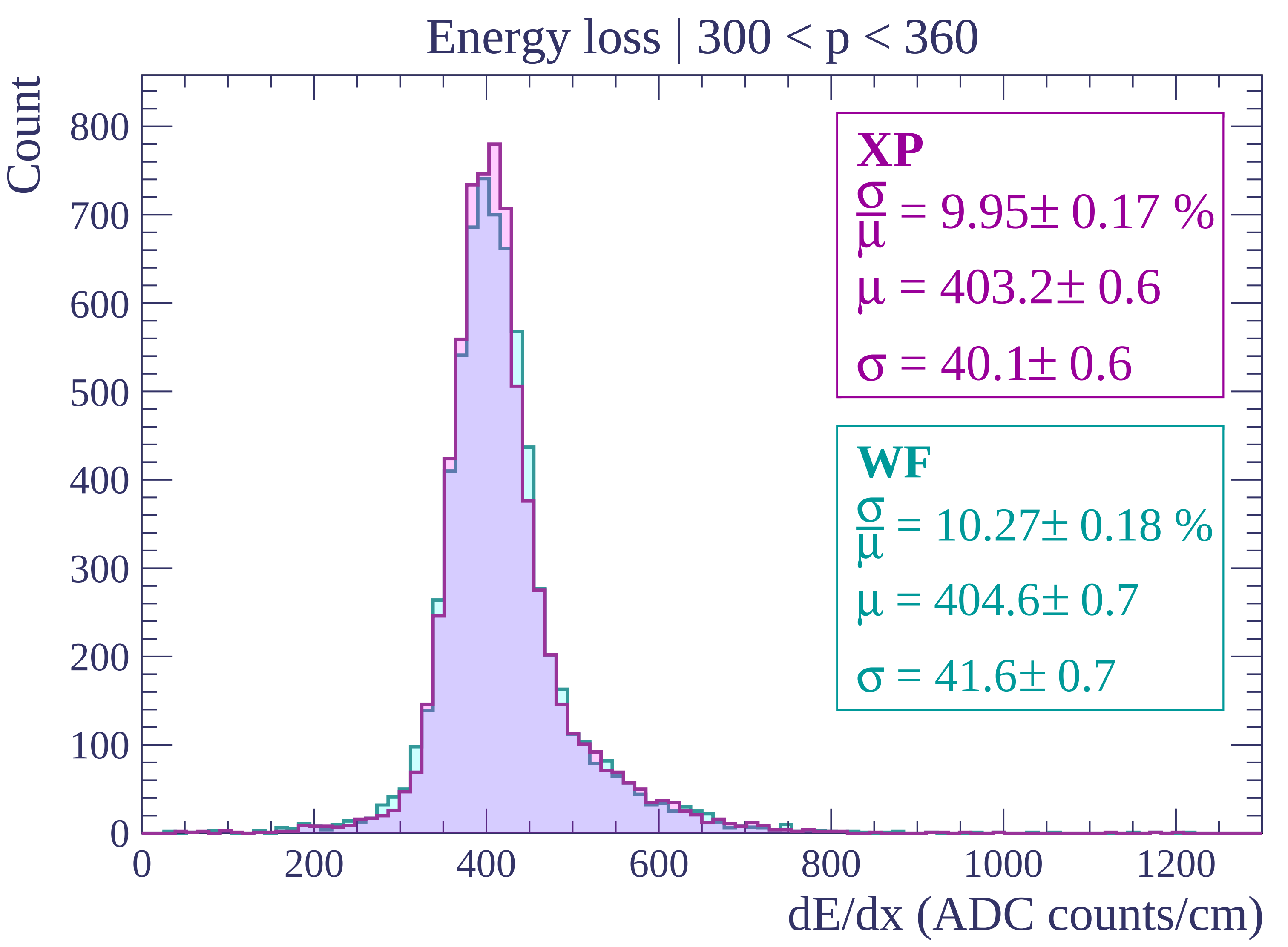

Energy loss | 60 < p < 120
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.20± 0.28 %
μ = 551.2± 1.3
σ = 56.2± 1.4
WF
$\frac{\sigma}{\mu}$ = 11.27± 0.33 %
μ = 545.4± 1.5
σ = 61.5± 1.6
0
50
100
150
200
250
0
200
400
600
800
1000
1200


Energy loss | 120 < p < 180
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 23.88± 0.75 %
μ = 506.8± 3.0
σ = 121.0± 3.1
WF
σ/μ = 25.17± 0.80 %
μ = 503.4± 3.3
σ = 126.7± 3.2

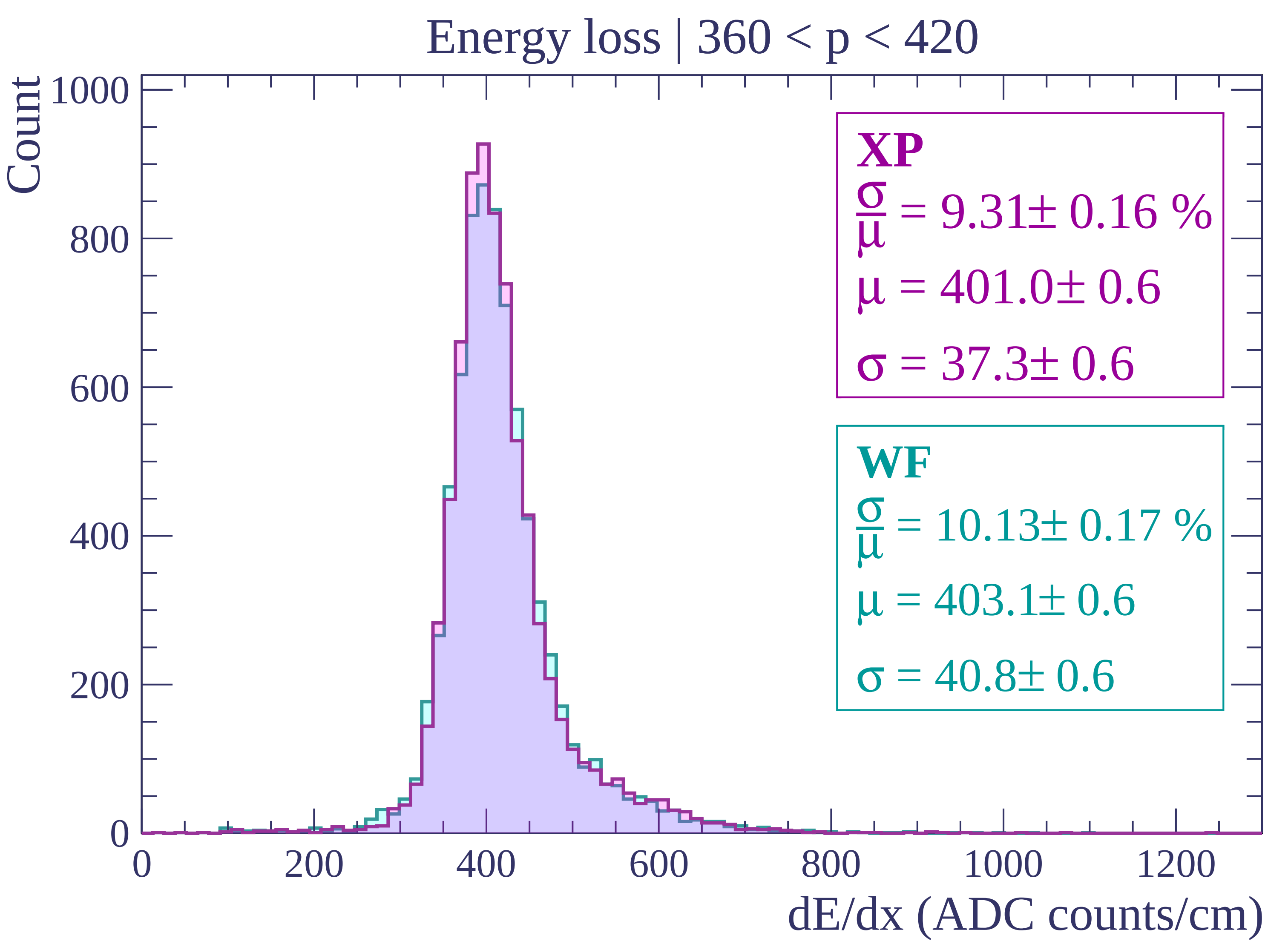


Energy loss | 180 < p < 240
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.95 ± 0.34 %
μ = 451.0 ± 1.3
σ = 62.9 ± 1.3
WF
$\frac{\sigma}{\mu}$ = 14.52 ± 0.35 %
μ = 454.6 ± 1.3
σ = 66.0 ± 1.4

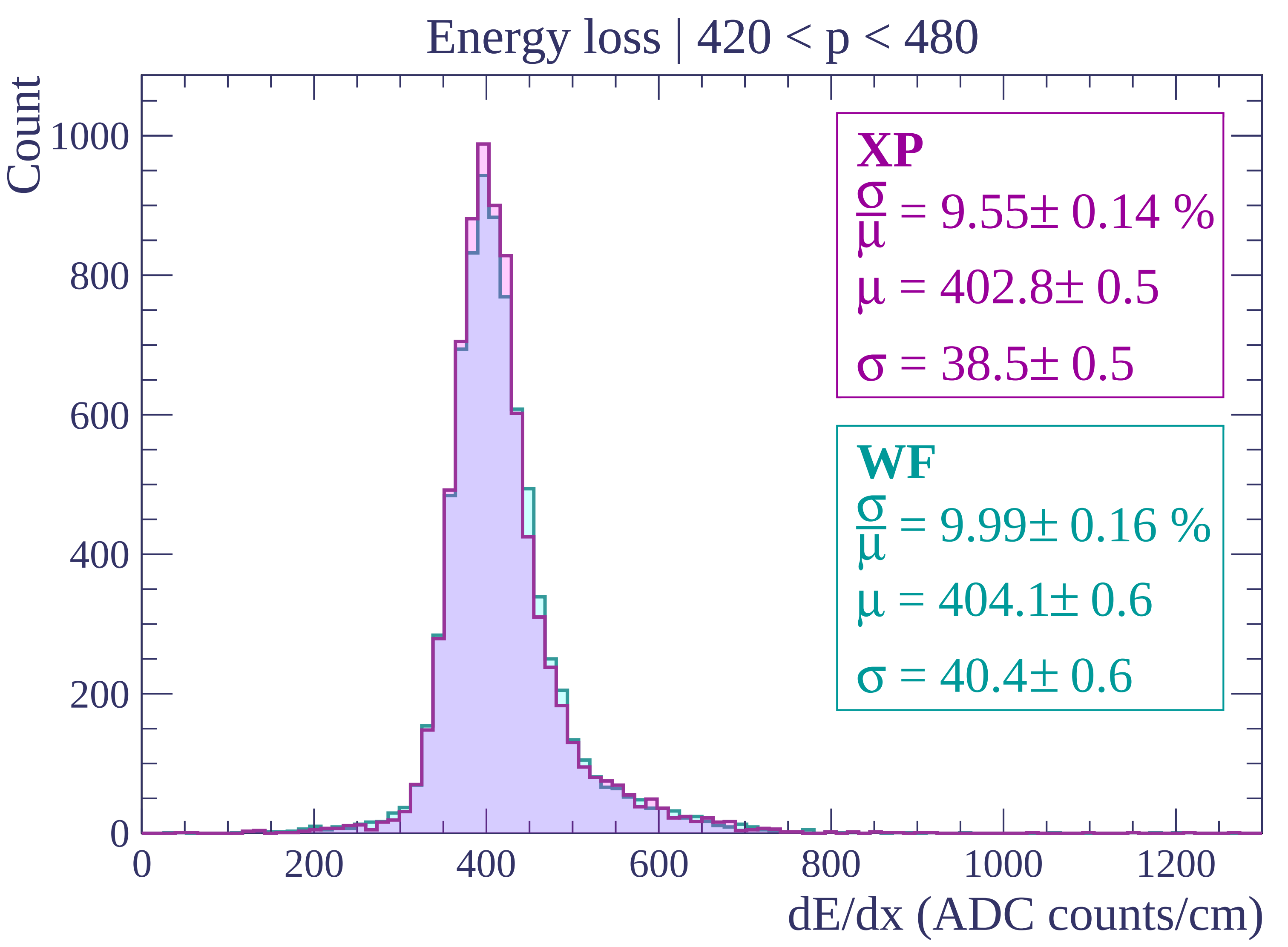

Energy loss | 240 < p < 300
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 10.86± 0.21 %
μ = 416.4± 0.8
σ = 45.2± 0.8
WF
σ/μ = 11.53± 0.25 %
μ = 417.0± 0.9
σ = 48.1± 0.9



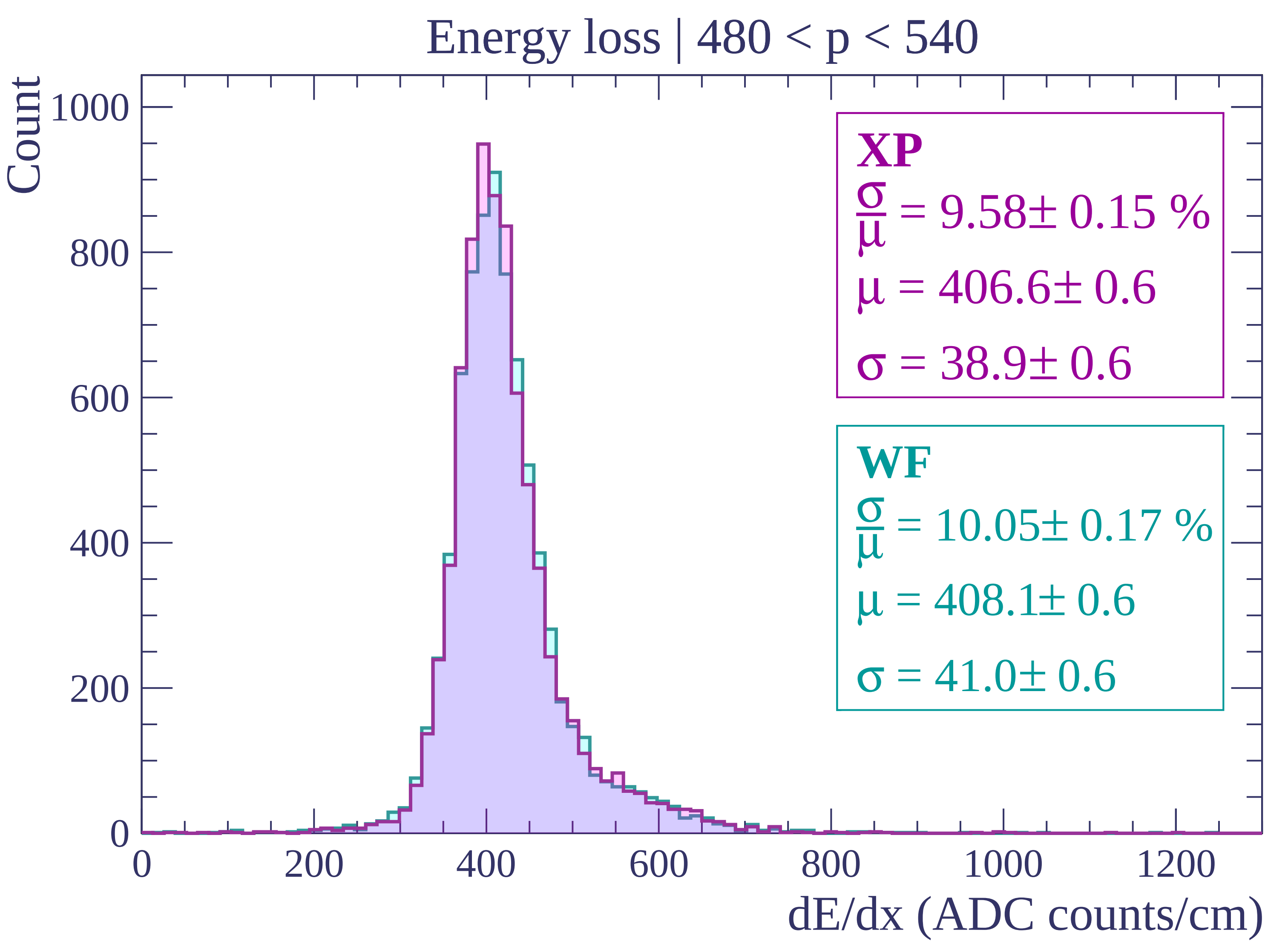
Energy loss | 300 < p < 360
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.95± 0.17 %
μ = 403.2± 0.6
σ = 40.1± 0.6
WF
σ/μ = 10.27± 0.18 %
μ = 404.6± 0.7
σ = 41.6± 0.7
0
200
400
600
800
1000
1200
100
300
500
700



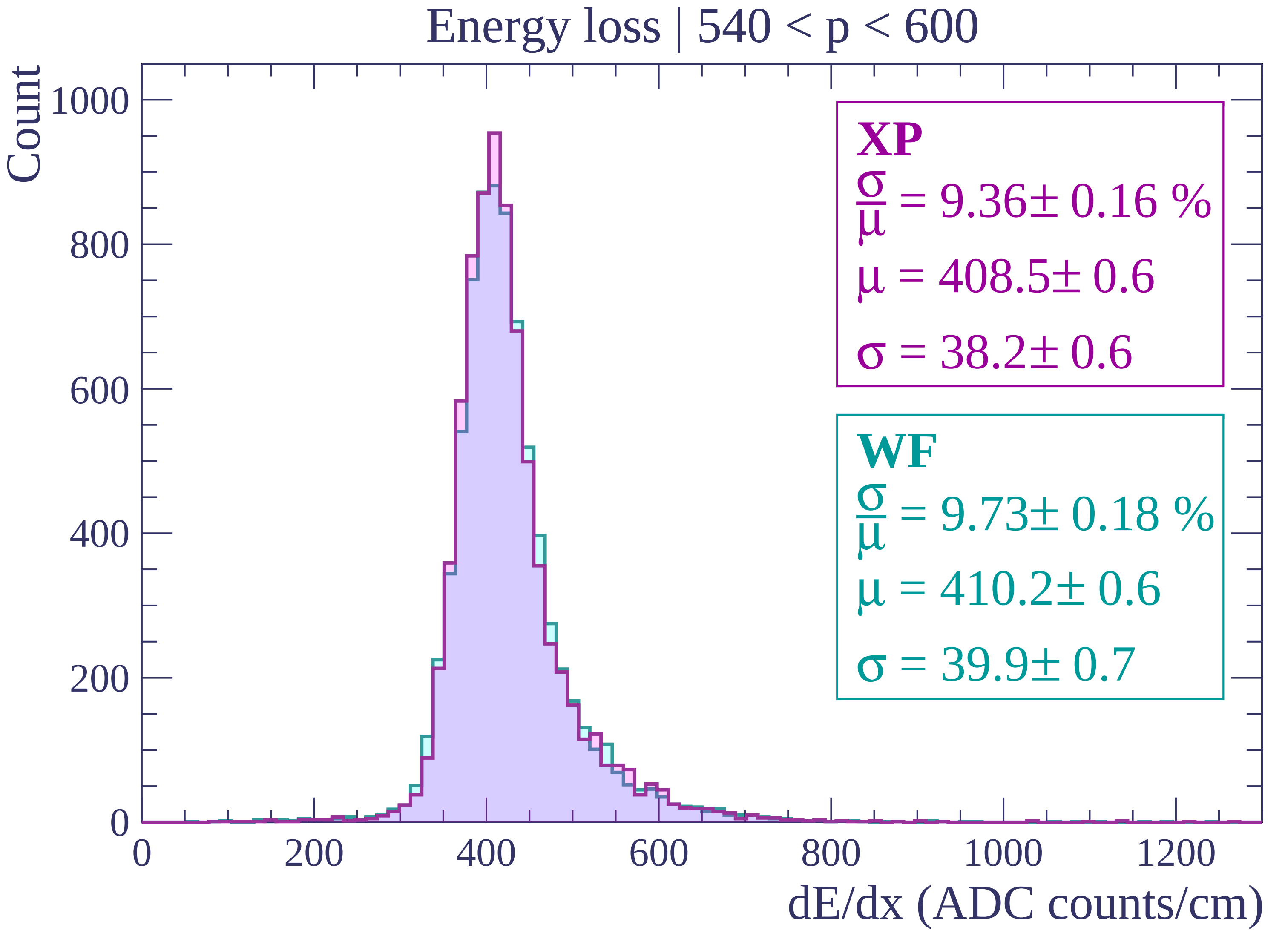
Energy loss | 360 < p < 420
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.31± 0.16 %
μ = 401.0± 0.6
σ = 37.3± 0.6
WF
σ/μ = 10.13± 0.17 %
μ = 403.1± 0.6
σ = 40.8± 0.6

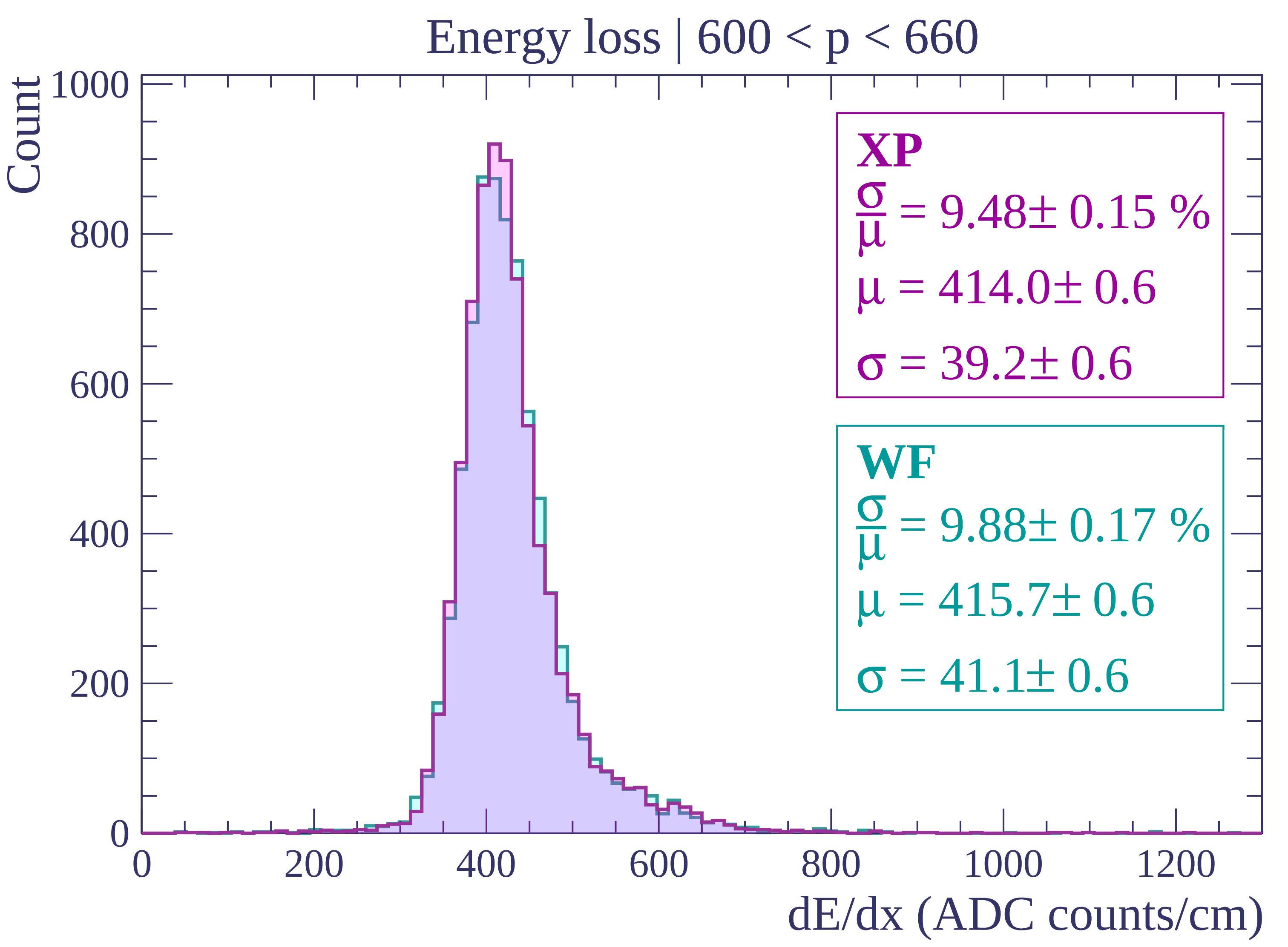


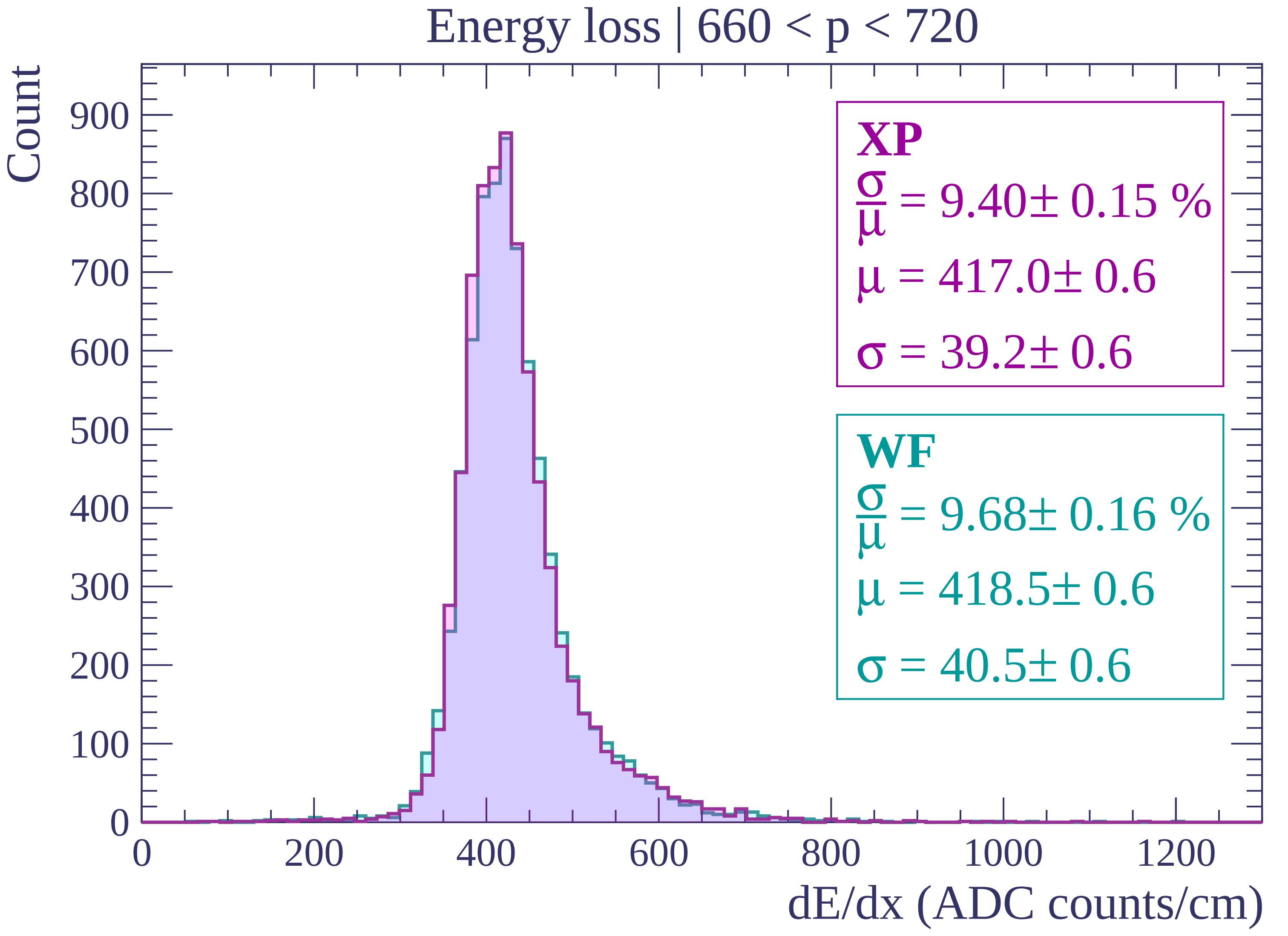
Energy loss | 420 < p < 480
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.55 \pm 0.14$ %
$\mu = 402.8 \pm 0.5$
$\sigma = 38.5 \pm 0.5$
WF
$\frac{\sigma}{\mu} = 9.99 \pm 0.16$ %
$\mu = 404.1 \pm 0.6$
$\sigma = 40.4 \pm 0.6$



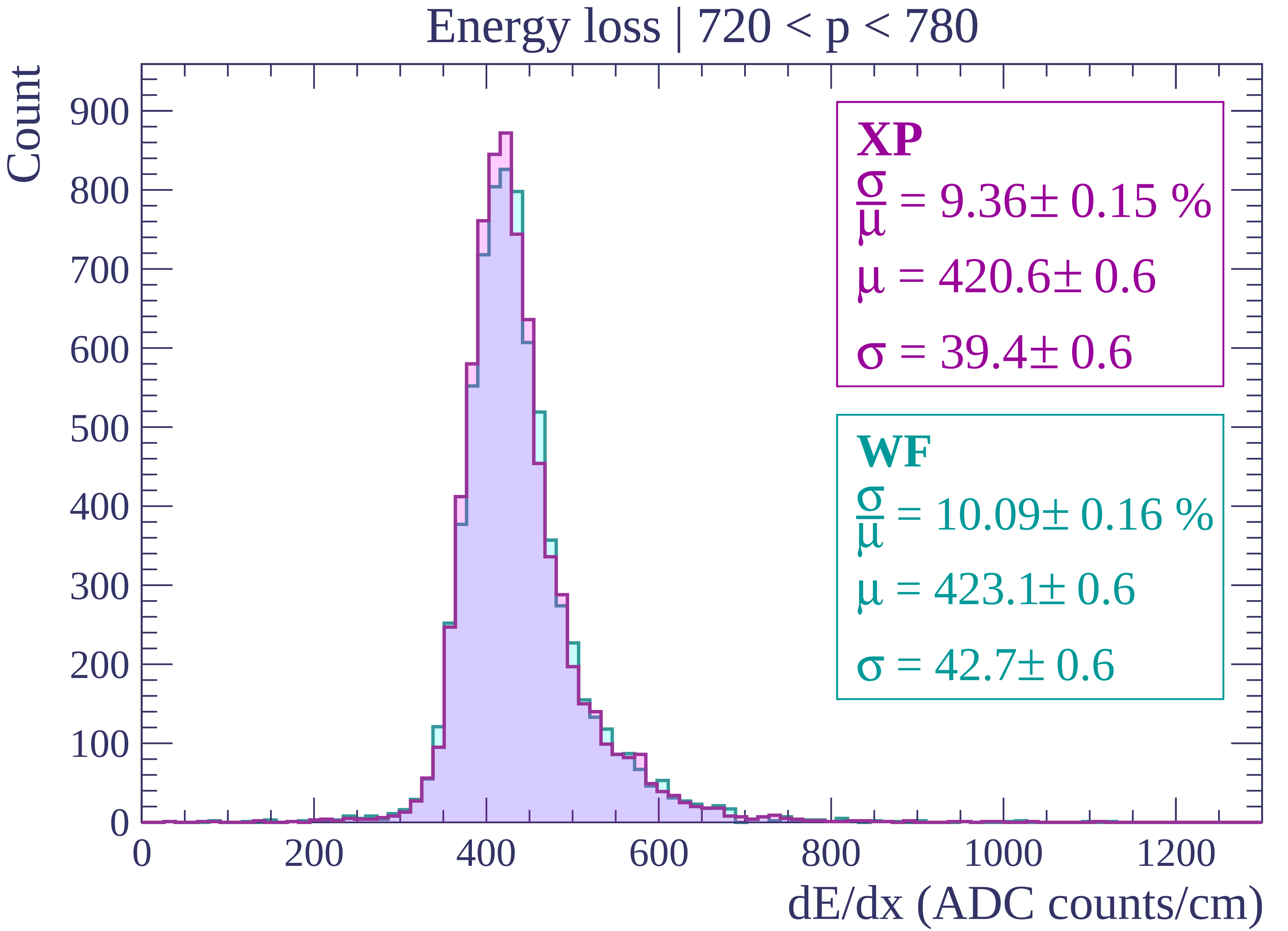
Energy loss | 480 < p < 540
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.58± 0.15 %
μ = 406.6± 0.6
σ = 38.9± 0.6
WF
$\frac{\sigma}{\mu}$ = 10.05± 0.17 %
μ = 408.1± 0.6
σ = 41.0± 0.6


Energy loss | 540 < p < 600
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.36± 0.16 %
μ = 408.5± 0.6
σ = 38.2± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.73± 0.18 %
μ = 410.2± 0.6
σ = 39.9± 0.7

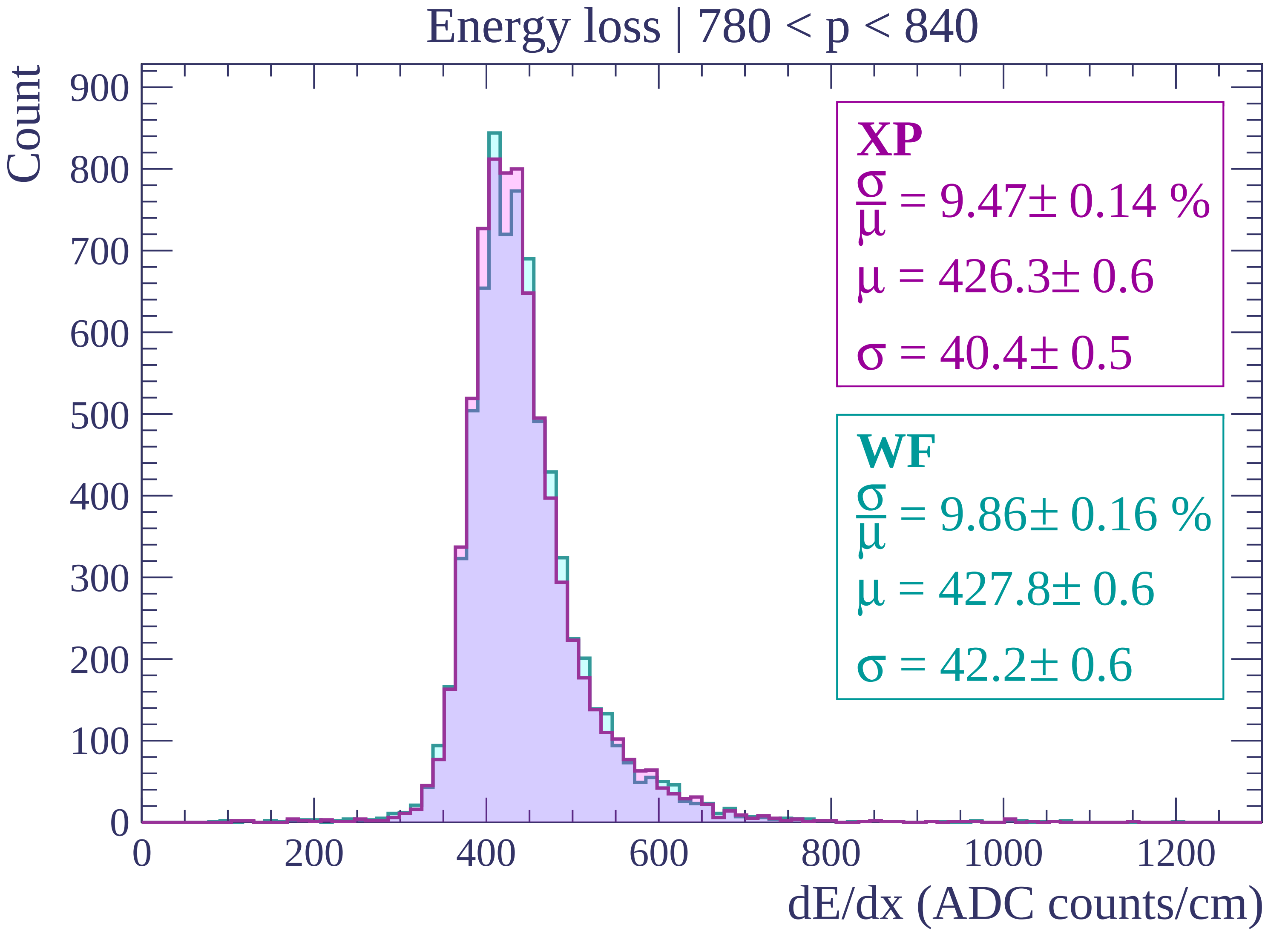

Energy loss | 600 < p < 660
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.48± 0.15 %
μ = 414.0± 0.6
σ = 39.2± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.88± 0.17 %
μ = 415.7± 0.6
σ = 41.1± 0.6



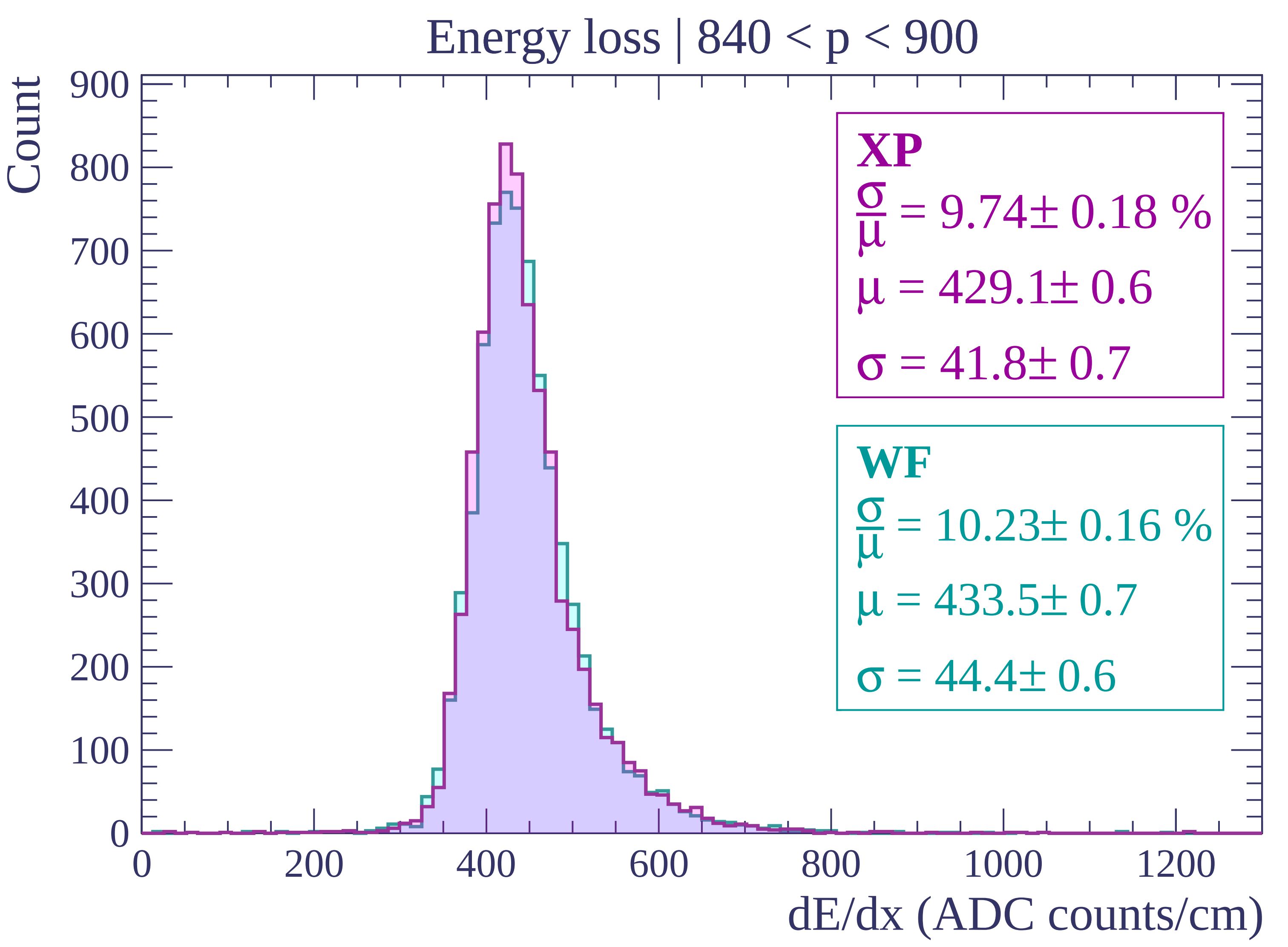
Energy loss | 660 < p < 720
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.40 \pm 0.15$ %
$\mu = 417.0 \pm 0.6$
$\sigma = 39.2 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 9.68 \pm 0.16$ %
$\mu = 418.5 \pm 0.6$
$\sigma = 40.5 \pm 0.6$



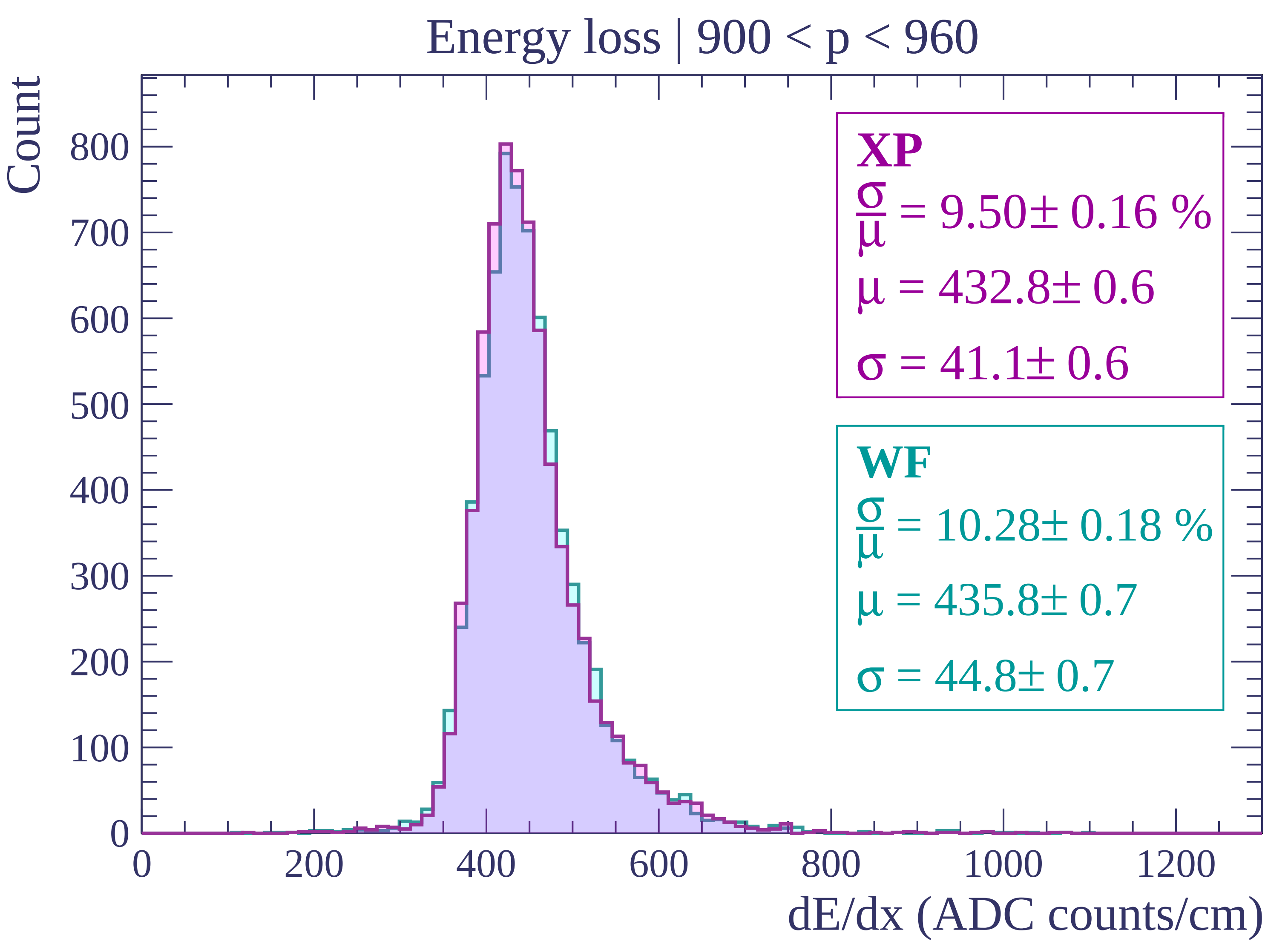
Energy loss | 720 < p < 780
Count
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.36 ± 0.15 %
μ = 420.6 ± 0.6
σ = 39.4 ± 0.6
WF
$\frac{\sigma}{\mu}$ = 10.09 ± 0.16 %
μ = 423.1 ± 0.6
σ = 42.7 ± 0.6



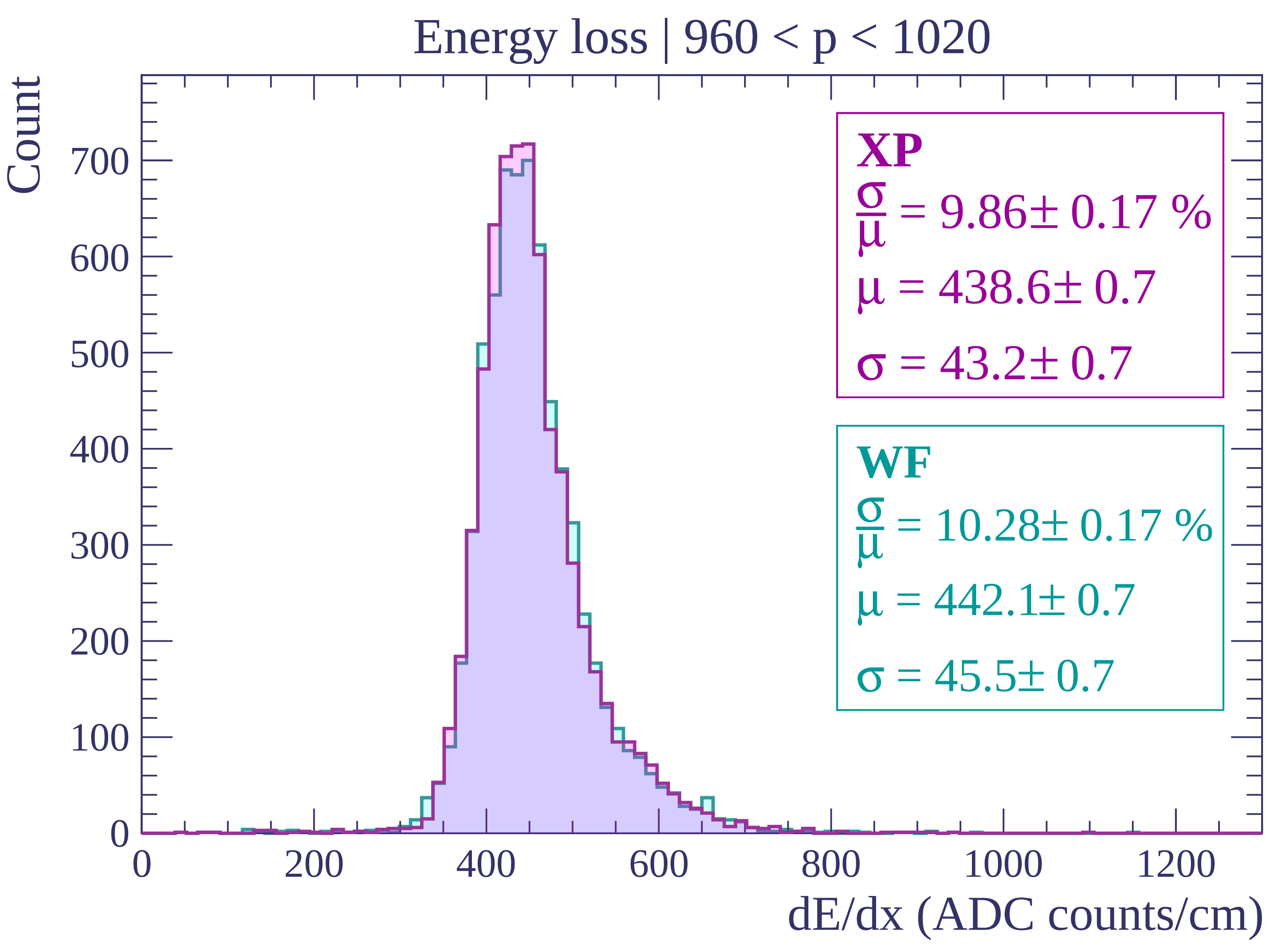
Energy loss | 780 < p < 840
Count
900
800
700
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.47± 0.14 %
μ = 426.3± 0.6
σ = 40.4± 0.5
WF
$\frac{\sigma}{\mu}$ = 9.86± 0.16 %
μ = 427.8± 0.6
σ = 42.2± 0.6



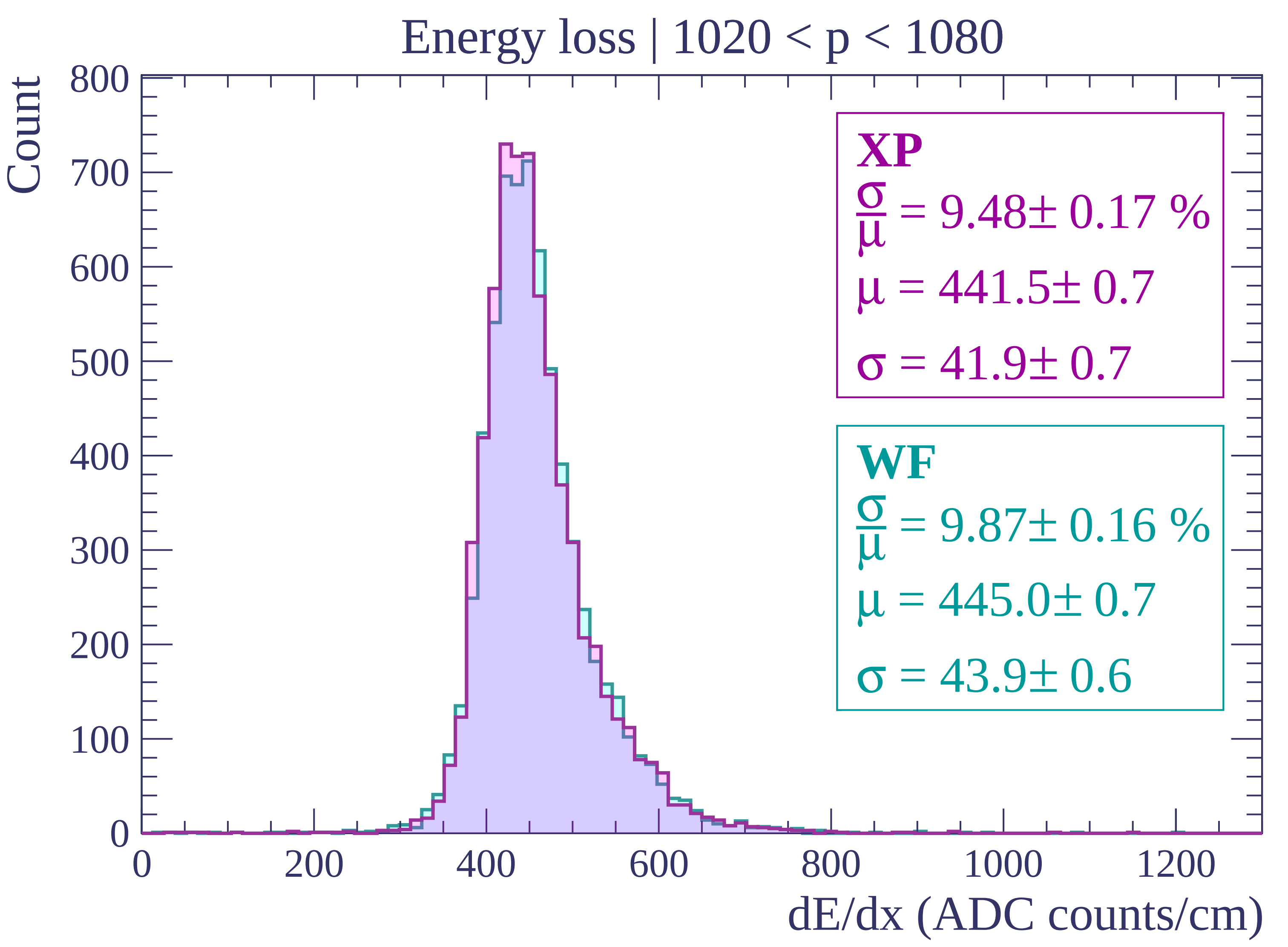
Energy loss | 840 < p < 900
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 9.74± 0.18 %
μ = 429.1± 0.6
σ = 41.8± 0.7
WF
σ/μ = 10.23± 0.16 %
μ = 433.5± 0.7
σ = 44.4± 0.6

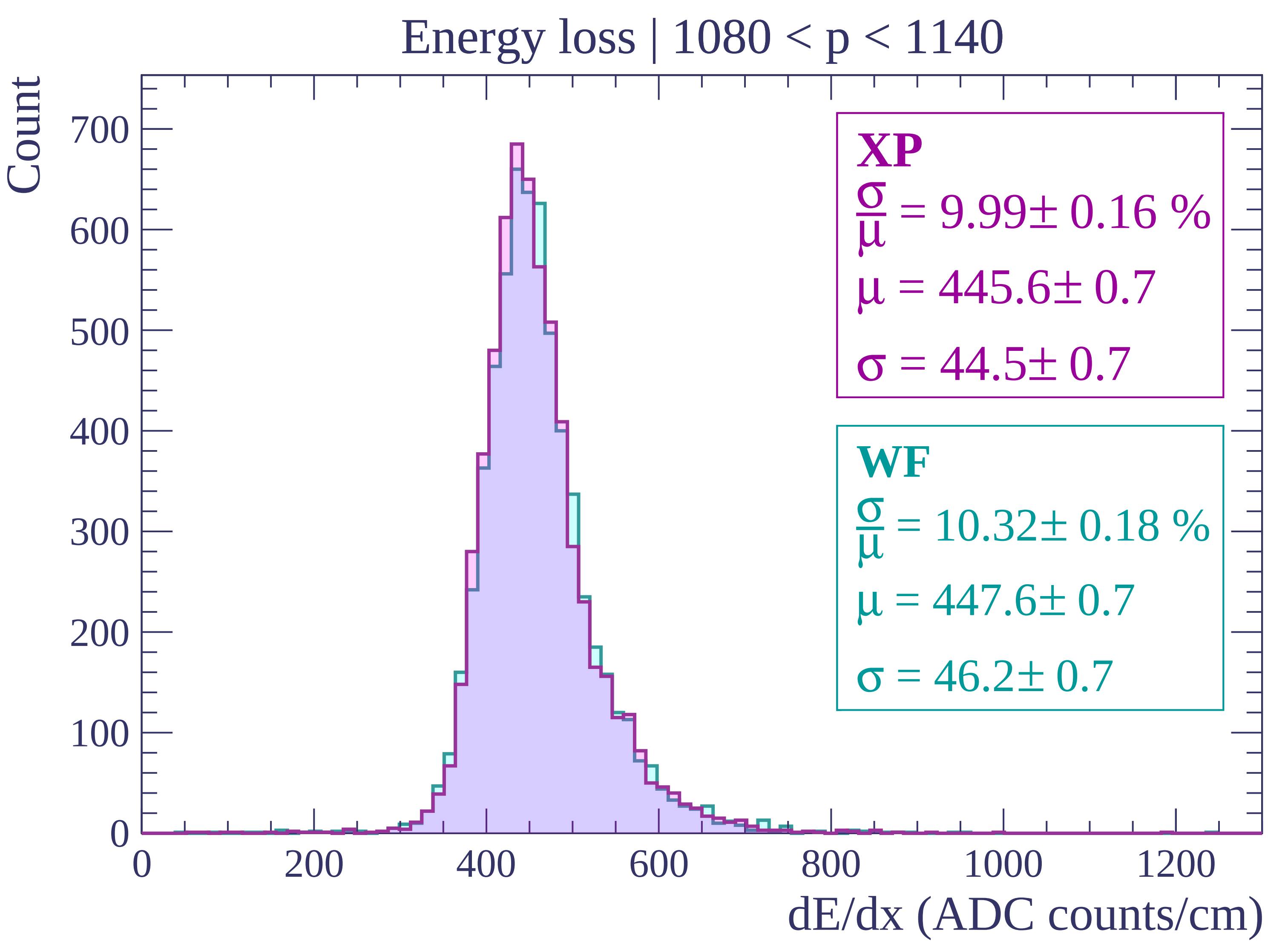


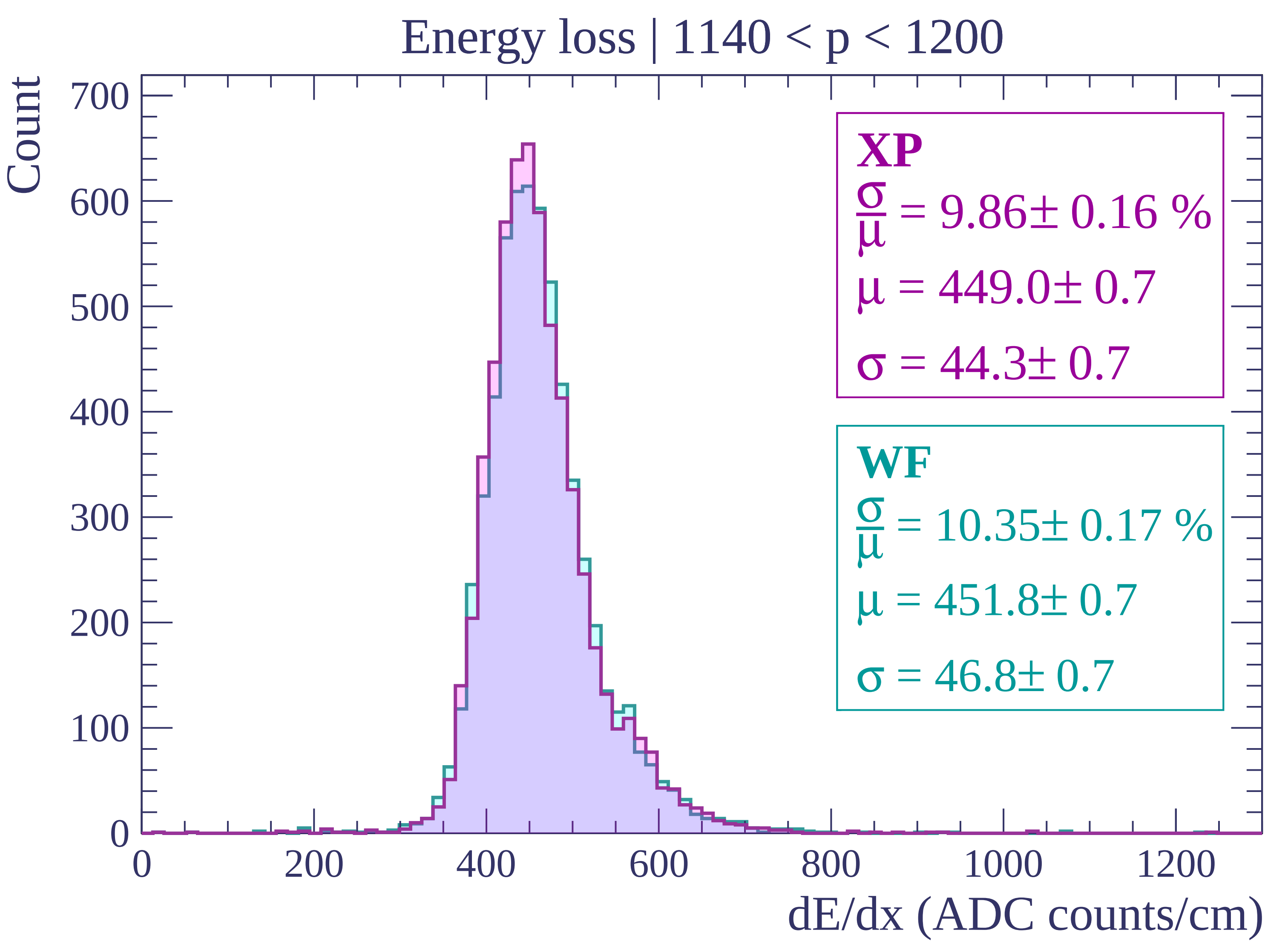
Energy loss | 900 < p < 960
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 9.50± 0.16 %
μ = 432.8± 0.6
σ = 41.1± 0.6
WF
σ/μ = 10.28± 0.18 %
μ = 435.8± 0.7
σ = 44.8± 0.7

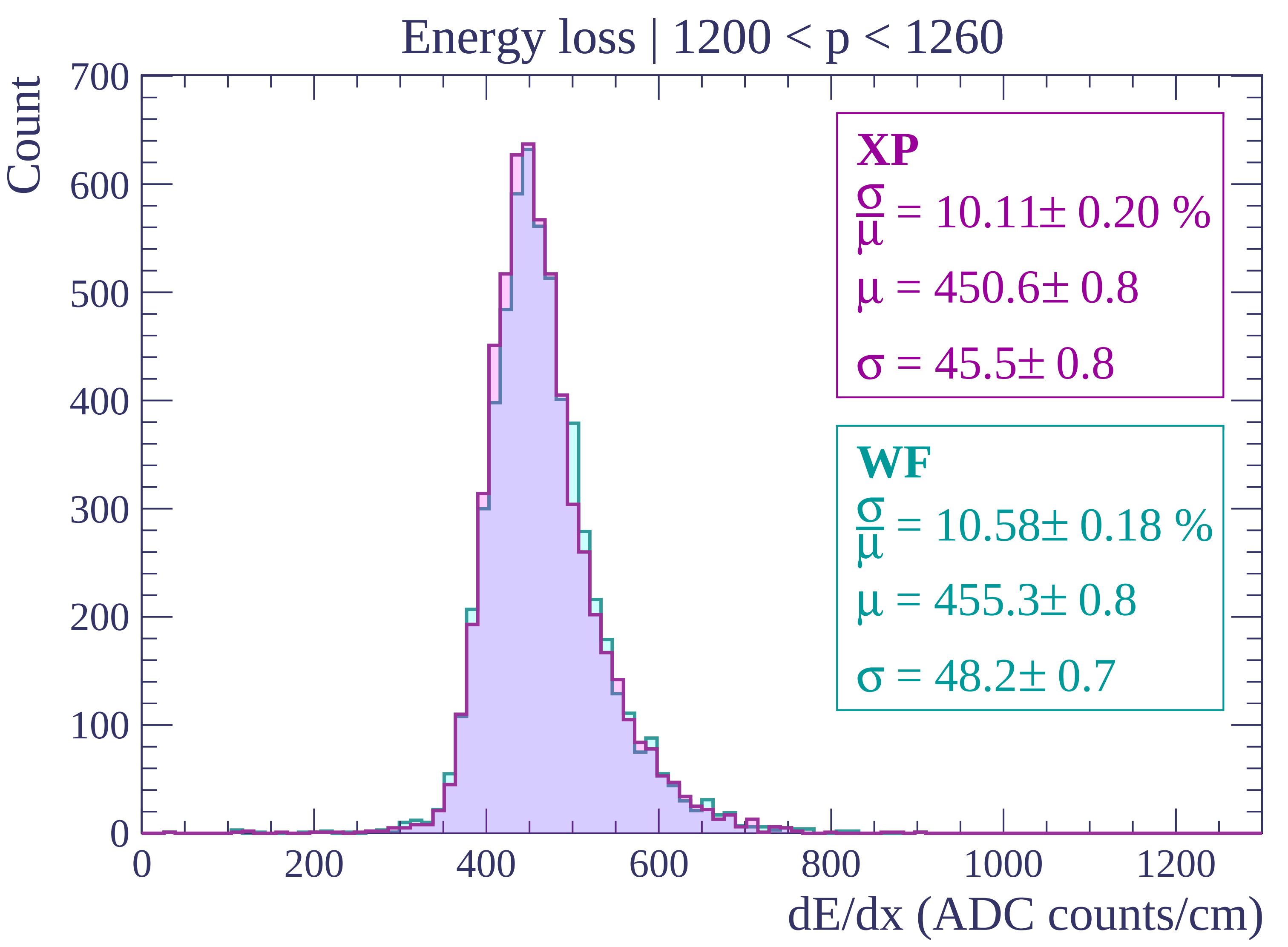


Energy loss | 960 < p < 1020
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.86± 0.17 %
μ = 438.6± 0.7
σ = 43.2± 0.7
WF
σ/μ = 10.28± 0.17 %
μ = 442.1± 0.7
σ = 45.5± 0.7

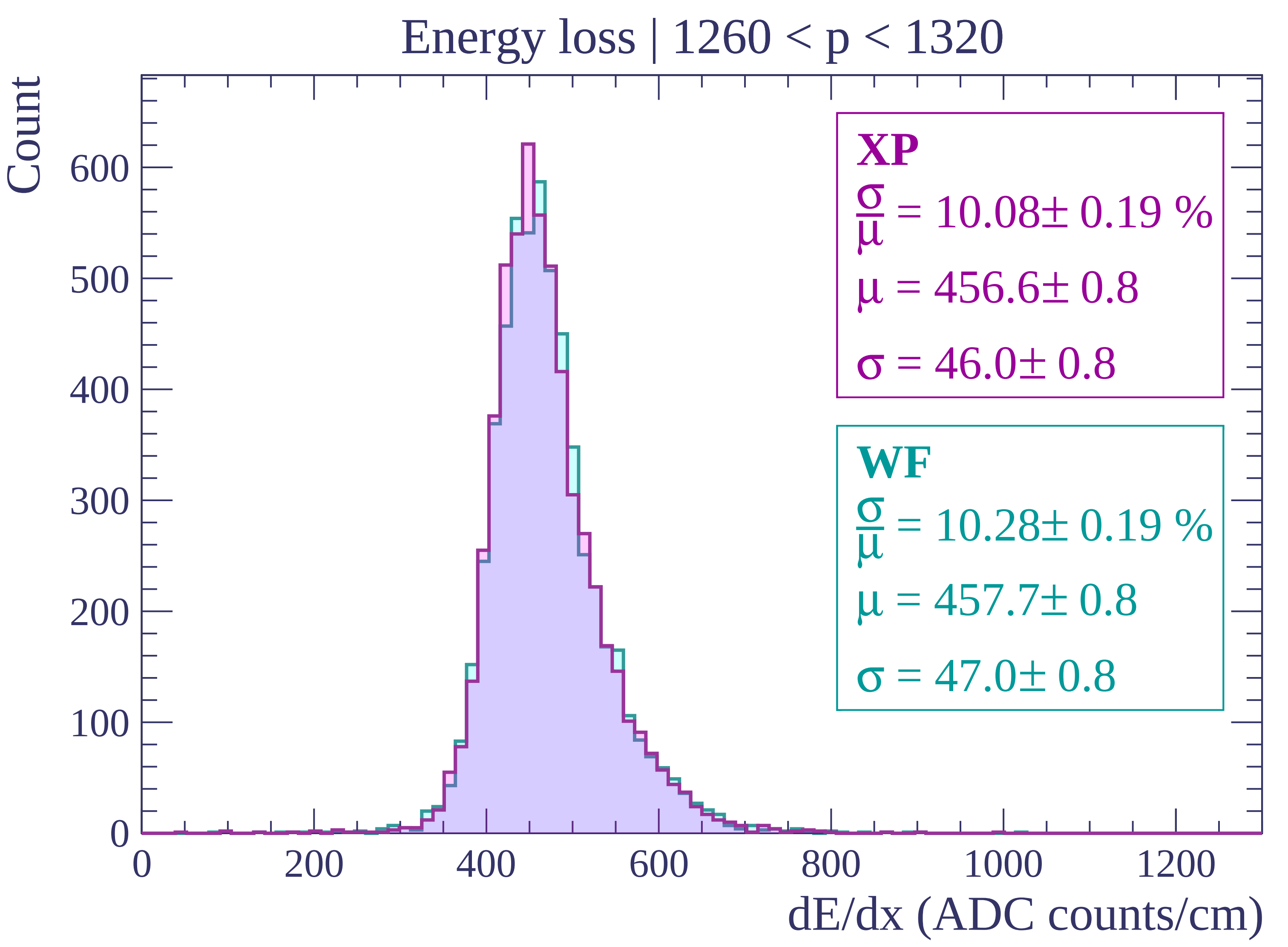

Energy loss | 1020 < p < 1080
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
700
XP
$\frac{\sigma}{\mu}$ = 9.48± 0.17 %
μ = 441.5± 0.7
σ = 41.9± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.87± 0.16 %
μ = 445.0± 0.7
σ = 43.9± 0.6

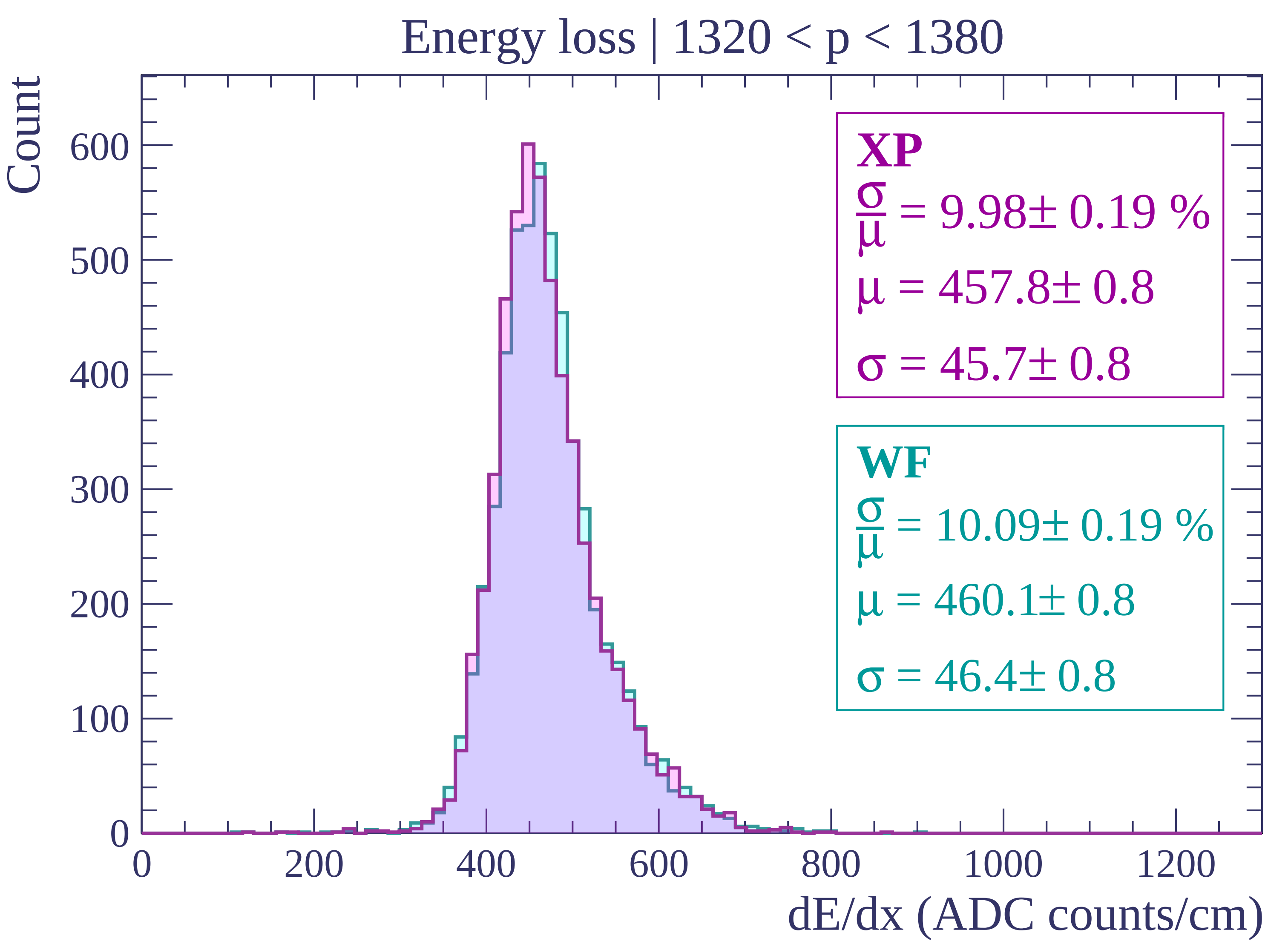

Energy loss | 1080 < p < 1140
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.99± 0.16 %
μ = 445.6± 0.7
σ = 44.5± 0.7
WF
σ/μ = 10.32± 0.18 %
μ = 447.6± 0.7
σ = 46.2± 0.7


Energy loss | 1140 < p < 1200
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.86± 0.16 %
μ = 449.0± 0.7
σ = 44.3± 0.7
WF
σ/μ = 10.35± 0.17 %
μ = 451.8± 0.7
σ = 46.8± 0.7

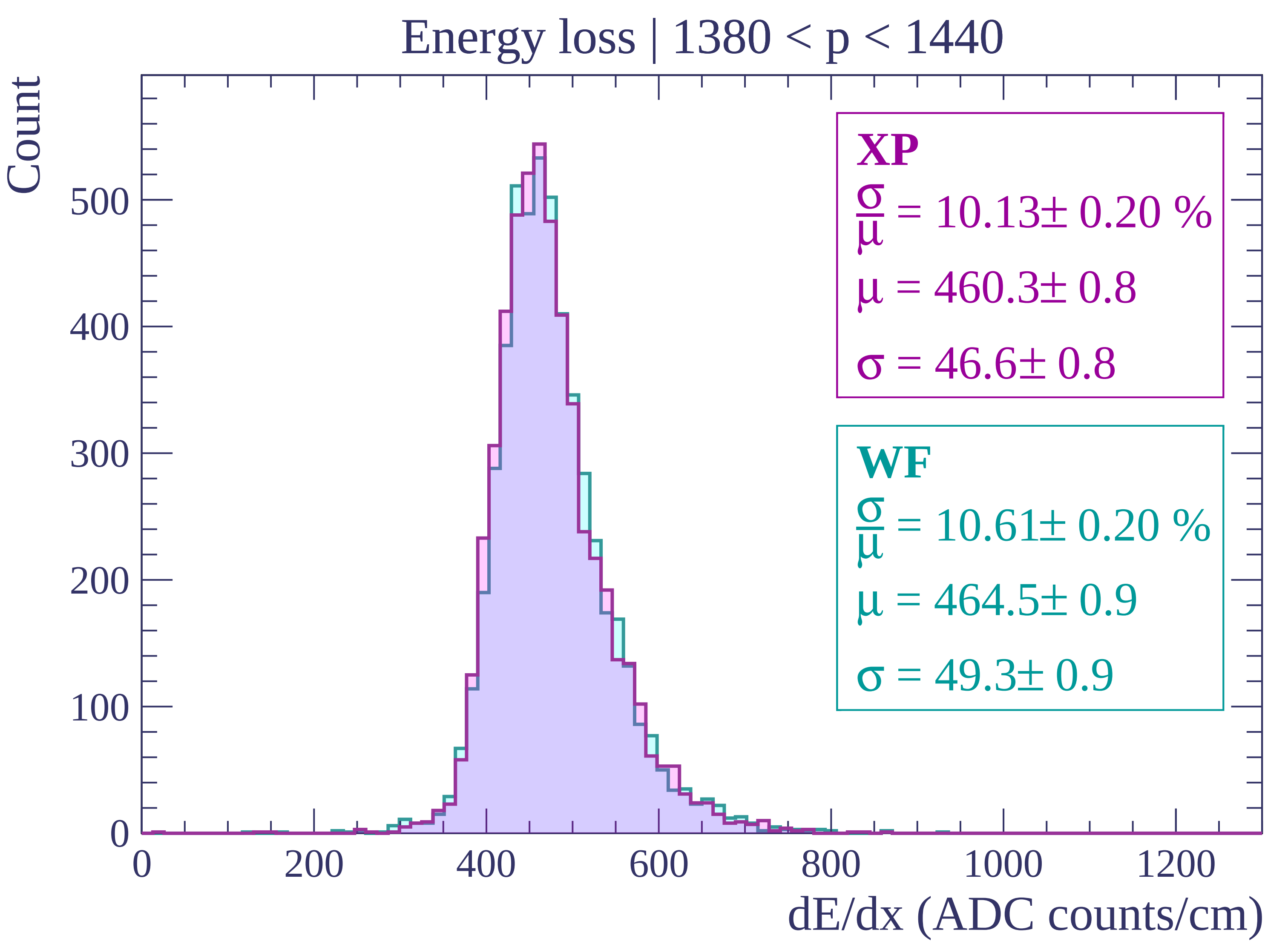


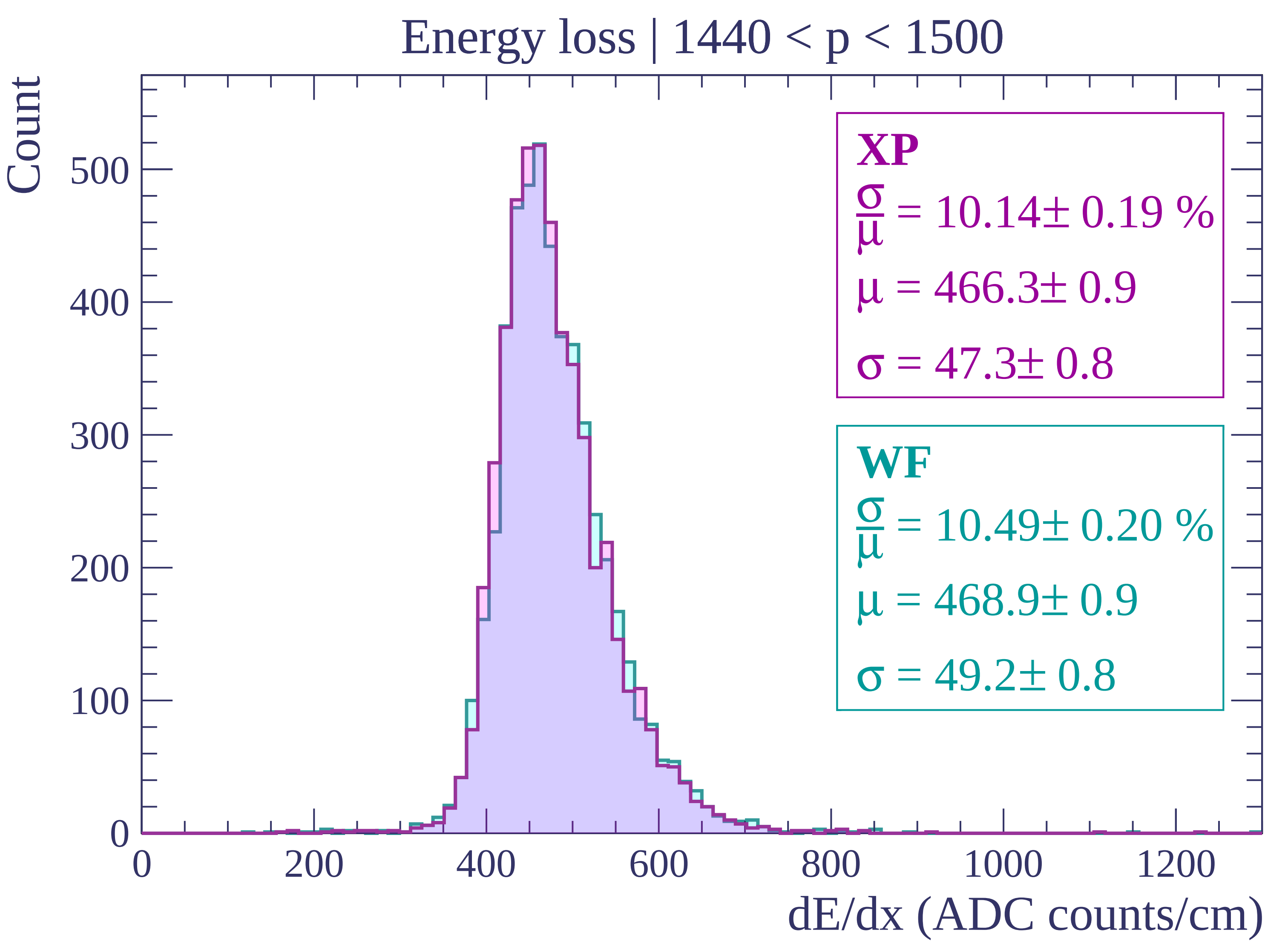
Energy loss | 1200 < p < 1260
Count
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.11± 0.20 %
μ = 450.6± 0.8
σ = 45.5± 0.8
WF
σ/μ = 10.58± 0.18 %
μ = 455.3± 0.8
σ = 48.2± 0.7


Energy loss | 1260 < p < 1320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.08 ± 0.19 %
μ = 456.6 ± 0.8
σ = 46.0 ± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.28 ± 0.19 %
μ = 457.7 ± 0.8
σ = 47.0 ± 0.8

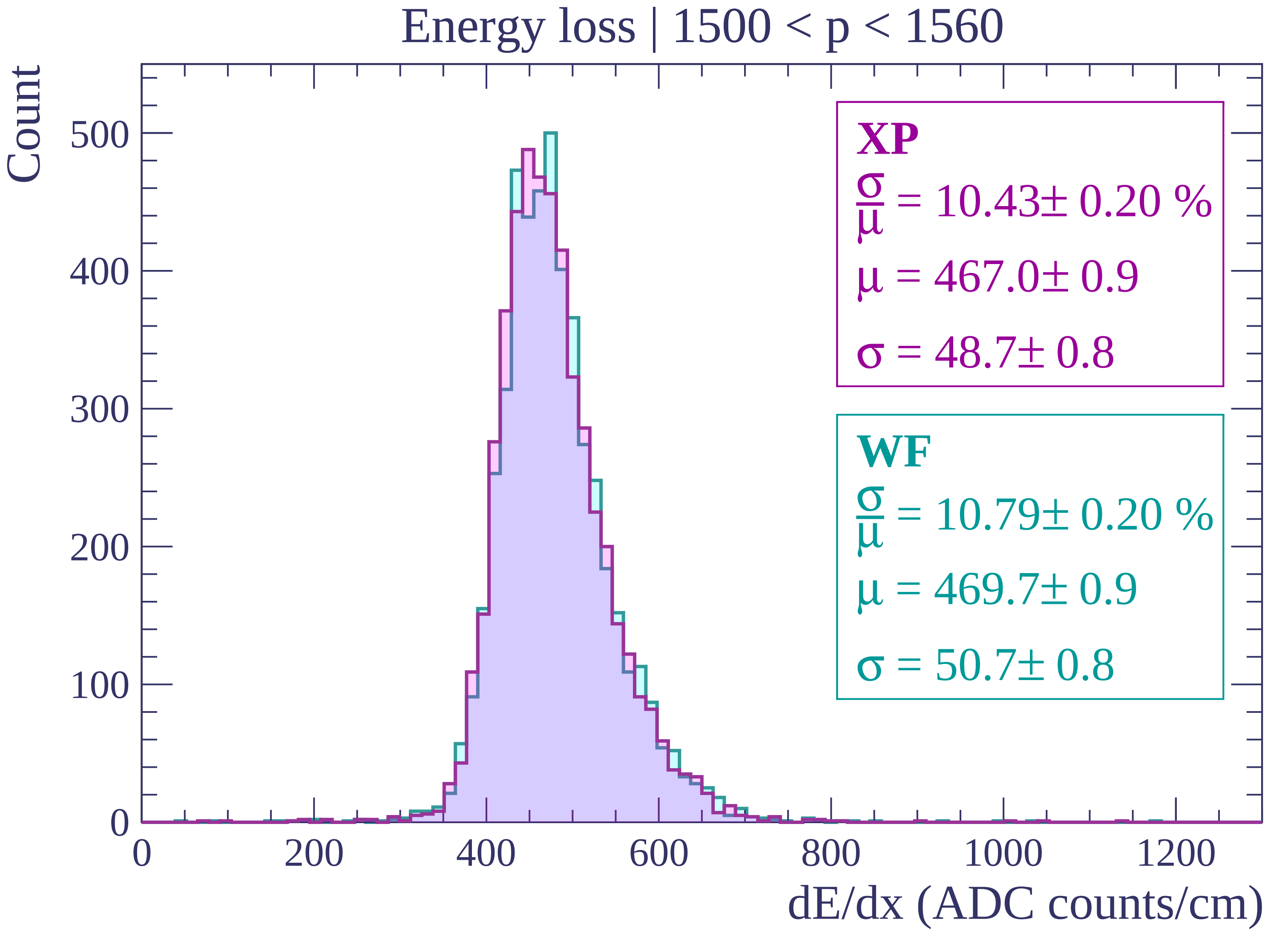


Energy loss | 1320 < p < 1380
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.98± 0.19 %
μ = 457.8± 0.8
σ = 45.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.09± 0.19 %
μ = 460.1± 0.8
σ = 46.4± 0.8

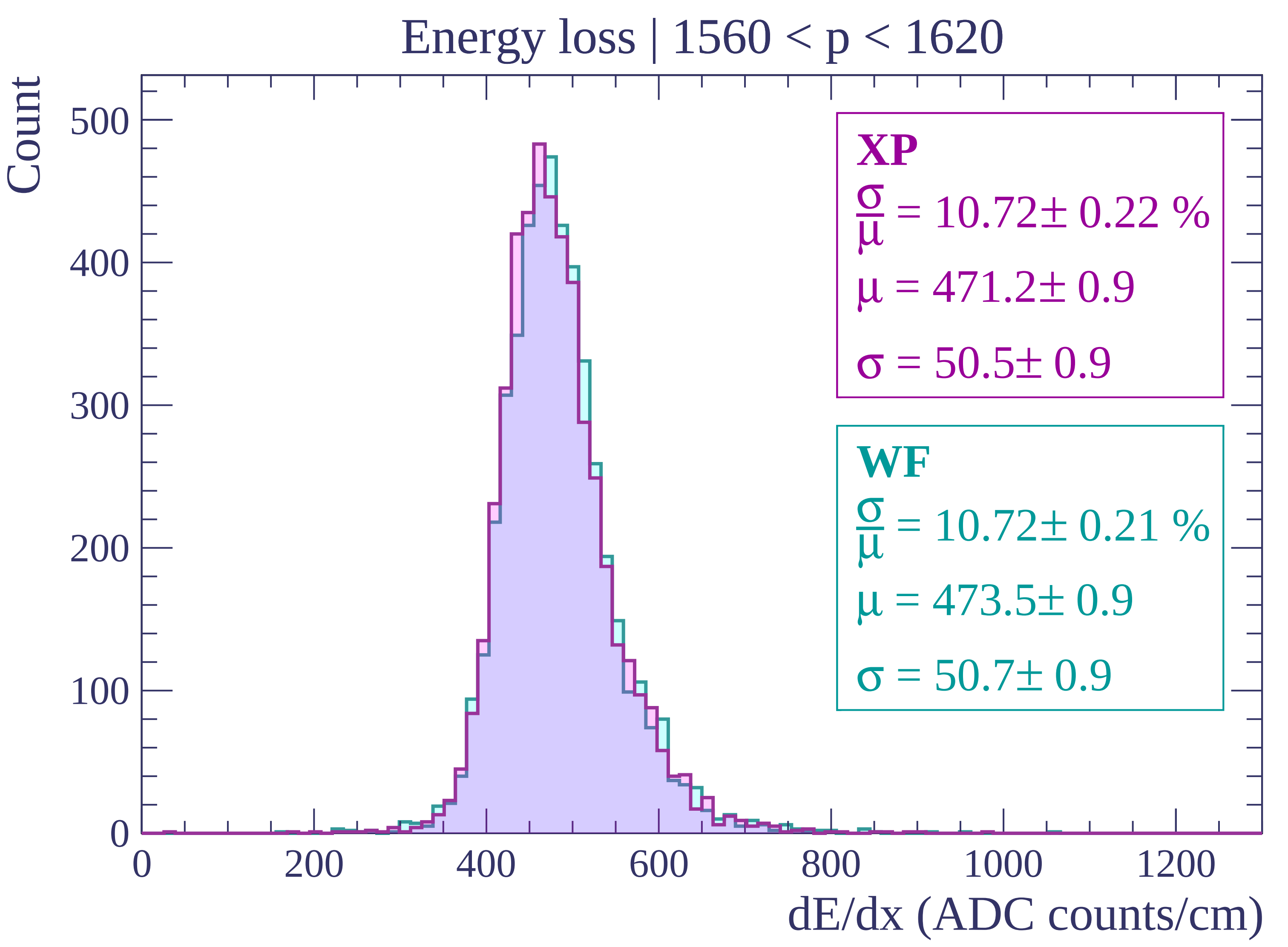

Energy loss | 1380 < p < 1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.13± 0.20 %
μ = 460.3± 0.8
σ = 46.6± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.61± 0.20 %
μ = 464.5± 0.9
σ = 49.3± 0.9



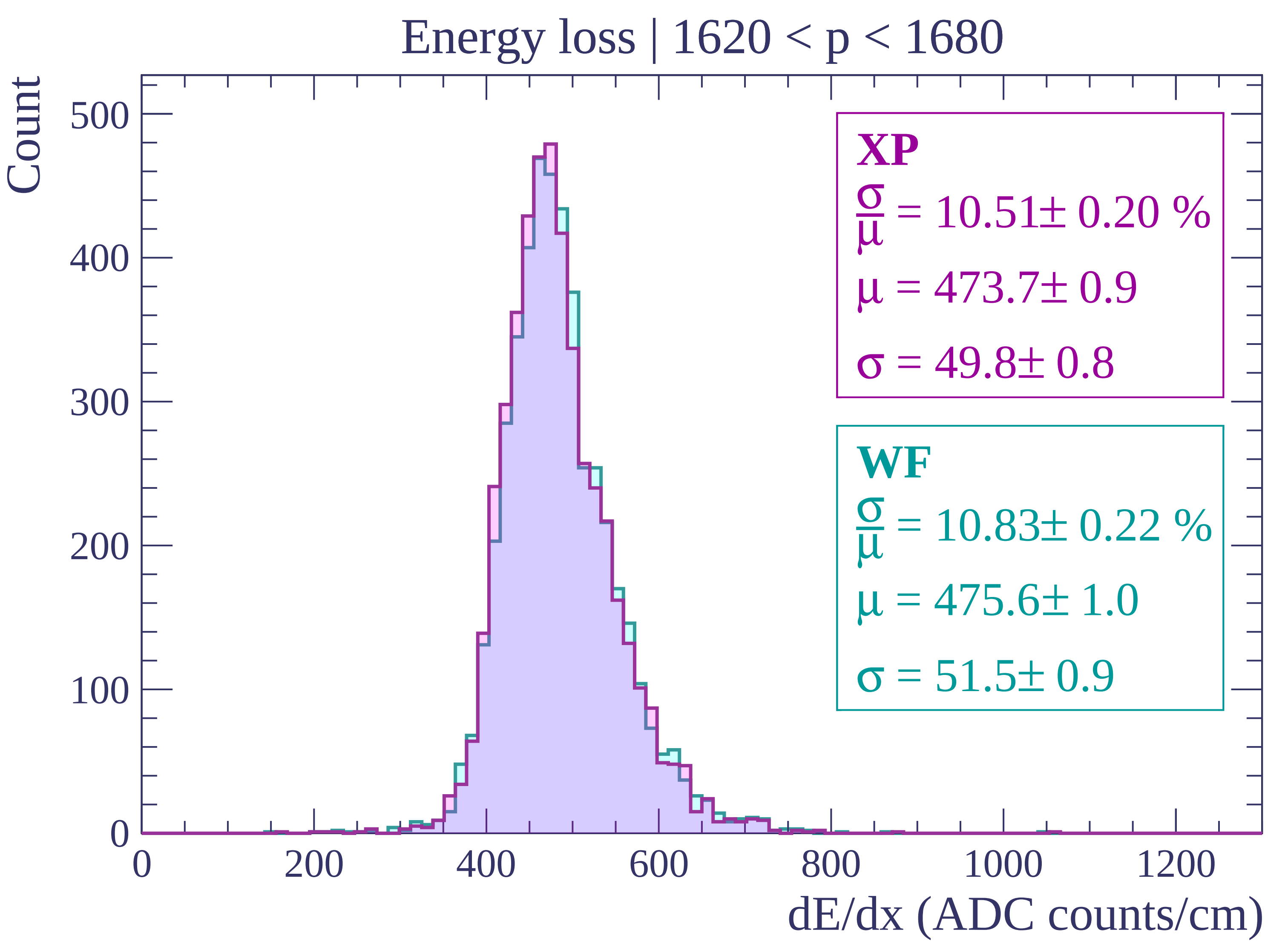
Energy loss | 1440 < p < 1500
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.14± 0.19 %
μ = 466.3± 0.9
σ = 47.3± 0.8
WF
σ/μ = 10.49± 0.20 %
μ = 468.9± 0.9
σ = 49.2± 0.8



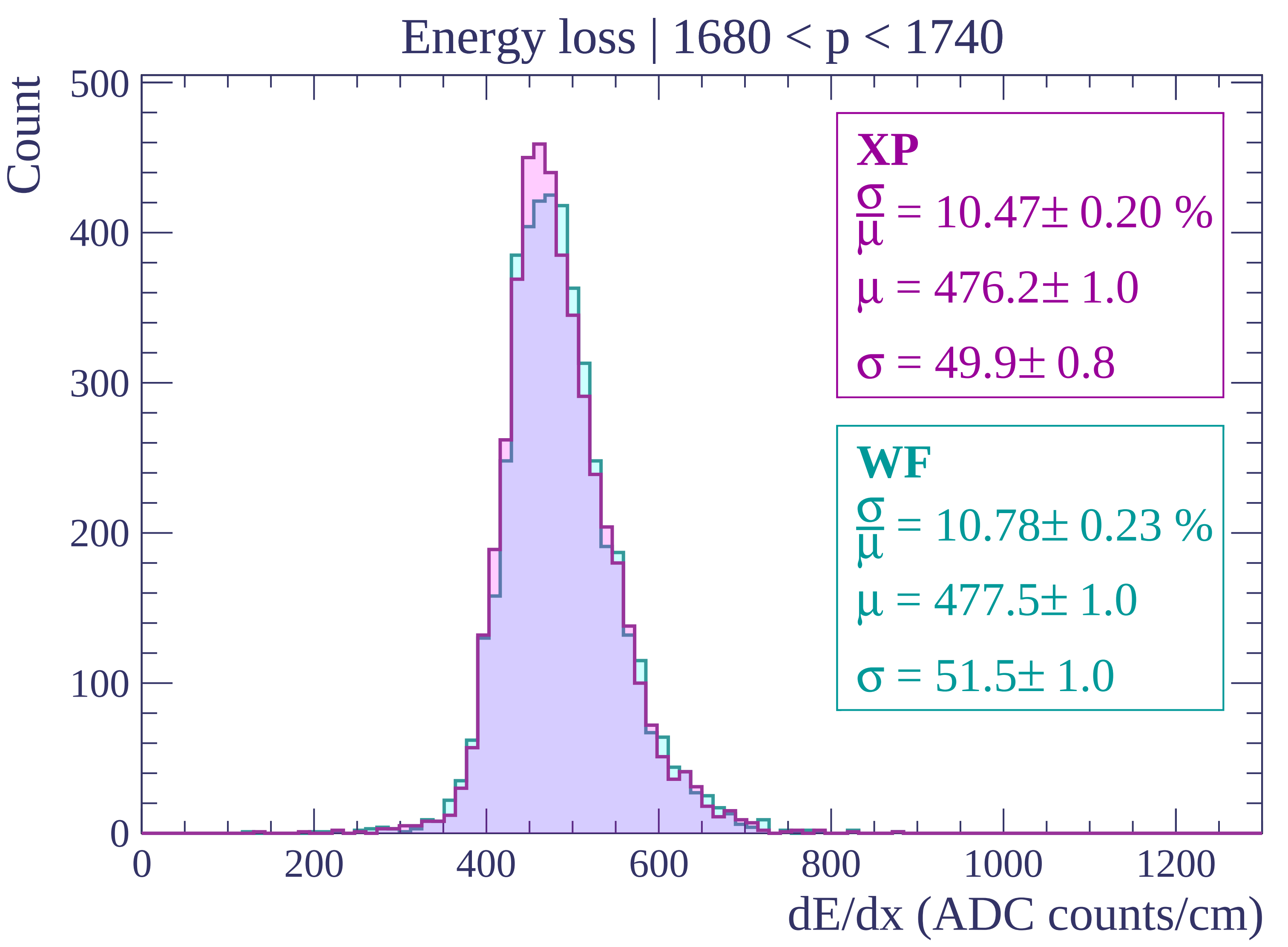
Energy loss | 1500 < p < 1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.43± 0.20 %
μ = 467.0± 0.9
σ = 48.7± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.79± 0.20 %
μ = 469.7± 0.9
σ = 50.7± 0.8

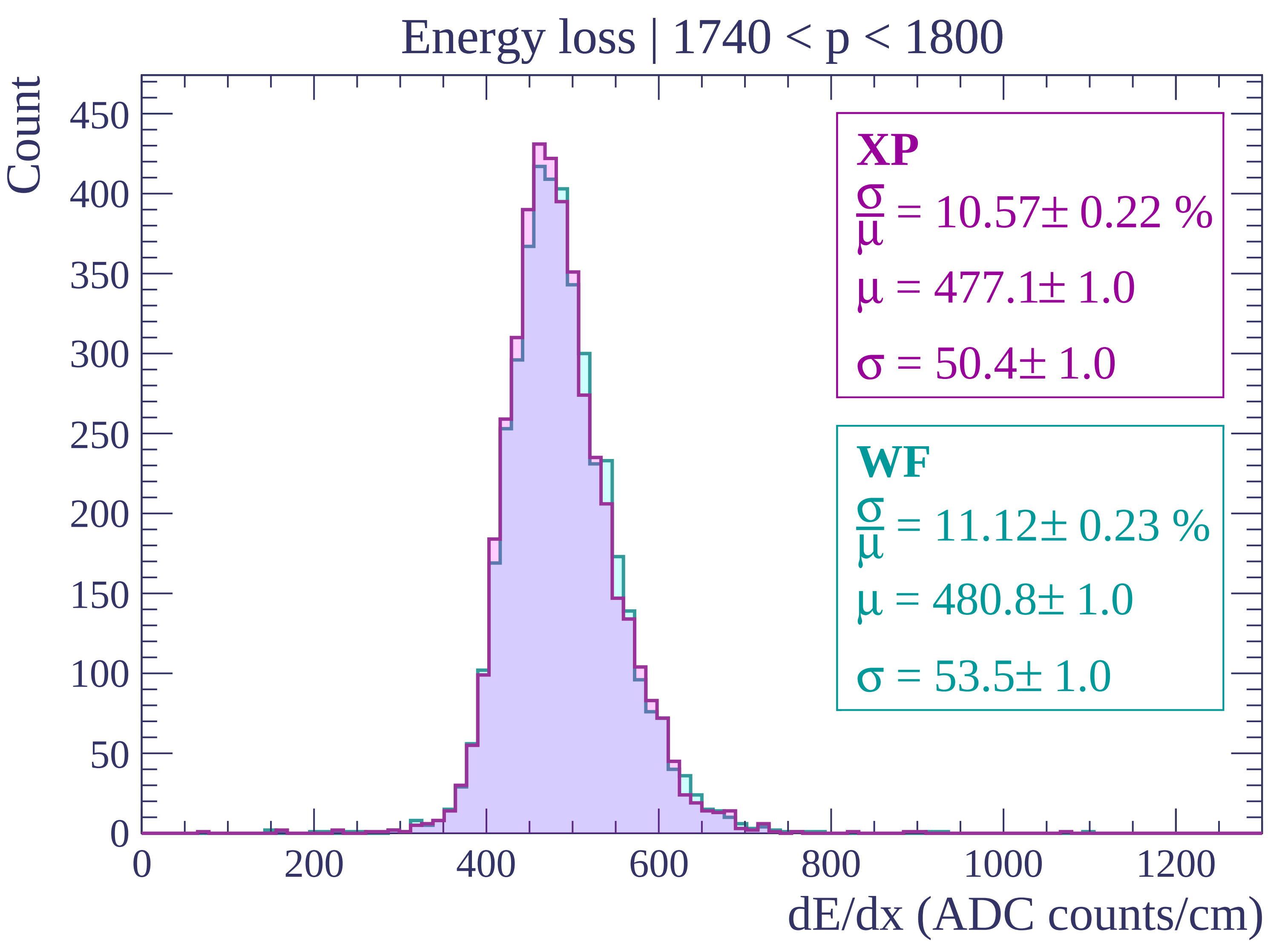


Energy loss | 1560 < p < 1620
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.72± 0.22 %
μ = 471.2± 0.9
σ = 50.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.72± 0.21 %
μ = 473.5± 0.9
σ = 50.7± 0.9

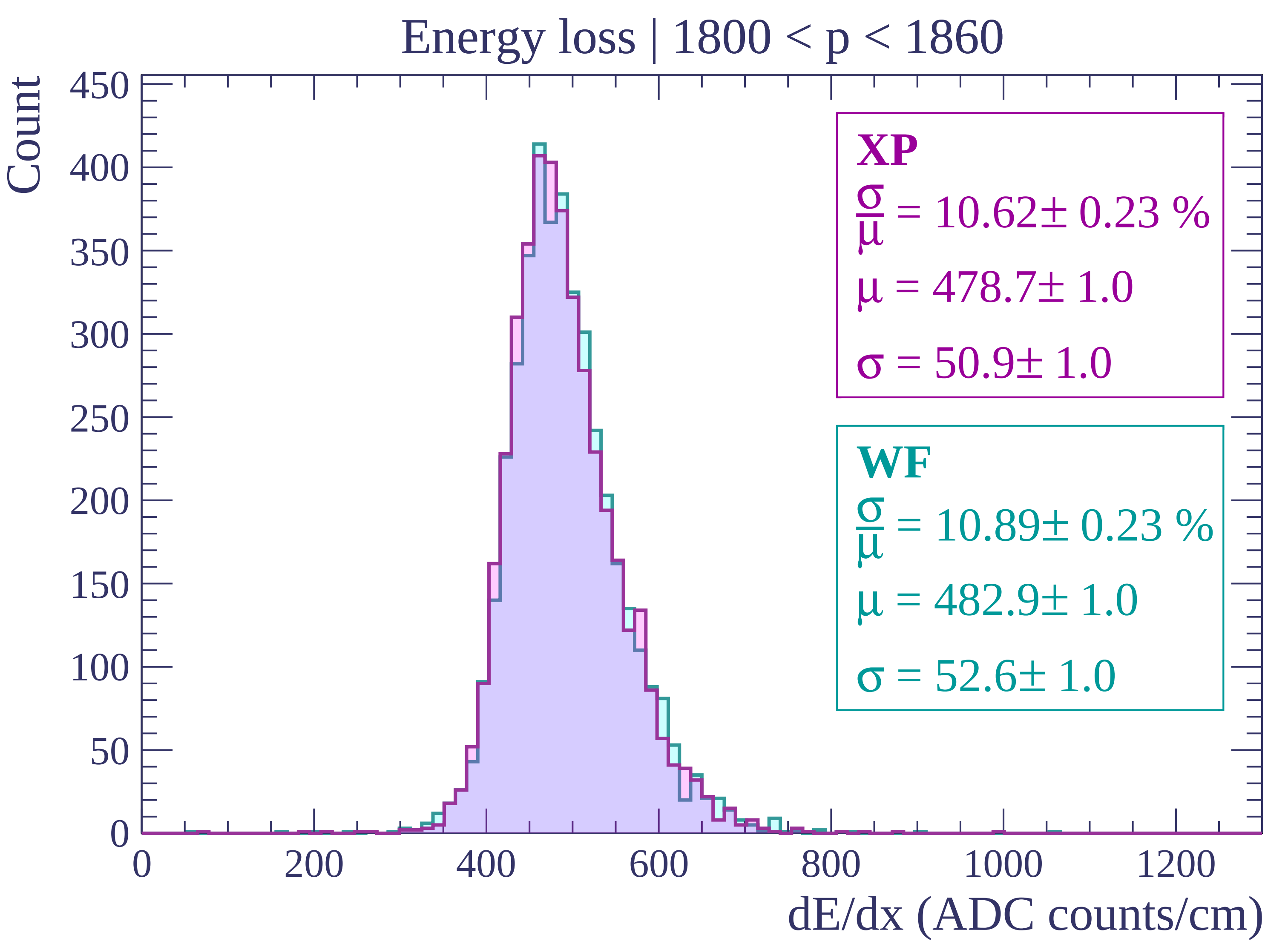


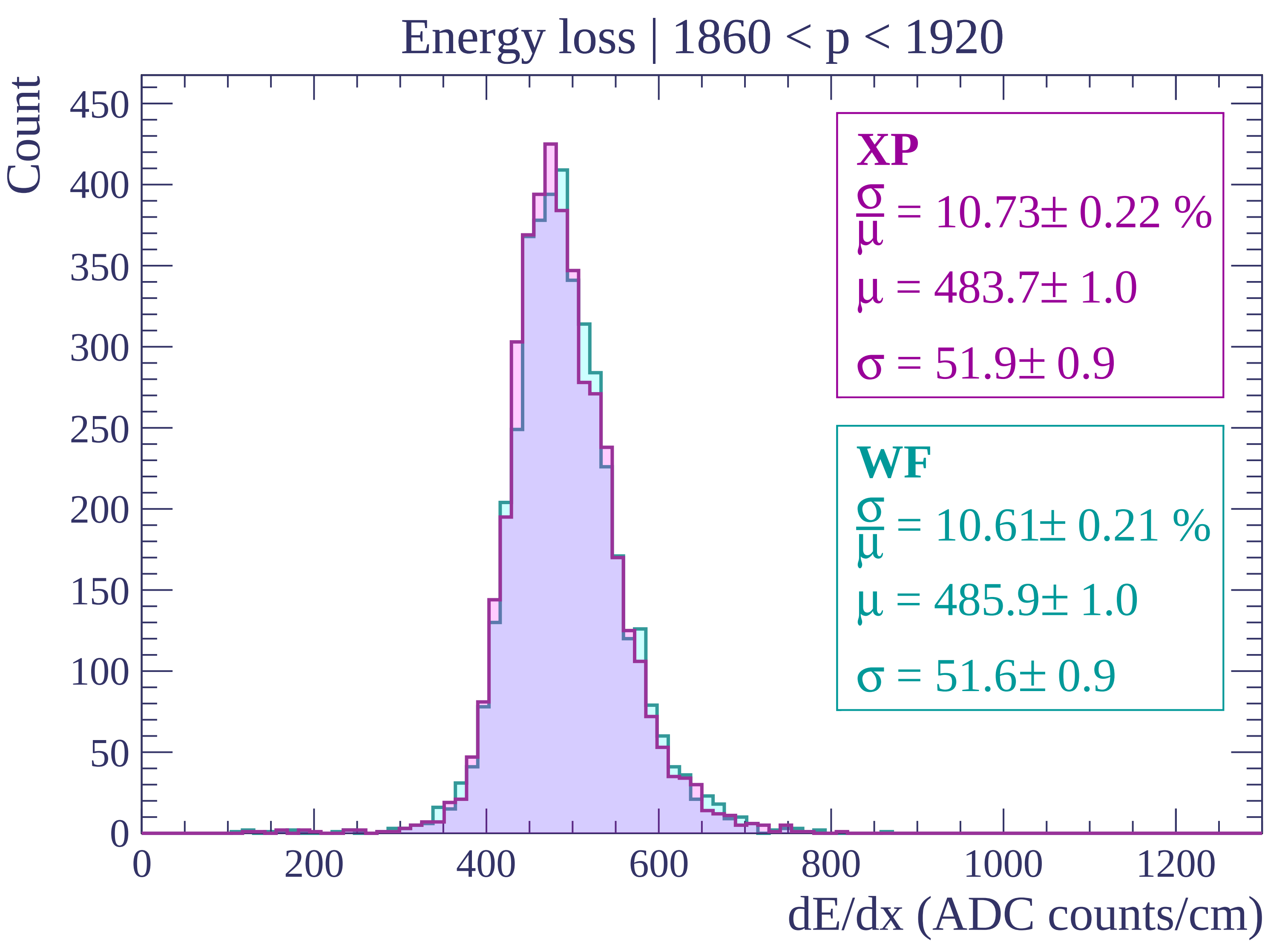
Energy loss | 1620 < p < 1680
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.51± 0.20 %
μ = 473.7± 0.9
σ = 49.8± 0.8
WF
σ/μ = 10.83± 0.22 %
μ = 475.6± 1.0
σ = 51.5± 0.9
0
100
200
300
400
500
0
200
400
600
800
1000
1200


Energy loss | 1680 < p < 1740
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 10.47± 0.20 %
μ = 476.2± 1.0
σ = 49.9± 0.8
WF
σ/μ = 10.78± 0.23 %
μ = 477.5± 1.0
σ = 51.5± 1.0

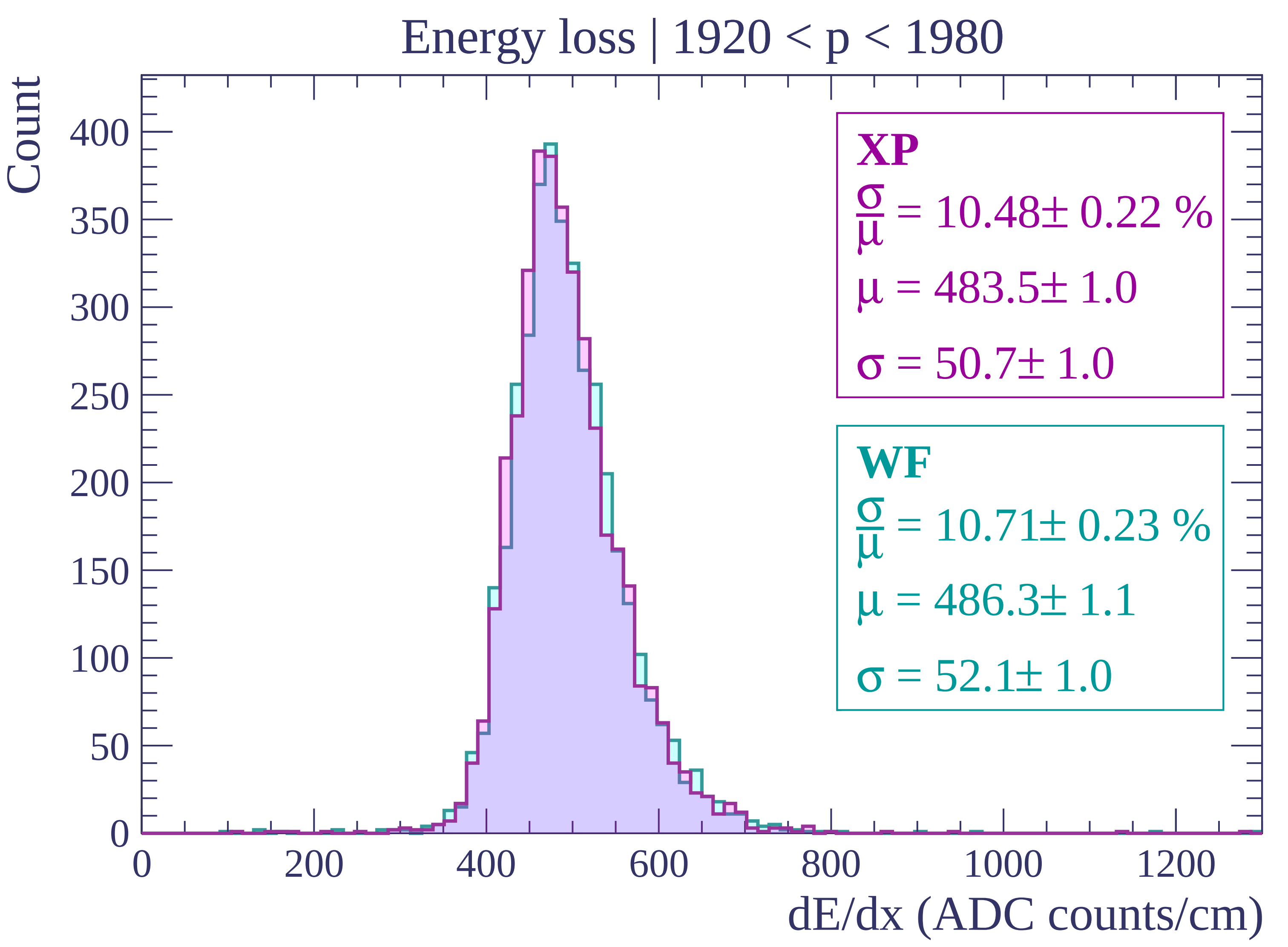


Energy loss | 1740 < p < 1800
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.57± 0.22 %
μ = 477.1± 1.0
σ = 50.4± 1.0
WF
$\frac{\sigma}{\mu}$ = 11.12± 0.23 %
μ = 480.8± 1.0
σ = 53.5± 1.0

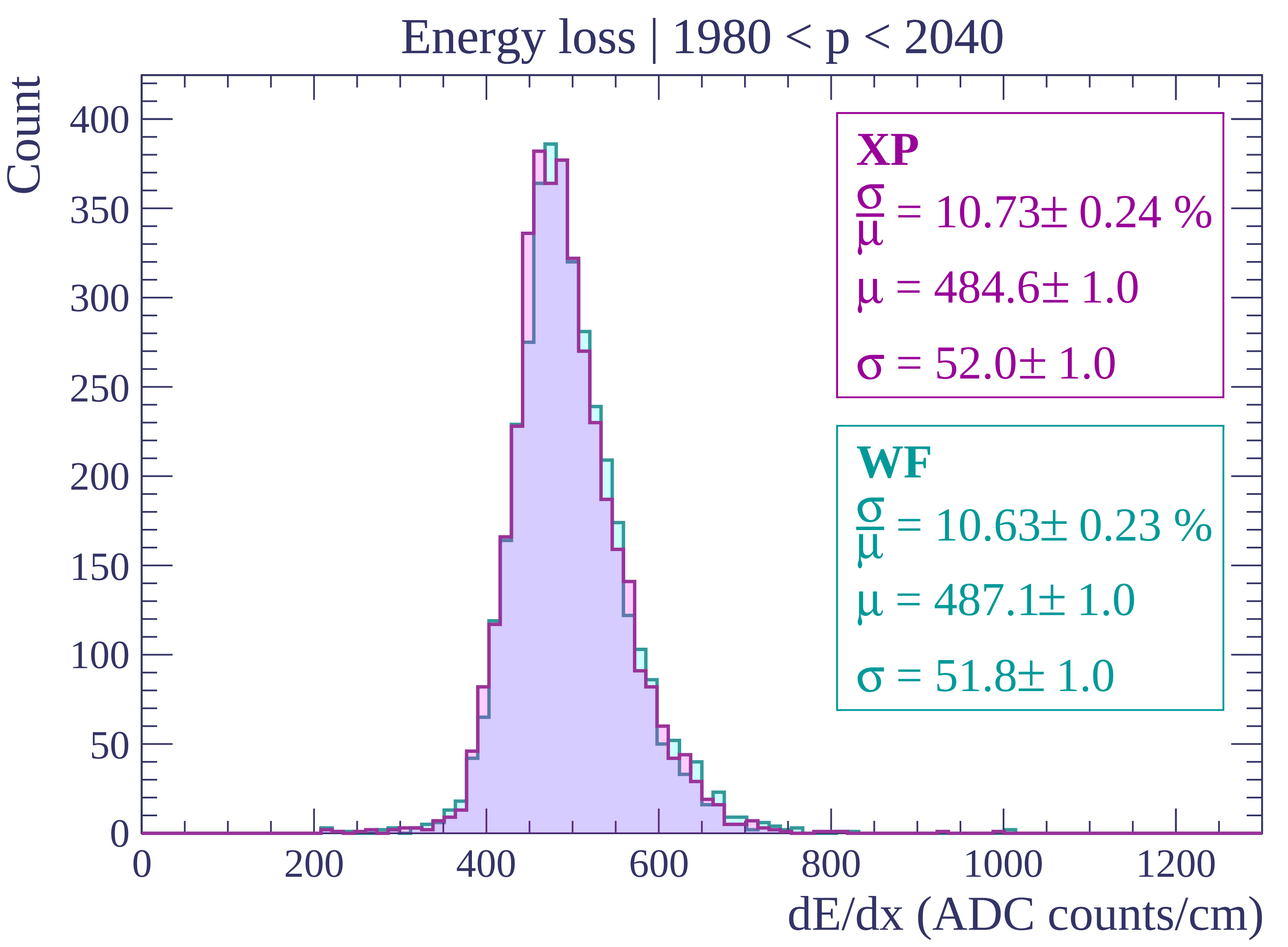

Energy loss | 1800 < p < 1860
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
450
XP
σ/μ = 10.62± 0.23 %
μ = 478.7± 1.0
σ = 50.9± 1.0
WF
σ/μ = 10.89± 0.23 %
μ = 482.9± 1.0
σ = 52.6± 1.0



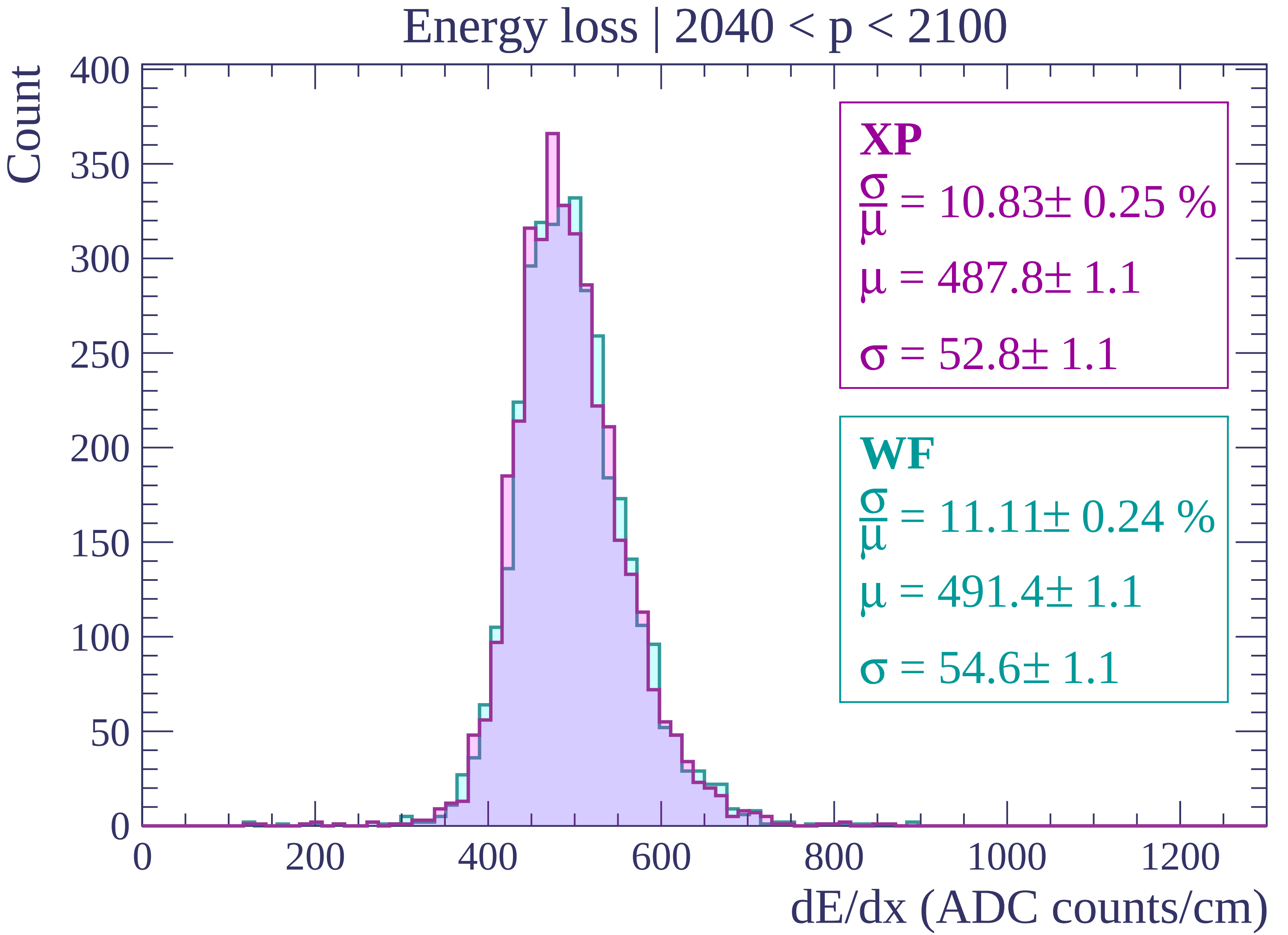
Energy loss | 1860 < p < 1920
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.73± 0.22 %
μ = 483.7± 1.0
σ = 51.9± 0.9
WF
$\frac{\sigma}{\mu}$ = 10.61± 0.21 %
μ = 485.9± 1.0
σ = 51.6± 0.9



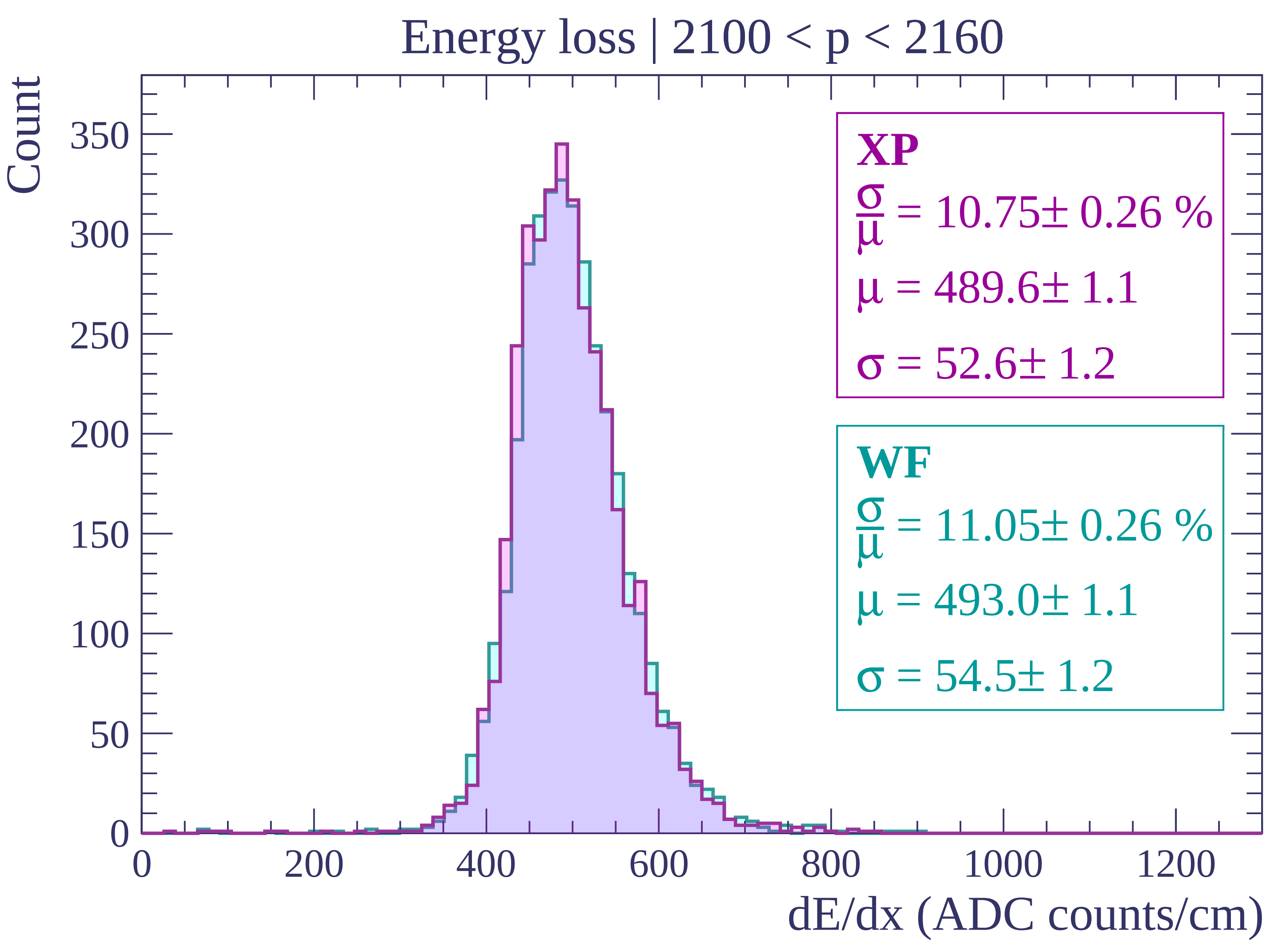
Energy loss | 1920 < p < 1980
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
350
400
XP
σ/μ = 10.48± 0.22 %
μ = 483.5± 1.0
σ = 50.7± 1.0
WF
σ/μ = 10.71± 0.23 %
μ = 486.3± 1.1
σ = 52.1± 1.0

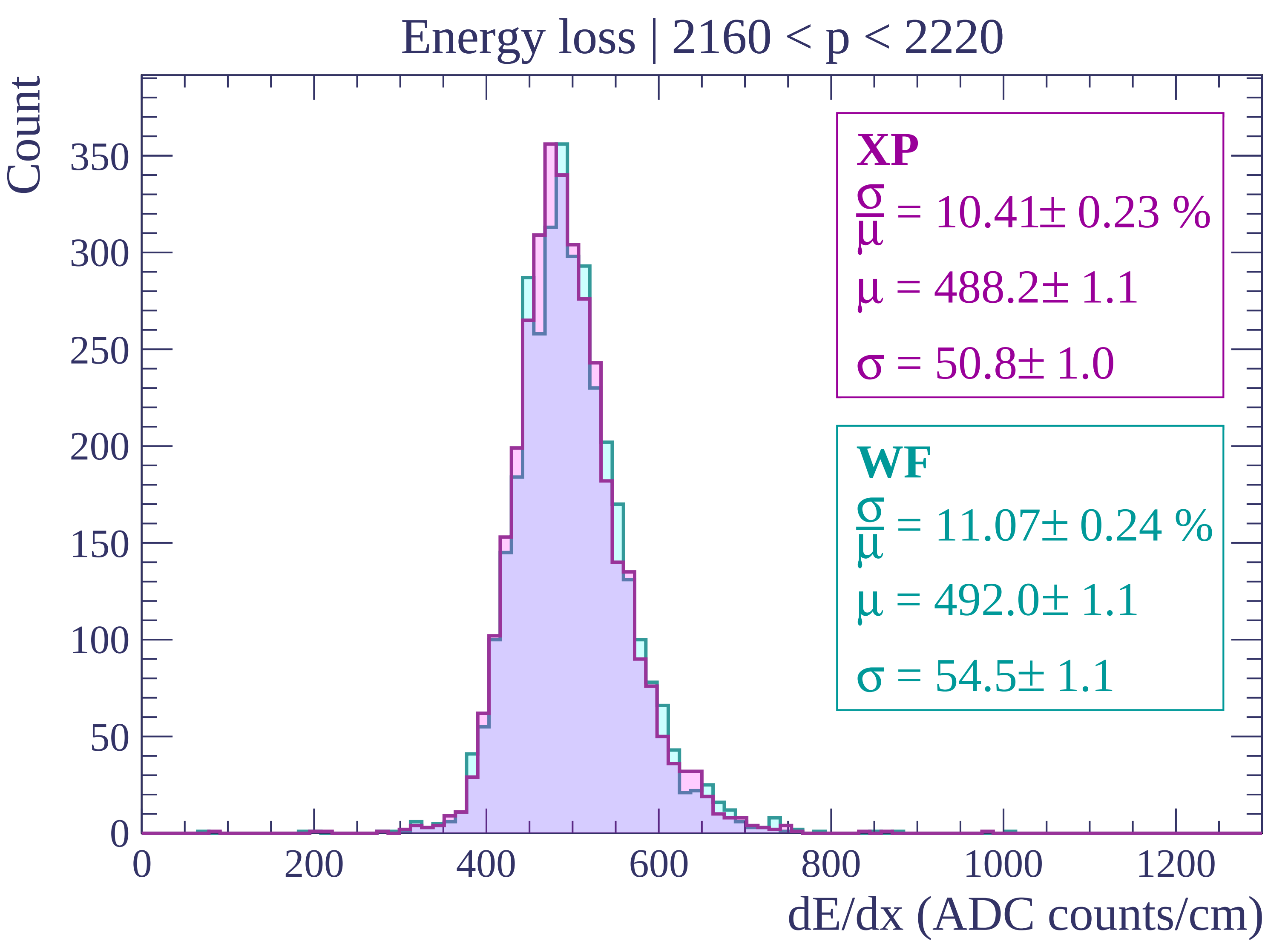


Energy loss | 1980 < p < 2040
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 10.73± 0.24 %
μ = 484.6± 1.0
σ = 52.0± 1.0
WF
σ/μ = 10.63± 0.23 %
μ = 487.1± 1.0
σ = 51.8± 1.0

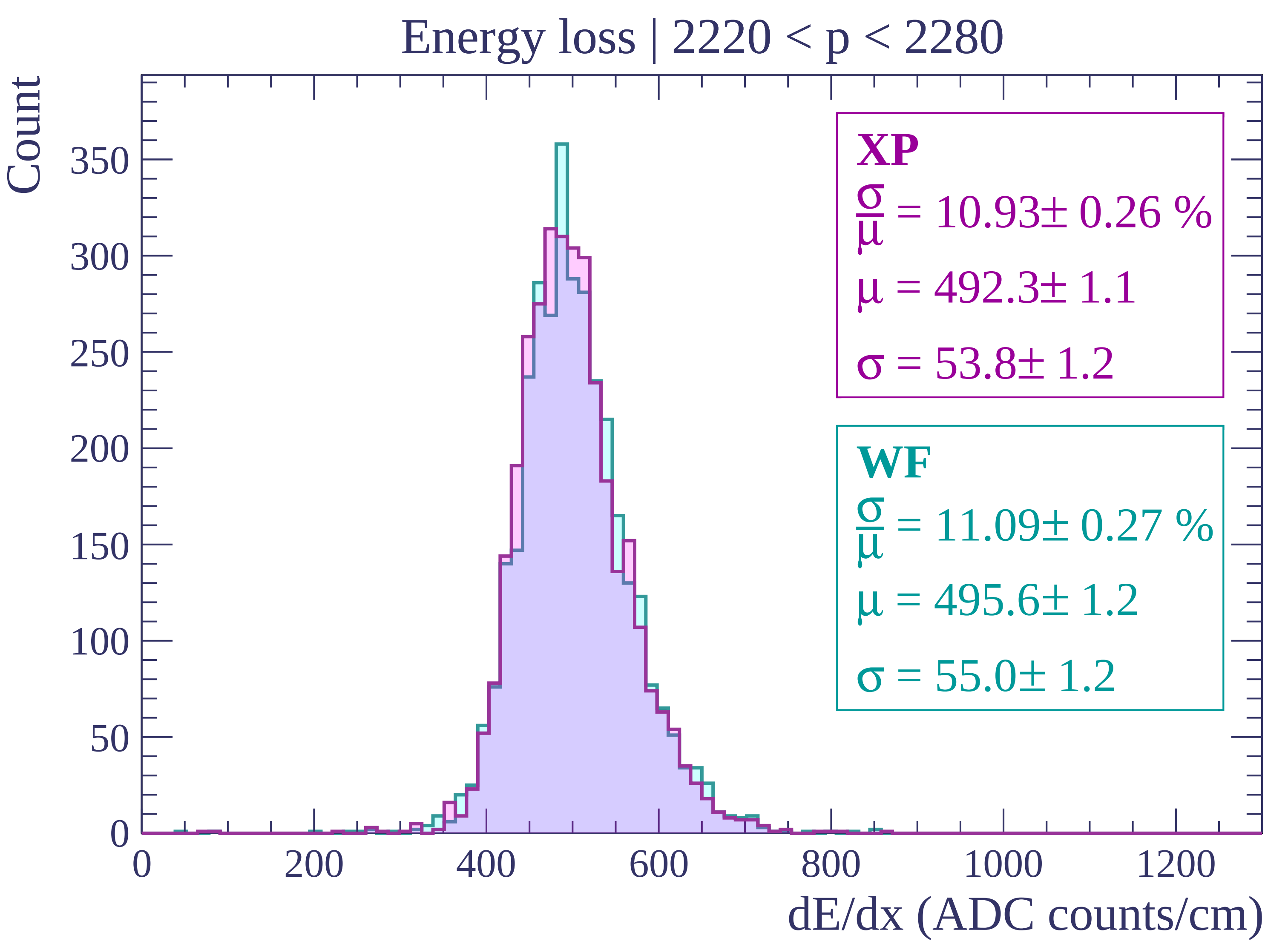


Energy loss | 2040 < p < 2100
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
σ/μ = 10.83± 0.25 %
μ = 487.8± 1.1
σ = 52.8± 1.1
WF
σ/μ = 11.11± 0.24 %
μ = 491.4± 1.1
σ = 54.6± 1.1

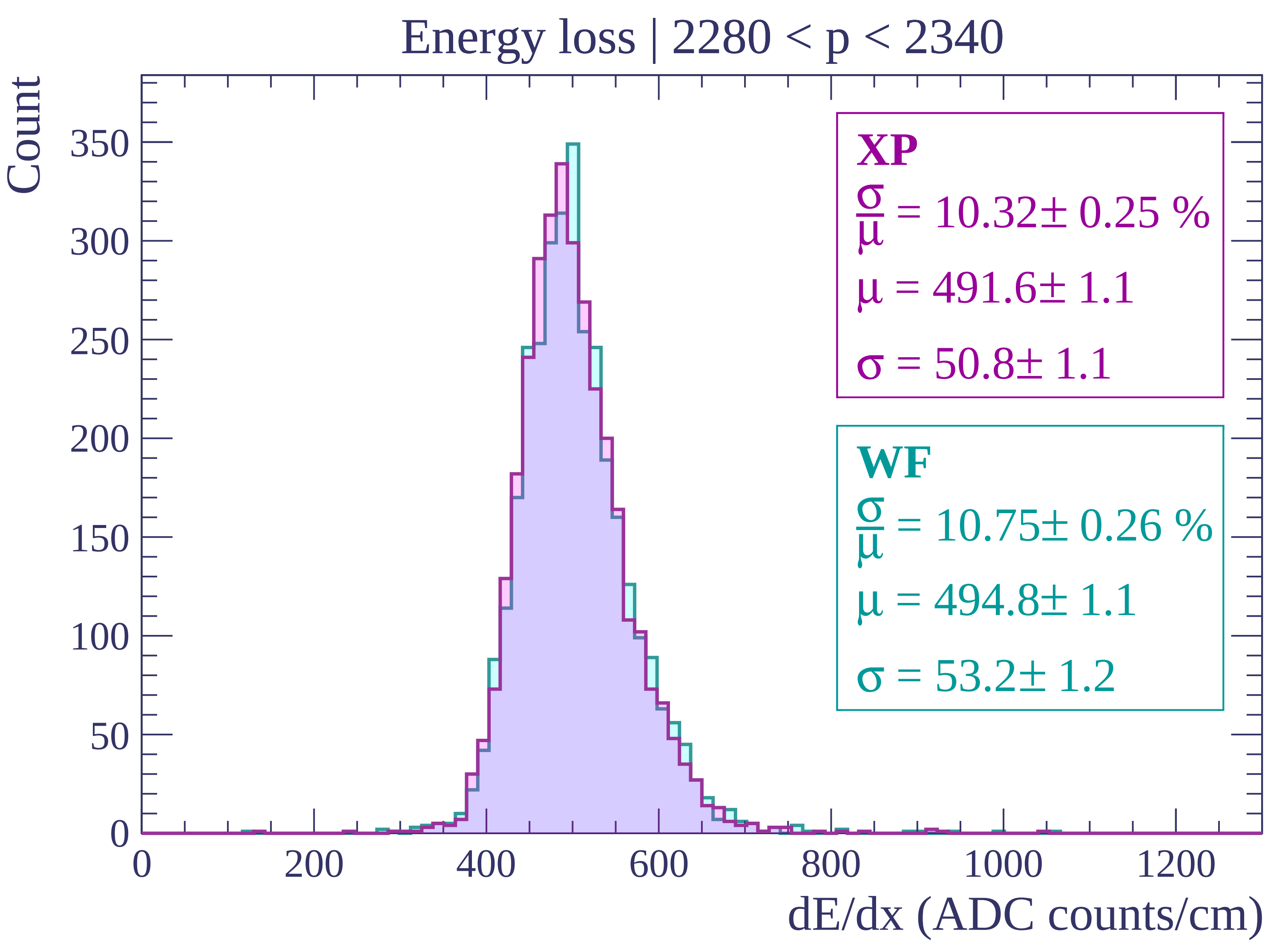

Energy loss | 2100 < p < 2160
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.75± 0.26 %
μ = 489.6± 1.1
σ = 52.6± 1.2
WF
$\frac{\sigma}{\mu}$ = 11.05± 0.26 %
μ = 493.0± 1.1
σ = 54.5± 1.2

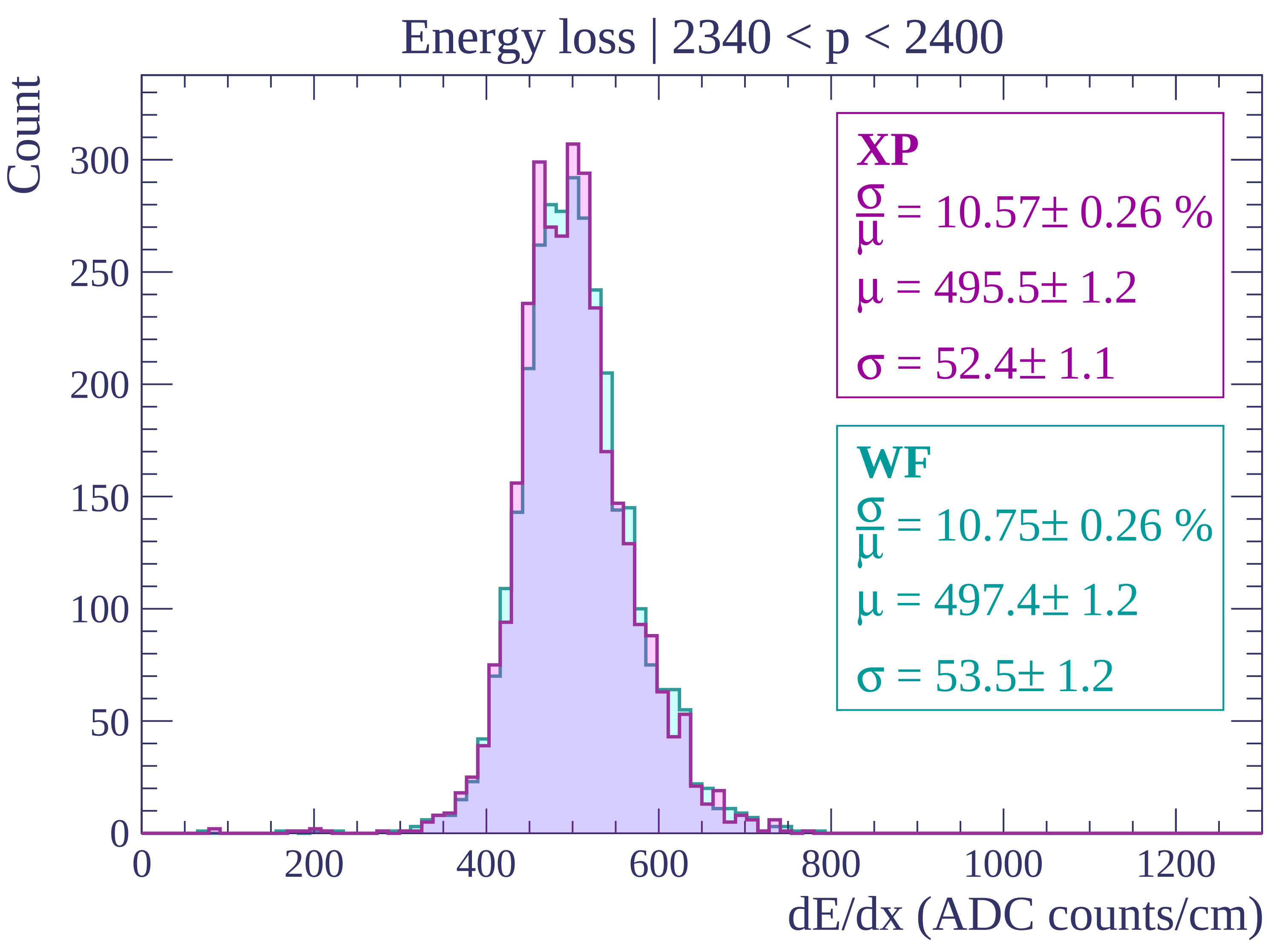


Energy loss | 2160 < p < 2220
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.41± 0.23 %
μ = 488.2± 1.1
σ = 50.8± 1.0
WF
$\frac{\sigma}{\mu}$ = 11.07± 0.24 %
μ = 492.0± 1.1
σ = 54.5± 1.1

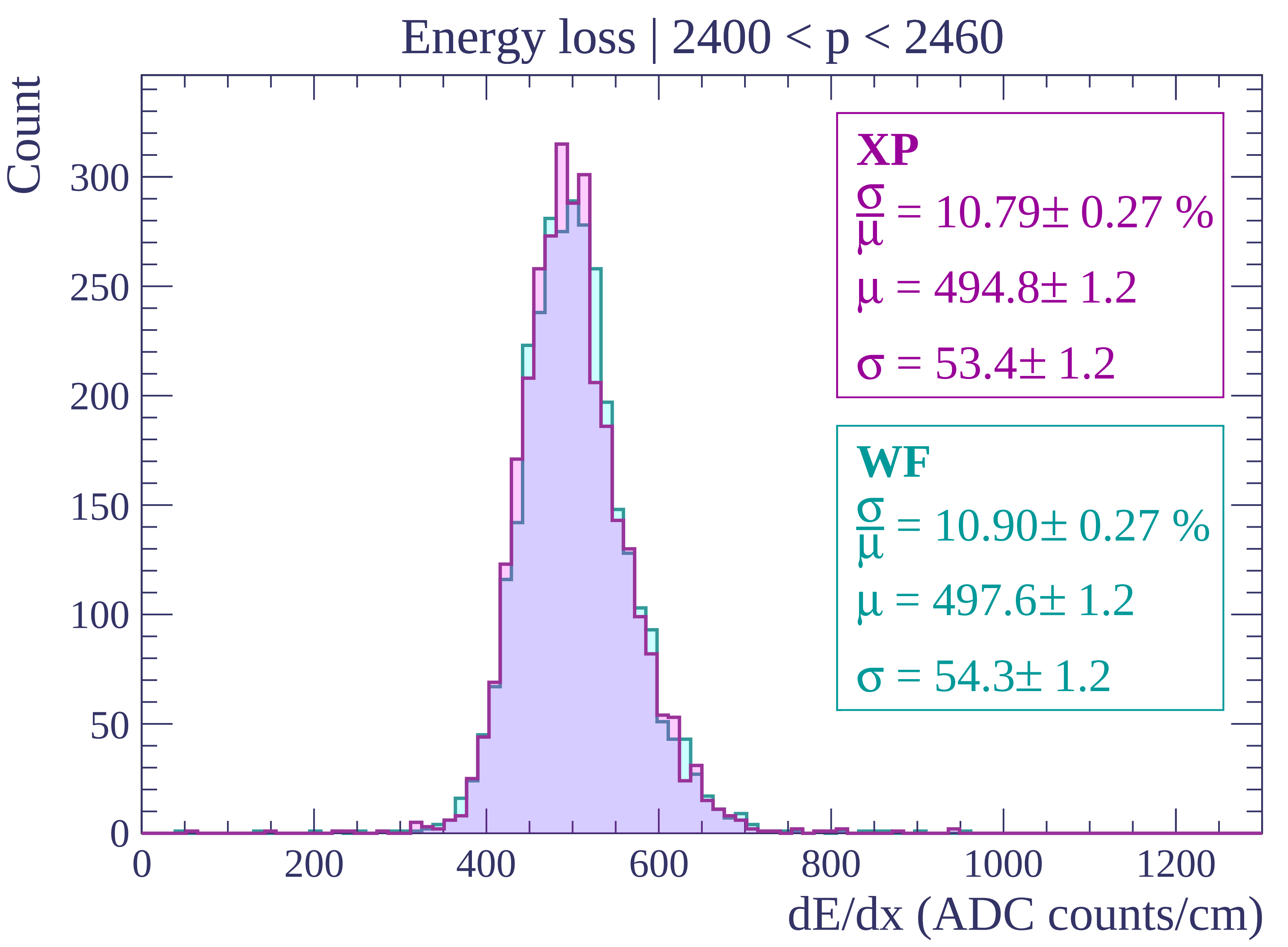


Energy loss | 2220 < p < 2280
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.93± 0.26 %
μ = 492.3± 1.1
σ = 53.8± 1.2
WF
$\frac{\sigma}{\mu}$ = 11.09± 0.27 %
μ = 495.6± 1.2
σ = 55.0± 1.2

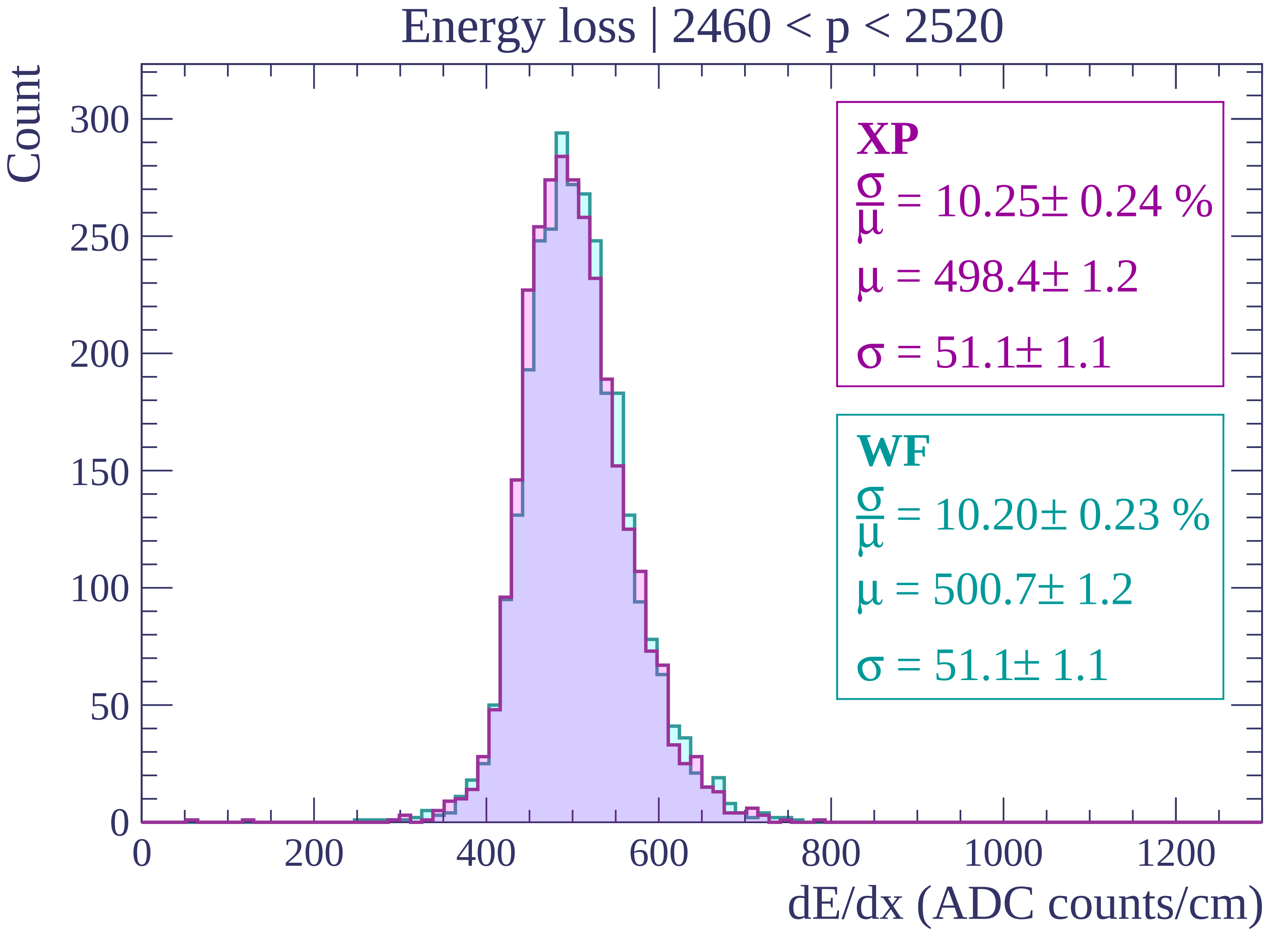

Energy loss | 2280 < p < 2340
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.32± 0.25 %
μ = 491.6± 1.1
σ = 50.8± 1.1
WF
$\frac{\sigma}{\mu}$ = 10.75± 0.26 %
μ = 494.8± 1.1
σ = 53.2± 1.2



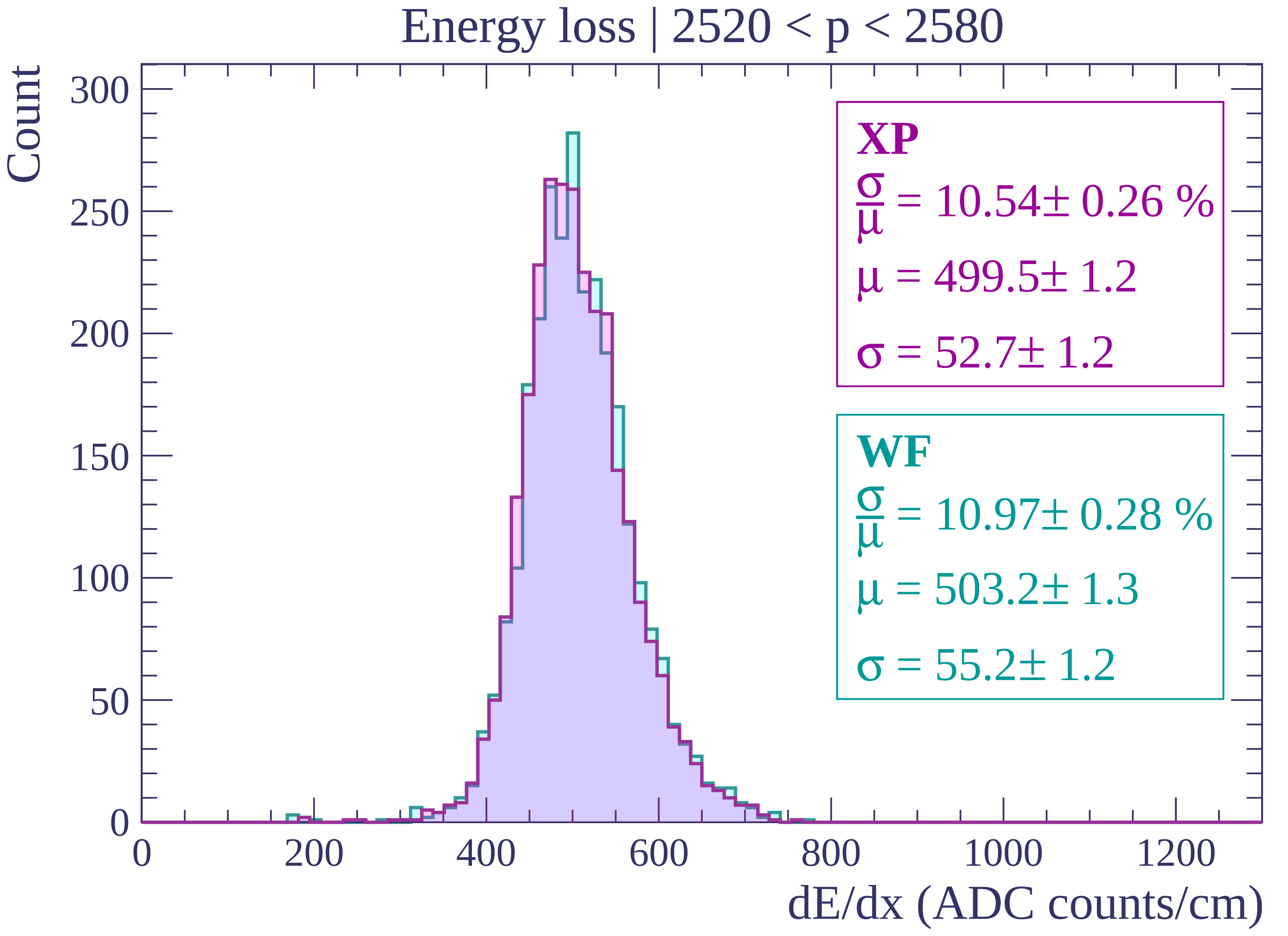
Energy loss | 2340 < p < 2400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 10.57± 0.26 %
μ = 495.5± 1.2
σ = 52.4± 1.1
WF
σ/μ = 10.75± 0.26 %
μ = 497.4± 1.2
σ = 53.5± 1.2



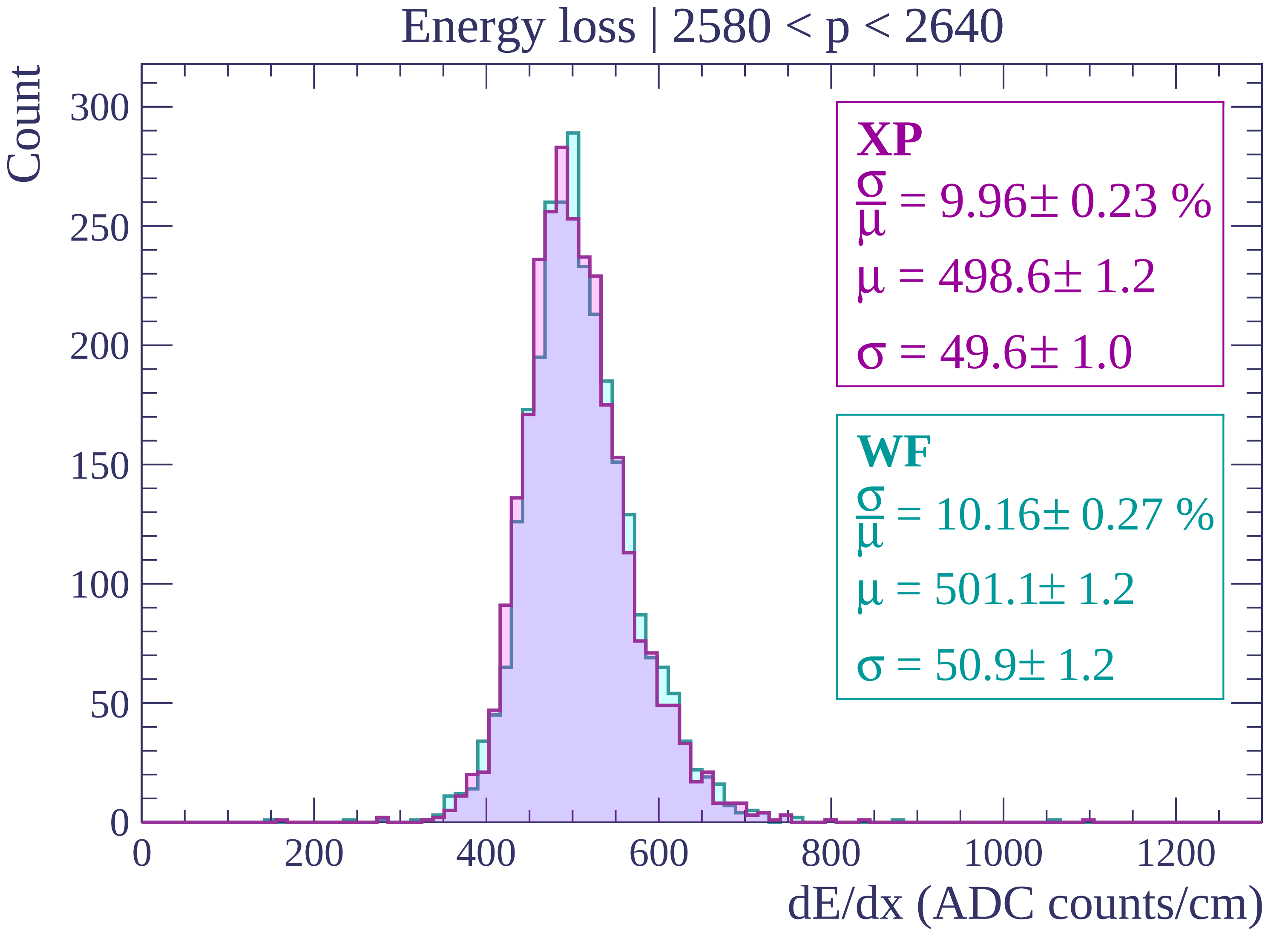
Energy loss | 2400 < p < 2460
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.79± 0.27 %
μ = 494.8± 1.2
σ = 53.4± 1.2
WF
$\frac{\sigma}{\mu}$ = 10.90± 0.27 %
μ = 497.6± 1.2
σ = 54.3± 1.2



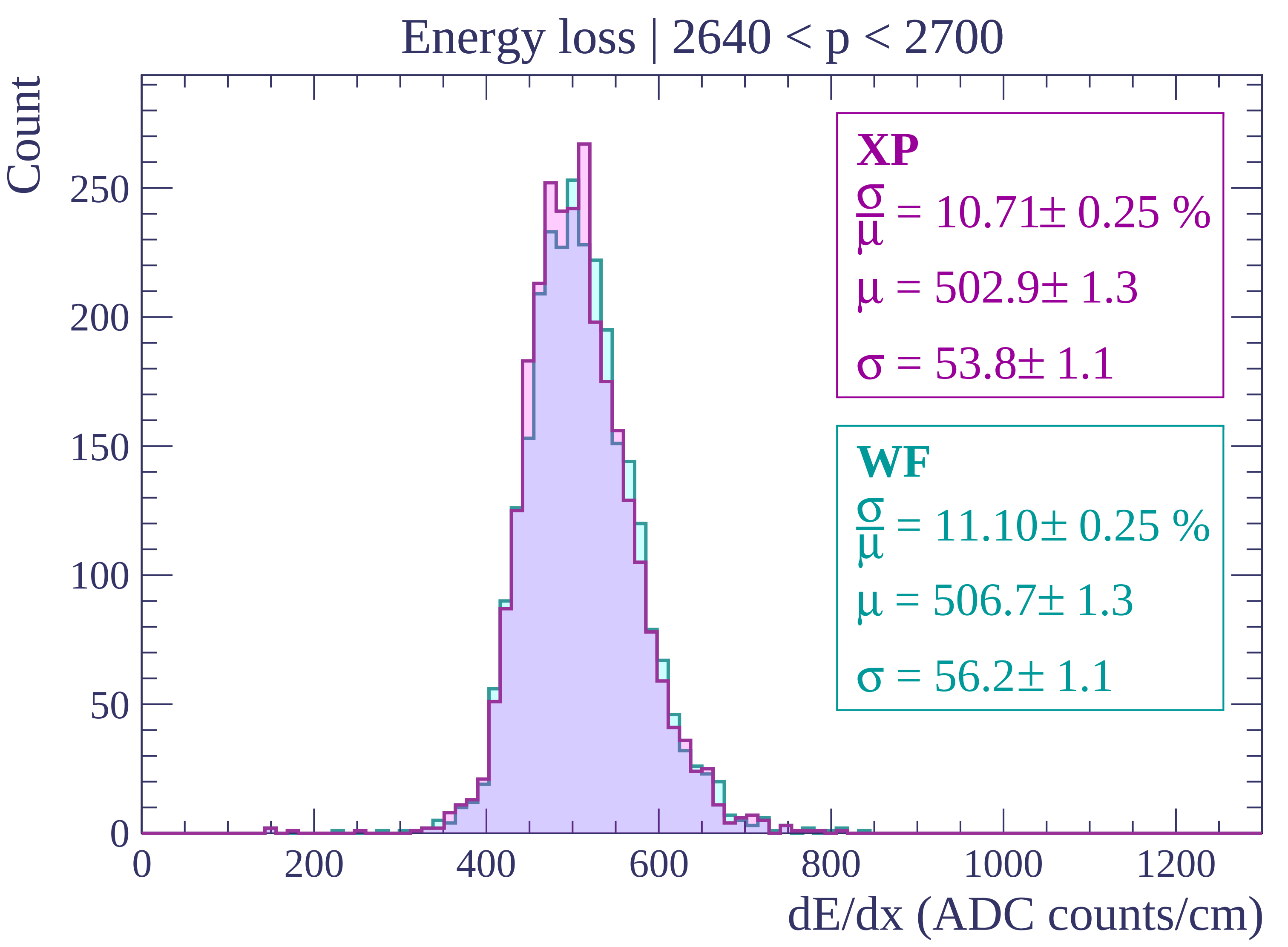
Energy loss | 2460 < p < 2520
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.25± 0.24 %
μ = 498.4± 1.2
σ = 51.1± 1.1
WF
$\frac{\sigma}{\mu}$ = 10.20± 0.23 %
μ = 500.7± 1.2
σ = 51.1± 1.1



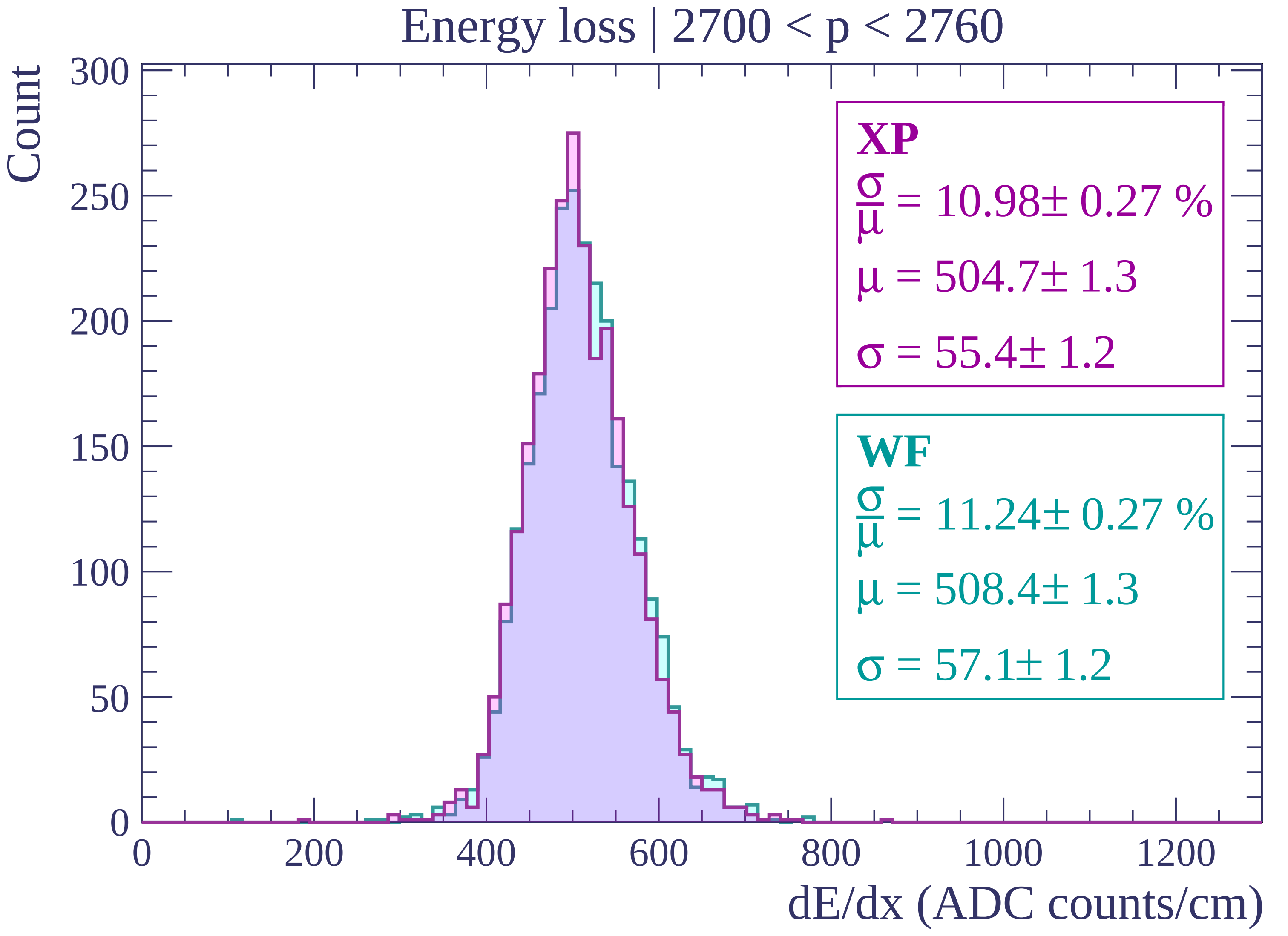
Energy loss | 2520 < p < 2580
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
300
XP
$\frac{\sigma}{\mu}$ = 10.54± 0.26 %
μ = 499.5± 1.2
σ = 52.7± 1.2
WF
$\frac{\sigma}{\mu}$ = 10.97± 0.28 %
μ = 503.2± 1.3
σ = 55.2± 1.2



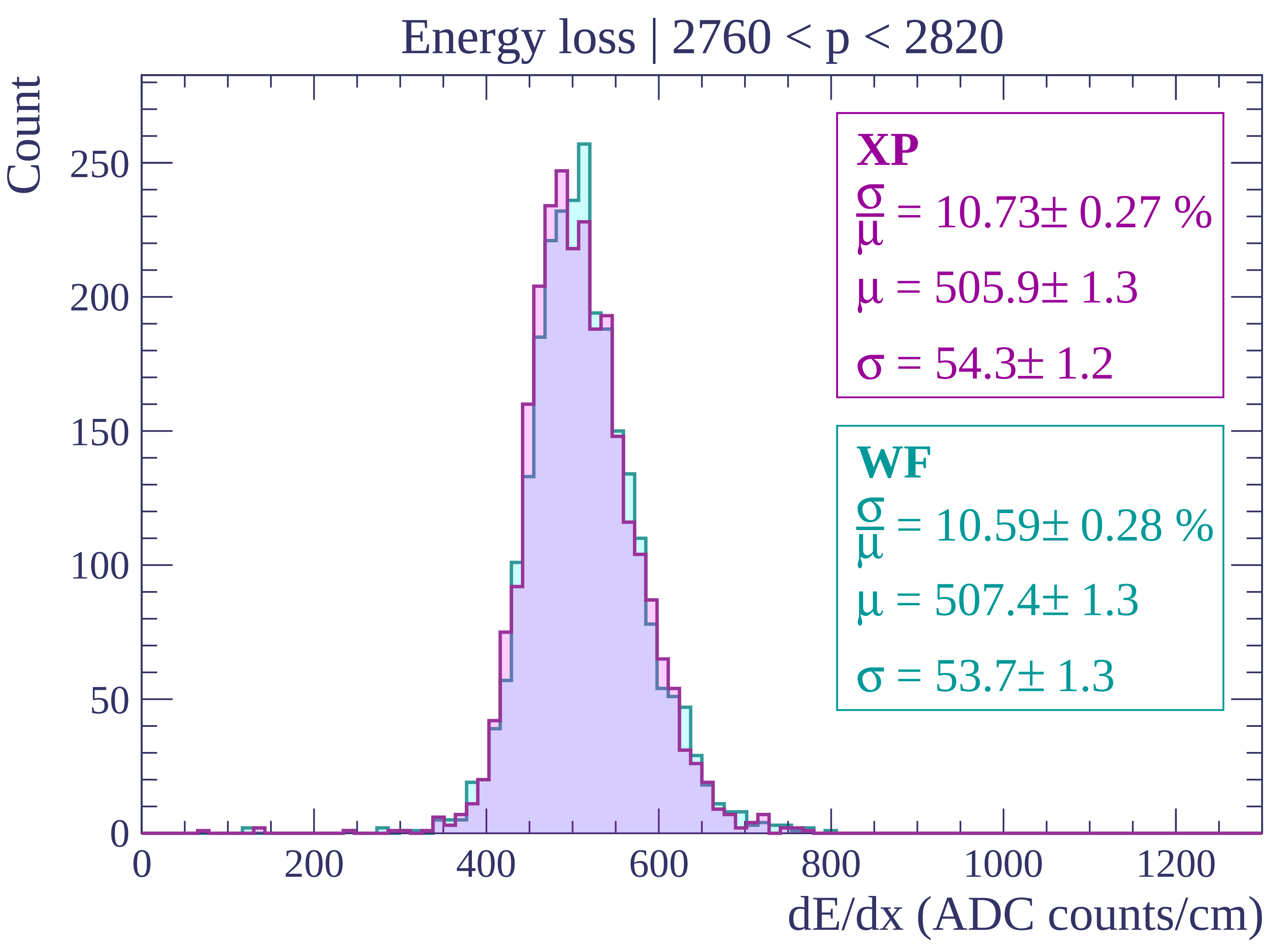
Energy loss | 2580 < p < 2640
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
σ/μ = 9.96± 0.23 %
μ = 498.6± 1.2
σ = 49.6± 1.0
WF
σ/μ = 10.16± 0.27 %
μ = 501.1± 1.2
σ = 50.9± 1.2

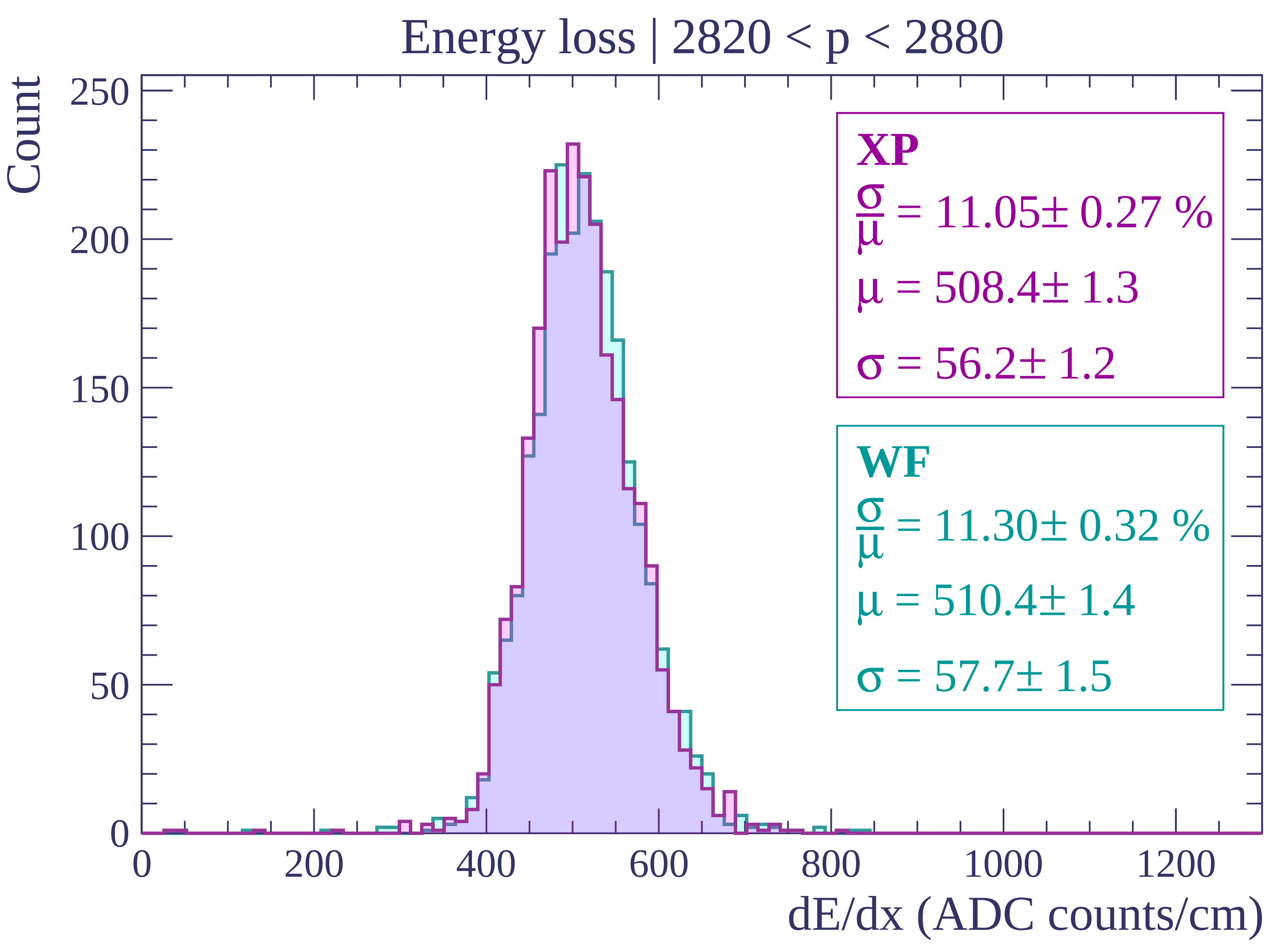


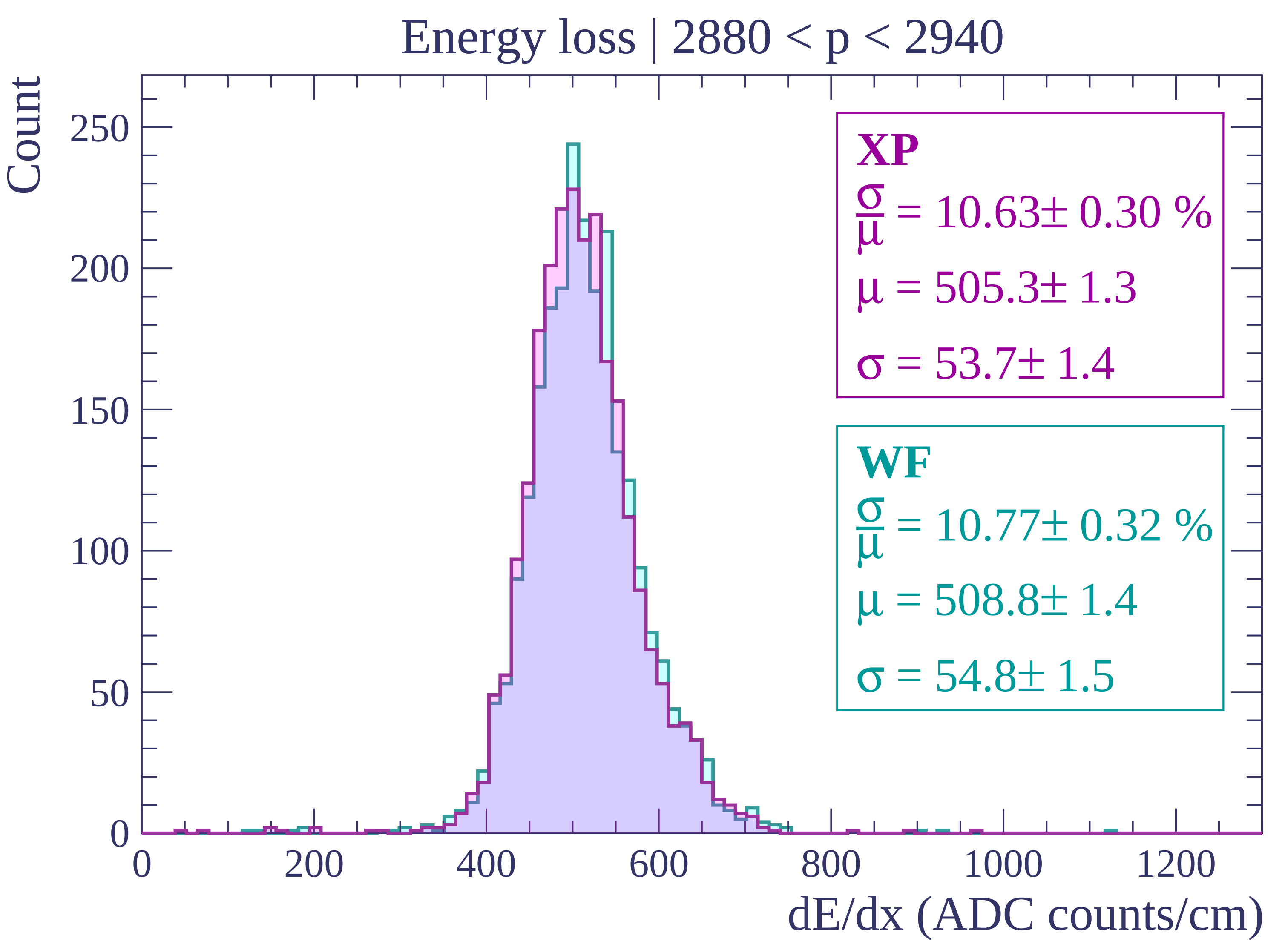
Energy loss | 2640 < p < 2700
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 10.71± 0.25 %
μ = 502.9± 1.3
σ = 53.8± 1.1
WF
$\frac{\sigma}{\mu}$ = 11.10± 0.25 %
μ = 506.7± 1.3
σ = 56.2± 1.1



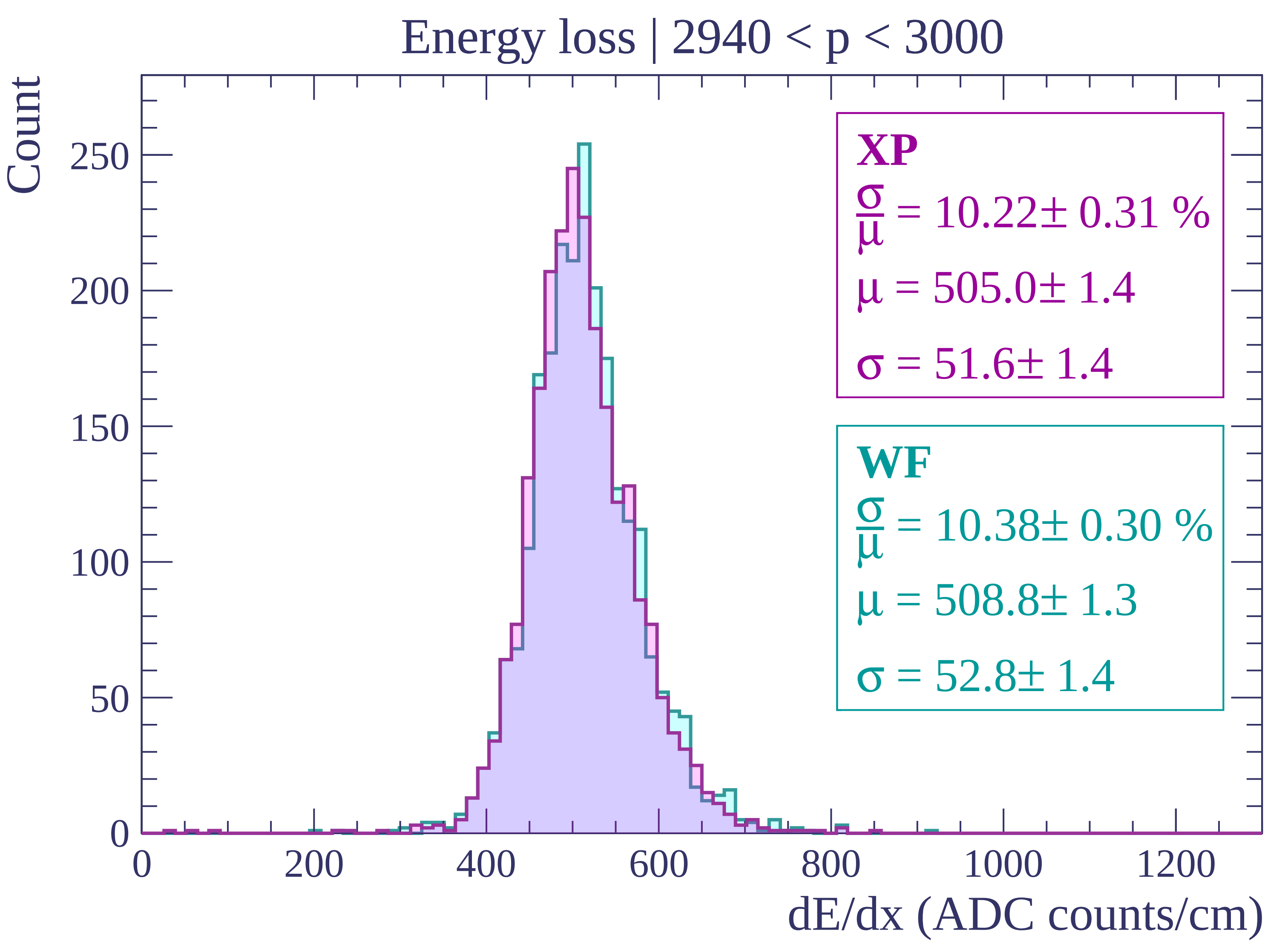
Energy loss | 2700 < p < 2760
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.98± 0.27 %
μ = 504.7± 1.3
σ = 55.4± 1.2
WF
$\frac{\sigma}{\mu}$ = 11.24± 0.27 %
μ = 508.4± 1.3
σ = 57.1± 1.2

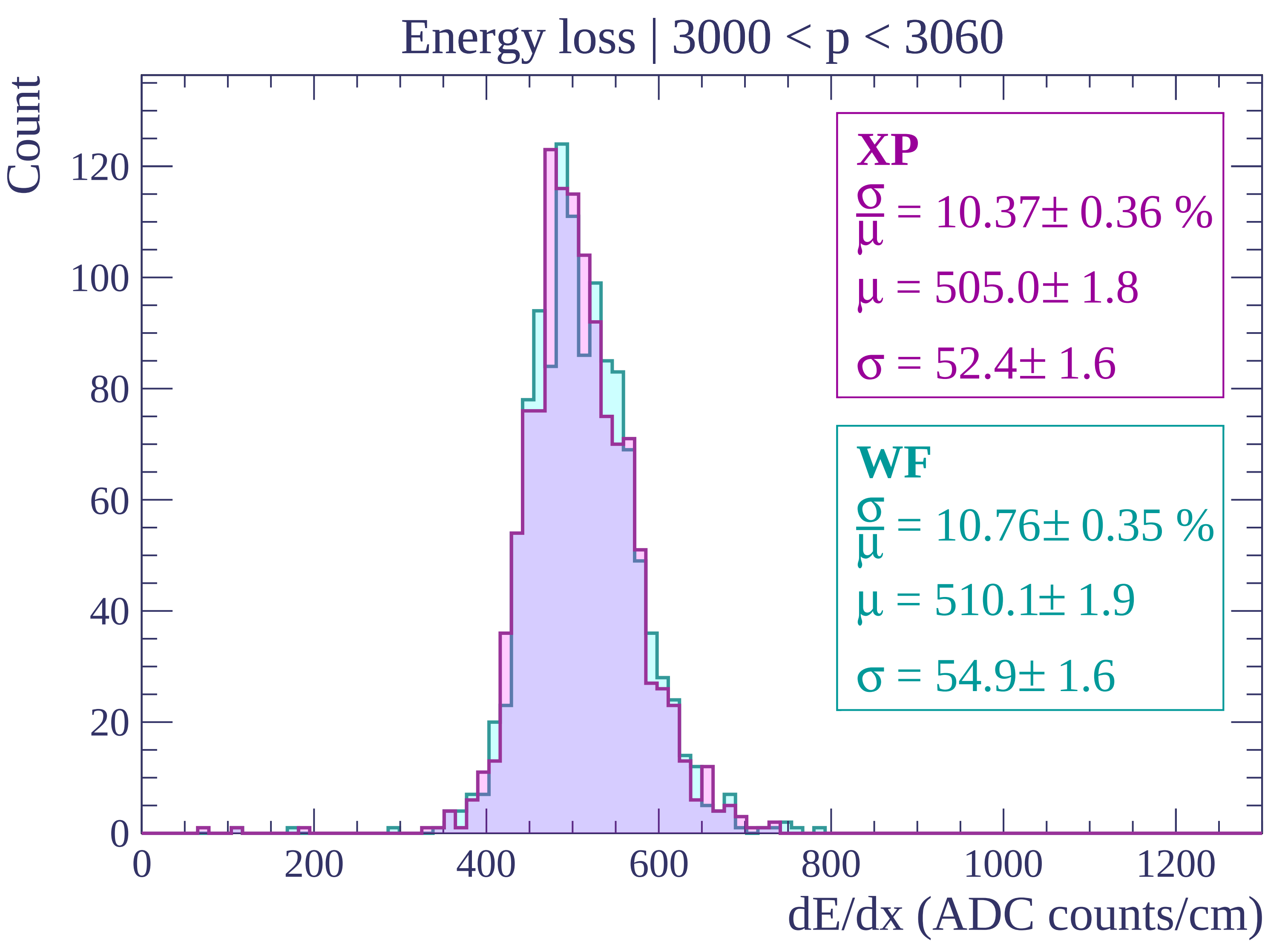

Energy loss | 2760 < p < 2820
Count
0
50
100
150
200
250
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.73± 0.27 %
μ = 505.9± 1.3
σ = 54.3± 1.2
WF
σ/μ = 10.59± 0.28 %
μ = 507.4± 1.3
σ = 53.7± 1.3


Energy loss | 2820 < p < 2880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 11.05± 0.27 %
μ = 508.4± 1.3
σ = 56.2± 1.2
WF
σ/μ = 11.30± 0.32 %
μ = 510.4± 1.4
σ = 57.7± 1.5

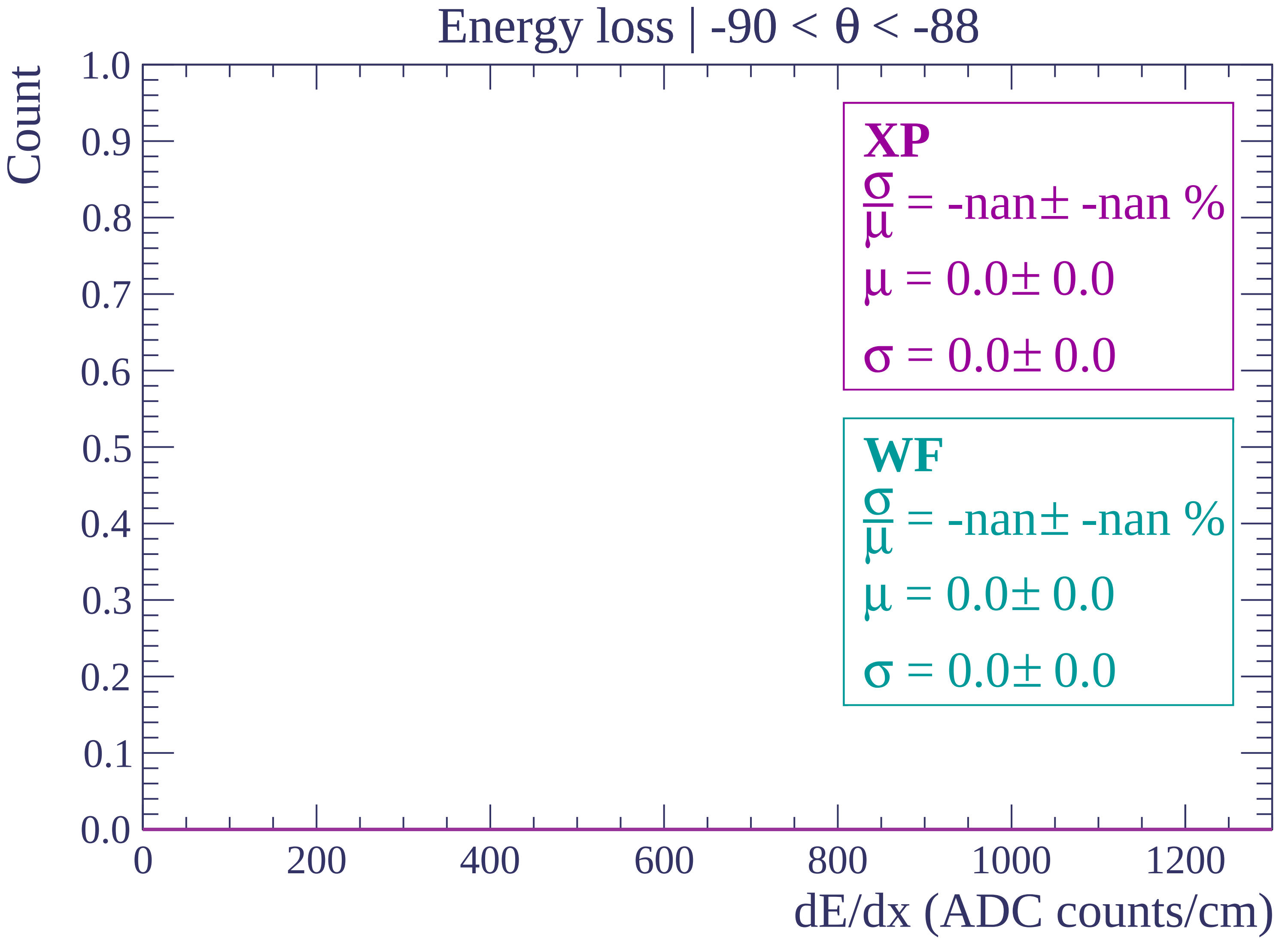


Energy loss | 2880 < p < 2940
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 10.63± 0.30 %
μ = 505.3± 1.3
σ = 53.7± 1.4
WF
σ/μ = 10.77± 0.32 %
μ = 508.8± 1.4
σ = 54.8± 1.5

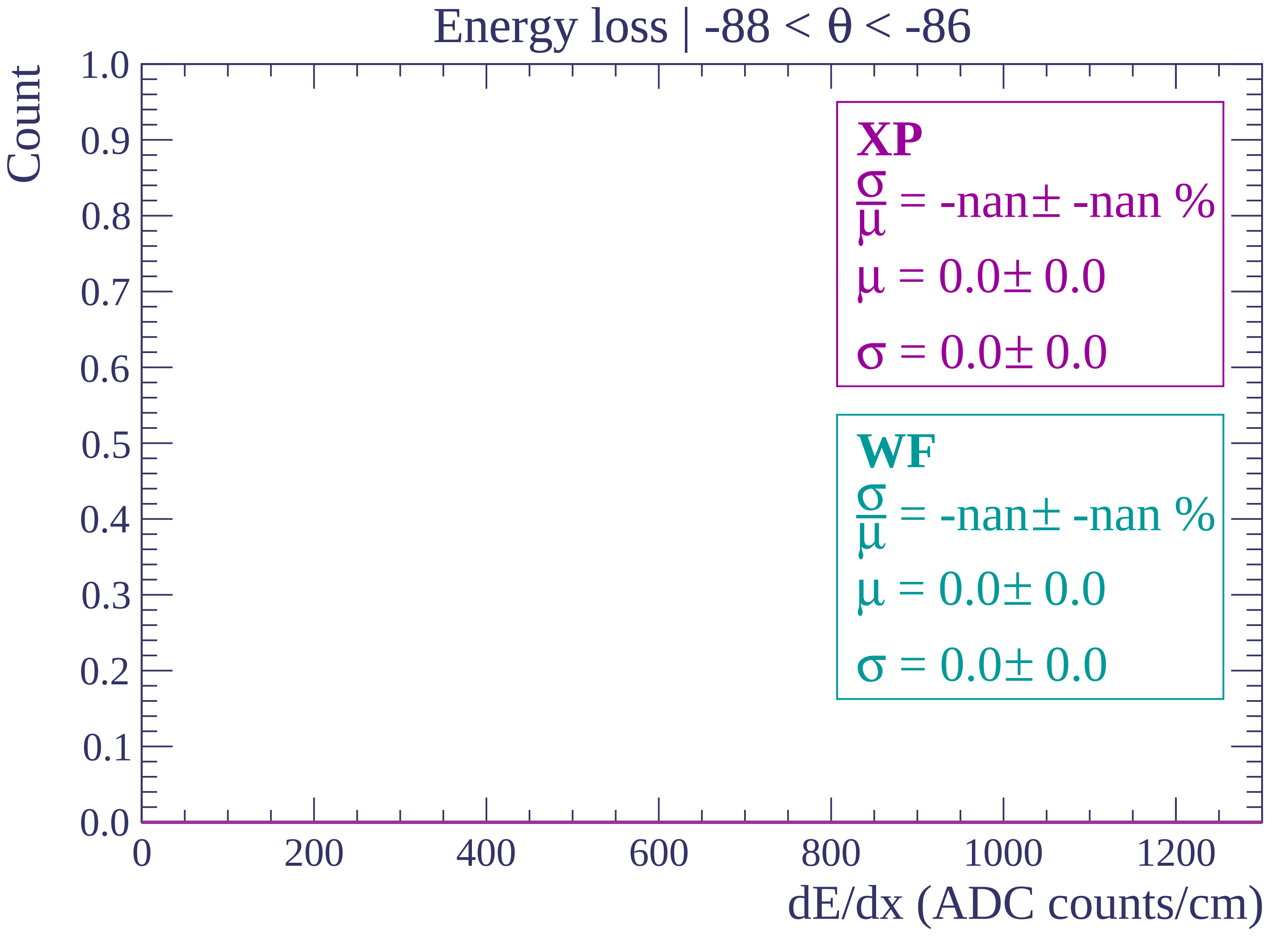

Energy loss | 2940 < p < 3000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
50
100
150
200
250
XP
σ/μ = 10.22± 0.31 %
μ = 505.0± 1.4
σ = 51.6± 1.4
WF
σ/μ = 10.38± 0.30 %
μ = 508.8± 1.3
σ = 52.8± 1.4



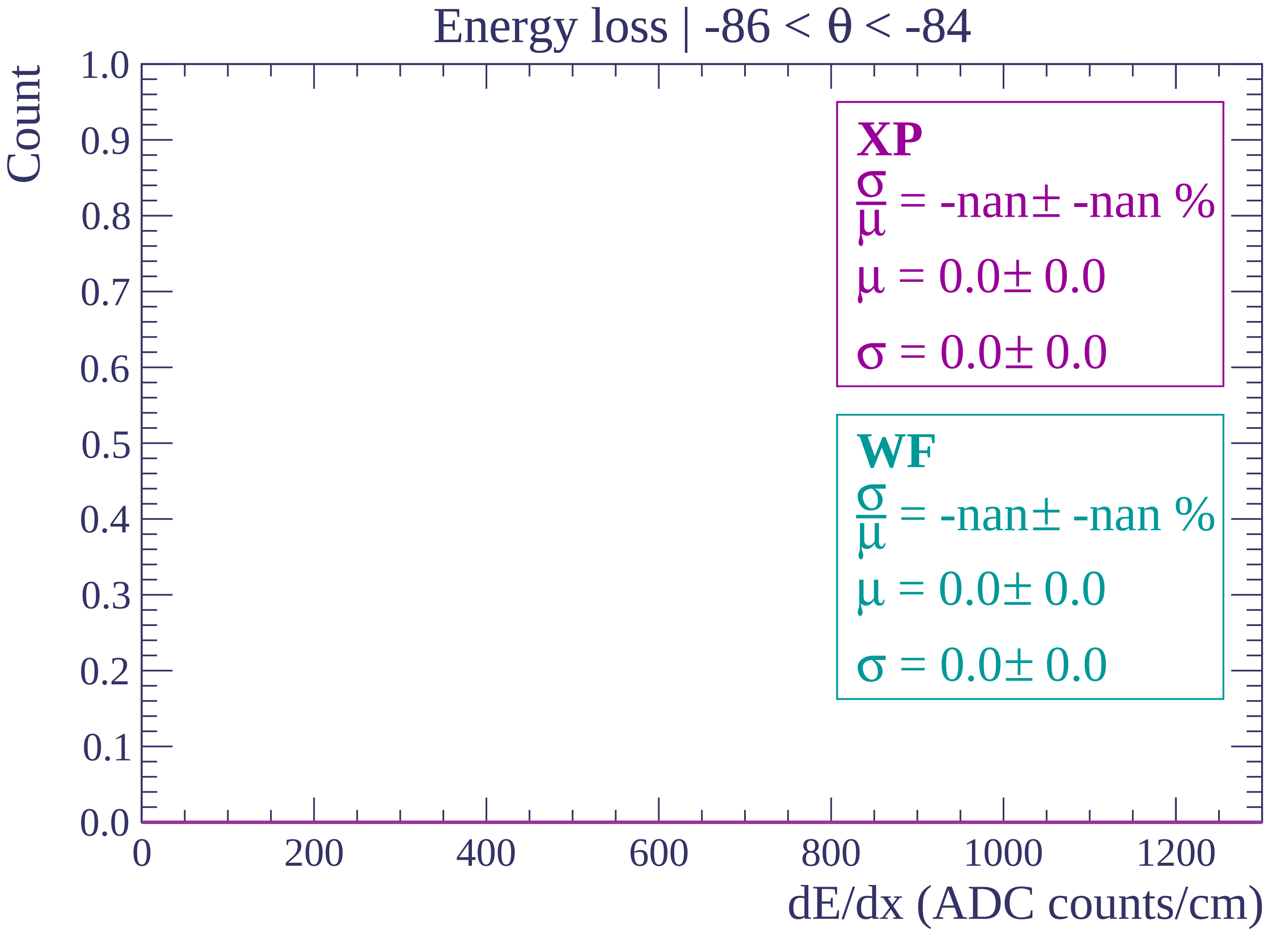
Energy loss | 3000 < p < 3060
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 10.37± 0.36 %
μ = 505.0± 1.8
σ = 52.4± 1.6
WF
σ/μ = 10.76± 0.35 %
μ = 510.1± 1.9
σ = 54.9± 1.6

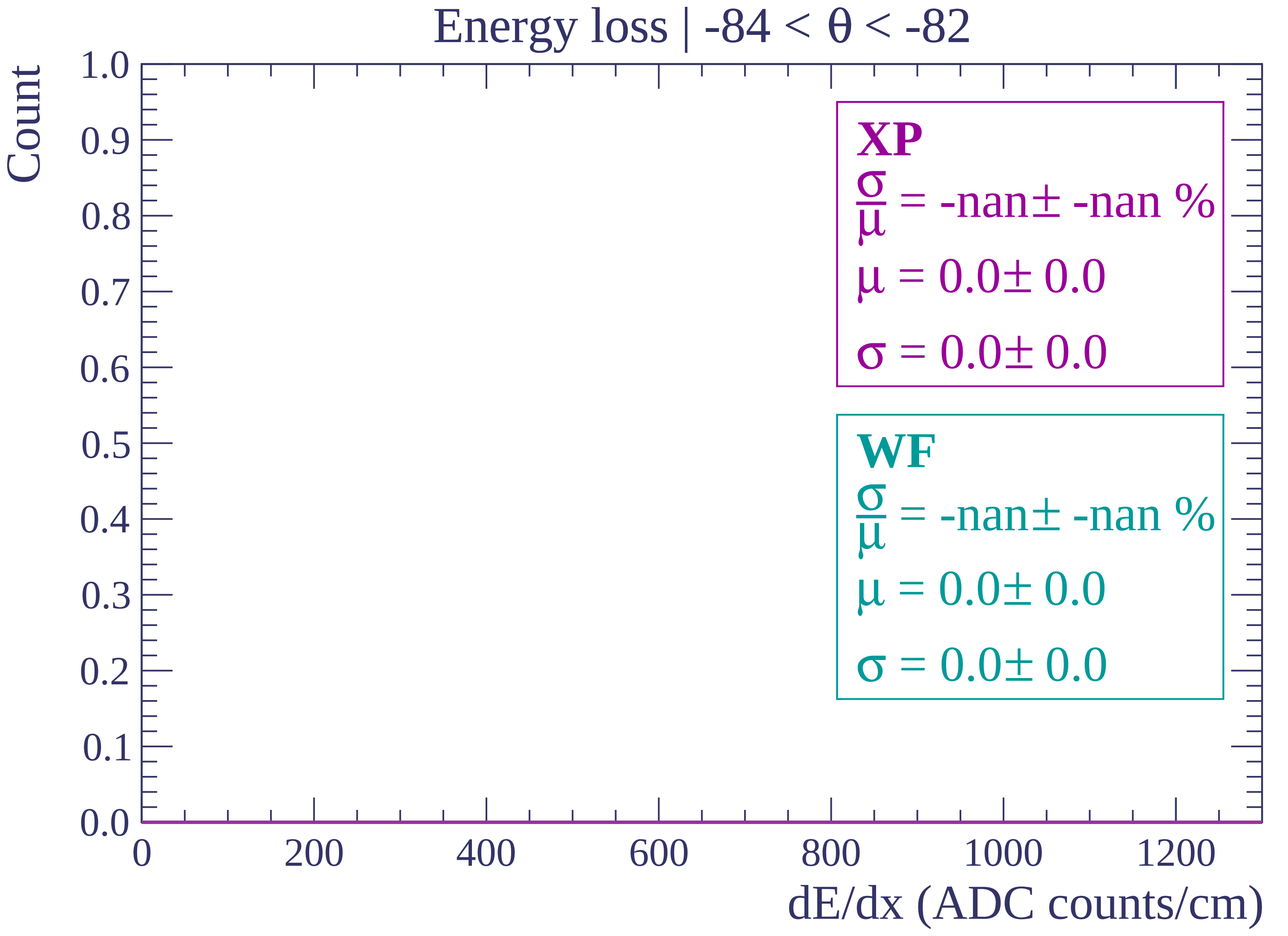


Energy loss | -90 < θ < -88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

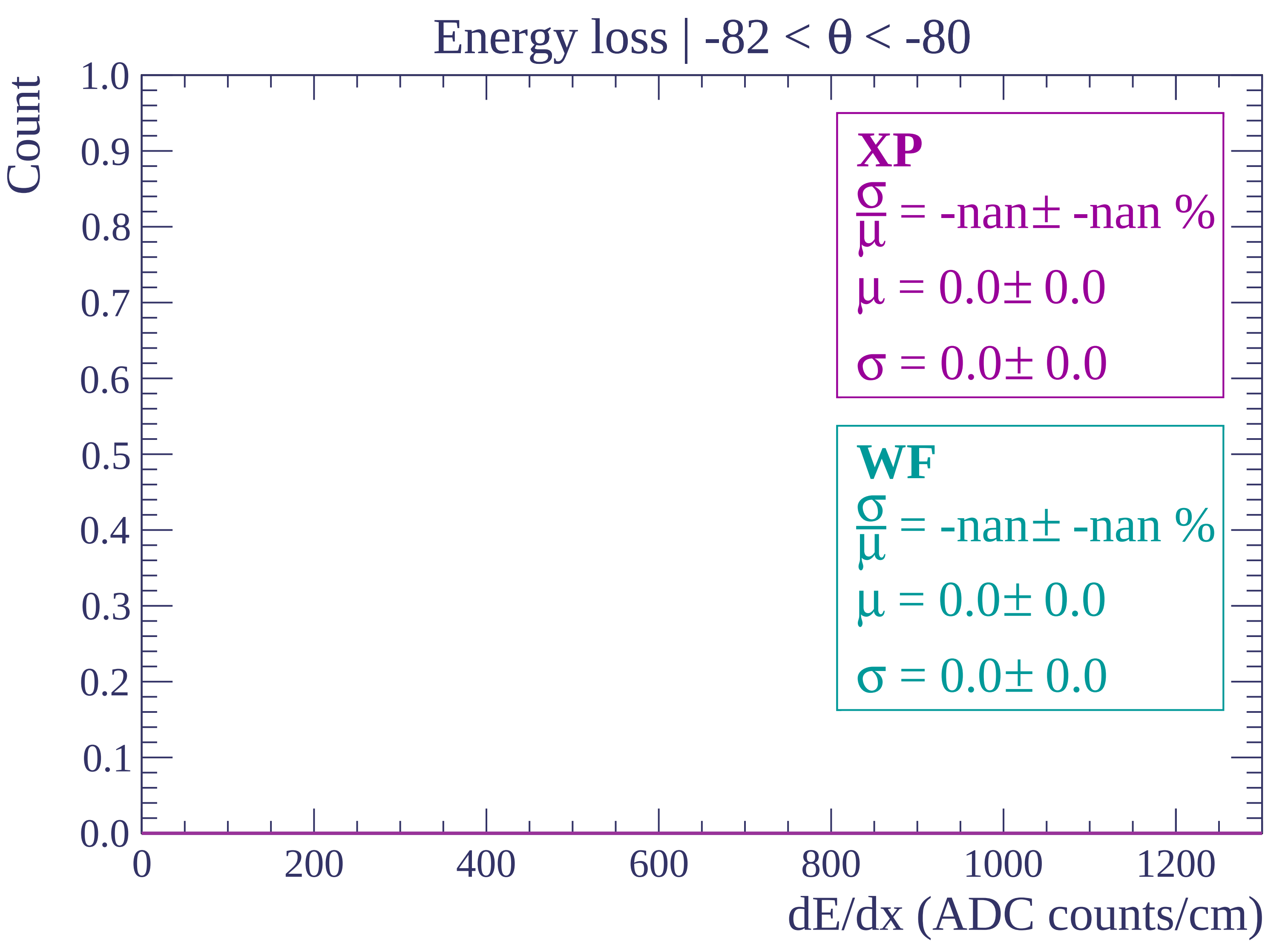


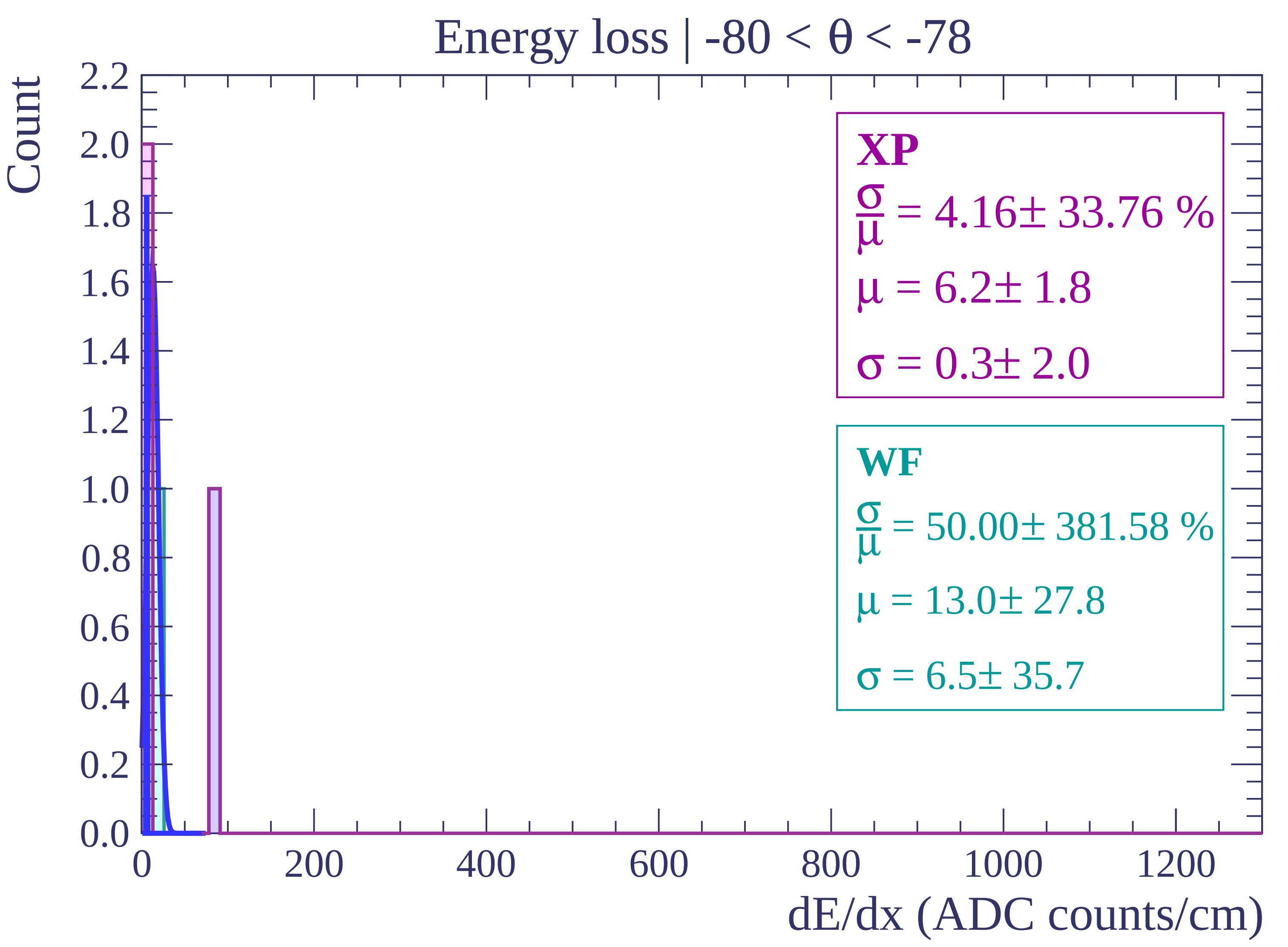
Energy loss | -88 < θ < -86
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

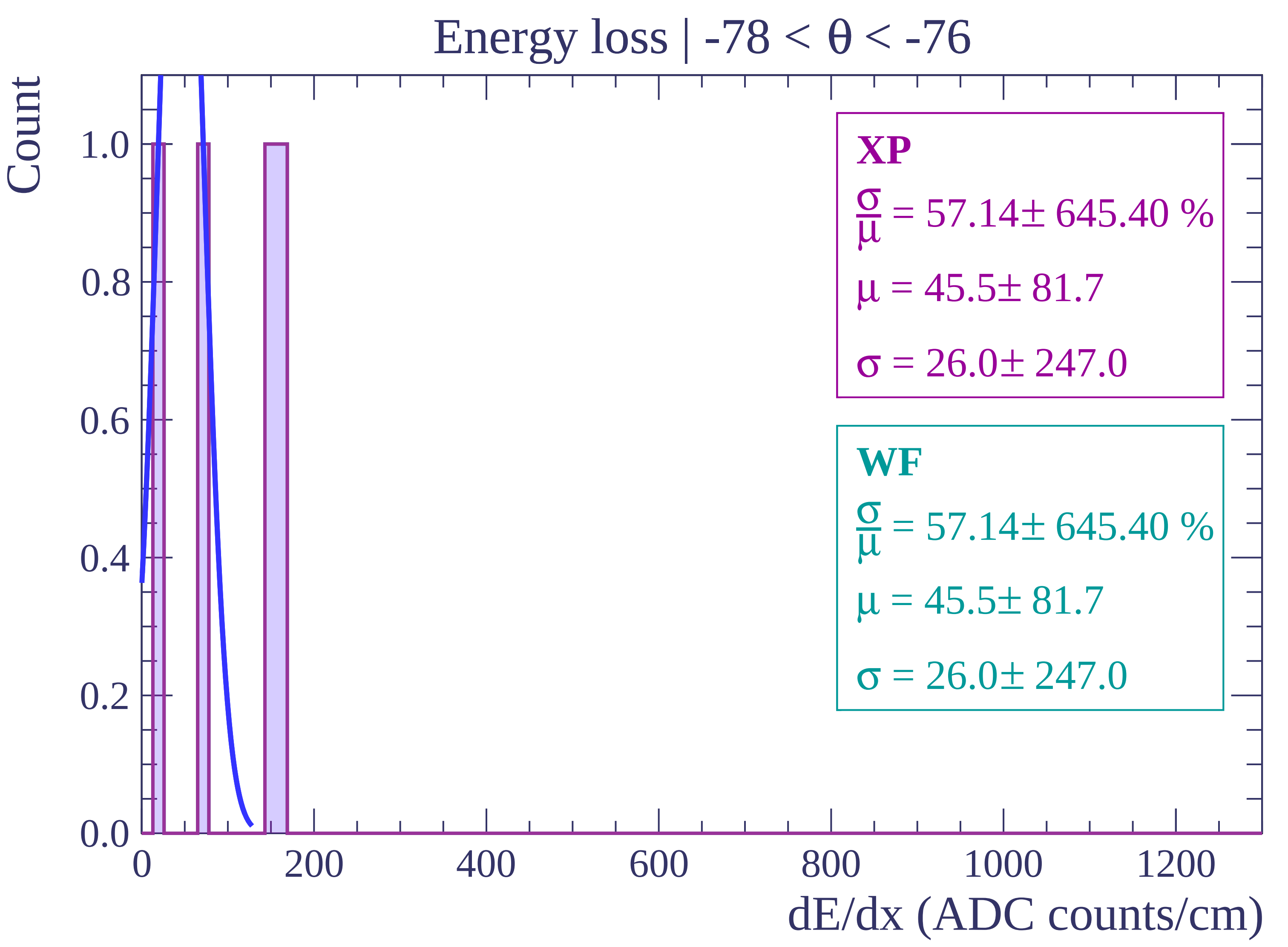

Energy loss | -86 < θ < -84
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



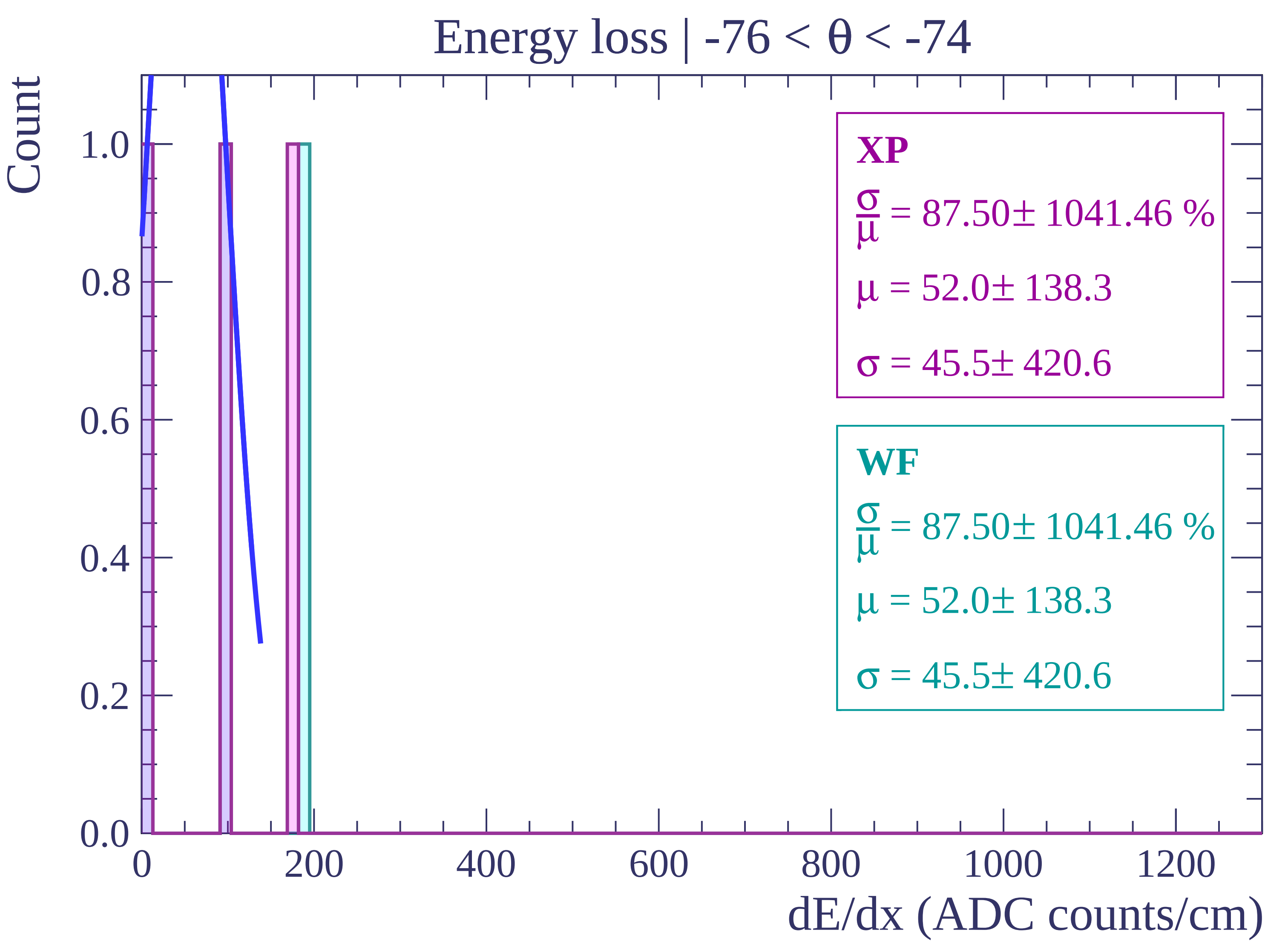
Energy loss | -84 < θ < -82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

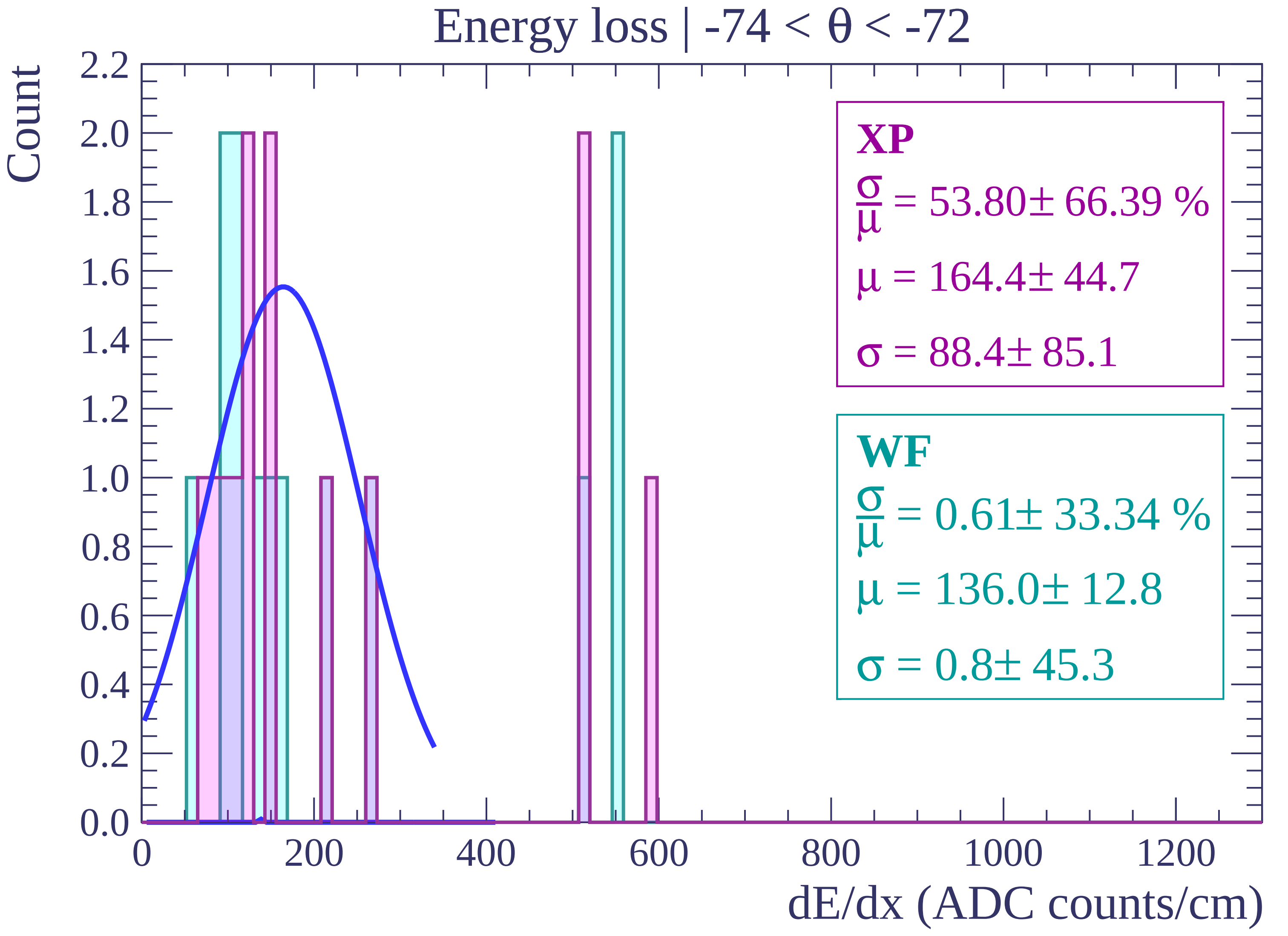

Energy loss | -82 < θ < -80
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -80 < θ < -78
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
XP
σ/μ = 4.16± 33.76 %
μ = 6.2± 1.8
σ = 0.3± 2.0
WF
σ/μ = 50.00± 381.58 %
μ = 13.0± 27.8
σ = 6.5± 35.7

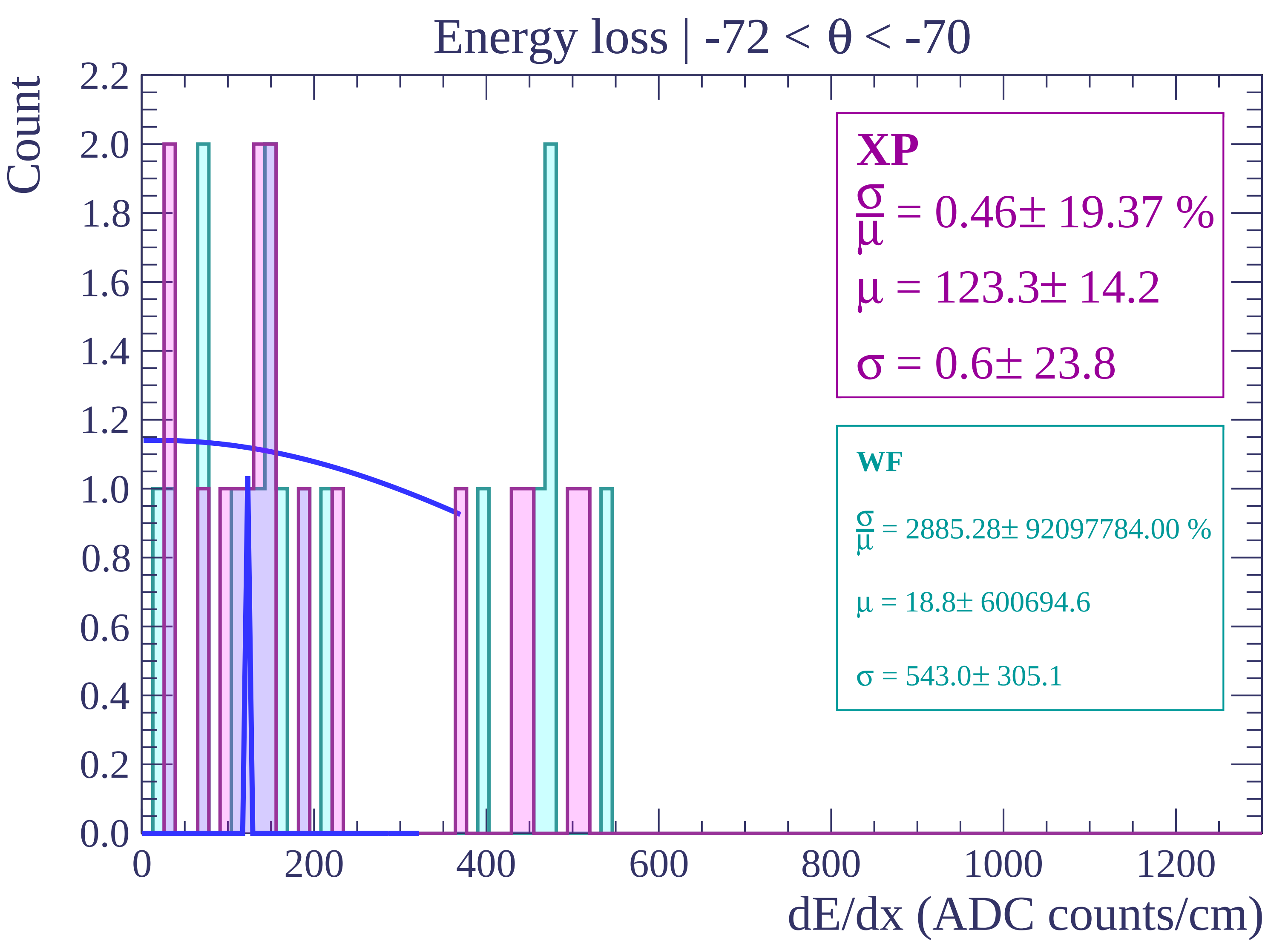


Energy loss | -78 < θ < -76
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 57.14± 645.40 %
μ = 45.5± 81.7
σ = 26.0± 247.0
WF
$\frac{\sigma}{\mu}$ = 57.14± 645.40 %
μ = 45.5± 81.7
σ = 26.0± 247.0

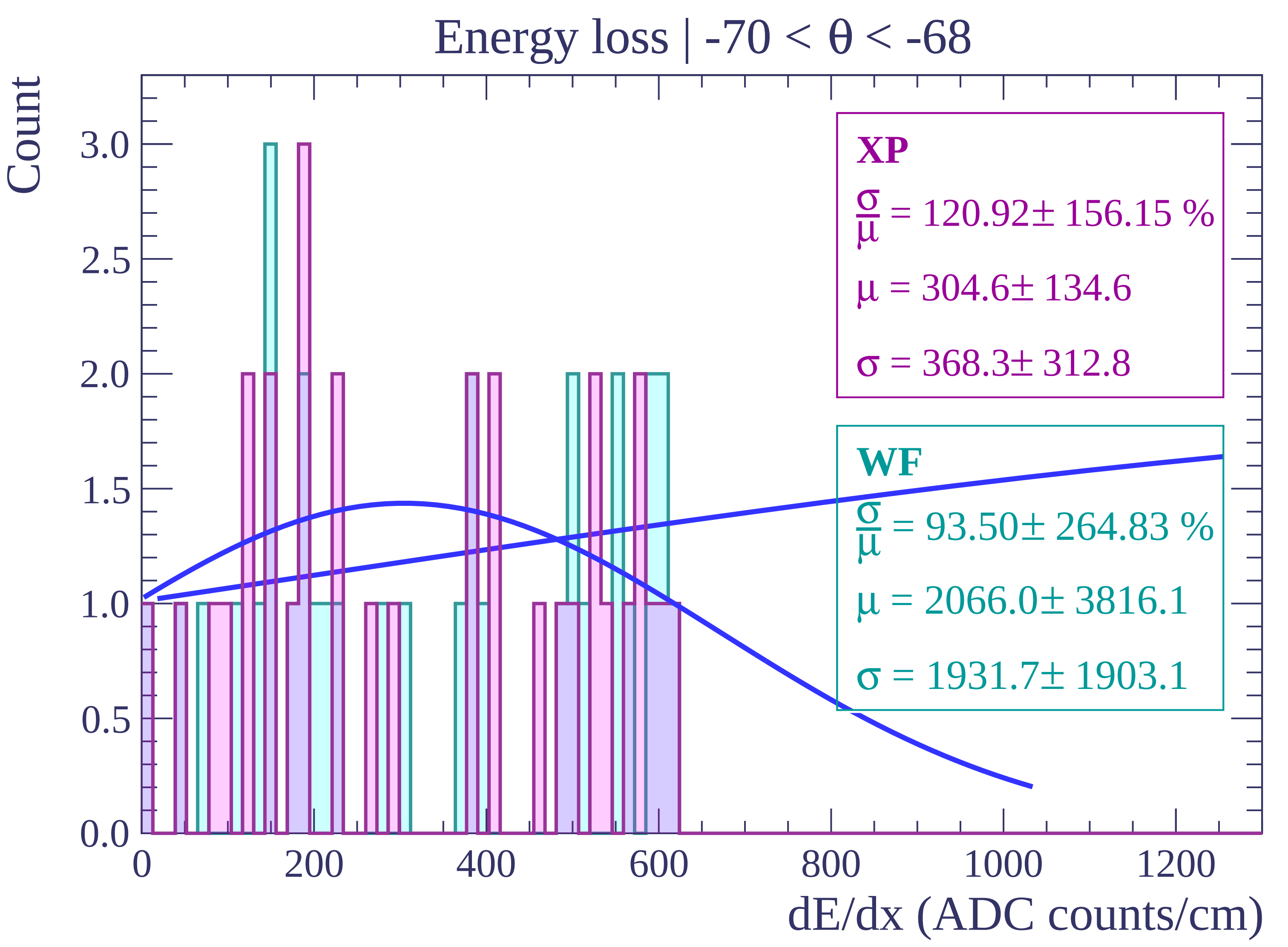

Energy loss | -76 < θ < -74
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 87.50± 1041.46 %
μ = 52.0± 138.3
σ = 45.5± 420.6
WF
σ/μ = 87.50± 1041.46 %
μ = 52.0± 138.3
σ = 45.5± 420.6



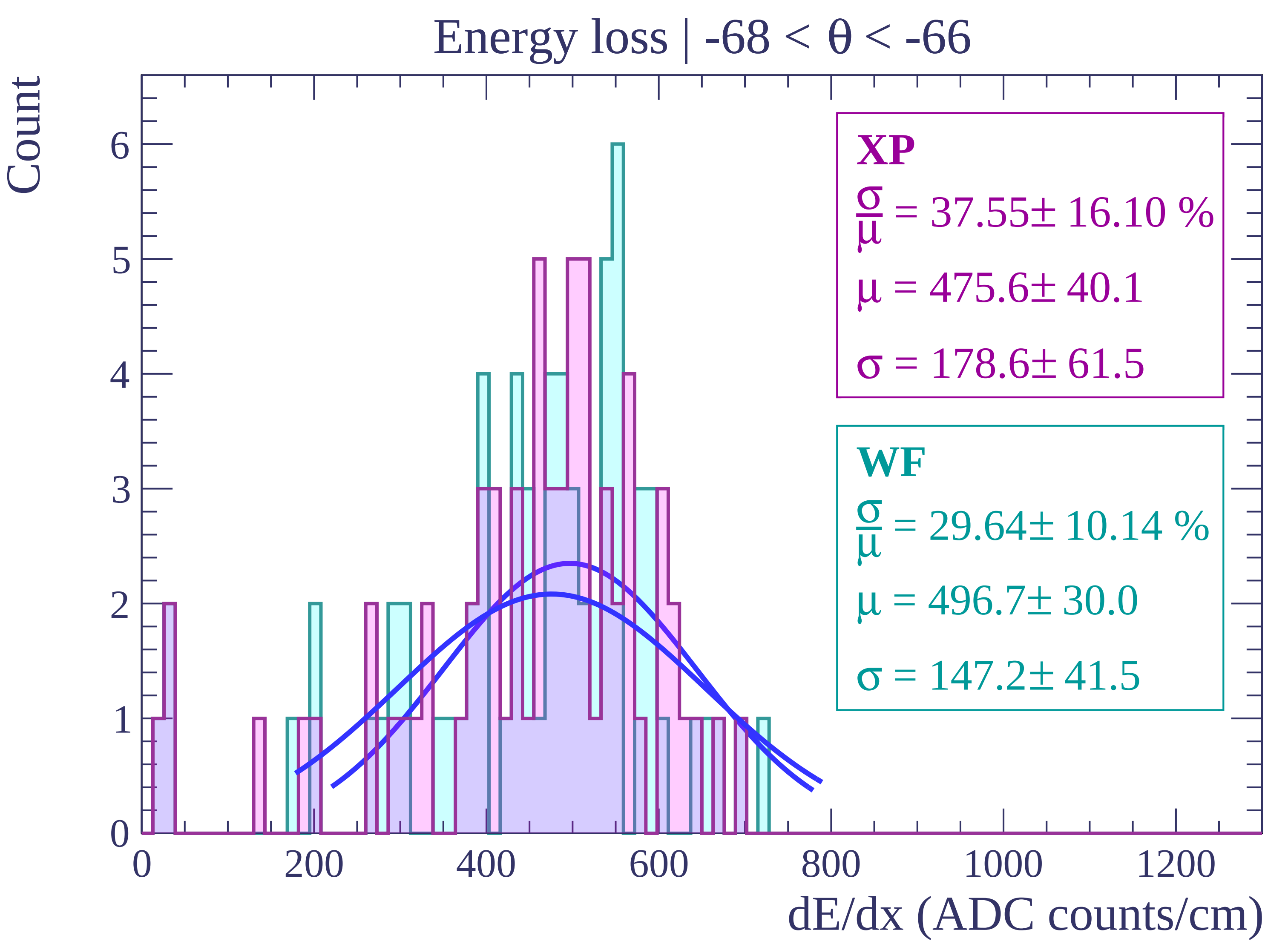
Energy loss | -74 < θ < -72
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 53.80± 66.39 %
μ = 164.4± 44.7
σ = 88.4± 85.1
WF
$\frac{\sigma}{\mu}$ = 0.61± 33.34 %
μ = 136.0± 12.8
σ = 0.8± 45.3
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200



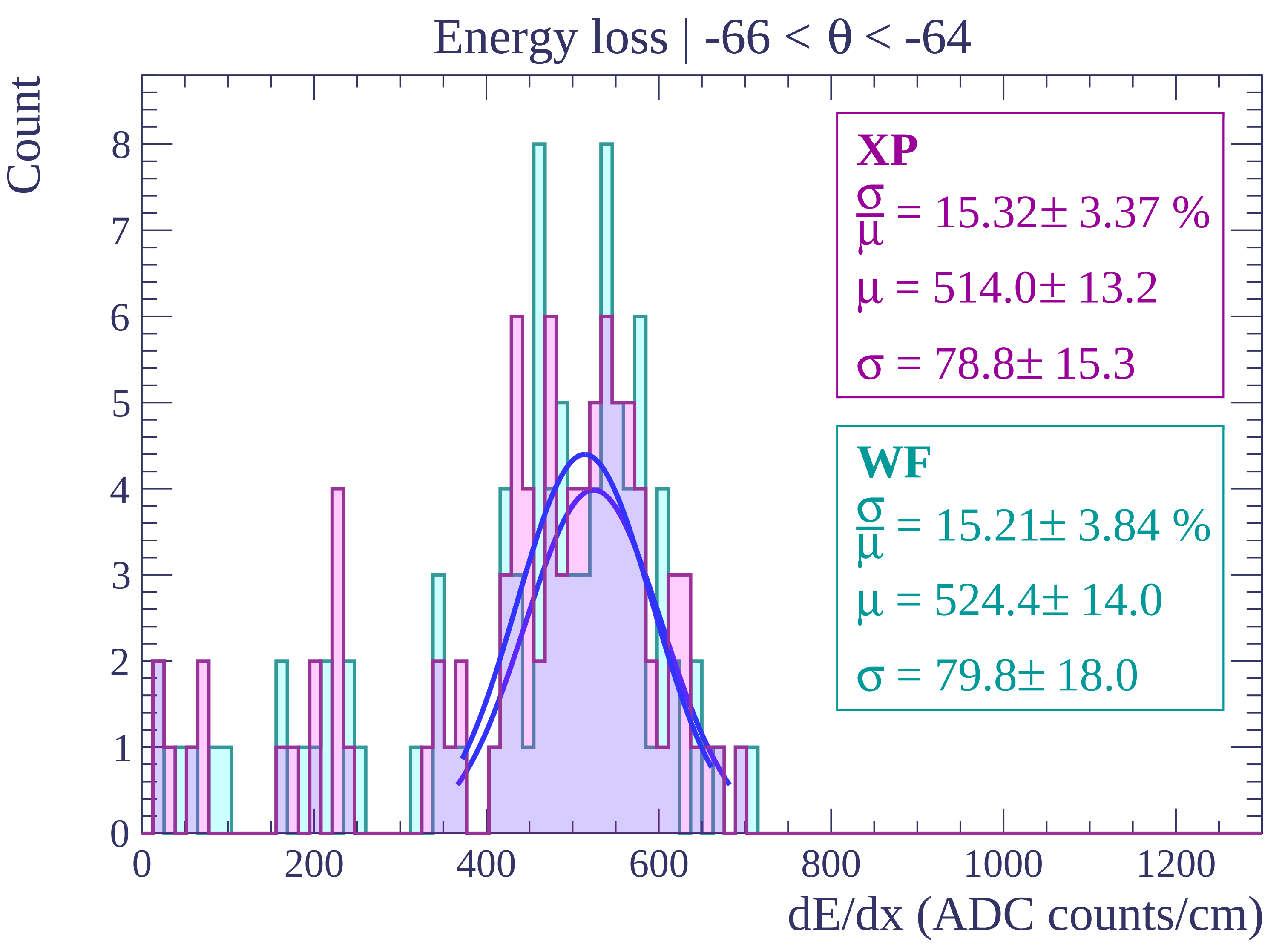
Energy loss | -72 < θ < -70
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.46± 19.37 %
μ = 123.3± 14.2
σ = 0.6± 23.8
WF
$\frac{\sigma}{\mu}$ = 2885.28± 92097784.00 %
μ = 18.8± 600694.6
σ = 543.0± 305.1

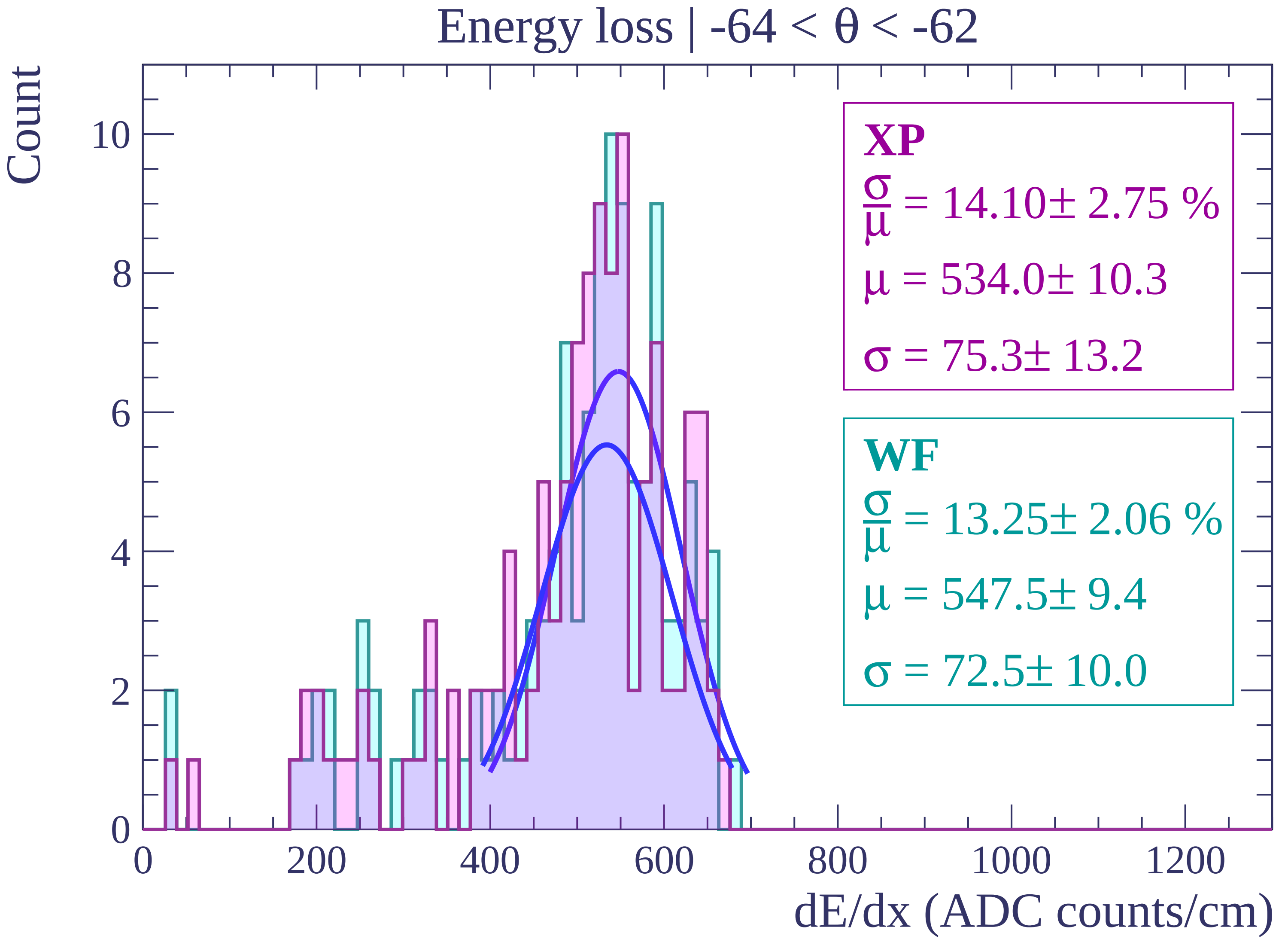


Energy loss | -70 < θ < -68
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 120.92± 156.15 %
μ = 304.6± 134.6
σ = 368.3± 312.8
WF
$\frac{\sigma}{\mu}$ = 93.50± 264.83 %
μ = 2066.0± 3816.1
σ = 1931.7± 1903.1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200

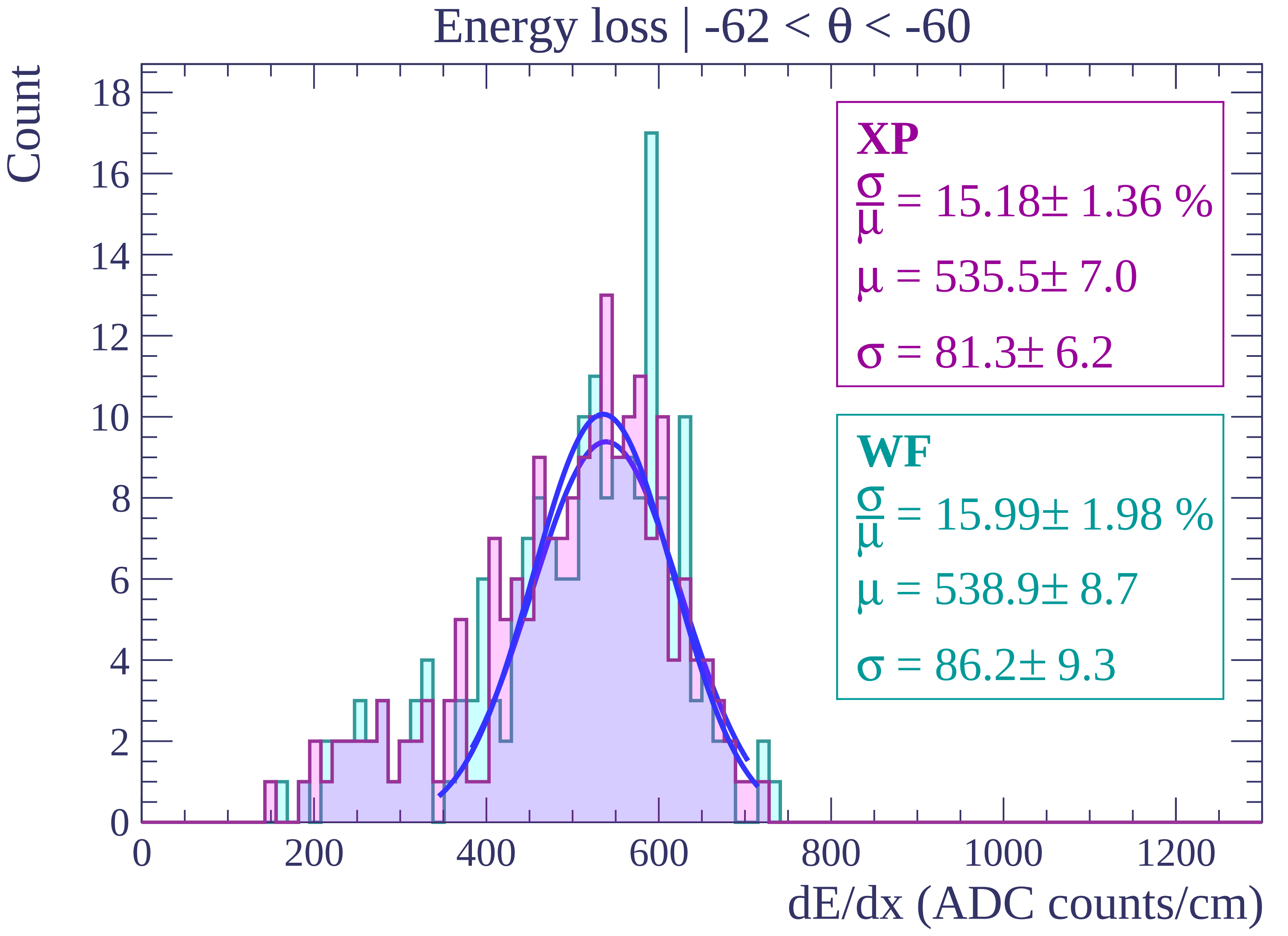


Energy loss | -68 < θ < -66
Count
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 37.55± 16.10 %
μ = 475.6± 40.1
σ = 178.6± 61.5
WF
σ/μ = 29.64± 10.14 %
μ = 496.7± 30.0
σ = 147.2± 41.5

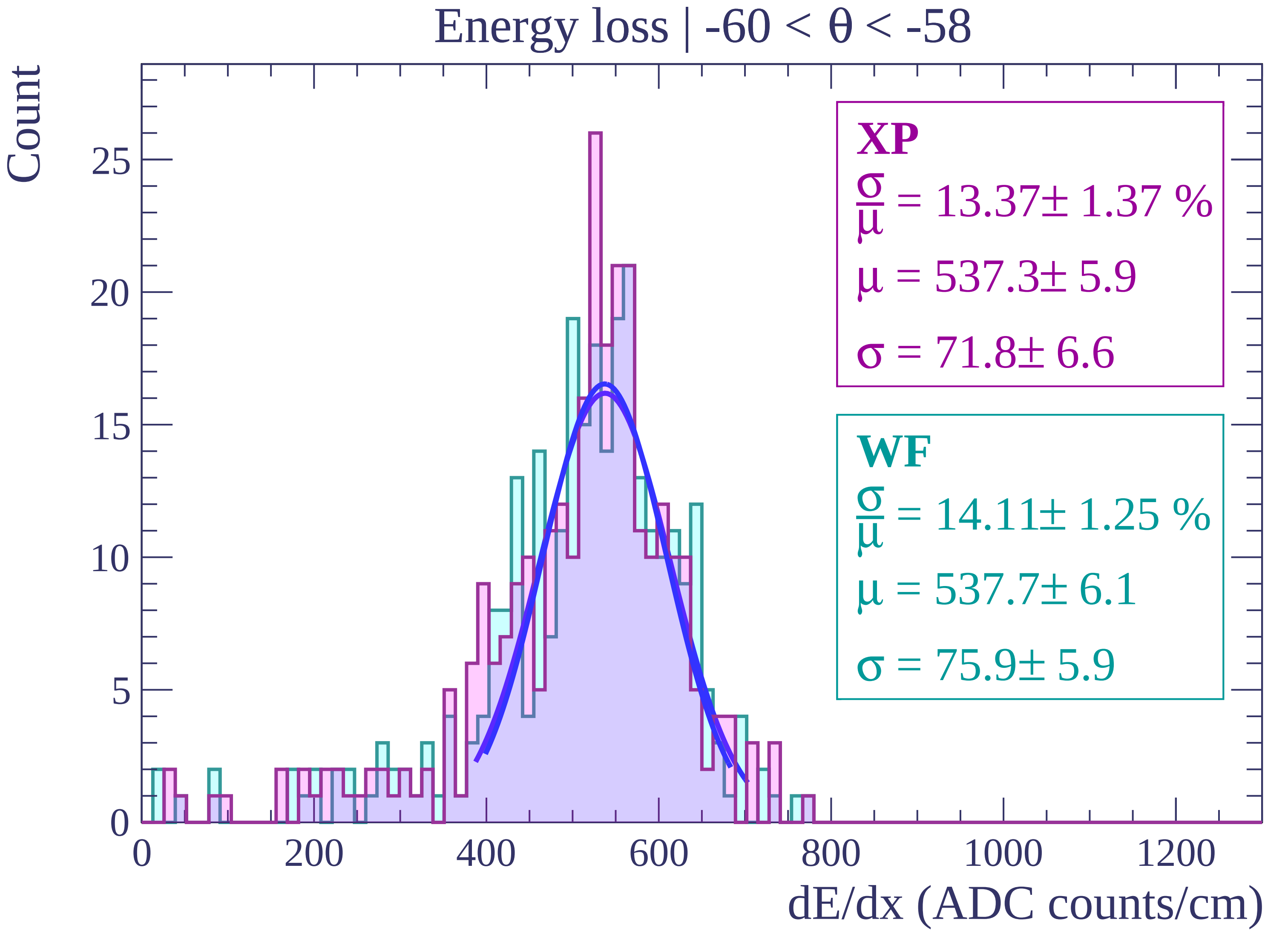

Energy loss | -66 < θ < -64
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.32± 3.37 %
μ = 514.0± 13.2
σ = 78.8± 15.3
WF
σ/μ = 15.21± 3.84 %
μ = 524.4± 14.0
σ = 79.8± 18.0
0
1
2
3
4
5
6
7
8
0
200
400
600
800
1000
1200

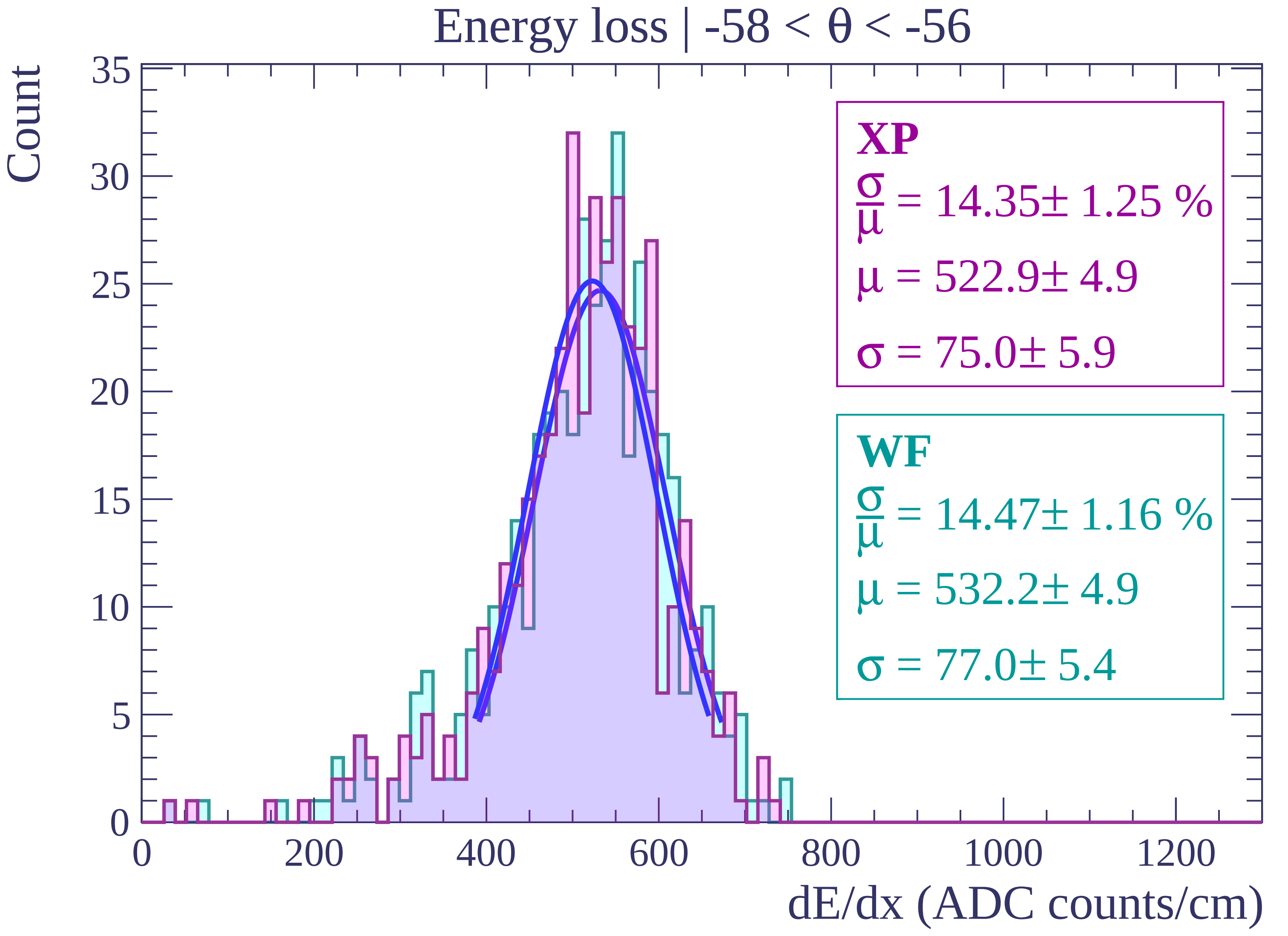


Energy loss | -64 < θ < -62
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.10± 2.75 %
μ = 534.0± 10.3
σ = 75.3± 13.2
WF
σ/μ = 13.25± 2.06 %
μ = 547.5± 9.4
σ = 72.5± 10.0
0
2
4
6
8
10
0
200
400
600
800
1000
1200

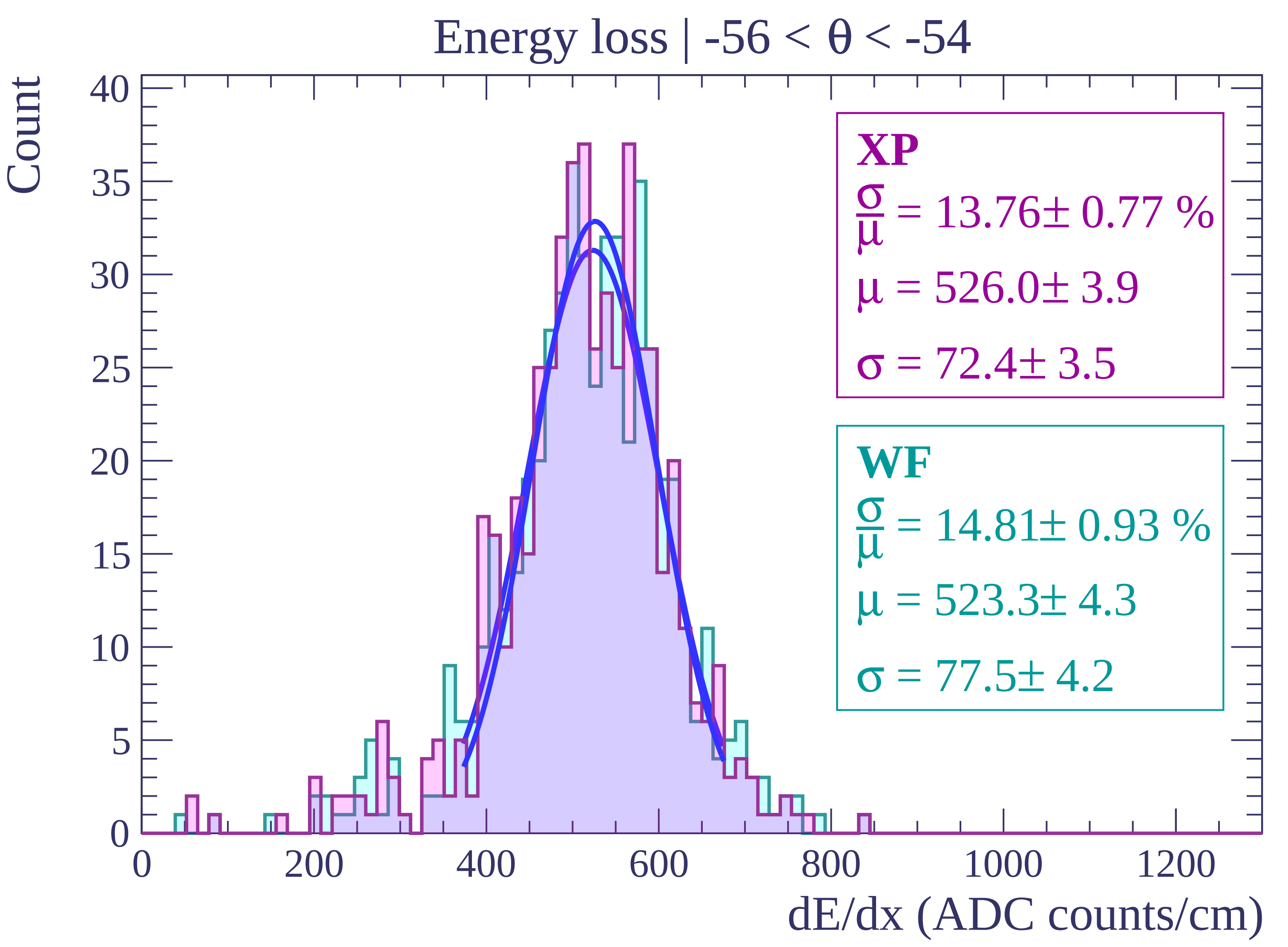


Energy loss | -62 < θ < -60
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
σ/μ = 15.18± 1.36 %
μ = 535.5± 7.0
σ = 81.3± 6.2
WF
σ/μ = 15.99± 1.98 %
μ = 538.9± 8.7
σ = 86.2± 9.3

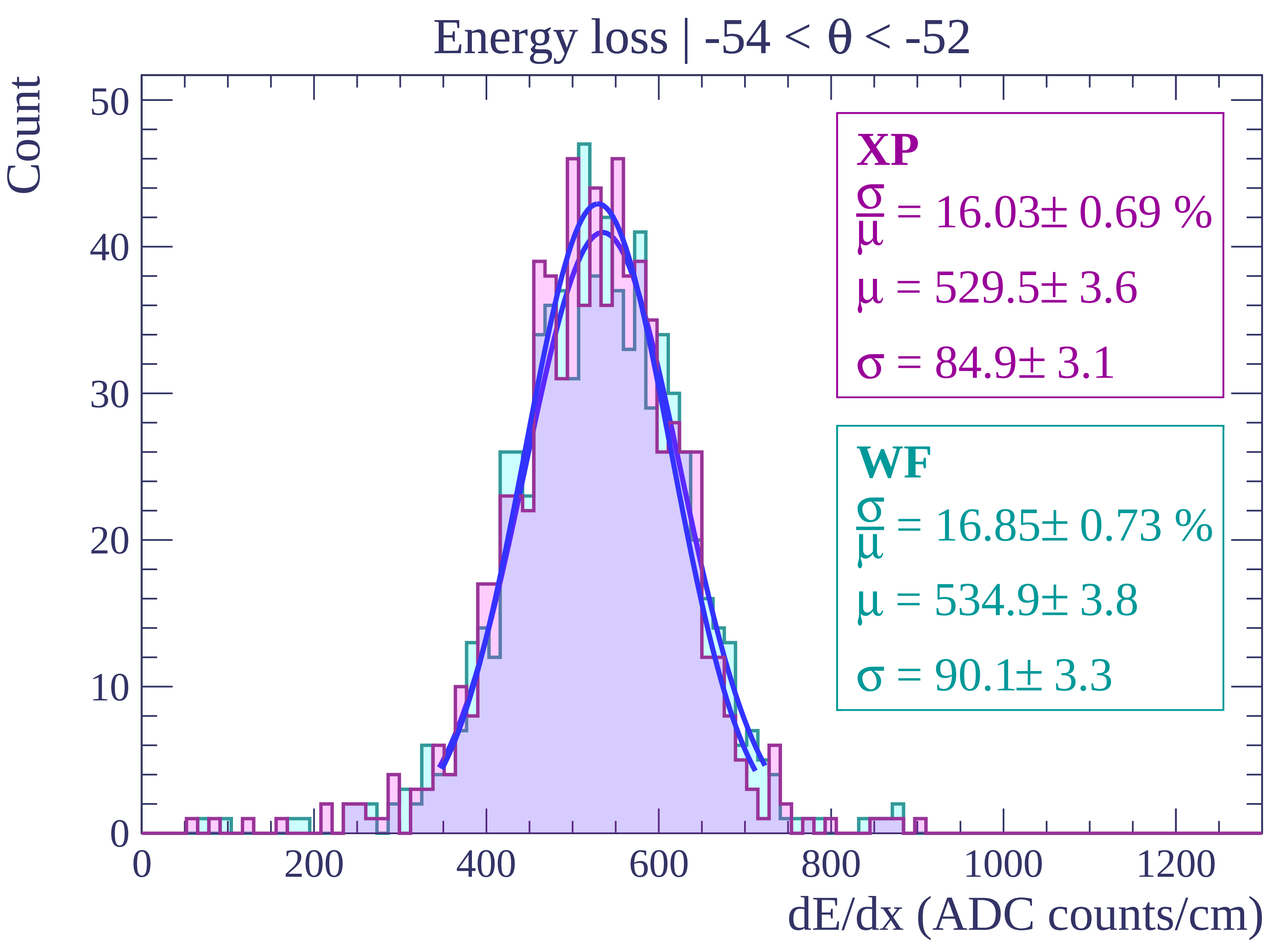

Energy loss | -60 < θ < -58
Count
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 13.37± 1.37 %
μ = 537.3± 5.9
σ = 71.8± 6.6
WF
σ/μ = 14.11± 1.25 %
μ = 537.7± 6.1
σ = 75.9± 5.9

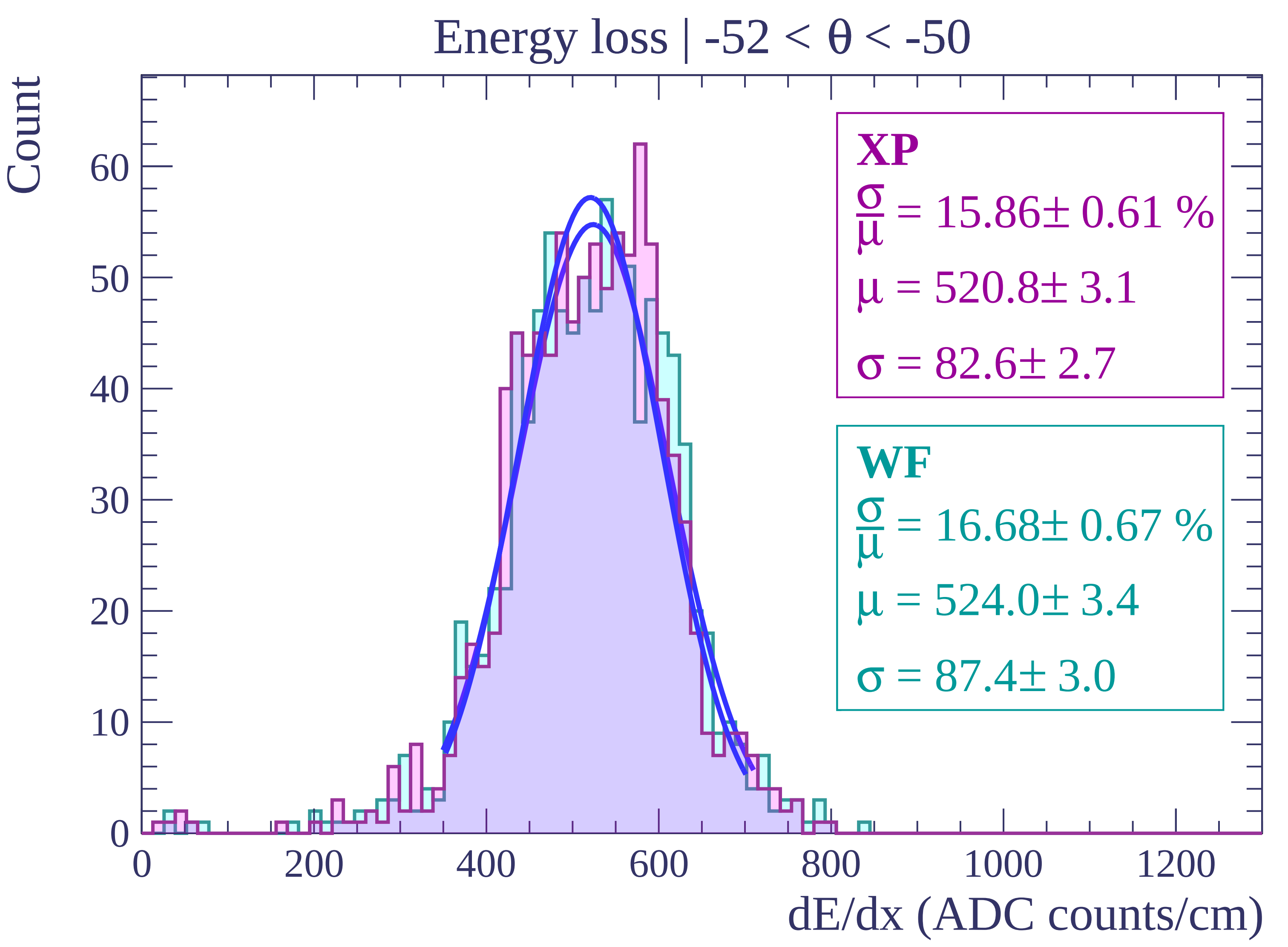


Energy loss | -58 < θ < -56
Count
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.35± 1.25 %
μ = 522.9± 4.9
σ = 75.0± 5.9
WF
σ/μ = 14.47± 1.16 %
μ = 532.2± 4.9
σ = 77.0± 5.4

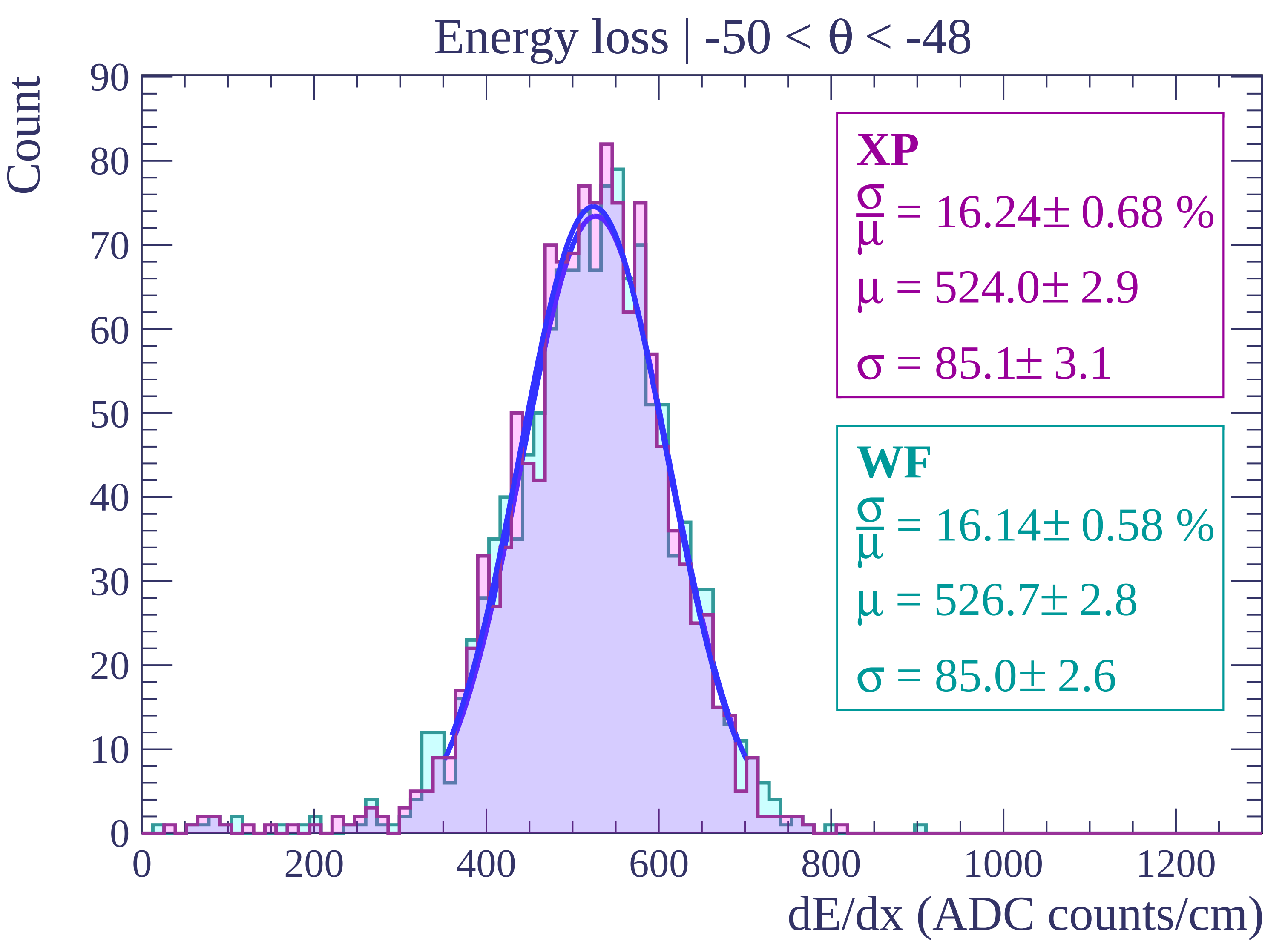


Energy loss | -56 < θ < -54
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
40
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 13.76± 0.77 %
μ = 526.0± 3.9
σ = 72.4± 3.5
WF
$\frac{\sigma}{\mu}$ = 14.81± 0.93 %
μ = 523.3± 4.3
σ = 77.5± 4.2

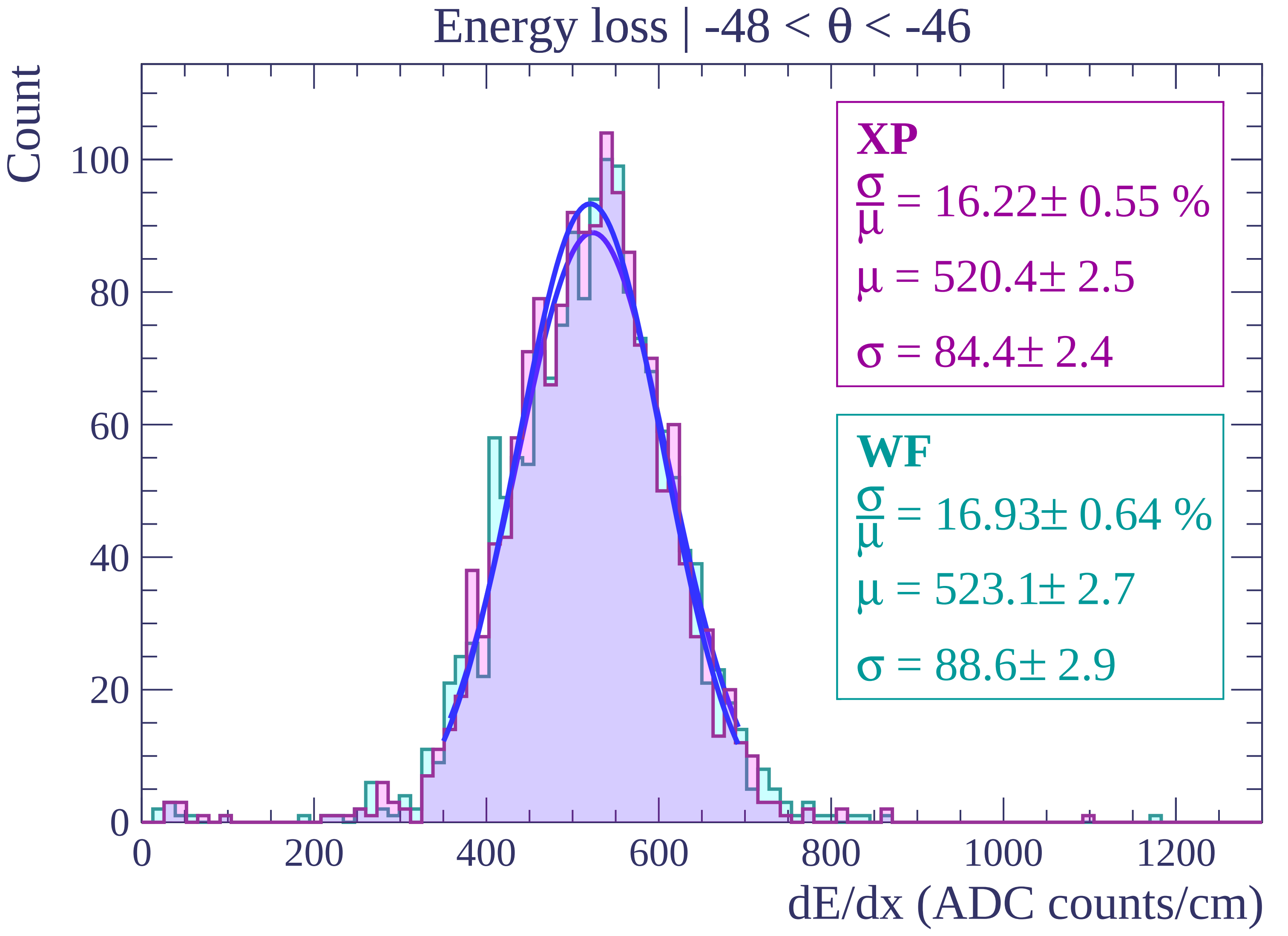

Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
XP
σ/μ = 16.03± 0.69 %
μ = 529.5± 3.6
σ = 84.9± 3.1
WF
σ/μ = 16.85± 0.73 %
μ = 534.9± 3.8
σ = 90.1± 3.3



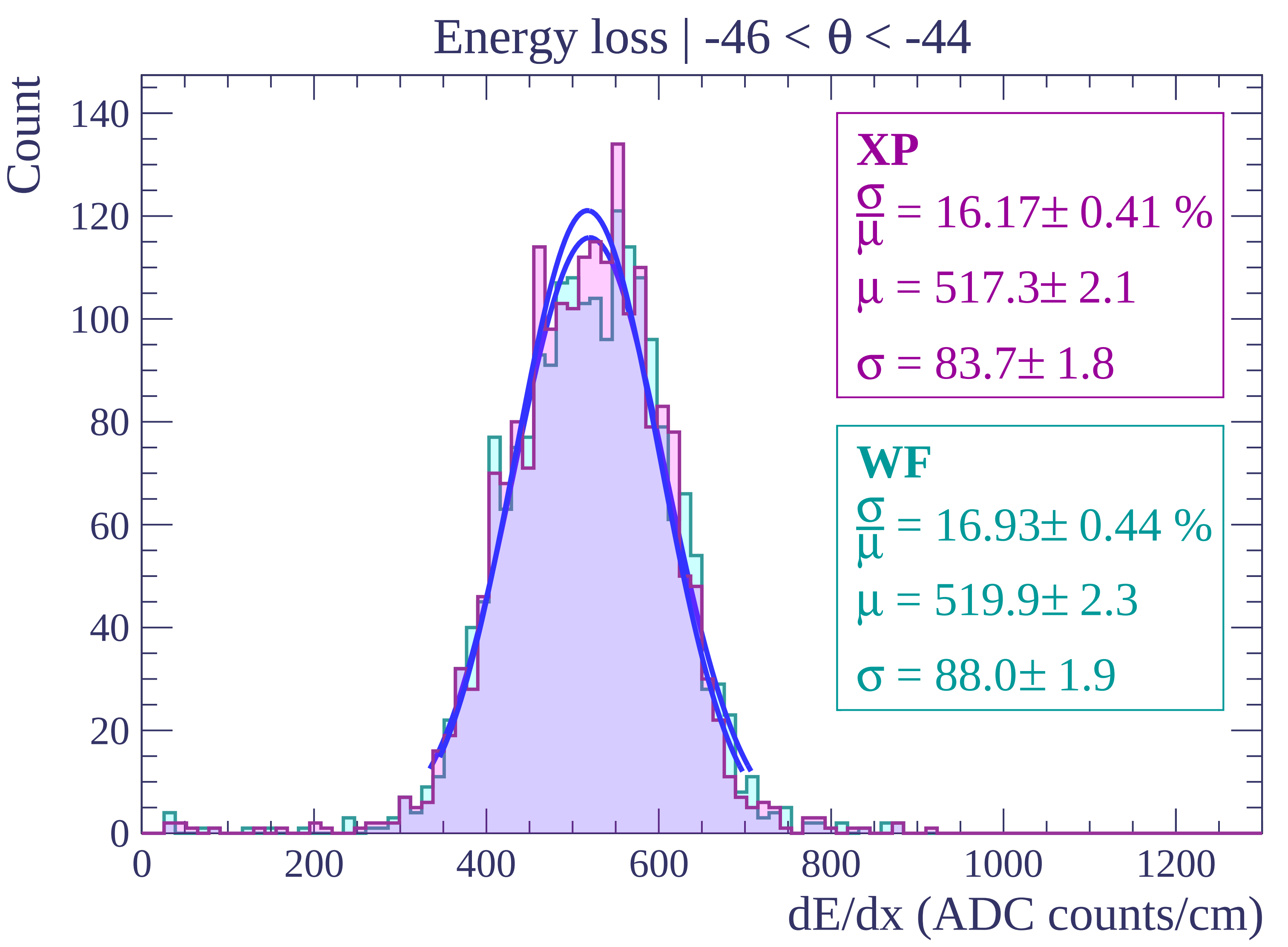
Energy loss | -52 < θ < -50
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 15.86± 0.61 %
μ = 520.8± 3.1
σ = 82.6± 2.7
WF
σ/μ = 16.68± 0.67 %
μ = 524.0± 3.4
σ = 87.4± 3.0

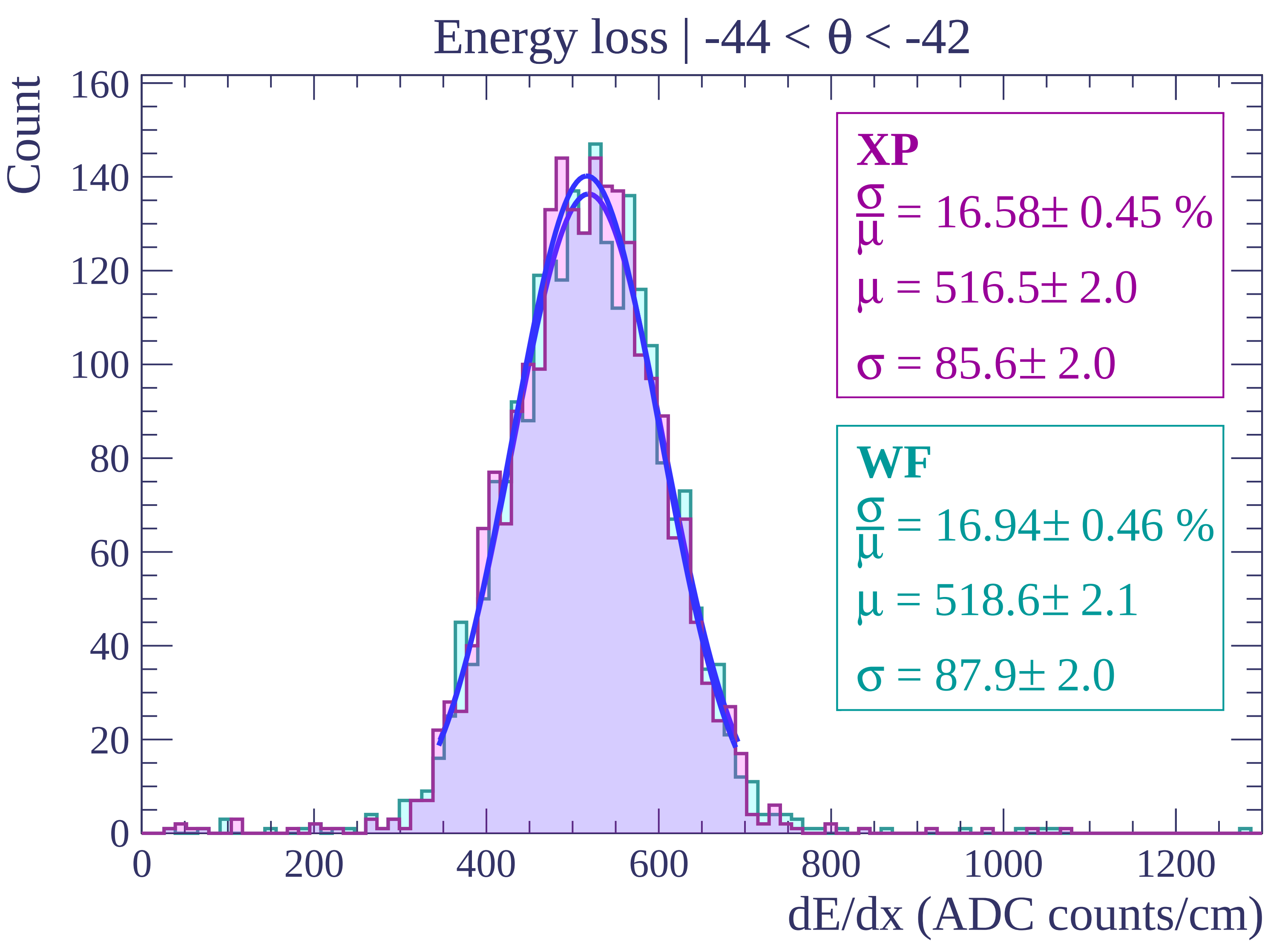


Energy loss | -50 < θ < -48
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 16.24± 0.68 %
μ = 524.0± 2.9
σ = 85.1± 3.1
WF
$\frac{\sigma}{\mu}$ = 16.14± 0.58 %
μ = 526.7± 2.8
σ = 85.0± 2.6

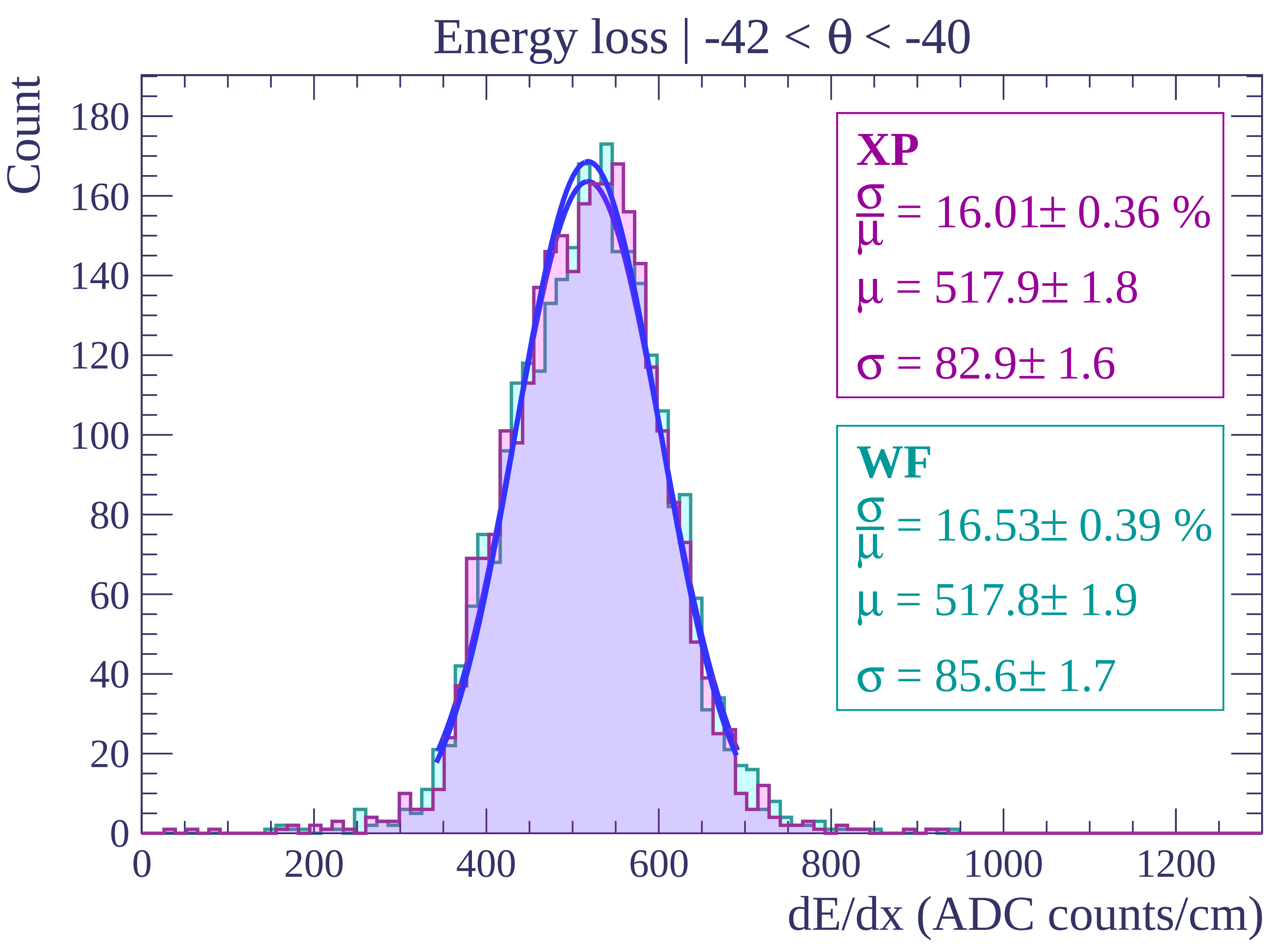


Energy loss | -48 < θ < -46
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
σ/μ = 16.22± 0.55 %
μ = 520.4± 2.5
σ = 84.4± 2.4
WF
σ/μ = 16.93± 0.64 %
μ = 523.1± 2.7
σ = 88.6± 2.9

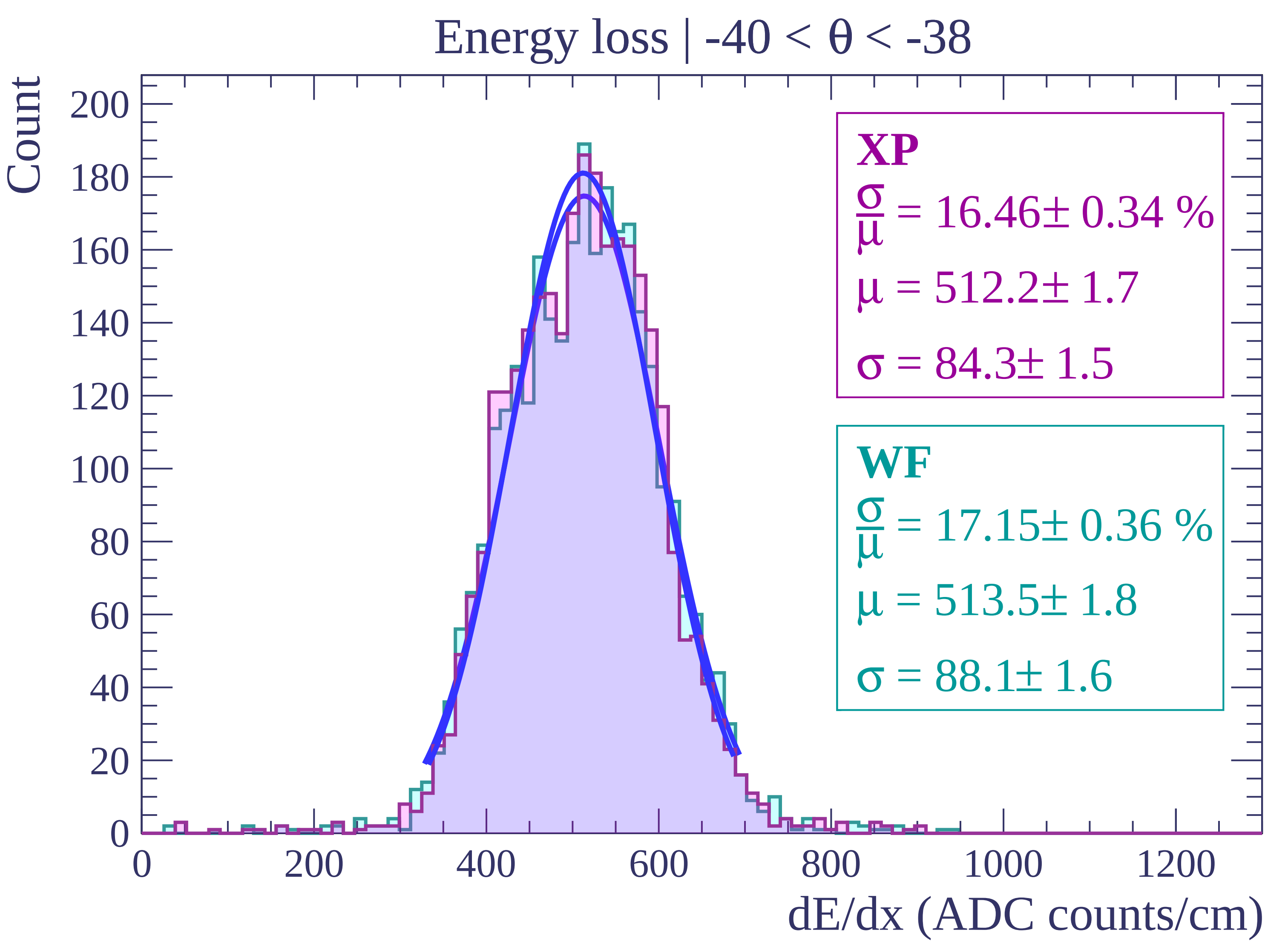

Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 16.17± 0.41 %
μ = 517.3± 2.1
σ = 83.7± 1.8
WF
σ/μ = 16.93± 0.44 %
μ = 519.9± 2.3
σ = 88.0± 1.9



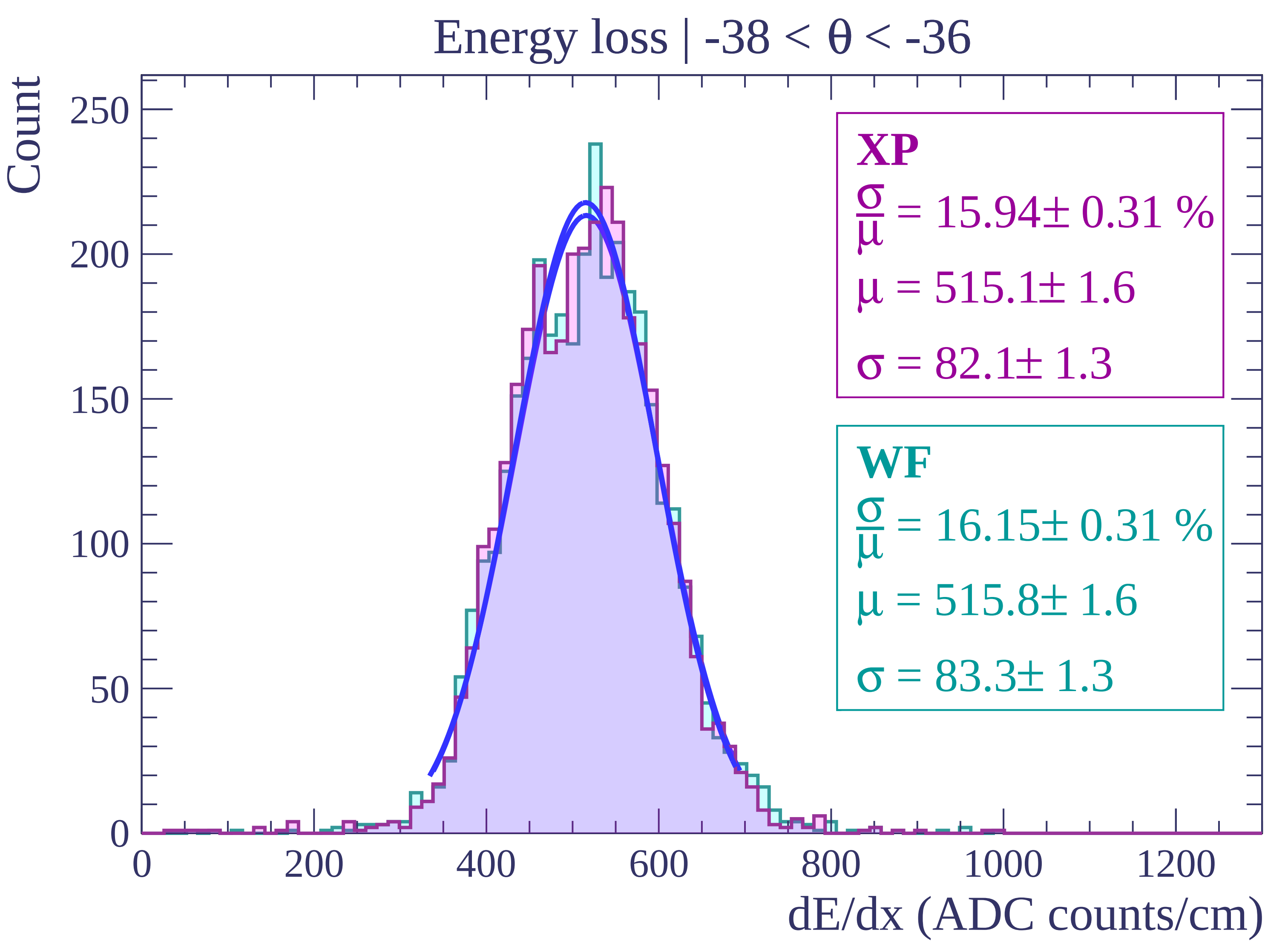
Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.58± 0.45 %
μ = 516.5± 2.0
σ = 85.6± 2.0
WF
σ/μ = 16.94± 0.46 %
μ = 518.6± 2.1
σ = 87.9± 2.0

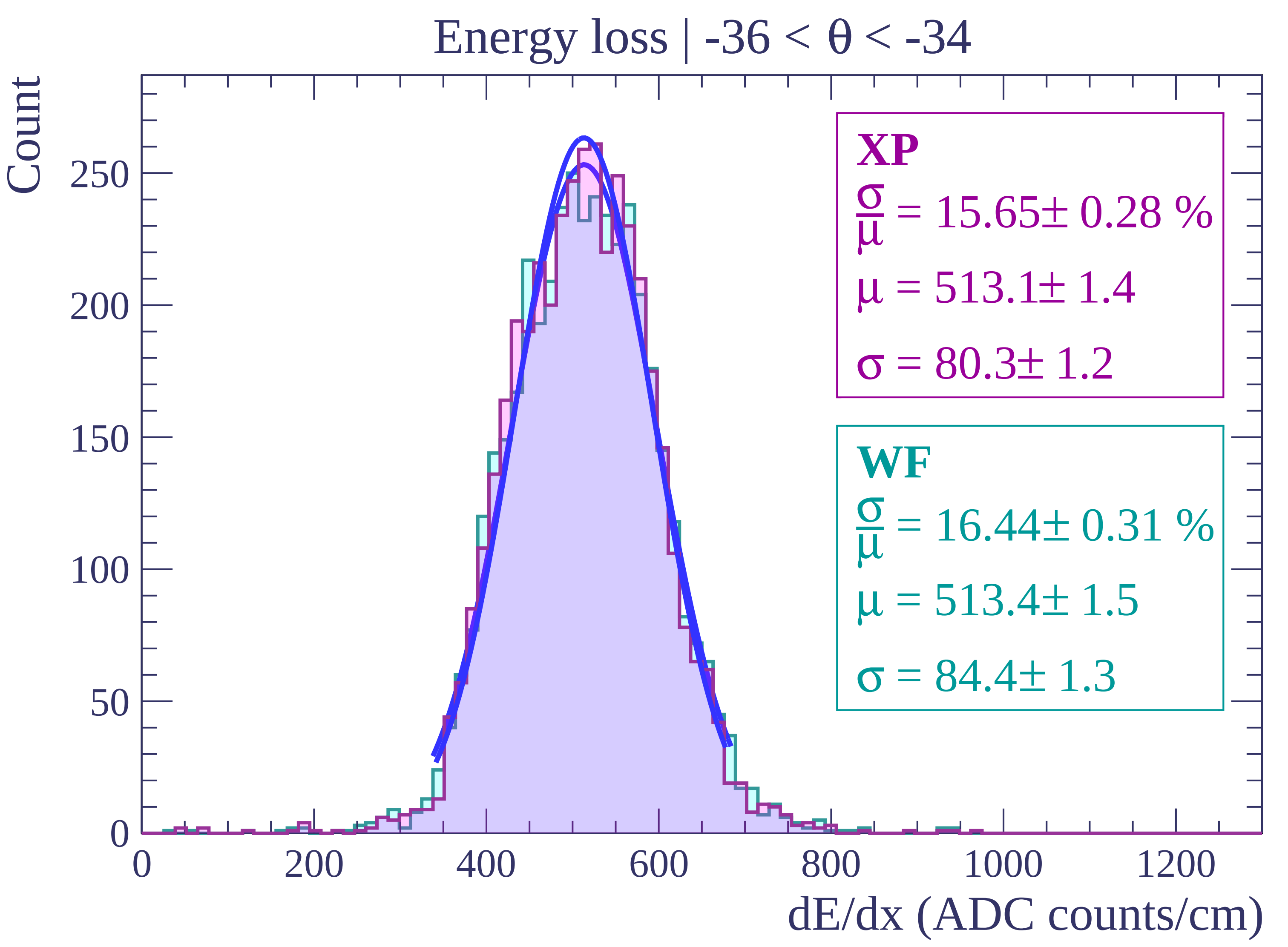


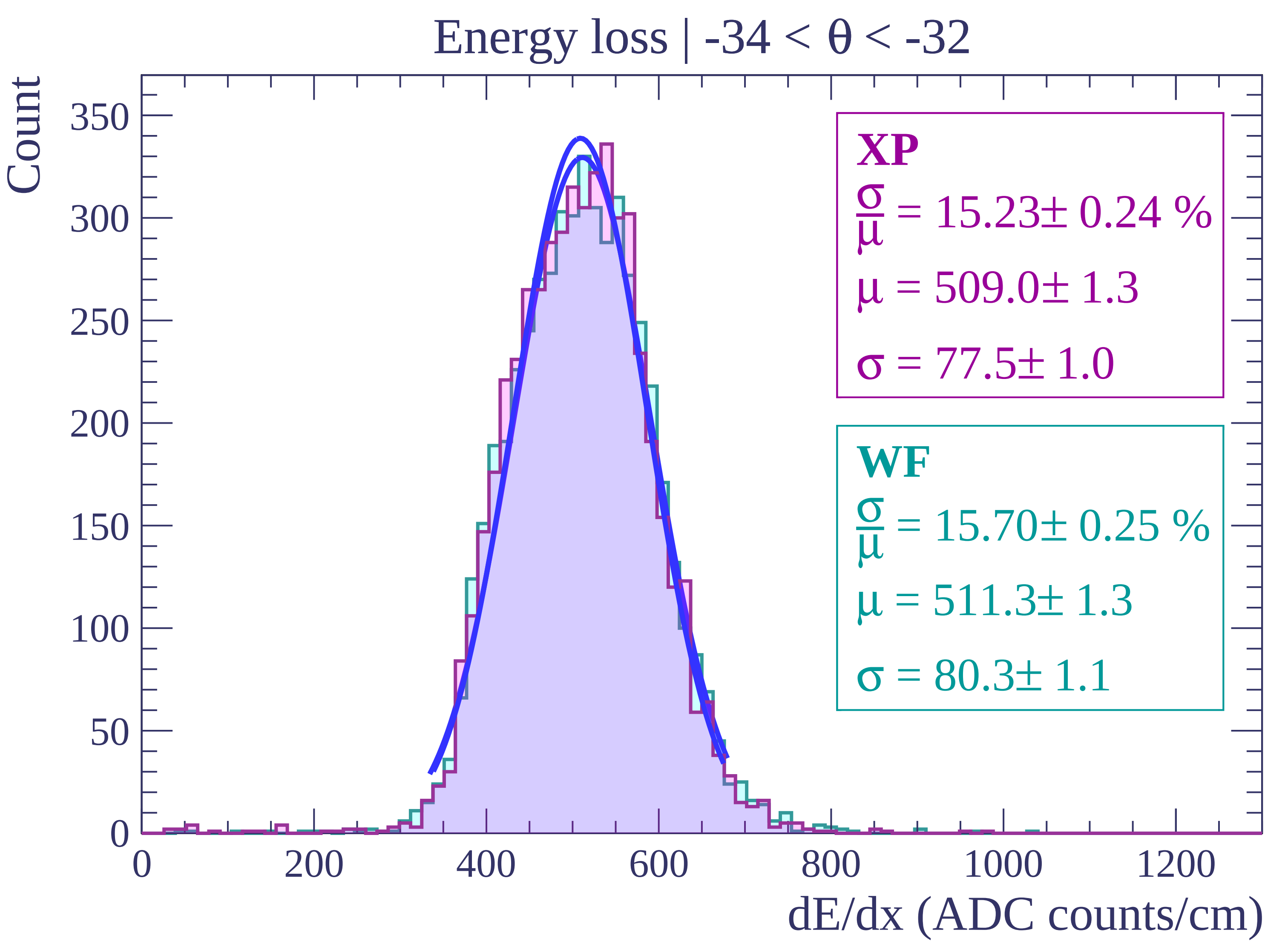
Energy loss | -42 < θ < -40
Count
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 16.01± 0.36 %
μ = 517.9± 1.8
σ = 82.9± 1.6
WF
$\frac{\sigma}{\mu}$ = 16.53± 0.39 %
μ = 517.8± 1.9
σ = 85.6± 1.7

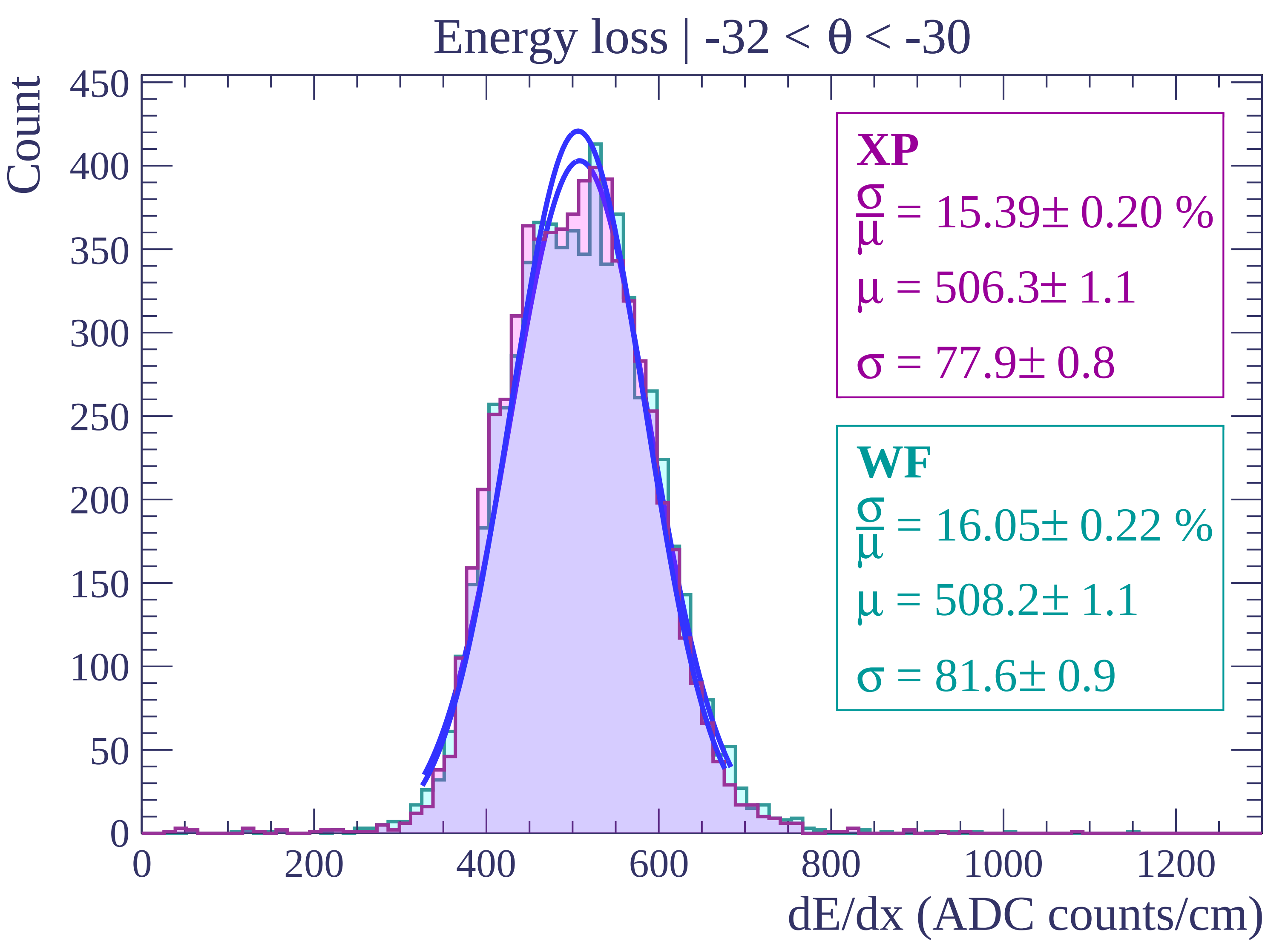


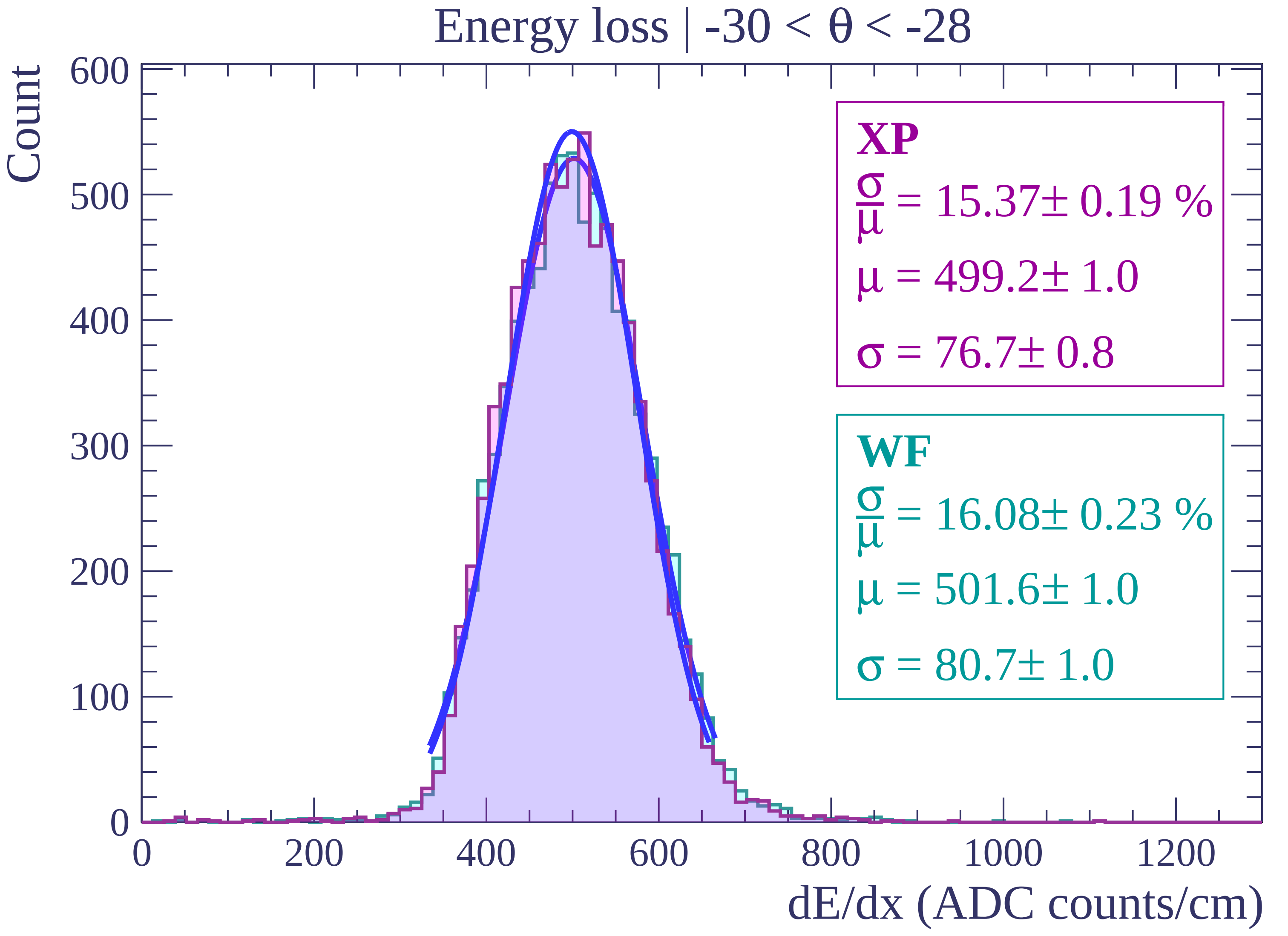
Energy loss | -40 < θ < -38
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
$\frac{\sigma}{\mu}$ = 16.46± 0.34 %
μ = 512.2± 1.7
σ = 84.3± 1.5
WF
$\frac{\sigma}{\mu}$ = 17.15± 0.36 %
μ = 513.5± 1.8
σ = 88.1± 1.6


Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
XP
σ/μ = 15.94± 0.31 %
μ = 515.1± 1.6
σ = 82.1± 1.3
WF
σ/μ = 16.15± 0.31 %
μ = 515.8± 1.6
σ = 83.3± 1.3

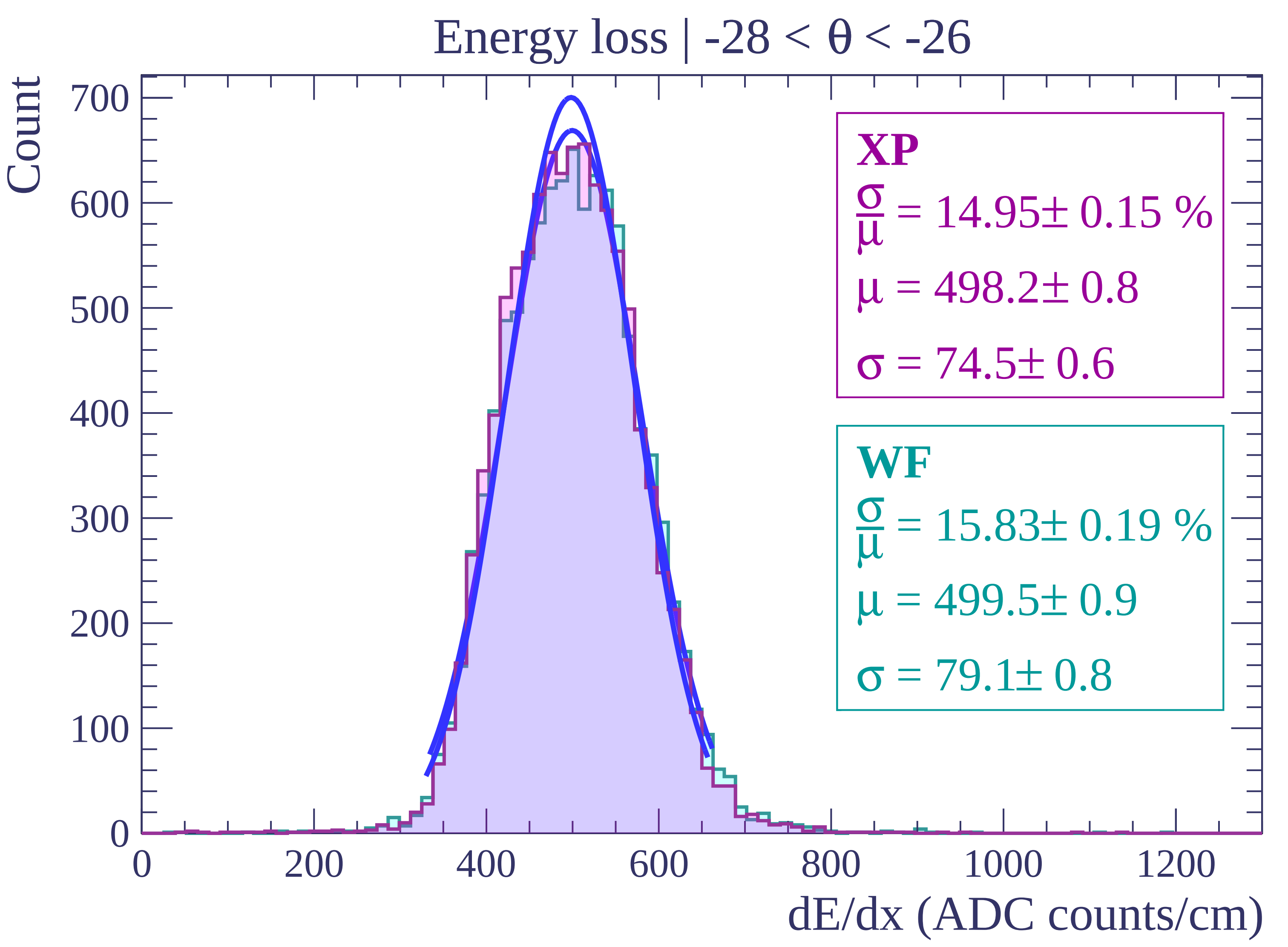

Energy loss | -36 < θ < -34
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.65± 0.28 %
μ = 513.1± 1.4
σ = 80.3± 1.2
WF
σ/μ = 16.44± 0.31 %
μ = 513.4± 1.5
σ = 84.4± 1.3


Energy loss | -34 < θ < -32
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.23± 0.24 %
μ = 509.0± 1.3
σ = 77.5± 1.0
WF
σ/μ = 15.70± 0.25 %
μ = 511.3± 1.3
σ = 80.3± 1.1
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200

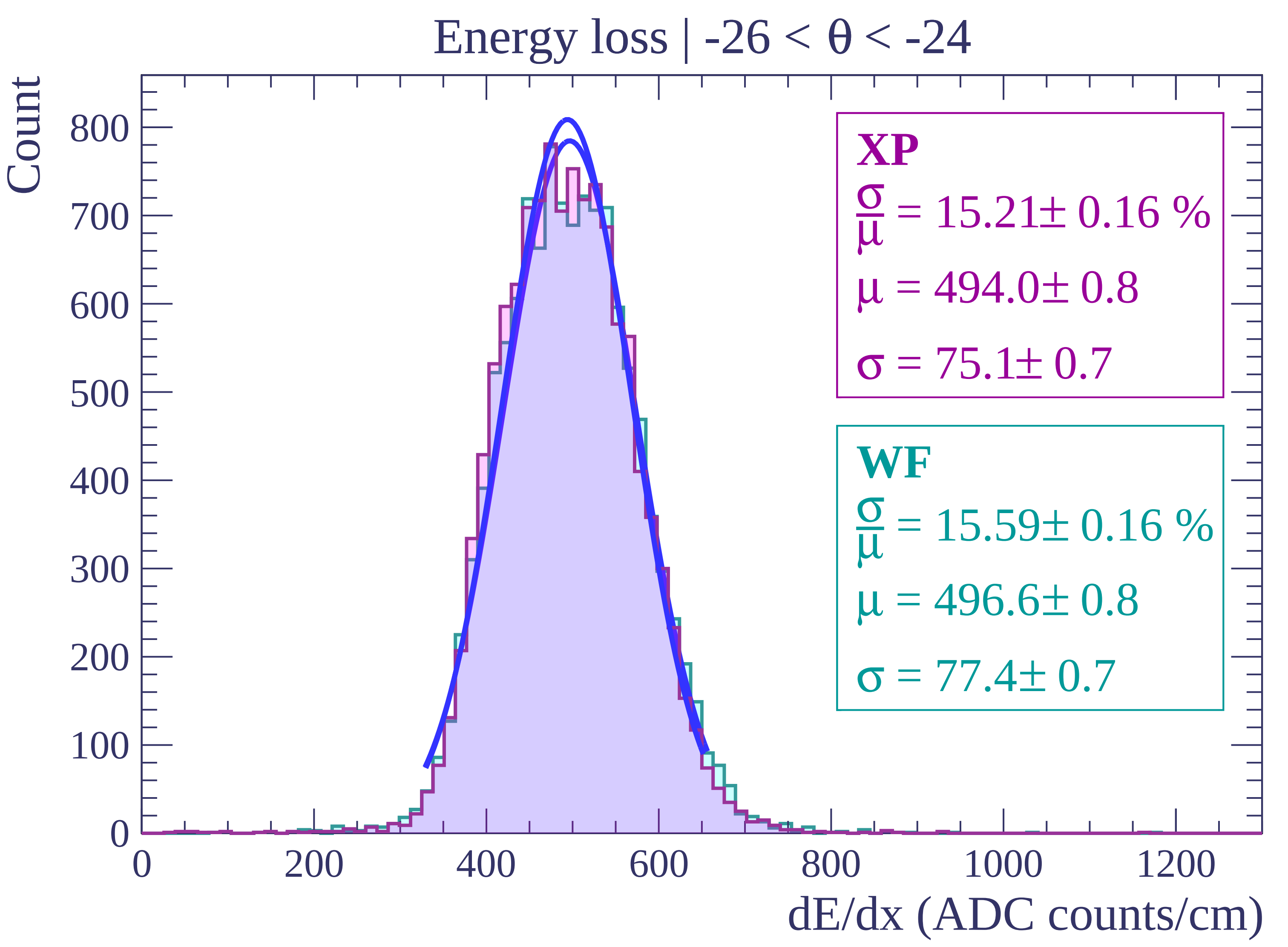

Energy loss | -32 < θ < -30
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 15.39± 0.20 %
μ = 506.3± 1.1
σ = 77.9± 0.8
WF
σ/μ = 16.05± 0.22 %
μ = 508.2± 1.1
σ = 81.6± 0.9



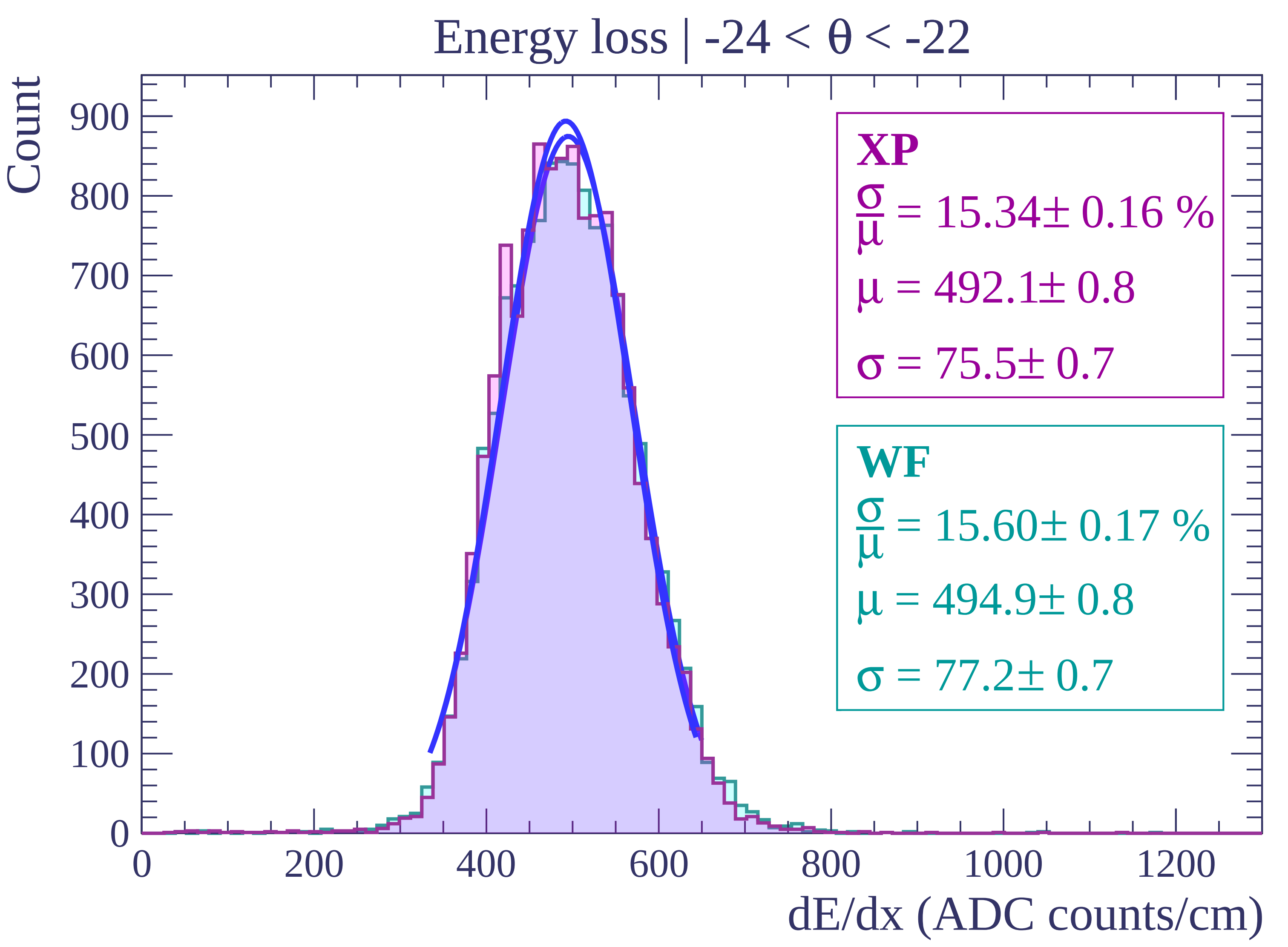
Energy loss | -30 < θ < -28
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.37± 0.19 %
μ = 499.2± 1.0
σ = 76.7± 0.8
WF
σ/μ = 16.08± 0.23 %
μ = 501.6± 1.0
σ = 80.7± 1.0



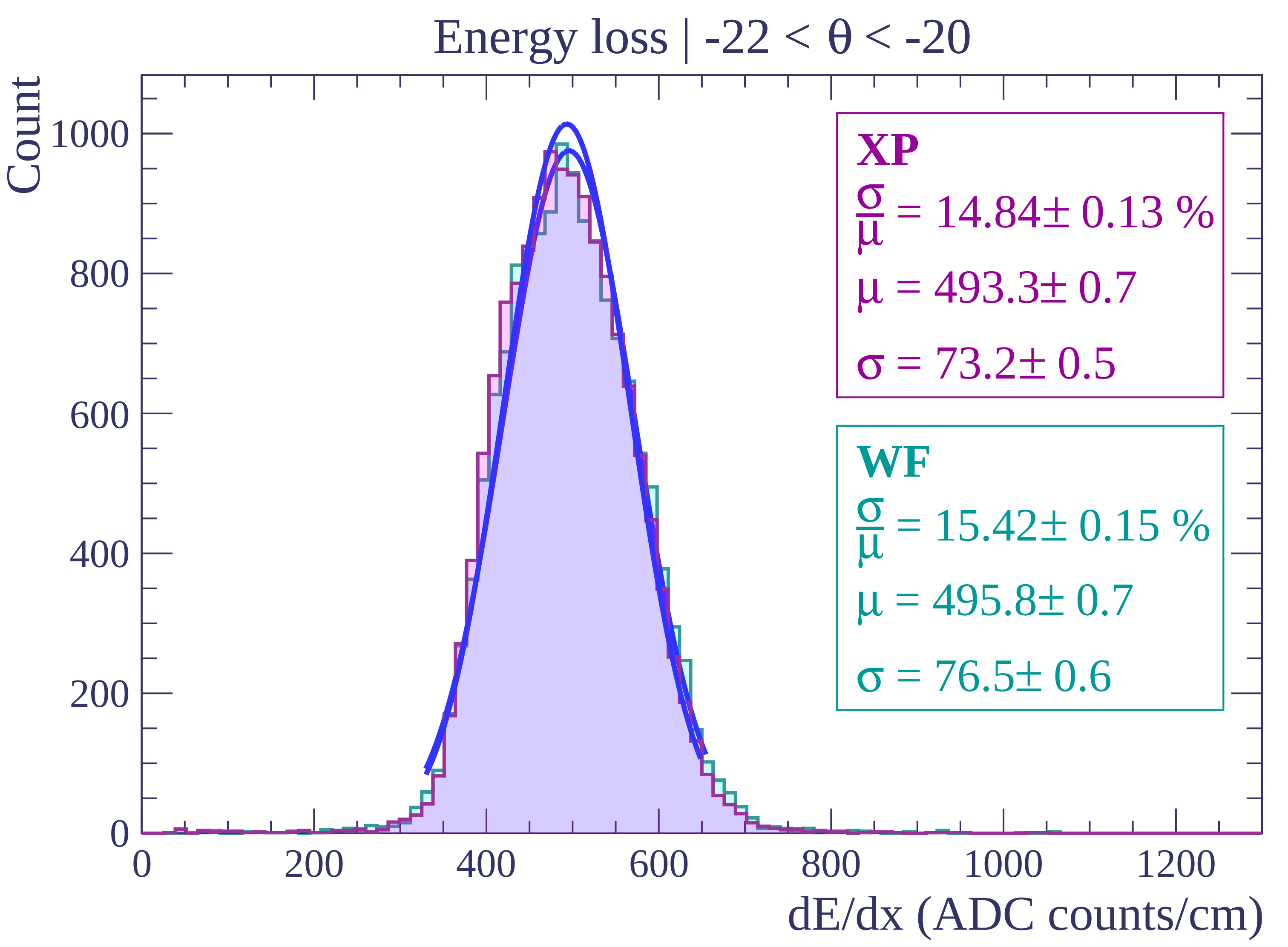
Energy loss | -28 < θ < -26
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 14.95± 0.15 %
μ = 498.2± 0.8
σ = 74.5± 0.6
WF
σ/μ = 15.83± 0.19 %
μ = 499.5± 0.9
σ = 79.1± 0.8



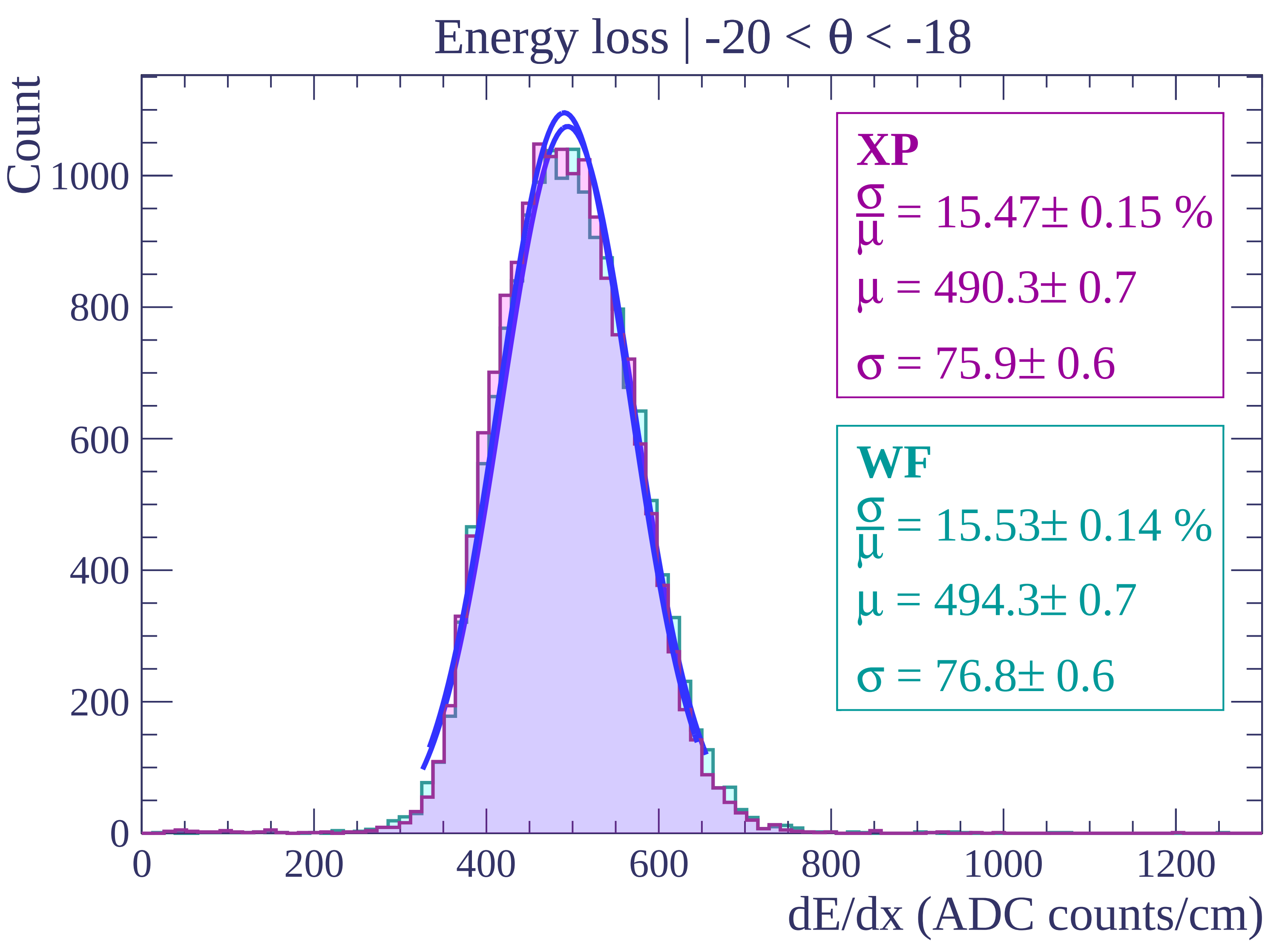
Energy loss | -26 < θ < -24
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.21± 0.16 %
μ = 494.0± 0.8
σ = 75.1± 0.7
WF
σ/μ = 15.59± 0.16 %
μ = 496.6± 0.8
σ = 77.4± 0.7
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200

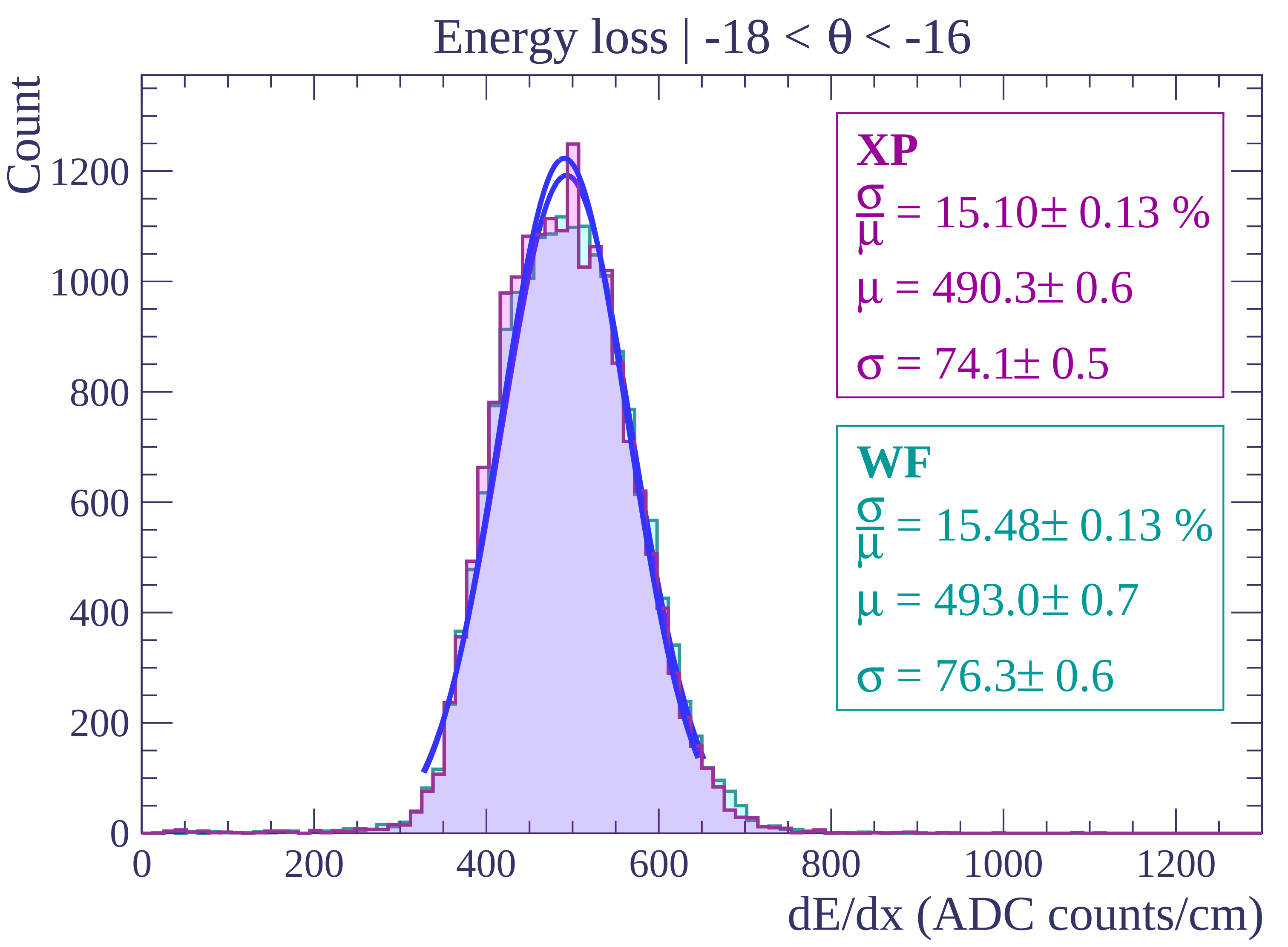


Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
σ/μ = 15.34± 0.16 %
μ = 492.1± 0.8
σ = 75.5± 0.7
WF
σ/μ = 15.60± 0.17 %
μ = 494.9± 0.8
σ = 77.2± 0.7

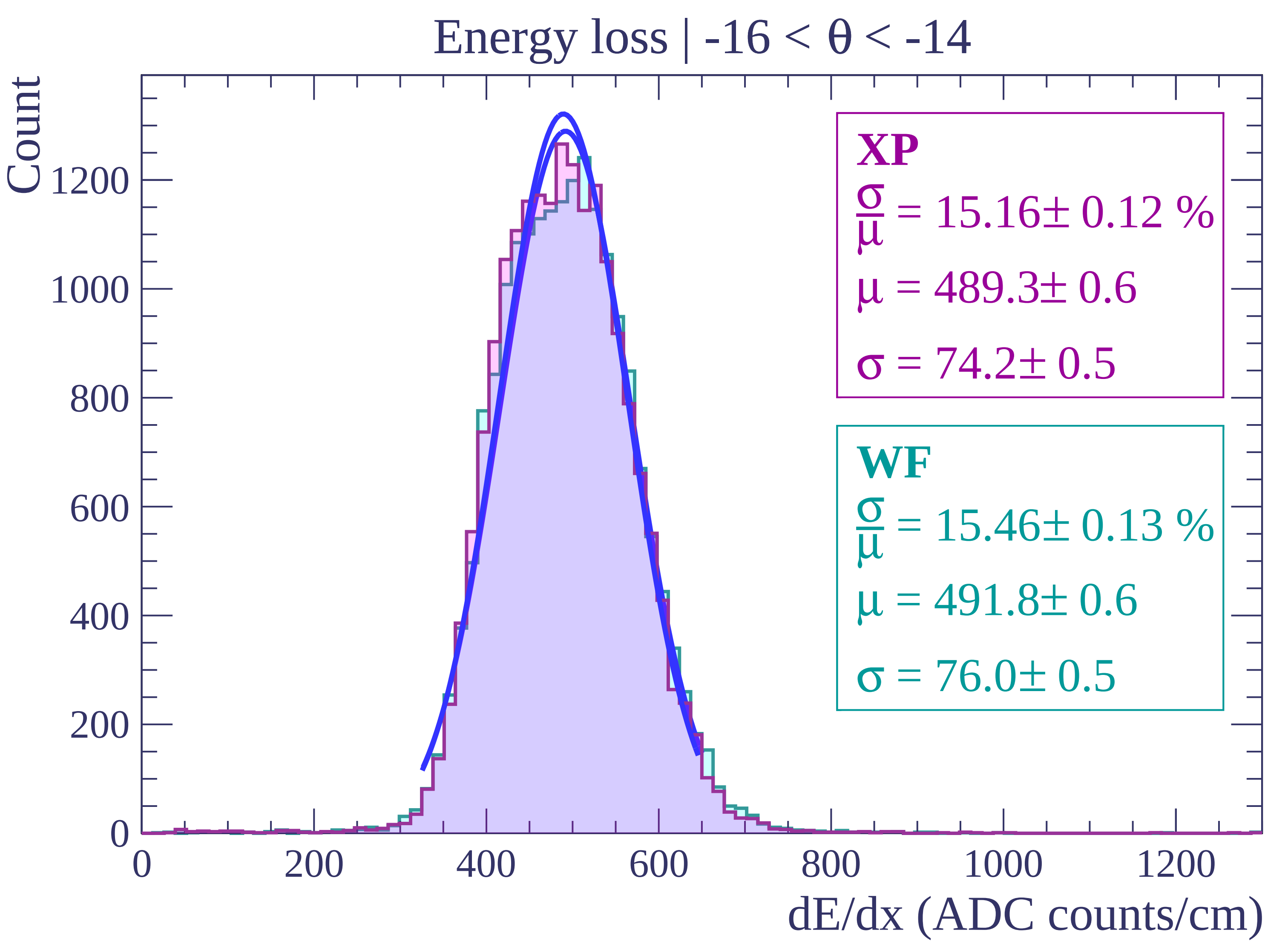


Energy loss | -22 < θ < -20
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.84± 0.13 %
μ = 493.3± 0.7
σ = 73.2± 0.5
WF
σ/μ = 15.42± 0.15 %
μ = 495.8± 0.7
σ = 76.5± 0.6

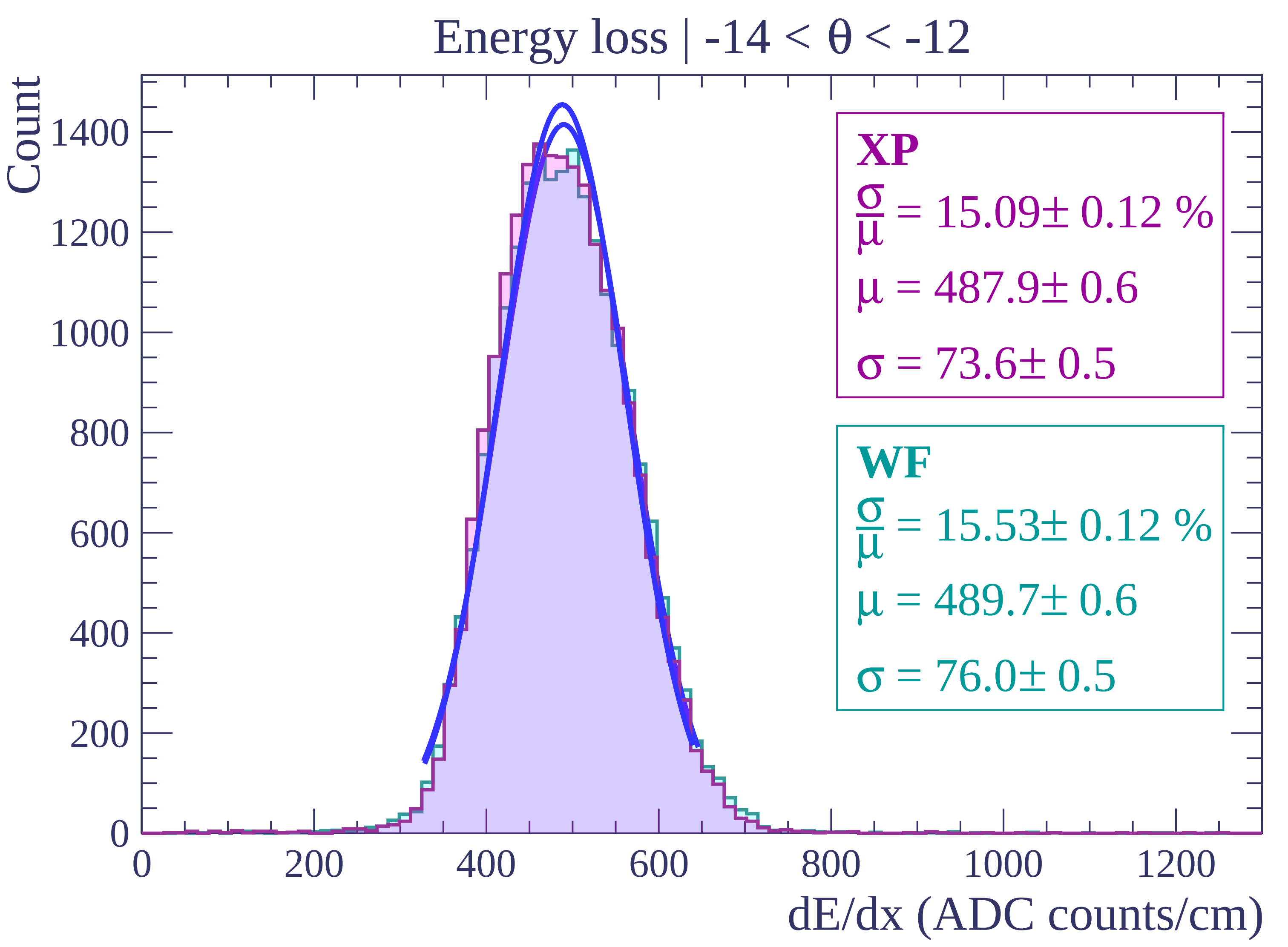

Energy loss | -20 < θ < -18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.47± 0.15 %
μ = 490.3± 0.7
σ = 75.9± 0.6
WF
σ/μ = 15.53± 0.14 %
μ = 494.3± 0.7
σ = 76.8± 0.6



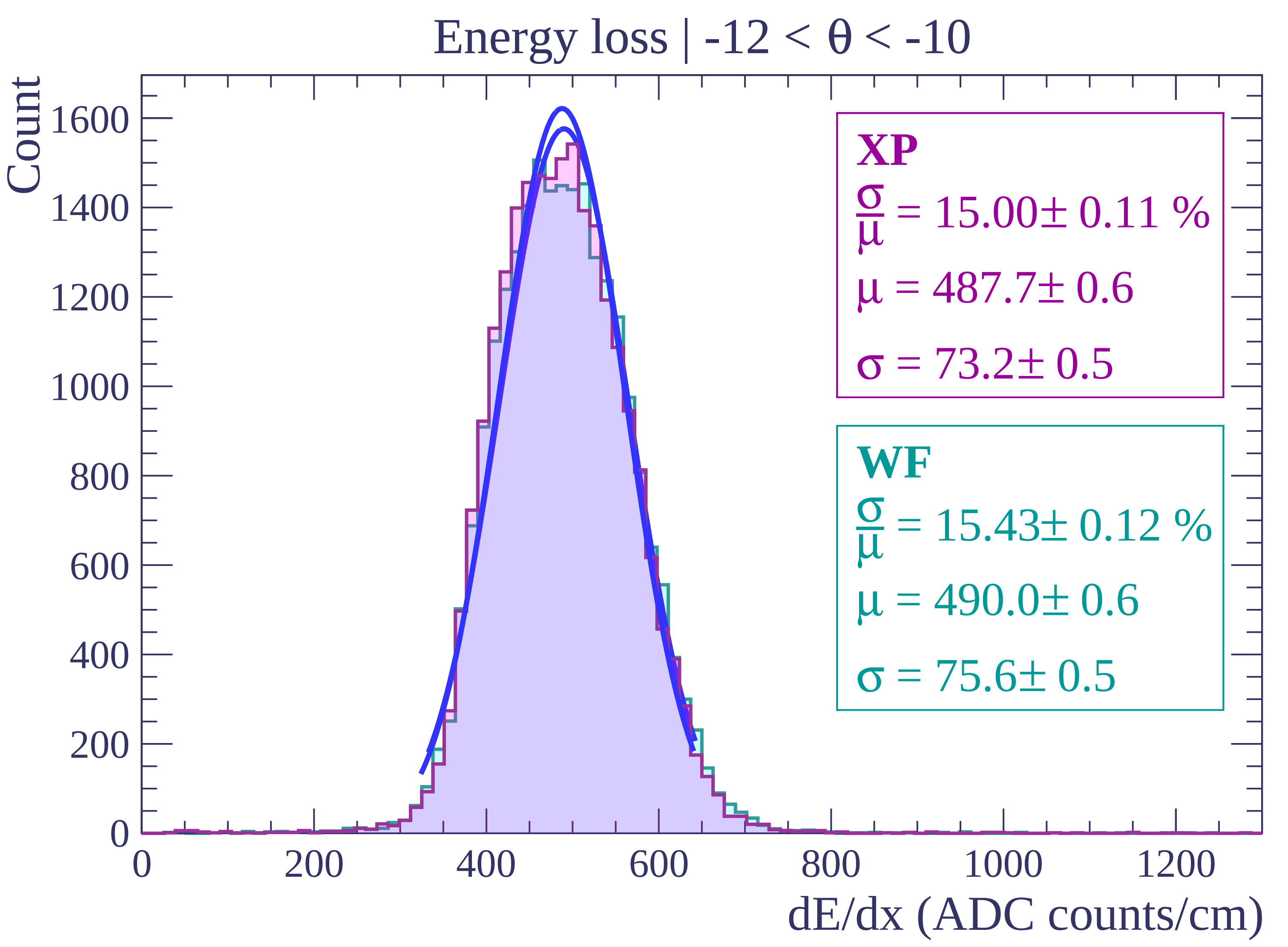
Energy loss | -18 < θ < -16
Count
0
200
400
600
800
1000
1200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.10± 0.13 %
μ = 490.3± 0.6
σ = 74.1± 0.5
WF
σ/μ = 15.48± 0.13 %
μ = 493.0± 0.7
σ = 76.3± 0.6



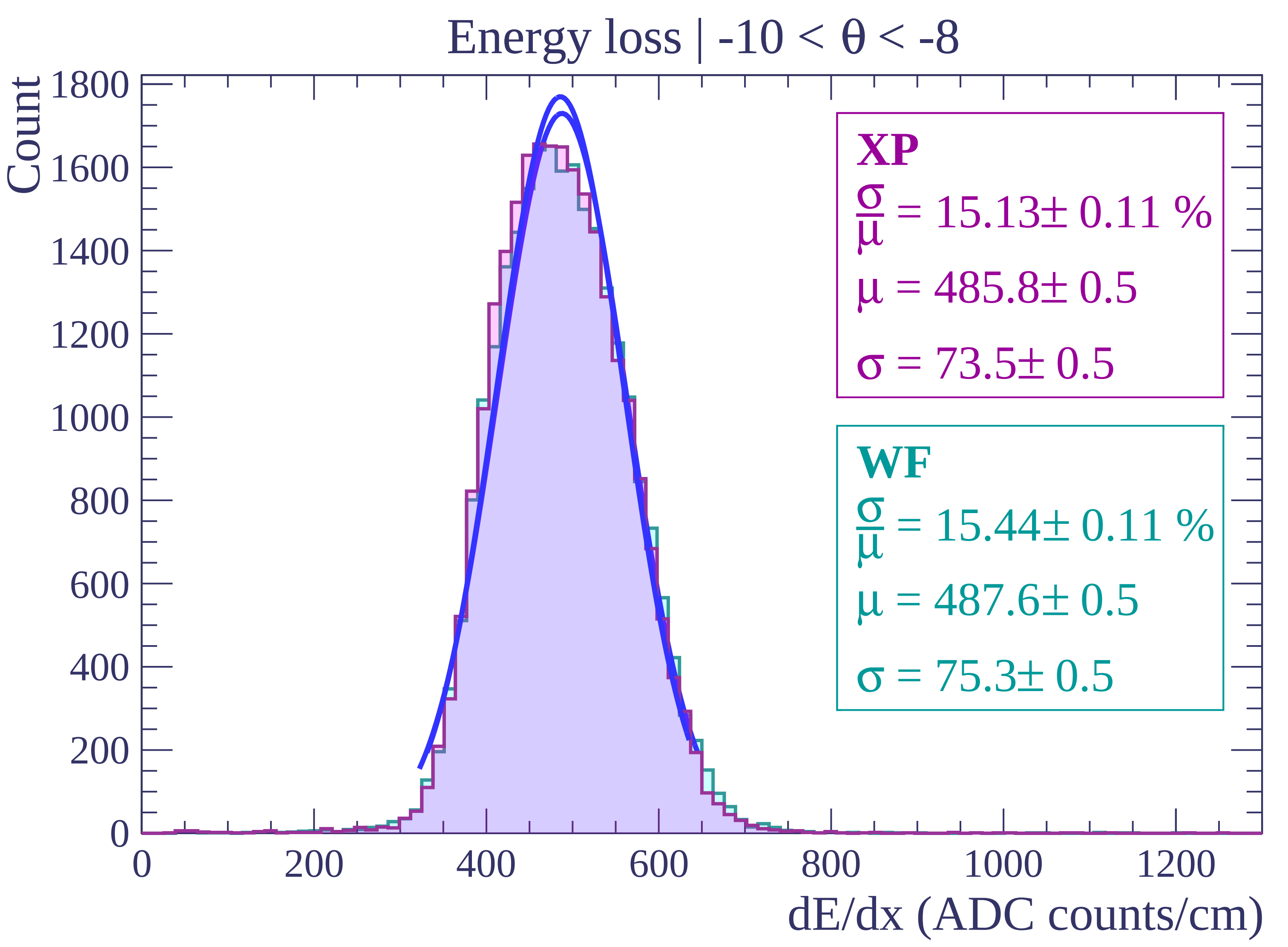
Energy loss | -16 < θ < -14
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.16± 0.12 %
μ = 489.3± 0.6
σ = 74.2± 0.5
WF
σ/μ = 15.46± 0.13 %
μ = 491.8± 0.6
σ = 76.0± 0.5

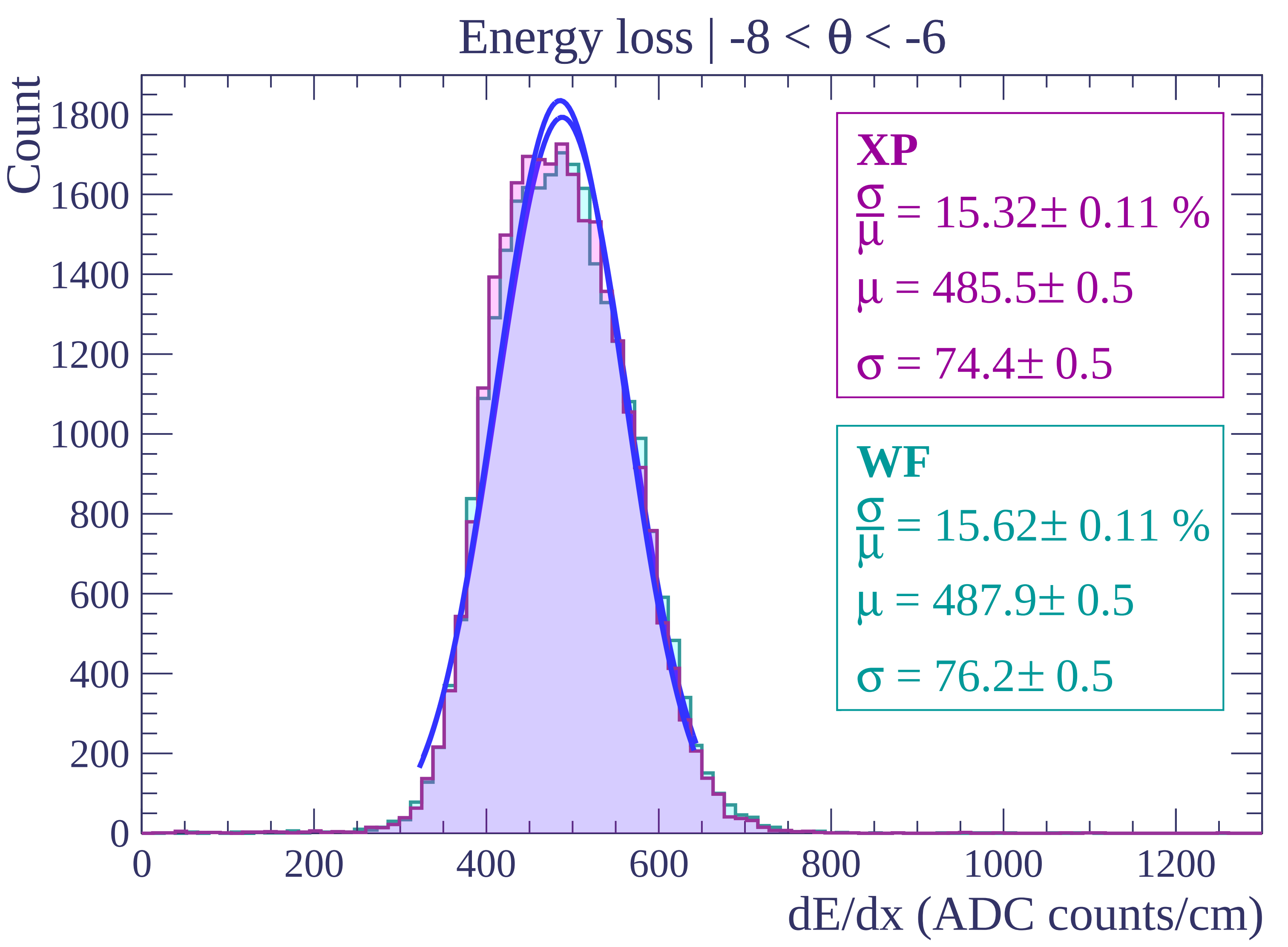


Energy loss | -14 < θ < -12
Count
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.09± 0.12 %
μ = 487.9± 0.6
σ = 73.6± 0.5
WF
σ/μ = 15.53± 0.12 %
μ = 489.7± 0.6
σ = 76.0± 0.5

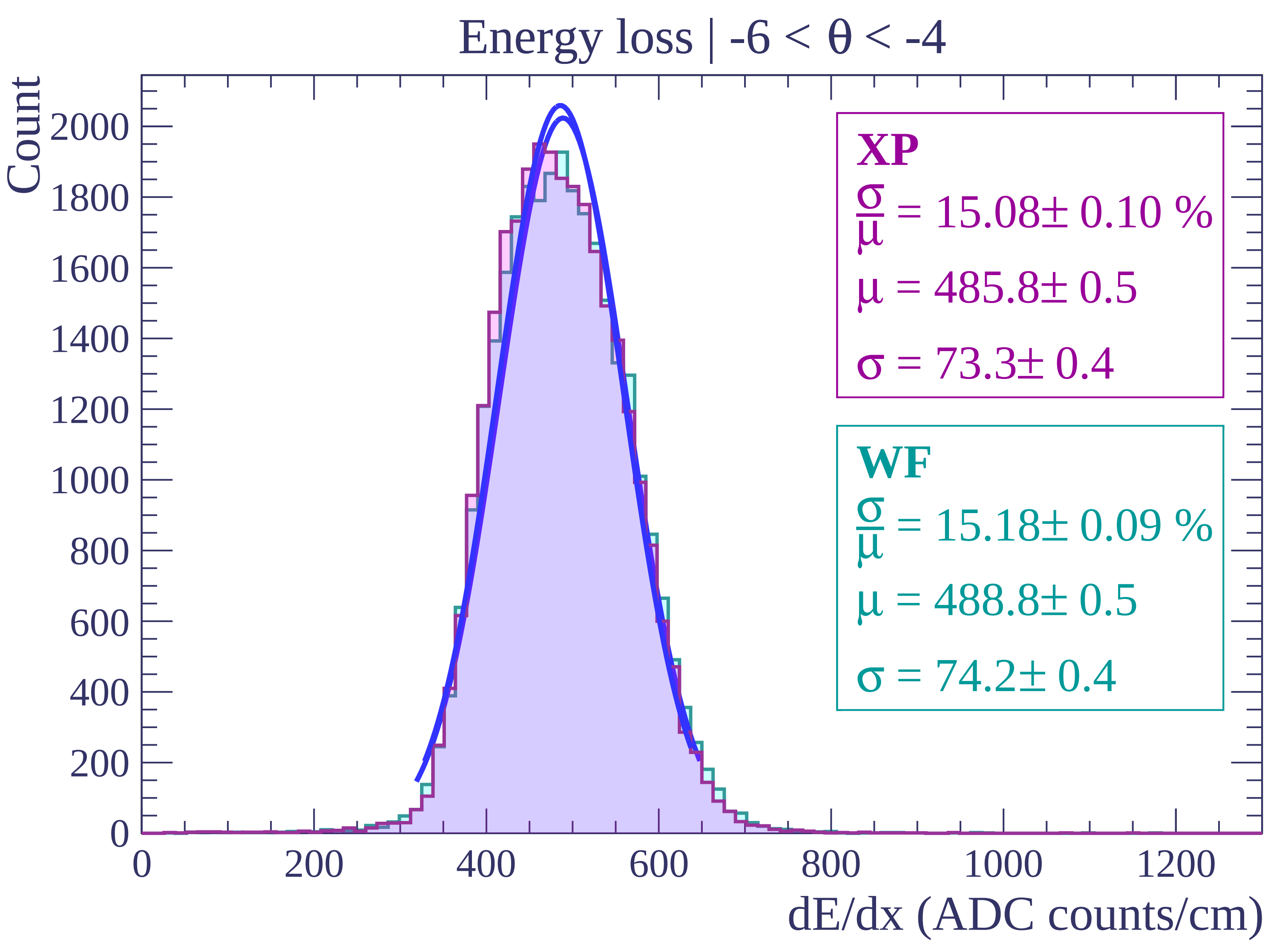


Energy loss | -12 < θ < -10
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.00± 0.11 %
μ = 487.7± 0.6
σ = 73.2± 0.5
WF
$\frac{\sigma}{\mu}$ = 15.43± 0.12 %
μ = 490.0± 0.6
σ = 75.6± 0.5

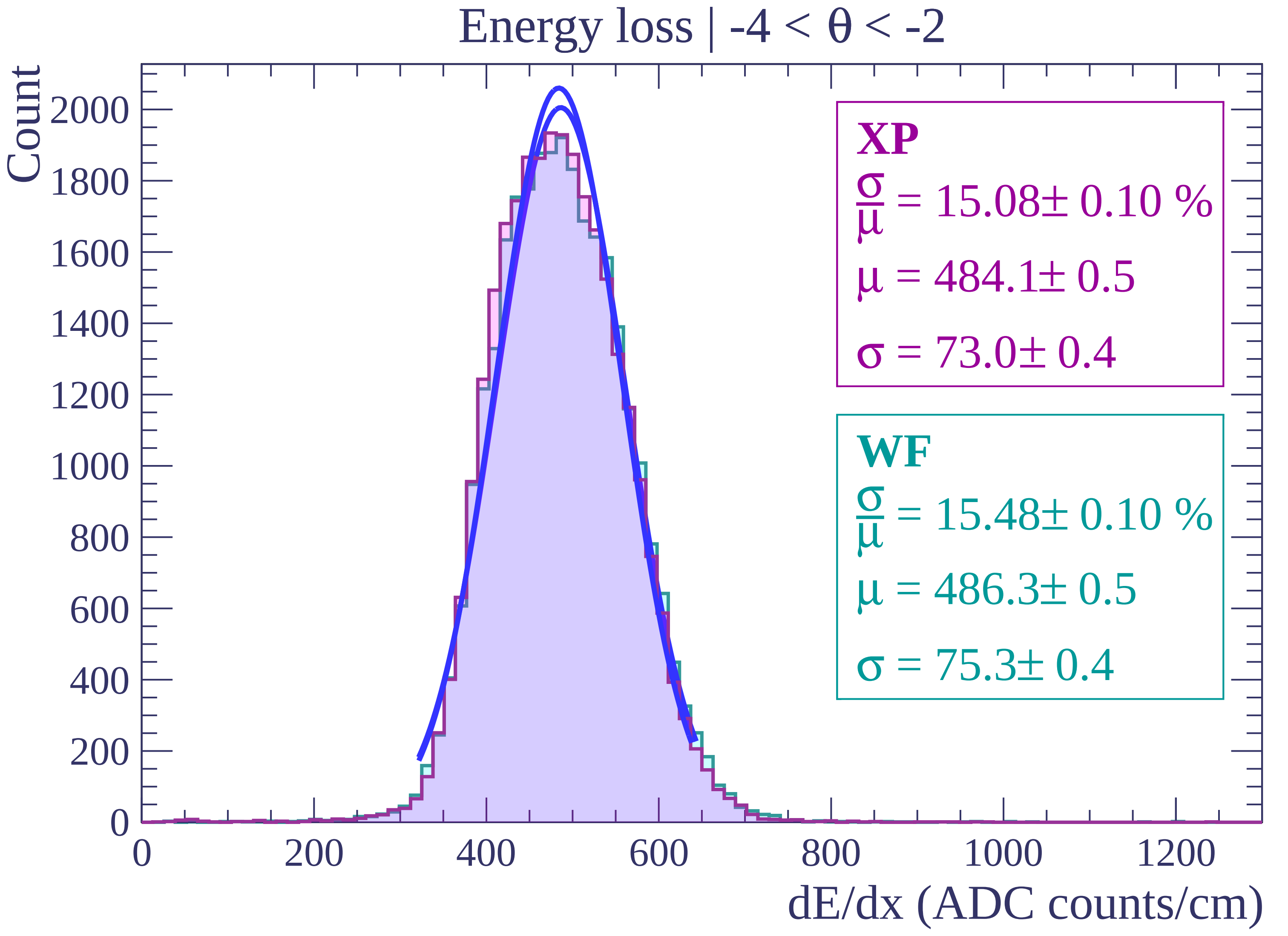

Energy loss | -10 < θ < -8
Count
0
200
400
600
800
1000
1200
1400
1600
1800
dE/dx (ADC counts/cm)
XP
σ/μ = 15.13± 0.11 %
μ = 485.8± 0.5
σ = 73.5± 0.5
WF
σ/μ = 15.44± 0.11 %
μ = 487.6± 0.5
σ = 75.3± 0.5



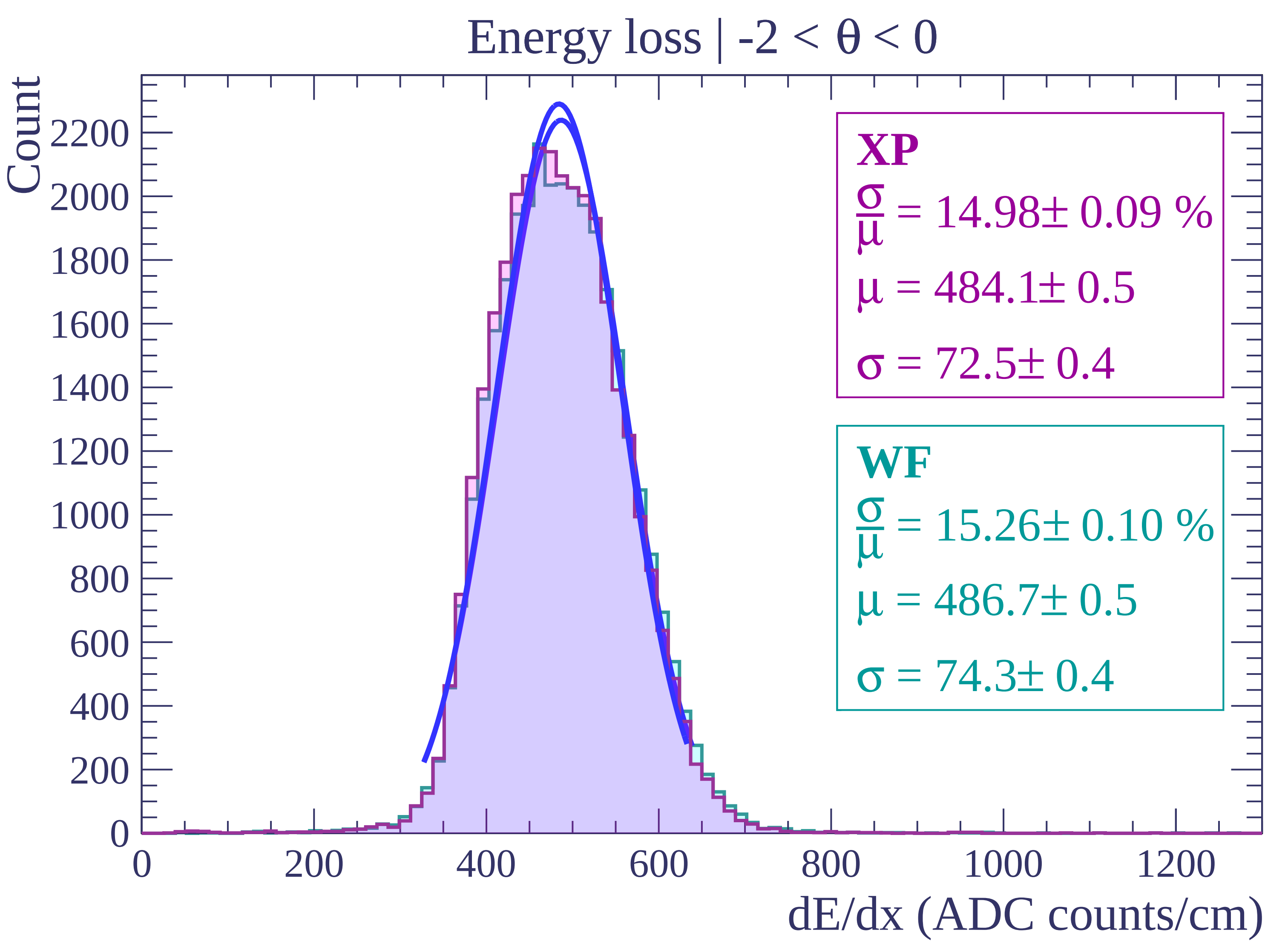
Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.32± 0.11 %
μ = 485.5± 0.5
σ = 74.4± 0.5
WF
$\frac{\sigma}{\mu}$ = 15.62± 0.11 %
μ = 487.9± 0.5
σ = 76.2± 0.5



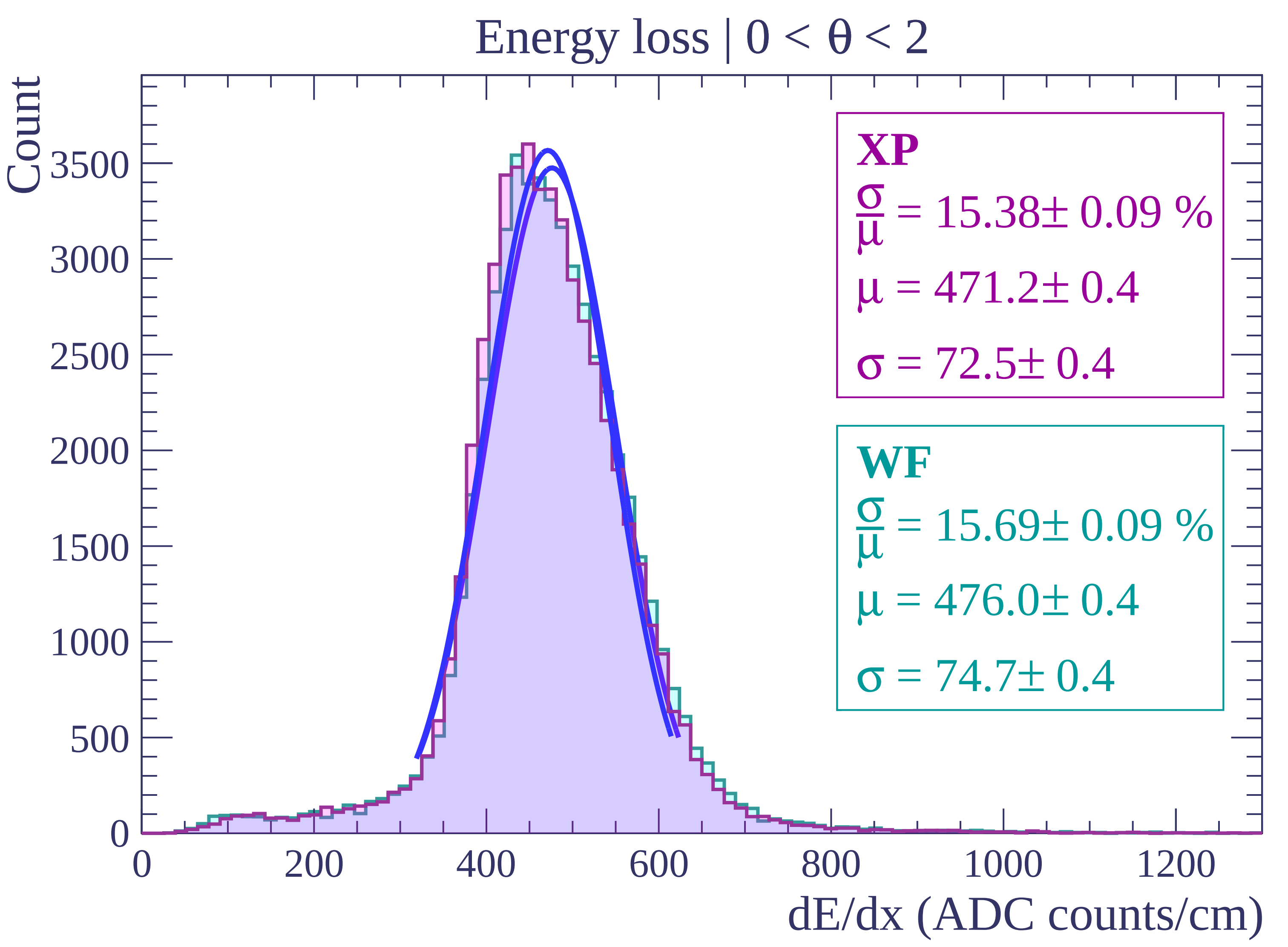
Energy loss | -6 < θ < -4
Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.08± 0.10 %
μ = 485.8± 0.5
σ = 73.3± 0.4
WF
σ/μ = 15.18± 0.09 %
μ = 488.8± 0.5
σ = 74.2± 0.4



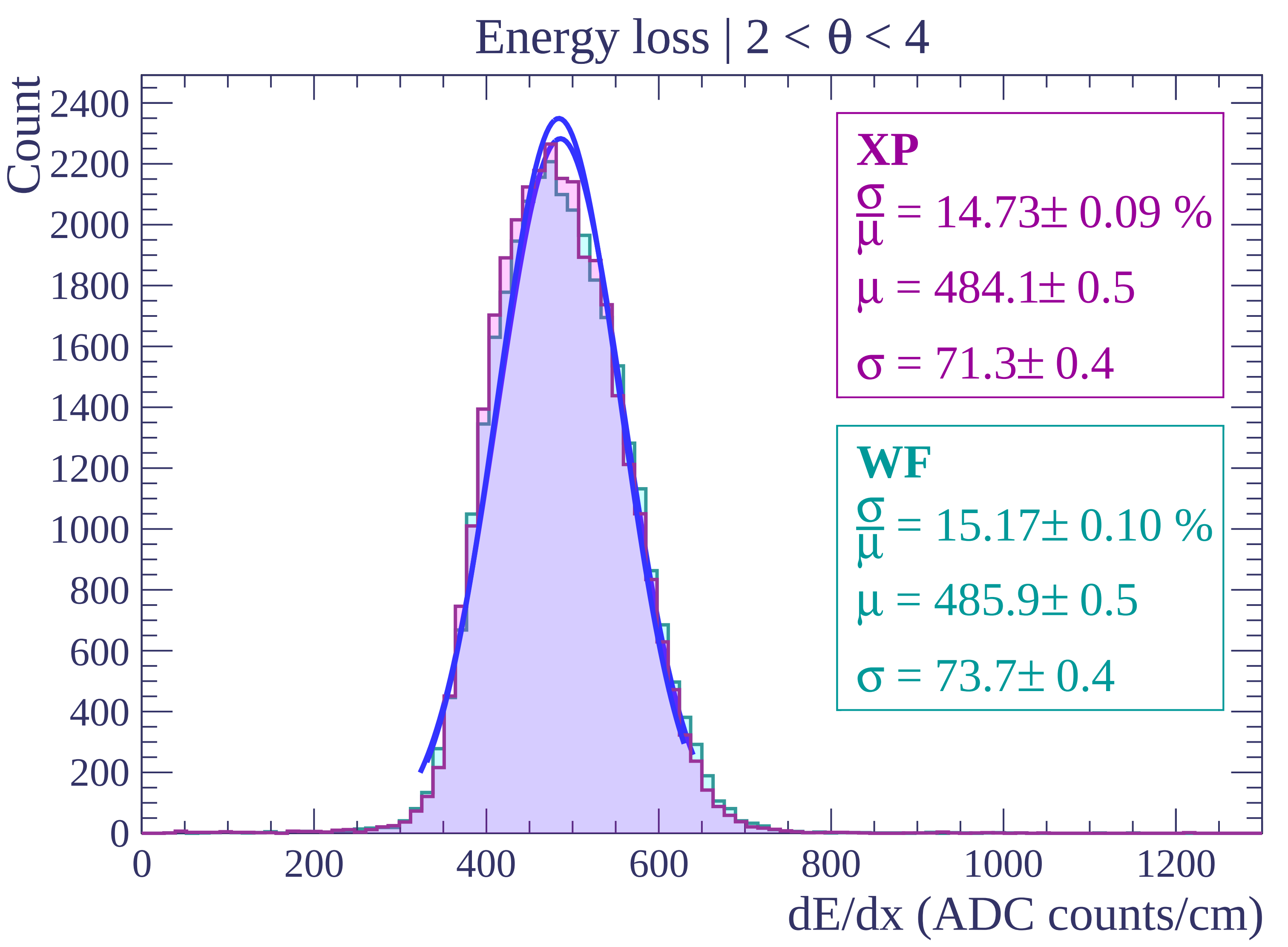
Energy loss | -4 < θ < -2
Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.08± 0.10 %
μ = 484.1± 0.5
σ = 73.0± 0.4
WF
$\frac{\sigma}{\mu}$ = 15.48± 0.10 %
μ = 486.3± 0.5
σ = 75.3± 0.4

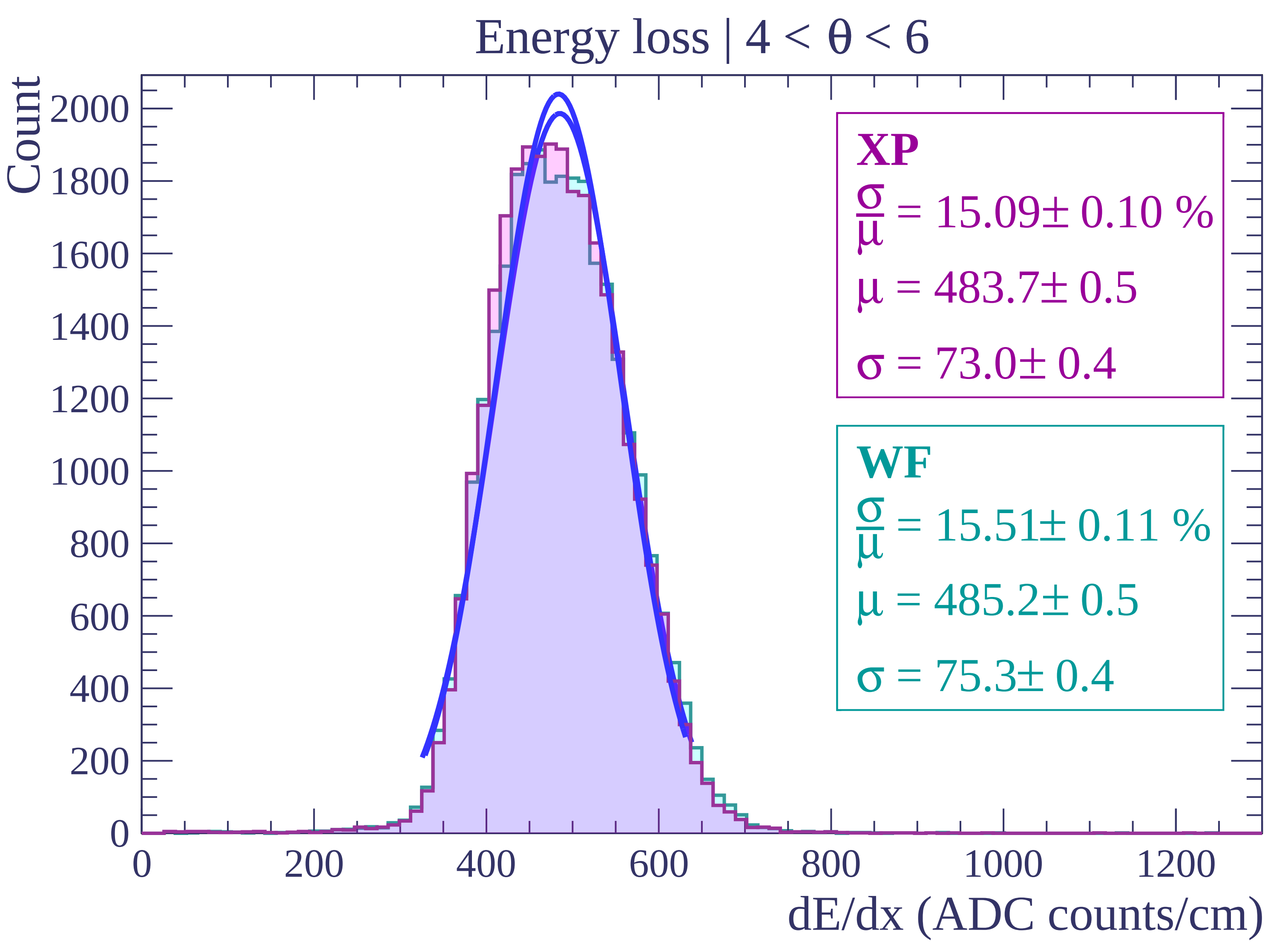


Energy loss | -2 < θ < 0
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
XP
σ/μ = 14.98± 0.09 %
μ = 484.1± 0.5
σ = 72.5± 0.4
WF
σ/μ = 15.26± 0.10 %
μ = 486.7± 0.5
σ = 74.3± 0.4

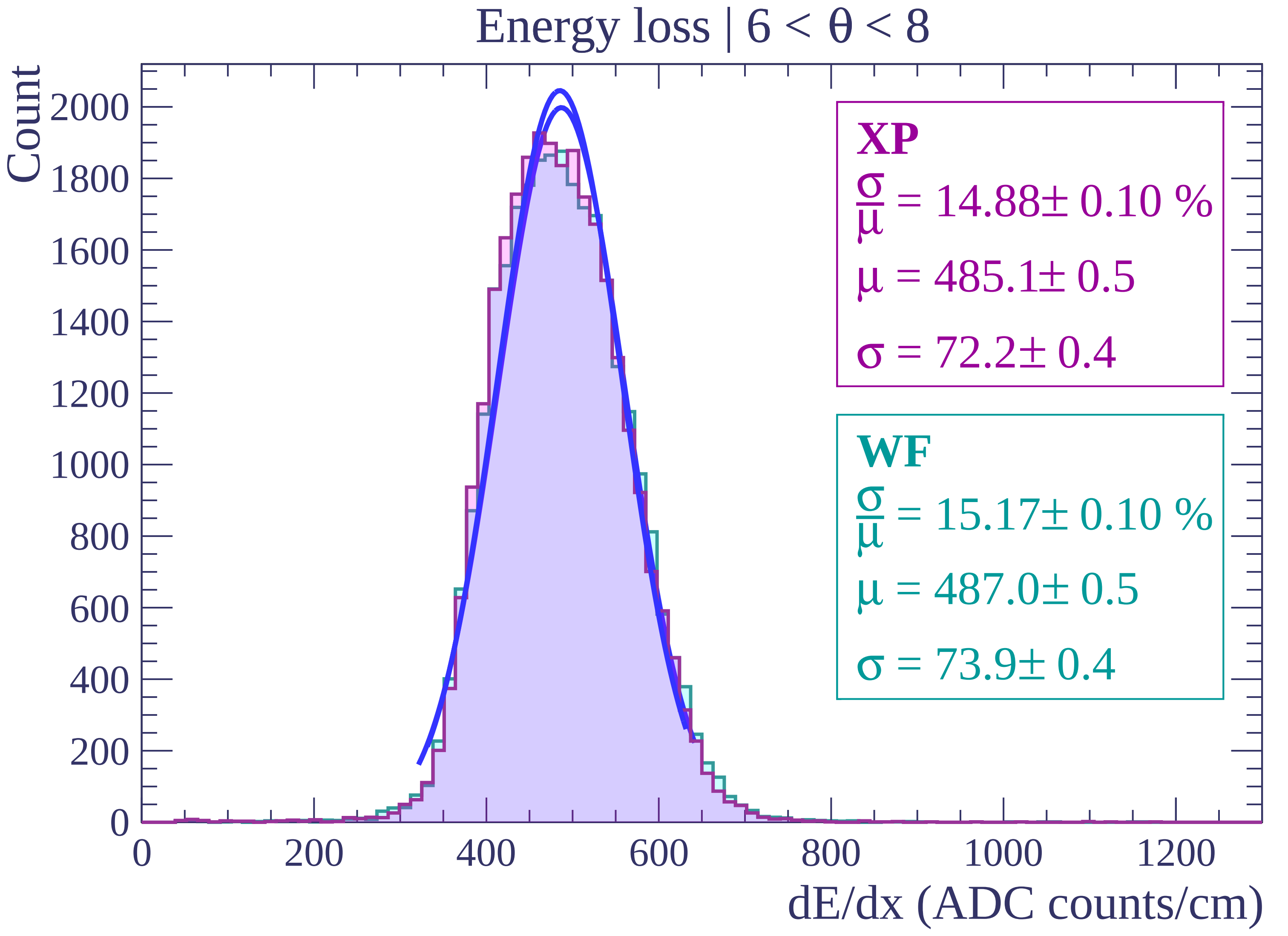


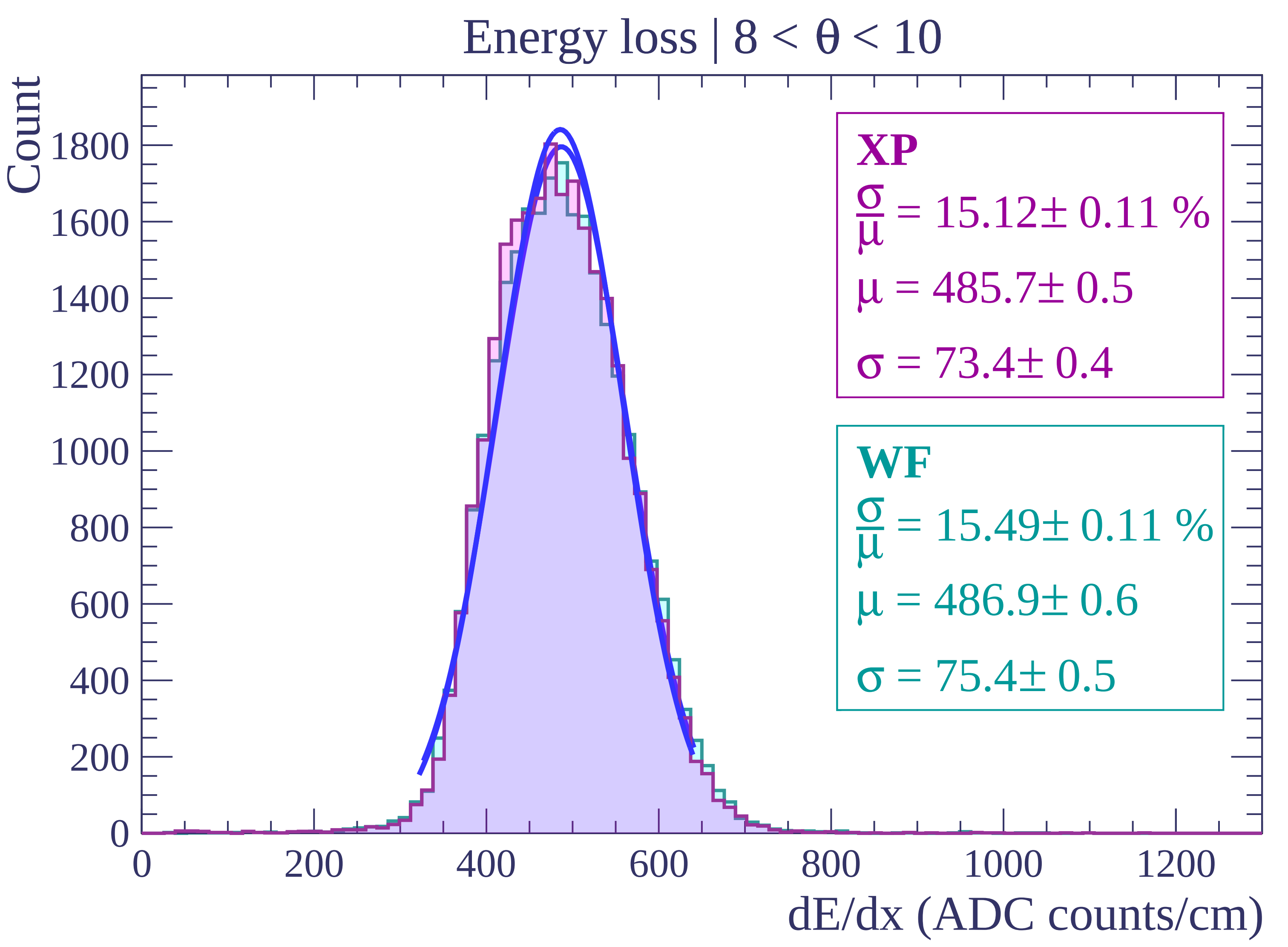
Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
500
1000
1500
2000
2500
3000
3500
XP
$\frac{\sigma}{\mu}$ = 15.38± 0.09 %
μ = 471.2± 0.4
σ = 72.5± 0.4
WF
$\frac{\sigma}{\mu}$ = 15.69± 0.09 %
μ = 476.0± 0.4
σ = 74.7± 0.4


Energy loss | 2 < θ < 4
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 14.73± 0.09 %
μ = 484.1± 0.5
σ = 71.3± 0.4
WF
$\frac{\sigma}{\mu}$ = 15.17± 0.10 %
μ = 485.9± 0.5
σ = 73.7± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
2400

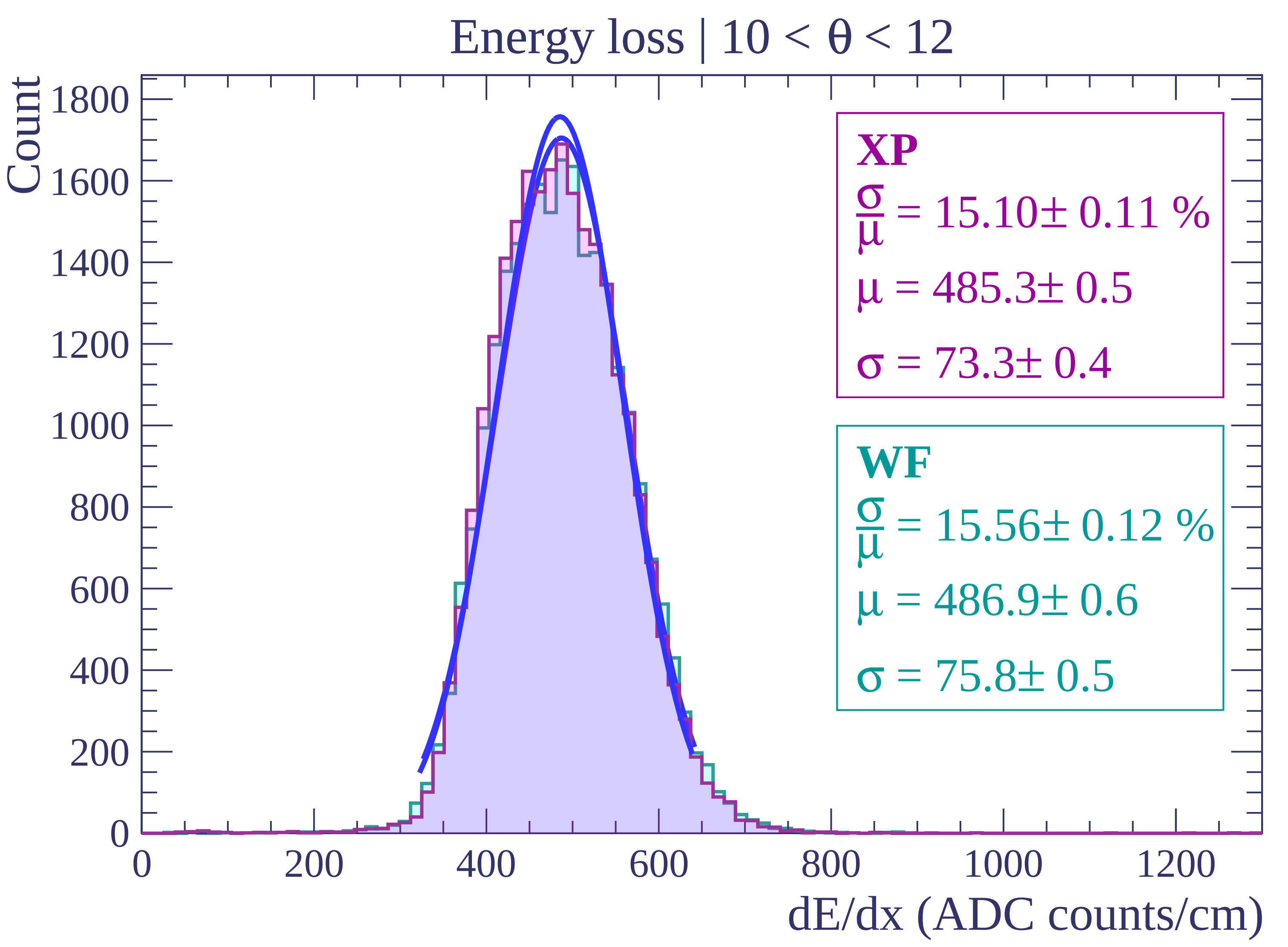


Energy loss | 4 < θ < 6
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.09± 0.10 %
μ = 483.7± 0.5
σ = 73.0± 0.4
WF
σ/μ = 15.51± 0.11 %
μ = 485.2± 0.5
σ = 75.3± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800
2000

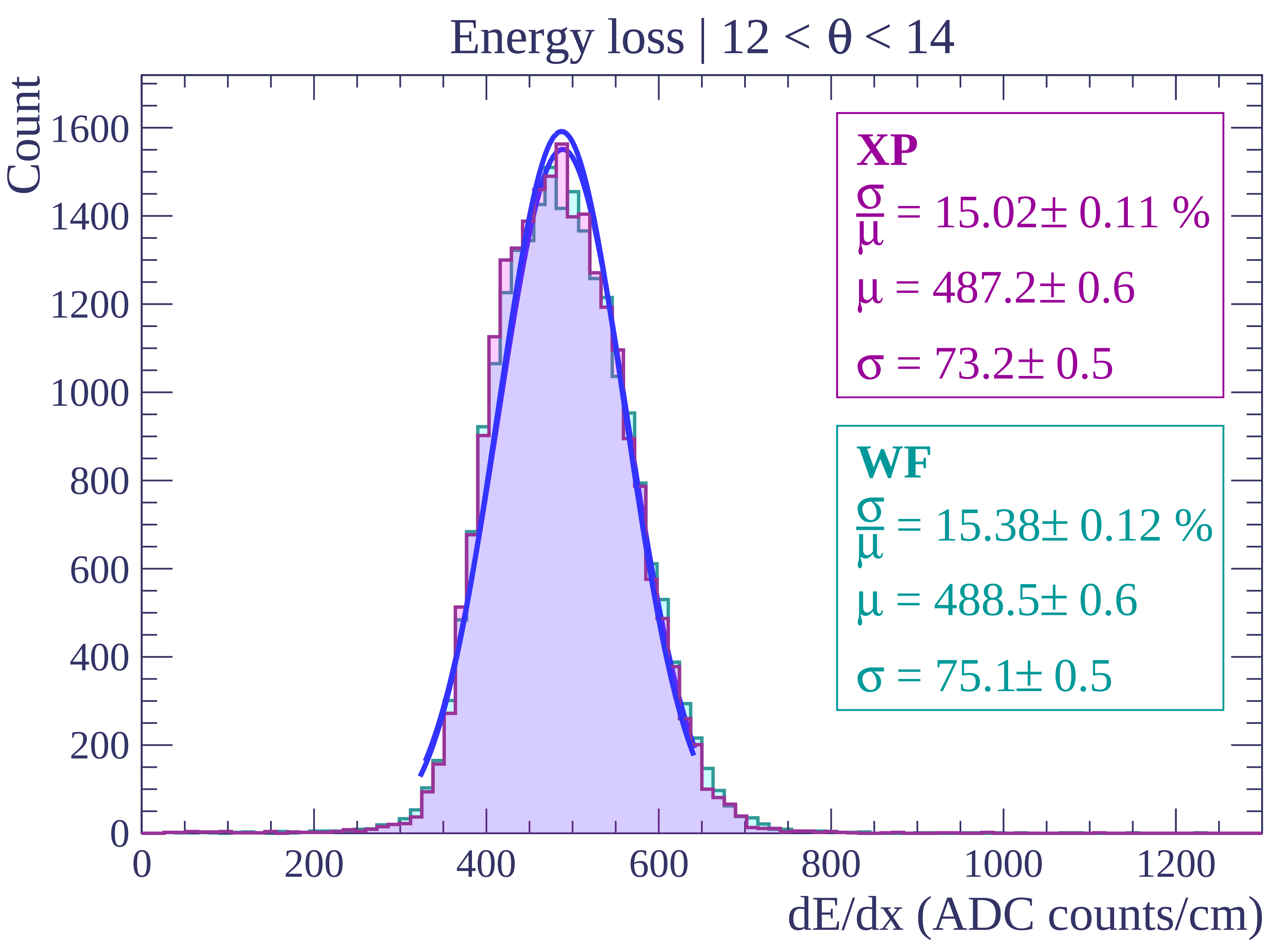

Energy loss | 6 < θ < 8
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
XP
σ/μ = 14.88± 0.10 %
μ = 485.1± 0.5
σ = 72.2± 0.4
WF
σ/μ = 15.17± 0.10 %
μ = 487.0± 0.5
σ = 73.9± 0.4



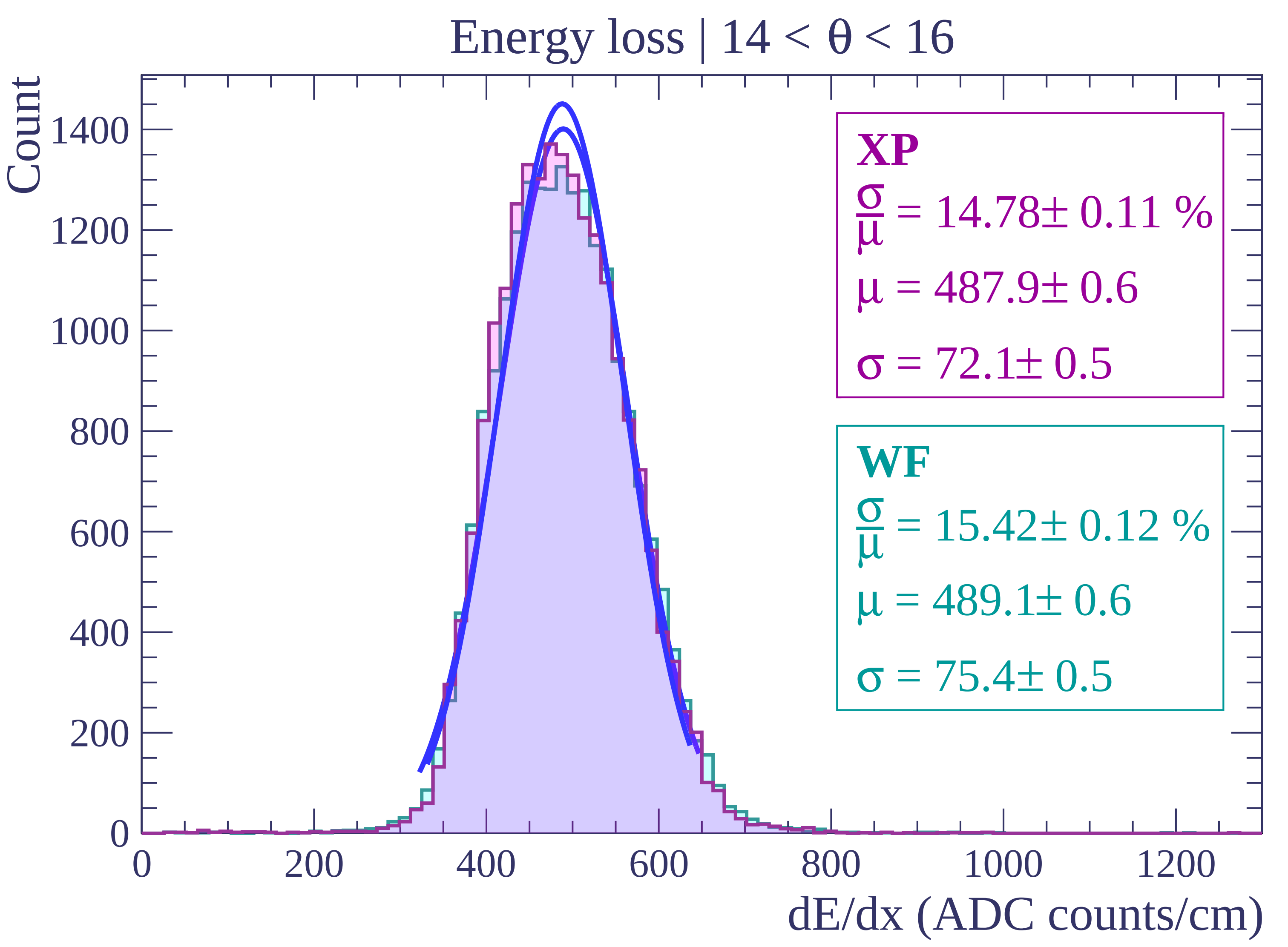
Energy loss | 8 < θ < 10
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
XP
σ/μ = 15.12± 0.11 %
μ = 485.7± 0.5
σ = 73.4± 0.4
WF
σ/μ = 15.49± 0.11 %
μ = 486.9± 0.6
σ = 75.4± 0.5



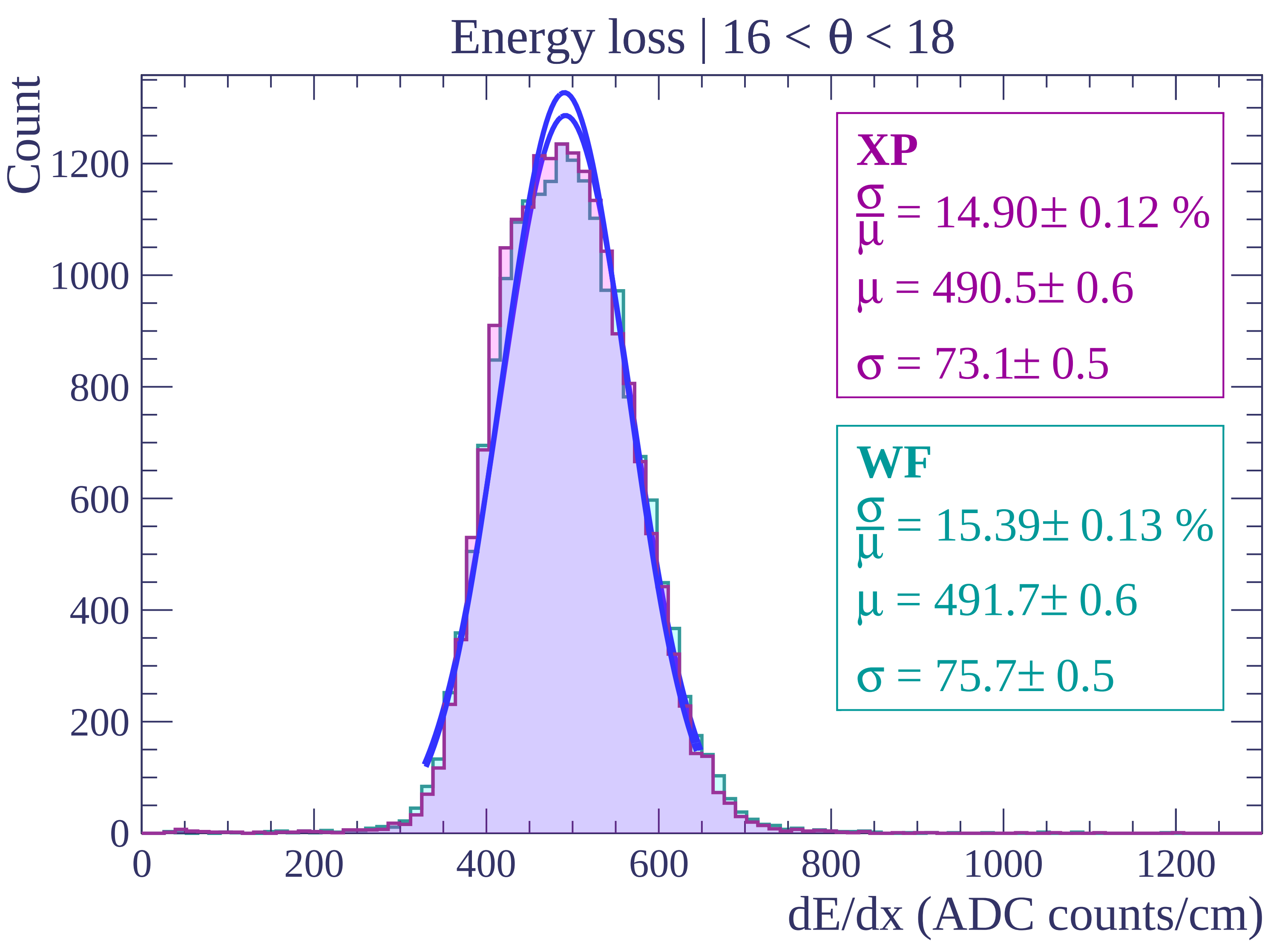
Energy loss | 10 < θ < 12
Count
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.10± 0.11 %
μ = 485.3± 0.5
σ = 73.3± 0.4
WF
$\frac{\sigma}{\mu}$ = 15.56± 0.12 %
μ = 486.9± 0.6
σ = 75.8± 0.5

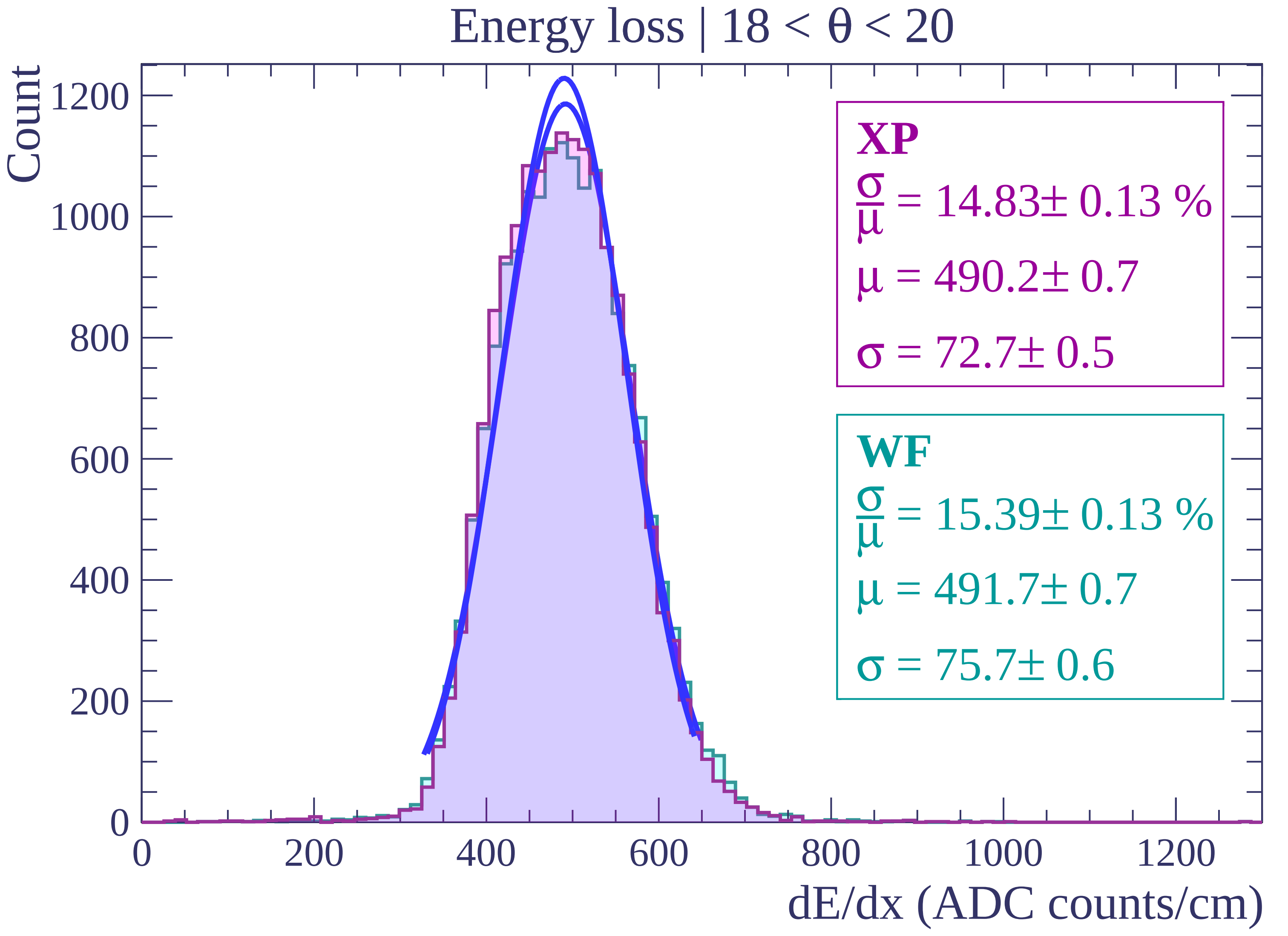


Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.02± 0.11 %
μ = 487.2± 0.6
σ = 73.2± 0.5
WF
σ/μ = 15.38± 0.12 %
μ = 488.5± 0.6
σ = 75.1± 0.5

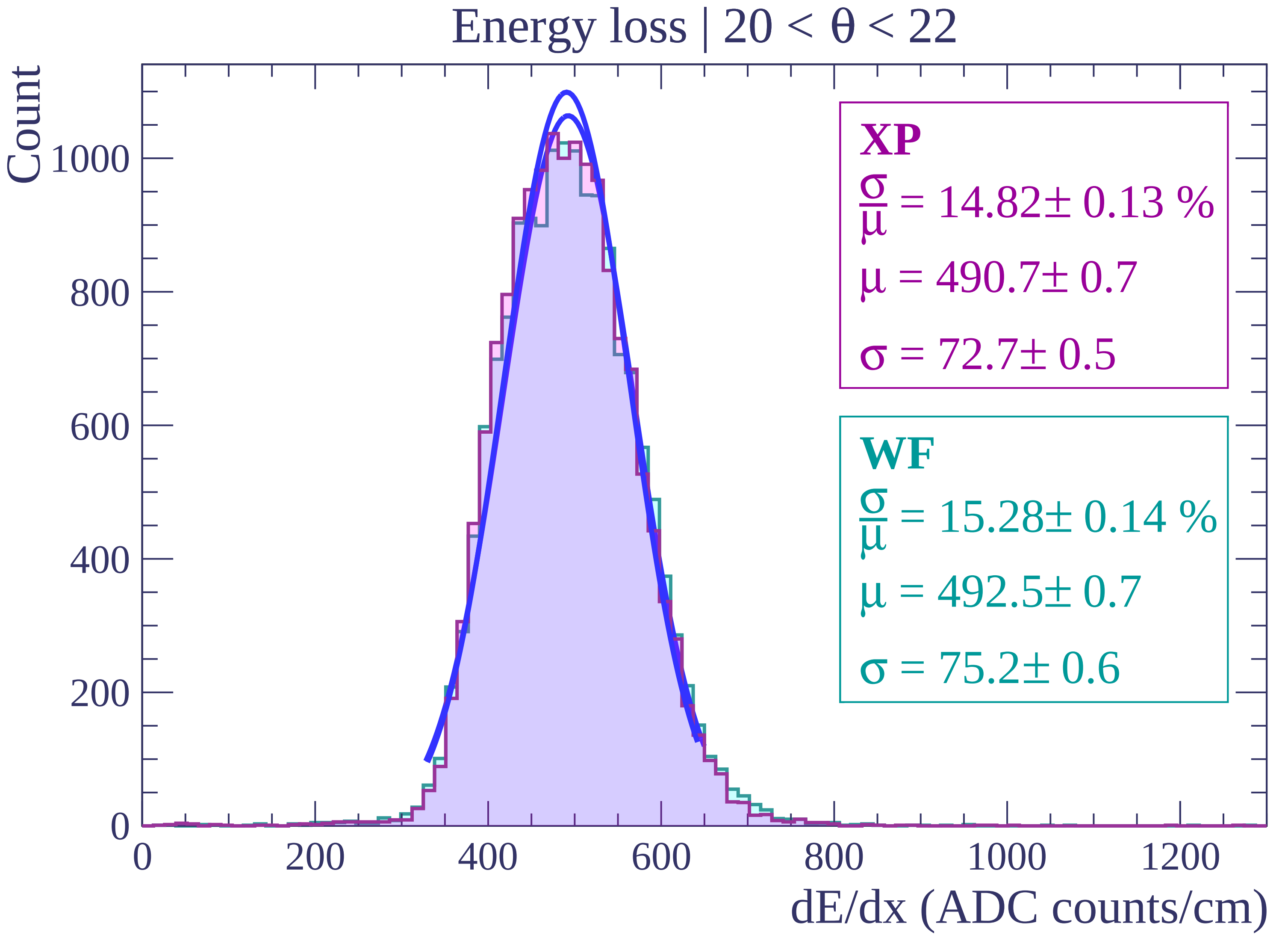


Energy loss | 14 < θ < 16
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
XP
σ/μ = 14.78± 0.11 %
μ = 487.9± 0.6
σ = 72.1± 0.5
WF
σ/μ = 15.42± 0.12 %
μ = 489.1± 0.6
σ = 75.4± 0.5

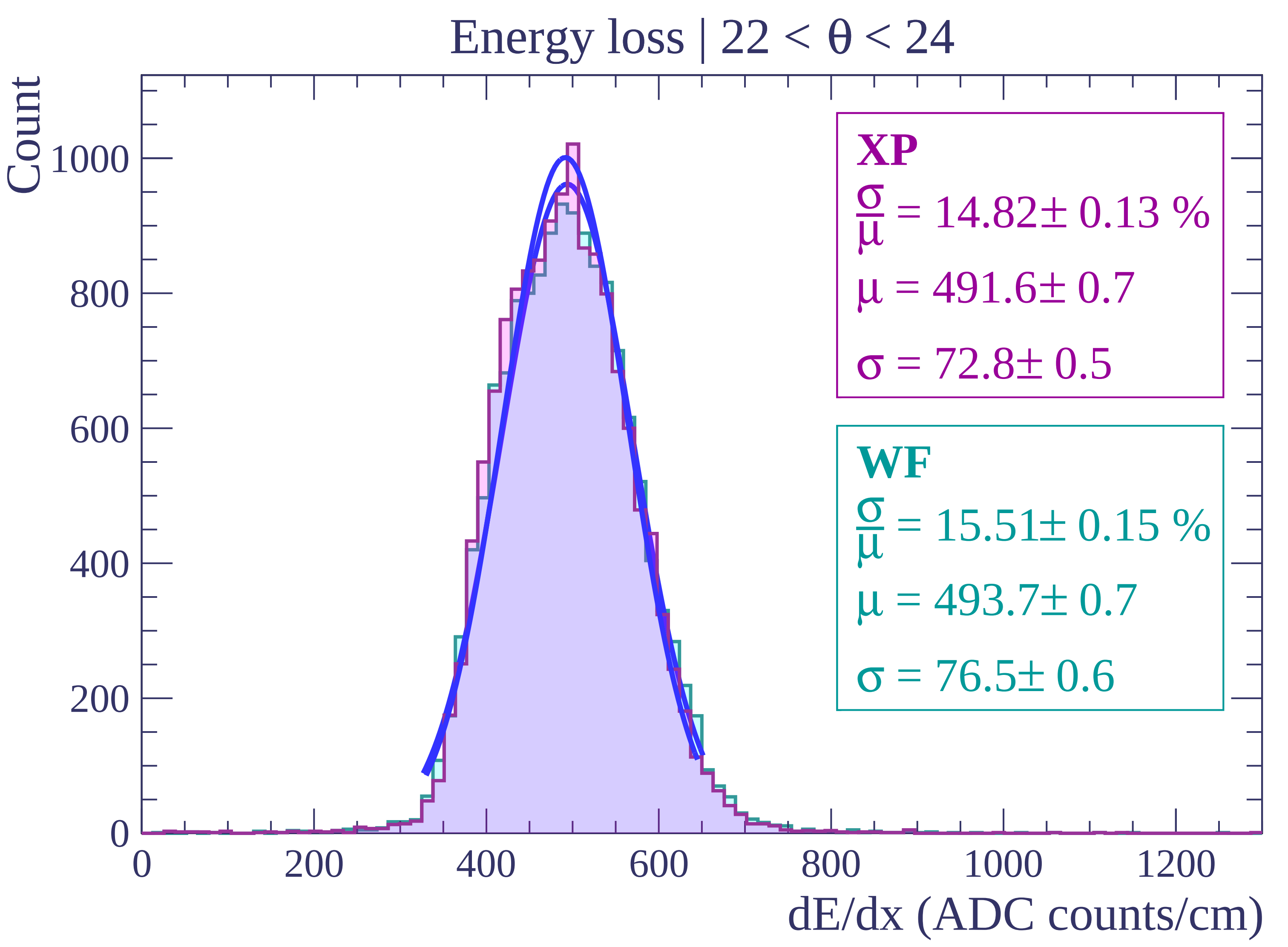

Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.90± 0.12 %
μ = 490.5± 0.6
σ = 73.1± 0.5
WF
σ/μ = 15.39± 0.13 %
μ = 491.7± 0.6
σ = 75.7± 0.5



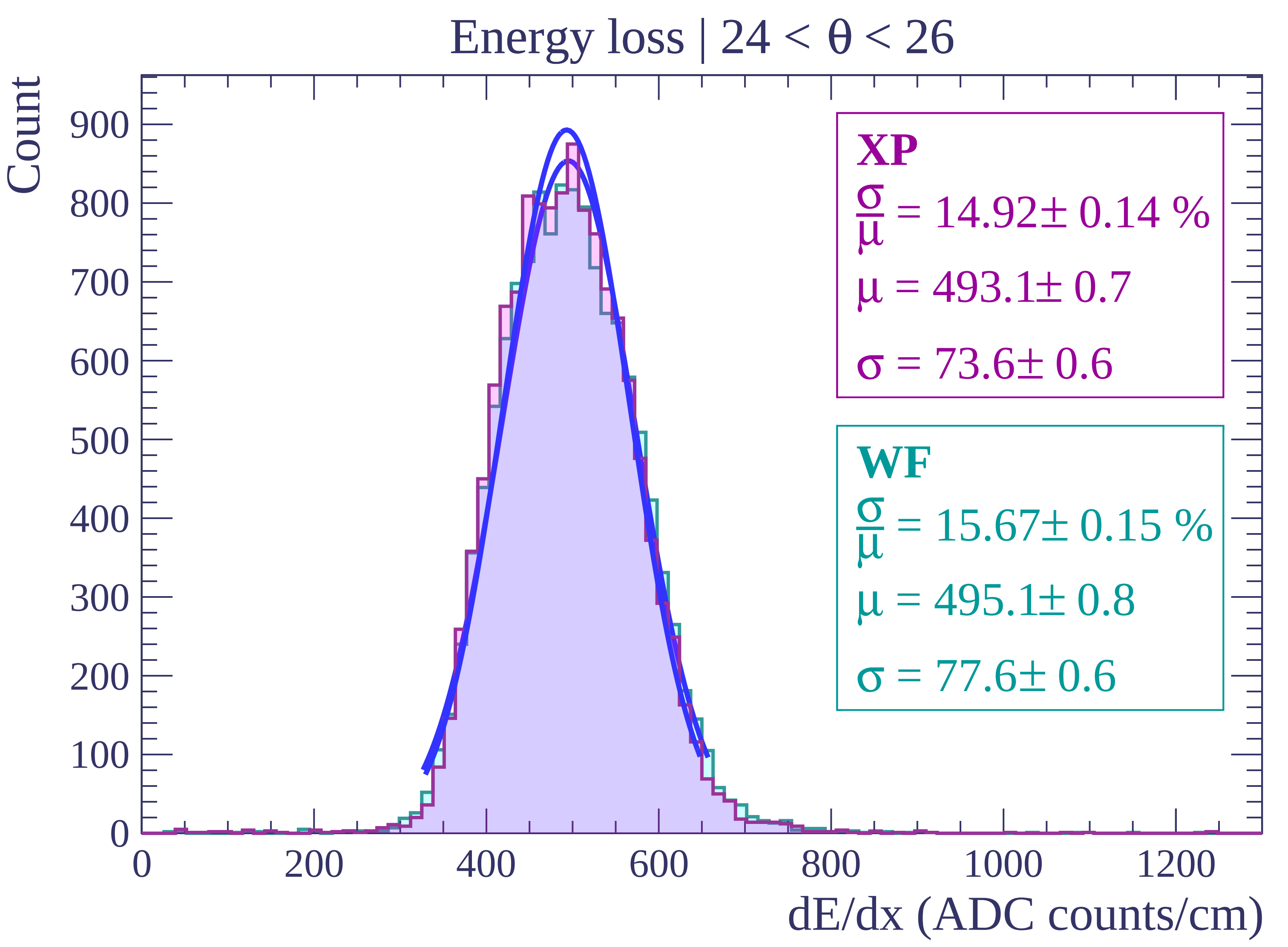
Energy loss | 18 < θ < 20
Count
0
200
400
600
800
1000
1200
XP
σ/μ = 14.83± 0.13 %
μ = 490.2± 0.7
σ = 72.7± 0.5
WF
σ/μ = 15.39± 0.13 %
μ = 491.7± 0.7
σ = 75.7± 0.6
dE/dx (ADC counts/cm)



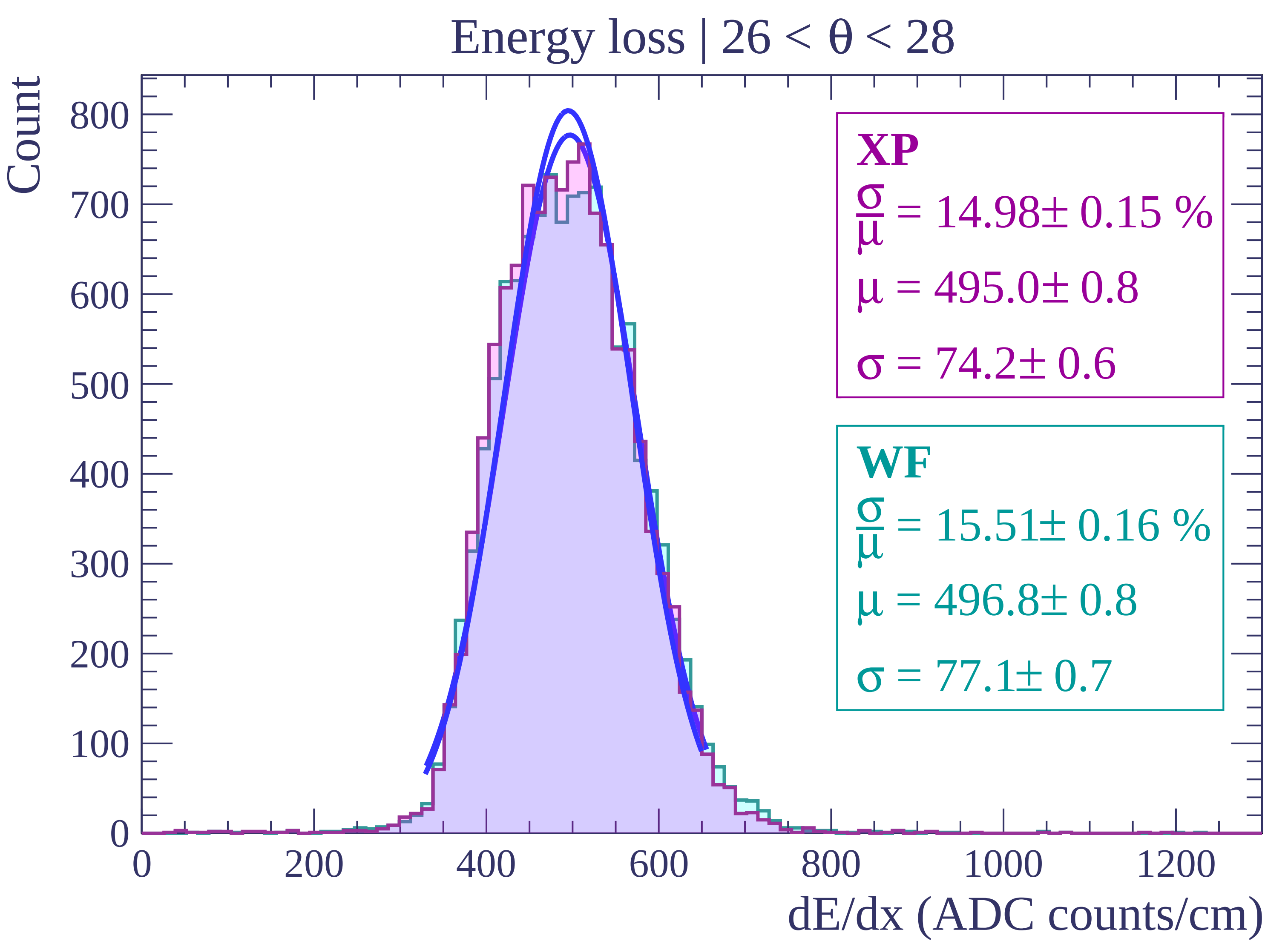
Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.82± 0.13 %
μ = 490.7± 0.7
σ = 72.7± 0.5
WF
σ/μ = 15.28± 0.14 %
μ = 492.5± 0.7
σ = 75.2± 0.6



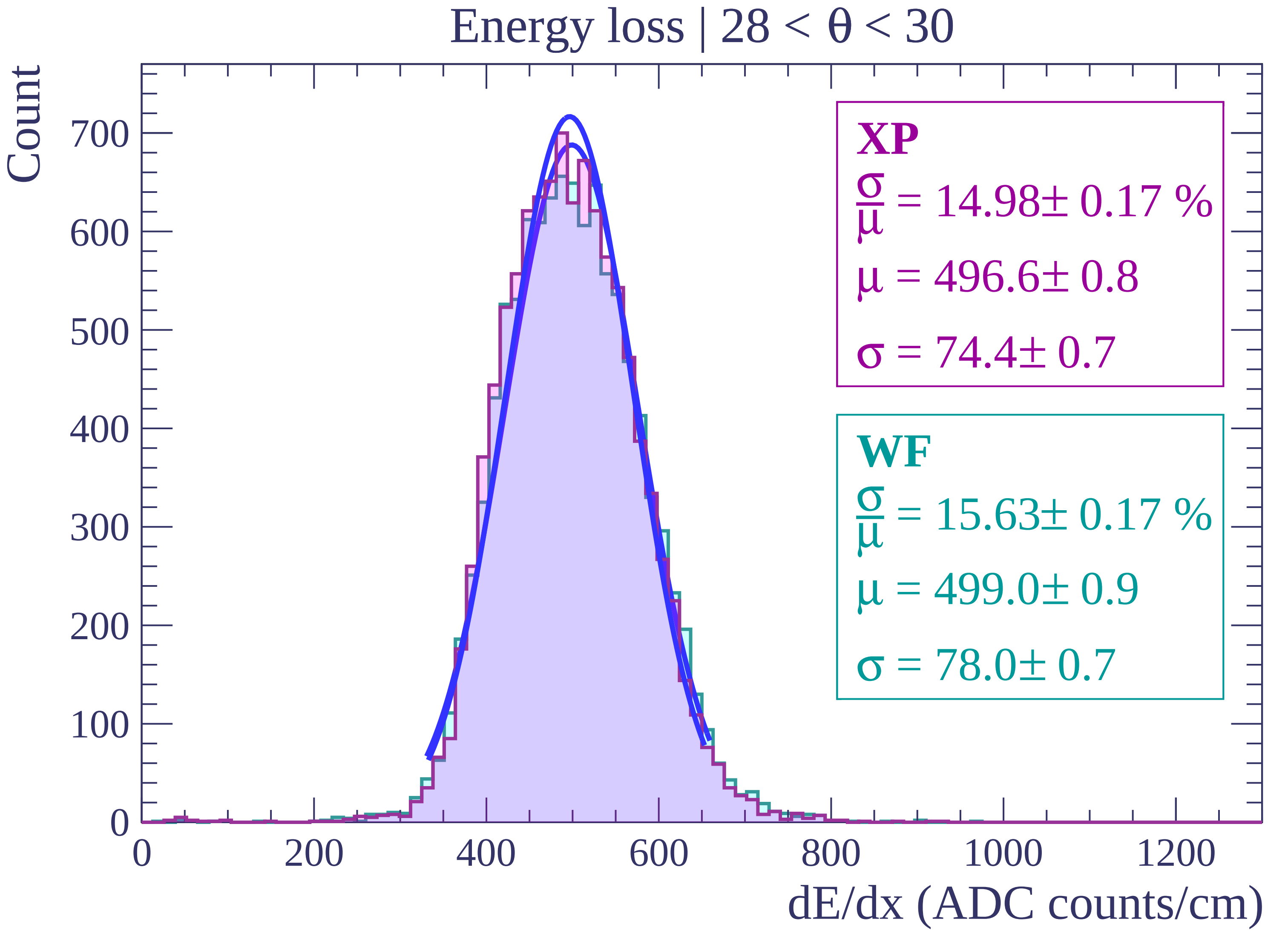
Energy loss | 22 < θ < 24
Count
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 14.82± 0.13 %
μ = 491.6± 0.7
σ = 72.8± 0.5
WF
σ/μ = 15.51± 0.15 %
μ = 493.7± 0.7
σ = 76.5± 0.6

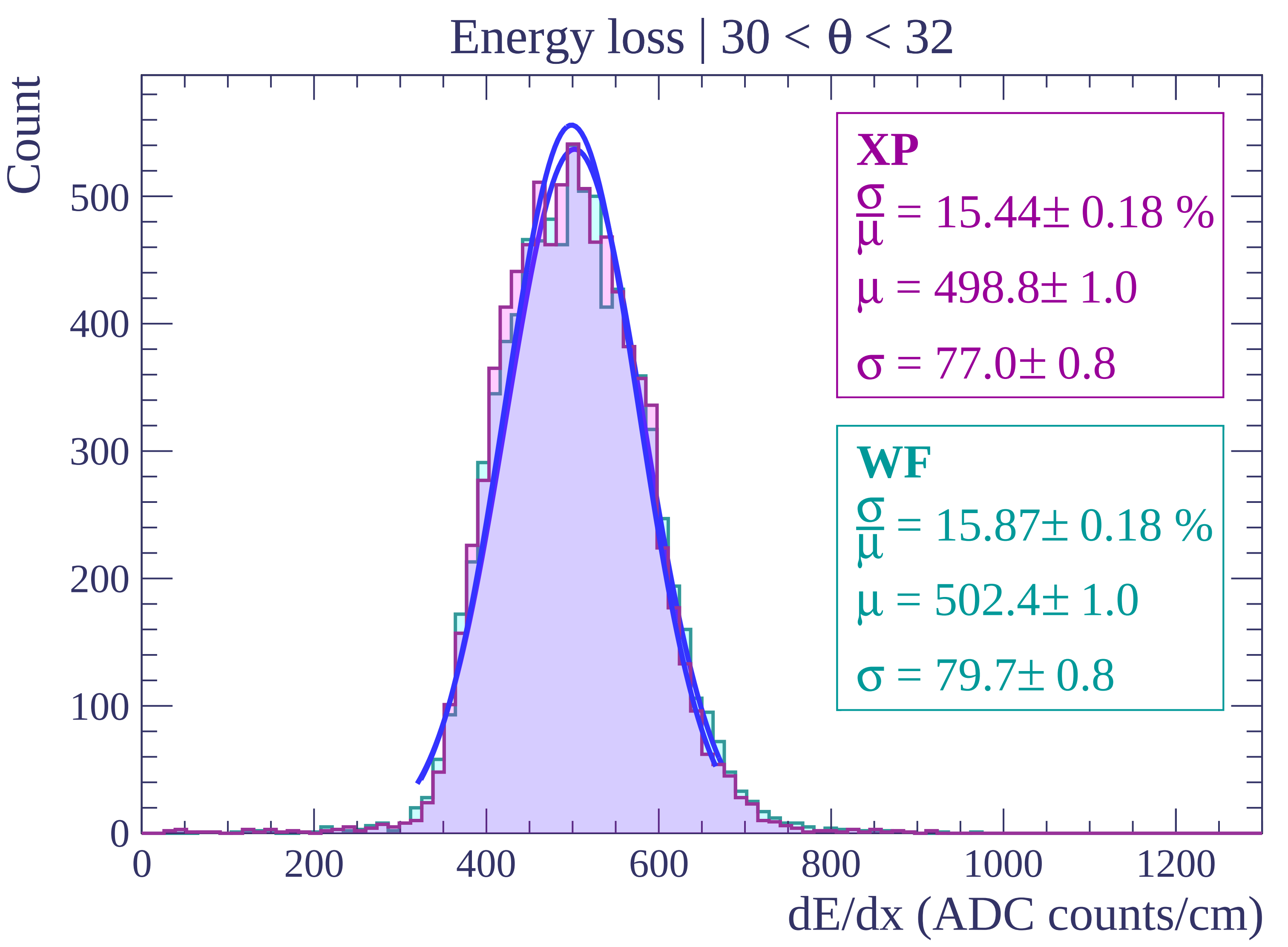


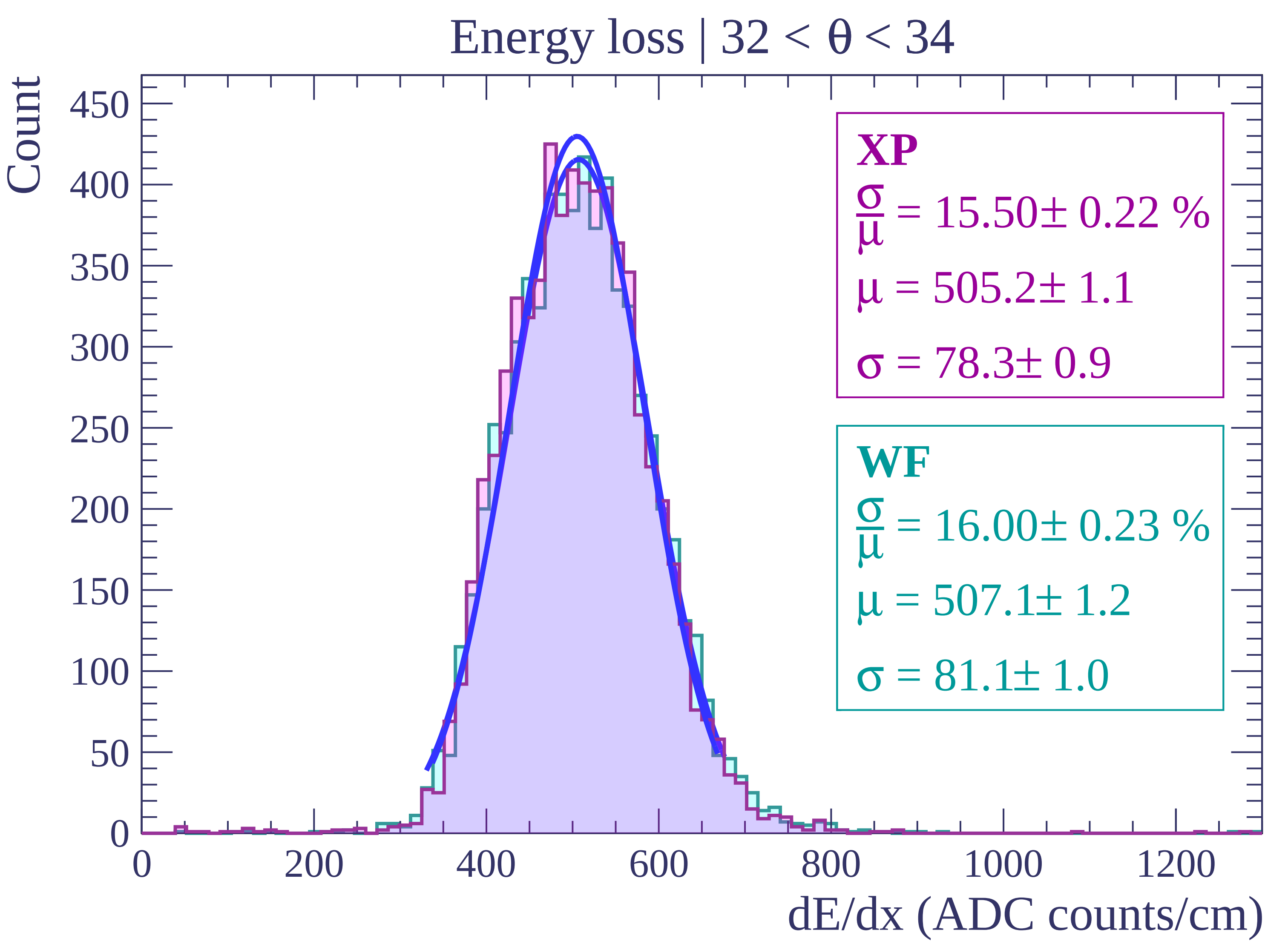
Energy loss | 24 < θ < 26
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 14.92± 0.14 %
μ = 493.1± 0.7
σ = 73.6± 0.6
WF
$\frac{\sigma}{\mu}$ = 15.67± 0.15 %
μ = 495.1± 0.8
σ = 77.6± 0.6

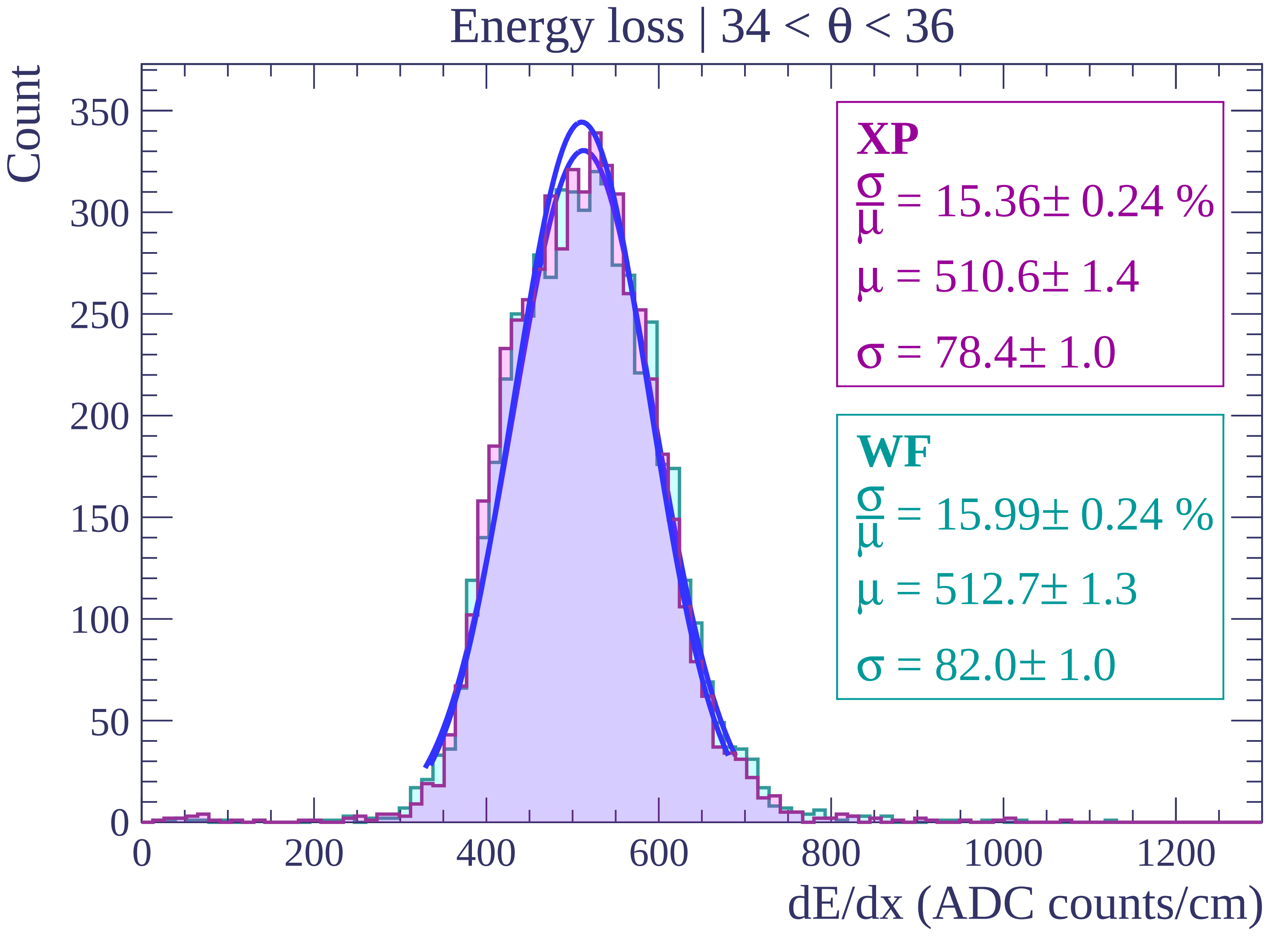


Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.98± 0.15 %
μ = 495.0± 0.8
σ = 74.2± 0.6
WF
σ/μ = 15.51± 0.16 %
μ = 496.8± 0.8
σ = 77.1± 0.7

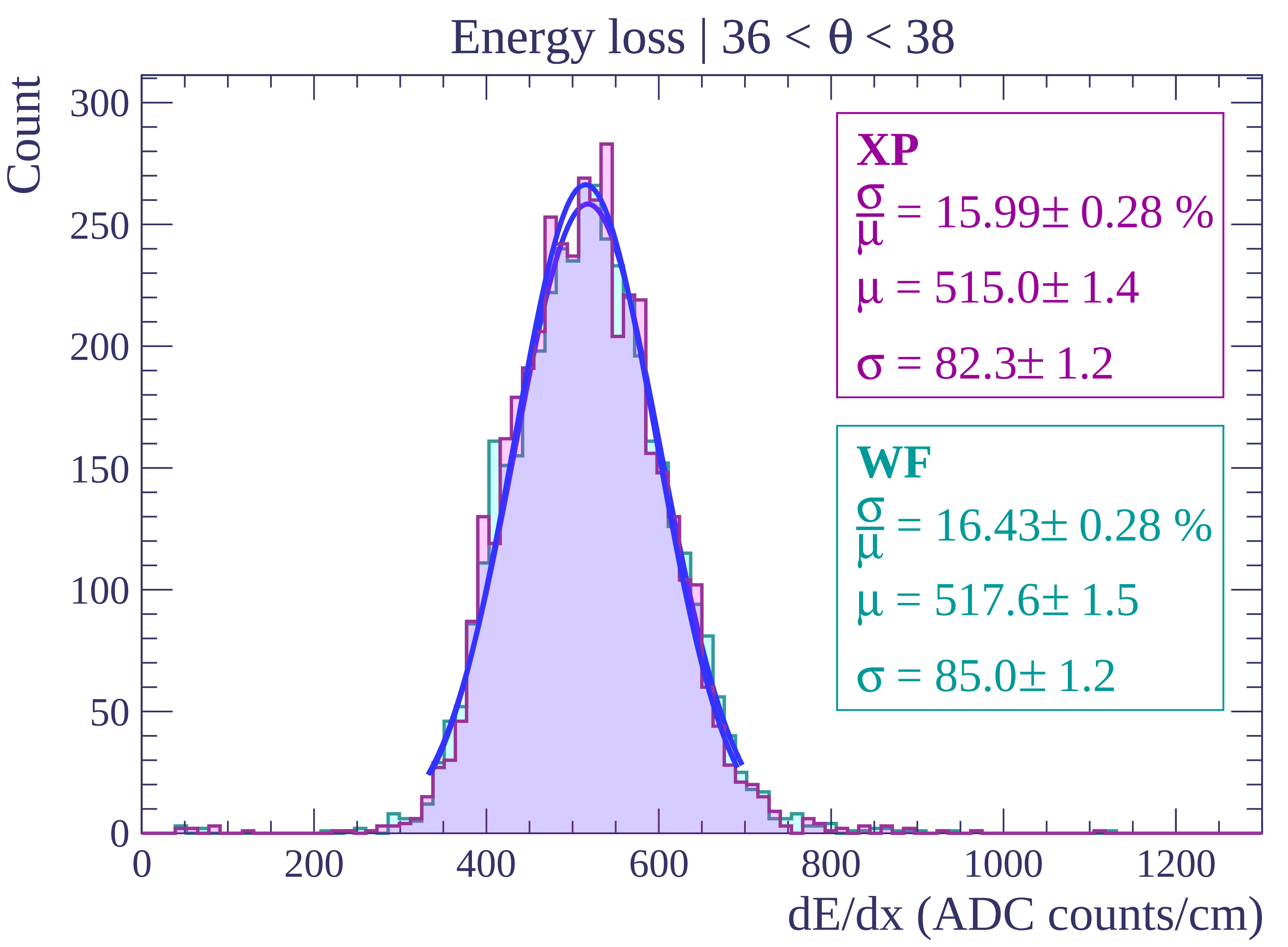

Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 14.98± 0.17 %
μ = 496.6± 0.8
σ = 74.4± 0.7
WF
$\frac{\sigma}{\mu}$ = 15.63± 0.17 %
μ = 499.0± 0.9
σ = 78.0± 0.7


Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 15.44± 0.18 %
μ = 498.8± 1.0
σ = 77.0± 0.8
WF
σ/μ = 15.87± 0.18 %
μ = 502.4± 1.0
σ = 79.7± 0.8

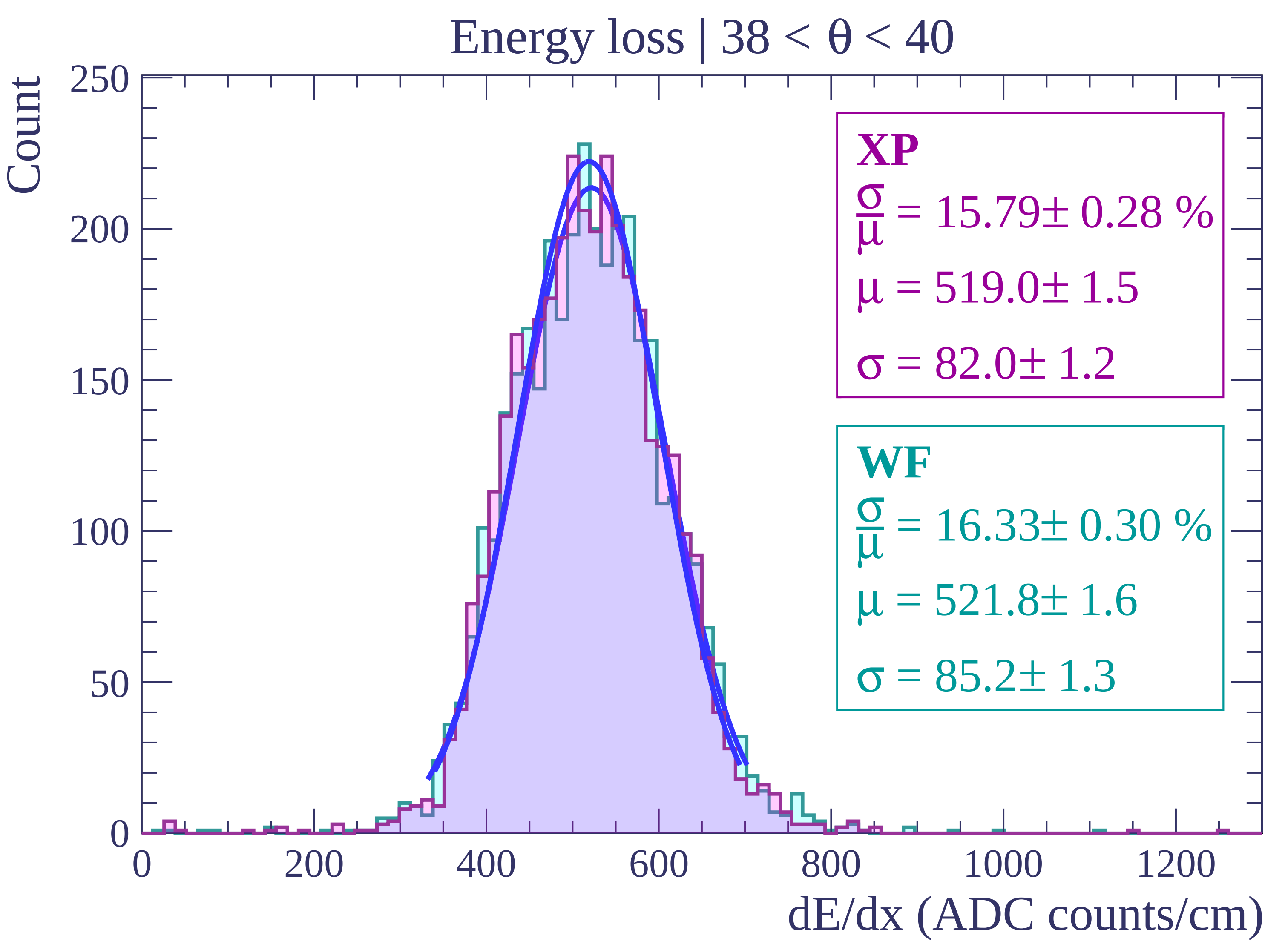

Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.50± 0.22 %
μ = 505.2± 1.1
σ = 78.3± 0.9
WF
σ/μ = 16.00± 0.23 %
μ = 507.1± 1.2
σ = 81.1± 1.0
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200



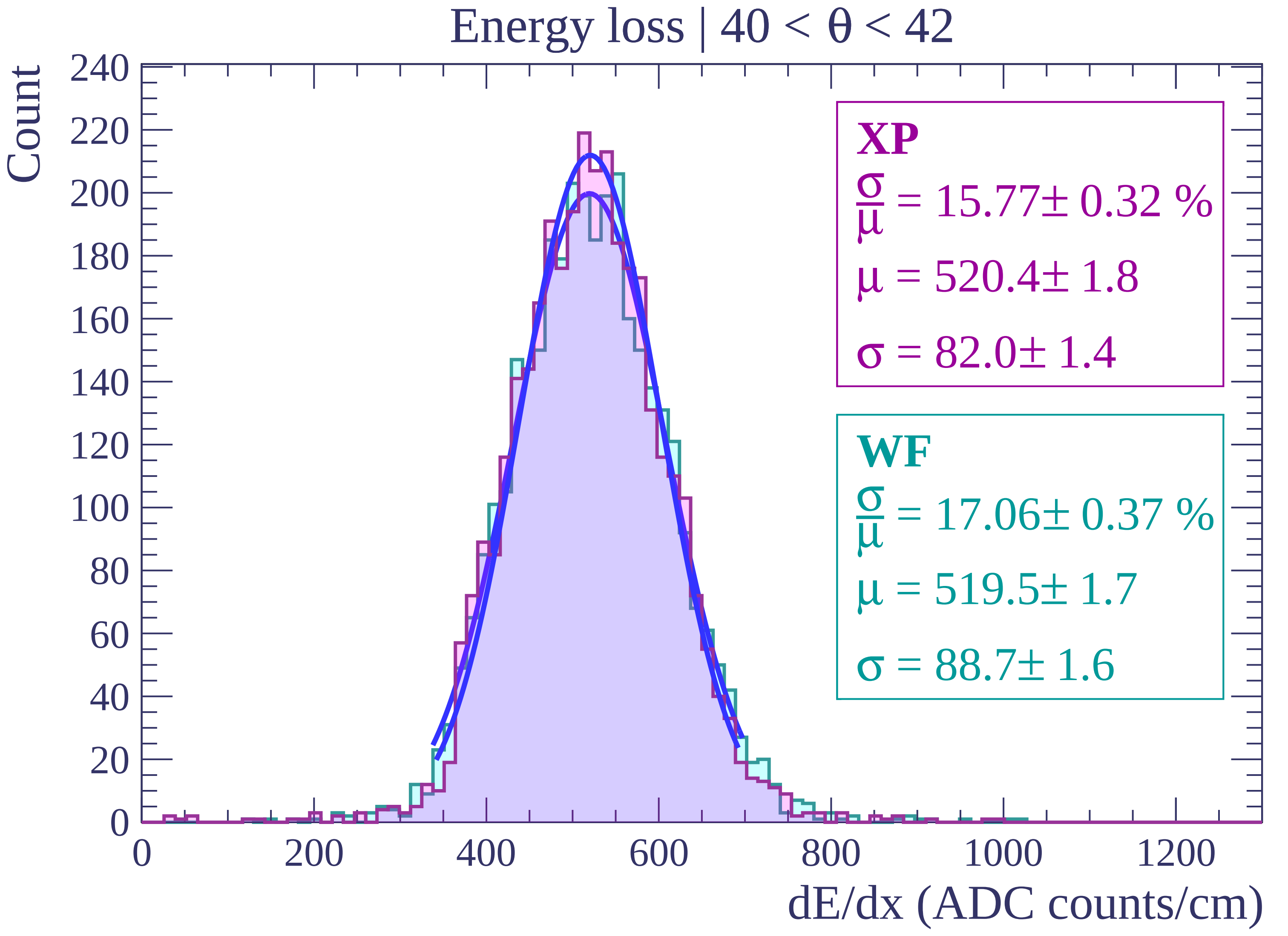
Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 15.36± 0.24 %
μ = 510.6± 1.4
σ = 78.4± 1.0
WF
σ/μ = 15.99± 0.24 %
μ = 512.7± 1.3
σ = 82.0± 1.0

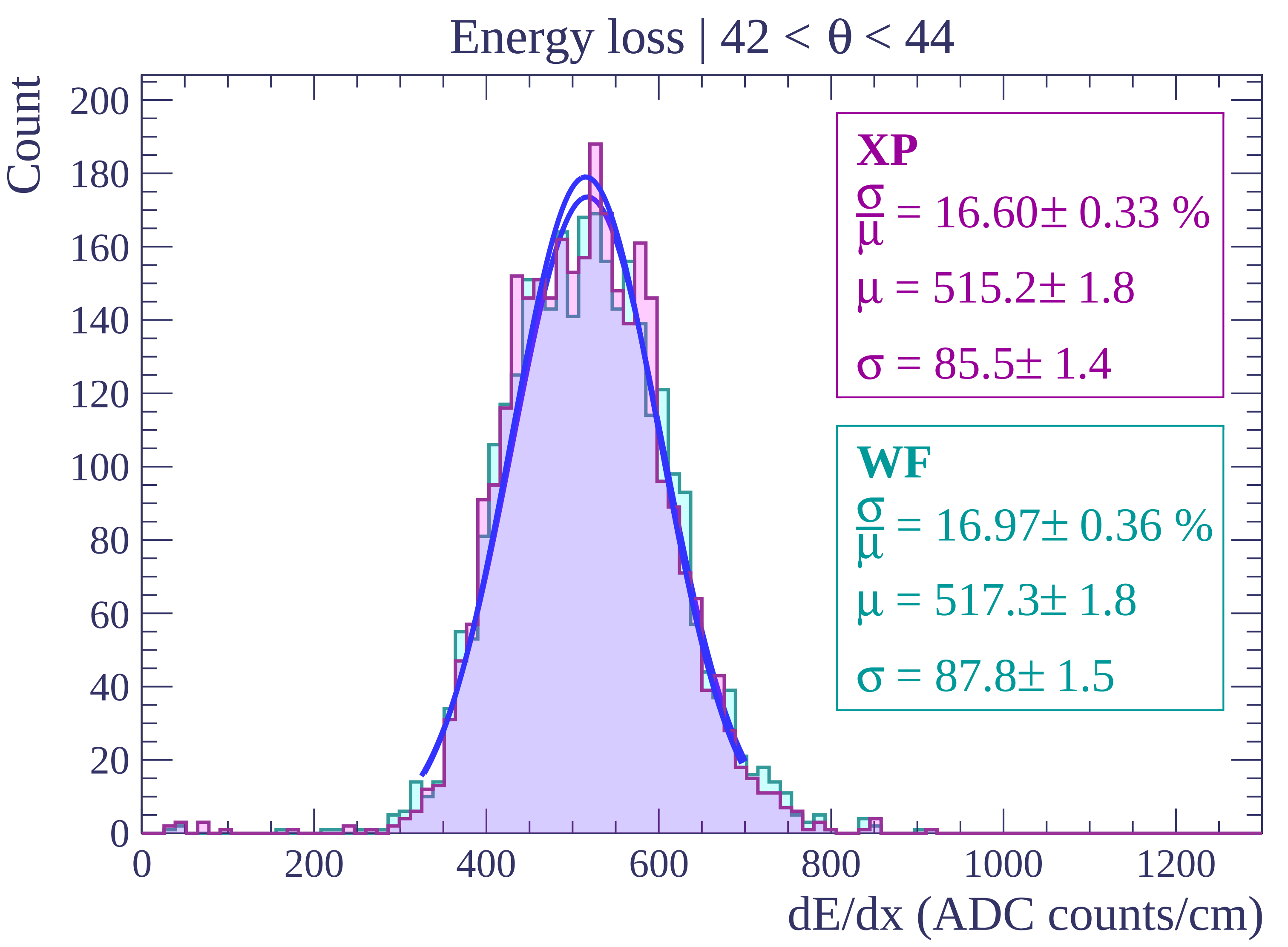


Energy loss | 36 < θ < 38
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.99± 0.28 %
μ = 515.0± 1.4
σ = 82.3± 1.2
WF
$\frac{\sigma}{\mu}$ = 16.43± 0.28 %
μ = 517.6± 1.5
σ = 85.0± 1.2

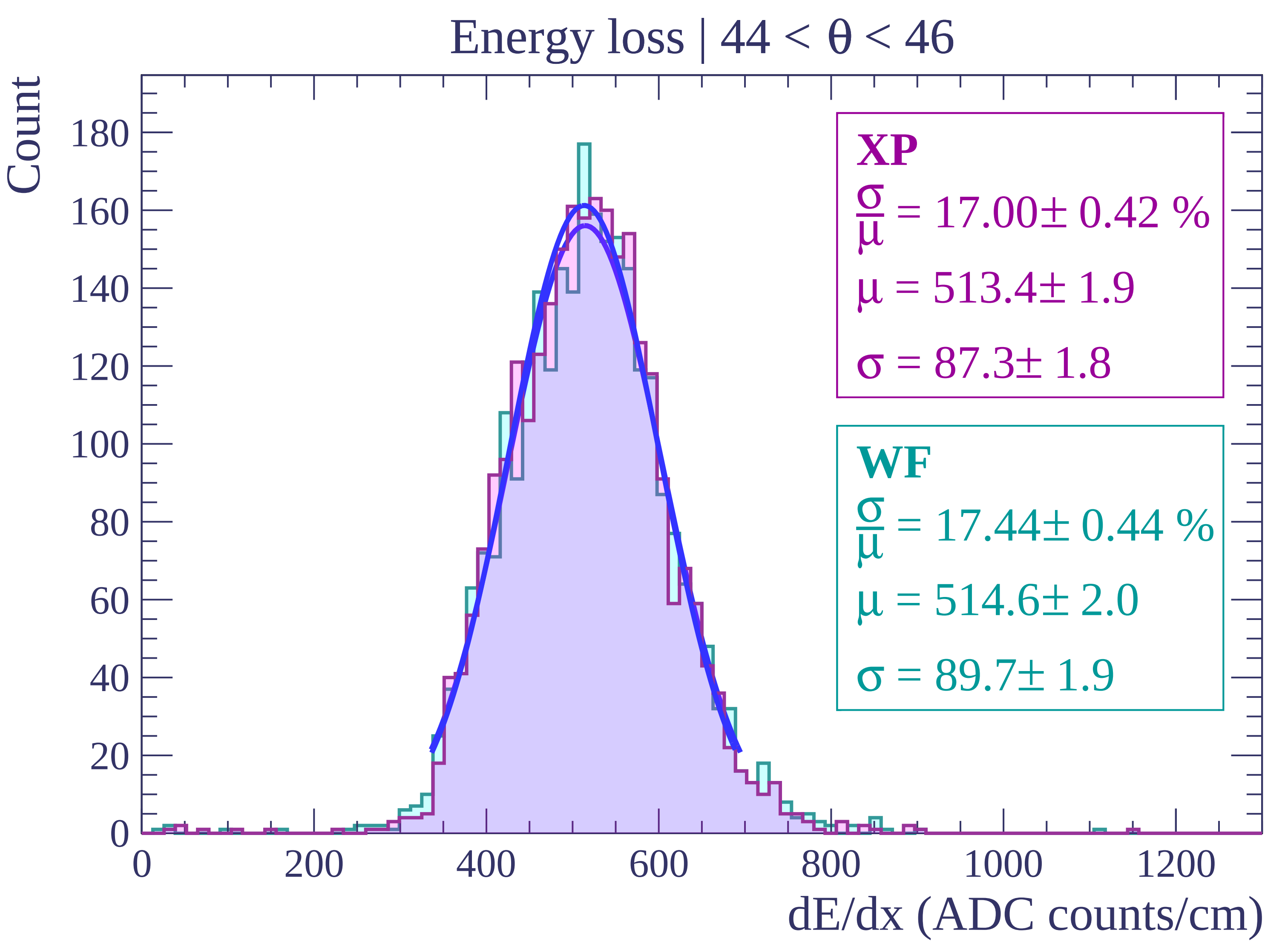


Energy loss | 38 < θ < 40
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.79± 0.28 %
μ = 519.0± 1.5
σ = 82.0± 1.2
WF
σ/μ = 16.33± 0.30 %
μ = 521.8± 1.6
σ = 85.2± 1.3

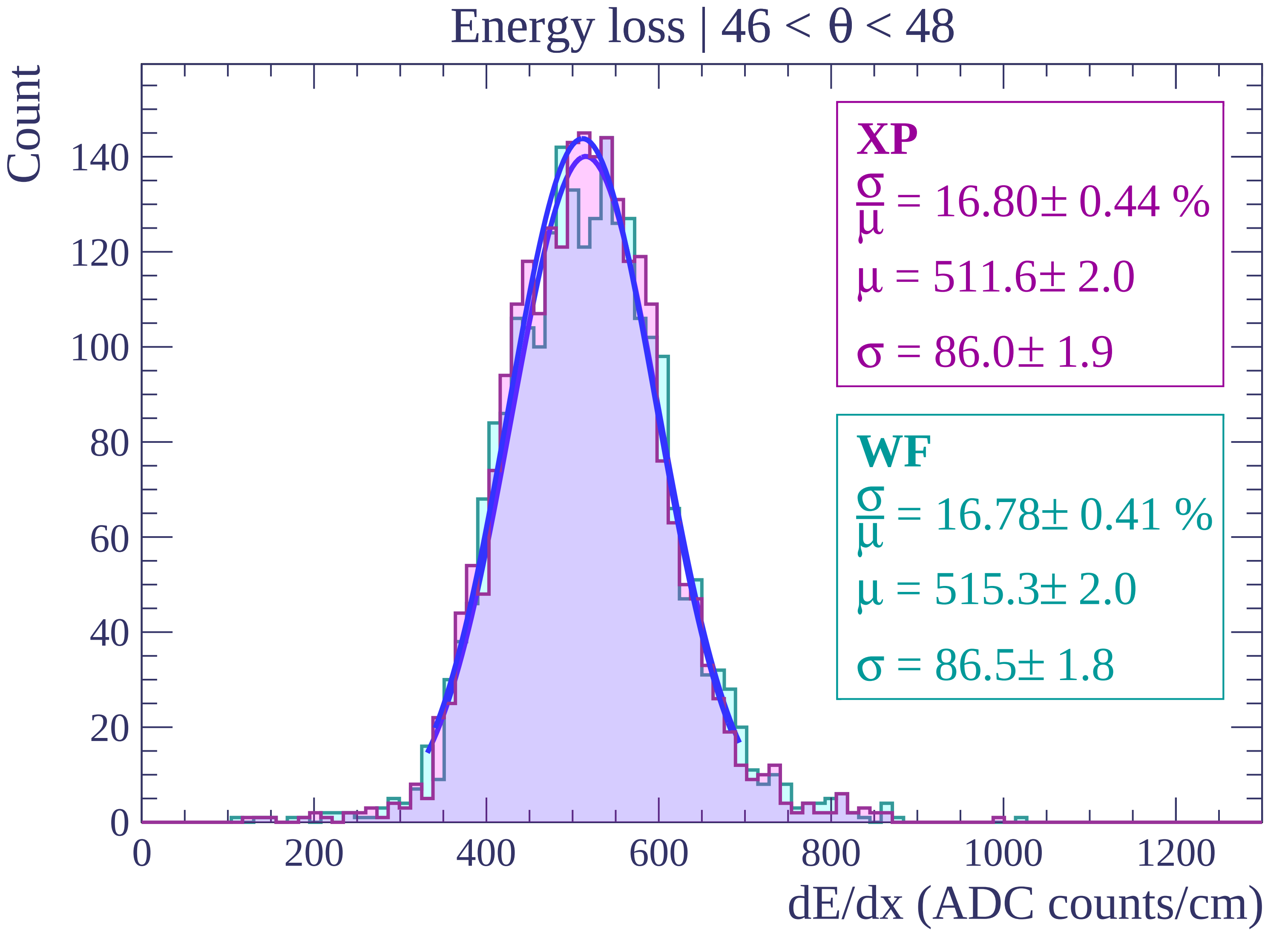

Energy loss | 40 < θ < 42
Count
240
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.77± 0.32 %
μ = 520.4± 1.8
σ = 82.0± 1.4
WF
σ/μ = 17.06± 0.37 %
μ = 519.5± 1.7
σ = 88.7± 1.6

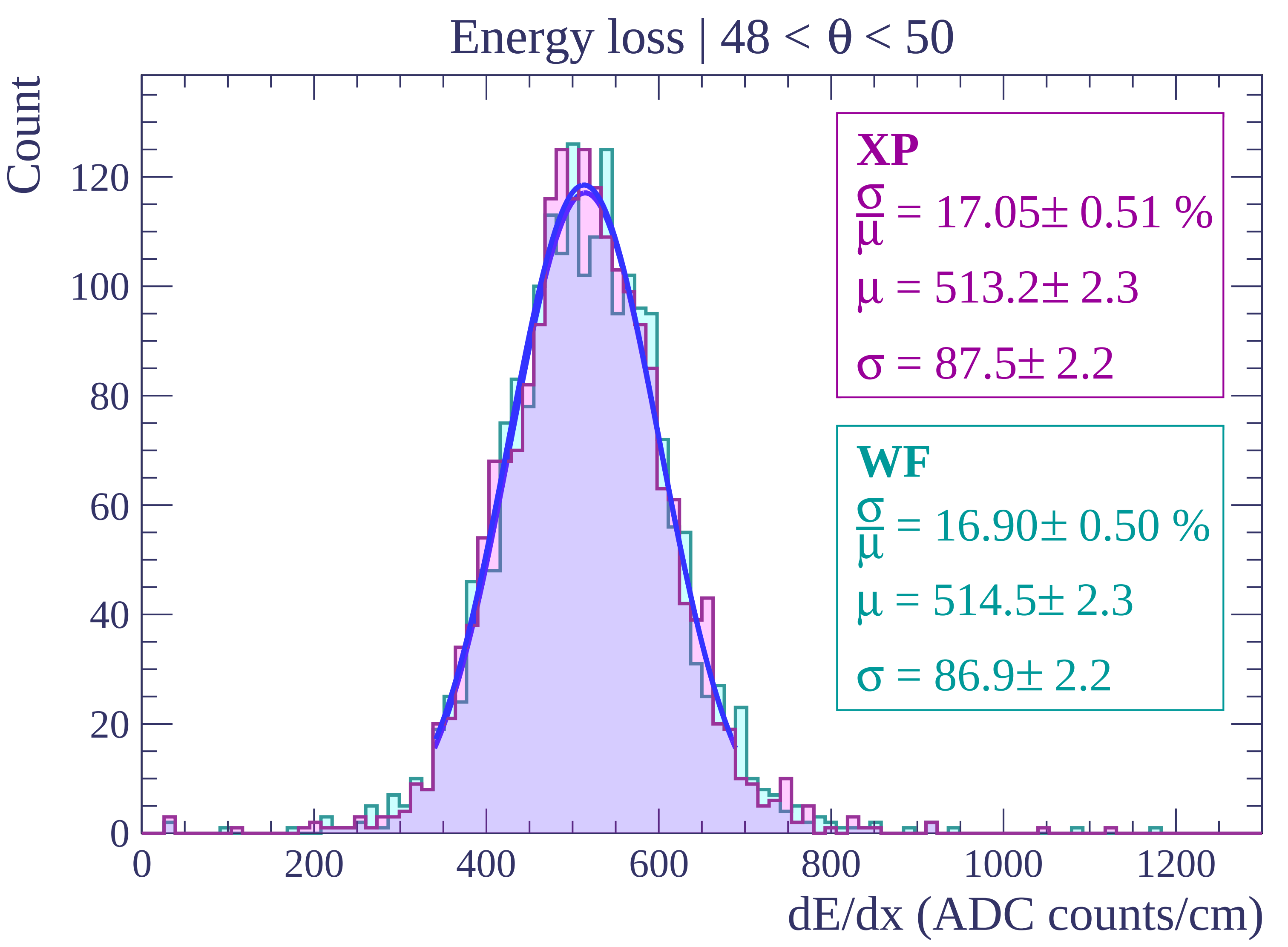


Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 16.60± 0.33 %
μ = 515.2± 1.8
σ = 85.5± 1.4
WF
$\frac{\sigma}{\mu}$ = 16.97± 0.36 %
μ = 517.3± 1.8
σ = 87.8± 1.5

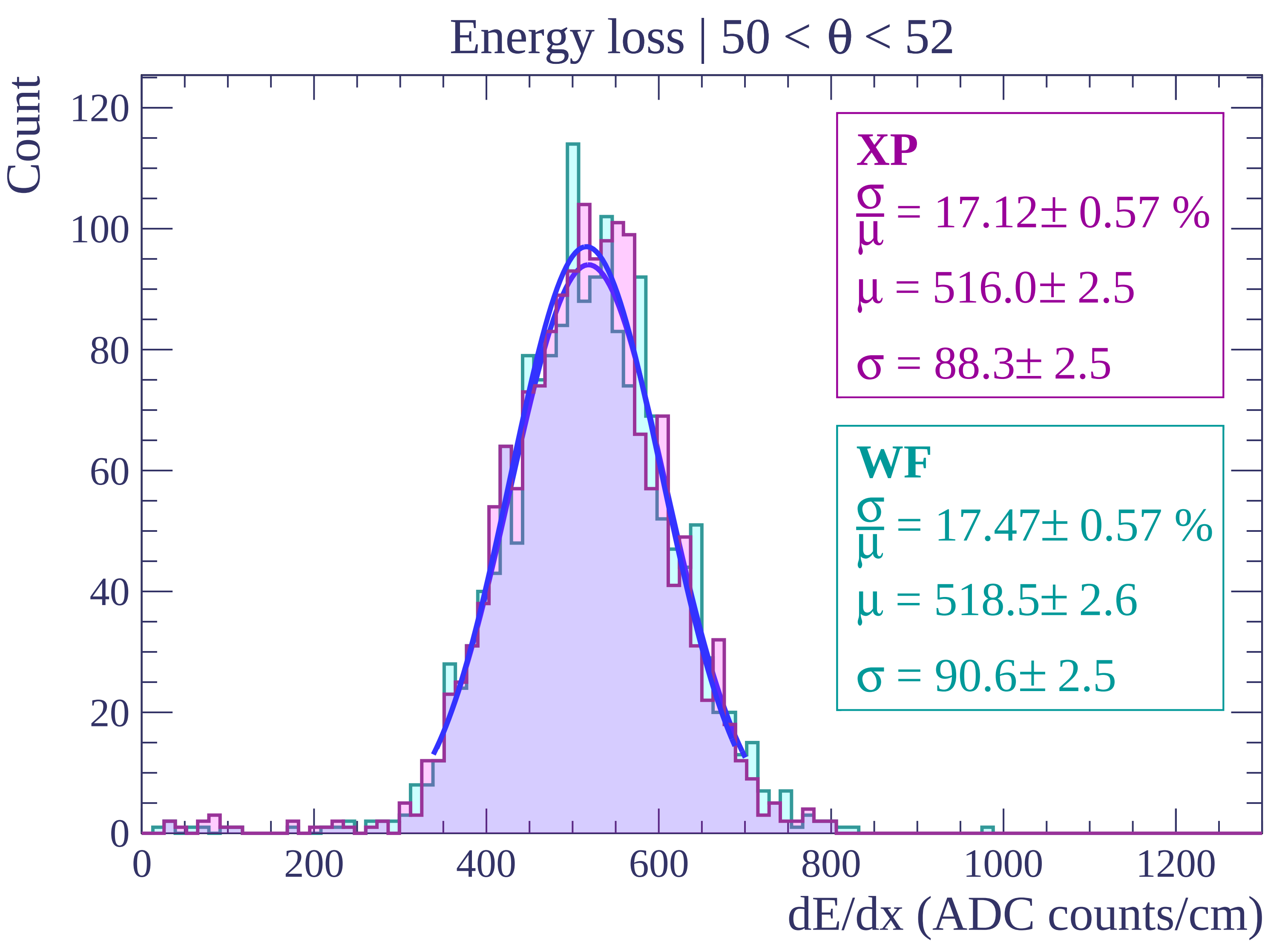


Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
σ/μ = 17.00± 0.42 %
μ = 513.4± 1.9
σ = 87.3± 1.8
WF
σ/μ = 17.44± 0.44 %
μ = 514.6± 2.0
σ = 89.7± 1.9

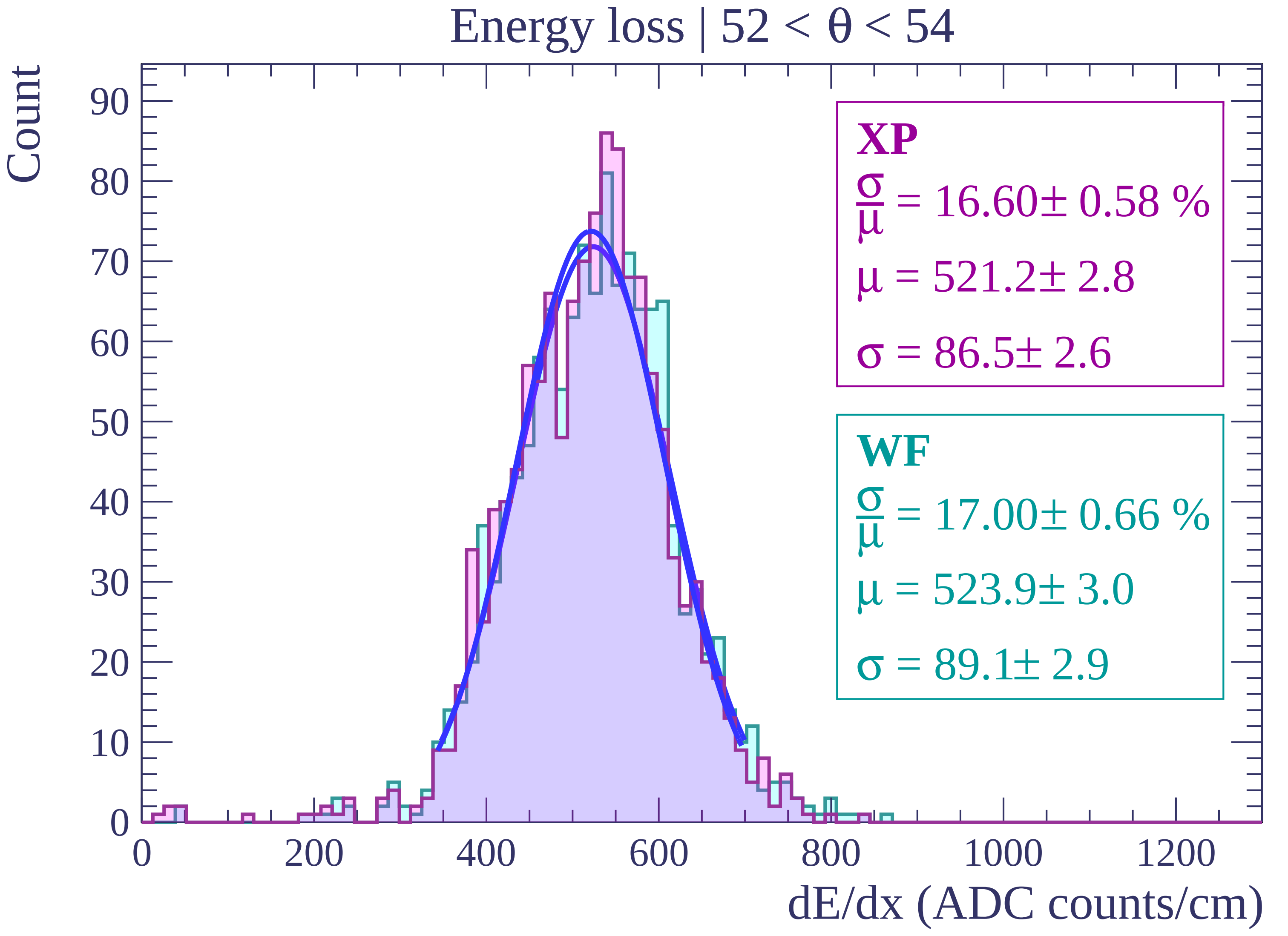

Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 16.80± 0.44 %
μ = 511.6± 2.0
σ = 86.0± 1.9
WF
σ/μ = 16.78± 0.41 %
μ = 515.3± 2.0
σ = 86.5± 1.8



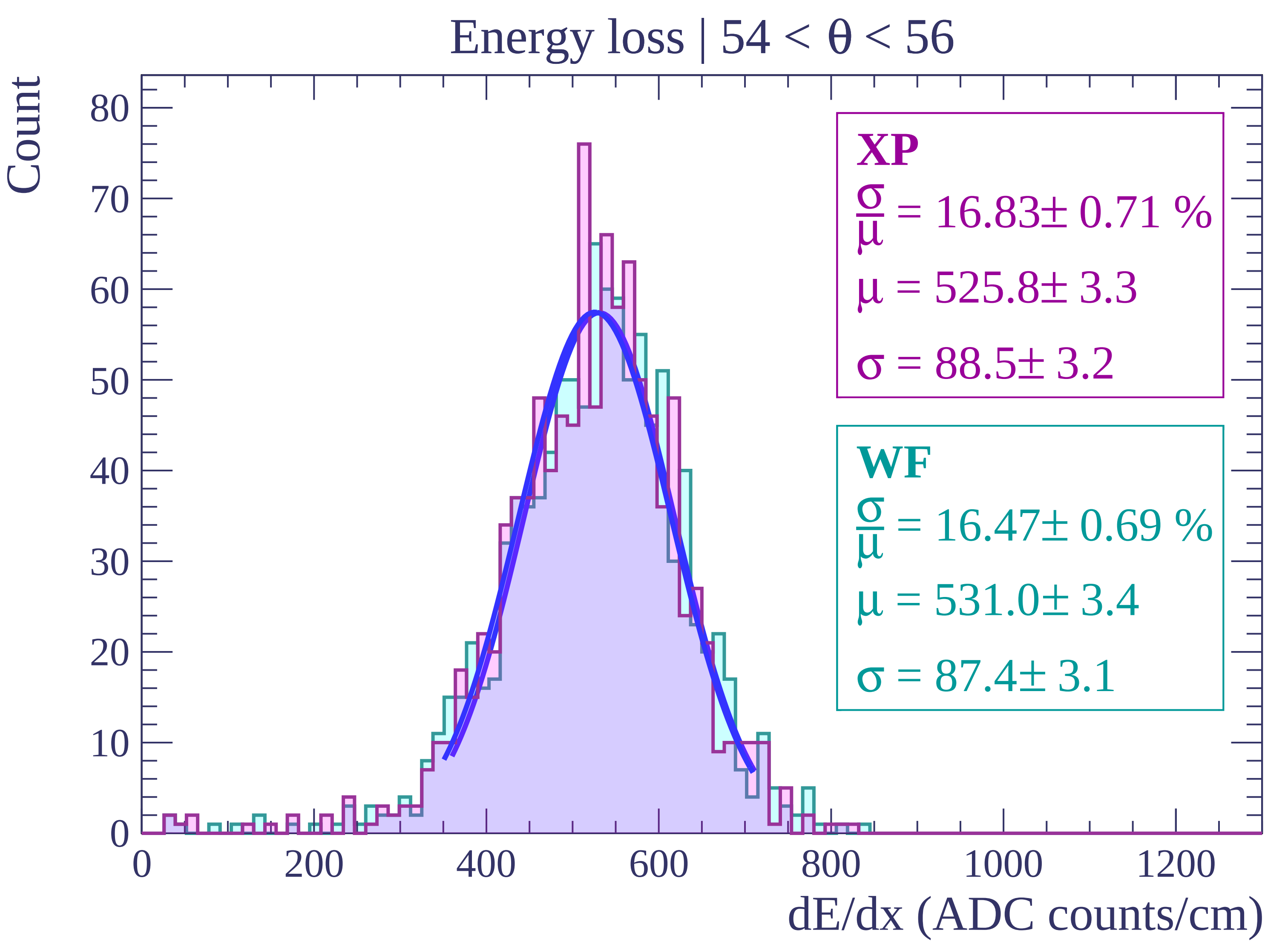
Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 17.05± 0.51 %
μ = 513.2± 2.3
σ = 87.5± 2.2
WF
σ/μ = 16.90± 0.50 %
μ = 514.5± 2.3
σ = 86.9± 2.2
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200



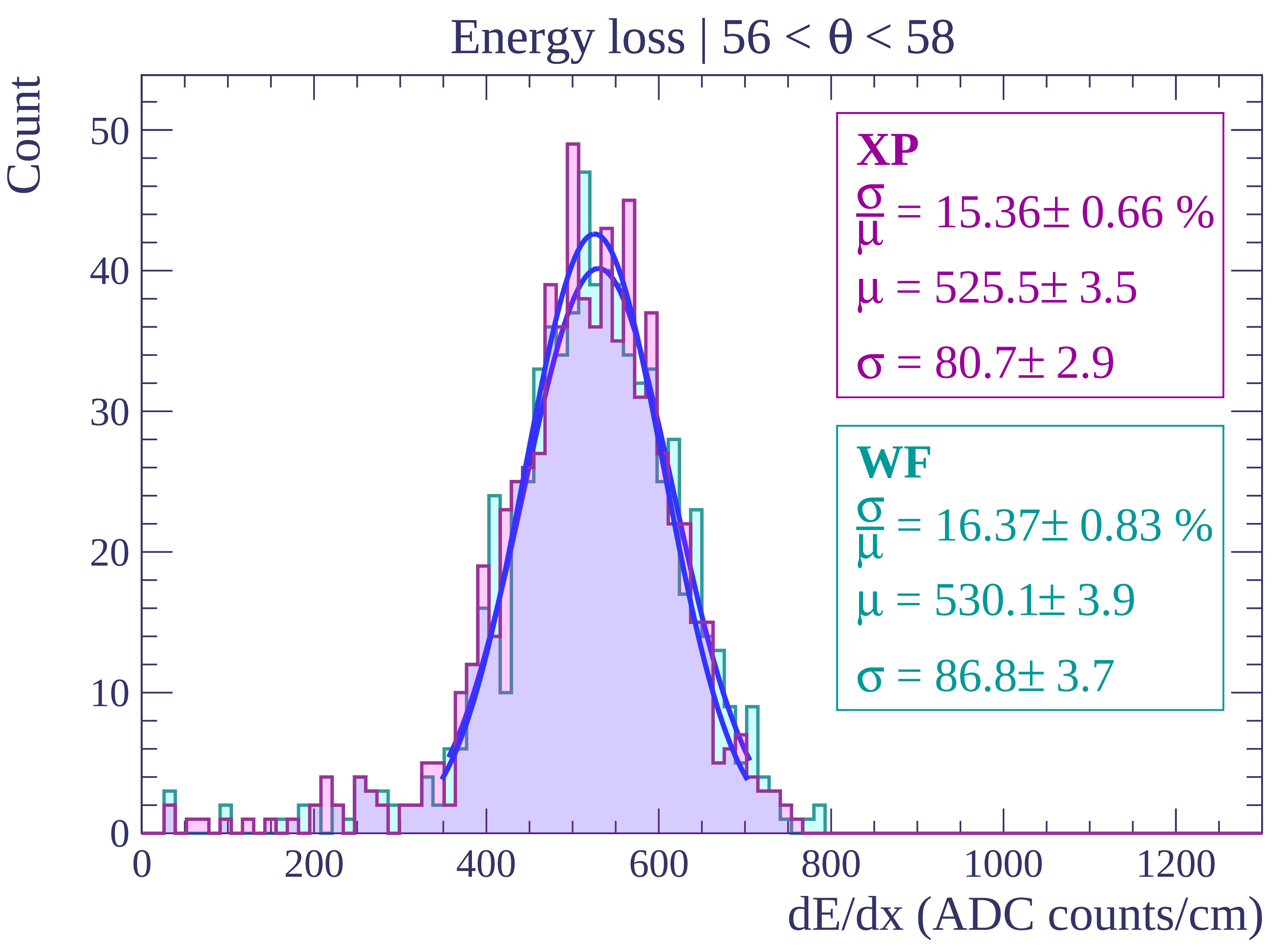
Energy loss | 50 < θ < 52
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
XP
σ/μ = 17.12± 0.57 %
μ = 516.0± 2.5
σ = 88.3± 2.5
WF
σ/μ = 17.47± 0.57 %
μ = 518.5± 2.6
σ = 90.6± 2.5



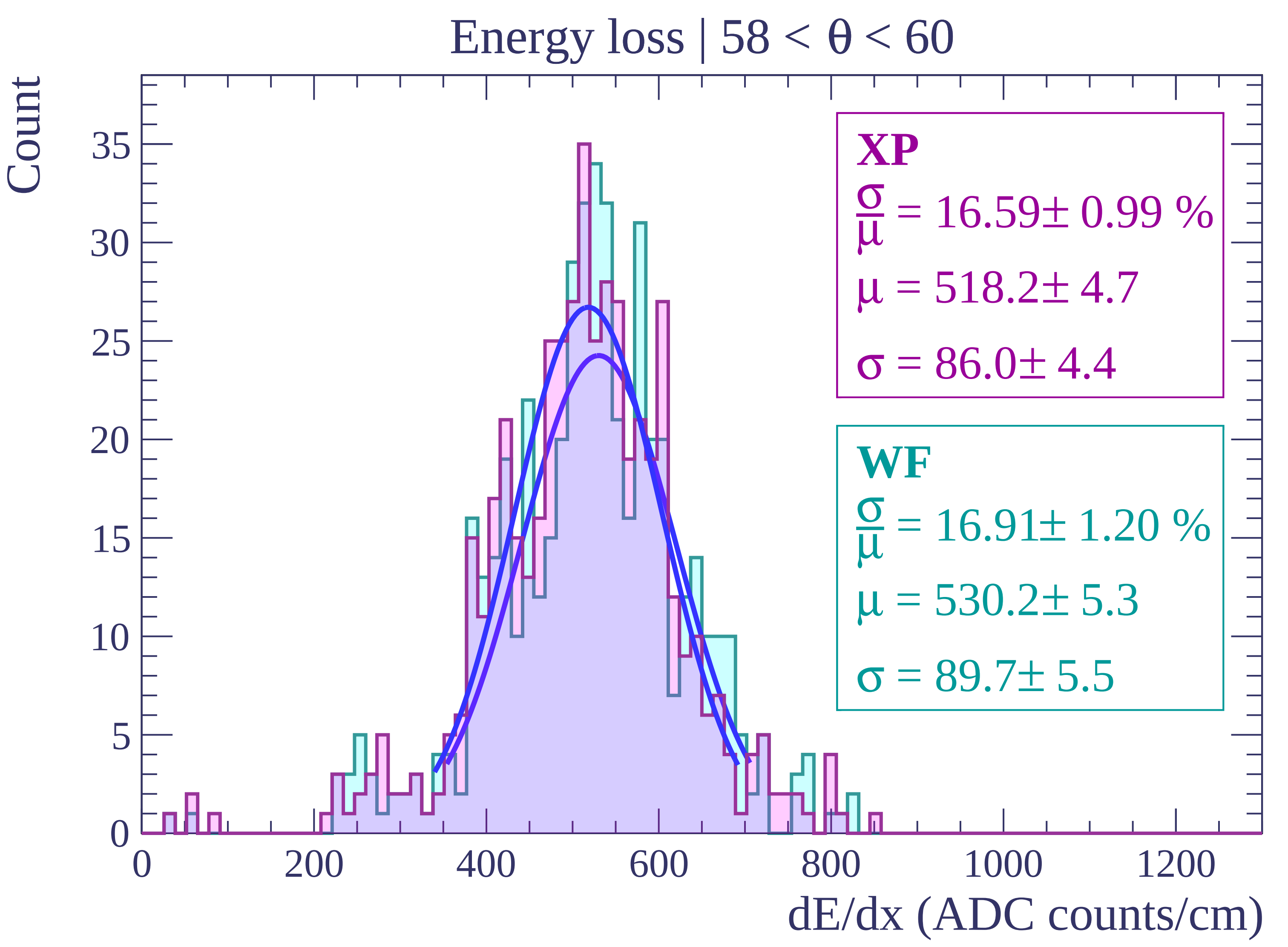
Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
70
80
90
XP
σ/μ = 16.60± 0.58 %
μ = 521.2± 2.8
σ = 86.5± 2.6
WF
σ/μ = 17.00± 0.66 %
μ = 523.9± 3.0
σ = 89.1± 2.9

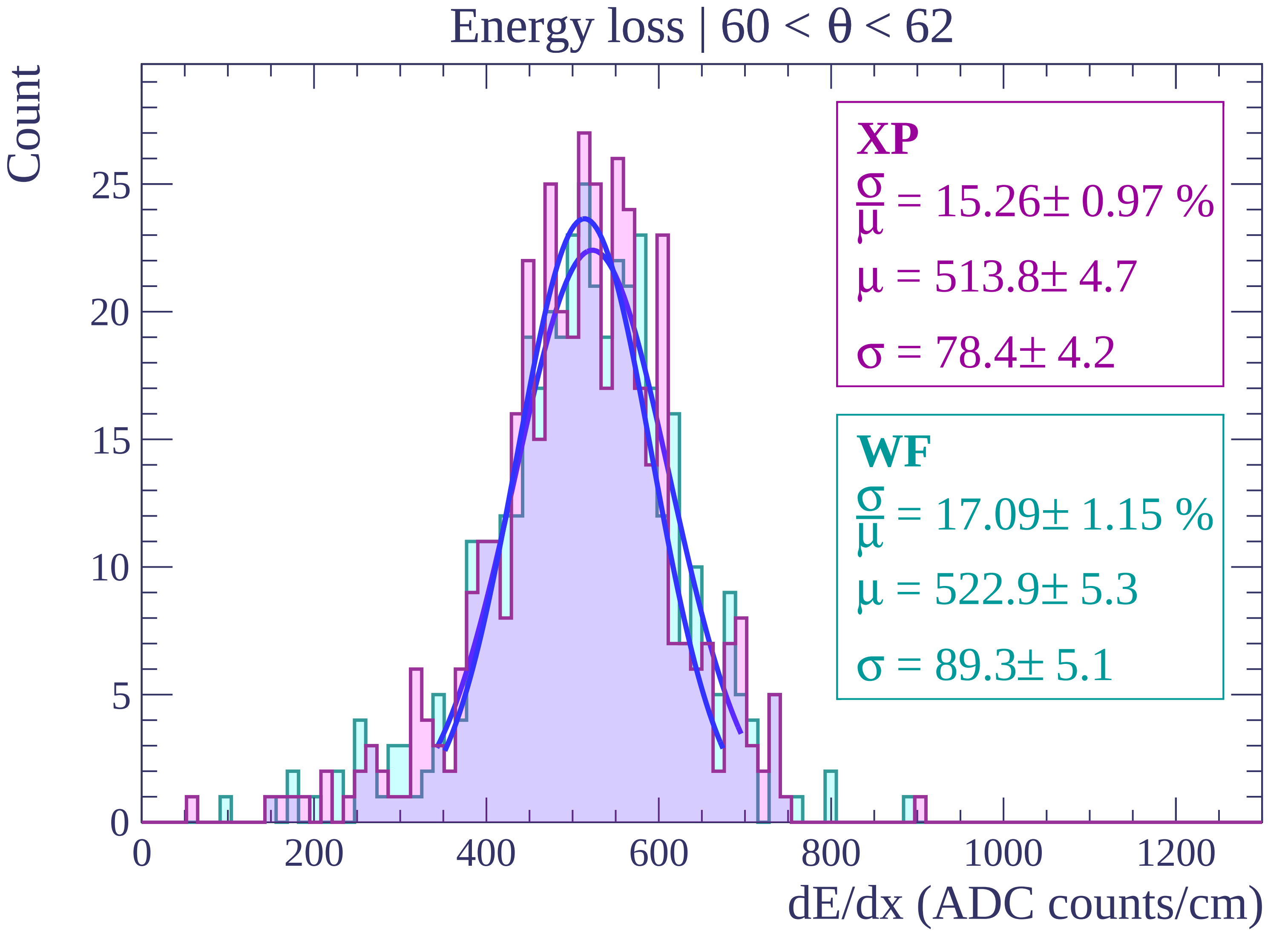


Energy loss | 54 < θ < 56
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.83± 0.71 %
μ = 525.8± 3.3
σ = 88.5± 3.2
WF
σ/μ = 16.47± 0.69 %
μ = 531.0± 3.4
σ = 87.4± 3.1
0
10
20
30
40
50
60
70
80
0
200
400
600
800
1000
1200

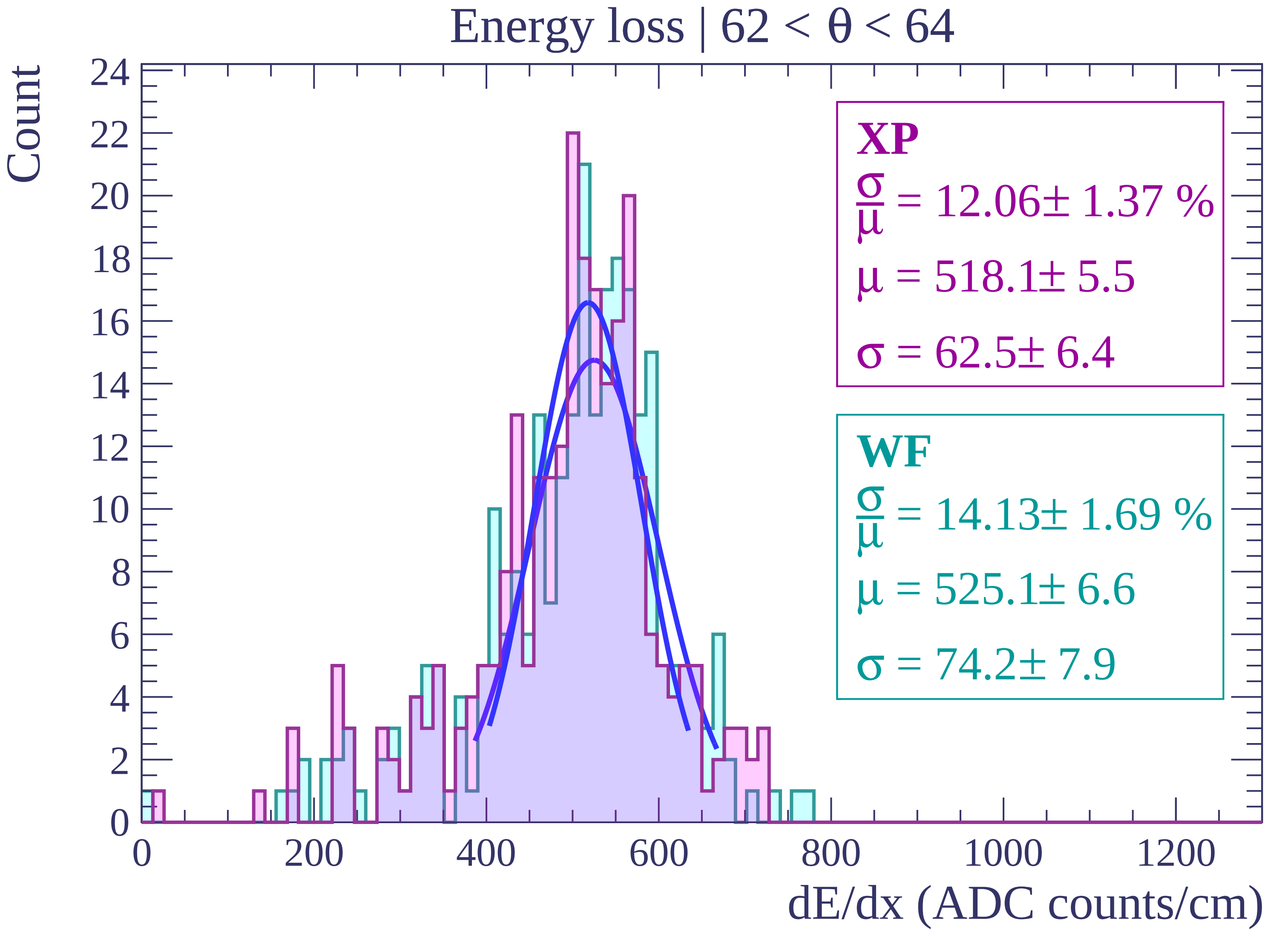


Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.36± 0.66 %
μ = 525.5± 3.5
σ = 80.7± 2.9
WF
$\frac{\sigma}{\mu}$ = 16.37± 0.83 %
μ = 530.1± 3.9
σ = 86.8± 3.7

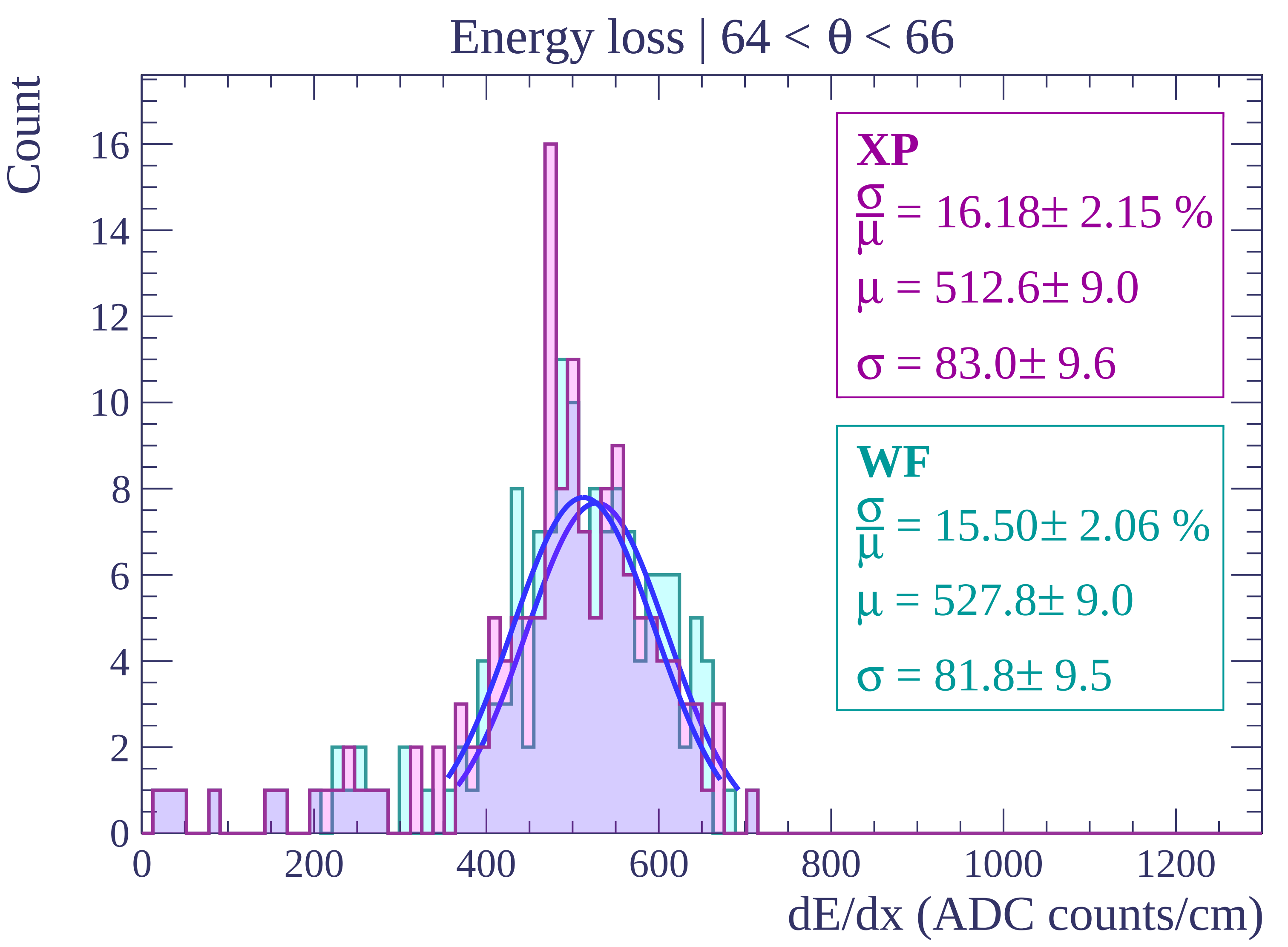

Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
35
0
200
400
600
800
1000
1200
XP
σ/μ = 16.59± 0.99 %
μ = 518.2± 4.7
σ = 86.0± 4.4
WF
σ/μ = 16.91± 1.20 %
μ = 530.2± 5.3
σ = 89.7± 5.5



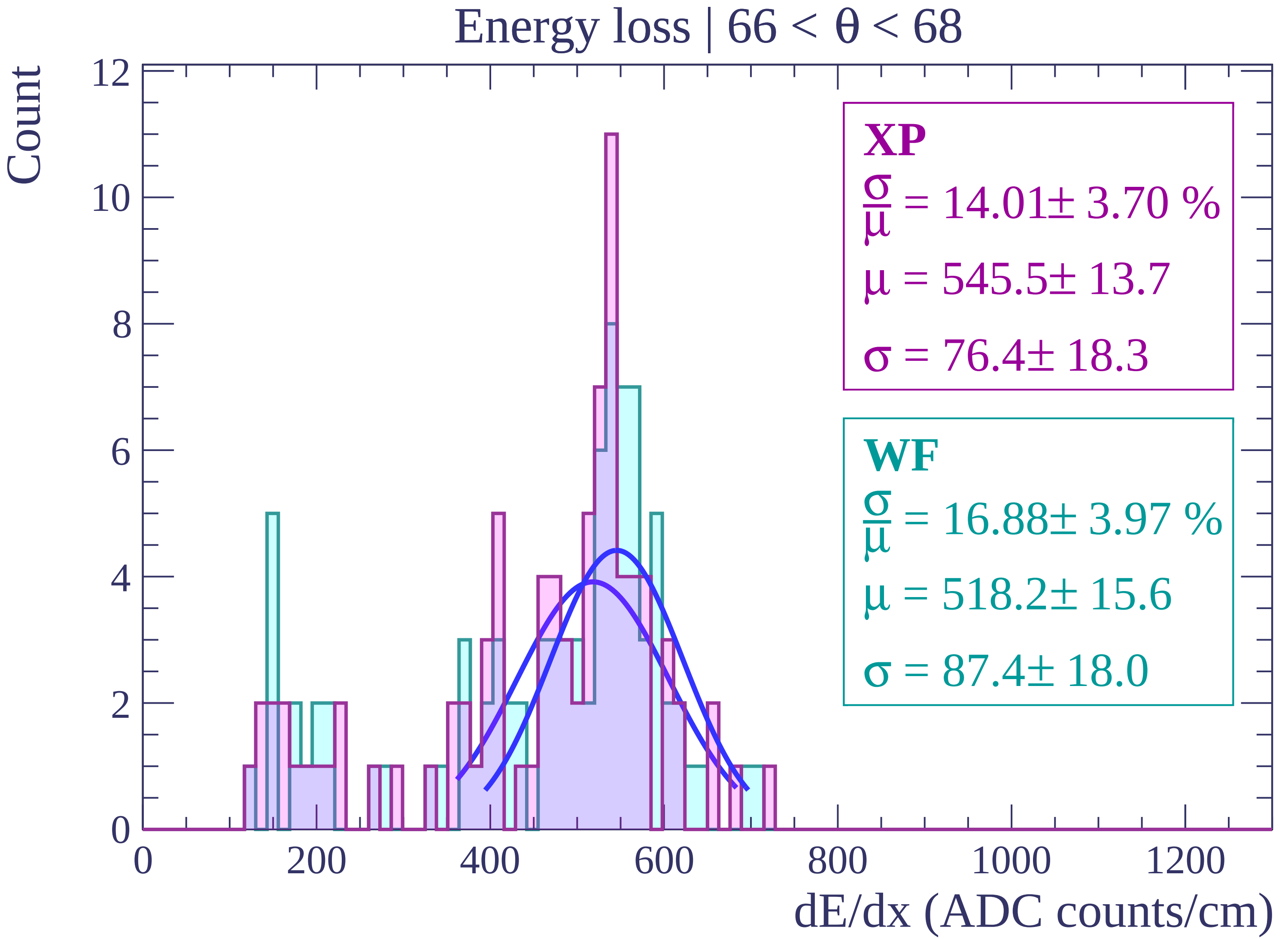
Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
5
10
15
20
25
XP
σ/μ = 15.26± 0.97 %
μ = 513.8± 4.7
σ = 78.4± 4.2
WF
σ/μ = 17.09± 1.15 %
μ = 522.9± 5.3
σ = 89.3± 5.1



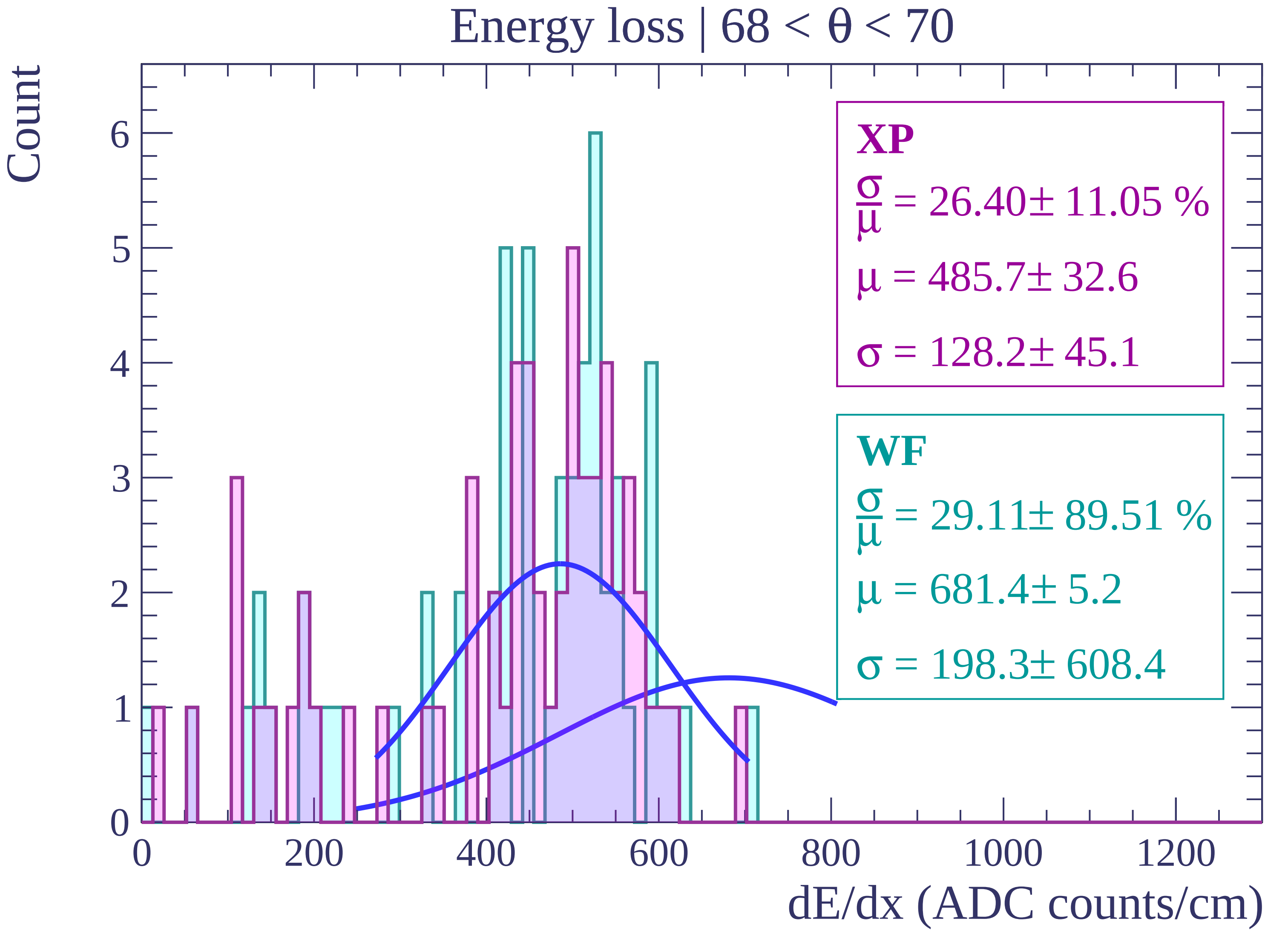
Energy loss | 62 < θ < 64
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
18
20
22
24
XP
σ/μ = 12.06± 1.37 %
μ = 518.1± 5.5
σ = 62.5± 6.4
WF
σ/μ = 14.13± 1.69 %
μ = 525.1± 6.6
σ = 74.2± 7.9



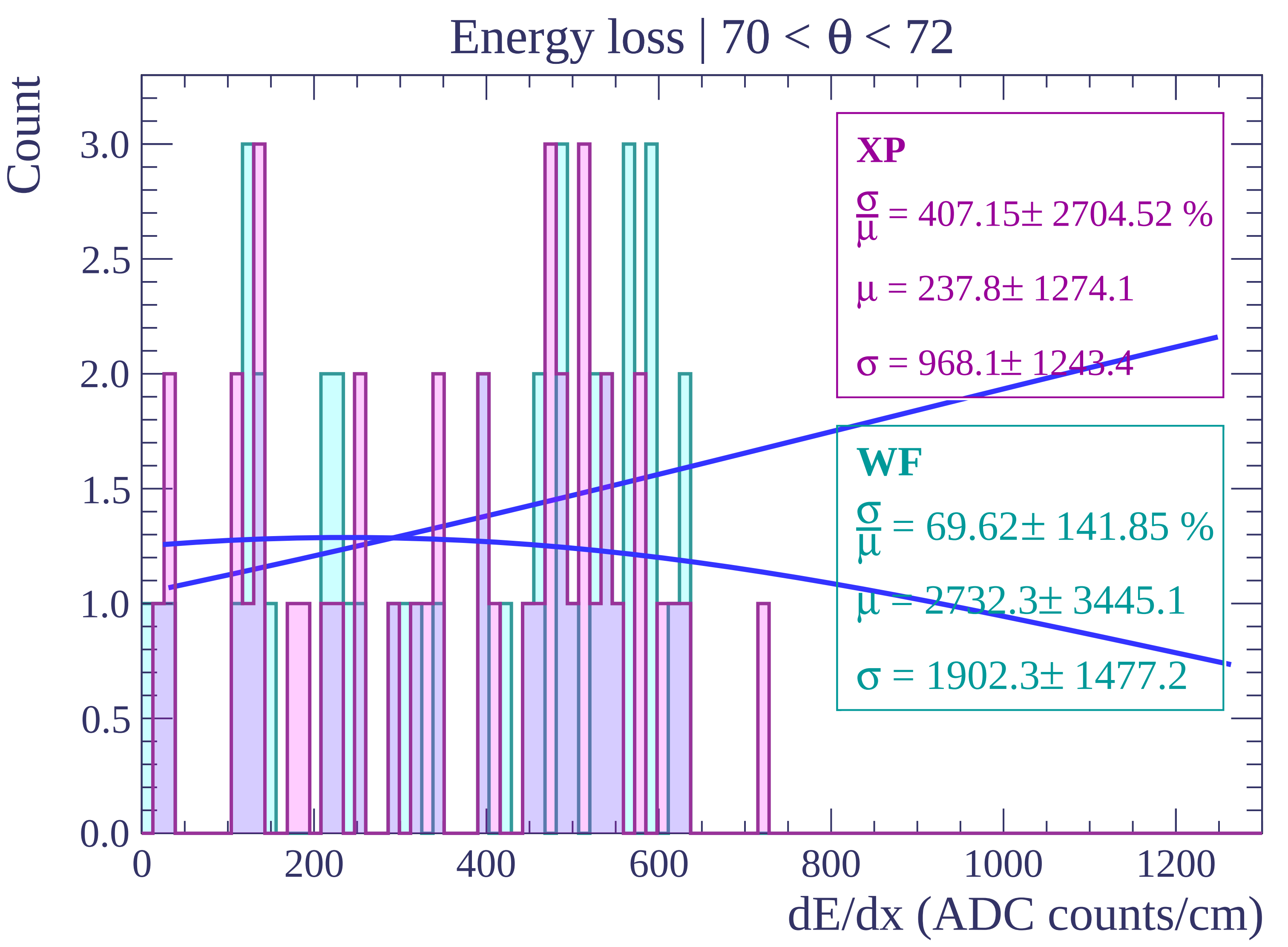
Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 16.18± 2.15 %
μ = 512.6± 9.0
σ = 83.0± 9.6
WF
$\frac{\sigma}{\mu}$ = 15.50± 2.06 %
μ = 527.8± 9.0
σ = 81.8± 9.5

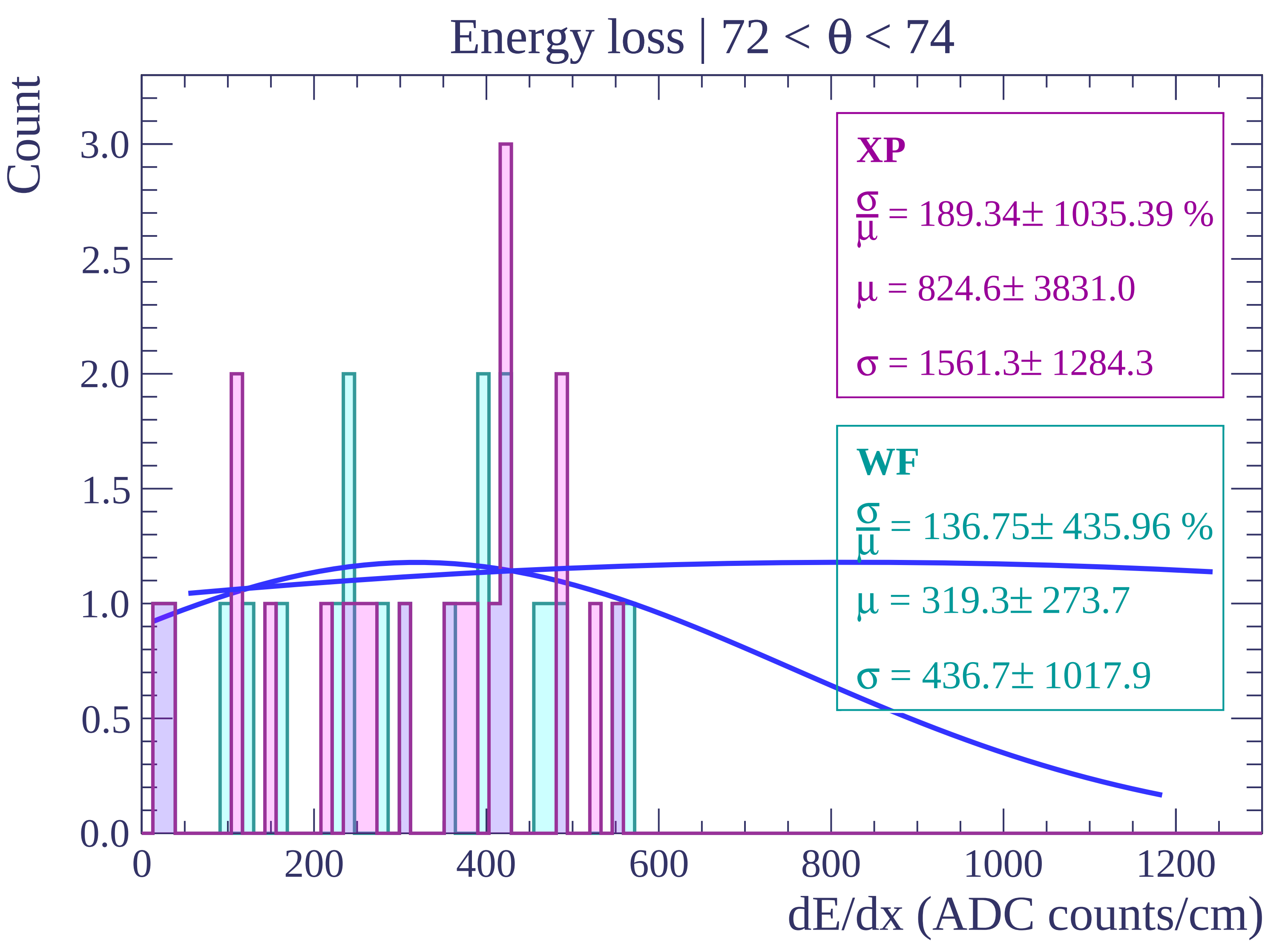


Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 14.01± 3.70 %
μ = 545.5± 13.7
σ = 76.4± 18.3
WF
σ/μ = 16.88± 3.97 %
μ = 518.2± 15.6
σ = 87.4± 18.0

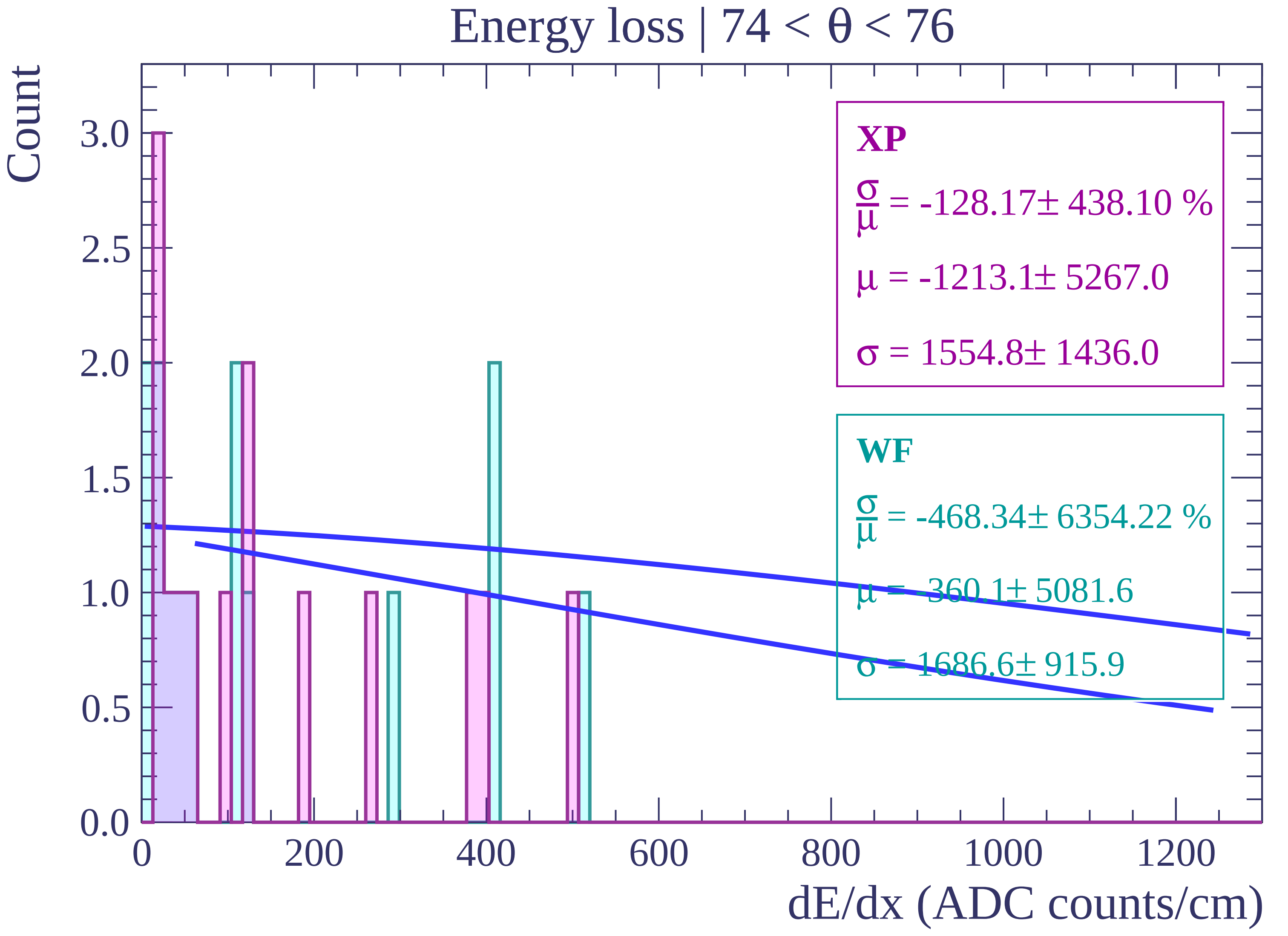


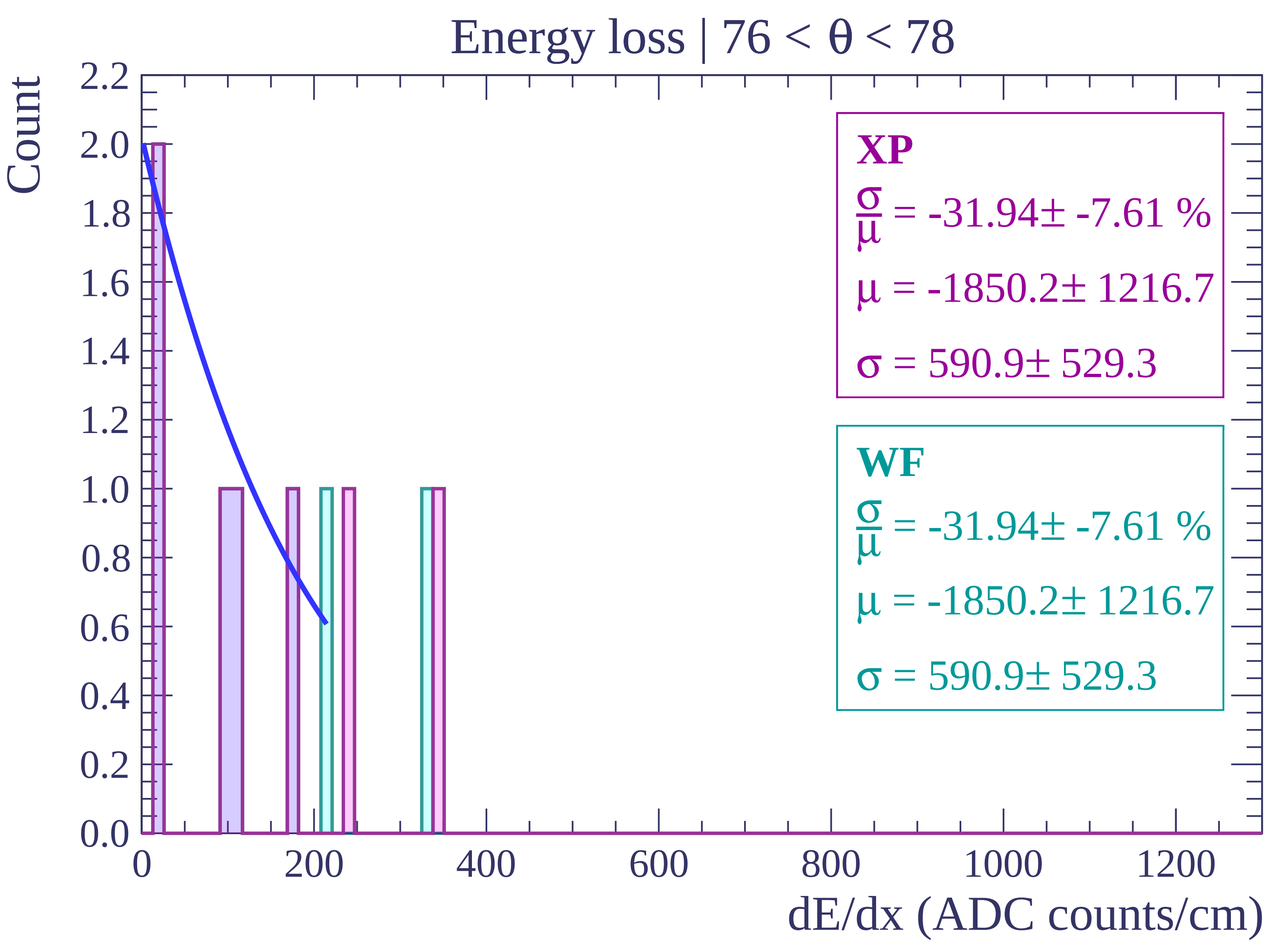
Energy loss | 68 < θ < 70
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
6
0
200
400
600
800
1000
1200
XP
σ/μ = 26.40± 11.05 %
μ = 485.7± 32.6
σ = 128.2± 45.1
WF
σ/μ = 29.11± 89.51 %
μ = 681.4± 5.2
σ = 198.3± 608.4


Energy loss | 70 < θ < 72
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 407.15± 2704.52 %
μ = 237.8± 1274.1
σ = 968.1± 1243.4
WF
σ/μ = 69.62± 141.85 %
μ = 2732.3± 3445.1
σ = 1902.3± 1477.2

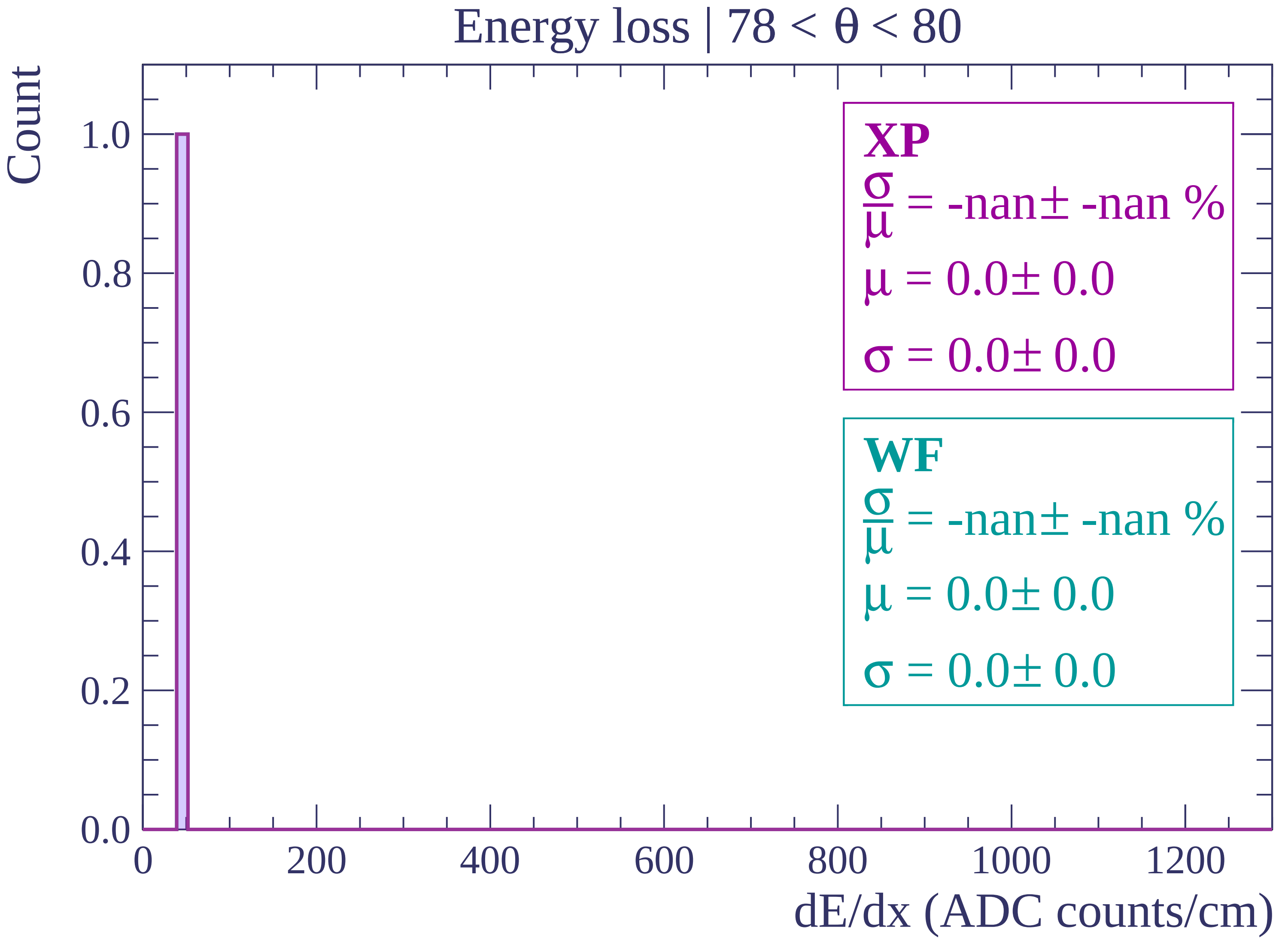


Energy loss | 72 < θ < 74
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 189.34± 1035.39 %
μ = 824.6± 3831.0
σ = 1561.3± 1284.3
WF
σ/μ = 136.75± 435.96 %
μ = 319.3± 273.7
σ = 436.7± 1017.9

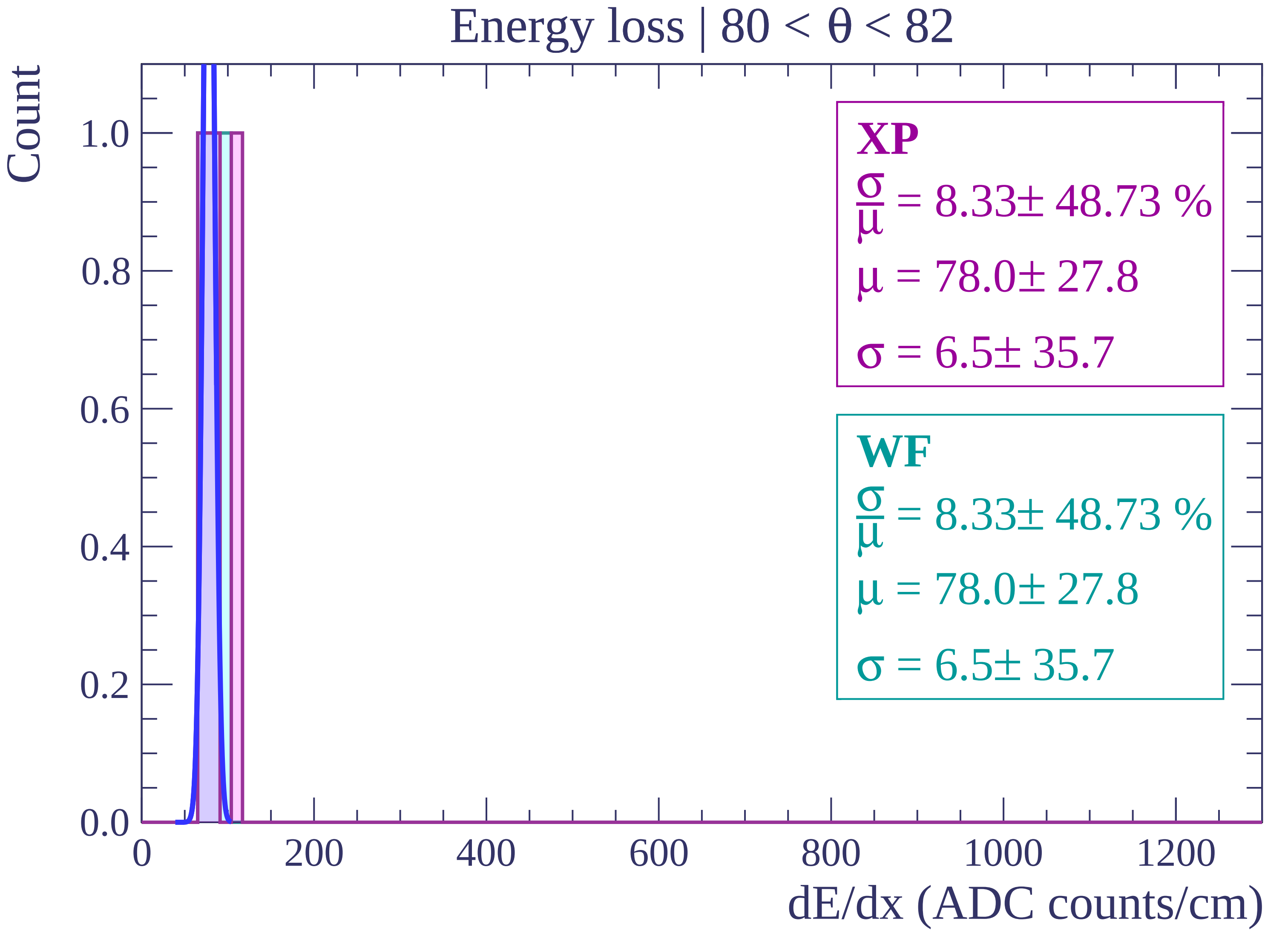

Energy loss | 74 < θ < 76
Count
dE/dx (ADC counts/cm)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
0
200
400
600
800
1000
1200
XP
σ/μ = -128.17± 438.10 %
μ = -1213.1± 5267.0
σ = 1554.8± 1436.0
WF
σ/μ = -468.34± 6354.22 %
μ = -360.1± 5081.6
σ = 1686.6± 915.9



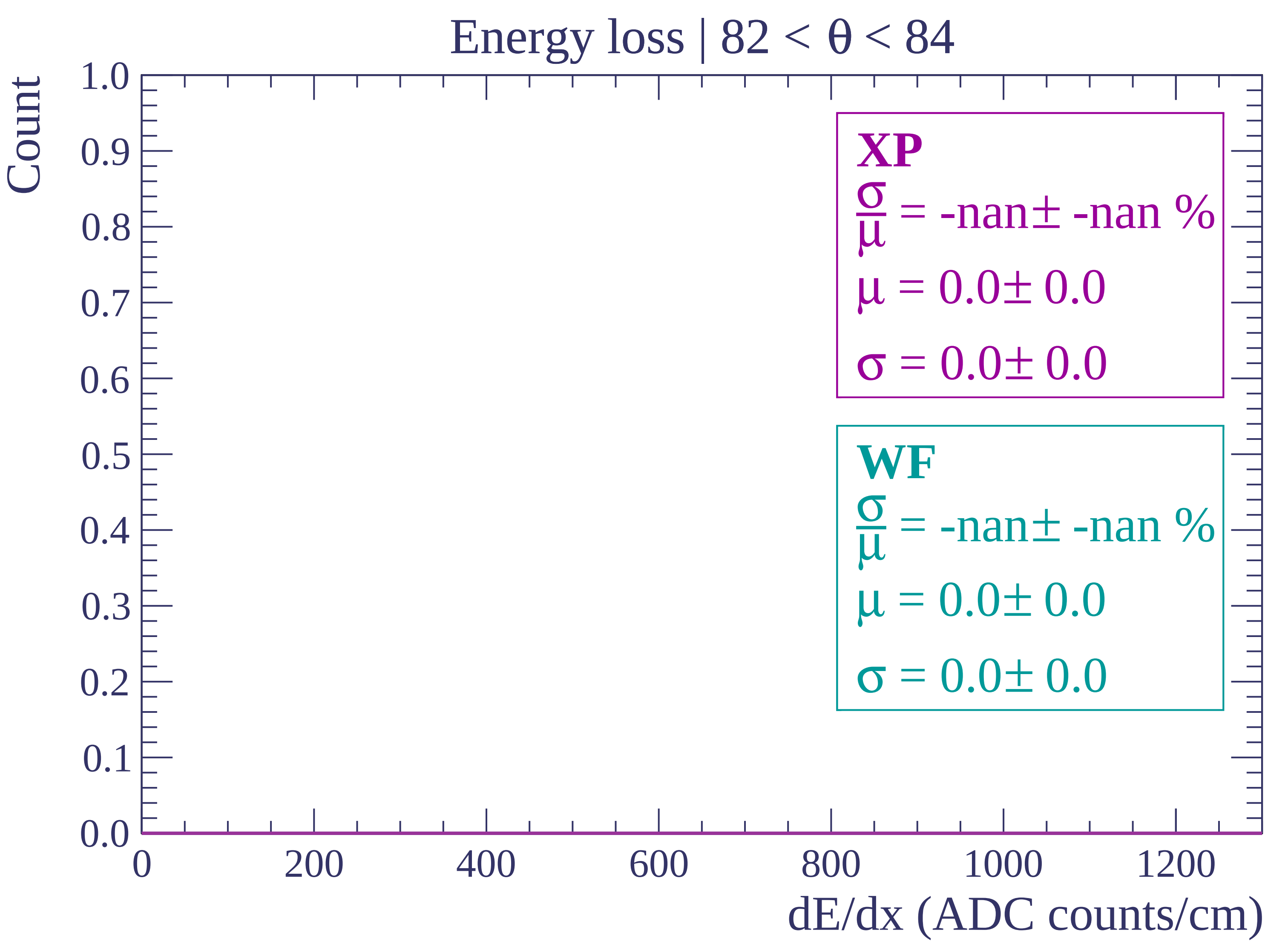
Energy loss | 76 < θ < 78
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -31.94± -7.61 %
μ = -1850.2± 1216.7
σ = 590.9± 529.3
WF
$\frac{\sigma}{\mu}$ = -31.94± -7.61 %
μ = -1850.2± 1216.7
σ = 590.9± 529.3

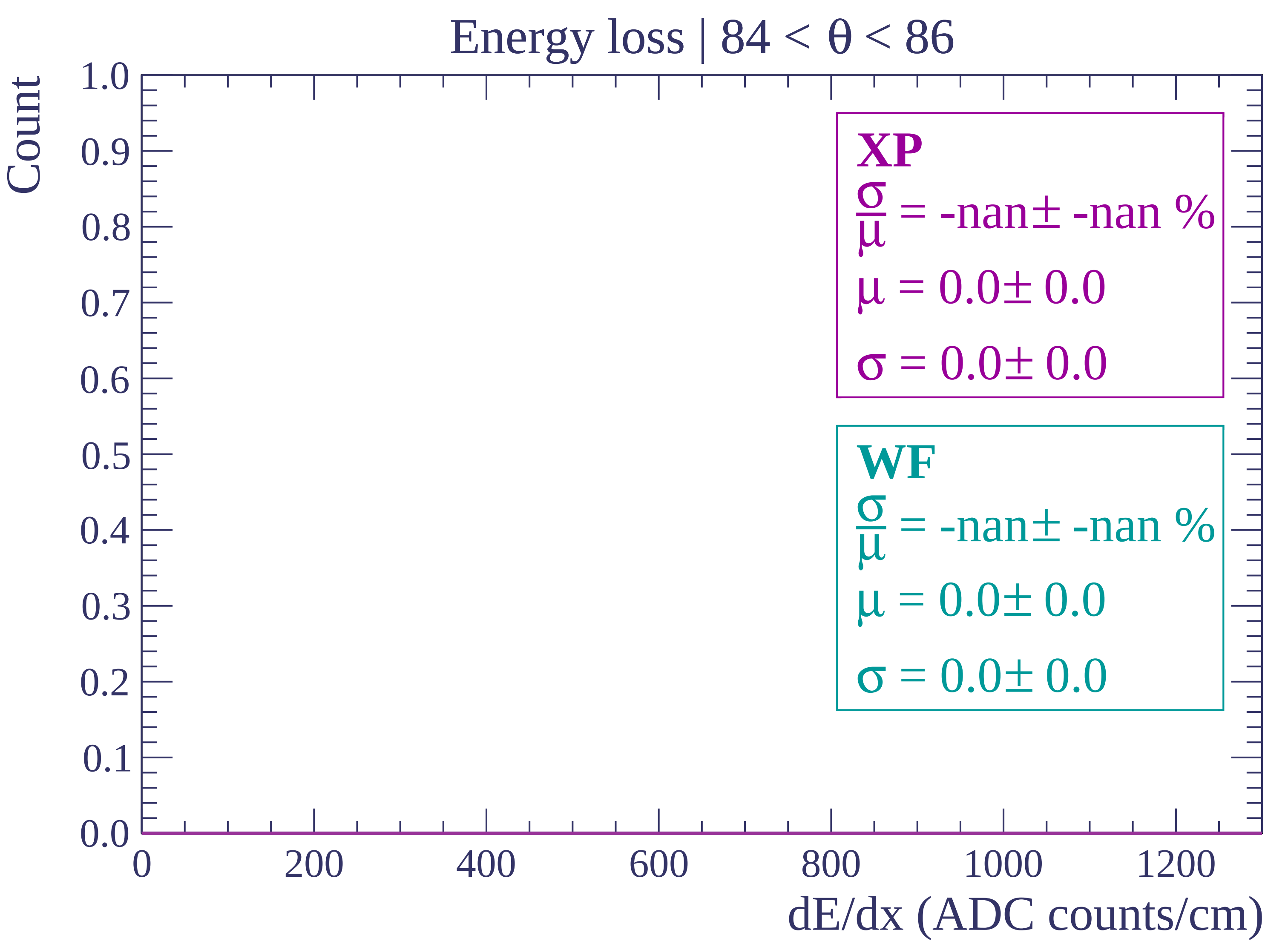


Energy loss | 78 < θ < 80
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

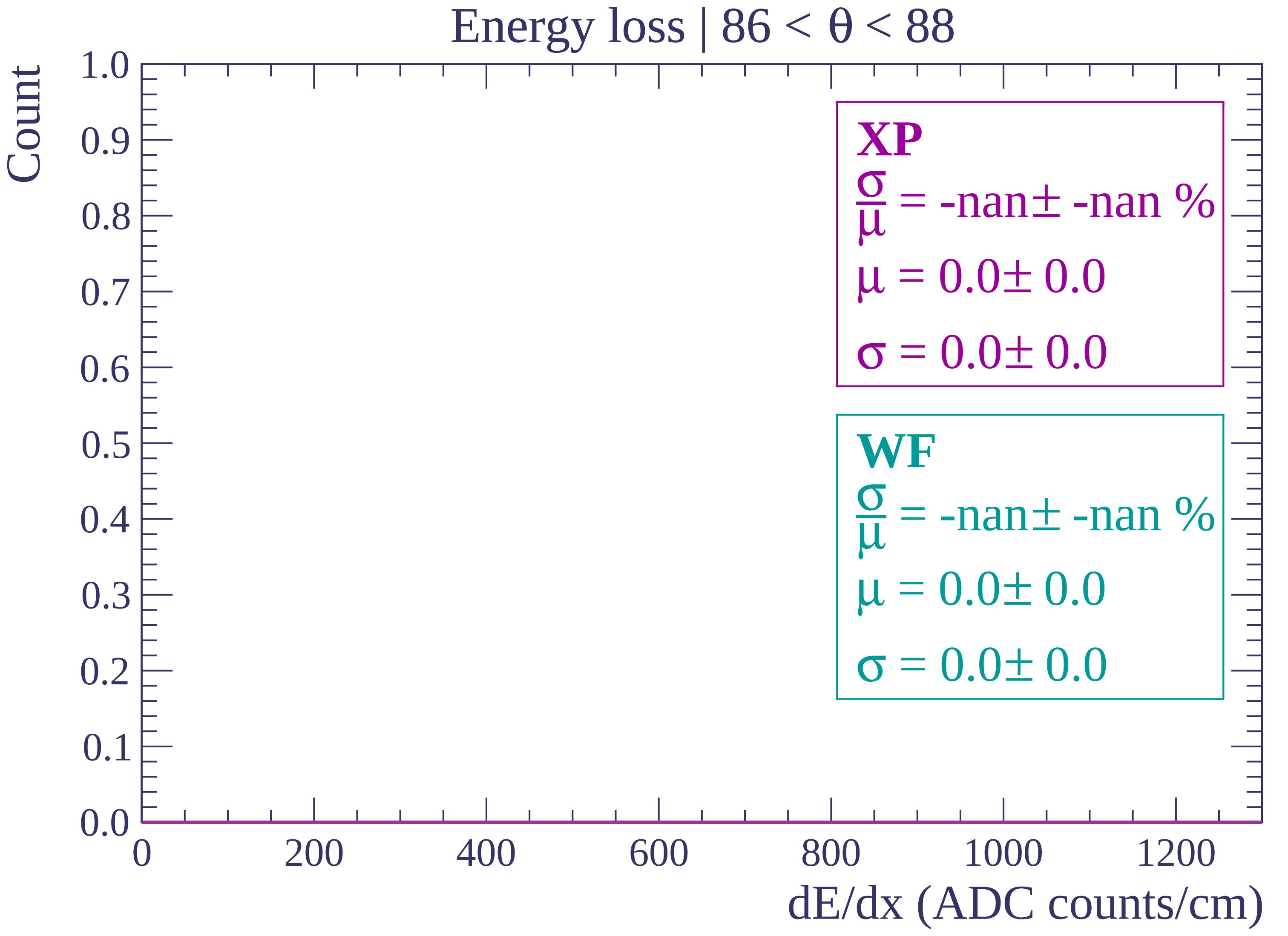


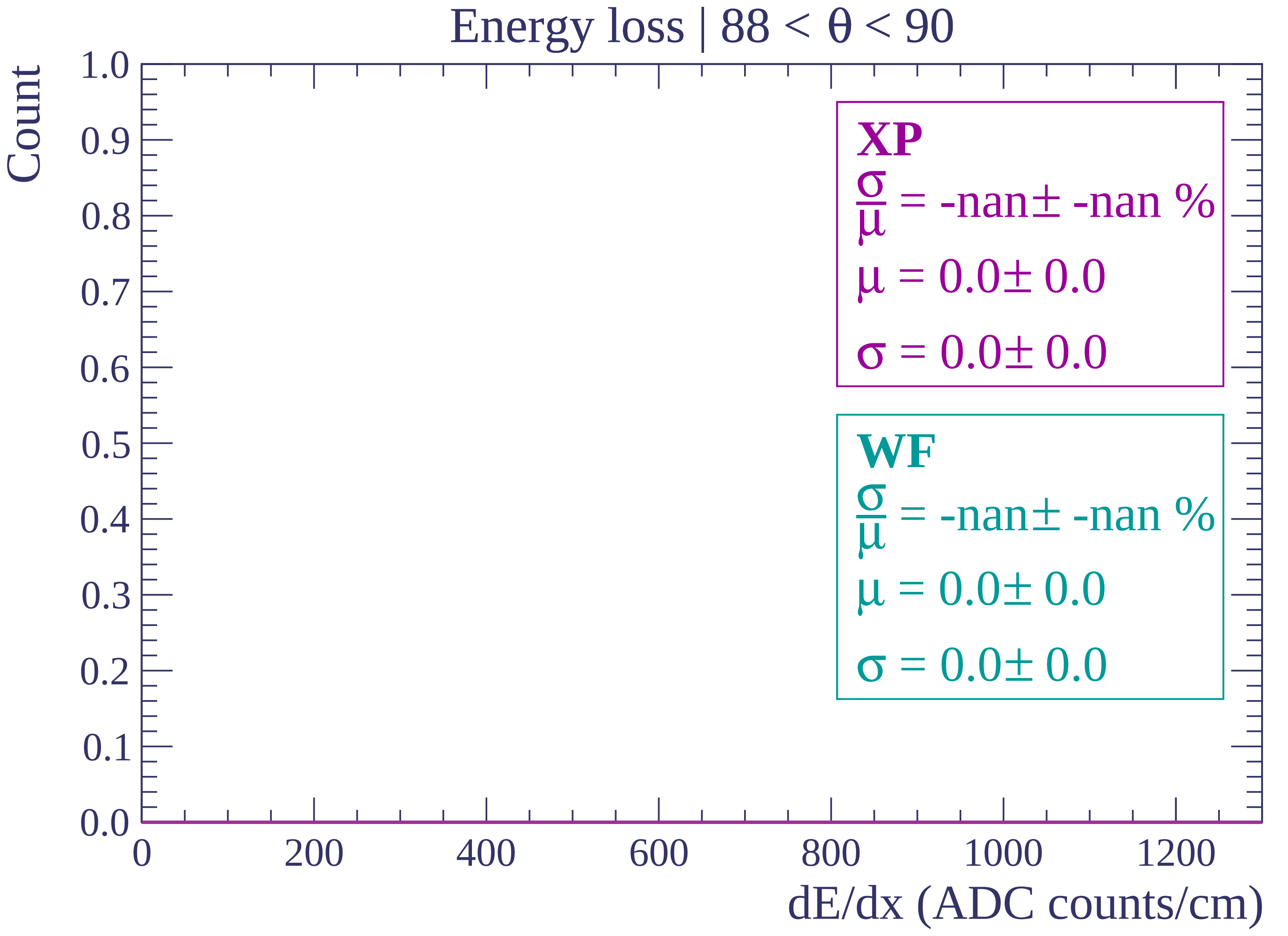
Energy loss | 80 < θ < 82
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.33± 48.73 %
μ = 78.0± 27.8
σ = 6.5± 35.7
WF
σ/μ = 8.33± 48.73 %
μ = 78.0± 27.8
σ = 6.5± 35.7

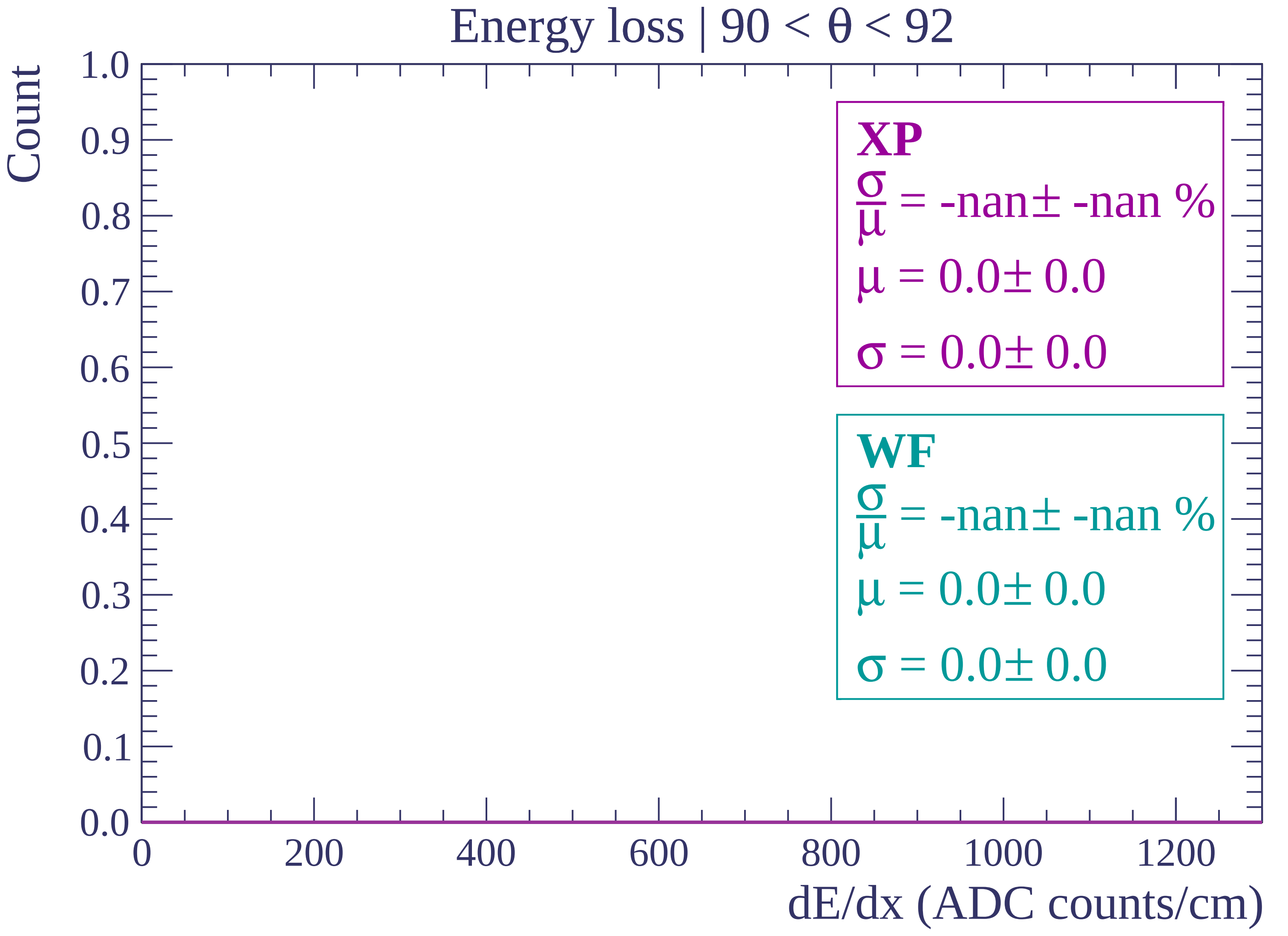

Energy loss | 82 < θ < 84
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

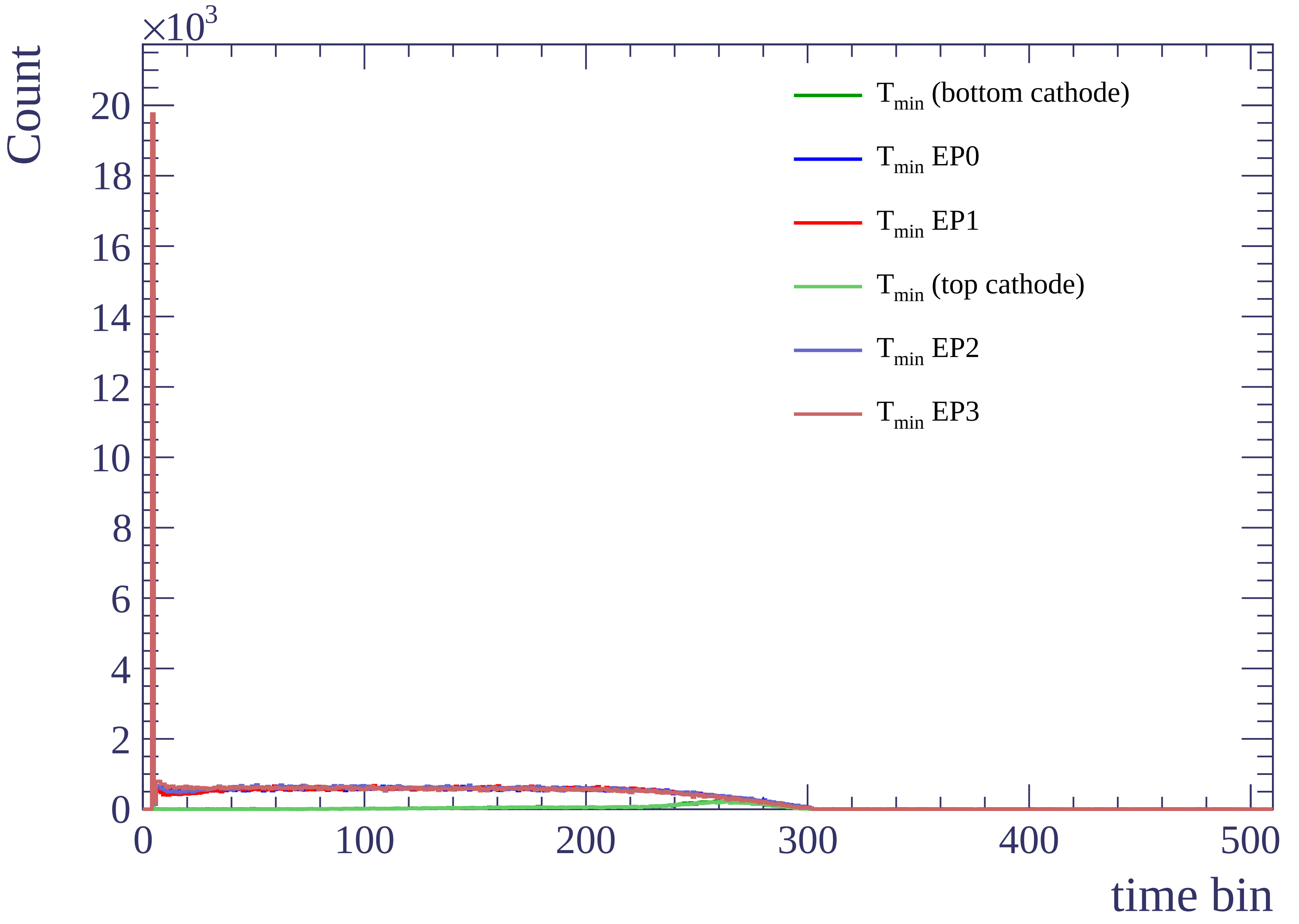


Energy loss | 84 < θ < 86
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

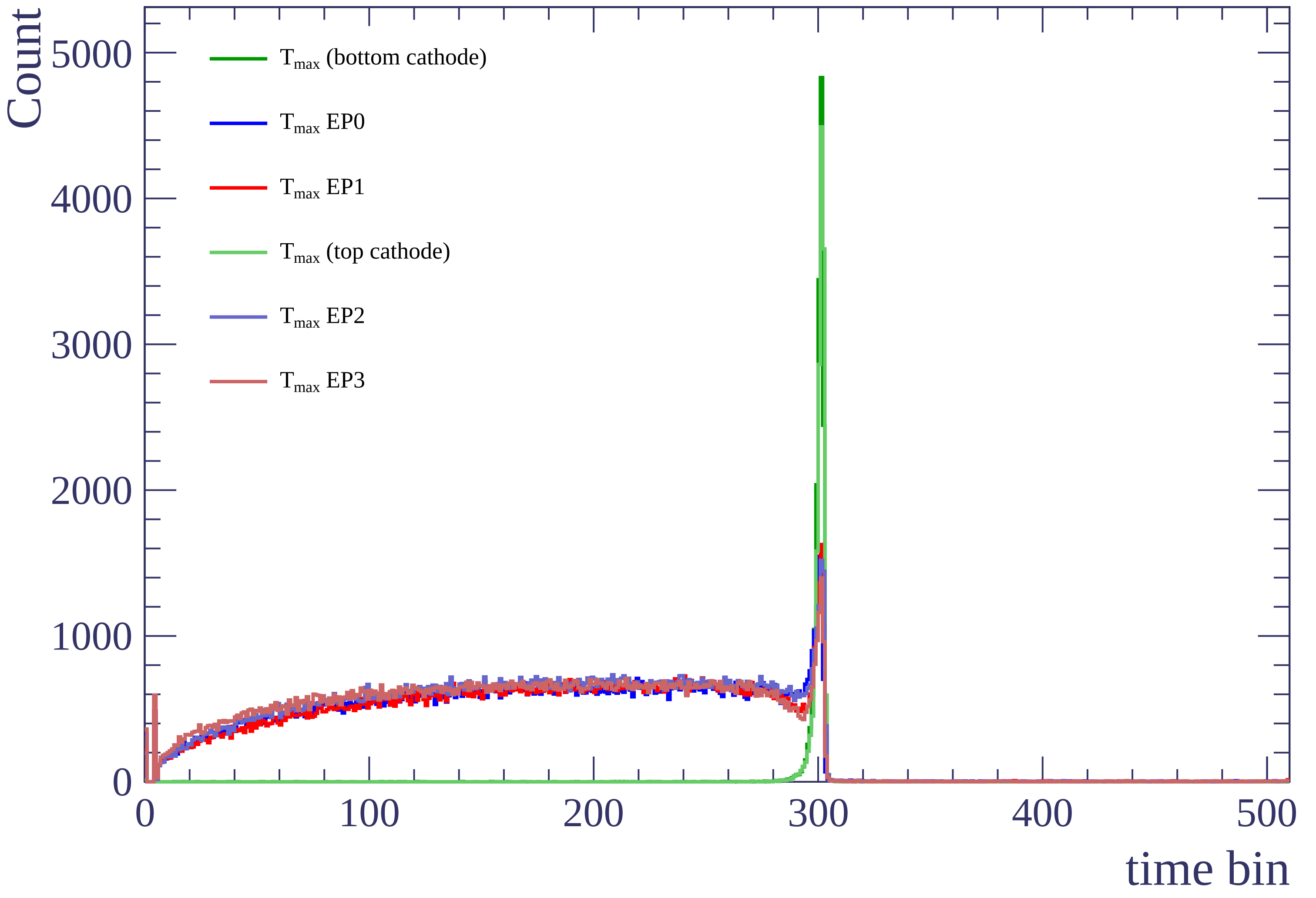

Energy loss | 86 < θ < 88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 88 < θ < 90
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 90 < θ < 92
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



$\times 10^3$
Count
0
2
4
6
8
10
12
14
16
18
20
0
100
200
300
400
500
time bin
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
1000
2000
3000
4000
5000
0
100
200
300
400
500